994. The total settlement of a compressible soil stratum 2 m deep and having a coefficient of volume compressibility of 0.02 cm^2/kg under a pressure increment of 2 kg/cm^2 will be

(a) 2 cm (b) 4 cm
(c) 8 cm (d) 10 cm

APPSC AEE 2012

Ans. (c) : Coefficient of volume compressibility
$m_v = 0.02\ cm^2/kg$
$\Delta p = 2 kg/cm^2$
Soil depth (H) = 2m
Total settlement of soil $\Rightarrow$
$\Delta h = H m_v \Delta p$
$\Delta h = 2 \times 0.02 \times 2$
$\Delta h = 0.08$ m
$\Delta h = 8$ cm

995. Which one of the following clays behaves like dense sand ?

(a) Over – consolidated clay with a high over – consolidation ratio
(b) Over – consolidated clay with a low over – consolidation ratio
(c) Normally consolidated clay
(d) Under - consolidated clay

APPSC AEE 2012
ESE 1998

Ans. (a) : Over-consolidated clay with a high over consolidation ratio behaves like dense sand

$$\text{over consolidation ratio} = \frac{\text{Pre consolidation pressure}}{\text{current effective stress}}$$

996. A clay soil, tested in a consolidometer, showed a decrease in void ratio from 1.20 to 1.10 in application of pressure 0.25 to 0.5 Kgf/cm^2. If the sample tested at the size was taken from a clay layer 3m in thickness, then consolidation settlement is :

(a) 0.136 m (b) 0.145 m
(c) 0.175 m (d) 0.081 m

TRB Poly. Lect. Tech. 2017

Ans. (a) : Given,
Thickness of clay layer (H_0) = 3 m
Initial void ratio, $e_0 = 1.2$
Final void ratio $e_f = 1.1$

$$\frac{\Delta H}{H_0} = \frac{\Delta e}{1+e_0}$$

$$\frac{\Delta H}{3} = \frac{1.2-1.1}{1+1.2}$$

Consolidation settlement, $\Delta H = 0.136$ m

997. An embankment is proposed to construct on 10m thick clay deposit having the coefficient of volume compressibility as $2.7 \times 10^{-4}\ m^2/kN$. Determine the settlement of clay by the increase in vertical stress of 60 kN/m^2 due to the construction of embankment.

(a) 162 mm (b) 160 mm
(c) 164 mm (d) 163 mm

TRB Poly. Lect. Tech. 2017

Ans. (a) : Given that,
$m_v = 2.7 \times 10^{-4}\ m^2/kN$
$\Delta\sigma = 60\ kN/m^2$
$H_o = 10$ m
Settlement of clay (ΔH)
$\Delta H = m_v \Delta\sigma\ H_0$
$\Delta H = 2.7 \times 10^{-4} \times 60 \times 10$
$\Delta H = 0.162$ m
$\Delta H = 162$ mm

998. Dimensional unit of co-efficient of consolidation is :

(a) $L^2 T^{-1}$ (b) $L^{-1} T^2$
(c) $L^3 T$ (d) LT^{-3}

TRB Poly. Lect. Tech. 2017

Ans. (a)

Coefficient of consolidation $C_v = \frac{T_v d^2}{t} m^2/sec$

$$\frac{m^2}{sec} = \frac{L^2}{T} = L^2T^{-1}$$

T_v = Time factor
d = length of drainage path
t = Time required for any stage of consolidation (sec)
Dimensional unit of coefficient of consolidation is L^2T^{-1}

999. Which of the following is not an assumption in Boussinesq's solution for stress distribution?

(a) Soil mass is finite
(b) Soil mass is isotropic
(c) Soil mass is homogeneous
(d) Soil mass is an elastic medium

UPPCL AE 2015

Ans. (a) : Assumption is Boussinesq's theory for stress distribution–
(a) Soil mass is homogeneous and isotropic
(b) Soil mass is an elastic medium.

1000. A soil is having a liquid limit of 35%. What will be the estimated compression index as per Skempton's equation for a remoulded sample of the soil?

(a) 0.125 (b) 0.175
(c) 0.225 (d) 0.181

UPPCL AE 2015

Ans. (b) : Given liquid limit = 35%
Skempton's equation for remolded clay, compression index, $(C_c) = 0.007 (w_L - 10)$
$= 0.007 (35 - 10)$
$\boxed{C_c = 0.175}$

1001. When a soil, whose present effective overburden pressure is the maximum pressure that it was subjected to in the past, then it is:

UPRVUNL AE 2016

OR

Which of the following is a clayey soil that has never been subjected to an effective pressure greater than existing overburden pressure and which is also completely consolidated by the existing overburden pressure?

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(a) normally consolidated soil
(b) pre-consolidated soil
(c) under-consolidated soil
(d) over consolidated soil

Kerala PSC AE 2015

Ans. (a) : When a soil, whose present effective overburden pressure is the maximum pressure that it was subjected to in the past, then it is normally consolidated.

Over Consolidation Ratio (OCR) = $\frac{\bar{\sigma}_0}{\bar{\sigma}}$

where, $\bar{\sigma}_0$ = pre consolidation stress

$\bar{\sigma}$ = existing effective stress

If existing effective stress , $\bar{\sigma} > \bar{\sigma}_0$ then the soil is over consolidated soil i.e.

for normally consolidated soil, OCR = 1

for preconsolidated soil, OCR > 1

for under consolidated soil, OCR < 1

1002. When foundations are constructed on very compressible clays, the consolidation settlement can be x the elastic settlement.

(a) Several times smaller than
(b) Several times greater than
(c) Equal to
(d) Unaffected by

UPRVUNL AE 2016

Ans.: (b) When foundations are constructed on very compressible clays, the consolidation settlement can be several times greater than the elastic settlement.

1003. The time required for attaining a certain degree of consolidation of clay layer is proportional to

(a) H^2 and C_v (b) H^2 and $\frac{1}{C_v}$

(c) $\frac{1}{H^2}$ and C_v (d) $\frac{1}{H^2}$ and $\frac{1}{C_v}$

Arunachal Pradesh Poly. Tech. Lect. 2019
GPSC AE June 2019
WBPSC Poly Lect. 2019
BPSC AE 2019 Paper (V) Section-I

Ans. (b) :

Solution, $t = \frac{T_v.H^2}{C_v}$

Where,

t = time required for attaining certain degree of consolidation.

C_V = Coefficient of consolidation

H = Length of drainage path

i.e $t \propto H^2$

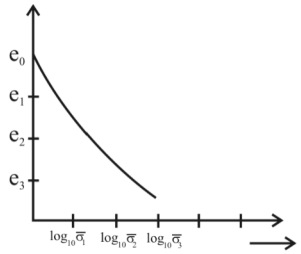
$t \propto \frac{1}{C_v}$

1004. The following represents the slope of the linear portion of the pressure void ratio curve in the consolidation test.

(a) Coefficient of compressibility
(b) Coefficient of volume compressibility
(c) Compression Index
(d) Coefficient of consolidation

GPSC AE June 2019
NPCC AE 2017

Ans. (c) : The slope of virgin curve on semi-log scale is called coefficient of compression or compression index.



Eff. stress on semi-log scale ($\bar{\sigma}$)

$$C_C = \frac{e_1 - e_2}{\log\left(\frac{\bar{\sigma}_2}{\bar{\sigma}_1}\right)}$$

1005. Consider the following statements:

1. Secondary consolidation occurs at constant effective stress.
2. Secondary consolidation is related to dissipation of pore water pressure
3. Secondary consolidation occurs due to readjustment of soil skeleton after disturbance during primary consolidation.
4. Rate of secondary consolidation is controlled by viscous absorbed layer surrounding clay particle in soil.

Which of the above statements are correct?

(a) 1, 2 and 3 (b) 2, 3 and 4
(c) 1, 3 and 4 (d) All of the above

GPSC AE Dec 2018

Ans. (c) : Secondary consolidation occurs at constant effective stress and it occurs due to readjustment of soil skeleton after disturbance during primary consolidation.

Rate of secondary consolidation is controlled by viscous absorbed layer surrounding clay particle.

1006. Assertion (A): Terzaghi considered only primary consolidation and neglected secondary consolidation in his theory of consolidation.
Reason (R): Secondary consolidation begins after the end of the primary consolidation.

(a) A is true but R is false
(b) A is false but R is true
(c) Both A and R are true but R is not the correct explanation of A
(d) Both A and R are true but R is the correct explanation of A

GPSC AE Dec 2018

Ans. (c) Terzaghi considered only primary consolidation because consolidation is mainly due to dissipation of pore water from voids.

Secondary consolidation begins after the end of the primary consolidation.

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1007. For which of the following soils will the Westergard's analysis produce better results than the Boussinesq's analysis?

GPSC AE Dec 2018

OR

Westergaard's analysis for stress distribution beneath loaded area is applicable to-

(a) sandy soils (b) clayey soils
(c) stratified soils (d) silty soils

OPSC AE Paper - II 2019
GPSC AE January 2018
Gujarat PSC AE 2017
Punjab PSC SDE 2016
UTTRAKHAND AE 2013
UK Combined AE Paper Ist 2012
APPSC AEE 2012
UKPSC (Urban engineering)-2011
UKPSC AE paper Ist 2007
JPSC AE 2009

Ans. (c) : Westergaard's analysis for stress distribution beneath loaded area is applicable for stratified soils.
For Stratified soil, westergard analysis are used.

$$\sigma_z = \frac{Q}{\pi z^2}\left[\frac{1}{1+2\left(\frac{r}{z}\right)^2}\right]^{3/2}$$

1008. To prevent the possibility of erosion and piping, following approaches are used:

1. By providing cut off wall at downstream side
2. Increasing the flow path by providing impervious blanket
3. Use or protective filter

Which of these statements are correct?

(a) 1, 2 and 3 (b) 1 and 2
(c) 2 and 3 (d) 1 and 3

GPSC AE Dec 2018

Ans. (c) : To prevent the possibility of erosion and piping, following approaches are used:
(i) Increasing the flow path by providing impervious blanket.
(ii) Use or protective filter.

1009. For a vertical concentrated load acting on the surface of a semi-infinite elastic mass, the vertical normal stress at a depth z is proportional to :

(a) z^2 (b) z
(c) $\frac{1}{z^2}$ (d) $\frac{1}{z}$

BHEL ET 2019
ESE 1997

Ans. (c) : By using formula,

$$\sigma = \frac{3}{2\pi}\frac{Q}{z^2}\left[\frac{1}{1+\left(\frac{r}{z}\right)^2}\right]^{5/2}$$

$$\sigma \propto \frac{1}{z^2}$$

1010. A clay layer 10 m thick in the field takes 250 days to attain 50% consolidation with the condition of double drainage. If the same clay layer is underlain by a hard rock, then the time taken to attain 50% consolidation is :

(a) 1000 days (b) 62.5 days
(c) 125 days (d) 500 days

BHEL ET 2019

Ans. (a) : H = 10 m, t = 250 days, U = 50%
In both the case consolidation is 50%

$$\frac{C_v}{T_v} = \frac{d^2}{t}$$

d_1 = H/2 (double drainage)

d_1 = H (single drainage)

$$\frac{d_1^2}{t_1} = \frac{d_2^2}{t_2}$$

$$\frac{(5)^2}{250} = \frac{(10)^2}{t_2}$$

t_2 = 1000 days

1011. The compression index of a soil

(a) decreases with an increase in the liquid limit
(b) increases with an increase in the liquid limit
(c) decrease with an increase in the plastic limit
(d) is not at all related to the liquid limit

Haryana PSC AE Paper–II 2018
GPSC Assistant Prof. 2016

Ans. (b) :
$C_C = 0.009\,(w_L - 10) \Rightarrow$ For Undisturbed soil

$$\boxed{C_C \propto w_L}$$

$w_L \rightarrow$ liquid limit
$C_C \rightarrow$ compression coefficient
$C_C = 0.007\,(w_L - 10) \rightarrow$ for remoulded soil

1012. The natural void ratio of a saturated clay strata, 3 m thick is 0.90. The final void ratio of the clay at the end of consolidation is expected to be 0.71 The total consolidation settlement of the clay state is

(a) 30 cm (b) 20 cm
(c) 25 cm (d) 15cm

UJVNL AE 2016
UPPSC Civil Eng. A.E. 2013(I)
Haryana PSC AE 2010
ESE 1995

Ans. (a) : Given,
height, H = 3m = 300 cm
initial void ratio, $e_0 = 0.9$
final void ratio, e = 0.71
As, we know that

$$\frac{\Delta H}{H} = \frac{\Delta e}{1+e_0}$$

$$\frac{\Delta H}{300} = \frac{0.9-0.71}{1+0.9}$$

$$\frac{\Delta H}{300} = \frac{0.19}{1.9}$$

$\Delta H = 30$ cm

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1013. A square footing of size 1 m, transmits a uniform pressure of 400 kPa to the underlying sandy soil. Determine the vertical stress at a depth of 1 metre below the foundation using the 2:1 method.

(a) 400 kPa (b) 200 kPa
(c) 100 kPa (d) 80 kPa

AP TRANSCO AEE 2017

Ans. (c) Given,
Square footing size 1 m × 1 m
Pressure (q_o) = 400 kPa
Depth (z) = 1 m

Vertical stress $(\sigma_z) = \dfrac{q_0 LB}{(L+Z)(B+Z)}$

$(\sigma_z) = \dfrac{400\times1\times1}{(1+1)(1+1)}$

$(\sigma_z) = 100\ kN/m^2$

1014. The configuration of flow nets depends upon

(a) The permeability of soil
(b) The difference in the head between upstream and downstream sides
(c) The boundary condition of flow
(d) The amount of seepage that takes place

UPRVUNL AE 2015
ESE 1998

Ans. (c) : The configuration of flow nets depends upon the boundary condition of flow.

1015. A clay deposit suffers a total settlement of 5 cm with one-way drainage. With two-way drainage, the total settlement will be

(a) 10 cm (b) 2.5 cm
(c) 20 cm (d) 5 cm

MPPSC AE 2014

Ans. (d) : Final consolidation does not depend upon drainage condition, so the ultimate settlement will be 5cm.

1016. A laboratory sample of clay of 2 cm thick takes 14.6 min. to attain 45% consolidation under double drainage condition. Determine the required time to attain same degree consolidation for the clay of 3m thick under single drainage condition.

(a) 1 year (b) 4 year
(c) 3.5 year (d) 2.5 year

UPPCL AE 2018

Ans. (d) : Sample 1
given
Thick = 2 cm, t = 14.6 min, U = 45%
U < 60%

$T_V = \dfrac{\pi}{4}(0.45)^2$

$T_V = C_v \dfrac{t}{d^2}$

T_V = time factor, d = length of drainage path
t = time in 'sec'
C_v = coefficient of consolidation (cm^2/sec)

$\dfrac{\pi}{4}(0.45)^2 = \dfrac{C_v\times14.6\times60}{\left(\frac{2}{2}\right)^2}$

$C_v = 1.815\times10^{-4}\ cm^2/sec$

Sample -2
d = 300 cm

$\dfrac{\pi}{4}\times0.45^2 = 1.815\times10^{-4}\times\dfrac{t}{300\times300}$

$t = 7886.44\times10^4$ sec

$t = \dfrac{7886.44\times10^4}{365\times24\times60\times60} = 2.5$ years

1017. The general Boussinesq equations for determining stress below soil considers :

(a) Concentrated load
(b) Cyclic load
(c) Uniformly varying load
(d) Uniformly distributed load

UPPCL AE 2019

Ans. (a) The general Boussinesq equation for determining stress below soil considers concentrated load.

1018. In the analysis of stresses in flexible pavements layers, the expression for vertical deflection at the center of a flexible plate on the surface in homogeneous mass is given by Δ = _____. Take poisons ratio of homogeneous soil as 0.5.

(a) $\dfrac{pa}{E_s}$ (b) $\dfrac{1.5pa}{E_s}$

(c) $\dfrac{1.5p}{a * E_s}$ (d) 1.5 paE_s,

Where, p = uniform pressure,
a = radius of circular load

UPPCL AE 2019

Ans. (b) In flexible pavements layers, the expression for vertical deflection at the centre of a flexible plate on the surface in homogenous mass is given by

$$\boxed{\Delta = 1.5\frac{pa}{E_s}}$$

Where, p = uniform pressure
a = radius of circular load

1019. A 5 m thick compressible soil undergoes 90% consolidation four times faster under two way drainage when comparted to one-way drainage. In an identical soil layer of 10 m thickness, two way drainage will be faster as compared to one-way drainage by

(a) 800% (b) 400%
(c) 250% (d) 200%

LMRC AM 2015

Ans (b) For the same degree of consolidation (T_v is constant)

$\Rightarrow t_1 \propto H^2$ (under single drainage)

Length of drainage path = H

$t_2, \propto \left(\dfrac{H}{2}\right)^2$ (under double drainage)

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Under double drainage condition length of drainage path (d_2) = $\left(\frac{H}{2}\right)$

Under single drainage condition length of drainage path (d_1) = H

$$\frac{t_1}{t_2}=\frac{d_1^2}{d_2^2}=\frac{H^2}{(H/2)^2}$$

$$t_2=\frac{t_1}{4}$$

Hence, two way drainage will be faster as compared to one way drainage by 400%.

1020. According to IS 2720 (V), for clayey soils to ensure uniform distribution of moisture throughout the soil mass, after mixing water, the soil-water mixture shall be left to stand of X hours, where X is:

(a) 16 (b) 12
(c) 18 (d) 24

LMRC AM 2015

Ans (d) According to IS 2720 (V), for clayey soils to ensure uniform distribution of moisture throughout the soil mass, after mixing water the soil-water mixture shall be left to sand for 24 hours.

1021. The slope of *e* versus log σ plot for a soil mass gives : (notations : *e*- void ratio, σ– effective stress)

CIL MT 2020

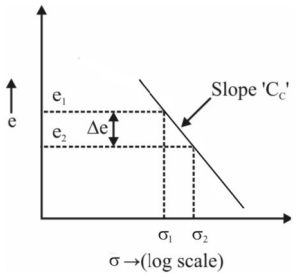
OR

The slope of the e-log p curve for a soil mass gives :

(a) Coefficient of permeability, k
(b) Coefficient of consolidation C_V
(c) Compression index, C_C
(d) Coefficient of volume compressibility, m_v

ISRO Scientist/Engineer 2013
OPSC Civil Services Pre 2011

Ans : (c) The slope of the e-log p curve for normally consolidated soil is referred to as the compression index C_c.



$$\left[C_c=\frac{e_1-e_2}{\log\sigma_2-\sigma_1}\right]$$

(1) Compression index (C_C) is given as -

$$Cc=\frac{-\Delta e}{\log_{10}\left(\frac{\sigma+\Delta\sigma}{\sigma}\right)}$$

It is slope of the linear portion of void ratio v/s logσ plot.

(2) Coefficient of volume change (m_v) ; it is defined as volumetric strain per unit increase in effective stress

$$m_v=\frac{-\Delta v/v}{\Delta\sigma}=\frac{\Delta e}{\Delta\sigma(1+e_0)}$$

Coefficient of consolidation (C_v)

$$C_v=\frac{k}{\gamma_w m_v}$$

1022. Two identical clay sample of the same size designated as 'A' and 'B' are subjected to consolidation test under identical loading conditions. Drainage takes place through one face in sample 'A' and through both the faces in sample 'B'. 50% consolidation of sample 'A' occurs in 10 min. The time required for 50% consolidation to occur in sample 'B' will be

(a) 40 min (b) 10 min
(c) 5 min (d) 2.5 min

Haryana PSC AE 2010
ESE 1995

Ans. (d) : $T_v=\frac{C_v\times t}{d^2}$

$$C_{V_1}=\frac{d_1^2}{t_1},\ C_{V_2}=\frac{d_2^2}{t_2}$$

$C_{V_1}=C_{V_2}=50\%$ $[d_1 = d, d_2 = d/2]$

$$\frac{d_1^2}{t_1}=\frac{d_2^2}{t_2}$$

$$t_2=10\times\left(\frac{d_2}{d_1}\right)^2=10\times\frac{1}{4}=2.5\text{ min}$$

1023. A stratum of clay 2 m thick will get consolidated 80% in 10 years. For the 80% consolidation of 8m thick stratum of the same clay, the time required will be:

(a) 100 years (b) 120 years
(c) 140 years (d) 160 years
(e) 180 years

Haryana PSC AE Paper–II 2018

Ans. (d) : Given,
t_1 = 10 year
d_1 = 2 m
d_2 = 8 m

$$\frac{t_1}{t_2}=\frac{d_1^2}{d_2^2},$$

t = time of consolidation
d = depth of stratum

$$\frac{10}{t_2}=\frac{(2)^2}{(8)^2}$$

$$t_2=\frac{640}{4}=160\ years$$

1024. Select the correct statement from the following:

(a) When stress decreases, void, ratio decrease
(b) When stress decreases, coefficient of permeability decreases

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(c) When stress decreases, coefficient of volume change decreases
(d) When stress decreases void ratio, coefficients of permeability and volume change decreases
(e) None of these options

Haryana PSC AE Paper–II 2018

Ans. (d) : Here $a_v = \frac{e_1 - e_2}{\sigma_1 - \sigma_2}$

If σ decreases then the coff. of compressibility (a_v) increases and void ratio ($e = e_1 - e_2$) decreases due to increase in stress.

If a_v increases then $m_v = \frac{a_v}{1+e}$, m_v also increases then the permeability $k = C_v m_v \gamma_w$. If m_v increases then k also increases.

1025. Over-consolidation of soils is caused due to:
(a) Erosion of over burden
(b) Melting of ice sheets after glaciations
(c) Permanent rise of water table
(d) All of the above
(e) None of these options

Haryana PSC AE Paper–II 2018

Ans. (d) : • Consolidation means slow expulsion of water from the soil mass and decrease in its volume due to some constant applied load.
• Over consolidated soil is that which has been more compacted than would be expected from the existing overburden pressure.

1026. Which one of the following statements regarding coefficient of consolidation C_v is correct?
(a) $C_v \propto k$ (b) $C_v \propto 1/k^2$
(c) $C_v \propto m_v$ (d) $C_v \propto a_v$

Haryana PSC AE 2017
ESE 2002

Ans. (a) : $C_v = \frac{k}{m_v \times \gamma_w}$ $\boxed{C_v \propto k}$

C_v = Coefficient of consolidation
m_v = Coefficient of volume compressibility
k = Coefficient of permeability
γ_w = Unit weight of water

1027. A clay layer 5m thick in field takes 300 days to attain 50% consolidation with condition of double drainage. If the same clay layer is underlain by hard rock then the time taken to attain 50% consolidation will be:
(a) 300 days (b) 600 days
(c) 900 days (d) 1200 days

BPSC AE Aug. 2019 Paper-V
UPPSC State Eng. A.E. 2011
Haryana PSC AE 2017
ESE 1996

Ans. (d) : Time (t_1) = 300 day,

Drainage is doubled $(d_1) = \frac{H}{2} = \frac{5}{2}$

Drainage is single $(d_2) = H = 5\,m$

$$\frac{t_1 = T_v \frac{d_1^2}{C_v}}{t_2 = T_v \frac{d_2^2}{C_v}}$$

$$\frac{300}{t_2} = \frac{\left(\frac{5}{2}\right)^2}{(5)^2} \Rightarrow t_2 = 1200\,days$$

1028. A clay layer 3 m deep is sandwiched by a pervious material. The soil reaches 90% consolidation in 3 years. At another site, the same clay layer is bounded by impervious boundary at the bottom and sand at the top. The time (in years) taken by this layer to reach 90% settlement would be :
(a) 9 (b) 21
(c) 18 (d) 12
(e) 6

CGPSC AE Shift - 2 2014
ESE 2012

Ans. (d) : $T_v = \frac{C_v t}{d^2}$ C_v , & T_v constant

Given,
t_1 = 3 year

$$\frac{t_1}{t_2} = \frac{(d_1)^2}{(d_2)^2}$$

$$d_1 = \frac{H}{2} = \frac{3}{2} = 1.5$$

$d_2 = H = 3$

$$\frac{3}{t_2} = \frac{(1.5)^2}{(3)^2} = \left(\frac{1}{2}\right)^2$$

$t_2 = 3 \times 4 = 12$ years

1029. Terzaghi's one dimensional consolidation theory assumes that -
(a) Unique relationship between the void ratio and effective stress is independent of time
(b) Unique relationship between the void ratio and effective stress is dependent on time
(c) Threre is no assumption regarding relationship between the void ratio and effective stress is independent of time
(d) None of the above

Rajasthan VP ITI 2018

Ans. (a) : Terzaghi's one dimensional consolidation theory assumes that unique relationship between the void ratio and effective stress is independent of time.

1030. A 6m thick clay layer is drained at both top and bottom. The co-efficient of consolidation of the soil is 8x 10^{-4} cm^2/sec. Time required for 60% consolidation of the layer due to an external load will be ---------- (in days)
(a) 336 (b) 337
(c) 369 (d) 435

UPRVUNL AE 2014

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Ans. (c) : Given U = 60%, $d = \frac{H}{2} = 3$

$C_v = 8 \times 10^{-4}$ cm^2/sec

$T_V = \frac{C_v \times t}{d^2} = \frac{\pi}{4}(0.60)^2$

$\frac{8\times10^{-4}\times t}{(6/2)^2} = \frac{\pi}{4}\times(0.6)^2$

t = 369 days

1031. Consider the following statements :

1. **Coefficient of consolidation normally increases with decreasing liquid limit of clay.**
2. **The larger the value of coefficient of consolidation, the longer it takes for full consolidation to occur.**

Which of the statement/s given above is/are correct?

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

UPRVUNL AE 2014

Ans. (a) Coefficient of consolidation normally increases with decreasing liquid limit of clay.

1032. If the liquid limit is increased then, the compression index of soil will:-

(a) Increase
(b) May increase or decrease
(c) Decrease
(d) Remain constant

UPPCL AE 2014

Ans. (a) : $C_c = 0.009 (w_L - 10)$
w_L = liquid limit
w_L increases, C_c will increase.

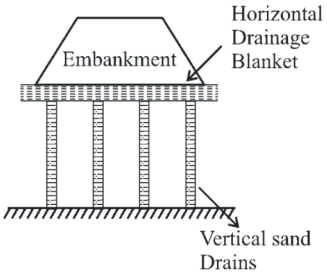
1033. Sand drains are used to

(a) Reduce the settlement
(b) Accelerate the consolidation
(c) Increase the permeability
(d) Transfer the load

RPSC AE (GWD) 2014

Ans : (b)

- Sand Drain is a process of Accelerate the consolidation which increase rate of drainage in the embankment by driving casing into the embankment and making vertical boreholes.
- A sand blanket is placed over the top of sand drains to connect all sand drains.

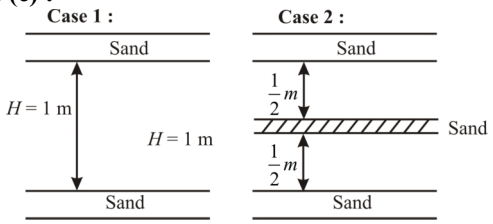


1034. A 1 m thick layer of saturated clay, drained at both faces, settles by 10 cm in one year. If a thin layer of pervious soil is introduced in the middle of this layer, then what will be the period during which the settlement to 10 cm will be compacted?

(a) 2 years (b) 0.75 years
(c) 0.25 years (d) 3 years

Karnataka PSC AE 2017 Paper - II

Ans. (c) :



$\therefore d = \frac{H}{2} = \frac{1}{2}m, T_v = \frac{C_v}{d^2}t$

Case – 2

$\therefore d_2 = \frac{H}{2} = \frac{1}{2}\times\frac{1}{2} = \frac{1}{4}$

$t_1 = 1yr \quad t \propto d^2$

$$\frac{t_1}{t_2} = \left(\frac{d_1}{d_2}\right)^2$$

$$\frac{t_1}{t_2} = \left(\frac{\frac{1}{2}}{\frac{1}{4}}\right)^2 = 4$$

$$t_2 = \frac{t_1}{4} = \frac{1}{4} = 0.25yr$$

$t_2 = 0.25$ yr

1035. In a Network's influence chart for stress distribution, there are 10 concentric circles and 50 radial lines. The influence factor of the chart is:

(a) 0.0002 (b) 0.002
(c) 0.02 (d) 0.2

Haryana PSC AE 2017

[UPPSC State Eng. A.E. 2003]

Ans. (b) : Network's influence

$$\text{Co-efficient} = \frac{1}{\text{No. of concentric circle} \times \text{No. of radial lines}}$$

$$= \frac{1}{10\times50} = 0.002$$

1036. The initial and final void ratio of a clay sample in a consolidation test are 1.0 and 0.5 respectively. If the initial thickness of the sample is 2.4 cm, then its final thickness will be

(a) 1.3 cm (b) 1.8 cm
(c) 1.9 cm (d) 2.2 cm

(UKPSC AE 2012 Paper–I)

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Ans : (b) $\frac{H_f}{H_i} = \frac{1+e}{1+e_0}$

H_f = Final thickness = ?

H_i = Initial thickness = 2.4

e = final void ratio = 0.5

e_0 = initial void ratio = 1.0

$$\frac{H_f}{2.4} = \frac{1+0.5}{1+1}$$

$$H_f = \frac{1.5 \times 2.4}{2} = 1.8 \text{ cm}$$

1037. The initial and final void ratios of a clay sample in a consolidation test are 1 and 0.5 respectively. If the initial thickness of the sample is 2.4 cm. then its final thickness will be:

(a) 1.3 cm (b) 1.8 cm

(c) 1.5 cm (d) 2.2 cm

HPPSC Poly. Lect. 2016

Ans : (b) Given,

$e_{initial} = 1$, $e_{final} = 0.5$

$H_0 = 2.4$ cm

$$\frac{\Delta H}{H} = \frac{\Delta e}{1+e_0}$$

$$\Delta H = \frac{(1-0.5)}{1+1} \times 2.4$$

$\Delta H = 0.6$ cm

thickness of layer = 2.4 – 0.6

= 1.8 cm

1038. Newmark developed influence charts to compute vertical stresses due to loaded area of any shape below any point either inside or outside of the loaded area. The radius of tenth circle for a Newmark's Chart, when it is divided into 20 sectors and if the influence factor is 0.005 will be:

(a) infinite (b) 2.54

(c) 3.54 (d) unity

JUVNL AE 2016

Ans. (a) :

Influence factor $I_f = 0.005$ (20 sectors divided)

The stress by Newmark chart (σ) = $(I_f)_q$ m

$$\sigma_z = \frac{1}{20} q \left[1 - \left\{\frac{1}{1+\left(\frac{R_i}{z}\right)^2}\right\}\right]$$

First radii = 0.270 z = R_1

R_2 = 0.40 z

R_3 = 0.77 z

.

.

R_{10} = infinity

1039. The Spring-Cylinder analogy is used in soil mechanics to explain _____.

(a) Time-dependent deformation of saturated clayey soils

(b) Compaction of clays

(c) Air removal from clayey soils

(d) Relative density of cohesion less soils

NPCC AE 2017

Ans. (a) : In spring cylinder analogy, if the spring has to share this addition load of soil, the spring water system undergoes deformation.

1040. In Boussinesq's solution to the problem of stress distribution, all the following assumptions are made, except:

(a) Soil is anisotropic

(b) Soil mass is semi-infinite

(c) Soil is homogeneous

(d) The elastic modulus of soil is constant

NPCC AE 2017

Ans. (a) : Soil is assumed to be homogeneous isotropic semi-infinite and elastic.

1041. Underload, the void ratio of a submerged saturated clay decreases from 1.00 to 0.92. What will be settlement of a 2 m thick clay of same type due to consolidation?

(a) 20 mm (b) 40 mm

(c) 80 mm (d) 100 mm

WBPSC AE 2016, 2007

Ans. (c) : Given, H = 2 m, $e_0 = 1$, Δe = 1-0.92

$e_f = 0.92$

$$\frac{\Delta H}{H} = \frac{\Delta e}{1+e_0} \Rightarrow \frac{\Delta H}{2} = \frac{1-0.92}{1+1}$$

$\Delta H = 0.08\text{m} = 80\text{mm}$

1042. If the time required for 50% consolidation of a remoulded sample of clay with single drainage is one minute, then the time required to consolidate the same sample of clay with same degree of consolidation with double drainage is

(a) $\frac{1}{2}$ minute (b) $\frac{1}{4}$ minute

(c) 2 minute (d) 4 minute

WBPSC AE 2014

Ans. (b) : Given, U = 50%,

For single drainage, $d_1 = d$,

For double drainage, $d_2 = d/2$

Relation between t and d

$$T_v = \frac{C_v \times t}{d^2}$$

$$\frac{t_1}{t_2} = \frac{d_1^2}{d_2^2}$$

$$\frac{1}{t_2} = \frac{d^2}{(d/2)^2} \Rightarrow t_2 = \frac{1}{4} \text{min}$$

1043. A normally consolidated clay settle 1 cm when effective stress was increased from 100 kN/m^2 to 200 kN/m^2. If the effective stress is further increased from 200 kN/m^2 to 400 kN/m^2 then settlement of the clay is

(a) 1 cm (b) 2 cm

(c) 4 cm (d) 8 cm

WBPSC AE 2014

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Ans. (a) : $\Delta H = \frac{C_c.H}{(1+e_0)} \times \log \frac{\bar{\sigma}_2}{\bar{\sigma}_1}$

$\Delta H = \log \frac{\bar{\sigma}_2}{\bar{\sigma}_1}$

Case 1,

$1 = \log \frac{200}{100} = \log 2$

case 2,

$\Delta H = \log \frac{400}{200} = \log 2$

$\Delta H = \log 2 = 1\text{cm}$

1044. In a homogeneous embankement type earth dam, the phreatic line is kept well within the body of the dam by

(a) Providing proper U/S slope protection
(b) Providing proper D/S slope protection
(c) Suitable increasing the top width
(d) Providing horizontal drainage filter at the D/S face

WBPSC AE 2003

Ans. (d) : In a homogeneous embankement type earth dam, the phreatic line is kept well within the body of the dam by providing horizontal drainage filter at the D/S face.

1045. For soil sampling boreholes should be stabilized for

(a) Prevention of entry of seepage water into borehole
(b) Prevention of collapse of side walls of borehole
(c) Quick advancement of borehole
(d) Lowering of ground water table

WBPSC AE (PHE) 1998

Ans. (b) : For soil sampling boreholes should be stabilized for prevention of collapse of side walls of borehole.

1046. The voiled ratio of a soil sample decreased from 1.59 to 1.25 when pressure was increased from 25 to 50 tonne/m^2. The coefficient at compressibility in m^2/tonne is

(a) 0.001 (b) 0.01
(c) 0.02 (d) 0.05

WBPSC AE (Common) 1998

Ans. (b) : Given, $e_0 = 1.59$, $e_f = 1.25$, $\bar{\sigma}_1 = 25\text{tonne}/\text{m}^2$, $\bar{\sigma}_2 = 50\ \text{tonne}/\text{m}^2$

Coefficient of compressibility (a_v)

$= \frac{\Delta e}{\Delta \bar{\sigma}} = \frac{1.59-1.25}{50-25} = 0.0136 = 0.01$

1047. According to Boussinesque's theory, under the application of a point load of 100 kN on the surface, the pressure bulb corresponding to an increment in vertical stress of 47.75 kN/sqm will extend to a depth of:

(a) 0.50 m (b) 1.00 m
(c) 2.00 m (d) 4.00 m

TSPSC Deputy Engineer 2015

Ans. (b) : Given, Q = 100 kN

$\sigma_z = 47.75\ \text{kN/m}^2$

$\sigma_z = 0.4775\frac{Q}{z^2}$

$z^2 = \frac{0.4775 \times 100}{47.75}$

$z^2 = 1 \Rightarrow z = 1\text{m}$

1048. The magnitude of total primary consolidation of a 6m thick clay with single drainage was estimated as 96 mm. Later it was found that, the medium has double drainage. Then, the magnitude of total primary consolidation settlement will be:

(a) 48 mm (b) 192 mm
(c) 384 mm (d) 96 mm

TSPSC Deputy Engineer 2015

Ans. (d) : $\Delta H = \frac{H.C_c}{1+e_0} \times \log \frac{\bar{\sigma}_2}{\bar{\sigma}_1}$

According to above equation, we can say that settlement is not depend on drainage path.

So, in single drainage path or double drainage path consolidation settlement is remain same.

ΔH = 96 mm (single drainage)

ΔH = 96 mm (double drainage)

1049. A 10 m thick clay layer with single drainage undergone 90% of primary consolidation settlement in 10 years. Considering the corresponding time factor as 0.848, the coefficient of consolidation is:

(a) 1.0 m^2/year (b) 10 m^2/year
(c) 8.48 m^2/year (d) 84.8 m^2/year

TSPSC AEE 2017

Ans. (c) : Given,

d = 10 m, $T_v = 0.848$, t = 10 year

$T_v = \frac{C_v.t}{d^2} \Rightarrow C_v = \frac{0.848 \times (10)^2}{10}$

$C_v = 8.48\ \text{m}^2/\text{year}$

1050. The time required for a consolidating medium with single drainage to undergo 90% of its primary consolidation settlement was estimated as 36 years. All other conditions remaining same, estimate the time required if the medium has double drainage.

(a) 3.6 years (b) 36 years
(c) 72 years (d) 9 years

TSPSC AEE 2017

Ans. (d) : $T_v = \frac{C_v.t}{d^2}$

$\frac{t_1}{t_2} = \frac{d_1^2}{d_2^2} \Rightarrow \frac{36}{t_2} = \frac{d^2}{(d/2)^2}$

$t_2 = 9$ year

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8. Shear Strength of Soil

1051. The Mohr's strength theory is based on the following fact:
(a) Material fails essentially by shear
(b) Ultimate strength of the material is determined by the stress in the plane of slip
(c) Failure criterion is independent of the intermediate principal stress
(d) All of the above
(e) None of these options

Haryana PSC AE Paper–II 2018

Ans. (d) : Mohr's strength theory
- According to Mohr. the failure is caused by a critical combination of normal and shear stress.
- The soil fails when the shear strength 'S' on the failure plane at failure is a unique function of normal stress acting on that plane.
- Failure of material occurs when the Mohr circle of stresses touches the Mohr envelope.

1052. If H and V are initial height and volume of the soil specimen. ΔH and ΔV are corresponding changes in the values of height and volume in a Triaxial Compression test. The area of cross-section A at failure or during any stage of Triaxial compression test is given
(a) $A = \frac{V + \Delta V}{H - \Delta H}$ (b) $A = \frac{V - \Delta V}{H + \Delta H}$
(c) $A = \frac{V + \Delta V}{H + \Delta H}$ (d) $A = \frac{V - \Delta V}{H - \Delta H}$

TNPSC AE 2018
WBPSC AE 2014

Ans. (d) : The calculation of deviator stress must be done on basis of the changed area of cross-section at failure or during any stage of the test.

$$A = \frac{V - \Delta V}{H - \Delta H}$$

1053. A consolidated drained triaxial test was conducted on a granular soil. At failure $\frac{\sigma_1}{\sigma_3} = 4.6$, the value of 'ϕ' of the soil is-
(a) 40° (b) 45°
(c) 53° (d) 65°

MPPSC AE 2017

Ans. (a) : For granular soil $C = 0$

$$\sigma_1 = \sigma_3 \tan^2\left(45^o + \frac{\phi}{2}\right) + 2C\tan\left(45^o + \frac{\phi}{2}\right)$$

$$\sqrt{4.6} = \tan\left(45^o + \frac{\phi}{2}\right)$$

$$\frac{\phi}{2} + 45^o = 65^o$$

$$\phi = 40^o$$

1054. The appropriate field test to determine the undrained shear strength of soft clay is
(a) Plate load test
(b) Static cone penetration test
(c) Standard penetration test
(d) Vane shear test

RPSC AE (GWD) 2014

Ans : (d) It suitable for sensitive clays– Vane shear test is a quick test, used either in laboratory or in the field, to determine the undrained shear strength of cohesive soil.
Shear strength of a clay sample t.
Case – (i) When only one end of the vane takes part in shearing.

$$\tau_f = \frac{T}{\pi d^2\left(\frac{h}{2} + \frac{d}{12}\right)}$$

Case - (ii) When both ends take part in shearing

$$\tau_f = \frac{T}{\pi d^2\left(\frac{h}{2} + \frac{d}{6}\right)}$$

1055. The phenomenon when soil loses its shear strength due to oscillatory motion is known as
(a) Consolidation (b) Shear failure
(c) Liquefaction (d) Sloughing

RPSC AE (GWD) 2014

Ans : (c) In loose saturated sands due to decrease in volume during disturbance or vibration, positive pore water pressure develops hence effective stress reduces and soil loses its shear strength. This phenomenon is known as liquefaction. It is generally observed near river and sea during the earthquake.

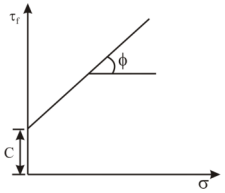
1056. The shear strength of a soil
(a) Is directly proportional to the angle of internal friction of the soil
(b) Is inversely proportional to the angle of internal friction of soil
(c) Decreases with increase in normal stress
(d) Decreases with decrease in normal stress

Chandigarh SDE 2017
RPSC AE (GWD) 2014

Ans : (d) The shear strength of a soil decreases with decrease in normal stress.
- Mohr coulomb failure criterion can be expressed in the form of :

$$\tau_f = C' + \bar{\sigma}\tan\phi'$$

$$\{\bar{\sigma} = \sigma - u\}$$



Where,
ϕ = total friction angle
$\bar{\sigma}_n$ = Total normal stress on critical plane.
C' = Effective cohesion.

1057. Which one of the following statements is incorrect in respect to shear strength of granular soils
(a) It is affected largely by the initial void ratio.

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(b) It is not affected by the effective stresses.
(c) The cohesion value of uncemented granular soil is approximately zero.
(d) Undrained strength is insignificant except during the earthquake.

RPSC VPITI 2012

Ans : (b) Shear strength of the granular soil. It is effected largely by the initial void ratio. The cohesion value of uncemented granular soil is approximately zero and undrained strength is insignificant except during the earthquake.

1058. If ϕ be the angle of shearing resistance, the angle of the failure plane with the major principal plane is equal to :

(a) $45^0 + \phi/2$ (b) $45^0 - \phi/2$
(c) $45^0 + \phi$ (d) $45^0 - \phi$

[RPSC AE (GWD) 2014]

Ans : (a) Angle of failure can be expressed in terms of angle of shearing resistance ϕ

$$2\phi_f = 90 + \phi$$

$$\phi_f = 45^o + \frac{\phi}{2}$$

1059. In a triaxial shear test a sample of C-ϕ soil fails making an angle of α_c^o with horizontal, then value of ϕ can be determined as:

(a) $2\left(\alpha_c^o - 45^o\right)$ (b) $\left(\alpha_c^o + 45^o\right)$
(c) $2\left(45^o - \alpha_c^o\right)$ (d) $2\alpha_c^o$

UPPSC AE 12.04.2016 Paper–I

Ans : (a) In a triaxial shear test a sample of C-ϕ soil fails making an angle of α_c^o with horizontal, then value of ϕ can be determined as $2\left(\alpha_c^o - 45^o\right)$.

1060. A soil has an angle of shearing of 30^0 and cohesion of 35 kN/m^2 . If the specimen of this soil is subjected to a tri-axial compression test, then the value of lateral pressure in the cell for failure to occur at total stress of 300 kN/m^2 will be

(a) 243.21 kN/m^2 (b) 44.41 kN/m^2
(c) 103.21 kN/m^2 (d) 59.59 kN/m^2

MPPSC AE 2016

Ans : (d)
Soil is subjected to a tri-axial testing machine :

$\phi = 30^0$, C = 35 kN/m^2, $\sigma_1 = 300$ kN/m^2

The value of lateral pressure = (σ_3)
Then

$$\sigma_1 = \sigma_3\left(\frac{1+\text{Sin}\,\phi}{1-\text{Sin}\,\phi}\right) + 2C\sqrt{\frac{1+\text{Sin}\,\phi}{1-\text{Sin}\,\phi}}$$

$$300 = \sigma_3\left(\frac{1+\text{Sin}\,30^0}{1-\text{Sin}\,30^0}\right) + 2\times 35\sqrt{\frac{1+\text{Sin}\,30^0}{1-\text{Sin}\,30^0}}$$

$$300 = 3\sigma_3 + 2\times 35\times\sqrt{3}$$

$$300 - 2\times 35\times\sqrt{3} = 3\sigma_3$$

$$\sigma_3 = \frac{300 - 121.24}{3} = 59.59\,\text{kN/m}^2$$

1061. In a direct shear test on sand (C = 0), area of circular box was 50 cm^2, normal load was 750 kN and failure load was 460 kN, ϕ will be :

(a) 31.5^0 (b) 29.3^0
(c) 35^0 (d) 28.5^0

[UPPSC State Eng. A.E. 2011 Ist Paper]

Ans : (a) Given, C = 0, A = 50 cm^2, P = 750 kN, P_f = 460 kN
We know,

$$S = C + \bar{\sigma}_n \tan\phi$$

$\because$ $C = 0$

$\therefore$ $S = \bar{\sigma}_v \tan\phi$ .....(i)

Shear strength at failure,

$$S = \frac{460}{A}\,\text{kN/m}^2$$

Vertical effective stress,

$$\bar{\sigma}_v = \frac{750}{A}\,\text{kN/m}^2$$

At failure,

$$\frac{460}{A} = \frac{750}{A}\tan\phi$$

$$\tan\phi = \frac{460}{750}$$

$$\phi = \tan^{-1}\left(\frac{46}{75}\right)$$

$$\phi = 31^o31' \approx 31.5^o$$

1062. Drained conditions are properly controlled in :

(a) Vane Shear Test (b) Direct Shear Test
(c) Triaxial Shear Test (d) None of these

[UPPSC State Eng. A.E. 2011 Ist Paper]

Ans : (c) Triaxial Test:
- This is the most widely used shear strength test and is suitable for all type of soil.
- Drainage can be controlled, whatever be the soil.
- Pore water pressure can be measured.
- Volume change can also be measured
- Failure plane is not predetermined.
- Stress distribution on failure plane is fairly uniform.

1063. A soil specimen is subjected to an all around pressure of 'P_a' in a triaxial cell. The specimen fails at a plunger pressure of 'P_v'. The radius of Mohr's circle of stress shall be :

(a) $\frac{P_v}{2}$ (b) $\frac{P_a}{2}$
(c) $\frac{P_v - P_a}{2}$ (d) $\frac{P_v + P_a}{2}$

[UPPSC State Eng. A.E. 2007(I)]

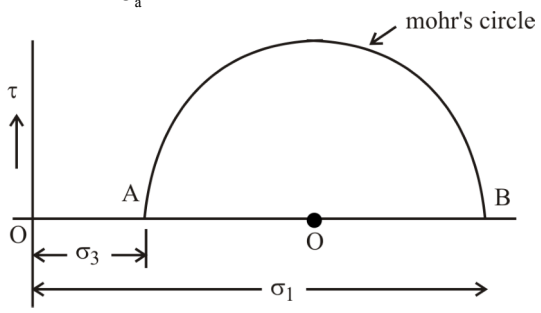
Ans : (a) A soil specimen is subjected to an all around pressure of 'P_a' in a triaxial cell the specimen fails at a plunger pressure of 'P_v'. Then the radius of Mohr's circle of stress shall be $\frac{P_v}{2}$.

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$\sigma_3 = \sigma_c = P_a$

$\sigma_1 = \sigma_3 - \Delta\sigma_d$

$= \sigma_a + R$



radius of Mohr's circle,

$$R = \frac{\sigma_1 - \sigma_3}{2} = \frac{(P_a + P_v) - P_a}{2}$$

$$R = \frac{P_v}{2}$$

1064. Which of the following statements is correct for direct shear test?

(a) test is useful for all types of soil
(b) drainage conditions can be fully controlled
(c) the failure plane is always horizontal
(d) all of the above

GPSC AE March 2018
APPSC AEE (Mains) 2016
[UPPSC State Eng. A.E. 2007(I)

Ans : (c) Direct shear test : The soil specimen that is to be tested is confined in a metal box of square cross section that is split into two halves horizontally, a small clearance being maintained between the two halves of the box.

In this test, drainage cannot be controlled hence rate of loading be such that pore water pressure does not develop i.e. it limit be a drained condition test.

This test is performed for clay because drainage can not be controlled.

Effective stress and total stress are same in this test. Volume expansion is measured using dial gauge.

1065. In a tri–axial test, the deviator stress is given by:

(a) $\sigma_1 + \sigma_3$ (b) $\frac{\sigma_1 + \sigma_3}{2}$

(c) $\sigma_1 - \sigma_3$ (d) $\frac{\sigma_1 - \sigma_3}{2}$

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (c) The triaxial test, deviator stress is given by

$\sigma_1 - \sigma_3$

$\Delta\sigma_d = \sigma_1 - \sigma_c$

$\Delta\sigma_d = \sigma_1 - \sigma_3$

σ_1 = final stress at failure

σ_3 or σ_c = Cell pressure

Deviator stress, produces, shear stresses with in the soil sample on all planes except the horizontal and vertical planes.

1066. For a sandy soil, the angle of internal friction is 30^0. If the major principal stress is 50 kN/m^2 at failure, then the corresponding minor principal stress (in kN/m^2) will be :

(a) 12.20 (b) 20.80
(c) 16.66 (d) 27.20

BPSC AE 2019 Paper (V) Section-I
UPPSC State Eng. A.E. 2004 (I)

Ans : (c) Given $\phi = 30^0$

$C = 0$ (sandy soil)

$\sigma_1 = 50\,kN/m^2$

$\sigma_3 = ?$

$$\sigma_1 = \sigma_3 \tan^2(45 + \theta/2) + 2C\tan\left(45 + \frac{\theta}{2}\right)$$

$$50 = \sigma_3 \tan^2\left(45 + \frac{30^o}{2}\right) + 0$$

$$50 = \sigma_3 \tan^2(45 + 15) + 0$$

$$\sigma_3 = \frac{50}{\tan^2(60)} = \frac{50}{(\sqrt{3})^2} = \frac{50}{3}$$

$\sigma_3 = 16.66$ kN/m^2

1067. Which one of the following planes is most likely to be the failure plane in sandy soils?

(a) Plane carrying maximum shear stress
(b) Plane carrying maximum normal stress
(c) Plane with maximum angle of obliquity
(d) Principal plane

BPSC AE 2012 Paper–V

Ans : (c) Planes is most likely to be the failure plane is sandy soil which is plane with maximum angle of obliquity.

1068. Expansion of soil under shear is called-

(a) liquefaction
(b) Volumetric deformation
(c) critical expansion
(d) dilatancy

BPSC AE 2012 Paper–V

Ans : (d) Soil dilation (dilatancy is measured in degrees) as an increase in soil volume when soil is being sheared. It is common to dense sands to increase in volume when sheared, due to particle inter locking. Expansion of soil under shear is called dilatancy.

1069. Most of the shear tests are done on equipment which are

(a) stress-controlled
(b) strain-controlled
(c) drainage-controlled
(d) volume - controlled

BPSC AE 2012 Paper–V

Ans : (b) Most of the shear tests are done on equipment which are strain controlled.

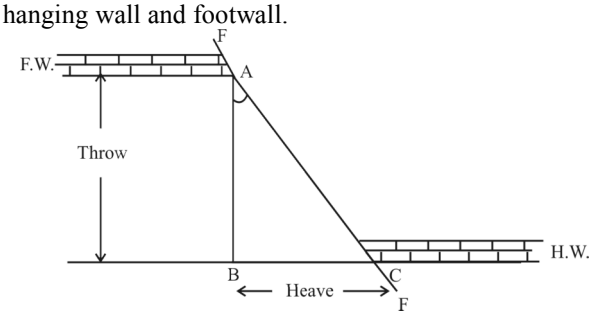
1070. In a fault, inclination of the fault plane with the vertical is called

(a) hang (b) hade
(c) throw (d) none of these

(BPSC AE Paper V 2001)

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Ans : (b) **Hade** – It is the inclination of the fault plane that is vertical.
Throw – It is the vertical displacement between the hanging wall and footwall.
Heave – It is the horizontal displacement between the hanging wall and footwall.



∠BAC = Hade
AB = Throw
BC = Heave
FF = Fault plane
FW = Foot wall
HW = hanging wall

1071. The shear strength of a soil sample is determined in the laboratory by conducting the
(a) triaxial shear test (b) vane shear test
(c) direct shear test (d) none of these
(BPSC AE Paper V 2001)

Ans : (a) **Triaxial shear test–** The strength test more commonly used in a research laboratory today is the triaxial compression test.
- Most versatile of all the shear testing methods, as it can be performed on all soils, be it sand, silt or clay.
- Pore water measurement can be done accurately.
- Volume changes can also be measured.
- Based on drainage and undrainage conditions of shear test.

1072. Unconsolidated undrained test is also called
(a) slow test (b) C-U test
(c) quick test (d) none of these
(BPSC AE Paper V 2001)

Ans : (c) **Unconsolidated undrained Test–**
- Suitable for clayey soils.
- Not suitable for sandy soils.
- Drainage not permitted at any stage.
- 5 to 10 minutes are required for whole test, thus called quick test.

1073. Shear failure indicate
(a) Lateral displacement of soil
(b) Upward displacement of soil
(c) Both lateral und upward displacement
(d) None of these
(BPSC AE Paper V 2006)

Ans : (a) Shear failure indicate the lateral displacement of soil.

1074. The shear strength in plastic undrained clay results from :
(a) internal friction (b) Cohesion
(c) Intergrannular friction (d) None of these
(BPSC AE Paper V 2006)

Ans : (b) The shear strength in plastic undrained clay results from cohesion soil. Plastic undrained clay does not possess internal friction.

1075. Consider the following short comings of direct shear test :
(1) Volume changes can not be measured
(2) Cannot be used for gravels
(3) Difficult to control drainage
(4) Direction of failure plane is fixed
of these statements
(a) 1, 2 and 3 are correct
(b) 2, 3 and 4 are correct
(c) 3, 4 and 1 are correct
(d) 1 and 2 are correct
(JPSC Combined AE Paper V 2013)

Ans : (c) **Direct shear test–**
- Oldest shear test still in used, quite simple to perform.
- Also called shear box test.
- Generally, shear is applied at constant rate of strain.
- Two types of application of shear are possible one in which the shear stress is controlled and the other in which the shear strain is controlled.
- Volume change can not be measured.
- Difficult to control drainage.
- Direction of failure plane is fixed.

1076. Which of the following statements are correct?
(1) The sand with its void ratio higher than its critical void ratio increases in volume when sheared
(2) The sand with its void ratio less than its critical void ratio increase in volume when sheared
(3) For the sand at critical void ratio, the volume change during shear is minimum
(a) 1, 2 and 3 (b) 1 and 2
(c) 2 and 3 (d) 1 and 2
(JPSC Combined AE Paper V 2013)

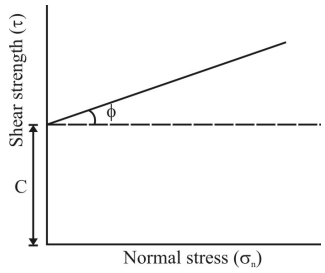
Ans : (c)
- The sand with its void ratio less than its critical void ratio increase in volume when sheared.
- For the sand at critical void ratio, the volume change during shear is minimum.

1077. Coulomb's equation for shear strength can be represented by (symbols have their usual meanings)
(a) $C = S + \sigma \tan \phi$ (b) $S = C - \sigma \tan \phi$
(c) $C = \sigma + C \tan \phi$ (d) $S = C + \sigma \tan \phi$
TNPSC AE 2018
GPSC R&B AE 2016
Karnataka PSC AE 2015 Paper - II
UKPSC AE 2012 Paper–I

Ans : (d) **Coulomb's law–** According to these theory shear strength of soil depends upon of cohesion between particle on critical plane.

$$S = \tau = C + \sigma_n \tan \phi$$

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Where,
σ_n = Normal Stress
ϕ = Total frictional Angle
C = Unit cohesion
τ = S = Shear Strength

1078. Un-confined compression test is generally applicable to
(a) Non-cohesive soil (b) Saturated clays
(c) Silt (d) Sandy soil

Kerala PSC AE 2015
UKPSC AE 2012 Paper–I

Ans : (b) The unconfined compressive strength test is a quick and simple test and is often used to determine the in situ. Strength of soft, saturated, fine-grained, soil deposits. It can not be conducted on coarse grained soils such as sands and gravels.

1079. Which one of the following tests can not be done without undisturbed sampling?
(a) Shear strength of sand
(b) Shear strength of clay
(c) Compaction parameters
(d) Atterberg limits

UKPSC AE 2012 Paper–I

Ans : (b) Shear strength of clay tests can not be done without undisturbed sampling.

1080. The water content of a soil remains unchanged during the entire test in–
(a) Drained test
(b) Consolidated undrained test
(c) Unconsolidated undrained test
(d) None of these

UKPSC AE 2013 Paper–I

Ans : (c) Unconsolidated undrained test–
- suitable for clayey soils.
- water content of soil remains unchanged.
- 5 to 10 minutes are required for whole test, thus called quick test.
- Drainage not permitted at any stage.

1081. An isobar is a curve which joins points of equal:
(a) horizontal stress (b) vertical stress
(c) shear stress (d) none of the above

(UKPSC AE (Paper–I) 2007)

Ans : (b) An isobar is a curve joining the points of equal vertical stress intensity.
The pressure inside an isobar is greater than the pressure present on that isobar.

1082. In an undrained tri-axial test on a saturated clay, the Poisson's ratio is
(a) $\frac{\sigma_3}{\sigma_1+\sigma_3}$ (b) $\frac{\sigma_3}{\sigma_1-\sigma_3}$
(c) $\frac{\sigma_1-\sigma_3}{\sigma_3}$ (d) $\frac{\sigma_1+\sigma 3}{\sigma_3}$

CGPSC AE Shift - 2 2014
UKPSC AE 2012 Paper–I

Ans : (a) In undrained tri-axial test on a saturated clay volumetric strain is zero.
For solve the problem we take lateral strain equal to zero.

$$\varepsilon_v = 0 = \frac{\sigma_3}{E} - \frac{\mu\sigma_1}{E} - \frac{\mu\sigma_2}{E}$$

and $\sigma_3 = \sigma_2$

$$= \mu(\sigma_3 + \sigma_1) = \sigma_3$$

Poisson's ratio (μ)= $\frac{\sigma_3}{\sigma_1+\sigma_3}$

1083. In–situ vane shear test is conducted to determine the shear strength of :
(a) cohesive soil (b) non–cohesive soil
(c) silty soil (d) sandy soil

(UKPSC AE (Paper-I) 2007)

Ans : (a) Vane shear test–
- suitable for cohesive soil such as clay.
- vane shear test is primarily used to find shear parameters of clayey soil.
- It can also be used to determine liquid limit of silty clay.

1084. In the triaxial test the major principal stress is:
(a) deviator stress
(b) cell pressure
(c) deviator stress + cell pressure
(d) none of the above

(UKPSC AE (Paper-I) 2007)

Ans : (c) Tri-axial test the major principal stress is deviator stress + cell pressure.
$\left[\sigma_1 = (\sigma_d + \sigma_C)\right]$
σ_C = cell pressure
σ_d = deviator stress

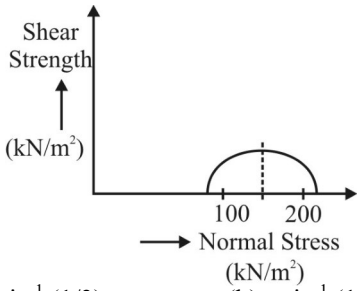
1085. If d represents the grain size, then the critical shear stress, at which the incipient motion of sediment takes place is proportional to:
(a) $d^{2/3}$ (b) d
(c) $d^{1/2}$ (d) $d^{1/3}$

(LMRC AE 2017 I–Shift)

Ans : (b) If d represents the grain size, then the critical shear stress, at which the incipient motion of sediment takes place is proportional to d.

1086. The results of a consolidated drained triaxial shear test on a normally consolidated clay are shown in Figure. The angle of internal friction will be

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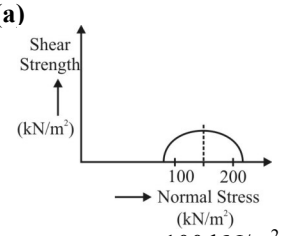


(a) $\sin^{-1}(1/3)$ (b) $\sin^{-1}(1/2)$

(c) $\sin^{-1}(2/3)$ (d) $\sin^{-1}(1/\sqrt{2})$

(UKPSC AE 2012 Paper–I)

Ans : (a)



$\sigma_3 = 100\ kN/m^2$, $\sigma_1 = 200\ kN/m^2$

$$\sigma_1 = \sigma_3\left(\frac{1+\sin\phi}{1-\sin\phi}\right) + 2C\sqrt{\frac{1+\sin\phi}{1-\sin\phi}}$$

for $C = 0$

$$200 = 100\frac{(1+\sin\phi)}{(1-\sin\phi)}$$

$2 - 2\sin\phi = 1 + \sin\phi$

$3\sin\phi = 1 \Rightarrow \sin\phi = 1/3$

$$\boxed{\phi = \sin^{-1}\left(\frac{1}{3}\right)}$$

1087. With the increase in the moisture content, the strength of compacted clayey soils generally

(a) Decreases

(b) Remains constant

(c) Increases

(d) Does not depend on the moisture content

(UP RVNL AE 2016)

Ans.: (a) Increase in the moisture content, then decrease the strength of compacted clayey soil.

1088. The process by which a mass of saturated soil is caused by external forces due to suddenly lose its shear strength and to behave as a fluid is called

(a) Piping (b) Slide

(c) Quick condition (d) Fluidity

UKPSC AE paper Ist 2007

Ans : (c) Quick condition is defined as a mass of saturated soil is caused by external forces due to suddenly lose its shear strength and to behave as a fluid.

1089. In direct shear test, proving ring is used to measure :

(a) displacement (b) shear load

(c) compressive load (d) all (a), (b) and (c)

(UKPSC AE (Paper-I) 2007)

Ans : (b) A proving ring is used to indicate the shear load taken by the soil initiated in the shearing plane.

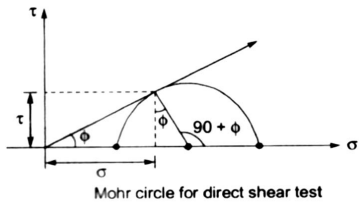
1090. A sample of dry sand was tested in direct shear test apparatus under a normal load of 72 kg. The shear load required to fail the sample was found to be 36 kg. The angle of internal friction (ϕ) will be –

(a) $\tan^{-1}\left(\frac{72+36}{36}\right)$ (b) $\tan^{-1}\left(\frac{72+36}{72}\right)$

(c) $\tan^{-1}\left(\frac{36}{72}\right)$ (d) $\tan^{-1}\left(\frac{72}{36}\right)$

[MPSC (Mains) 2015 Paper-II]

Ans. (c) :



Mohr circle for direct shear test

$$tan\phi = \left(\frac{\tau_f}{\sigma}\right)$$

$$\Rightarrow \quad \tan\phi = \frac{36}{72}$$

$$\phi = \tan^{-1}\left(\frac{36}{72}\right)$$

1091. A soft saturated clayey soil tested unconfined gave an axial stress of 50 kN/m² at failure. The shear strength of the soil is-

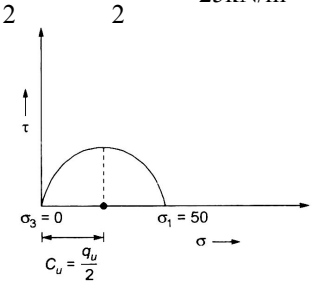
(a) 50 kN/m² (b) 100 kN/m²

(c) 25 kN/m² (d) None of the above

MPSC (Mains) 2017 Paper–II

Ans. (c) : Shear strength of soil

$$= \frac{UCS}{2} = \frac{50kN/m^2}{2} = 25kN/m^2$$



1092. The suitable method of finding the shear strength of very plastic cohesive soils is by means of-

(a) Cone test (b) Peretration test

(c) Vane shear test (d) Torsional shear test

OPSC AE -2016 (II)

OPSC AEE 2015 PAPER–I

Karnataka PSC AE 2015 Paper - II

Ans. (c) : Vane Shear Test is used to measure undrained shear strength of soft clay in the field itself. It is suitable for soft saturated clay (marine clay) to find undrained shear strength and it can also be used to find sensitivity of clays.

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- Shear strength of such soil may be significantly affected during sampling and handling.
- For such soil, vane shear test can be done in the field.
- Plastic cohesive soils are very sensitive and hence it is very difficult to obtain undisturbed specimen.

1093. A clear dry sand sample is tested in direct shear test. The normal stress and the shear stress at failure are both equal to 120 kN/m^2. The angle of shearing resistance of the sand will be

(a) 25 degree (b) 35 degree
(c) 45 degree (d) 55 degree

BPSC AE 2019 Paper (V) Section-I

Ans. (c) : Given, $\bar{\sigma}_n = 120 kN/m^2$

$\tau = 120$ kN/m^2

We know that, $\tau = C + \bar{\sigma}_n \tan\phi$

$C = 0$ for sand.

So, $\tau = \bar{\sigma}_n \tan\phi$

$120 = 120 \tan\phi$

$\tan\phi = 1$

$\boxed{\phi = 45^0}$

1094. An initial cross-sectional area of a clay sample was 15 cm^2. The failure strain was 25% in an unconfined compression test. The corrected area of the sample at failure would be

(a) 15 cm^2 (b) 20 cm^2
(c) 25 cm^2 (d) 30 cm^2

BPSC AE 2019 Paper (V) Section-I

Ans. (b) : Given, $A_i = 15 cm^2$

$\varepsilon_L = 0.25$

Corrected area $(A_f) = \dfrac{A_i}{1-\varepsilon_L}$

$= \dfrac{15}{1-0.25}$

$\boxed{A_f = 20 \text{ cm}^2}$

1095. A clay soil sample is tested in a triaxial apparatus in consolidation-drained conditions at a cell pressure of 100 kN/m^2. What will be the pore water pressure at a deviator stress of 40 kN/m^2.

(a) 0 kN/m^2 (b) 30 kN/m^2
(c) 40 kN/m^2 (d) 80 kN/m^2

[HPPSC 2016-II]

Ans. (a) : In consolidation drained (CD) test, drainage is allowed in both stages. Pore pressure is zero as drainage is allowed during the application of pressure.

1096. σ_z is the vertical stress at a depth equal to z in the soil mass due to surface point load Q. The vertical stress at depth equal to 2 z will be :

(a) 0.25 σ_z (b) 0.50 σ_z
(c) 1.0 σ_z (d) 2.0 σ_z

[HPPSC 2016-II]

Ans. (a) : As per Boussinesq's equation

$$\sigma_Z = \frac{3Q}{2\pi z^2} \cdot \left[\frac{1}{1+\left(\frac{r}{z}\right)^2}\right]^{5/2}$$

$$\sigma_Z \propto \frac{1}{z^2}$$

$$\sigma_{Z_1} = \sigma_Z \times \frac{z^2}{z_1^2} = \sigma_Z \times \frac{z^2}{(2z)^2} = \sigma_Z = 0.25\sigma_Z$$

1097. If a sample of clay has a cohesive strength of 80 kPa and an angle of shearing resistance of 10^0, the shear strength of clay at a normal stress of 100 kPa will be :

(a) 97.63 kPa (b) 98.48 kPa
(c) 78.78 kPa (d) 95.34 kPa

[HPPSC AE 2016]

Ans. (a) : Given angle of shearing resistance, $\phi = 10^o$
cohesive strength, C = 80 kPa
normal stress, σ = 100 kPa
According to Mohr's c-ϕ relationship of soil
$\tau = C + \sigma \tan\phi$
$\tau = 80 + 100 \tan 10^o = 97.63$ kPa

1098. Drainage is not permitted during application of cell pressure (before application of deviator stress) in :
1. Unconsolidated undrained test
2. Consolidated undrained test
3. Consolidated drained test
Out of these statements :

(a) 1 and 2 are correct (b) 1 and 3 are correct
(c) 2 and 3 are correct (d) Only 1 is correct

[HPPSC AE 2016]

Ans. (d) : Two stages of triaxial test are consolidation and shear respectively. If consolidation is allowed in first stage then it is drained test if not allowed then undrained test.

1099. The normal stress and shear stress at failure on the failure plane are 10 kPa and 4 kPa respectively, then the angle of internal friction of the soil and the angle of the inclination of the failure plane to the major principal plane are .....

(a) $21^0 48'$, $55^0 54'$ (b) $55^0 54'$, $21^0 48'$
(c) $10^0 54'$, $50^0 27'$ (d) $50^0 27'$, $10^0 54'$

[HPPSC AE 2016]

Ans. (a) :
$C - \phi$ = relation of soil is
$\tau = C + \sigma\tan\phi$
for cohesionless soil C = 0
$\tau = \sigma\tan\phi$

$$\therefore \tan\phi = \frac{\tau}{\sigma} = \frac{4}{10}$$

$\Rightarrow \phi = \tan^{-1} 0.4 = 21°48'$

$$\alpha = 45 + \phi/2 = 45 + \frac{21°48'}{2} = 55°54'$$

$\phi \rightarrow$ angle of internal friction
$\alpha \rightarrow$ inclination of failure plane

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1100. A normally consolidated clayey soil failed under a major principle stress of 300 kPa with a corresponding minor principal stress of 100 kPa. If, for the same soil, the minor principal stress had been 200 kPa, the major principal stress $\phi = 30^0$ will be :

(a) 300 kPa (b) 400 kPa
(c) 600 kPa (d) 900 kPa

[HPPSC AE 2016]

Ans. (c) : For normally consolidated clay, $C = 0$

$$\sigma_1 = \sigma_3 \tan^2\left(45^o + \frac{\phi}{2}\right)$$

$$300 = 100\tan^2\left(45^o + \frac{\phi}{2}\right) \Rightarrow \phi = 30^o$$

New, $\sigma_3 = 200$

$$\therefore \sigma_1 = 200\tan^2\left(45^o + \frac{30^o}{2}\right) = 600\text{kPa}$$

1101. In a triaxial shear test conducted on a soil sample is having cohesion of 12 kPa and the angle of shearing resistance of 36^0, if the cell pressure is 200 kPa, the deviator stress at failure will be :

(a) 617.5 kPa (b) 817.5 kPa
(c) 770.37 kPa (d) 47.1 kPa

[HPPSC AE 2016]

Ans. (a) : Given, Cell pressure (σ_3) = 200 kPa
Angle of shearing resistance (ϕ) = 36°
Cohesion (C) = 12 kPa

$$\sigma_1 = \sigma_3 \tan^2\left(45 + \frac{\phi}{2}\right) + 2C\tan\left(45 + \frac{\phi}{2}\right)$$

$$\sigma_1 = 200\tan^2\left(45^o + \frac{36^o}{2}\right) + 2\times12\times\tan\left(45^o + \frac{36^o}{2}\right)$$

= 817.5 kPa

Now, $\sigma_1 = \sigma_c + \sigma_d$
σ_c = cell pressure
σ_d = deviator stress
$\sigma_d = 817.5 - 200$
$\sigma_d = 617.5$ kPa

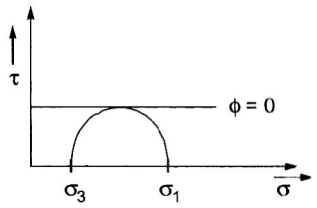
1102. Which of the following is true ?

(a) The large the coefficient of consolidation, the longer it takes for consolidation to occur
(b) Pore pressure parameter A is a constant for a soil
(c) A saturated loose sand sample reduces in volume when sheared under undrained conditions
(d) Saturated sand can exhibit an angle of shearing resistance of zero

[HPPSC AE 2016]

Ans. (d) : $t = \frac{d^2}{C_v}.T_v$; As coefficient of consolidation increase the time required (t) for consolidation decreases. Pore pressure parameters varies with the stress change in soil. for saturated loose sand there is no volume change when sheared under undrained condition.

For saturated sands, angle of shearing resistance can be zero, it can be obtained in case of UU testing.

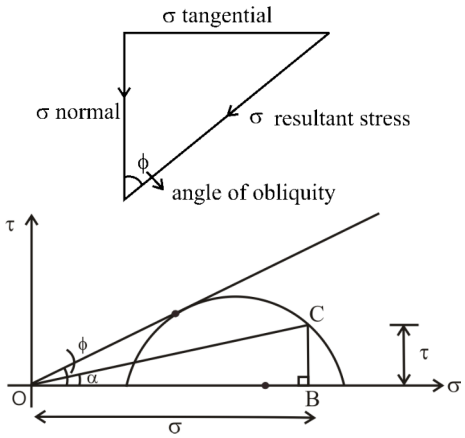


1103. Shear failure of soils takes place, when :

(a) the angle of obliquity is maximum
(b) maximum cohesion is reached in cohesive soils
(c) ϕ reaches its maximum value in cohesionless soils
(d) residual strength of the soil is exhausted

Karnataka PSC AE 2015 Paper - II
RRB SSE (Bilaspur) Yellow Paper : 21.12.2014

Ans. (a) Angle of obliquity– It is the angle made by normal stress with the tangential stress.



α = angle of obliquity = $\tan^{-1}\frac{\tau}{\sigma}$

where, α becomes ϕ, failure occurs.

1104. Unconfined compressive strength test is :

(a) Consolidated drained test
(b) Consolidated undrained test
(c) Undrained test
(d) Drained test

Rajasthan JEN (Degree) Shift - II 2016

Ans. (c) :

Soil Properties	**Unconfined compression strength (kg/cm^2)**
very soft	< 0.25
soft	0.25 – 0.50
firm	0.5 – 1
stiff	1–2
very stiff	2–4
Hard	> 4

1105. Which of following tests is conducted to assess shear strength parameter of the soil :

(a) Hygrometer test (b) Compaction test
(c) Triaxial test (d) vane shear test

GPSC AE Class (1&2) Paper - 2 2017

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Ans. (c) : The shear strength for soil can be calculated by triaxial $\boxed{\tau = S = C + \bar{\sigma}_n \tan\phi}$

Shear stress = τ, S= shear strength & soil

C = cohesion of soil, $\bar{\sigma}_n$ = Effective stress

ϕ = friction angle

1106. In a Mohr's diagram, a point above Mohr's envelope indicates :

(a) an imaginary condition
(b) a safe condition
(c) an imminent failure condition
(d) a condition of maximum obliquity

GPSC AE Class (1&2) Paper - 2 2017
ESE 1998

Ans. (a) A point above the mohr's envelope is imaginary condition be more than shear strength.

1107. For assessing the long-term stability of a slope in stiff fissured clay which type of the following laboratory test has to be done :

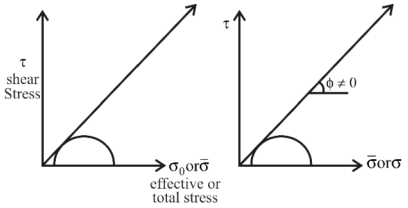
(a) Undrained triaxial test
(b) Drained triaxial test
(c) Consolidated undrained test
(d) Quick vane shear test

GPSC AE Class (1&2) Paper - 2 2017

Ans. (b) : For long term stability, we go for consolidation drain test (C-D), because as in this (Pore water) drain condition is allow flow out & consolidation is slow but give stability is settlement + to the structure.

For OCC (behave like as dens sand) & NCC (as sand)

$C \neq 0, \phi \neq 0$ $C \neq 0, \phi \neq 0$



Test :-

UU → for short term stability (Fast process test)

CU → for stability for earth dam during sudden shock condition

1108. A shear test was conducted on a soil sample. At failure, the ratio of $(P_1 - P_3) / 2$ to $(P_1 + P_3) / 2$ was equal to 1. Which one of the following shear test represents this condition?

(a) Drained triaxial compression test
(b) Undrained triaxial compression test
(c) Undrained shear test
(d) Unconfined compression test

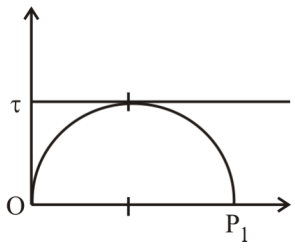
GPSC AE Class (1&2) Paper - 2 2017

Ans. (d) :

$\left[\dfrac{P_1 - P_3/2}{P_1 + P_3/2} = 1\right]$ (unconfined compressive test)

Put $P_3 = 0 \Rightarrow \dfrac{P_1/2}{P_1/2} = 1$

P_3 confined pressure is zero (no lateral pressure) only axial stress (deviate pressure or stress act) P_1



In mohr's Circle –

Centre $\dfrac{\sigma_1 + \sigma_2}{2} = \dfrac{P_1 + 0}{2}$

Shear stress $\dfrac{\sigma_1 - \sigma_3}{2} = \dfrac{P_1 - 0}{2}$

1109. Considerable loss of shear strength due to shock or disturbance is exhibited by :

(a) Under-consolidated clays
(b) Normally consolidated clays
(c) Over consolidated clays
(d) Organic soil

GPSC AE Class (1&2) Paper - 2 2017
ESE 2010

Ans. (a) : Under - consolidated clays soil losses of its shear strength due to shock or disturbance.

1110. Shear strength of a soil is its

(a) minimum resistance to shear stresses just before the failure
(b) minimum resistance to shear stresses just after the failure
(c) maximum resistance to shear stresses just before the failure
(d) maximum resistance to shear stresses just after the failure

GPSC AE (CLASS 1 & 2) 2019

Ans : (c) : Shear strength is maximum soil can resist against shear deformation (shear stress) so that the failure can't be happen before.

1111. In a Mohr's circle, the shear stress τ_f on the plane of maximum obliquity is

(a) less than the maximum shear stress τ_{max}
(b) more than the maximum shear stress τ_{max}
(c) equal to the maximum shear stress τ_{max}
(d) numerically equal to $(\sigma_1 - \sigma_3)/2$

GPSC AE (CLASS 1 & 2) 2019

Ans : (a) : for stable condition when the maximum angle of obliquity is equal the maximum angle of friction, the shear stress is should be less than maximum shear stress so that failure can't be happen

$S < \tau_{max} = C + \bar{\sigma} \tan\phi$

1112. Which of the following test is not used to measure the shear strength of soil?

(a) Triaxial Compression Test
(b) Standard Proctor Test
(c) Unconfined Compression Test
(d) Vane Shear Test

GPSC AE (CLASS 1 & 2) 2019

Ans : (b) Standard proctor test is used to determine the compaction of different types of soil and the properties of soil with a change in moisture content.

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1113. Among which of the following tests conducted for measurement of shear strength of soil, no excess pore pressure is set up at any stage of the test?
(a) drained test
(b) undrained test
(c) consolidated undrained test
(d) quick test

Kerala PSC AE 2015

Ans. (a) : Drained tests are conducted to allow the water of drain freely and hence no excess pore pressure is developed at any stage.
→ Consolidated undrained test is compression test in which soil specimen is first consolidated under all around pressure in triaxial cell before failure.

1114. Which of the following is a field test?
(a) vane shear test
(b) direct shear test
(c) triaxial compression test
(d) unconfined compression test

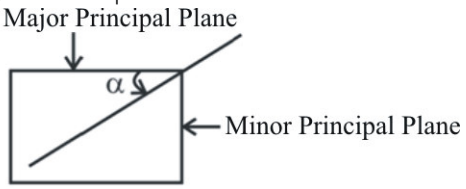
Kerala PSC AE 2015

Ans. (a) : → Vane shear test is a field test to measure unconfined compressive strength of very soft clay.
→ Direct shear test is done for cohesion less soil to determine shear strength of soil.
→ Tri-axial test is done for all type of soil to determine shear strength of soil and intermediate principal stress equal to confining pressure.

1115. If α is the angle between the direction of the failure and major principal plane, and ϕ is the angle of shearing resistance. Then the relationship between this two is
(a) $\alpha = 90^o - \frac{\phi}{2}$ (b) $\alpha = 90^o + \frac{\phi}{2}$
(c) $\alpha = 45^o - \frac{\phi}{2}$ (d) $\alpha = 45^o + \frac{\phi}{2}$

TNPSC AE 2019

Ans. (d) $\alpha = 45 + \phi/2$



1116. The shear strength for a saturated clay from unconfined compression test is
(a) Twice the unconfined compression strengt
(b) Half the unconfined compression strength
(c) Half the unconfined compression strength
(d) Equal to the unconfined compression strength

TNPSC AE 2019

Ans. (b) Since it is Clay (ϕ=0) [Shear parameter=0]
$q_u = 2C$, q_u = unconfined compressive length
$S = \tau = C + \sigma_n \tan\phi$

$\tau = C + 0 \quad \tau = \frac{q_u}{2}$

1117. The strength of a soil is usually identified by
(a) direct tensile stress
(b) direct compressive stress
(c) ultimate shear stress
(d) effective stress

GUJRAT PSC AE - II 2017

Ans : (c) The strength of a soil is usually identified by ultimate shear stress.

1118. What can be said for soil in which failure envelope for the the total stress mohr's circles becomes a horizontal line?
(a) Angle of friction is zero
(b) Angle of friction is greater than 15
(c) Angle of friction is greater than 30
(d) Angle of friction is greater than 45

APPSC Poly. Tech. Lect. 2020

Ans. (a) : When angle of friction is zero, then the failure envelope for the total stress mohr's circle becomes a horizontal line for soil.

1119. The angle of friction for purely cohesive soil is
(a) 45^0 (b) 30^0
(c) 15^0 (d) zero

ASSAM PSC CCE (PRE) 2015
UJVNL AE 2012

Ans : (d) : In compact soil, the friction angle is zero. Shear strength is compressible soil is due to the cohesion of the soil.
for C-ϕ soil $\boxed{\tau = C + \sigma\tan\phi}$
for ϕ soil $\boxed{\tau = \sigma\tan\phi}$
for C soil $\boxed{\tau = C}$

1120. Which of the following corrections is/are required while using standard penetration test values?
(a) Overburden pressure
(b) Dilatancy
(c) Ground water
(d) All of the above

ASSAM PSC CCE (PRE) 2015

Ans : (d) Overburden pressure, Dilatancy, Ground water are required using standard penetration test values.

1121. An undrained triaxial compression test is carried out on a saturated clay sample under a cell pressure of 200 kN/m^2. The sample failed at a deviator stress of 400 kN/m^2. The cohesion of' the given clay sample is :
(a) 50kN/m^2 (b) 200kN/m^2
(c) 300kN/m^2 (d) 400kN/m^2

KPSC AE 2020

Ans. (b) : Given,
$\sigma_3 = 200$ kN/m^2 (cell pressure)
$\sigma_d = 400$ kN/m^2 (deviator stress)
$\sigma_1 = \sigma_3 + \sigma_d = 200 + 400 = 600$ kN/m^2
undrained triaxial test
$C_u = \frac{600 - 200}{2} = 200$ kN/m^2

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1122. A vane 20 cm long and 10 cm in diameter was pressed into soft marine clay at the bottom of a bore hole. Torque was applied gradually and failure occurred at 2000 kg/cm, the cohesion of the clay in kg/cm^2 is :

(a) $\frac{5}{\pi\times 7}$ (b) $\frac{7}{\pi\times 6}$

(c) $\frac{12}{\pi\times 7}$ (d) $\frac{7}{\pi\times 12}$

KPSC AE 2020

Ans. (c) : Given, H = 20 cm
D = 10 cm
T = 2000 kg/cm
When both take part in shearing

$$\tau = \frac{T}{\pi D^2\left[\frac{H}{2}+\frac{D}{6}\right]}$$

$$\tau = \frac{2000}{\pi(10)^2\left[\frac{20}{2}+\frac{10}{6}\right]}$$

$$\tau = \frac{12}{\pi\times 7}$$

1123. For the determination of shear strength parameters C and ϕ, of the soil in the laboratory the test to be conducted will be :

(a) Tri-axial compression test
(b) Penetration test
(c) Sieve analysis
(d) Compaction test

Nagaland PSC Paper-II 2017

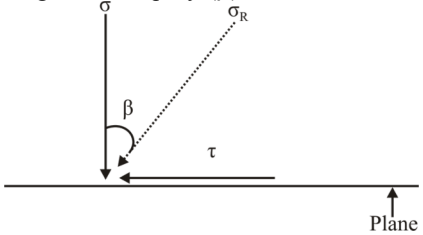
Ans. (a) : Tri-axial compression test is used for the determination of shear characteristics of all type of soil under different drainage condition.

1124. Statement (I): The inclination of the resultant stress with normal can exceed the angle of repose (adopting old terminology).
Statement (II): The ratio of the difference between greatest and least intensities of pressure to their sum cannot exceed the sine of the angle of repose (adopting old terminology).

(a) Both Statements-I and Statement-II are individually true and Statement-II is the correct explanation of Statement-I.
(b) Both Statements-I and Statement-II are individually true but Statement-II is NOT the correct explanation of Statement-I
(c) Statement-I is true but Statement-II is false.
(d) Statement-I is false but Statement-II is true.

ESE 2019

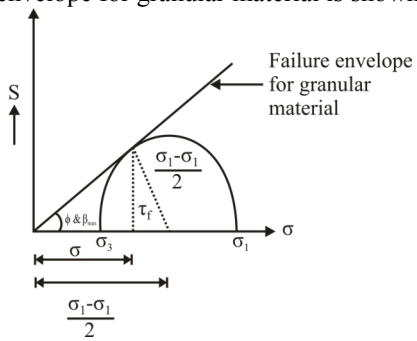
Ans. (d) The inclination of resultant stress with normal is called angle of obliquity (β) as shown below.



σ_R = Resultant stress = $\sqrt{\sigma^2+\tau^2}$

$$\beta = \tan^{-1}\left[\frac{\tau}{\sigma}\right]$$

The angle of repose of a granular material is the steepest angle at which the soil can be piled without slumping.
The term "angle of repose" is generally used with respect to granular material like sand. In the case of loose granular material, the angle of friction (ϕ) and angle of repose are equal to each other.
Failure envelope for granular material is shown below.



For failure plane, the β will be maximum

$$\beta_{max} = \tan^{-1}\left[\frac{\tau_f}{\sigma}\right]$$

From the above, $\beta_{max} = \tan^{-1}\left[\frac{\tau_f}{\sigma}\right]$

$\therefore$ Statement I is false

$$\sin\phi = \frac{\frac{\sigma_1-\sigma_3}{2}}{\frac{\sigma_1+\sigma_3}{2}} \quad \text{(or)} \quad \sin\phi = \frac{\sigma_1-\sigma_3}{\sigma_1+\sigma_3}$$

$\therefore$ Statement II is true

1125. Consider the following statements with regard to Soil Testing:

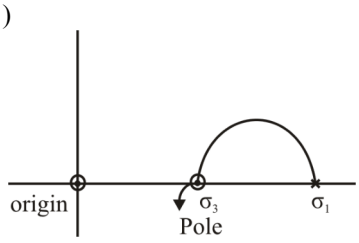
1. **The origin and pole are at the same point in a Mohr's circle**
2. **The shear stress is maximum on the failure plane**
3. **Mohr's circle drawn with data from an unconfined compression test passes through the origin**
4. **Maximum shear stress occurs on a plane inclined at 45^0 to the principal plane**

Which of the above statements are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 3 and 4 only (d) 1 and 4 only

ESE 2018

Ans. (c) (1)



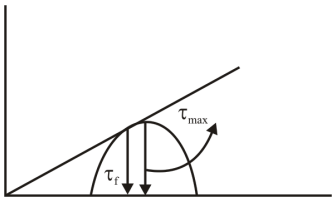
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Pole: Origin of planes (point of intersection of major & minor principal plane).

∵ So origin & pole need not be same point on mohr circle.

(2) Shear stress need not be maximum at every time

$\tau_f \le \tau_{max}$



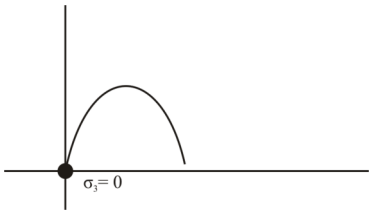
In this case $\tau_f < \tau_{max}$

So option '2' wrong.

(3) Unconfined compression test

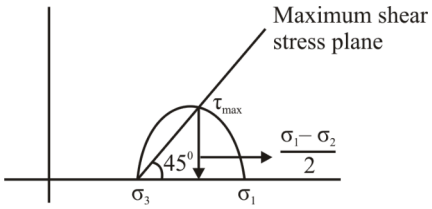
$\sigma_c = 0$

$\sigma_3 = 0$



∴ So Mohr's circle passes through origin

(4)



$$\tau_{max} = \frac{\sigma_1 - \sigma_3}{2}$$

Maximum shear stress plane makes always an angle of 45^0 w.r.t principal planes.

1126. A dry sand specimen is put through a tri-axial test. The cell pressure is 50 kPa and the deviator stress at failure is 100 kPa. The angle of internal friction for the sand specimen is

(a) 15^0 (b) 30^0

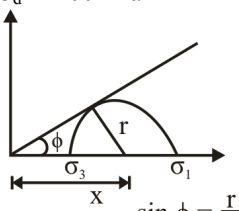
(c) 45^0 (d) 55^0

ESE 2018, 1997
OPSC Civil Services Pre 2011
Haryana PSC AE 2010
UPPSC State Eng. A.E. 2003

Ans. (b)

Cell pressure, $\sigma_c = 50$ kPa

Deviator stress 'σ_d' = 100 kPa



$\sin\phi = \frac{r}{x}$

For Dry Sand C = 0

$$\sin\phi = \frac{\sigma_1 - \sigma_3}{\sigma_1 + \sigma_3}$$

$$\phi = \sin^{-1}\left(\frac{\sigma_1 - \sigma_3}{\sigma_1 + \sigma_3}\right)$$

$\sigma_c = \sigma_3 = 50$ kPa

$\sigma_1 = \sigma_d + \sigma_3$ $(\sigma_d = \sigma_1 - \sigma_3)$

$[\sigma_1 = 100+50 = 150$ kPa$]$

$$\phi = \sin^{-1}\left(\frac{150 - 50}{150 + 50}\right)$$

$$\phi = \sin^{-1}\left(\frac{1}{2}\right)$$

Angle of Internal Friction 'ϕ' = 30^0

1127. Statement (I): The stresses and strains in a soil mass depend on the stress-deformation characteristics, anisotropy and non-homogeneity of the soil and also on the boundary conditions.
Statement (II): Boussinesq's theory of stress distribution in soils deals with layered soils only.

(a) Both Statements-I and Statement-II are individually true and Statement-II is the correct explanation of Statement-I.

(b) Both Statements-I and Statement-II are individually true but Statement-II is NOT the correct explanation of Statement-I

(c) Statement-I is true but Statement-II is false.

(d) Statement-I is false but Statement-II is true.

ESE 2017

Ans. (c)

■ Boussinesq's theory of stress distribution in soils is applicable to homogenous and isotropic soils only. The layered soil doesn't represent the homogeneity and isotropic condition. Hence for layered soils, Boussinesq's theory is not applicable. For layered soils, Westergaard's theory is appropriate.

1128. Statement (I): Cohesion and angle of internal friction are shear strength parameters of soils.
Statement (II): Cohesion is zero for pure sand and angle of internal friction is zero for pure clay.

(a) Both Statements-I and Statement-II are individually true and Statement-II is the correct explanation of Statement-I.

(b) Both Statements-I and Statement-II are individually true but Statement-II is NOT the correct explanation of Statement-I

(c) Statement-I is true but Statement-II is false.

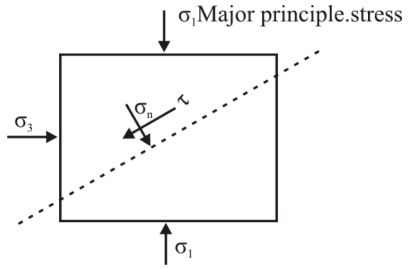
(d) Statement-I is false but Statement-II is true.

ESE 2017

Ans. (b)

■ The cohesion and internal friction angle are usually called shear strength parameter of soils.

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$[\tau = C + \sigma' \tan\phi]$

τ = shear strength
C = cohesion
ϕ = angle of internal friction.

■ Pure Sand, C = 0
Pure clay, $\phi = 0$

1129. An undrained triaxial compression test is carried out on a saturated clay sample under a cell pressure of 50 kN/m^2. The sample failed at a deviator stress of 100 kN/m^2. The cohesion of this clay sample would be

(a) 25 kN/m^2 (b) 50 kN/m^2
(c) 75 kN/m^2 (d) 100 kN/m^2

ESE 2017

Ans. (b)

Cell pressure = 50 kN/m^2
Deviator stress 'σ_d' = 100 kN/m^2.
For undrained test; $\phi = 0$

$$\sigma_1 = \sigma_3 \tan^2\left(45 + \frac{\phi}{2}\right) + 2C \tan\left(45 + \frac{\phi}{2}\right)$$

$$\sigma_1 = \sigma_3 + 2C$$

$$C = \frac{\sigma_1 - \sigma_3}{2}$$

$$C = \frac{\sigma_d}{2} = \frac{100}{2}$$

$\boxed{C = 50 kPa}$ Cohesion

1130. In an unconfined compression test, a sample clay 100 mm long and 50 mm in diameter fails under a load of 150 N at 10% strain. What is the shearing resistance taking into account the effect of change in cross section of the sample?

(a) 0.068 N/mm^2 (b) 0.088 N/mm^2
(c) 0.034 N/mm^2 (d) 0.044 N/mm^2

ISRO Scientist/Engineer 2020

Ans. (c) : $A_o = \frac{\pi}{4} \times d^2$

Initial Area, $A_0 = 1963.50 mm^2$
Strain, $\varepsilon = 10\% = 0.10$
failure load (P) = 150 N

Correct area at failure $(A_C) = \frac{A_o}{1-\varepsilon}$

$$= \frac{1963.50}{1-0.1}$$

$$A_C = 2181.66 mm^2$$

Unconfined compressive strength $(q_u) = \frac{P}{A_C}$

$$= \frac{150}{2181.66}$$

$$= 0.068 N/mm^2$$

Shearing resistance $\phi = \frac{q_u}{2} = 0.034 N/mm^2$

1131. The angle of internal friction is least for

(a) Angular grained loose sand
(b) Round grained loose sand
(c) Silts
(d) Clays

ISRO Scientist/Engineer 2018

Ans : (d) : Angle of internal friction for a given soil is the angle on the graph (Mohr's Circle) of the shear stress and normal effective stresses at which shear failure occurs.

1132. A clay soil sample is tested in tri-axial apparatus in consolidated drained conditions at a cell pressure of 100KN/m^2. What will be the pore water pressure at a deviator stress of 400KN/m^2.

(a) 0KN/m^2 (b) 20KN/m^2
(c) 40KN/m^2 (d) 60KN/m^2

ISRO Scientist/Engineer 2014

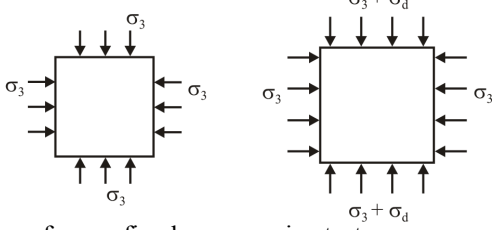
Ans : (a)

1133. Which one of the following conditions is valid in case of unconfined compression test in comparison to triaxial test?

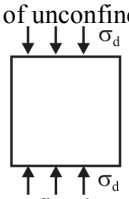
(a) Minor principal stress = 0
(b) Minor principal stress = 0.5 × major principal stress
(c) Minor principal stress = major principal stress
(d) Major principal stress = 3 × minor principal stress

ISRO Scientist/Engineer 2013
ESE 2009

Ans : (a) Triaxial Test



In case of unconfined compressive test



[σ_3= 0] Minor principal stress

■ Unconfined compression test is a special case of triaxial test in which the confining pressure is zero. This test may be conducted on undisturbed or remoulded cohesive soils. It cannot be conducted on coarse-grained soils such as sands and gravel as these cannot stand without lateral support.

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1134. The direct shear test suffers from the following disadvantage :

(a) Drain condition cannot be controlled
(b) Pore water pressure cannot be measured
(c) Shear stress on the failure plane is not uniform
(d) The area under the shear and vertical loads does not remain constant throughout the test

OPSC AE Paper- II 2019

Ans. (*) : (All options are correct)
Disadvantages of direct shear test

- Drainage condition cannot be controlled and pore water pressure can not be measured.
- Failure plane is always horizontal and pre-determined which may not be the weakest plane.
- Non-uniform stress distribution on shear plane failure starts at edge and progresses towards centre.
- Area of specimen under normal and shear does not remain constant during the test. Hence calculation of normal and shear stresses are done on the basis of nominal area (original area) which is not correct.
- Direction of principal plane are not known at every stage of the test it is only when ..... failure envelope is known that direction of principal stress will be known.

1135. The equation $\tau = C + \sigma \tan \phi$ is given by

(a) Rankine (b) Terzaghi
(c) Mohr (d) Coulomb

OPSC AE Paper- II 2019

Ans. (d)

$\tau = C + \sigma \tan \phi$
τ = Shear strength of soil
σ = Normal stress on the plane of rupture
ϕ = Angle of internal friction

initially coulomb believed that σ = total stress. Later after the knowledge of effective stress it was realised that 'σ' is actually the effective stress, hence new definition of shear stress come out.

$\tau = C' + \bar{\sigma} \tan \phi'$
$\bar{\sigma} = (\sigma - u)$ = effective stress
σ = Total stress
u = Pore water pressure

C' and ϕ' are effective stress shear parameters

Note :

C and ϕ are known as shear strength parameters of soil.
C and ϕ are not the inherent properties of soil.

These are related to the type of test and the condition under ..... they are measured.

1136. The stability or shear strength of fine-grained soils can be increased by draining them with the passage of direct current through them. This process is known as

(a) Electro-osmosis
(b) Zeta potential
(c) Electro-chemical hardening
(d) Consolidation

ESE 2020

Ans. (a)

- **Electro-Osmosis–** It is a method of drainage of cohesive soil in which a direct current (DC) is passed through a saturated soil between a positive electrode and a negative electrode, as a result pore water migrates to cathode.
  As the water is drained out, the consolidation process can be upto a hundred times faster rate than mechanical consolidation that is why substantial improvement in the shear strength of soil is achieved quickly.
- **Zeta Potential–** Zeta potential is defined as the potential difference between the dispersion medium and the stationary layer of fluid attached to the particle and it is measured by a zeta potential analyzer.
- **Electrochemical Hardening–** When a direct current (DC) potential of from 20 to 200 volts is applied across two aluminium rods embeded in clay soil, aluminium goes into solution at both electrodes, water is transported, and the intervening soil is hardened.
- **Consolidation–** It is the process in which reduction in volume takes place by expulsion of water under long term static load.

1137. A sample of saturated sand tested in a drained Triaxial compression test showed an angle of internal friction of 30^0. Deviatroic stress at failure for the sample at a confining pressure of 200 kPa is equal to :

(a) 200 kPa (b) 400 kPa
(c) 600 kPa (d) 800 kPa

PSPCL AE 2012

Ans. (b) : We know
σ_d = deviator stress
σ_c = confining stress $= \sigma_3 = 200$kPa
σ_1 = Axial stress
internal friction, $\phi = 30^o$

$\therefore \sigma_1 = \sigma_3 \tan^2 (45+\phi/2) + 2C \tan (45+\phi/2)$
For sand (C = 0)

$$\sigma_1 = \sigma_3 \tan^2 \left(45 + \frac{\phi}{2}\right) + 0$$

$$\sigma_1 = \sigma_3 \tan^2 60$$

$$\sigma_1 = \sigma_3 \left(\sqrt{3}\right)^2$$

$\sigma_1 = 200 \times 3 = 600$ kPa
$\sigma_1 = \sigma_d + \sigma_c$
$600 = \sigma_d + 200$
$\sigma_d = 400$ kPa

1138. Unconfined compression test is suitable for determining the :

(a) Sensitivity of clays
(b) Settlement of embankments
(c) Strength of partially saturated clay
(d) None of the above

PSPCL AE 2012

Ans. (d) : Unconfined compression test is suitable for determining the shear strength for clay soil only.

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1139. A direct shear test was conducted on a sand specimen under a normal stress of 200 kN/Sqm which failed at a shear stress of 100 kN/sqm. Then angle of internal friction of the soil is :

(a) 26.6 (b) 29.5
(c) 30.0 (d) 32.6

PSPCL AE 2012
JPSC AE 2009

Ans. (a) : $\tau = C + \sigma_n \tan\phi$
$100 = 0 + 200 \tan\phi$ (sand C = 0)
$\tan\phi = 0.5$
$\phi = 26.67^0$ [tan 26.67 = 0.5]

1140. In an undrained triaxial compression test, the sample failed at a deviator stress of 200 kN/m^2 when the cell pressure was 100 kN/m^2. The cohesion intercept in this case would be

(a) 200 kN/m^2 (b) 100 kN/m^2
(c) 300 kN/m^2 (d) 50 kN/m^2

GPSC AE March 2018

Ans. (b) : $\sigma_d = 200$ kN/m^2
$\sigma_3 = 100$ kN/m^2
In undrained triaxial compression test
$\phi = 0^o$
$\sigma_1 = \sigma_d + \sigma_3$
$\sigma_1 = 100 + 200$
$\sigma_1 = 300$ kN/m^2
From stress relationship
$\sigma_1 = \sigma_3 \tan^2(45 + \phi/2) + 2C\tan(45 + \phi/2)$
$300 = 100 \times 1^2 + 2C \times 1$
$C = 100$ kN/m^2

1141. If the cohesion of a pure clay found in an unconfined compressive strength test is 1 kg/cm^2, then its unconfined compressive strength in kg/cm^2 is

(a) 0.5 (b) 2
(c) 1 (d) 4

GPSC AE March 2018

Ans. (b) : Cohesion of a pure clay (C) = 1 kg/cm^2

$$C = \frac{\text{Unconfined compressive strength}}{2}$$

(q_u) Unconfined compressive strength = C × 2
= 1 × 2
$q_u = 2kg/cm^2$

1142. For a gravel of median size 16 mm, the critical shear stress is about

(a) 24 Pa (b) 15 Pa
(c) 0.7 Pa (d) 0.3 Pa

UPSC JWM 2017

Ans. (c) : Gravel are having 40-70% of shear stress/strength about critical shear stress = 0.7 Pa.

1143. Consider the following statements:
1. Undrained shear strength of saturated clay is independent of total stress
2. Unconfined compression test is classified as unconsolidated undrained test
3. Unconfined compression test is applicable to soil specimen which is fissured
Which of the above statements are correct?

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 1 and 3 only (d) 2 and 3 only

UPSC JWM 2017

Ans. (b) : Undrained shear strength of saturated clay depends on normal stress on critical plane.
Unconfined compression test is classified as confined undrain test
Unconfined compression test is applicable for clay, fissured soil, etc.

1144. If correct value of cohesion of highly soft clay is to be determines, choose the correct type of test that should be carried out ……

(a) Field vane shear test
(b) Triaxial shear test
(c) Direct shear test
(d) Laboratory unconfined compression test

GPSC Poly. Tech. Lect. 2016

Ans. (a) : The field vane shear test is a method of measuring the undrained shear strength of cohesive soil of highly soft clays.

1145. Stress path equation for tri-axial test upon application of deviatoric stress is, q 10 3 0.5p.The respective values of cohesion, C (in kPa) and angle of internal friction, ϕ are :

(a) 20 and 30degree (b) 30and20degree
(c) 20and20degree (d) 30and30degree

KPTCL AE 2016

Ans. (a) :
Stress path equation –

$$\frac{\delta_1 - \delta_1}{2} = \cos\phi + \frac{\delta_1 - \delta_3}{2}\sin\phi$$

Given equation is
$C \times \cos\phi = 10\sqrt{3}$
$\sin\phi = 0.5$
$\phi = 30^0$
$\therefore\ C \times \cos 30^0 = 10\sqrt{3}$
$C = 20, \quad \phi = 30^0$

1146. The shear strength of loamy soil depends upon

(a) internal friction
(b) cohesion
(c) both internal friction and cohesion
(d) neither internal friction nor cohesion

Maharashtra PSC Mains 2012 Paper -2

Ans. (c) : Loam soil composed mostly of sand, silt and smaller amount of clay. So, it has both internal friction and cohesion.

1147. The useful method of finding the shear strength of soft sensitive clayey soils is by means of :

(a) Cone penetration test
(b) SPT test
(c) Vane shear test
(d) Tensional shear test

Nagaland PSC Paper-II 2017
OPSC Civil Services Pre 2011

Ans. (c) : Vane shear test is used to determine the undrained shear strength of soils especially soft clays.

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1148. An undrained triaxial compression test is carried out on saturated clay sample under a cell pressure of 100 kN/m^2. The sample failed at a deviator stress of 200 kN/m^2. The cohesion of the given sample of clay is

(a) 100 kN/m^2 (b) 150 kN/m^2
(c) 200 kN/m^2 (d) 50 kN/m^2

TRB Poly. Tech. Lect. 2012

Ans. (b) Deviator stress = $\sigma_1 - \sigma_3$

Cell pressure $(\sigma_3) = 100$ kNm^2

deviator stress = 200 kN/m^2

$\sigma_1 = 200 + 100$

$\sigma_1 = 300$ kN/m^2

$\sigma_1 = 2C$ (for C – soil)

$300 = 2C$

$C = \frac{300}{2} = 150$ kN/m^2

1149. Which one of the following planes is not most likely to be the failure plane in Sandy soil?

(a) Maximum shear stress plane
(b) Planes carrying maximum normal stress
(c) Principal plane
(d) Planes with maximum angle of obliquity

TRB Poly. Tech. Lect. 2012

Ans. (d) : Planes with maximum Angle of obliquity is not most likely to be the failure plane in sandy soil.

1150. Shear strength of the soil is

(a) Directly proportional to angle of internal friction of soil
(b) Inversely proportional to angle of internal friction of soil
(c) Decrease with increase in normal stress
(d) Increase with increase in normal stress

TSGENCO AE 2015

Ans. (d) : Shear strength (τ) of soil is given by

$\tau = C + \sigma_n \tan \phi$

(τ) shear strength increases with increase in normal stress (σ_n)

C = cohesion of soil

ϕ = angle of internal friction

1151. Consistency as applied to cohesive soils is an indicator of its.

(a) Density (b) Shear strength
(c) Moisture content (d) Porosity

UJVNL AE 2016

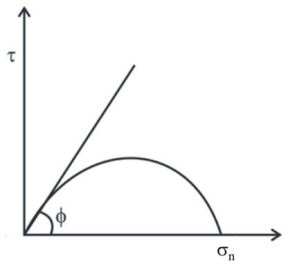
Ans. (b) Soil consistency is the strength with which soil materials are held together or the resistance of soils to deformation and rupture, soil consistency is measured for wet, moist and dry soil samples. So, consistency as applied to cohesive soils is an indicator of its shear strength.

1152. The angle between abscissa and the tangent to the curve representing the relationship of shearing resistance to normal stress acting within the soil is called

(a) Angle of repose
(b) Angle of internal friction
(c) Cohesiveness
(d) Hydraulic gradient

Punjab PSC SDE 2016

Ans. (b) :



where ϕ = angle of internal friction

τ = shearing resistance

σ_n = normal stress

1153. For determination of shear strength parameters, C and ϕ of soil in the laboratory the test to be conducted will be

(a) triaxial compression test
(b) sieve analysis
(c) compaction test
(d) relative density test

Arunachal Pradesh Poly. Tech. Lect. 2019
ESE 2012

Ans. (a)

Test	Application
Triaxial compression test	Shear strength parameter
Sieve analysis	Particle size analysis coarse soil
Compaction test	optimum moisture content
Relative density test	Measures of compaction of cohesionless soil

1154. The effective stress friction angle of a saturated cohesionless soil is 38^0. The ratio of shear stress to normal effective stress on the failure plane is

(a) 0.781 (b) 0.616
(c) 0.488 (d) 0.438

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (a) Given, Friction angle, $\phi = 38^o$

Shear stress $(\tau) = \sigma_n \tan \phi$

$$\frac{\tau}{\sigma_n} = \tan 38^0 = 0.781$$

1155. The object of triaxial compression test is to determine the shear strength of soil under :

(a) Lateral displacement
(b) Longitudinal displacement
(c) Lateral Confinement
(d) Longitudinal confinement

TRB Poly. Lect. Tech. 2017

Ans. (c) : The object of triaxial compression test is to determine the shear strength of soil under lateral confinement.

1156. In which of the following tests is the solid specimen of soil subjected to direct stress in three mutually perpendicular directions?

(a) Direct Shear Test
(b) Unconfined Compression Test

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(c) Vane Shear Test
(d) Triaxial Compression Test

SJVNL ET 2019

Ans. (d) : Triaxial Compression Test is the solid specimen of soil subjected to direct stress in three mutually perpendicular direction.

1157. In general, plastic undrained clay has negligible or zero :
(a) Cohesion
(b) Shear strength
(c) Angle of internal friction
(d) Structural resistance

UPPCL AE 2015

Ans. (c) : For plastic undrained clay
$\phi = 0^o$ always
ϕ = internal friction angle

1158. The length/diameter ratio of cylindrical specimens used in triaxial test is generally
(a) 1 (b) 2
(c) 3 (d) 4

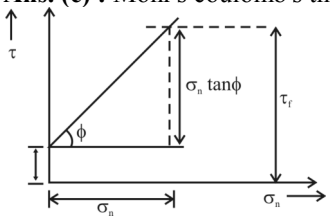
GPSC AE June 2019

Ans. (b) : The length/diameter ratio of cylindrical specimens used in triaxial test is the generally 2.

1159. If C is cohesion, σ is normal stress and ϕ is angle of internal friction of soil, then Coulomb's equation of shear strength (S) can be represented as
(a) $C = S + \sigma \tan\phi$ (b) $S = \sigma + c \tan\phi$
(c) $C = S - \sigma \tan\phi$ (d) $S = C - \sigma \tan\phi$

GPSC AE June 2019

Ans. (c) : Mohr's coulomb's theory



$\tau_f = C + \sigma_n \tan\phi$

τ_f = Shear strength of soil
C = Cohesion
σ_n = normal stress
ϕ = Angles of internal friction of soil

1160. A clear dry and sample was tested in a direct shear test. If the normal stress and the shear stress at failure were equal to 120 kN/m^2, then the angle of shearing resistance of the sand would be:
(a) 30º (b) 45º
(c) 60º (d) 90º

GPSC AE Dec 2018

Ans. (b) : $\sigma_n = 120$ kN/m^2
$\tau = 120$ kN/m^2
$\tau = \sigma_n \tan\phi$
$120 = 120 \tan\phi$
$1 = \tan\phi$,
$\tan 45 = \tan\phi$
$\phi = 45^o$

1161. In an unconfined compression test on stiff clay, if the failure plane made an angle of 53º to the horizontal, what would be the angle of shearing resistance?
(a) 12º (b) 14º
(c) 16º (d) 18º

GPSC AE Dec 2018

Ans. (c) : Given, angle of failure plane (ϕ_f) = 53º

$$\phi_f = \frac{\phi}{2} + 45 \Rightarrow (53 - 45) \times 2 = \phi$$

$\phi = 16^o$

1162. The height to diameter ratio of a laboratory vane, in vane shear test is
(a) 1.0 (b) 1.5
(c) 2.0 (d) 3.0

GPSC Assistant Prof. 2016

Ans. (c) : Vane shear test

	Lab size	Field size
Height of vane (H)	20 mm	10 to 20 cm
Dia of vane (D)	12 mm	5 to 10 cm
Thickness of vane (t)	0.5 to 0.1 mm	2 to 3 cm

$$\frac{H}{D} = \frac{20mm}{12mm} = 1.67 \approx 2$$

1163. If drainage is permitted throughout, in a shear strength test during the application of both and shear stress, so that full consolidation occurs and no excess pore water pressures develop at any stage of the test, is known as
(a) consolidated undrained test
(b) consolidated drained test
(c) unconsolidated undrained test
(d) unconfined compression test

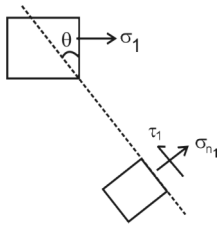
GPSC Assistant Prof. 2016

Ans. (b) : If drainage is permitted throughout, in a shear strength test during the application of both and shear stress, so that full consolidation occurs and no excess pore water pressures develop at any stage of the test, is known as consolidated drained test.

1164. The maximum shear stress in a uniaxial loading occurs at a plane inclined at
(a) 45^0 and 135^0 (b) 45^0
(c) 135^0 (d) 90^0 and 180^0

GPSC Assistant Prof. 2016

Ans. (a)



$$\tau_1 = -\left(\frac{\sigma_1 - 0}{2}\right)\sin 2\theta$$

If $\theta = 45^0$ or 135^0
then

$$\tau_1 = \tau_{max} = -\left(\frac{\sigma_1 - 0}{2}\right)$$

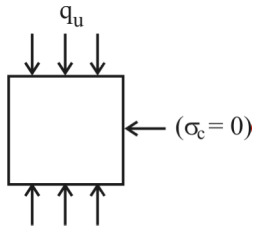
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1165. The shear strength of a saturated clay from unconfined compression test is equal to

(a) half the unconfined compressive strength
(b) unconfined compressive strength
(c) twice the unconfined compressive strength
(d) four times the unconfined compressive strength

GPSC Assistant Prof. 2016

Ans. (a) Unconfined compression test (no confining pressure)



q_u = unconfined compressive strength

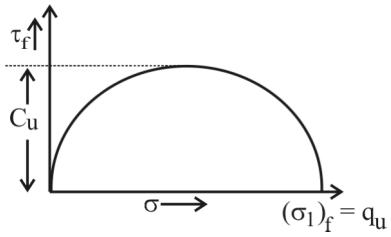
Unconfined compress strength $(q_u) = (\sigma_1)_f$

here,

$\sigma_3 = 0$

For C-ϕ soil

$$(\sigma_1)_f = 2C \tan\left(45 + \frac{\phi}{2}\right)$$



for C-soil

$(\sigma_1)_f = 2C$

For clay's or C-soil

$$\tau_f = S = C_u = \frac{q_u}{2}$$

1166. A direct shear test was conducted on a cohesionless soil specimen under a normal stress of 200 kN/m^2. The specimen failed at a shear stress of 200 kN/m^2. The angle of internal friction of the soil is

(a) 45^0 (b) 30^0
(c) 60^0 (d) 75^0

UJVNL AE 2016

Ans. (a) : Normal stress, $\sigma = 200$ kN/mm^2
failure shear stress, $\tau = 200$ kN/m^2,
$C = 0$
As, we know that,
$\tau = C + \sigma \tan \phi$
$200 = 0 + 200 \tan \phi$
$\tan \phi = 1$
$\phi = 45^0$

1167. Undrained shear strength C_u of a saturated clay tested in unconfined compression is given in terms of unconfined compressive strength q_u as

(a) $C_u = 0.5\ q_u$ (b) $C_u = q_u$
(c) $C_u = 0.66\ q_u$ (d) $C_u = 2\ q_u$

UJVNL AE 2016

Ans. (a) : Undrained shear strength C_u of a saturated clay tested in unconfined compression is given in terms of unconfined compressive strength q_u as

$$C_u = 0.5\ q_u$$

1168. A cohesionless soil, in a direct shear test, under a normal stress of 100 kPa yields a maximum shearing resistance of 100 kPa. The value of angle of internal friction is

(a) 45° (b) 30°
(c) 60° (d) 0

APTRANSCO 2017
AP TRANSCO AEE 2017

Ans. (a) : Given
$\bar{\sigma}_n = 100$ kPa
$\tau = 100$ kPa
$\tau = c + \bar{\sigma}_n \tan \phi$
100 kPa = 0 + 100 kPa tan ϕ
1 = tan ϕ
$\phi = 45^0$ Angle of internal friction

1169. In-situ vane shear test is used to measure shear strength of:-

(a) Sandy soil
(b) Stiff clays
(c) Very soft and sensitive clays
(d) Gravel

UPPCL AE 2014
MPPSC AE 2014

Ans. (c) : In-situ vane shear test is used to measure shear strength of very soft and sensitive clays (like marine clay).

1170. In Unconfined compression test, minor principal stress is equal to :

(a) Deviator stress (b) Axial stress
(c) Major principal stress (d) Zero

UPPCL AE 2019

Ans. (d) Unconfined compression test is a special case of triaxial test in which the confining pressure is zero.
$\sigma_3 = 0$
σ_3 is minor principal stress
unconfined compression test is suitable only for saturated clay.

1171. In an unconfined compression test on a saturated clay, the undrained shear strength was found to be 6t/m^2. If a sample of the same soil is tested in an undrained condition, in triaxial compression at a cell pressure of 20 t/m^2, the major principal stress at failure will be

(a) 48 t/m^2 (b) 32 t/m^2
(c) 24 t/m^2 (d) 12 t/m^2

Haryana PSC AE 2010
ESE 1995

Ans. (b) : Given, cell pressure = 20 t/m^2
Major principal stress in undrained test
= cell pressure + undrained shear strength × 2
= 20 + 6 × 2 = 32 t/m^2

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1172. Select correct statement from the following:

(a) Failure plane does not carry maximum shear stress
(b) Failure plane carries maximum shear stress
(c) Failure plane does not carry equal to maximum shear stress
(d) All of the above statements
(e) None of these statements

Haryana PSC AE Paper–II 2018

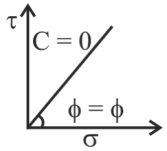
Ans. (a) : Only in the case of pure clay where (angle of internal friction) is zero failure plane carries max. Shear stress, in every other case fails at a shear stress less than max. Shear stress.

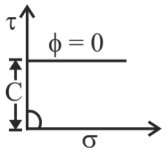
$$S = C + \bar{\sigma}_n \tan\phi$$

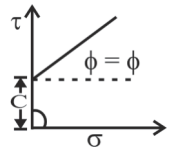
C = 0 for cohensionless soil

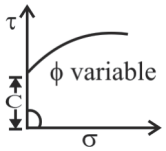
$\phi = 0$ For cohesive soil

1173. Which one of the following figures gives the failure envelope for normally consolidated saturated clay sample tested in triaxial test under drained conditions?

(a) 

(b) 

(c) 

(d) 

Haryana PSC AE 2017

Ans. (a) : j

1174. In a triaxial tests carried out on a sand sample with cell pressure 20 kPa, the observed value of applied stress at the point of failure is 4 kPa. The angle of internal friction on the soil is :

(a) 10 degree (b) 15 degree
(c) 20 degree (d) 45 degree
(e) 30 degree

CGPSC AE Shift - 2 2014

Ans. (e) : for sand C = 0

$\sigma_3 = 20$ kPa

$\sigma_1 - \sigma_3 = 4$ kPa

$\sigma_1 = 20 + 4 = 24$ kPa

$$\sigma_1 = \sigma_3 \tan^2\left(45 + \frac{\phi}{2}\right) + 2C \tan\left(45 + \frac{\phi}{2}\right)$$

$$24 = 20\tan^2\left(45 + \frac{\phi}{2}\right)$$

$$\boxed{\phi = 30^0}$$

1175. Laboratory vane shear test can also be used to determine-

(a) Shear parameter of silty sand
(b) Shear parameter of sandy clay
(c) Liquid limit of silty clay
(d) Plastic limit of clayey silt

Rajasthan VP ITI 2018

Ans. (b) : Laboratory vane shear test can also be used to determine Shear parameter of sandy clay.

1176. When a loose saturated sand deposit is subjected to a shock or dynamic load, its shear strength will

(a) increase
(b) decrease
(c) remains same
(d) increases initially and decreases later

GPSC AE Dec 2018

Ans. (b) : $\tau = \sigma_n \tan\phi$

Due to shock or dynamic load, shear strength decreases.

1177. Consider the following statements in respect of a direct shear test:

1. **The Mohr circle can be drawn for stress conditions prior to failure**
2. **The failure plane may not be the weakest plane**
3. **Control on drainage is possible during this test**
4. **The stresses are greater towards the centre of the sample**
5. **The side walls of the shear box cause lateral restraint on the specimen**

Which of the above statements are correct?

(a) 1 and 2 (b) 3 and 4
(c) 2 and 5 (d) 4 and 5

UPSC JWM 2017

Ans. (c) : Limitation of direct shear test –
1) No control on the drainage
2) Failure plane is predetermined
3) Pore pressure can not be measured
4) Side walls of the shear box cause lateral restraint on the specimen.

1178. Which of the following factors are associated with the behavior of sand mass during earthquake of cause liquefaction?

1. **Number of stress cycle**
2. **The frequency and amplitude of vibration of waves generated by an earthquake**
3. **Characteristics of sand**
4. **Relative density**

Select the correct answer using the choice given below:

(a) 1 and 2 (b) 2 and 4
(c) 3 and 4 (d) 1, 2, 3 and 4

GPSC AE Dec 2018

Ans. (d) : (i) Number of stress cycle
(ii) The frequency and amplitude of vibration of waves generated by an earthquake
(iii) Characteristics of sand
(iv) Relative density.

1179. Which of the following shear test can be conveniently used to determine the sensitivity of the soils?

(a) Unconfined compression test
(b) Direct shear test
(c) Vane shear test
(d) Triaxial compression test

CIL MT 2020
SJVNL ET 2019

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Ans. (c) : Vane shear test is primarily to used to find shear parameters of clayey soil it can also be used to determine sensitivity of soils.

1180. Cohesion and angle of internal friction are:

A. Shear parameters of soil
B. Plasticity characteristics of soil

(a) Only B (b) Neither A nor B
(c) Both A and B (d) Only A

IRCON AE 2017

Ans. (d) : The plasticity of soil is its ability to undergo-deformation without cracking.
The shear parameters cohesion (c) and friction angle (ϕ) can be determined by direct shear test and unconfined compression test.

1181. If a sample of dry sand tested in direct shear test gives failure shear stress as 100 kN/m^2 at a normal stress of 200 kN/m^2, then the angle of internal friction is given by

(a) $\tan^{-1}(2)$ (b) $\tan^{-1}(1)$
(c) $\tan^{-1}\left(\frac{1}{4}\right)$ (d) $\tan^{-1}\left(\frac{1}{2}\right)$

WBPSC AE 2016

Ans. (d) : Given, Shear stress, (τ) = 100 kN/m^2
Normal stress (σ) = 200 kN/m^2
$\tau = C + \sigma \tan \phi$ (C) = 0
$100 = 200 \tan \phi$
$\phi = \tan^{-1}\left(\frac{1}{2}\right)$

1182. The angle of internal friction of clays as per UU tests is usually

(a) 0º to 5º (b) 5º to 20º
(c) 20º to 30º (d) 30º to 45º

WBPSC AE 2014

Ans. (b) : The angle of internal friction of clays as per UU tests is usually 5º to 20º.

1183. Length to diameter ratio for cylindrical specimen of soil for triaxial shear test is normally

(a) 1.0 (b) 1.5
(c) 2.0 (d) 2.5

WBPSC AE 2012

Ans. (c) : Length to diameter ratio for cylindrical specimen of soil for triaxial shear test is (L/D) 2.0.

1184. A soil having, $C = \frac{\sqrt{3}}{2}, \phi = 30^\circ$ is subjected to cell pressure of 1 unit. The deviator stress at failure will be

(a) 3 unit (b) 4 unit
(c) 5 unit (d) 6 unit

WBPSC AE 2003

Ans. (c) : $\sigma_1 = \sigma_3 \tan^2\left(45 + \frac{\phi}{2}\right) + 2C\tan\left(45 + \frac{\phi}{2}\right)$

$= 1 \times \tan^2\left(60^\circ\right) + 2 \times \frac{\sqrt{3}}{2} \times \tan 60^\circ$

$\sigma_1 = 3 + 3 = 6$

deviator stress, $(\sigma_d) = \sigma_1 - \sigma_3 = 6 - 1 = 5$ unit

1185. 'Sensitivity' of soil can be accessed from

(a) standard penetration test
(b) static cube penetration test
(c) field density test
(d) vane shear test

WBPSC AE (PHE) 1998

Ans. (d) : 'Sensitivity' of soil can be accessed from vane shear test.

1186. In an unconfined compression test the specimen fails at a compressive stress of 14 t/m^2. The shear strength of the soil is

(a) $7\frac{t}{m^2}$ (b) $14\frac{t}{m^2}$
(c) $28\frac{t}{m^2}$ (d) unknown

WBPSC AE (PHE) 1998

Ans. (a) : Given, unconfined compression test C_u = 14 t/m^2

Shear strength = $\frac{C_u}{2} = \frac{14}{2} = 7\ t/m^2$

1187. For estimating bearing capacity of a saturated clay deposit the shear strength perimeters should be determined from

(a) direct shear test
(b) undrained triaxial test
(c) drained triaxial test
(d) consolidated undrained triaxial test

WBPSC AE (Common) 1998

Ans. (a) : For estimating bearing capacity of a saturated clay deposit the shear strength perimeters should be determined from direct shear test.

1188. The shear strength of a pure clay specimen when tested in Unconfined compression Test was found to be 100 kPa. If the same specimen was tested in Tri-axial Compression Test, the deviatoric stress at which specimen will undergo failure when the confining stress was 50 kPa, will be:

(a) 50 kPa (b) 100 kPa
(c) 200 kPa (d) 400 kPa

TSPSC Deputy Engineer 2015

Ans. (c) : Given, $\phi = 0$,
Shear strength, $S = \tau = 100$ kPa
Confining stress, $\sigma_3 = \sigma_c = 50$ kPa
$S = C + \sigma \tan \phi$ ($\phi = 0$)
$100 = C$
$C = 100$ kPa

$\sigma_1 = \sigma_3 \tan^2\left(45 + \frac{\phi}{2}\right) + 2C\tan\left(45 + \frac{\phi}{2}\right)$

$= 50 \tan^2\left(45^0\right) + 2c \tan 45^0$

$= 50 + 2 \times 100 = 250$ kPa

deviator stress, $\sigma_d = \sigma_1 - \sigma_3 = 250 - 50 = 200$ kPa

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1189. In a tri-axial compression test, a specimen failed at a deviatoric stress of 180 kPa when the cell pressure was 100 kPa. Determine the major and minor principal stresses

(a) 180, 100 (b) 80, 20
(c) 280, 100 (d) 100, 80

TSPSC AEE 2017

Ans. (c) : Major principal stress, $\sigma_1 = \sigma_c + \sigma_d$
$= 100 + 180 = 280$ kPa
Minor principal stress, $\sigma_3 = \sigma_c = 100$ kPa

1190. In a direct shear test on a dry clean sandy soil specimen with C = 0 kPa, $\phi = 45^o$, determine the shear stress at failure when the normal stress was 100 kPa.

(a) 45 kPa (b) 75 kPa
(c) 145 kPa (d) 100 kPa

TSPSC AEE 2017

Ans. (d) : Given, C = 0 kPa, $\phi = 45^o$
Shear stress, $\tau = C + \sigma \tan \phi$
$= 0+100 \tan 45$
$= 100\times1$
$= 100$ kPa

1191. The shear strength of a Clayey soil ($\phi = 0$) is found to be 60 kPa when tested in and unconfined compressive strength test. Then, the major and minor principal stresses at failure respectively are

(a) 120 kPa, 60 kPa (b) 120 kPa, 0 kPa
(c) 60 kPa, 120 kPa (d) 120 kPa, 120 kPa

TSPSC AEE 2017

Ans. (b) : Given, shear strength = 60 kPa
Major principal stress $\sigma_1 = 2\times$ shear strength
$= 2\times60$
$= 120$ kPa
Minor principal stress, $\sigma_3 = 0$

1192. Which of the following statements is/are correct?

(a) Quick condition and liquefaction of saturated sands are based on similar phenomenon
(b) Quick condition is with only earth dams
(c) Liquefaction is possible in dry sand also
(d) All of the above

Chandigarh AE 2017

Ans. (a) : Quick condition and liquefaction of saturated sands are based on similar phenomenon.

1193. Which of the following statements is/are correct?

(a) Liquefaction is generally association with sandy sand
(b) Liquefaction is not possible in normal clays
(c) Highly sensitive clays may undergo liquefaction under vibrations
(d) All of the above

Chandigarh SDE 2017

Ans. (d) :

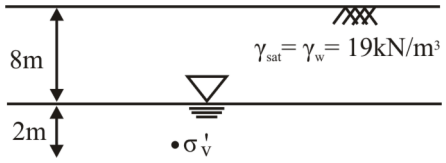
1194. The representative liquid limit and plastic limit values of a saturated consolidated clay deposit are 60% and 30%, respectively. The saturated unit weight of the soil is 19 kN/m^3. The water table is at 8 m below ground level. At a depth of 10 m from the ground surface, the undrained shear strength of the soil will be nearly

(a) 37.7 kN/m^2 (b) 33.5 kN/m^2
(c) 29.3 kN/m^2 (d) 25.1 kN/m^2

ESE 2019

Ans. (a)

For saturated consolidate clay–
Liquid Limit $w_L = 60\%$
Plastic Limit $w_p = 30\%$
$\gamma_w = 19$ kN/m^3



For clays soil; $\tau = C_u$
$[I_P = w_L - w_P = 60-30 = 30\%]$
$[\sigma_v' = 8\gamma + 2\gamma']$
$\therefore$ $\boxed{\gamma = \gamma_{sat} = \gamma_w}$
$\sigma_v' = 8 \times 19 + 2\,[19-9.81]$
$[\sigma_v' = 170.4\text{kPa}]$
Now, from skempton and Henkel.

$$\sigma_v' = \frac{C_u}{0.11 + 0.0037\ I_P}$$

$$\frac{C_u}{\sigma_v'} = 0.11 + 0.0037 \times 30$$

$$\frac{C_u}{\sigma_v'} = 0.221$$

$C_u = 0.221 \times 170.4$
$\boxed{C_u = 37.7\text{kPa}}$ Shear Strength

1195. Which one of the following statement provides the best argument that direct shear tests are not suited for determining shear parameters of a clay soil?

(a) Failure plane is not the weakest plane
(b) Pore pressures developed can not be measured
(c) Satisfactory strain levels cannot be maintained
(d) Adequate consolidation can not be ensured

TNPSC AE 2018

Ans. (b) : In direct shear test measurement of pore water pressure is not possible, due to which effective stress acting cannot be determined.

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9. Earth Pressure and Retaining Wall

1196. While making the vertical excavation in soft saturated clay, soil caved in at a depth of 4 m. If the unit weight of soil is 20000 N/m^3, what is the cohesion of the soil?

(a) 26.67 kN/m^2 (b) 40 kN/m^2
(c) 20 kN/m^2 (d) 10 kN/m^2

BHEL ET 2019

Ans. (c) : For clay,
$K_a = 1,\ h_c = 4$ m, $\gamma = 20$ kN/m^3

Depth of unsupported excavation $= \dfrac{4C}{\gamma\sqrt{K_a}}$

$$4 = \frac{4C}{20\times\sqrt{1}}$$

$C = 20$ kN/m^2

1197. An excavation is to be made up to a depth of 6m in cohesive soil having an unit weight of 20 kN/m^3, what is the required cohesive strength of clay to keep the sides of earth remains vertical without any support.

(a) 29.3 kN/m^2 (b) 28.9 kN/m^2
(c) 30.0 kN/m^2 (d) 31.0 kN/m^2

TRB Poly. Lect. Tech. 2017

Ans. (c) : Given,
$H = 6$m, $\gamma = 20$ kN/m^3
$\phi = 0$

$$H = \frac{4C}{\gamma\sqrt{K_a}}$$

$$K_a = \frac{1-\sin\phi}{1+\sin\phi}$$

$$K_a = \frac{1-0}{1+0} = 1$$

$$6 = \frac{4\times C}{20}$$

$C = 30$ kN/m^2

1198. Excavation was being done for a foundation trench in a plastic clay having unit of 2.3 gm/cc. When the depth of excavation reaches 4 meters, it failed. On the assumption that $\phi = 0$, the value of cohesion of the clay is :

(a) 2.3 kg/cm^2 (b) 0.23 kg/cm^2
(c) 23 kg/cm^2 (d) 230 kg/cm^2

[HPPSC 2016-II]

Ans. (b) : Given, $z_c = 4$m
$\gamma = 2.3$ gm/cc
$\phi = 0$

Critical height for excavation in clay is given as $z_c = \dfrac{4C}{\gamma\sqrt{K_a}},\ \phi = 0,\ K_a = 1$

$$\frac{4C}{\gamma} = 4 \text{ m}$$

$$\Rightarrow \quad \frac{4\times C}{2.3\text{g/cc}} = 4 \text{ m} = 400 \text{ cm}$$

$$\Rightarrow \quad C = 230 \text{ g/cm}^2 = 0.23 \text{ Kg/cm}^2$$

1199. The earth pressure of a soil at rest, is proportional to

(a) tan (45 - θ) (b) tan (45 + θ)
(c) (1 - sin θ) (d) (1 + sin θ)

ISRO Scientist/Engineer 2013

Ans : (c) When the retaining wall is at rest then the ratio between the lateral earth pressure and the vertical pressure is called the co-efficient of the earth pressure at rest.

$$\left[K_o = \frac{\text{Lateral pressure}}{\text{Vertical pressure}}\right]$$

$\left[K_o = 1 - \sin\phi\right]$ for ϕ soil.

ϕ = Angle of internal friction.

1200. Cohesive soil are :

(a) Good for backfill because of low lateral pressure
(b) Poor for backfill because of large lateral pressure
(c) Good for backfill because of high shear strength
(d) Ideal for backfill soil

OPSC Civil Services Pre 2011

Ans. (b) : Cohesive soils are poor for back-fill because of large lateral pressure.

1201. A vertical cut is to be made in saturated clay with C = 14 kN/m^2, $\phi = 0^0$ and $\gamma = 21$ kN/m^3. Without side collapse, what is the theoretical depth to which the clay can be excavated :

(a) 1.67 m (b) 2.67 m
(c) 3.67 m (d) 4.33 m

Rajasthan JEN (Degree) Shift - II 2016

Ans. (b) : Given that -
cohesion, $C = 14$ kN/m^2
friction angle, $\phi = 0^0$
unit weight of soil, $\gamma = 21$ kN/m^3
Theoretical depth (H_C) –

$$H_C = \frac{4C}{\gamma}\tan\left(45^0 + \frac{\phi}{2}\right)$$

$$H_C = \frac{4\times 14}{21}\tan\left(45^0 + \frac{0^0}{2}\right)$$

$$H_C = \frac{56}{21}$$

$H_C = 2.67$ m

1202. A vertical cut in a clayey soil with unit weight of 19 kN/m^3 failed when the depth of cut 4.0 m. The cohesive strength of clay is-

(a) 76.0 kN/m^2 (b) 8.5 kN/m^2
(c) 38 kN/m^2 (d) 19.0 kN/m^2

OPSC AE -2016 (II)
OPSC AEE 2015 PAPER–I

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Ans. (d) : Critical depth for vertical cut in clayey soil is given as $\left(\frac{4C}{\gamma\sqrt{k_a}}\right)$ *i.e.* $\frac{4C}{\gamma\sqrt{k_a}} = 4m$

For clayey soil $\phi = 0$, $k_a = 1$

$$\frac{4\times C}{19\times\sqrt{1}} = 4$$

$$C = \frac{\sqrt{1}\times 19\times 4}{4}$$

$\therefore C = 19kN/m^2$

1203. The ratio of the horizontal effective stress σ_h' to the vertical effective stress, σ_v' is termed as :

(a) Safe Bearing Capacity
(b) Coefficient of lateral earth pressure
(c) Friction
(d) Optimum ratio

Karnataka PSC AE 2017 Paper - II

Ans. (b) : $\frac{\sigma h'}{\sigma v'}$ = coefficient of lateral earth pressure

it is of 3 types
(i) Earth pressure coefficient at rest, $k_{o'}$
(ii) Active earth pressure coefficient, $k_{a'}$
(iii) Passive earth pressure coefficient, $k_{p'}$

1204. The Poisson ratios of soil sample 1 & 2 are μ_1 and μ_2 respectively and the coefficient of earth pressure at rest for soil sample 1 and 2 are k_1 and k_2 respectively. If $\mu_1/\mu_2 = 1.5$ and $(1-\mu_1)/(1-\mu_2) = 0.875$, then k_1/k_2 will be

(a) 1.3125 (b) 1.7143
(c) 1.8213 (d) 1.9687

ISRO Scientist/Engineer 2014

Ans : (b) Poisson ratio is μ_1, μ_2
Coefficient of earth pressure at rest is k_1, k_2
Given :

$$\frac{\mu_1}{\mu_2} = 1.5, \quad \frac{1-\mu_1}{1-\mu_2} = 0.875$$

$$k_1 = \frac{\mu_1}{1-\mu_1}, \quad k_2 = \frac{\mu_2}{1-\mu_2}$$

$$\frac{k_1}{k_2} = \frac{\mu_1}{\mu_2}\times\frac{(1-\mu_2)}{(1-\mu_1)} = \frac{1.5}{0.875} = 1.7143$$

1205. For a soil having angle of shearing resistance is 30^0, the coefficient of active earth pressure is

(a) $\frac{1}{3}$ (b) 3
(c) $\frac{1}{2}$ (d) 2

GPSC AE (CLASS 1 & 2) 2019
GUJRAT PSC AE-II 2017
ASSAM PSC CCE (PRE) 2015

Ans : (a) : Given, angle of shearing resistance $\phi = 30^o$

Coefficient of active earth pressure $(k_a) = \frac{1-\sin\phi}{1+\sin\phi}$

$$k_a = \frac{1-\sin 30^0}{1+\sin 30^0}$$

$$k_a = \frac{1-\frac{1}{2}}{1+\frac{1}{2}}$$

$$\boxed{k_a = \frac{1}{3}}$$

1206. Consider the following assumptions regarding Coulomb's Wedge Theory:

1. **There is equilibrium of every element within the soil mass of the material**
2. **There is equilibrium of the whole of the material**
3. **Backfill is wet, cohesive, and ideally elastic**
4. **The wall surface is rough**

Which of the above assumptions are correct?

(a) 1 and 3 only (b) 1 and 4 only
(c) 2 and 3 only (d) 2 and 4 only

ESE 2018

Ans. (d)
According to columb's wedge theory
- Soil sliding wedge itself acts as rigid body, limit equilibrium is taken by considering sliding wedge as whole of material.
- Back of wall is rough & soil should be dry & cohesion less.

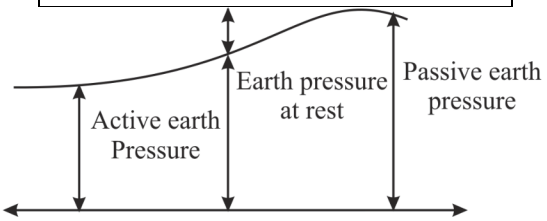
1207. The correct sequence of the given parameters in descending order of earth pressure intensity is

(a) Active, passive, at rest
(b) Passive, active, at rest
(c) Passive, at rest, active
(d) At rest, passive, active

GPSC AE Class (1&2) Paper - 2 2017
UPRVUNL AE 2015
ESE 2000

Ans : (c)
- Soil has a tendency to exert a lateral pressure against any object in contact.
- Passive earth pressure is much greater than active earth pressure. An intermediate situation between active and passive earth pressure when the retaining structure is perfectly stationary and does not move in either direction is called earth pressure at rest.

Passive pressure > At rest > Active pressure



$K_P > K_0 > K_a$

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1208. Assertion (A): The state of earth pressure at rest is the state of equilibrium with zero strain condition.
Reason (R): In rest condition neither the wall nor the soil moves
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not a correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

(UPRVUNL AE 2015)

Ans : (a) The state of earth pressure at rest is the state of equilibrium with zero strain condition. The stress condition for an element relating to earth pressure at rest condition neither the wall nor the soil moves.

$$(K_0) = \frac{\mu}{1-\mu}$$

$$\boxed{K_0 = 1 - \sin\theta}$$

where, μ = poisson's ratio
K_0 = coefficient of earth

1209. For a sand having an internal friction of 30°, the ratio of active to passive lateral earth pressure will be–
(a) 9 (b) 1
(c) 1/3 (d) 1/9

Gujarat PSC AE (N.W.R.) 2020

Ans. (d) : Internal friction angle $\phi = 30^o$

$$\text{active lateral earth pressure } (K_a) = \frac{1-\sin\phi}{1+\sin\phi}$$

$$\text{passive lateral earth pressure } (K_p) = \frac{1+\sin\phi}{1-\sin\phi}$$

$$\frac{\text{active lateral earth pressure}}{\text{passive lateral earth pressure}} = \frac{\frac{1-\sin\phi}{1+\sin\phi}}{\frac{1+\sin\phi}{1-\sin\phi}}$$

$$= \frac{(1-\sin\phi)^2}{(1+\sin\phi)^2}$$

$$= \frac{(1-\sin 30)^2}{(1+\sin 30)^2}$$

$$= \frac{\left(1-\frac{1}{2}\right)^2}{\left(1+\frac{1}{2}\right)^2} = \frac{(1/2)^2}{(3/2)^2} = \frac{1}{9}$$

1210. A gravity retaining wall has an overturning moment of 200 kN-m about the left hand bottom edge. If the resisting moment is 500 kN-m, what is the factor of safety for overturning? Is it safe or unsafe for overturning only?
(a) 2.5 unsafe (b) 2.5 safe
(c) 0.4 unsafe (d) 1.5 safe

APPSC Poly. Tech. Lect. 2020

Ans. (b) : Given
overturning moment = 200 kN-m
resisting moment = 500 kN-m

$$\text{factor of safety for overturning} = \frac{\text{resisting moment}}{\text{overturning moment}}$$

$$\text{F.O.S} = \frac{500}{200} = 2.5$$

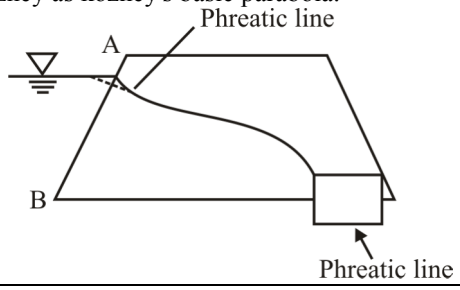
and, resisting moment 500kN-m is grater than the over turning moment 200 kN-m hence the gravity retaining wall is safe.

1211. A phreatic line is defined as the line within a dam section below which there are
(a) Positive equipotential lines
(b) Atmospheric pressure
(c) Positive hydrostatic pressure
(d) Negative hydrostatic pressure

OPSC AE Paper- II 2019

Ans. (c) : A phreatic line is the top flow line at which water pressure is zero (only atmospheric pressure).

It is perpendicular to upstream face AB which is 100% equipotential line its shape is approximately a parabola (correction at u/s face, rest a parabola) given by kozney as kozney's basic parabola.



1212. Rankine's theory of active earth pressure assumes
(a) Soil mass is homogeneous, dry and cohesionless.
(b) Ground surface is a plane which may be horizontal or inclined.
(c) Back of the wall is vertical and smooth.
(d) All of these

OPSC AE Paper- II 2019

Or

Rankine's theory of earth pressure assumes that back of the wall is
(a) plane and smooth
(b) plane and rough
(c) vertical and rough
(d) vertical and smooth

GPSC AE June 2019
Gujarat PSC AE 2017
UPRVUNL AE 2014

Ans. (d) : Rankine's theory–Rankine's theory considers stress in soil mass when it attains plastic equilibrium.
By plastic equilibrium we refer that every point in the soil mass experience shear failure, under the effects of shear stress developed.
Assumption in Rankine theory.

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1. Soil is semi-infinite, homogeneous, isotropic dry and cohesion less.
2. Soil is in state of plastic condition at the time of active and passive pressure generation.
3. The backfill soil is horizontal.
4. Back of wall is vertical and smooth.
5. Rupture surface is a plane surface which is obtained by considering the plastic equilibrium of soil.

1213. If the retaining wall moves away from the backfill, it is known as :

[UPPSC State Eng. A.E. 2004(I)]
[UPPSC State Eng. A.E. 2003]

Or

When retaining wall moves away from the back fill, the pressure exerted on wall is termed as-

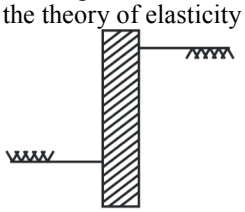
(a) Active state condition
(b) Passive state condition
(c) At rest condition
(d) None of the above

JPSC AE 2009
WBPSC AE 2001

Ans : (a) Types of Lateral Earth Pressure: Lateral earth pressure can be divided into three categories depending upon the movement of retaining wall with respect to back fill soil.

1. Earth pressure at rest

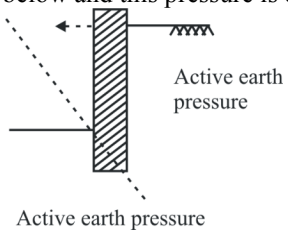
- Wall does not move at all, soil elements is not subjected to any strain, the element in this condition is known as at rest condition.
- It is possible to evaluate earth pressure at rest using the theory of elasticity



At rest

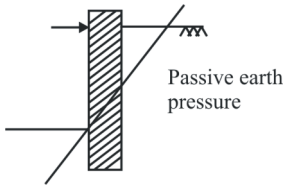
2. Active Earth pressure

Pressure exerted on wall due to soil where wall moves away from soil. Developed in the direction opposite to movement of wall, active earth pressure on wall will fall below and this pressure is called active earth pressure.



3. Passive earth pressure

Pressure developed on wall when wall moves towards the soil.



1214. Depth below the ground surface is C-ϕ soil, where active earth pressure intensity is zero, is given by:

or

A tension crack usually forms in the soil nearer the top of the wall up to a depth Z, given by

(a) $\frac{2C\sqrt{K_a}}{\gamma}$ (b) $\frac{2C\gamma}{\sqrt{K_a}}$

(c) $\frac{2C}{\gamma\sqrt{K_a}}$ (d) $\frac{2C\gamma}{K_a}$

WBPSC AE 2012
CIL MT 2020
UPPSC AE 12.04.2016 Paper–I

Ans : (c) Depth, where the active earth pressure intensity is zero are –

$$\left[Z_C = \frac{2C}{\gamma\sqrt{K_a}}\right]$$

for purely cohesive soil ($\phi = 0$)

$$\left[Z_C = \frac{2C}{\gamma}\right]$$

1215. If the coefficient of passive earth pressure is 4, then what will be the value of active earth pressure ?

(a) 1/4 (b) 1/16
(c) 16 (d) 4

RPSC AE (GWD) 2014

Ans : (a) The relation between active earth pressure (K_a) and passive earth pressure (K_p) is –

$$K_a \times K_P = 1$$

$$K_a = \frac{1}{K_P}$$

$$\boxed{K_a = \frac{1}{4}}$$

1216. Coulomb's theory of earth pressure is based on

(a) The theory of eleasticy
(b) The theory of plasticity
(c) Empirical rules
(d) Wedge theory

RPSC AE (GWD) 2014

Ans : (d) The wedge theory of earth pressure is based on the concept of a sliding wedge which is turn off from the rest of the backfill on movement of the wall. In the case of active earth pressure, the sliding wedge moves downwards and outwards on a slip surface relative to the intact backfill and in the case of passive earth pressure, the sliding wedge moves upwards and inwards.

1217. A vertical retaining wall retains a C-ϕ backfill with a surcharge of uniform intensity q per unit area. The depth Z_o where the active earth pressure is zero, is given by

(a) q/γ (b) $\frac{2C'}{\gamma}\tan\alpha' - q/\gamma$

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(c) $\frac{2C'}{\gamma}\tan\alpha' + q/\gamma$ (d) $\frac{2C'}{\gamma}\tan\alpha'$

RPSC AE (GWD) 2014

Ans : (b) Direction away from Active earth pressure–

$$P_a = k_a\sigma_z - 2C\sqrt{k_a}$$

$$\sigma_z = q + \gamma H$$

$$P_a = k_a(q + \gamma H) - 2C\sqrt{k_a}$$

$P_a = 0, H = Z_0$

$$0 = k_a(q + \gamma Z_0) - 2C\sqrt{k_a}$$

$$k_a(q + \gamma Z_o) = 2C\sqrt{k_a}$$

$$q + \gamma Z_o = \frac{2C}{\sqrt{k_a}}$$

$$Z_o = \frac{2C}{\gamma\sqrt{k_a}} - \frac{q}{\gamma} = \frac{2C}{\gamma}\tan\alpha^1 - \frac{q}{\gamma}$$

1218. The wall friction of the retaining wall

(a) Decrease active earth pressure but increase passive earth pressure

(b) Decrease passive earth pressure but increase active earth pressure

(c) Decreases both active and passive earth pressure

(d) Increases both active and passive earth pressure

JPSC Combined AE Paper V 2013
RPSC POLY. TECH. LECT. 2011
ESE 2005

Ans : (a) The magnitude of active or passive earth pressure depends not only on the ϕ but also on friction between the soil and back of wall represented by the angle δ. The value of δ may range between $\frac{1}{2}\phi$ to $\frac{3}{4}\phi$ in most cases and it can be assumed as $\frac{2}{3}\phi$ in the absence of precise data.

The wall friction decrease active earth pressure and increase passive earth pressure.

1219. The effect of cohesion on a soil is to

(a) Reduce both active and passive earth pressure intensities

(b) Increase both active and passive earth pressure intensities

(c) Reduce active earth pressure intensity but to increase passive earth pressure intensity

(d) Increase active earth pressure intensity but to reduce passive earth pressure intensity

RPSC ACF 2011
WBPSC AE 2007

Ans : (c) The effect of cohesion on a soil is to reduce active earth pressure intensity but to increases the passive earth resistance of a soil.

1220. What is the critical depth of a vertical cut in a saturated cohesive soil with C = 1.5 t/m^2 and γ = 1.5 t/m^3.

(a) 0.83 m (b) 1.67 m

(c) 3.33 m (d) 4.00 m

[UPPSC State Eng. A.E. 2007(II)]

Ans : (d) Given,

cohesion, C = 1.5 t/m^2,

unit weight, γ = 1.5 t/m^3

critical depth,

$$h_c = \frac{4c}{\gamma}\tan(45 + \phi/2)$$

$$= \frac{4\times1.5}{1.5}.\tan(45 + 0/2) = 4.0 \text{ m}$$

1221. For 6m deep excavation in soft clay, γ = 18kN/m^3, C = 26kN/m^2, Taylor's stability number, S_n = 0.172, factor of safety (F_c) against sliding will be:

(a) 1.2 (b) 1.3

(c) 1.4 (d) 1.5

UPPSC AE 12.04.2016 Paper–I

Ans : (c) Given, Cohesion, C = 26 kN/m^2

Unit weight γ = 18 kN/m^3

Taylor's stability number (S_n) = 0.172

Taylor's stability No.

$$S_n = \frac{C_u}{F_c\gamma H}$$

$$0.172 = \frac{26}{18\times6\times F_c}$$

$$F_c = \frac{26}{18\times6\times0.172} = 1.399$$

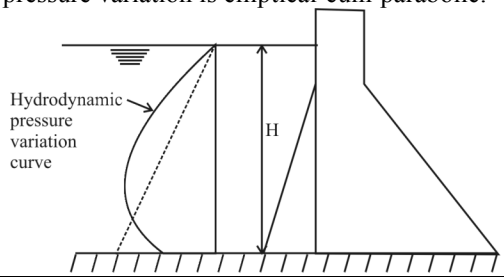
1222. In case of gravity dam subjected to earthquake, the hydrodynamic pressure variation curve is taken to be:

(a) Parabolic (b) Elliptical

(c) Triangular (d) Elliptical cum parabolic

[UPPSC Civil Eng. A.E. 2013(II)]

Ans : (a) Hydrodynamic forces are due to horizontal acceleration of the foundation and dam there is an instantaneous hydrodynamic pressure (or suction) exerted against the dam in addition to hydrostatic force. The pressure variation is elliptical-cum-parabolic.



1223. A retaining wall retains a sand strata with ϕ = 30^0 up to its top. If a uniform surcharge of 12 t/m^2 is subsequently put on the sand strata, then the increase in the lateral earth pressure intensity on the retaining wall will be

(a) 0 t/m^2 (b) 4 t/m^2

(c) 6 t/m^2 (d) 12 t/m^2

MPPSC AE 2016
UPPSC State Eng. A.E. 2003
ESE 1995

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Ans : **(b)** Given, Friction angle, $\phi = 30^o$
Uniform surcharge, $q = 12\ t/m^2$
Retaining walls are designed for Active earth pressure

$$K_a = \frac{1-\sin\phi}{1+\sin\phi} = \frac{1-\sin 30^o}{1+\sin 30^o} = \frac{1}{3}$$

increase in lateral earth pressure intensity due to uniform surcharge (q)

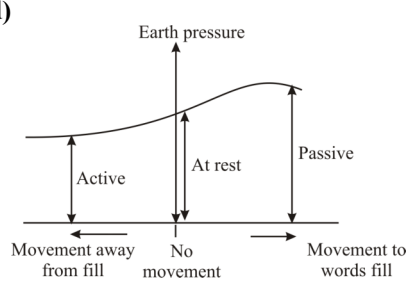
$$\Delta P = q.K_a$$

$$\Delta P = 12 \times \frac{1}{3} = 4\ t/m^2$$

1224. An earth–retaining structure may be subjected to the following lateral earth pressures :
1. Earth pressure at rest;
2. Passive earth pressure;
3. Active earth pressure.
The correct sequence of the increasing order of magnitudes of these pressures in
(a) 3, 2, 1 (b) 1, 3, 2
(c) 1, 2, 3 (d) 3, 1, 2

MPPSC AE 2016

Ans : (d)



Hence, passive earth pressure is greater than earth pressure at rest and active earth pressure. Actuating elements are the elements causing instability. increasing order

$$\boxed{K_a < K_0 < K_p}$$

1225. An unsupported vertical cut may be made in cohesive soil to a height of :
(a) $\frac{2C}{\gamma}\tan\left(45^0 + \frac{\phi}{2}\right)$ (b) $\frac{2q_u}{\gamma}\tan\left(45^0 + \frac{\phi}{2}\right)$
(c) $\frac{4q_u}{\gamma}\tan\left(45^0 - \frac{\phi}{2}\right)$ (d) $\frac{4C}{\gamma}\tan\left(45^0 + \frac{\phi}{2}\right)$

WBPSC AE 2016
APPSC AEE 2012
[UPPSC State Eng. A.E. 2011 Ist Paper]

Ans : **(d)** An unsupported vertical cut may be made in cohesive soil to a height of

$$H_c = 2Z_C$$

$$H_c = 2 \times \frac{2C}{\gamma}\tan\left(45 + \frac{\phi}{2}\right)$$

$$H_c = \frac{4C}{\gamma}\tan\left(45 + \frac{\phi}{2}\right)$$

1226. The down stream slope of an earth dam is critical :
(a) when reservoir is full
(b) when steady seepage occurs
(c) at the end of construction
(d) combination of (a), (b) and (c)

[UPPSC State Eng. A.E. 2011 Ist Paper]

Ans : **(b)** The steady seepage condition under five reservoir is critical for the downstream slope. Due to the seepage, a seepage, force acts in direction of flow which add to the gravity forces and causing condition of unstability to the downstream face on the other hand, upstream face is kept stable as the seepage force acts towards the face.

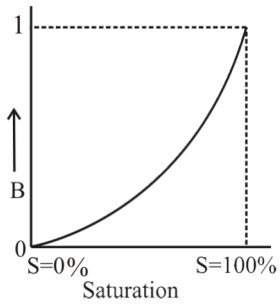
1227. For a saturated clayey soil, skempton's Pore pressure parameter 'B' is :
(a) 0.1 (b) 0.5
(c) 1.0 (d) 2.0

[UPPSC State Eng. A.E. 2013(I)]
[UPPSC State Eng. A.E. 2007(II)]

Ans : **(c)** The pore pressure parameter 'B' is the response of the ratio of change in pore pressure to change in cell pressure.

$$\therefore \quad B = \frac{\Delta V_c}{\Delta \sigma_c}$$

ΔV_c = change in pore pressure
$\Delta \sigma_c$ = change in cell pressure
$0 \le B \le 1$



Hence for saturated clayey soil B = 1.0.

1228. To stabilize a concrete retaining wall against sliding, the ratio of sliding force to resisting force should be :
(a) ≤ 1.55 (b) ≥ 1.55
(c) ≥ 0.645 (d) ≤ 0.645

[UPPSC State Eng. A.E. 2007(II)]

Ans : **(d)** For no sliding of retaining wall,

$$FOS = \frac{\text{Resisting force}}{\text{Sliding force}} > 1.5$$

A minimum FOS against safety is 1.55

$$1.55 = \frac{F_{resisting}}{F_{Sliding}} = 1.55$$

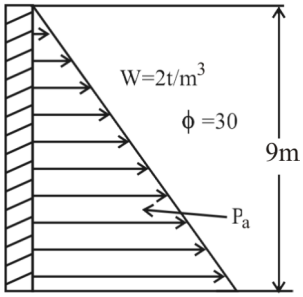
$$\frac{F_{Sliding}}{F_{resisting}} \le \frac{1}{1.55} \le 0.645$$

1229. The active earth pressure per meter length on a retaining wall of 9 m height with a smooth vertical base of sand having unit weight of 2 T/m^3 and angle of internal friction of 30^0 is :
(a) 1T (b) 2T
(c) 27T (d) 81T

[UPPSC State Eng. A.E. 2007(II)]

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Ans : (c)



$$k_a = \frac{1-\sin\phi}{1+\sin\phi} = \frac{1-\sin 30^o}{1+\sin 30^o} = \frac{1}{3}$$

$$P_a = \frac{1}{2}k_a\gamma H^2 = \frac{1}{2}\times\frac{1}{3}\times 2\times 9\times 9$$

$P_a = 27$ T

1230. As per Jacky formula K_0 (coefficient of Earth pressure at rest) is determined as :

(a) $K_0 = 1-\cos\phi'$ (b) $K_0 = 1-\sin\phi'$

(c) $K_0 = 1-\tan\phi'$ (d) $K_0 = 1-\cot\phi'$

Where ϕ' is the angle of shearing resistance of soil.

GPSC Assistant Prof. 2016
UKPSC AE (Urban) 2011
UPPSC State Eng. A.E. 2007(II)

Ans : (b) Coefficient of earth pressure at rest, $K_0 = 1-\sin\phi'$ where ϕ' is the angle of shearing resistance of soil.

1231. For a gravity retaining wall, the maximum eccentricity of the base reaction for "no tension" condition to be satisfied is :

(a) one–third of the base width
(b) two–third of the base width
(c) one–sixth of the base width
(d) none of the above

[UPPSC State Eng. A.E. 2007(I)]

Ans : (c) In gravity retaining wall, for the no tension condition.

$$e_{max} \le \frac{B}{6}$$

1232. To have zero active earth pressure intensity at the top of a wall in cohesive soil, the required intensity of uniform surcharge is :

(a) $2C\cot\alpha$ (b) $2C\tan\alpha$

(c) $-2C\cot\alpha$ (d) $-2C\tan\alpha$

[UPPSC State Eng. A.E. 2007(I)]

Ans : (b) Active earth pressure

$P_a = k_a\sigma_z - 2C\sqrt{k_a}$

$P_a = k_a(q+\gamma z) - 2C\sqrt{k_a}$

Given, at $z = 0$, $P_a = 0$

$$0 = k_a q - 2C\sqrt{k_a}$$

$$q = \frac{2C}{\sqrt{k_a}}$$

$$= \frac{2C}{\sqrt{\frac{1}{\tan^2\left(45+\frac{\phi}{2}\right)}}}$$

$$q = 2C\tan\left(45+\frac{\phi}{2}\right)$$

$q = 2C\tan\alpha$

where, α = Angle of failure plane with major principal plane.

$$\alpha = 45+\frac{\phi}{2}$$

1233. As per IS code, the frequency ratio (ratio of operating frequency of machine to the natural frequency of soil) should not be within the range of

(a) 0.5 to 1.5 (b) 1.0 to 2.5

(c) 1.5 to 3.0 (d) 3.0 to 6.0

BPSC AE 2012 Paper–V

Ans : (a) As per IS code, the frequency ratio (ratio of operating frequency of machine to the natural frequency of soil) should not be within the range of 0.5 to 1.5.

1234. Under dynamic loading, the soil most likely to be affected is

(a) loose saturated clay
(b) loose saturated sand
(c) dense saturated sand
(d) loose saturated silt

BPSC AE 2012 Paper–V

Ans : (b) Under dynamic loading, the soil most likely to affected is loose saturated sand. For loose sands, the first two conclusions remain valid, but the horizontal compression to reach full passive resistance may be as large as 15%.

1235. The yield of retaining wall required to reach failure is

(a) More for the active state
(b) More for the passive state
(c) Equal for both active and passive
(d) No yield necessary

(BPSC AE Paper V 2006)

Ans : (b) The yield of retaining wall required to reach failure is more for the passive state.

1236. Consider the following statements :

(1) Coulomb earth pressure theory does not take the roughness of wall into consideration
(2) Active earth pressure on a retaining wall decreases due to increase in wall friction
(3) Rankine theory of earth pressure assumes that back of wall is vertical and smooth

Of these statements–

(a) 1 and 2 are correct
(b) 3 and 1 are correct
(c) 2 and 3 are correct
(d) 1, 2 and 3 are correct

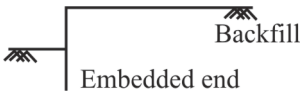
(JPSC Combined AE Paper V 2013)

Ans : (c) Active earth pressure on a retaining wall decrease due to increase in wall friction. Masonary

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retaining wall founded on compressible clay will move away from the backfill. So active earth pressure on wall.

→ Rankine assumed the retaining wall as smooth and vertical.



In other words, there are no shearing stress as between the wall and the soil and the stress relationship for any element adjacent to the wall is the same as for any other element for away from the wall.

1237. Shallow pits should be provided with lateral support when the depth exceeds :

(a) 2m (b) 2.5m
(c) 3.0 m (d) 3.5m

(UKPSC AE (Paper-I) 2007)

Ans : (c) If shallow pits should be provided with lateral support when the depth exceeds 3.0 m.

1238. Active earth pressure of a soil is defined as the lateral pressure exerted by the soil when:

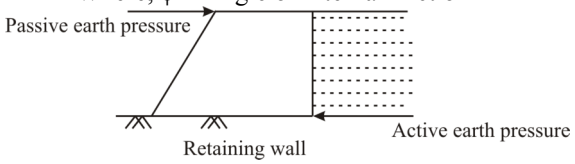
(a) the retaining wall is at rest
(b) the retaining wall tends to move away from the backfill
(c) the retaining wall moves in the soil
(d) none of the above

(UKPSC AE (Paper-I) 2007)

Ans : (b) Soil exerts a push against the retaining wall by virture of its tendency to slip laterally and seek its natural slope or angle of repose. Thus making the wall to move slightly away from backfilled soil mass. This kind of pressure is known as "active earth pressure."

$$k_a = \frac{1-\sin\phi}{1+\sin\phi}$$

Where, ϕ = Angle of internal friction



1239. The minimum allowable factor of safety against sliding for a cantilever retaining wall is–

(a) 3.0 (b) 2.0
(c) 1.5 (d) 2.5

UKPSC AE 2013 Paper–I

Ans : (c) For no sliding to occur, Rn < Rv.μ
Where, μ tan δ' = coefficient of friction between the base of the wall and the soil.
The factor of safety F against sliding is given by

$$F = \frac{Rv.\mu}{Rn}$$

The minimum value of F should be 1.5.

1240. Given that for a soil deposit, K_o = earth pressure coefficient at rest; K_a = active earth pressure coefficient; μ = Poisson's ratio. The value of $\frac{(1-\mu)}{\mu}$ is given by-

(a) $\frac{K_a}{K_p}$ (b) $\frac{K_o}{K_a}$
(c) $\frac{K_p}{K_a}$ (d) $\frac{1}{K_o}$

MPPSC AE 2017
Rajasthan JEn (Degree) 2016
JPSC AE 2009
(ESE 1997)

Ans. (d) : Earth pressure at rest, K_0 is given as

$$K_0 = \frac{\mu}{1-\mu}$$

Hence, $\frac{1-\mu}{\mu} = \left(\frac{1}{K_o}\right)$

1241. If Δp is increment of pressure on a normally consolidated saturated soil mass, as per Terzahghi's theory at the instant of application of pressure increment i.e., When time t = 0, what is the pore pressure developed in the soil mass?

(a) Zero
(b) Very much less than Δp
(c) Equal to Δp
(d) Greater than Δp

MPPSC AE 2017

Ans. (c) : For saturated soil mass, $B = 1$

$\Delta u = B\ \Delta p$

for $B = 1$

$\Delta u = \Delta p$

1242. An all-around RCC peripheral retaining wall is other side. The retaining wall has RCC floor slab retaining wall will be analyzed in-

(a) Passive condition
(b) Active condition
(c) At rest condition
(d) Partially active and partially passive condition

[MPSC 2015 Paper-II Mains]

Ans. (c) : Passive condition- When the wall moves towards the backfill.
Active condition - When retaining wall moves away from backfill.
At rest condition - Retaining wall does not move.

$K_a = \frac{1-\sin\phi}{1+\sin\phi}$ $K_0 = \frac{\mu}{1-\mu}$

$K_p = \frac{1+\sin\phi}{1-\sin\phi}$ μ = poisson ratio

1243. *Statement* (A) : Coffer-dam is a structure to be constructed in standing water condition prior to the construction of bridge foundations.
***Statement* (B) : Cutting edge and steining are the two essential component parts of the cofferdam.**

(a) Both the statements (A) and (B) are true
(b) Both the statements (A) and (B) is false
(c) Statement (A) is true but (B) is false
(d) Statement (B) is true but (A) is false

[MPSC 2015 Paper-II Mains]

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Ans. (c) : A cofferdam also called as coffer, is an enclosure built within a body of water to allow the enclosed area to be pumped out.

$\Rightarrow$ A is correct and B is wrong statement.

1244. A wall 6 m high has a smooth vertical back and retained sand as a backfill which is submerged. The sand has γ_{sat} = 20 kN/m^3 and $\phi = 30^o$. The total active earth pressure is-

(a) 90 kN/m^2 (b) 60 kN/m^2

(c) 120 kN/m^2 (d) None of the above

MPSC (Mains) 2017 Paper–II

Ans. (b) : $\gamma_{sat} = 20$ kN/m^3, $\phi = 30^o$

$$\text{active earth pressure } (K_a) = \frac{1-\sin\phi}{1+\sin\phi} = \frac{1-\frac{1}{2}}{1+\frac{1}{2}} = \frac{1}{3}$$

$$P_a = \frac{1}{2} K_a \gamma' H^2$$

$$= \frac{1}{2} \times \frac{1}{3} (20-10) \times 6^2 = 60 \text{ kN/m}^2$$

1245. The coefficient of earth pressure at rest is given in-

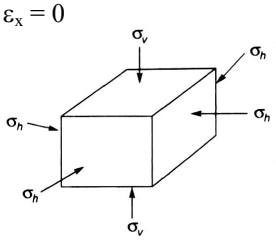
(a) $\frac{\mu}{(1-\mu)}$ (b) $\left(\frac{1+\mu}{\mu}\right)$

(c) $\frac{\mu}{(1+\mu)}$ (d) $\frac{(1-\mu)}{\mu}$

GUJRAT PSC AE - II 2017
APPSC AEE (Mains) 2016
OPSC AE -2016 (II)
OPSC AEE 2015 PAPER–I

Ans. (a) : For plain strain condition,

$\varepsilon_x = 0$



Here $\sigma_x = \sigma_y = \sigma_h$ and $\sigma_z = \sigma_v$

$$e_x = \left(\frac{-\sigma_x}{E}\right) - \mu\left(\frac{-\sigma_y}{E}\right) - \mu\left(\frac{-\sigma_z}{E}\right) = 0$$

$\sigma_h(1 - \mu) = \mu\sigma_v$

$$\frac{\sigma_h}{\sigma_v} = k_0 = \left(\frac{\mu}{1-\mu}\right)$$

1246. A cantilever retaining wall of 8 meter height retains sand. The properties of sand are; e = 0.4, $\phi = 30^0$, G = 2.65, γ_w = 9.8 kN/m^3. Using Rankine's theory, the active earth pressure at the base when the back fill is dry is :

(a) 45 kN/m (b) 52.5 kN/m

(c) 60.5 kN/m (d) 49.5 kN/m

Karnataka PSC AE 2015 Paper - II

Ans. (d) :

Given data. e = 0.4, $\phi = 30^0$, G = 2.65, γ_w = 9.8 kN/m^3, H = 8

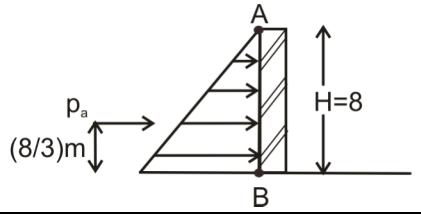
$$k_a = \frac{1-\sin\phi}{1+\sin\phi} = \frac{1}{3}$$

$$\gamma_t = \left(\frac{G+eS}{1+e}\right)\gamma_w$$

(Since back fill is dry S = 0)

$$\gamma_d = \frac{G\gamma_w}{1+e} = \frac{2.65 \times 9.81}{1+0.4} = 18.57 \text{ kN/m}^3$$

$$(P_a)_B = \frac{1}{3}\gamma_d . H = \frac{1}{3} \times 18.57 \times 8 = 49.5 \text{ kN/m}^2$$



1247. Coefficient of earth pressure at rest is given by:

(a) $k_0 = \frac{1-\sin\phi}{1+\sin\phi}$ (b) $k_0 = 1-\sin\phi$

(c) $k_0 = \frac{1+\sin\phi}{1-\sin\phi}$ (d) $k_0 = 1+\sin\phi$

Karnataka PSC AE 2015 Paper - II

Ans. (b) :

co-efficient of earth pressure at rest, $K_o = 1 - \sin\phi$.

$$K_o = \frac{\mu}{1-\mu}$$

co-efficient of active earth pressure

$$(K_a) = \frac{1-\sin\phi}{1+\sin\phi}$$

co-efficient of passive earth pressure

$$(K_p) = \frac{1+\sin\phi}{1-\sin\phi}$$

1248. The corresponding lateral earth pressure, when the retaining wall moves away from the back fill is :

(a) Passive earth pressure

(b) Active earth pressure

(c) Earth pressure at rest

(d) None of these

Karnataka PSC AE 2015 Paper - II

Ans. (b) : Lateral earth pressure can be divided into three condition depending upon the movement of retaining wall with respect to back fill soil.

(1) Earth pressure at rest – wall does not move at all.

(2) Active earth pressure – wall moves away from back fill soil.

(3) Passive earth pressure – wall moves towards back fill soil.

1249. In an earth dam critical condition for which the stability has to be checked during construction with or without partial pool is/are :

(a) downstream slope

(b) upstream slope

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(c) Both U/S & D/S slope
(d) None

JPSC AE (Pre) 2019

Ans. (c) : The following conditions are usually critical for the stability of earth dam–
case (i) construction condition with or without partial pool (for U/S and D/S slope)
case (ii) Reservoir partial (for U/S) slope
case (iii) Sudden drawdown (for U/S) slope
case (iv) steady seepage (for D/S slope)
case (v) Steady seepage with sustained rainfall (for D/S slope, where annual rain fall is 200cm or more)
case (vi) Earthquake condition (for U/S and D/S slope)

1250. Force considered for the analysis of an elementary profile of a gravity dam under empty reservoir condition are -
(a) Uplift pressure (b) Water pressure
(c) Self-weight (d) Wave pressure

RPSC AE 2018

Ans. (c) Since, reservoir is empty, so there shall be no water, now is absence of water, there will be no possibility of water/uplift/wave pressure.

1251. An unsupported excavation is made to the maximum possible depth in a clay soil having $\gamma_t = 18$ kN/m^3, C = 100 kN/m^2, $\phi = 30^0$. The active earth pressure, according to Rankine's theory, at the base level of excavation is -
(a) 115.47 kN/m^2 (b) 54.36 kN/m^2
(c) 27.18 kN/m^2 (d) 13.25 kN/m^2

RPSC AE 2018

Ans. (a) : $K_a = \frac{1-\sin\phi}{1+\sin\phi} = \frac{1-\sin 30^o}{1+\sin 30^o}$

$$K_a = \frac{1}{3}$$

Depth of unsupported cut, $(h) = \frac{4C}{\gamma\sqrt{K_a}}$

$$= \frac{4\times 100}{18\sqrt{1/3}} = 38.49 \text{ m}$$

Active earth pressure at this depth,

$P_a = K_a \gamma h - 2C\sqrt{K_a}$

$= \frac{1}{3}\times 18 \times 38.49 - 2\times 100\sqrt{1/3}$

$P_a = 115.47 \text{ KN/m}^2$

1252. The earth pressure behind a bridge abutment is-
(a) Active
(b) Passive
(c) At rest
(d) Constant always and everywhere

RPSC AE 2018
ESE 2013

Ans. (c) Since at bridge abutment, there is no movement of wall so earth pressure shall be is rest.

1253. Coefficient of earth pressure at rest is
(a) Less than coefficient of active earth pressure but greater than coefficient of passive earth pressure
(b) Greater than coefficient of active earth pressure but less than coefficient of passive earth pressure
(c) Less than both coefficient of active earth pressure and coefficient of passive earth pressure
(d) Greater than both coefficient of active earth pressure and coefficient of passive earth pressure

GPSC AE (CLASS 1 & 2) 2019
OPSC Civil Services Pre 2011

Ans : (b) : Coefficient of earth pressure at rest is greater than coefficient of active earth pressure but less than coefficient of passive earth pressure.

$$K_p > K_o > K_a$$

$$K_a = \frac{1-\sin\phi}{1+\sin\phi}, \quad K_p = \frac{1+\sin\phi}{1-\sin\phi}, \quad K_o = 1 - \sin\phi$$

1254. The factor of safety against sliding of structures which resist lateral forces (such as retaining walls) shall be not less than____when dead load, live load and earth pressure only are considered
(a) 1.50 (b) 1.75
(c) 2.00 (d) 2.25

GPSC AE (CLASS 1 & 2) 2019

Ans : (b) : The factor of safety against sliding of structure which resist lateral load such as retaining walls should not be less than 1.75 when dead load, live load and earth pressure are only considered.

1255. The angle between the horizontal and the maximum slope that a soil assumes through natural process is called:
(a) Angle of internal friction
(b) Cohesiveness
(c) Angle of repose
(d) Hydraulic gradient

KPSC AE 2016

Ans. (c) Angle of Repose - Angle between the horizontal and the maximum slope that a soil assumes through natural processes. For granular soils, the effect of height of slope is negligible; for cohesive soils, the effect of height of slope is so great that the angle of repose is meaningless.
Angle of Internal friction- Angle between the abscissa and tangent to the curve representing the relationship between the shearing the resistances at failure to the normal stress acting within a soil.
Hydraulic Gradient – This difference or drop of hydraulic head per unit distance of flow
Cohesiveness – Cohesiveness is the degree to which the group members are attracted to each other and are motivated to stay in the groups. Cohesiveness defines the degree of closeness that the members feel with the groups.

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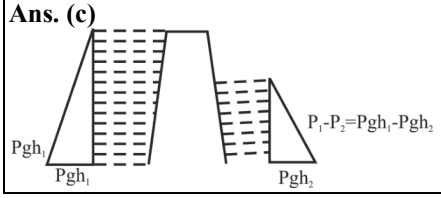
1256. The total pressure "P" acting on a sluice gate is (Where P_1, P_2–total pressure acting on the upstream and downstream sides of the sluice gate)

(a) $-P_1-P_2$ (b) P_1+P_2

(c) P_1+P_2 (d) $-P_1+P_2$

TNPSC AE 2019

Ans. (c)



1257. A retaining wall 10 m high with vertical back retains moist with a horizontal surface. If unit weight of sand is 15 kN/cum and the angle of internal friction is 30 degree, the total active pressure (in kN) experienced by the wall per m run is

(a) 25 (b) 2250

(c) 250 (d) 23

(e) 83.33

CGPSC AE 2014 Shift–I

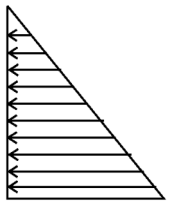
Ans. (c) :

$h = 10$ m

$\gamma_t = 15\ kN/m^3$

$\phi = 30^0$

$P_A = K_A \gamma_t \times h$



Active pressure diagram

$$K_A = \tan^2\left(45 - \frac{\phi}{2}\right)$$

$$= \tan^2 30^0$$

$$K_A = \frac{1}{3}$$

$$\therefore P_A = \frac{1}{3} \times 15 \times 10 = 50\,kN/m^2$$

Active pressure per meter run

$$F_A = \frac{1}{2} K_A \gamma_t \times H^2 = \frac{1}{2} \times \frac{1}{3} \times 15 \times 10^2$$

$$\boxed{F_A = 250 kN}$$

1258. A wall constructed for the stability of the excavated portion of the road on the hill side is known as

(a) Parapet wall (b) retaining wall

(c) breast wall (d) guide wall

TNPSC AE 2012

Ans. (c) Breast wall–A breast wall is constructed to protect the natural sloping ground from the cutting action of natural agents. Breast wall also prevent slides of unreliable soils.

1259. The condition for stability of a dam against over turning is :

(a) $\frac{P.h/3}{W(b-\bar{x})} \geq f.s.$ (b) $\frac{W(b-\bar{x})}{P \times h/3} \geq f.s.$

(c) $\frac{P.h/3}{Wb} \geq f.s.$ (d) $\frac{W.b}{P.h/3} \geq f.s.$

TNPSC AE 2015

Ans. (b) : Overturning moment $= P \times \frac{h}{3}$

Substitution moment $= W(b-\bar{x})$

$$\text{factor of safety} = \frac{W(b-\bar{x})}{P \times \frac{h}{3}}$$

$$\text{factor of safety (fos)} \leq \frac{W(b-\bar{x})}{P \times \frac{h}{3}}$$

The value of protection coefficient against over turning of dam is 1.5.

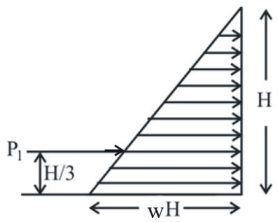
1260. If a retaining wall is the constructed to retain water of height 'H' and 'w' is unit weight of water, then water pressure acting on retaining wall will be

(a) $\frac{K_a wH^2}{2}$ (b) $\frac{K_p wH^2}{2}$

(c) $\frac{wH^2}{2}$ (d) wH^2

Maharashtra PSC AE (Mains) 2012 Paper–I

Ans : (c)



Height = H, Unit weight of water = w

So, the pressure $(P_1) = \frac{1}{2} \times wH \times H$

$$P_1 = \frac{wH^2}{2}$$

1261. Along a phreatic line in an earth dam

(a) pressure is atmospheric

(b) pressure is greater than atmospheric

(c) pressure is less than atmospheric

(d) pressure head is constant but not zero

ASSAM PSC CCE (PRE) 2015

Ans : (a) Phreatic line is the top flow line which separates saturated from unsaturated zone and below which positive hydrostatic pressure exists in the dam section. Along the phreatic line, the atmospheric pressure exists.

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1262. The silt pressure (P_s) exerted on a gravity dam is given by (Where γ_s = submerged unit weight, ϕ = angle of internal friction, and h_s = The height to which the silt is deposited)

(a) $P_s = \frac{1}{3}\gamma_s . h_s^2 . \frac{1-\sin\phi}{1+\sin\phi}$

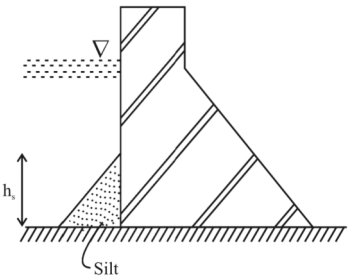
(b) $P_s = \frac{1}{2}\gamma_s . h_s^2 . \frac{1-\sin\phi}{1+\sin\phi}$

(c) $P_s = \frac{1}{2}\gamma_s . h_s^3 . \frac{1-\sin\phi}{1+\sin\phi}$

(d) $P_s = \frac{1}{2}\gamma_s . h_s \frac{1-\sin\phi}{1+\sin\phi}$

Rajasthan AE (Nagar Nigam) 2016

Ans. (b) :



$$P_S = \frac{1}{2}K_a \gamma_s \times h_s^2$$

$$P_S = \frac{1}{2}\gamma_s . h_s^2 . \frac{1-\sin\phi}{1+\sin\phi} \quad \because K_a = \frac{1-\sin\phi}{1+\sin\phi}$$

1263. A state of pressure that exists when the movement of a wall is such that the soil tends to compress horizontally:

(a) At rest pressure (b) Active pressure

(c) Passive pressure (d) Pore pressure

LMRC AM 2019

Ans. (c) Passive earth pressure: A state of pressure that exists when the movement of a wall is such that the soil tends to compress horizontally.

Earth pressure at rest: Wall does not move at all, soil elements is not subjected to any strain, the element in this condition is know as at rest condition it is possible to evaluate earth pressure at rest using the theory of elasticity.

Active earth pressure: Pressure exist on wall due to soil where wall moves away from soil.

1264. The coefficient of active earth pressure is given by :

(a) $k_a = \frac{1+\sin\phi}{1-\sin\phi}$ (b) $k_a = \frac{1-\sin\phi}{1+\sin\phi}$

(c) $k_a = \frac{1-\tan\phi}{1+\tan\phi}$ (d) $k_a = \frac{1+\tan\phi}{1-\tan\phi}$

Nagaland PSC Paper-II 2017

Ans. (b) : The coefficient of active earth pressure (K_a) is given by

$$\boxed{k_a = \frac{1-\sin\phi}{1+\sin\phi}}$$ $\phi \rightarrow$ Angle of internal friction

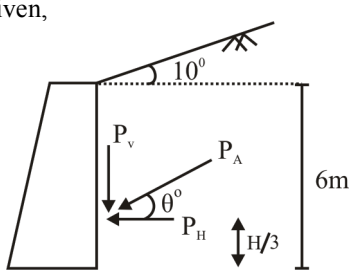
1265. A 6 m high retaining wall with a vertical back has a backfill of silty sand with a slope of 10^0 for the backfill. With values of K_H = 760 kg/m^2/m and K_V = 100 kg/m^2/m, the total active earth pressure will approximately be

(a) 128 kN/m (b) 134 kN/m

(c) 138 kN/m (d) 142 kN/m

ESE 2019

Ans. (c) Given,



Retaining wall–

Height = 6m

Slope = 10^0

K_H = 760 kg/m^2/m

K_v = 100 kg/m^2/m

Now, By Peck, Hanson and Thornburn method,

$$P_H = \frac{1}{2}K_H \times H^2 = \frac{1}{2}\times 760 \times (6)^2$$

$$\boxed{P_H = 13680 kg/m = 136.8 kN/m}$$

$$P_V = \frac{1}{2}K_V \times H^2 = \frac{1}{2}\times 100 \times (6)^2$$

[P_v=1800kg/m = 18kN/m]

So, Total Active Earth pressure 'P_A' = $\sqrt{P_H{}^2 + P_V{}^2}$

$$P_A = \sqrt{(136.8)^2 + (18)^2}$$

$$P_A = \sqrt{18714.24 + 324} = \sqrt{19038.24}$$

$$\boxed{P_A = 137.97 \simeq 138 kN/m}$$

1266. The coefficient of passive and active earth pressure for a loose sand having an angle of internal friction of 30° is

(a) 1/2, 2 (b) 2, 1/2

(c) 1/3, 3 (d) 3, 1/3

ISRO Scientist/Engineer 2019

Ans : (d) : $k_a = \frac{1-\sin\phi}{1+\sin\phi} = \frac{1-\sin 30^o}{1+\sin 30^o}$

$$= \frac{0.5}{1.5} = \frac{1}{3}$$

$$k_p = \frac{1+\sin\phi}{1-\sin\phi} = \frac{1+\sin 30^o}{1-\sin 30^o} = 3$$

1267. A retaining wall, 4.5m high has a smooth vertical back. The back fill has a horizontal surface in level with top of the wall. There is a uniformly distributed surcharge load of 2 t/m^2 intensity over the back fill. The density of the soil is 1.9 t/m^3,its angle of shearing resistance is 30^0 and cohesion is zero. Determine the magnitude of the total active pressure per metre length of wall.

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(a) 94.1t (b) 9.41t
(c) 941t (d) 0.941t

ISRO Scientist/Engineer 2014

Ans : (b) Given,

Height 'H' = 4.5 m

$\gamma = 1.9\ t/m^3$

$\phi = 30^0$

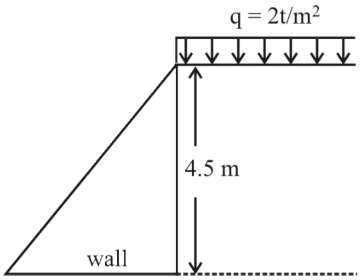
$q = 2t/m^2$

Now, $K_a = \frac{1-\sin\phi}{1+\sin\phi} = \frac{1-\sin 30^0}{1+\sin 30^0}$

$\left[K_a = \frac{1}{3}\right]$

Active pressure due to weight of the soil at the base of wall = $K_a\gamma H$

$= \frac{1}{3}\times 1.9\times 4.5 = 2.85$



Active pressure due to uniform surcharge = K_aq

$= \frac{1}{3}\times 2 = \frac{2}{3}$

Total active pressure per meter length of the wall–

$P_a = \frac{1}{2}K_a\gamma H^2 + K_aqH$

$P_a = \frac{1}{2}\times\frac{1}{3}\times 1.9\times(4.5)^2 + \frac{1}{3}\times 2\times 4.5$

$= 6.412 + 3$

$\boxed{P_a = 9.41t}$

1268. The critical vertical depth H_c of free standing soil can be upto passive earth pressure, is given by

(a) $H_c = \frac{\gamma h}{2}\tan^2\left(45-\frac{\phi}{2}\right)$

(b) $H_c = \frac{\gamma h^2}{2}\tan^2\left(45-\frac{\phi}{2}\right)$

(c) $H_c = \frac{4C}{\gamma\tan\left(45-\frac{\phi}{2}\right)}$

(d) $H_c = \frac{4C}{\gamma\tan\left(45+\frac{\phi}{2}\right)}$

Where 'c' is cohesion, 'γ' is the density of the soil, 'h' is the depth of filling and 'ϕ' is the angle of internal friction.

APTRASCO 2017

Ans : (d) : Critical depth (H_C)

The depth up to which the total earth pressure is zero is known as critical height–

$H_C = 2\ Z_C = 2\times\left(\frac{2C}{\gamma\sqrt{K_P}}\right)$

$= \frac{4C}{\gamma\sqrt{K_P}}$ $\quad\therefore K_P = \tan^2\left(45^0+\frac{\phi}{2}\right)$

$\left[H_C = \frac{4C}{\gamma\tan\left(45^0+\frac{\phi}{2}\right)}\right]$

where, C = cohesion of soil

γ = soil density

ϕ = angle of internal friction

k_P = passive earth pressure

1269. In case of cantilever retaining wall, the maximum bending moment is found out using the earth pressure coefficient k_a where

(a) $k_a = 1/(1+\sin\phi)$

(b) $k_a = 1(1-\sin\phi)$

(c) $k_a = (1+\sin\phi)/(1-\sin\phi)$

(d) $k_a = (1-\sin\phi)/(1+\sin\phi)$

APTRASCO 2017

Ans : (d) : Maximum bending moment of retaining wall

$$M = \frac{1}{6}k_a\gamma h^3$$

where, k_a = Coefficient of active earth pressure

$= \left(\frac{1-\sin\phi}{1+\sin\phi}\right)$

γ = density of soil

h = Back fill height of retaining wall

ϕ = angle of internal friction

1270. For upstream face of an earth dam, the most adverse condition for stability of slope, is:

(a) sudden draw down.

(b) steady seepage condition.

(c) during construction.

(d) during overflowing.

WBPSC Poly Lect. 2019

Ans. (a) : For upstream face of an earth dam, the most adverse condition for stability of slope, is sudden draw down.

1271. The active earth pressure of a soil is proportional to (where ϕ is the angle of friction of the soil).

(a) $\tan(45^0-\phi)$ (b) $\tan^2\left(45^0+\frac{\phi}{2}\right)$

(c) $\tan^2\left(45^0-\frac{\phi}{2}\right)$ (d) $\tan(45^0+\phi)$

OPSC AE Paper- II 2019

Ans. (c) : $P_a = K_a\,\sigma_a$

$$K_a = \frac{1-\sin\phi}{1+\sin\phi} = \tan^2\left(45-\frac{\phi}{2}\right)$$

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1272. The ratio of the horizontal stress to the vertical stress is called coefficient of

(a) Active earth pressure
(b) Passive earth pressure
(c) Earth pressure
(d) Plastic earth pressure

ESE 2020

Ans. (c) : **Coefficient of Earth Pressure–** It is defined as the ratio of horizontal stress to the vertical stress.

$$\sigma_h = k.\sigma_v$$

or, $$\boxed{k = \frac{\sigma_h}{\sigma_v}}$$

- **Coefficient of active earth pressure–**When the retaining wall is moving away from the backfill then the ratio between horizontal and vertical earth pressure is known as coefficient of active earth pressure.
- **Coefficient of passive earth pressure–**When the retaining wall is moving towards the backfill, then the ratio between the lateral and vertical earth pressure is known as coefficient of passive earth pressure.

1273. Which of the following assumptions of the Rankine theory of lateral earth pressure are correct?

1. The soil mass is semi-infinite, homogeneous, dry and cohesion-less
2. The ground surface is a plane which may be horizontal or inclined
3. The wall yields about the base and thus satisfies the deformation condition for plastic equilibrium

(a) 1 and 2 only (b) 1 and 3 only
(c) 1, 2 and 3 (d) 2 and 3 only

ESE 2020

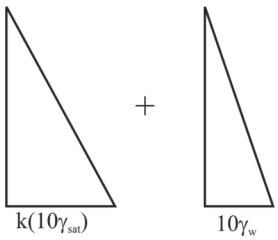
Ans. (c) : Rankine's theory assumptions are–
(1) Soil is homogeneous, semi-infinite, dry and cohesionless.
(2) Ground (or) top surface is plane which may be horizontal or inclined.
(3) For plastic equilibrium condition, failure should occur at all points in soil mass which is theoretical. So, it is assumed as local states of plastic equilibrium in a soil mass can be created by rotating a retaining wall about its base either away (or) into the backfill.

1274. Saturated unit weight of a soil is 20 kN/m^3 and unit weight of water is 10 kN/m^3. If the ground water table is at the surface of soil lateral earth pressure coefficient of soil is 0.4,effective lateral stress at 10 m depth will be-

(a) –20 kPa (b) 40 kPa
(c) 80 kPa (d) 180 kPa

JPSC AE 2009

Ans. (c) : $\gamma_{sat} = 20$ kN/m^3, $\gamma_w = 10$ kN/m^3
k (earth pr. coefficient) = 0.4
z = 10 m



Total Stress

Lateral stress at depth 10 m.

$$\sigma = k\gamma_{sat}z + \gamma_w z$$
$$\sigma = 0.4\times 20\times 10 + 10\times 10$$
$$\sigma = 180 \text{ kPa.}$$

Lateral effective stress-
$\sigma_{eff} = \sigma$ – pore water pr.
$\sigma_{eff} = 180 - 10 \times 10 = 80$ kPa

1275. Large movement of retaining structure is required to produce

(a) active earth pressure
(b) passive earth pressure
(c) both active and passive earth pressure
(d) at rest pressure

Maharashtra PSC Mains 2012 Paper -2

Ans. (b) : The passive earth pressure occurs when a wall or plane moves towards earth, this will create additional pressure so it requires large movement of retaining structure.

1276. If the coefficient of active earth pressure is 0.4, the coefficient of passive pressure is

(a) 0.6 (b) 2.5
(c) 4.0 (d) 0.4

Sikkim PSC AE 2017

Ans. (b) : Given, Active earth pressure (K_a) = 0.4
We know,
$K_a \times K_P = 1$
$\therefore \quad 0.4 \times K_P = 1$

$$K_P = \frac{1}{0.4}$$

$\boxed{K_P = 2.5}$

1277. Lateral earth pressure on a retaining wall is

(a) Equal to the mass of soil retained
(b) Proportional to the height of retaining wall
(c) Proportional to the internal friction of soil
(d) Equal to the square of the depth of soil

Sikkim PSC AE 2017

Ans. (c) : Lateral earth pressure on a retaining wall is proportional to the internal friction of soil.

1278. The pressure on a phreatic surface is

(a) Less than atmospheric pressure
(b) Equal to atmosheric pressure
(c) Greater than atmospheric pressure
(d) Not related to atmospheric pressure

TANGEDCO AE 2018

Ans. (b) : Phreatic surface is top flow line. It is a streamline. The pressure on this line is atmospheric (zero) and below is hydrostatic.

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1279. The horizontal strain required to produce passive earth pressure condition in a cohesionless soil is :

(a) More (b) Less
(c) Very less (d) Less than Zero

TRB Poly. Lect. Tech. 2017

Ans. (a) The horizontal strain required to preduce passive earth pressure.

1280. In the passive state of cohensionless soil, minor stress is

(a) horizontal (b) vertical
(c) 45° to horizontal (d) 30° to horizontal

TRB Poly. Tech. Lect. 2012

Ans. (b) : In Non-cohesive soil, in passive state of plastic equilibrium case, minor principal stress in vertical & major principal stress is horizontal, As wall moves forwards soil.

1281. For a sandy soil having an angle of internal friction 30°, the ratio of passive and active lateral earth pressure will be

(a) 2 (b) 4
(c) 3 (d) 9

GPSC AE Dec 2018
Haryana PSC AE 2017
TRB Poly. Tech. Lect. 2012
UPPSC State Eng. A.E. 2008

Ans. (d) : Given,
angle of internal friction, $\phi = 30^o$

We know $K_a = \tan^2\left(45 - \frac{\phi}{2}\right)$

$$K_P = \tan^2\left(45 + \frac{\phi}{2}\right)$$

$$\frac{P_P}{P_a} = \frac{\frac{1}{2}K_P.\gamma H^2}{\frac{1}{2}K_A.\gamma H^2} = \frac{K_P}{K_a} = \frac{\tan^2\left(45 + \frac{30^0}{2}\right)}{\tan^2\left(45 - \frac{30^0}{2}\right)}$$

$$= \frac{\tan^2(60^o)}{\tan^2(30^0)} = \frac{\left(\sqrt{3}\right)^2}{\left(\frac{1}{\sqrt{3}}\right)^2} = 3 \times 3 = 9$$

1282. The coefficient of earth pressure at rest for a sand having the Poisson's ratio value of 0.25 is :

(a) 1 (b) 3
(c) 1/3 (d) 1/2

TRB Poly. Lect. Tech. 2017

Ans. (c) : Given, Poisson's ratio (μ) = 0.25
Coefficient of earth pressure at rest.

$$K_o = \frac{\mu}{1-\mu}$$

$$K_o = \frac{0.25}{1-0.25} = \frac{0.25}{0.75}$$

$$K_o = \frac{1}{3}$$

1283. Hydrodynamic pressure due to earthquake acts at a height of

(a) $\frac{3H}{2\pi}$ (b) $\frac{3H}{5\pi}$
(c) $\frac{4H}{5\pi}$ (d) $\frac{4H}{3\pi}$

from the base.

BPSC Poly. Tech. Lect. 2016

Ans. (d) : Hydrodynamic pressure due to earthquake acts at a height of $\frac{4H}{3\pi}$ from the base.

1284. What will be the ratio of active earth pressure to passive earth pressure for a soil at a depth of 8 m and having an angle of internal friction as 30° and unit weight?

(a) 9 :1 (b) 8 : 1
(c) 1 : 9 (d) 9 : 8

UPPCL AE 2015

Ans. (c) :

(P_a) Active earth pr. = $\gamma h k_a = \gamma h \left(\frac{1-\sin\phi}{1+\sin\phi}\right)$

(P_p) Passive earth pr. = $\gamma h k_p = \gamma h \left(\frac{1+\sin\phi}{1-\sin\phi}\right)$

$$\frac{P_a}{P_p} = \left(\frac{1-\sin\phi}{1+\sin\phi}\right)^2$$

$$\frac{P_a}{P_p} = \left(\frac{1-\sin 30^o}{1+\sin 30^o}\right)^2$$

$$\frac{P_a}{P_p} = \frac{1}{9}$$

1285. An unrestrained slope can fail by any of the following mechanisms except.

(a) Flow (b) Spread
(c) Topple (d) Shrink

UPRVUNL AE 2016

Ans.: (d) An unrestrained slope can fail by flow, spread and topple.

1286. Which of the following is a flexible retaining wall-

(a) Cantilever (b) Gravity
(c) Semi-gravity (d) Mechanically stabilized

UPRVUNL AE 2016

Ans.: (d) Mechanically stabilized is a flexible retaining wall.

1287. If top of earth retained in horizontal, the coefficient of passive earth pressure for retaining wall become:

(a) $K_p = \frac{1-\sin\phi}{1+\sin\phi}$ (b) $K_p = \frac{1+\sin\phi}{1-\sin\phi}$
(c) $K_p = \frac{\sin\phi-1}{\sin\phi+1}$ (d) $K_p = \frac{\sin\phi+1}{\sin\phi-1}$

Maharashtra PSC (Mains) 2019 Paper–I
GPHC AE 2018

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Ans. (b) : Co-efficient of passive earth pressure, $K_p = \frac{1+\sin\phi}{1-\sin\phi}$

1288. In the design of retaining wall, both active earth pressure and passive earth pressure is considered due to soil available on both sides (with different heights of R.C.C. retaining wall. If angle of repose, $\phi = 30^o$, then what will be the relation between coefficient of active earth pressure (K_a) and passive earth pressure (K_p)?

(a) $K_a = \frac{1}{3}K_p$ (b) $K_a = 3K_p$

(c) $K_a = 9K_p$ (d) $K_a = \frac{1}{9}K_p$

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (d) : $K_a = \frac{1-\sin 30^o}{1+\sin 30^o} = \frac{1-1/2}{1+1/2} = \frac{1/2}{3/2} = \frac{1}{3}$

$K_p = \frac{1+\sin 30^o}{1-\sin 30^o} = \frac{1+1/2}{1-1/2} = \frac{3/2}{1/2} = 3$

$\frac{K_a}{K_p} = \frac{1/3}{3} = \left(\frac{1}{9}\right)$

$K_a = \frac{1}{9}K_p$

1289. A vertical cut is to be made in a saturated clay with C = 18 kN/m^2, $\phi = 0^o$ and $\gamma = 16$ kN/m^3. What is the theoretical depth to which the clay can be excavated without side collapse?

(a) 0.89 m (b) 2.5 m

(c) 4.5 m (d) 9 m

GPSC AE Dec 2018

Ans. (c) :

$H = \frac{4C}{\gamma\sqrt{K_a}}$

For $\phi = 0$, $K_a = 1$

$C = 18$ kN/m^2

$\gamma = 16$ kN/m^3

$H = \frac{4\times 18}{16\times 1} = 4.5$ m

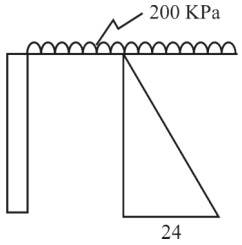
1290. A retaining wall is constructed on the side of a river bank with smooth vertical back 4m. The foundation is over an expansive collapsible soil and has a horizontal surface at the top of the wall and carries a uniformly distributed load of 200 kPa. The unit weight and angle of internal friction of the soil are 18000 N/m^3 and 30° respectively. Compute the total active earth pressure per metre length of the wall.

(a) 314.6 kN/m (b) 410.4 kN/m

(c) 848 kN/m (d) 266.4 kN/m

BHEL ET 2019

Ans. (a)



Given, H = 4 m

q = 200 kPa

$\phi = 30^0$, $\gamma = 18$ kN/m^3

$K_a = \frac{1-\sin\phi}{1+\sin\phi} = \frac{1-\sin 30^0}{1+\sin 30^0}$

$K_a = \frac{1}{3}$

$P_a = K_a \gamma H - 2C\sqrt{K_a}$

$P_a = \frac{1}{3}\times 18\times 4 = 24$

Total active earth pressure

$= K_a.H.q + \frac{1}{2}K_a\gamma.H^2$

$= 200\times\frac{1}{3}\times 4 + \frac{1}{2}\times 4\times 24$

$= 266.6 + 48$

$= 314.6$ kN/m

1291. Culmann's graphical method for active pressure calculation can be used for

(a) a broken backfill surface

(b) a plain inclined backfill surface

(c) a layered backfill with different densities

(d) all of the above

GPSC Assistant Prof. 2016

Ans. (d) : Culmann's graphical method for active pressure calculation can be used for

(a) A broken backfill surface

(b) A plain inclined backfill surface

(c) A layered backfill with different densities

1292. A pure clay deposit (having $\phi = 0$) behind a retaining wall has cohesion of 40 kN/m^2 and a unit weight of 20 kN/m^2; the depth of tension cracks will be

(a) 4.0 m (b) 2.0 m

(c) 1.0 m (d) 0.5 m

GPSC Assistant Prof. 2016

Ans. (a) : $P_a = K_a\gamma Z_C = 2C\sqrt{K_a}$

for $P_o = 0$

$Z_C = \frac{2C}{\gamma\sqrt{K_a}}$

Depth of tension crack = $\frac{2C}{\gamma\sqrt{K_a}} = \frac{2(40)\text{kN/m}^2}{20\left(\text{kN/m}^3\right)\times\sqrt{1}}$

$= 4.0$ m

$K_a = 1$, $\therefore \phi = 0$

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1293. For a cantilever sheet pile wall, the earth pressure below the dredge line are

(a) only active
(b) ony passive
(c) active and passive both
(d) neither active nor passive

GPSC Assistant Prof. 2016

Ans. (c) : For a cantilever sheet pile wall, the earth pressure below the dredge line are active and passive both.

1294. When movement of a wall under the earth pressures from backfill was prevented, the coefficient of earth pressure was recorded as 0.5. The ratio of the coefficient of passive and active earth pressure of the back fill

(a) 1/3 (b) 1/9
(c) 3 (d) 9

UJVNL AE 2016
ESE 2002

Ans. (d) : Coefficient of earth pressure = 0.5

$K_0 = 0.5$ (K_0 = Earth pressure at rest)

$1-\sin\phi = 0.5$

$\sin\phi = 0.5$

$\phi = 30^0$

Coefficient of active earth pressure $= \dfrac{1-\sin\phi}{1+\sin\phi}$

$$K_a = \frac{1-\sin 30^0}{1+\sin 30^0}$$

$$K_a = \frac{1/2}{3/2} = \frac{1}{3}$$

Coefficient of passive earth pressure $= = \dfrac{1+\sin\phi}{1-\sin\phi}$

$$= \frac{1+\sin 30}{1-\sin 30} = 3$$

Ratio of $\dfrac{K_p}{K_a} = \dfrac{3}{(1/3)} = 9$

1295. In a cohesionless soil deposit with a unit weight of 15 kN/m^2 and an angle of internal friction of 30^0, the active and passive earth pressures (in kN/m^2) at a depth of 10 m will be, respectively :

(a) 150 and 50 (b) 100 and 200
(c) 50 and 450 (d) 200 and 100

UJVNL AE 2016

Ans. (c) : Given, C = 0, $\gamma = 15$ kN/m^2

$\phi = 30^0$

$$K_a = \frac{1-\sin\phi}{1+\sin\phi} = \frac{1-\sin 30}{1+\sin 30} = \frac{1}{3}$$

$$K_p = \frac{1}{K_a} = \frac{1}{1/3} = 3$$

Active earth pressure = $K_a\gamma z = \dfrac{1}{3}\times 15\times 10 = 50$ kN/m^2

Passive earth pressure = $K_p\gamma z = 3\times 15\times 10 = 450$ kN/m^2

1296. The meaning of active earth pressure coefficient (K_a) is:-

(a) Passive stresses (b) Total stresses
(c) Neutral stresses (d) Effective stresses

UPPCL AE 2014

Ans. (d) : Active earth pressure coefficient means effective stresses.

1297. How much increase in pressure will be, if a uniform surcharge of 120 kN/m^2 is placed on the back fill with $\phi = 30°$

(a) 12 kN/m^2 (b) 120 kN/m^2
(c) 30 kN/m^2 (d) 40 kN/m^2

UPPCL AE 2014

Ans. (d) : Given, q = 120 kN/m^2

$\phi = 30^o$

increase in pressure, $\Delta\sigma = qK_a$

$$120\times\frac{1-\sin 30^o}{1+\sin 30^o} = 120\times\frac{\frac{1}{2}}{\frac{3}{2}}$$

= 40 kN/m^2

1298. The earth pressure theory that is used for the design of cantilever retaining wall is

(a) Meyerhof's theory (b) Rankine's theory
(c) Terzagi's theory (d) Skempton's theory

APPSC AEE 2016

Ans. (b) : Earth pressure theory used for design of cantilever retaining wall
(1) Rankine theory
(2) Coulomb's theory

1299. In active state of plastic equilibrium in a non cohesive soil with horizontal ground surface

(a) Major principal stress is horizontal
(b) Major principal stress is vertical
(c) Minor principal stress is inclined
(d) Minor and major principal stresses are equally inclined to horizontal

GPSC AE June 2019
GPSC AE January 2018

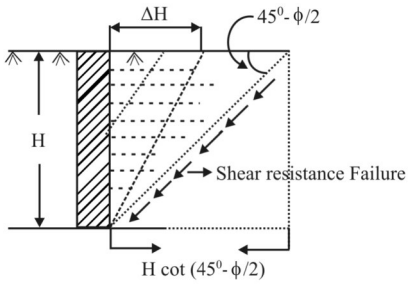
Ans. (b) : Major principal stress is vertical in active state of plastic equilibrium in a non-cohesive soil with horizontal ground surface.

1300. The inclination of failure plane behind a vertical wall in the passive pressure case is inclined to the horizontal at

(a) $45^0 - \phi/2$ (b) $45^0 - \phi$
(c) $45^0 + \phi/2$ (d) $45^0 + \phi$

GPSC AE Dec 2018
MPPSC AE 2014

Ans. (a)



Passive Earth Pressure.

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1301. Assuming elastic load-deformation behavior of a soil, determine the at-rest earth pressure coefficient of the soil having Poisson's ratio 0.31.

(a) 0.45 (b) 0.50
(c) 0.69 (d) 0.33

UPPCL AE 2018

Ans. (a) : Given, Poisson ratio, $\mu = 0.31$

At-rest earth pressure coefficient $K_o = \dfrac{\mu}{1-\mu}$

$K_o = \dfrac{0.31}{1-0.31} = 0.45$

1302. For normally consolidated soil with drained angle of internal friction of 30^0, the coefficient of "at rest earth pressure" (k_0) is:

(a) 0.75 (b) 0.6
(c) 0.5 (d) 1

UPPCL AE 2019

Ans. (c) The coefficient of earth pressure at rest (K_o) = $(1 - \text{Sin}\phi)$

ϕ = internal friction angle = 30^0

$k_0 = 1 - \text{Sin } 30^0$

$\boxed{k_0 = 0.5}$

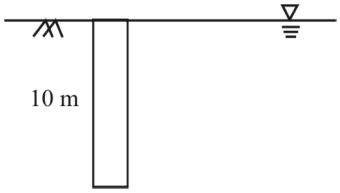
1303. The intensity of passive earth pressure (N/sq m) at a depth of 10 m in a dry cohesionless sand with an angle of internal friction of 30 degree and submerged unit weight of 21.81 kN/m^3, if the water table is at ground level, will be (assume the unit weight of water as 10 kN/m^3):

(a) 39.6 (b) 654.3
(c) 460.0 (d) 153.3

(LMRC AE 2017 I–Shift)

LMRC AM 2015

Ans (b)



Given,

$\phi = 30^0$

$K_p = \tan^2\left(45 + \dfrac{\phi}{2}\right) = 3$

$\gamma_{sub} = 21.81 \text{kN/m}^3$

at bottom intensity of passive earthen pressure

$= K_p.\gamma_{sub}.z$

$= 3 \times 21.81 \times 10 = 654.3 \text{ N/m}^2$

1304. For a sample of dry cohesionless soil with friction angle □, the failure plane will be inclined to the major principal plane by an angle equal to -

(a) $45^0 + \phi$ (b) $45^0 - \phi$
(c) $45^0 - \phi/2$ (d) $45^0 + \phi/2$

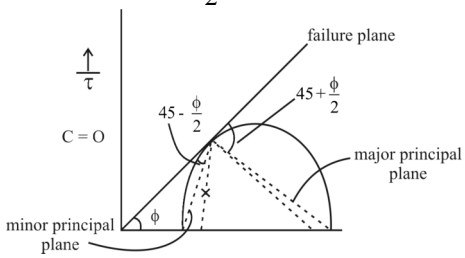
Rajasthan VP ITI 2018

Haryana PSC AE Paper–II 2018

Ans. (d) : Friction angle = ϕ

Angle between major principal plane & shearing plane $= 45 + \dfrac{\phi}{2}$

Angle between minor principal plane & shearing plane $= 45 - \dfrac{\phi}{2}$



1305. The angle between Coulomb's failure envelope and the horizontal is called :

(a) Angle of friction (b) Angle of repose
(c) Inclination angle (d) Cohesion angle

UPRVUNL AE 2014

Ans. (a) : The angle between coulomb's failure envelope and the horizontal is called angle of friction.

1306. In cohesive soils, the range of earth pressure coefficient from active to passive condition is :

(a) 0.14 - 14 (b) 0.5 - 2
(c) 2 - 4 (d) 4 - 8

UPRVUNL AE 2014

Ans. (b) : In cohesive soils, the range of earth pressure coefficient from active to passive condition is 0.5-2.

Coefficient of active earth pressure $(K_a) = \dfrac{1-\sin\phi}{1+\sin\phi}$

Coefficient of passive earth pressure $(K_p) = \dfrac{1+\sin\phi}{1-\sin\phi}$

ϕ = internal friction angle

1307. Classical Earth Pressure theory is referred to by :

(a) Coulomb's theory
(b) Rankine's theory
(c) Tresca's theory
(d) Both Coulomb's and Rankine's theories

UPRVUNL AE 2014

Ans. (d) : Classical Earth Pressure theory is referred to by both Coulomb's and Rankine's theories.

1308. The coefficient (Rankine's) for passive earth pressure for a sand having an angle of internal friction of 30º is

(a) $\dfrac{1}{3}$ (b) $\dfrac{1}{2}$
(c) 2 (d) 3

BPSC Poly. Tech. Lect. 2016

Ans. (d) : Coefficient of permeability for passive

$\phi = 30^o$ (Angle of internal friction)

$k_p = \dfrac{1+\sin\phi}{1-\sin\phi}$

$= \dfrac{1+\sin 30}{1-\sin 30} = \dfrac{1+0.5}{1-0.5} = \dfrac{1.5}{0.5} = 3$

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1309. Saturated unit weight of a soil is 20 kN/cum and the unit weight of water is 12 kN/cum. If the groundwater table is at the surface of soil and lateral earth pressure coefficient of soil is 0.4, effective lateral stress (kPa) at 10 m depth will be :

(a) 32 (b) 40
(c) 48 (d) 20
(e) 12

CGPSC AE Shift - 2 2014

Ans. (a) : Effective stress at 10 m depth = $(\gamma_{sat} - \gamma_w)$ d
$= (20–12) \times 10 = 80$
lateral stress = $\mu \times$ effective stress = $0.4 \times 80 = 32$

1310. The critical height of an unsupported vertical cut in a cohesive soil is given by

(a) $\frac{2C}{\gamma}\tan\left(45+\frac{\phi}{2}\right)$ (b) $\frac{4C}{\gamma}\tan\left(45+\frac{\phi}{2}\right)$
(c) $\frac{4C}{\gamma}\cot\left(45+\frac{\phi}{2}\right)$ (d) $\frac{2C}{\gamma}\cot\left(45+\frac{\phi}{2}\right)$

Rajasthan AE Nagar Nigam Shift-3, 2016

Ans. (b) : Vertical height of an unsupported vertical cut in a cohesive soil = $\frac{4C}{\gamma}\tan\left(45+\frac{\phi}{2}\right)$
C = cohesion of soil
γ = soil unit weight
ϕ = interval friction angle

1311. According to Coulomb's wedge theory, the active earth pressure slides the wedge:

(a) Up and inwards on a slip surface
(b) Down and outwards on a slip surface
(c) Horizontal upward and parallel to base
(d) Horizontal inward and parallel to base
(e) None of these options

Haryana PSC AE Paper–II 2018

Ans. (b) :
- Coulomb's wedge theory provides a method of analysis that gives the resultant horizontal force on a retaining system for any slope of wall, friction and slope of backfill provided.
- This theory assumes that soil shear resistance develops along the wall and failure plane.
- As per Coulomb's theory, a wedge of soil above a failure plane move outward and downward in the active case when the wall moves away from the backfill due to lateral earth pressure.

1312. Which one of the following statements regarding lateral earth pressure is correct?

(a) The lateral strain required to fully mobilize the soil passive pressure is considerably smaller than the lateral strain required to fully mobilize the soil active pressure
(b) The lateral strain required to fully mobilize the soil passive pressure is slightly smaller than the lateral strain required to fully mobilize the soil active pressure
(c) The lateral strain required to fully mobilize the soil passive pressure is slightly greater than the lateral strain required to fully mobilize the soil active pressure
(d) The lateral strain required to fully mobilize the soil passive pressure is considerably greater than the lateral strain required to fully mobilize the soil active pressure

HPPSC Poly. Lect. 2016

Ans : (d) In lateral earth pressure condition, the lateral strain required to fully mobilize the soil passive pressure is considerable greater than the lateral strain required to fully mobilize the soil active pressure.

1313. The coefficient of earth pressure is defined as the ratio of

(a) Effective stress to neutral stress
(b) Total stress to effective stress
(c) Vertical stress to horizontal stress
(d) Horizontal stress to vertical stress

WBPSC AE 2012

Ans. (d) : The coefficient of earth pressure is defined as the ratio of horizontal stress to vertical stress.

1314. The term counterfort is used in design as

(a) Intze tank (b) Friction pile
(c) Retaining wall (d) Grillage footing

WBPSC AE 2003

Ans. (c) : The term counterfort is used in design as Retaining wall.

1315. If is the single of slope with the horizontal, and 'ϕ' the angle of shearing resistance, F, the factor of safety against sliding for a mass of soil bounded by a sloping surface is given by

(a) tan 'ϕ' tan i (b) $\frac{\tan i}{\tan\phi}$
(c) $\frac{\tan\phi}{\tan i}$ (d) $\frac{\tan(\phi+i)}{\tan\phi}$

WBPSC AE 2001

Ans. (c) : If is the single of slope with the horizontal, and 'ϕ' the angle of shearing resistance, F, the factor of safety against sliding for a mass of soil bounded by a sloping surface is given by $\frac{\tan\phi}{\tan i}$.

1316. As per Rankine's formula the passive earth pressure at a depth 8 m in dry cohensionless and with angle a integral frication of 38° and unit weight of dry sand as 200 kg/m^3

(a) $67.2\frac{kN}{m^2}$ (b) $65.0\frac{kN}{m^2}$
(c) $61.2\frac{kN}{m^2}$ (d) $60.0\frac{kN}{m^2}$

take value of sin 38° as 0.6157

WBPSC AE (MCD) 1998

Ans. (a) : Passive earth pressure,

$$K_p = \frac{1+\sin\phi}{1-\sin\phi} = \frac{1+\sin 38^0}{1-\sin 38^o} = \frac{1+0.6157}{1-0.6157} = 4.2$$

Given,
Unit weight of dry sand, $\gamma = 200$ kg/m^3
$H = 8$ m
$P_p = K_p.\gamma.H$
$= 4.2 \times 200 \times 8 = 6720$ kg/m^2
$P_P = 67.2$ kN/m^2

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1317. Passive earth pressure decreases with

(a) decrease of angle of internal friction
(b) increase of height of wall
(c) movement of wall towards the backfill
(d) increase of surcharge on backfill

WBPSC AE (PHE) 1998

Ans. (a) : Coefficient of passive earth pressure

$$K_p = \frac{1+\sin\phi}{1-\sin\phi}$$

$K_p \propto \phi$

ϕ = internal friction angle

ϕ decrease then K_p is decreases.

$P_P = K_p.\gamma.H$

$P_P \propto K_p$

K_p decrease then P_P is decreases.

1318. Intensity of active earth pressure at a depth of 10 m in dry cohesionless spill with an angle of internal friction of 30° and with unit weight of 1.80 t/m^3

(a) 3 t/m^2 (b) 6 t/m^2
(c) 7 t/m^2 (d) 9 t/m^2

WBPSC AE (PHE) 1998

Ans. (b) : Given,

Depth (H) = 10 m

Unit weight, γ = 1.8 t/m^3

$$K_a = \frac{1-\sin\phi}{1+\sin\phi} = \frac{1-\sin 30^o}{1+\sin 30^o} = \frac{1}{3}$$

intensity of active earth pressure,

$$P_a = K_a.\gamma.H = \frac{1}{3}\times 1.8\times 10 = 6\ t/m^2$$

1319. Active earth pressure of a soil is proportionate to

(a) $\tan^2\left(45^o-\frac{\theta}{2}\right)$ (b) $\sin^2\left(45^o-\frac{\theta}{2}\right)$

(c) $\cos^2\left(45^o+\frac{\theta}{2}\right)$ (d) $\tan^2\left(45^o-\frac{\theta}{2}\right)$

where θ = angle of internal friction of the soil

WBPSC AE (PHE) 1998

Ans. (a) : Active earth pressure (K_a) = $\frac{1-\sin\phi}{1+\sin\phi} = \tan^2\left(45-\frac{\phi}{2}\right)$

ϕ = angle of internal friction.

1320. Rankine's earth pressure theory is least violated in

(a) cantilever retaining wall
(b) gravity retaining wall
(c) sheet pile coffer dam
(d) anchored bulk head

WBPSC AE (PHE) 1998

Ans. (d) : Rankine's earth pressure theory is least violated in anchored bulk head.

1321. Coefficient of earth pressure at rest is generally less than one but it can be greater than 1, if the soil is

(a) normally consolidated
(b) highly organic
(c) heavily over consolidated
(d) elastic

WBPSC AE (PHE) 1998

Ans. (c) : Coefficient of earth pressure at rest is generally less than one but it can be greater than 1, if the soil is heavily over consolidated.

1322. The total active earth pressure on a retaining wall due to dry sandy back fill acts at H/3 above the base of the wall and is directly proportional to

(a) 1/H (b) H
(c) H^2 (d) H^3

WBPSC AE (Common) 1998

Ans. (c) : Total active earth pressure (P_a)

$$= \frac{1}{2}K_a.\gamma.H^2$$

Act at $\frac{H}{3}$ above the base, $\boxed{P_a \propto H^2}$

1323. A 3 m high retaining wall with vertical face is resisting a moist back fill with horizontal top surface having γ = 20 kN/cum. The percentage increase in Total Active Earth Pressure, if the back fill gets submerged with γ_{sat} = 22 kN/cum and γ_W = 10 kN/cum, is:

(a) 20 (b) 40
(c) 60 (d) 100

TSPSC Deputy Engineer 2015

Ans. (c) : Given,

H = 3m, γ = 20 kN/m^3

γ_w = 10 kN/m^3 γ_{sat} = 22 kN/m^3

Total active earth pressure $P_a = \frac{1}{2}K_a.\gamma.H^2$

$$P_a = \frac{1}{2}\times 1\times 20\times(3)^2 = 90 kN/m^2$$

$$(P_a)_1 = \frac{1}{2}K_a\gamma_{sat}H^2 + \frac{1}{2}\gamma_w.K_a.H^2$$

$$= \frac{1}{2}\times 1\times 22\times(3)^2 + \frac{1}{2}\times 10\times 1\times(3)^2 = 144\ kN/m^2$$

Increase in total active earth pressure =

$$= \frac{(P_a)_1 - (P_a)}{P_a}\times 100 = \frac{144-90}{90}\times 100 = 60\%$$

1324. What of the following statements is incorrect?

(a) Even the slight movement of wall away from backfill develops full active earth pressure.
(b) Earth pressure at rest will be present when the wall does not move relative to the back fill.
(c) Full passive earth pressure develops only after considerable movement of wall towards the backfill.
(d) For a given wall and backfill conditions, the magnitude of active earth pressure will always be higher than passive earth pressure.

TSPSC AEE 2017

Ans. (d) : These options are correct :

- Even the slight movement of wall away from backfill develops full active earth pressure.
- Earth pressure at rest will be present when the wall does not move relative to the back fill.
- Full passive earth pressure develops only after considerable movement of wall towards the backfill.

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10. Shallow and Deep Foundation

1325. Sheet piles are used primarily for
(a) to carry vertical load from structure
(b) to prevent seepage of water
(c) retaining side of cuts
(d) to increase bearing capacity of soil

WBPSC AE (PHE) 1998

Ans. (b) : Sheet piles are used primarily for to prevent seepage of water.

1326. Sand pile are primarily used for
(a) increasing the bearing capacity of soil
(b) accelerating the total settlement
(c) to reduce the settlement
(d) to increase shear strength of soil

WBPSC AE (PHE) 1998

Ans. (d) : Sand pile are primarily used for to increase shear strength of soil.

1327. The net ultimate bearing capacity of a 2m wide square footing laid on pure cohesive soil is estimated at 400 kPa. All other parameters remaining same what will be the net ultimate bearing capacity if size of the footing is doubled?
(a) 100 kPa (b) 200 kPa
(c) 400 kPa (d) 50 kPa

TSPSC AEE 2017

Ans. (c) :

1328. For foundation on clayey soil, the maximum differential settlement is limited
(a) 20 mm (b) 30 mm
(c) 40 mm (d) 50 mm

WBPSC AE (MCD) 1998

Ans. (c) : For foundation on clayey soil, the maximum differential settlement is limited 40 mm.

1329. The foundation on weaker soil may be done by
(a) Grillage footings (b) Column footings
(c) Raft footings (d) Any of the above

WBPSC AE (MCD) 1998

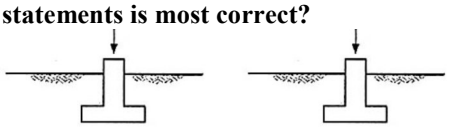
Ans. (d) : The foundation on weaker soil may be done by Grillage footings, Column footings, Raft footings.

1330. For heavy structure on sandy soil, the type of foundation generally provided is
(a) Strap footing (b) Pier footing
(c) Raft footing (d) Combined footing

WBPSC AE (MCD) 1998

Ans. (b) : For heavy structure on sandy soil, the type of foundation generally provided is Pier footing.

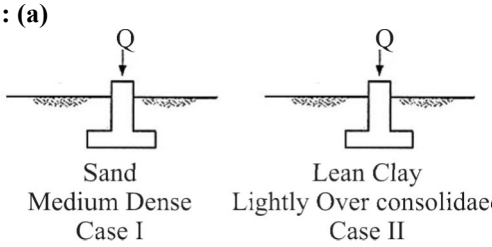
1331. The figure shows two identical building footings with the same load but constructed in two different soil types. Which of the following statements is most correct?



(a) The long-term settlement for case I is less than case II
(b) The long-term settlement for case II is less than case I
(c) The long-term settlements are the same for both cases
(d) Settlement is not a concern for either case

HPPSC Poly. Lect. 2016

Ans : (a)



Clay have ultimate settlement but sand have initial settlement. The long term settlement for case-I is less than case-II.

1332. A wooden pile is being driven with a drop hammer weighing 20 kN and having a free fall of 1.0m. The penetration in the last blow is 5 mm. The load carrying capacity of the pile as per engineering news Formula is :
(a) 33.33 kN (b) 66.66 kN
(c) 222.2 kN (d) 111.1 kN

TRB Poly. Lect. Tech. 2017

Ans. (d) : W = 20 kN
h = 1 m = 100 cm
S = 5 mm = 0.5 cm
As per engineering news formula.

$$Q_{allowable} = \frac{W.h}{fos(S+C)}$$

C = 2.5 cm for drop hammer
fos = 6

$$Q_{allowable} = \frac{20\times100}{6\times(0.5+2.5)}$$

$$Q_{allowable} = 111.1 \text{ kN}$$

1333. Net ultimate bearing capacity of a soil is 25 t/m^2 and density 1.7 t/m^3. The safe bearing capacity at 1 m below the ground surface taking a factor of safety 2.5 will be
(a) 10 t/m^2 (b) 25 t/m^2
(c) 11.7 t/m^2 (d) 62.5 t/m^2

ASSAM PSC CCE (PRE) 2015

Ans : (c) : Given
ultimate bearing capacity = 25 t/m^2
density γ = 1.7 t/m^3
D = 1m
factor of safety = 2.5

$$\text{net bearing capacity} = \frac{25}{2.5}$$

$$= 10 \text{ t/m}^2$$

safe bearing capacity = 10 + γ.D
= 10 + 1.7 × 1
= 11.7 t/m^2

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1334. For a saturated soil, skempton's B-parameter is

(a) nearly zero (b) nearly 0.5
(c) nearly 1.0 (d) very high

MPPSC AE 2014

Ans. (c)

$$B = \frac{\Delta V_C}{\Delta \sigma_C} = \frac{\Delta U_C}{\Delta \sigma_3}$$

$B \rightarrow$ Pore Pressure Parametor

ΔU_C = Change in Pore Prossure

$\Delta \sigma_C = \Delta \sigma_3$ = Change in cell pressure

- $0 \leq B \leq 1$
- $\boxed{B = 0}$ for dry soil
- $\boxed{B = 1}$ for saturated soil

1335. Which of the following is not an assumption in terzhaghi's formulation of bearing capacity equation?

(a) Plain strain condition exists
(b) Base of the footing is smooth
(c) The width of the base of the footing is less than or equal to the founding depth
(d) Shear strength is governed by Mohr-Coulomb failure criteria

UPRVUNL AE 2016

Ans.: (b) Assumption in terzhaghi's formula:
(1) footing is located at a depth D_f below the ground surface such that $D_f \leq B$, where B is the width of the footing. This means he considered the foundation to be shallow.
(2) Footing is continuous, this makes the analysis two dimensional.
(3) Base of foundation is rough
(4) The angle of roughness between base of footing and soil is considered equal to friction angle of the soil.
(5) failure is general shear failure and at the time of failure soil reaches into plastic state.
(6) the soil above the base of the footing is replaced by on equivalent surcharge ($q = \gamma D_f$). This means that shearing resistance of the soil located above the base is neglected however for dip foundation it can not be neglected.
(7) the stress zone of the soil extends upto foundation level and not upto ground level.

1336. If width of a square footing resting on clay is doubled, conferred to original footing, ultimate bearing capacity will be increased by:

(a) 200% (b) 400%
(c) 300% (d) Zero per cent

WBPSC Poly Lect. 2019

Ans. (d) : Ultimate bearing capacity as per Terzaghi's equation for square footing.

$$q_u = 1.3\, CN_C + q\, N_q + 0.4x\, B\, N_x \quad (q = \gamma D_f)$$

for purely cohesive soil,

$N_q = 1$ and $N_x = 0$

squre footing resting on clay so, $D_f = 0$

hence, $q_u = 1.3\, CN_C$

Hence, ultimate bearing capacity is independent of the size of footing or width of the footing.

1337. A purely cohesive soil was tested by unconfined compression test. The mean unconfined compression strength was obtained as 50 kN/sq.m. The net ultimate bearing capacity of the soil adopting terzaghi's concept will be (adopt bearing capacity factor = 5.7, 1 kg approximately equal to 10 N).

(a) 90 kN/sq.m (b) 120 kN/sq.m
(c) 142.50 kN/sq.m (d) 162.50 kN/sq.m

ISRO Scientist/Engineer 2020

Ans. (c) :

1338. For local shear failure (if ϕ = angle of internal friction)

(a) $\phi > 28^0$ (b) $\phi > 36^0$
(c) $\phi < 28^0$ (d) $\phi < 36^0$

ASSAM PSC CCE (PRE) 2015

Ans : (c) For local shear failure $\rightarrow \phi < 28^0$

1339. The net ultimate bearing capacity of a purely cohesive soil

(a) depends on the width of the footing and is independent of the depth of the footing
(b) depends on the width as well as the depth of the footing
(c) depends on the depth but is independent of the width of the footing
(d) is independent of both the width and depth of the footing

GPSC AE (CLASS 1 & 2) 2019

Ans : (d) : As per Terzaghi, net ultimate bearing capacity–

$$q_{nu} = CN_C + q\,(Nq - 1) + \frac{1}{2} B\gamma N\gamma$$

for pure cohesive clay ($\phi = 0$)

$N_C = 5.7$, $N_q = 1$, $N\gamma = 0$

$[q_{nu} = 5.7\ C]$

Thus, net ultimate bearing capacity for pure clay soil ($\phi = 0$) is independent of width and depth of footing.

1340. The minimum gross pressure intensity at the base of the foundation at which the soil fails in shear is called as :

(a) Ultimate bearing capacity
(b) Net safe bearing capacity
(c) Allowable bearing capacity
(d) Safe bearing capacity

(RRB SSE (shift-III), 02.09.2015)

Ans : (a) The ultimate bearing capacity is defined as the minimum gross pressure at base of the foundation at which the soil fails in shear.

$$\text{safe bearing capacity} = \frac{\text{ultimate bearing capacity}}{\text{factor of safety}}$$

ultimate bearing capacity = safe bearing capacity × factor of safety.

1341. The ultimate bearing capacity of a sandy soil when tested with 750 mm diameter plate as determined from a field load tests is 300 kN/sq.m. Estimate the ultimate bearing capacity (kN/sq.m) of a circular footing with a diameter of 1.5 m.

(a) 600 (b) 150

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(c) 450 (d) 750
(e) 300

CGPSC AE Shift - 2 2014

Ans. (a) :

1342. Settlement occurs in short time for the foundation resting on
(a) Coarse grained soil
(b) Fine grained soil
(c) Mixed soil
(d) Not related to type of soil

OPSC AE Paper- II 2019

Ans. (a) : Settlement occurs in short time for the foundation resting on coarse grained soil.

1343. The safe bearing capacity of the soil is equal to
(a) Normal strength × Factor of safety
(b) Ultimate bearing power/ Factor of safety
(c) Ultimate tensile strength/ Factor of safety
(d) Ultimate compressive strength/Factor of safety

Gujarat PSC AE 2017

Ans : (b) The safe bearing capacity of the soil is equal to ultimate bearing power/ Factor of safety.

1344. The minimum bearing capacity of a soil under a given footing occurs when the groundwater table at the location is at-
(a) The base of the footing
(b) The ground level
(c) Depth equal to one-half the width of the footing
(d) A depth equal to the width of the footing

MPPSC AE 2017

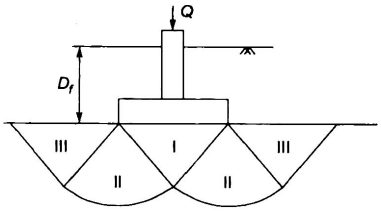
Ans. (b) : $q_{nu} = cN_c + qN_q + \frac{1}{2}B\gamma N_\gamma$

At ground q is least, hence q_{nu} is least.

1345. In Terzaghi's bearing capacity analysis, the soil wedge immediately below the footing remains in state of-
(a) Plastic equilibrium
(b) Radial shear
(c) Elastic equilibrium
(d) Linear shear

MPPSC AE 2017

Ans. (c) :



According to Terzaghi's the failure zone is divided into 3 zones. Zone 1 represents active Rankine zone and is in a state of elastic equilibrium. Zone 2 is known as zone of radial shear.
Zone 3 represents passive zone.

1346. The ultimate bearing capacity of a soil, is-
(a) Load at which soil consolidates
(b) Load at which soil fails
(c) Total load on the bearing area
(d) Safe load on the bearing area

OPSC AE -2016 (II)

Ans. (b) : Ultimate bearing capacity of a soil is the load at which the soil fails.

1347. According to Terzaghi, the net ultimate bearing capacity of clay is given by
(a) CN_Q (b) CN_1
(c) CN_C (d) $1.3\ CN_C$

OPSC AE Paper- II 2019

Ans. (c) : $q_{nu} = CN_C$ for stripfooting

1348. Terzaghi's bearing capacity factors, N_c, N_q and N_γ are functions of:
(a) cohesion of the soil only.
(b) angle of shearing resistance of the soil only.
(c) dimension of the footing.
(d) both cohesion and angle of shearing resistance of the soil.

WBPSC Poly Lect. 2019

Ans. (b) : Terzaghi's bearing capacity factors, N_c, N_q and N_γ are functions of angle of shearing resistance of the soil only.

1349. The weight of the hammer used in the standard penetration test is :
(a) 50 kg (b) 60 kg
(c) 65 kg (d) 75 kg

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (c) The standard weight of hammer used in SPT is 63.5 Kg and height of free fall in 75 cm.

1350. In a plate load test on a sandy soil, the test plate of 60 cm × 60 cm undergoes a settlement of 5 mm at a pressure of 12 × 10^4 N/m^2. What will be the expected settlement of 3m × 3m footing under the same pressure?
(a) 9 mm (b) 15 mm
(c) 20 mm (d) 25 mm

Gujarat PSC AE (N.W.R.) 2020

Ans. (a) : for sandy soil

$$\frac{S_f}{S_p} = \left[\frac{B_f(B_p + 0.3)}{B_p(B_f + 0.3)}\right]^2$$

Given,
S_p = 5mm, B_p= 60cm, B_f = 3 m

$$S_f = S_p\left[\frac{3(0.6+0.3)}{0.6(3+0.3)}\right]^2 = 5\left[\frac{2.7}{1.98}\right]^2$$

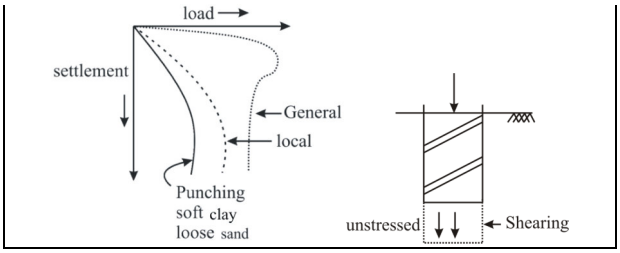
$$S_f = 9.2 \approx 9 \text{ mm}$$

1351. Which of the following exhibits maximum deformation?
(a) Local shear failure
(b) General shear failure
(c) Punching shear failure
(d) Composite failure

BPSC AE 2012 Paper–V

Ans : (c) Punching shear failure soil below the foundation gets cut off from adjecent soil by shearing and excessive/large settlement or maximum deformation will recorded.

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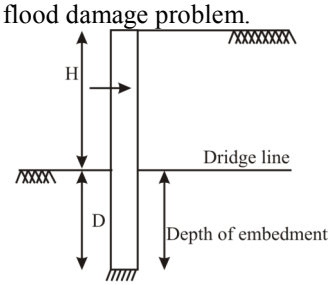


1352. Due to large leakage and flood damage problems, following type of coffer dam is not preferred–

(a) Braced type
(b) Cantilever sheet pile type
(c) Cellular type
(d) Double wall type

UKPSC AE 2013 Paper–I

Ans : (b) Cantilever sheet pile is not preferred to large leakage and flood damage problem.



1353. For determining the ultimate bearing capacity of soil, the recommended size of a square bearing plate to be used in plate load test should be 30 to 75 cm square with a minimum thickness of

(a) 10 mm (b) 16 mm
(c) 25 mm (d) 32 mm

OPSC AE Paper- II 2019
GPSC AE January 2018

Ans. (c) : Circular or square bearing plates of weld steels are used of thickness not less than 25 mm and varying in size from 300 mm to 750 mm (300, 450, 600, 750) are used smaller size plates are used in dense or stiff soils where as larger plates are being used in loose or soft soil.

1354. for multi-storeye buildings having isolated foundation on sand, the maximum permissible settlement is

(a) 59 mm (b) 60 mm
(c) 75 mm (d) 100 mm

GPSC AE (CLASS 1 & 2) 2019

Ans : (b) :
- Multi-storeye buildings having isolated foundation on sand, the maximum permissible settlement $\Rightarrow$ 60 mm.
- Permissible total settlement isolated foundation clay $\Rightarrow$ 65 mm.
- Permissible total settlement of isolated foundation sand $\Rightarrow$ 40 mm.

1355. The factor of safety for shallow foundation against overturning shall be not less than___ when dead load, live load and earth pressures are considered together with wind load or seismic forces.

(a) 1.5 (b) 2.0
(c) 2.5 (d) 3.0

GPSC AE (CLASS 1 & 2) 2019

Ans : (a) : The factor of safety against sliding of structures which resist lateral forces shall be not less than or equal to 1.5 when, dead load, live load and earth pressure are considered together with wind load or seismic force.

1356. Terzaghi's bearing capacity factors namely N_c, N_q, N_γ are functions of

(a) Shape of the foundation
(b) Size of the foundation
(c) Angle of shearing resistance
(d) Cohesion

GPSC AE (CLASS 1 & 2) 2019
(ESE 2005)

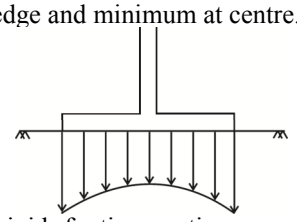
Ans : (c) : N_c, N_q, N_γ are called 'bearing capacity factors' for shallow continuous footing. The parameters are interms of effective stresses and N_c, N_q, N_γ are function of angle of shearing resistance.

1357. Contact pressure for a rigid footing resting on clay at the edge and the centre are _____ and _____, respectively.

(a) Zero; maximum
(b) Minimum; maximum
(c) Maximum; minimum
(d) Maximum; zero

LMRC AM 2019

Ans. (c) For cohesive soil (clay) contact pressure is maximum at edge and minimum at centre.



Note : For rigid footing resting on cohesive soils, settlement is uniform but contact pressure varies at edges contact pressure is maximum and at centre it is minimum which forms inverted bowl shape as shown in below figure.

1358. Who made the assumption that 'a strip footing placed on the ground surface sinks vertically downwards into the soil at failure'?

(a) Rankine (b) Hogentogler
(c) Prandtl (d) Terzaghi

Gujarat PSC AE (N.W.R.) 2020

Ans. (c) : Prandtl made the assumption that a strip footing placed on the ground surface sinks. Vertically downwards into the soil at failure.

1359. The diameter of enlarged bore or bulb for under reamed piles used in expansive soils is generally taken as

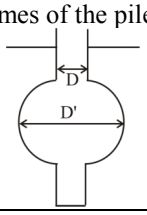
(a) 0.3 – 0.6 times the length of the pile
(b) 1.5 – 1.2 times the pile shaft dia
(c) 2 to 3 times the pile shaft dia
(d) 1000 mm

APPSC AEE (Mains) 2016

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Ans. (c) : Under reamed pile

D' = 2 to 3 times of the pile shaft dia.



1360. Standard Penetration Test is carried out on which type of soils?

(a) Cohesionless Soils (b) Medium Clays
(c) Gravely Soils (d) Very Stiff Clays

Gujarat PSC AE (N.W.R.) 2020

Ans. (a) : Standard Penetration Test is carried out on Cohesionless Soils.

1361. The advantage of concrete pile over a timber pile is

(a) No decay due to termites
(b) No restriction on length
(c) Higher bearing capacity
(d) All the above

RPSC ACF 2011

Ans : (d) Advantage of concrete piles–

(i) No decay due to termites
(ii) No restriction on length
(iii) Higher bearing capacity
(iv) Better quality control can be implemented as compared to bored cast in situ piles.
(v) These piles can be constructed in various cross-section shape such as circular, octagonal or square.

1362. If water table is encountered in the standard pit while conducting plate load test.

(a) The pit is considered unsafe
(b) The load test should be abandoned
(c) The bearing capacity of soil cannot be determined in this condition
(d) Test should be conducted with complete dewatering continuously throughout the test duration

RPSC AE (GWD) 2014

Ans : (d)

- Plate load test is a field test to determine the ultimate bearing capacity of soil, and the probable settlement under a given loading. This test should be conducted with complete dewatering continuously throughout the test duration.
- The test essentially consist in loading a rigid plate at the foundation level, and determining the settlements corresponding to each load increment.

1363. A 600 mm square bearing plate settles by 15 mm in plate load test on a cohesionless soil under an intensity of loading of 0.2 N/mm^2. The settlement of a prototype shallow footing 1 m square under the same intensity of loading

(a) 15 mm (b) Between 15 mm to 25 mm
(c) 25 mm (d) Greater than 25 mm

TRB Poly. Lect. Tech. 2017
RPSC LECTURE 2011

Ans : (b) B_p = Square of the Test plate = 0.6 m
B_f = square of the foundation =1.0 m
S_p = settlement of the test plate = 0.015 m
S_F = settlement of foundation = ?

Plate and settlement of foundation is following relation :

$$\frac{S_f}{S_p} = \left(\frac{B_f}{B_p}\left[\frac{B_p + 0.3}{B_f + 0.3}\right]\right)^2$$

$$\frac{S_f}{S_p} = \left(\frac{1.0}{0.6}\left[\frac{0.6+0.3}{1+0.3}\right]\right)^2 = \left(\frac{0.9}{0.6\times 1.3}\right)^2$$

$$S_f = \left(\frac{0.9}{0.6 \times 1.3}\right)^2 \times 0.015 = 0.01997\,\text{m}$$

$$\boxed{S_f = 19.97\ \text{mm}}$$

Note : The bearing plate is either circular or square, made of mild steel of not less than 25 mm in thickness and varying in size from 300 to 750 mm with chequered or grooved bottom.

1364. The failure of foundation of a building is due to

(a) Withdrawal of subsoil moisture
(b) Unequal settlement of soil
(c) Lateral escape of supporting material
(d) All of the above

RPSC AE (GWD) 2014

Ans : (d) The failure of foundation of a building is due to–

- Settlement of soil foundation which is called settlement failure.
- With drawal of subsoil moisture.
- Unequal settlement of soil.
- Lateral escape of supporting material.
- Sliding or sliping of foundation soil such failure is called shear failure.

1365. A black cotton soil is unsuitable for foundation because it

(a) Undergoes volumetric changes with change of atmospheric conditions
(b) Swells excessively when wet
(c) Shrinks excessively when dry
(d) All of the above

RPSC ACF 2011

Ans : (d) Black cotton soil have been formed from basalt or trap and contain clay mineral montmorillonite, which is responsible for the excessive swelling and shrinkage characteristics of the soil.

Note– Under reamed piles should be used in foundations in black cotton soils.

1366. The lowest part of a structure which transmits the load to the soil is known as

(a) Super structure (b) Plinth
(c) Foundation (d) Basement

RPSC AE (GWD) 2014

Ans : (c) The lowest part of a structure which transmits the load of the structure together with live loads, seismic and wind pressure to the soil or ground surface on which the structure rests, ensuring its safe bearing capacity its called foundation.

{Shallow foundation (D ≤ B)
{deep foundation (D > B)

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1367. The ultimate bearing capacity of a soil is 300 kN/m^2, The depth of foundation is 1 m and unit weight of soil is 20kN/m$^{2.}$ Choosing a factor of safety of 2.5, the net safe bearing capacity is

(a) 100 kN/m^2 (b) 112 kN/m^2
(c) 80 kN/m^2 (d) 100.5 kN/m^2

RPSC AE (GWD) 2014

Ans : (b) Ultimate bearing capacity of a soil and net safe bearing capacity is following relation-

$$q_{nu} = q_u - \gamma D_f$$

Given that, $q_u = 300\ N/m^2$ $\gamma = 20\ KN/m^3$ $D_f = 1m$

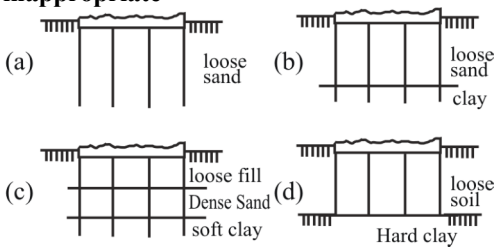
$$q_{nu} = 300 - 20 \times 1$$
$$= 280\ N/m^2$$

Net safe bearing capacity $q_{ns} = \dfrac{q_{nu}}{F.O.S}$

$$q_{ns} = \frac{280}{2.5} \qquad (\text{Given that FOS} = 2.5)$$

$$\boxed{q_{ns} = 112\,kN/m^2}$$

1368. In which one of the following conditions, is the pile system as shown in figure highly inappropriate



RPSC AE (GWD) 2014

Ans : (c)

- In loose and medium dense sand, the driving of pile results into compaction of about 3d space around pile.
- For dense sand, the driving will cause loosing of soil and there by group efficiency comes out to be less than one.

1369. Well foundation are commonly used as foundation for the following structure

(a) Water tanks
(b) Bridge
(c) Buildings
(d) Reciprocating machines

RPSC AE (GWD) 2014

Ans : (b) Well foundation are known as pier foundation or deep foundation. Pier foundation are some what similar to pile foundations but are typically larger in area than piles. An opening is drilled to the desired depth and concrete is poured to make a pier foundation.
Usually, pier foundation are used for bridges.

1370. The maximum differential settlement in isolated footings on sandy soil shall not exceed–

(a) 40 mm (b) 100 mm
(c) 65 mm (d) 25 mm

(RPSC ACF 2011)

Ans: (d) The maximum differential settlement in isolated footing on sandy soil shall not to be exceed is 25 mm.

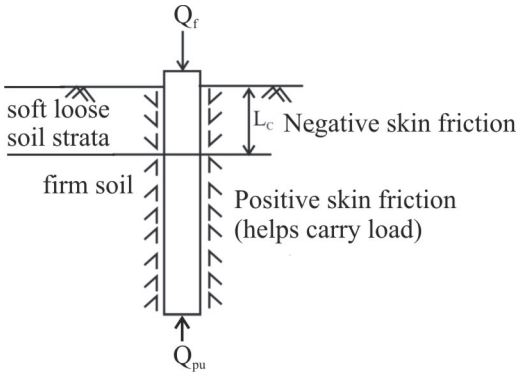
1371. Negative skin friction in a soil is considered when the pile is constructed through a

(a) Fill material
(b) Dense coarse sand
(c) Over consolidated stiff clay
(d) Dense flux sand

RPSC AE (GWD) 2014
WBPSC AE 2001

Ans : (a) Negative skin friction or downward drag is a phenomenon which occurs when soil layer surrounding to the pile settles more than the settlement of pile, this condition occurs when soil surrounding to the pile is loose/soft.

- A fill material placed over the clay stratum.
- Lowering of the ground water table.
- Reconsolidation occurring due to disturbance caused by pile driving in sensitive clay stratum, etc.



1372. Two footings, one circular and the other square, are found on the surface of a purely cohesionless soil. Diameter of the circular footing and width of square footing is the same. Ratio of ultimate bearing capacity of circular to square footing is

Gujarat PSC AE (N.W.R.) 2020
GPSC AE Class (1&2) Paper-2, 2017
Karnataka PSC AE 2017 Paper-II
ASSAM PSC CCE (PRE) 2015
JPSC Combined AE Paper V 2013
UPPSC State Eng. A.E. 2008

(a) 1.00 (b) 1.20
(c) 0.75 (d) 1.33

Ans : (c) for square footing

$$q_{nu} = 1.3\ CN_c + q\ (N_q - 1) + 0.4\ B\gamma N_\gamma$$

for circular footing

$$q_{nu} = 1.3\ CN_c + q\ (N_q - 1) + 0.3\ B\gamma N_\gamma$$

for cohesionless soil $C = 0$
for foundation on surface,
surcharge, $q = 0$

$$\frac{q_{nu(circular)}}{q_{nu(square)}} = \frac{0.3}{0.4} = \frac{3}{4} = 0.75$$

1373. The critical damping for a single degree of freedom is given by the expression :

(a) $2\sqrt{km}$ (b) $2\pi\sqrt{km}$
(c) $2\pi\sqrt{\dfrac{k}{m}}$ (d) $\pi\sqrt{\dfrac{k}{m}}$

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k = stiffness coefficient
m = mass of machine and foundation

[UPPSC State Eng. A.E. 2007(II)]

Ans : (a) The critical damping for a single degree of freedom is given by the expression.

$$C_C = 2\sqrt{km} \quad \text{or} \quad C_C = 2m\omega_n \quad \text{or} \quad C_c = 2m\sqrt{\frac{k}{m}}$$

k = Stiffness coefficient
M = Mass of machine and foundation
ω_n = circular natural frequency.

1374. Bearing capacity increase with the
(a) increase in eccentricity of load
(b) increase in inclination of load
(c) increase in depth of water table below footing twice the width of footing
(d) decrease in unit weight of soil

(JPSC Combined AE Paper V 2013)

Ans : (c) With increase in depth, there is increase in bearing capacity for given type of soil.

1375. Ultimate bearing capacity (q_f) of a square footing 2.5m wide resting at 1.5m depth in a sandy soil having unit weight $\gamma = 18kN/m^3$, $N_q = 33$, $N_r = 48$, using factor of safety as 3, will be:
(a) 487 kN/m^2 (b) 585kN/m^2
(c) 741 kN/m^2 (d) 512 kN/m^2

UPPSC AE 12.04.2016 Paper–I

Ans : (b) Given, F.O.S = 3
Unit weight $\gamma = 18$ kN/m^3
width (B) =2.5 m, depth (D_f) = 1.5 m
for sandy soil C = 0

$q_f = 1.3\,CN_c + \gamma D_f N_q + 0.4\gamma B N_r$
$= 1.3\times0\times N_c + 18\times1.5\times33 + 0.4\times18\times2.5\times48$
$q_f = 1755$ kN/m^2

using factor of safety $(q_f) = \frac{1755}{3}$ kN/m^2

$$\boxed{q_f = 585 \text{ kN/m}^2}$$

1376. A 30cm diameter friction pile is embedded 10m into a homogeneous consolidated deposit. Unit cohesion developed between clay and pile shaft is 4t/m^2 and adhesion factor is 0.7. The safe load for safety 2.5 will be:
(a) 21.50 t (b) 11.57 t
(c) 10.55 t (d) 6.85 t

UPPSC AE 12.04.2016 Paper–I

Ans : (c) The friction load capacity
$q_f = \alpha \times C_u \times PL$
Given $\alpha = 0.7$, $C_u = 4$ t/m^2, $P = \pi \times 0.3$ m, $L = 10$ m
$q_f = 0.7 \times 4 \times \pi \times 0.3 \times 10 = 26.4$

safe load $= \frac{26.4}{2.5} = 10.55$ t

1377. Using Engineering News formula, the allowable load (Q_{up}) of a wooden pile hammered with a drop hammer (W = 30kN), having free fall (H) 1.5 m, penetration in last blow (S) 5mm, using empirical constant (C) as 2.5 and factor of safety as 6, will be:
(a) 185 kN (b) 250 kN
(c) 212 kN (d) 231 kN

UPPSC AE 12.04.2016 Paper–I

Ans : (b) Engineering News formula–

$$Q_{up} = \frac{W_h.H}{F(S+C)}$$

Given,
$W_h = 30$ kN
S = 5 mm = 0.5 cm
C = 2.5
H = 1.5 m = 150 cm
F = 6

$$Q_{up} = \frac{30\times150}{6(0.5+2.5)}$$

$$Q_{up} = \frac{4500}{18}$$

$Q_{up} = 250$ kN

1378. For a proposed building, raft foundation, isolated footings and combined footings are being considered. These foundations are to be listed in the decreasing order of preference in terms of performance. Which one of the following is the correct order of listing?
(a) Raft foundation – Combined footings – Isolated footings
(b) Isolated footings – Raft foundation – Combined footings
(c) Combined footings – Raft foundation – Isolated footings
(d) Combined footings – Isolated footings – Raft foundation

MPPSC AE 2016

Ans : (a)
1. Raft foundation is a large slab supporting a number of columns and wall under the entire structure or a large part of structure.
2. Combined footing is used when two columns are so close to each other that their individual footing would overlap. Also when property line is so close to one column that a spread footing would be eccentrically loaded when kept entirely within. So, by combining it with that of an interior column the load is evenly distributed.
3. Isolated footing is provided to support an individual column.

1379. Consider the following statements regarding negative skin friction in piles :
1. It is developed when the pile is driven through a recently deposited clay layer.
2. It is developed when the pile is driven through a layer of dense sand.
3. It is developed due to a sudden drawdown of the water table.
Which of these statements is/are correct?
(a) 1 alone (b) 2 alone
(c) 2 and 3 (d) 1 and 3

JPSC Combined A.E. Paper V 2013
MPPSC Civil Eng. A.E. 2016
ESE 1995

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Ans : (d) Following condition may cause negative skin friction–

→ It is developed due to a sudden "drawdown or down ward drag" lowering of the ground water table.

→ Negative skin friction increase gradually as the consolidation of the clay layer proceeds since the effective. overburden pressure increases due to dissipation of excess pore pressure.

1380. A Square pipe of section 30 cm × 30 cm and length 10 m penetrates a deposit of clay having C = 50 kN/m^2 and the adhesion factor = 0.8. The load carried by the pile shaft only is

(a) 1920 kN (b) 750 kN
(c) 600 kN (d) 480 kN

MPPSC AE 2016

Ans : (d) Square pile = 30 cm × 30 cm
length of pile (L) = 10 m
Adhesion factor = (α) = 0.80
deposit of clayey soil (C) = 50 kN/m^2
Load carried by skin friction

$$P = F_S.A_S$$
$$= \alpha.C \text{ (Perimeter} \times \text{length)}$$
$$= 0.8 \times 50 \times 4 \times 0.3 \times 10$$
$$\boxed{Q_P = 480\,kN}$$

1381. Consider the following statements associated with local shear failure of soils :

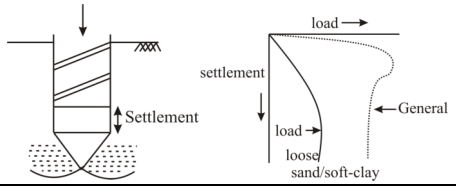
1. Failure is sudden with well–defined ultimate load.
2. This failure occurs in highly compressible soils.
3. Failure is preceded by large settlement. Which of these statements are correct?

(a) 1, 2 and 3 (b) 1 and 2
(c) 2 and 3 (d) 1 and 3

MPPSC AE 2016

Ans : (c) Local shear failure–

- It occurs in highly compressible soils.
- Failure is preceded by large settlement.
- It occurs in loose sand soft clay in shallow foundation.
- It load settlement curve is plotted progressive failure will occur.



1382. In a plate load test, plate width is 60 cm. Footing width is 2.5m. Plate settles by 6 mm under 25 kPa. Find settlement of footing (S_F) under same intensity. Soil is sandy :

(a) S_F = 15.25 mm (b) S_F = 10.76 mm
(c) S_F = 12.76 mm (d) S_F = 09.51 mm

[UPPSC State Eng. A.E. 2011 Ist Paper]

Ans : (b) Plate (P), Footing (f)

$B_P = 60cm$, $B_f = 2.5m$
$S_P = 6mm$, $S_f = ?$

$$\frac{S_f}{S_P} = \left[\frac{B_f(B_P+0.3)}{B_P(B_f+0.3)}\right]^2 \text{ for sandy soil}$$

$$S_f = S_P\left[\frac{B_f(B_P+0.3)}{B_P(B_f+0.3)}\right]^2$$

$$S_f = 6\left[\frac{2.5(0.6+0.3)}{0.6(2.5+0.3)}\right]^2$$

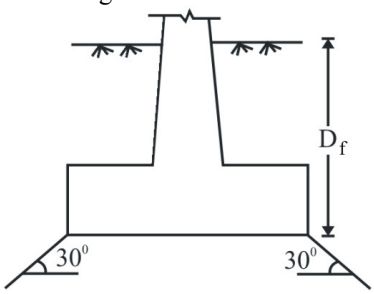
$$S_f = 6\times1.793 = 10.76\,mm$$

1383. In the settlement analysis of pile groups, the load is assumed to be transferred at an angle of:

(a) 20^0 (b) 30^0
(c) 40^0 (d) 45^0

[UPPSC State Eng. A.E. 2011 Ist Paper]
[UPPSC State Eng. A.E. 2004(I)]

Ans : (b) In settlement analysis load is assumed to be distributed at 30^0 angle.

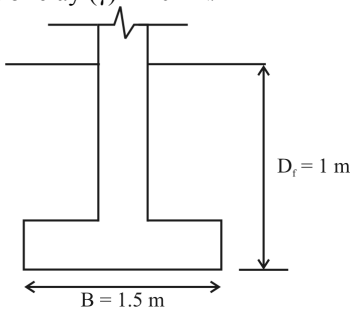


1384. The gross bearing capacity of a footing is 450 kN/m^2. If the footing is 1.5 m wide at a depth of 1 m in clayey soil with unit weight of 20 kN/m^3, then the net bearing capacity (kN/m^2) will be :

(a) 400 (b) 430
(c) 435 (d) 440

TNPSC AE 2018
ESE 2014
[UPPSC State Eng. A.E. 2011 Ist paper]
[UPPSC State Eng. A.E. 2003]

Ans : (b) Given:
Gross bearing capacity (q_s) = 450 kN/m^2
Width of footing, (B) = 1.5 m
Unit weight of clay (γ) = 20 kN/m^3



Net bearing capacity (q_n) = gross bearing capacity – over burden pressure

$$q_n = q_{gross} - \gamma D_f$$
$$= 450 - (1\times 20)$$
$$= 430\ kN/m^2$$

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1385. According to Rankine's formulae, the minimum depth (h) of foundation is given by :

(a) $h = \frac{p}{w}\left(\frac{1-\sin\phi}{1+\sin\phi}\right)^2$ (b) $h = \frac{w}{p}\left(\frac{1-\sin\phi}{1+\sin\phi}\right)^2$

(c) $h = \frac{p}{w}\left(\frac{1-\sin\phi}{\tan\phi}\right)^2$ (d) $h = \frac{w}{p}\left(\frac{1-\sin\phi}{1+\tan\phi}\right)^2$

[UPPSC State Eng. A.E. 2011 Ist Paper]

Ans : (a) According to Rankine's formula.
Ultimate bearing capacity,

$$q_n = k_p^2\ \gamma D_f$$

$$\therefore \quad D_f \min = \frac{q_n}{k_p^2\gamma}$$

where, k_p = coefficient of passive earth pressure

$$k_p = \left(\frac{1+\sin\phi}{1-\sin\phi}\right)$$

γ = unit weight of soil

$$D_p \min = \frac{p}{w}\left(\frac{1-\sin\phi}{1+\sin\phi}\right)^2$$

$$D_p \min = \frac{q_n}{\gamma}\left(\frac{1-\sin\phi}{1+\sin\phi}\right)^2$$

1386. The minimum centre to centre spacing of friction piles of diameter D as per Bureau of Indian Standards is :

(a) 2.5 D (b) 3 D
(c) 3.5 D (d) 4.0 D

[UPPSC State Eng. A.E. 2008]

Ans : (b) As per Indian standard IS : 2911 the minimum centre to centre spacing for friction pile is,
(i) for point bearing pile = 2.5 D
(ii) for friction pile = 3D
(iii) In loose type sand =2D

1387. In the case of a pile foundation negative skin friction may occur at a load, which is :

(a) equal to the design load
(b) higher than the design load
(c) lower than the design load
(d) of any magnitude of load

GPSC AE (CLASS 1 & 2) 2019
UPPSC State Eng. A.E. 2008
ESE 2001

Ans : (c) Skin friction reduces the load carrying capacity of the pile. Hence pile failure can take place at load lower than design load.

1388. In under–reamed pile foundation, the ratio of the shaft diameter to the bulb diameter is :

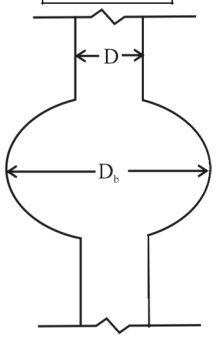
(a) $\frac{1}{1.25}$ (b) $\frac{1}{2.25}$

(c) $\frac{1}{2.50}$ (d) $\frac{1}{3.00}$

Haryana PSC AE 2017
PTCUL AE 2016
[UPPSC State Eng. A.E. 2008]
ESE 2001

Ans : (c) In under reamed pile foundation, the ratio of shaft diameter to the bulb diameter is

$$\frac{D}{D_b} = \frac{1}{2\text{ to }3}$$



Generally value, 2.5 is used.

1389. In the Engineering News record Formula for determining the safe carrying capacity of a pile, the factor of safety used is :

(a) 2.0 (b) 2.5
(c) 3.0 (d) 6.0

GPSC AE Dec 2018
UPPSC State Eng. A.E. 2008
ESE 2001

Ans : (d) In the engineering new record formula for determining the safe carrying capacity of a pile, the factor of safety used is 6.

1390. One square footing and other circular footing are constructed in pure clay. The size of the square footing is the same as the diameter of the circular footing. The ratio of the net ultimate bearing capacity of square and circular footing is :

(a) 1.0 (b) 1.30
(c) 2.0 (d) 4.0

[UPPSC State Eng. A.E. 2007(II)]

Ans : (a) The ratio of the net ultimate bearing capacity of square and circular footing is 1.

$$q_{unet} = \frac{\left(\gamma D N_q + 0.3\gamma B N_\gamma\right)_{circular}}{\left(\gamma D N_q + 0.4\gamma B N_\gamma\right)_{square}}$$

$$q_{unet} \simeq 1.0$$

1391. For finding the factor of safety in case of pile foundation, negative skin friction is :

(a) substracted from the working load
(b) multiply to the working load
(c) added to the working load
(d) not considered

[UPPSC State Eng. A.E. 2007(II)]

Ans : (c) FOS for pile footing when negative skin friction occurs.

$$FOS = \frac{Q_{uP}}{\text{working load} + Q_{nf}}$$

Q_{up} = Ultimate bearing capacity of pile
Q_{nf} = Negative skin friction resistance.

1392. Cyclic pile load tests are carried out to determine :

(a) cohension of the soil

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(b) friction angle of the soil
(c) negative skin friction
(d) skin friction and point bearing component of pile

[UPPSC State Eng. A.E. 2007(II)]

Ans : (d) Cyclic pile load tests are carried out to determine skin friction and point bearing component of pile.

1393. Franki pile is a :
(a) cast in–situ concrete cased pile
(b) cast in–situ concrete uncased pile
(c) pre cast concrete pile
(d) steel pile

[UPPSC State Eng. A.E. 2007(II)]

Ans : (b) Franki pile is a cast in-situ concrete uncased pile.
It is suitable for granular type of soils and for soils bearing stratum of limited, thickness is located at reasonable depth. This type of pile has mushroom shape bottom which acts like spread footing.

1394. Angular distortion limits for building in which cracking is not permissible is given by :
(a) $\frac{1}{1000}$ (b) $\frac{1}{2000}$
(c) $\frac{1}{100}$ (d) $\frac{1}{500}$

[UPPSC State Eng. A.E. 2007(II)]

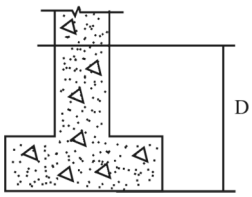
Ans : (a) Angular distortion limits for building in which cracking is not permissible is given by $\frac{1}{1000}$.

1395. The factor of safety of foundation with respect to shear failure is given by :
(a) $F = \frac{q_f - \gamma D}{q - \gamma D}$ (b) $F = \frac{q_f - \gamma D}{q - \gamma D}$
(c) $F = \frac{q_f}{q}$ (d) $F = \frac{q_f - \gamma D}{q}$

q_f = ultimate bearing capacity
q = maximum load at foundation level

[UPPSC State Eng. A.E. 2007(II)]

Ans : (d)



ultimate bearing capacity = q_u
Net ultimate bearing capacity
$q_{nu} = q_u -$ over burden
$q_{nu} = q_u - \gamma D$
Net safe bearing capacity

$$q_{ns} = \frac{q_{nu}}{FOS}$$

$q_{ns} = q$ and $q_u = q_f$

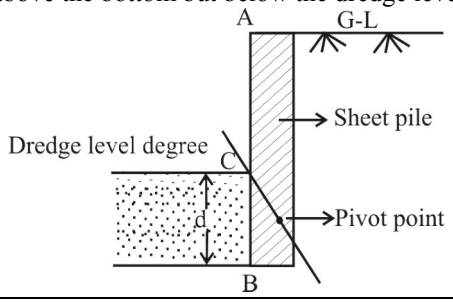
$$q = \frac{q_f - \gamma D}{FOS}$$

$$F = \frac{q_f - \gamma D}{q}$$

1396. In a cantilever sheet pile wall, the rotation is about :
(a) the bottom of the wall
(b) the dredge level
(c) above the bottom but below the dredge level
(d) none of the above

[UPPSC State Eng. A.E. 2007(I)]

Ans : (c) In a cantilever sheet pile wall, the rotation is about above the bottom but below the dredge level.



1397. An open caisson is one, which is during construction :
(a) open at the top and closed at the bottom
(b) open both at the top and at the bottom
(c) closed at the top and open at the bottom
(d) none of the above

[UPPSC State Eng. A.E. 2007(I)]

Ans : (b) The top and bottom of the caisson is open during construction, it may have any shape in plan i.e circular, rectangular etc. it has a cutting edge which is fabricated at the size and the first segment of shaft built on it. The soil inside the shaft in dredged by suitable means and other segment added to it.

1398. The efficiency of pile group depends on :
(a) soil type
(b) method of pile installation
(c) pile spacing
(d) all of the above

[UPPSC State Eng. A.E. 2007(I)]

Ans : (d) The efficiency of pile group (η) depends mainly on :
(i) spacing between piles
(ii) type of soil in which the piles are installed
(iii) manner of pile installation i.e. drived, bored or cast in situ.

1399. For a purely cohesive soil the bearing capacity at ground surface for a circular footing as per Terzaghi's analysis will be :
(a) 5.14 C (b) 6.2 C
(c) 7.2 C (d) None of the above

[UPPSC State Eng. A.E. 2007(I)]

Ans : (d) For pure clays

$N_c = 5.7$

$N_q = 1$

$N_\gamma = 0$

for circular footing on pure clay

$q_u = 1.3\ CN_c + \gamma.D_f$

at ground surface, $D_f = 0$

$q_u = 1.3\ C \times 5.7$

$q_u = 7.41C$

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1400. Which one of the following is the correct Terzaghi's general bearing capacity equation for soil :

(a) $q_u = CN_c + \gamma DN_q + \frac{1}{2}\gamma BN_\gamma$

(b) $q_u = CN_c - \gamma DN_q + \frac{1}{2}\gamma BN_\gamma$

(c) $q_u = CN_c + \frac{1}{2}\gamma DN_q + \gamma BN_\gamma$

(d) None of the above

[UPPSC State Eng. A.E. 2007(I)]

Ans : (a) Terzaghi bearing capacity equation

$$q_u = CN_c + \gamma DN_q + \frac{1}{2}\gamma BN_\gamma$$

Where N_c, N_q and N_γ are bearing capacity factors for pure clay ($\phi = 0$)

$N_c = 5.7$

$N_q = 1$

$N_\gamma = 0$

1401. The bearing capacity factors N_c, N_q, and N_γ depend on :

(a) cohesion of soil

(b) angle of internal friction of soil

(c) unit weight of soil

(d) density of soil

GUJRAT PSC AE - II 2017

UPPSC State Eng. A.E. 2003, 2004(I)

Ans : (b) The bearing capacity forces N_c, N_q and N_γ depends upon angle of internal friction of soil and may be given as

$$N_c = \cot\phi\left(N_q - 1\right)$$

$$N_q = N_\phi . e^{\pi \tan\phi}$$

$$N_\phi = \text{ influence value } = \tan^2\left(45 + \frac{\phi}{2}\right) = k_P$$

$$N_\gamma = 1.8\tan\phi\left(N_q - 1\right)$$

1402. According to Skempton's formula for a surface footing of square shape, the net ultimate bearing capacity on a purely cohesive soil of cohesion 'C' is

(a) 1.4 C (b) 6.0 C

(c) 7.4 C (d) 9.0 C

BPSC AE 2012 Paper–V

Ans : (b) Skempton's method :

→ This theory is applicable for pure cohesive only clay.

→ Net ultimate bearing capacity (q_{nu}) : For square footing

$$\boxed{q_{nu} = CN_c}$$

Where N_c = Skempton bearing capacity factor which depends upon $\left(\frac{D_f}{B}\right)$ ratio.

$$N_C = 6\left[1 + 0.2\frac{D_f}{B}\right]$$

For surface footing [$D_f = 0$]

$N_C = 6\ [1+0] = 6$

$[q_{un} = 6C]$

1403. The maximum size of plate for plate load test is

(a) 30 cm square (b) 45 cm square

(c) 60 cm square (d) 75 cm square

BPSC AE 2012 Paper–V

Ans : (d) In plate load test the bearing plate is squared of minimum size 30 cm^2, and the maximum size is 75 cm^2. The thickness of the steel plate should not be less than 25 mm.

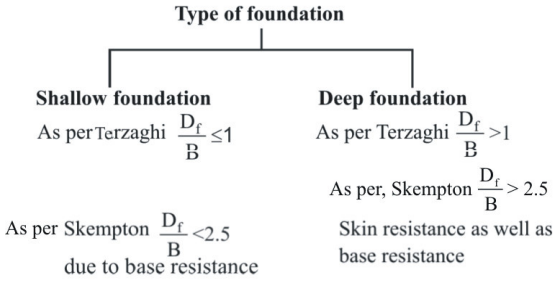
1404. A foundation may be called a shallow one if its depth to width ratio is

(a) greater than 2 (b) less than 2

(c) greater than 4 (d) less than 4

(BPSC AE Paper V 2001)

Ans : (b)



1405. The floating foundation is one in which

(a) The foundation floats in water

(b) The super structure rests on a number of the standing piles

(c) The foundation consists of cellular raft

(d) The net soil pressure beneath the foundation structure is zero

(BPSC AE Paper V 2006)

Ans : (d) The floating foundation is defind as the net soil pressure beneath the foundation structure is zero.

1406. In design of foundation the soil is loaded with a factor of safety on bearing capacity to ensure

(a) The soil behaviour to be elastic at working load

(b) soil parameters are not precisely known

(c) The soil is elasto-plastic

(d) None of these

(BPSC AE Paper V 2006)

Ans : (a) The design of foundation the soil is loaded with a factor of safety on bearing capacity to ensure the soil behaviour to be elastic working load.

1407. Consideration of both shallow and deep foundation depends

(a) Only on depth to width ratio

(b) Only on the predominant structural action

(c) Both of the above

(d) None of the above

(BPSC AE Paper V 2006)

Ans : (a) shallow and deep foundation depends upon depth to width ratio $\left(\frac{D_f}{B}\right)$.

$$\frac{D_f}{B} \le 1 \rightarrow \textit{shallow foundation}$$

$$\frac{D_f}{B} > 1 \rightarrow \textit{deep foundation}$$

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1408. If pile foundation is adopted in an expansive soil deposit, then the most suitable type of pile is

(a) Straight shaft driver pile
(b) Uniform diameter bored pile
(c) Multiple bulb under-reamed pile
(d) Cast-iron H-pile

(BPSC AE Paper V 2006)

Ans : **(c)** A pile foundation is adopted in an expansive soil deposit, the most suitable type of pile is multiple bulb under-reamed pile.

1409. A basement raft of 12 m square is placed at a depth of 3 m below ground level in a deposit of silty clay having C-4t/m^2, θ=0, γ=2t/m^3 and water table at ground level with a factor of safety 3, the approximate net bearing pressure is

(a) 14 t/m^2 (b) 8t/m^2
(c) 11 t/m^2 (d) 10 t/m^2

(BPSC AE Paper V 2006)

Ans : **(c)** Given,

$B = 12$ m, $D_f = 3$ m, $C = 4t/m^2$, $\theta = 0^o$

$\gamma = 2t/m^3$

F.O.S. = 3

safe bearing capacity $q_{ns} = \frac{q_{nu}}{F.O.S}$

Bearing capacity of shallow square footing

$$q_{nu} = 1.3CN_C + qN_q + 0.4\gamma BN_\gamma$$

$q = \gamma D_f$, $N_q = 0$

Skempton method : {for square & circular footing}

$$N_C = 6\left[1 + 0.2 \times \frac{D_f}{B}\right]$$

$$= 6\left[1 + 0.2 \times \frac{3}{12}\right] = 6.3$$

$q_{nu} = 1.3 \times 4 \times 6.3 + 2 \times 3 \times 0$

$q_{nu} = 32.76$ t/m^2

$$q_{ns} = \frac{q_{nu}}{FOS} = \frac{32.76}{3} = 10.92\,t/m^2$$

$\simeq 11t/m^2$

1410. Negative skin friction on piles may occur when:

(a) The soil is untrained
(b) The residual strength of soil is high
(c) There is constant seepage through the soil adjacent to pile
(d) The fill has consolidated around the pile after driving

(BPSC AE Paper V 2006)

Ans : **(d)** Negative skin friction increase gradually as the consolidation of the clay layer proceeds since the effective overburden pressure gradually increases due to dissipation of excess pore pressure.

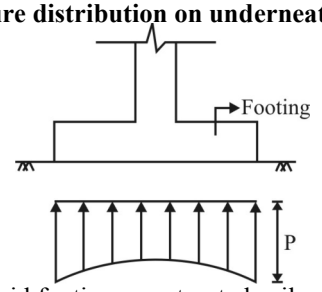
1411. In using a Newmark's chart, the loaded area is drawn to scales equal to

(a) The depth scale shown in the chart
(b) the width scale shown in the chart
(c) the area scale shown in the chart
(d) any convenient scale depending on intensity of load

JPSC Combined AE Paper V 2013

Ans : **(a)** In using a Newmark's chart, the loaded area is drawn to scales equal to depth scale shown in the chart.

1412. The figure given below represents the contact pressure distribution on underneath as:

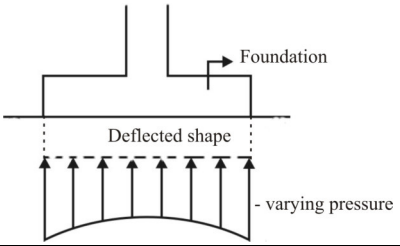


(a) rigid footing on saturated soil
(b) rigid footing on sand
(c) flexible footing on saturated clay
(d) flexible footing on sand

(UKPSC AE 2012 Paper–I)

Ans : (a)

- Rigid footing on the surface of clay soil contact pressure is maximum at the edges and minimum at the centre.
- rigid footing on saturated soil.



1413. A load of 4000 kN is uniformly distributed over an area of 3m× 3m. Average vertical stress at a depth of 2m below will be (assuming 2 vertical : 1 horizontal distribution)–

(a) 250 kN/m^2 (b) 160 kN/m^2
(c) 200 kN/m^2 (d) 220 kN/m^2

UKPSC AE 2013 Paper–I

Ans : (b) Average vertical stress at a depth of 2m below will be 160 kN/m^2.

1414. Negative skin friction in a soil is considered when the pile is constructed through a

(a) fill material
(b) dense coarse sand
(c) over consolidated stiff clay
(d) dense fine sand

GPSC Poly. Tech. Lect. 2016
UKPSC AE 2012 Paper–I

Ans : (a) In case of filled up soil material, soil settles downward relative to pile thus imparts drag in downward direction which is known as negative skin friction. It should be considered in design.

1415. The two criteria for determining of allowable bearing capacity of foundation are:

(a) Shear failure and settlement
(b) Bond failure and shear failure
(c) Tensile failure and settlement
(d) Tensile failure and compression failure

CGPSC AE Shift - 2 2014
UK Combined AE Paper I 2010

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Ans. (a) : Bearing capacity :
* The load carrying capacity of foundation soil or rock which enables it to bear and transmit loads from a structure.
* The load supporting power of a soil without shear failure is known as bearing capacity.

1416. According to IS code, allowable settlement of raft foundation on sand is
(a) 25 mm to 40 mm (b) 40 mm to 65 mm
(c) 75 mm to 100 mm (d) 100 mm to120 mm

(UKPSC AE 2012 Paper–I)
[UPPSC State Eng. A.E. 2007(II)]

Ans : (b) As per IS recommendation–

For sand		
	Maximum settlement	**max. difference settlement**
Isolated foundation	40 mm	25 mm
Raft foundation	40 to 65 mm	25 mm
For clay		
	maximum settlement	**max. difference settlement**
Isolated foundation	65 mm	40 mm
Raft foundation	65 to 100 mm	40 mm

1417. According to Terzaghi, the net ultimate bearing capacity of clay is given by where : N_q N_γ & N_c are bearing capacity factors, C-cohesion
(a) $C.N_q$ (b) $C.N_\gamma$
(c) $C.N_c$ (d) $1.2\ C.N_c$

UKPSC AE (Paper–I) 2012, 2007

Ans : (c) According to Terzaghi, the net bearing capacity of clay (q_{nu}) = CN_c

$\boxed{q_{nu} = CN_c}$

1418. The maximum unit pressure that a soil can withstand without rupture in shear or without excessive settlement of the structure is called :
(a) ultimate bearing capacity
(b) safe bearing capacity
(c) allowable bearing capacity
(d) none of the above

UKPSC AE (Paper-I) 2007

Ans : (a) It is that maximum gross pressure at the base of foundation which can be applied without shear failure.

or

It is the minimum at which soil just fail in shear.

1419. The gross bearing capacity of a 1.5 m wide strip footing at a depth of 1.0 m is 440 kN/m^2 . If γ = 20 kN/m^2. what is the net bearing capacity ?
(a) 410 kN/m^2 (b) 420 kN/m^2
(c) 460 kN/m^2 (d) none of the above

UKPSC AE (Paper-I) 2007

Ans : (b) Depth of foundation = 1 m

$\gamma = 20\ kN/m^3$

$q = \gamma D_f = 20 \times 1 = 20\ kN/m^2$

Net bearing capacity $q_n = q_u - q$

$q_n = 440 - 20 = 420\ kN/m^2$

1420. Which of the following is not an assumption in Terzaghi's formulation of bearing capacity equation?
(a) Plain strain condition exists
(b) Base of the footing is smooth
(c) The width of the base of the footing is less than or equal to the founding depth
(d) Shear strength is governed by Mohr-Coulomb failure criteria

(UP RVNL AE 2016)

Ans.: (b) Assumption of Terzaghi's Theory–
(1) foundation is shallow $\left(\frac{D_f}{B} \le 1\right)$
(2) Base of foundation is rough.
(3) footing is continuous/strip (L >>> B)
(4) It makes analysis 2 dimensional along the width & depth only.
(5) At the time of failure soil reaches into plastic equilibrium.
(6) failure is general shear failure.

1421. According to Terzaghi's equation, the bearing capacity of strip footing resting on cohesive soil (C = 10 kN/m^2) per unit depth and unit width (N_C = 5.7) will be–
(a) 47 kN/m^2 (b) 67 kN/m^2
(c) 57 kN/m^2 (d) 77 kN/m^2

TSGENCO_2015
UKPSC AE 2013 Paper–I
ESE 1999

Ans : (c) Given, C = 10 kN/m^2
For C-soil the value of $\boxed{N_c = 5.7}$
for strip footing

$$q_u = CN_C + \gamma D N_q + \frac{1}{2}\gamma B.N_\gamma$$
$$= 10 \times 5.7 + 0 \times 1 \times 1 + \frac{1}{2} \times 0 \times 1 \times 0$$
$$= 57\ kN/m^2$$

1422. The maximum differential settlement in isolated footings on sandy soil shall not exceed–
(a) 40 mm (b) 100 mm
(c) 65 mm (d) 25 mm

UKPSC AE 2013 Paper–I

Ans : (d) As per IS recommendation–

For sand : –		
	Maximum settlement	**max. difference settlement**
Isolated foundation	40 mm	25 mm
Raft foundation	40 to 65 mm	25 mm
For clay : –		
	maximum settlement	**max. difference settlement**
Isolated foundation	65 mm	40 mm
Raft foundation	65 to 100 mm	40 mm

1423. A cantilever sheet pile derives its stability from–
(a) Self–weight of sheet pile
(b) Lateral resistance of soil
(c) Anchor road

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(d) None of the above

BPSC AE 2019 Paper (V) Section-I

UKPSC AE 2013 Paper–I

Ans : (b) A cantilever sheet pile retains backfill at higher level on one side. The stability is entirely from the lateral passive resistance of the soil into which the sheet pile is driven, like that of a free cantilever sheet pile.

1424. H–Piles is a type of one of the following piles–

(a) Concrete Piles (b) Timber Piles
(c) Steel Piles (d) Composite Piles

UKPSC AE 2013 Paper–I

Ans : (c) Steel piles : There are usually H – piles (rolled H shape) piles, or sheet piles (rolled section of regular shapes). They may carry loads upto 100 tonnes or more.
Note- concrete filled steel piles and steel H-pile are used as long piles with high bearing capacity.

1425. Raft foundation is useful where–

(a) Soil having low bearing capacity
(b) Heavy loads are transferred through columns
(c) Super structure is sensitive to differential settlement
(d) All the above

UKPSC AE 2013 Paper–I

Ans : (d) Raft foundation is preferred where–

- The soil has a low bearing capacity.
- There is a possibility of differential settlement in case individual footing is used.
- Stress on soil needs to be reduced.
- Load of the structure has to be distributed over a large area.
- When soil strata are unpredictable and contain pockets of compressible soil.
- Heavy loads are transferred through columns.

1426. The load carrying capacity of a friction pile in a clayey strata is 180 kN. If a group of 7 piles are designed with an efficiency factor of 0.75, then the total load carrying capacity (kN) of the group will be:

(a) 1260 (b) 945
(c) 1680 (d) 1200

(LMRC AE 2017 I–Shift)

(ESE 2005)

Ans : (b) W (load) = 180 kN
Efficiency factor = 0.75
No. of pile = 7
Then

$$\text{Total load} = 180 \times 0.75 = 135$$
$$= 135 \text{ (one pile) (kN)}$$

Design a group pile (Total load) = 135 × 7 = 945 kN

1427. When heavy structural loads from columns are required to be transferred to a soil of low bearing capacity, the following foundation is not provided :

(a) Isolated footing (b) Pile foundation
(c) Raft foundation (d) Grillage foundation

Uttarakhand Combined State AE (Paper II) 2007

Ans : (a) Isolated footing are used when bearing capacity of soil is good. If soil has low bearing capacity raft foundation can also be used.

1428. A pile which obtains most of its load carrying capacity at the base of pile is known as

(a) Friction piles (b) Footing piles
(c) Compaction piles (d) End bearing piles

(UKPSC AE Paper I 2007)

Ans : (d) End bearing piles behaves as columns. These piles transfers the load to the end in the soil. These are used in soil having low bearing capacity.

- friction pile transfer the load in the soil through the friction from there surface.
- compaction piles are used to compact loose soil so that there bearing capacity can be increased.

1429. A shallow foundation is defined as a foundation which:

(a) has low bearing capacity
(b) has a depth of embedment less than its width
(c) is resting on the ground surface
(d) causes less settlement

(UKPSC AE Paper I, 2007)

Ans : (b)

- A shallow foundation transmits structural loads to the soil strata at a relatively small depth.
- Shallow foundation is one which is laid at a depth D_f not exceeding the width B of the foundation.

1430. A vertical cut is to be made in saturated and submerged clay with C = 15 kN/m^2, γ_{sat} = 18.5 kN/m^3 and γ_w = 10 kN/m^3. What is the theoretical depth to which the clay can be excavated without side collapse?

(a) 3.2 m (b) 3.5 m
(c) 6.0 m (d) 7.0 m

MPPSC AE 2017

Ans. (d) : C = 15 kPa

$$\gamma_{sat} = 18.5 \text{kN/m}^3$$
$$\gamma_w = 10 \text{ kN/m}^3$$
$$\sigma = K_a \gamma' Z - 2C\sqrt{K_a}$$
$$\sigma = 0. at \ Z_c = \frac{2C}{\gamma'\sqrt{K_a}} = \frac{2 \times 15}{(18.5 - 10)}$$
$$= 3.53$$

Maximum height for self stability

$$Z_c = 2 \times 3.53 \text{ m}$$
$$= 7.06 \text{ m}$$

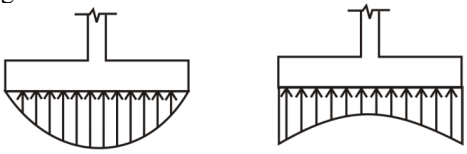
1431. *Statement* (A) : In Boussinesq's theory of stress computations, soil is considered to be unstressed before application of the load.
***Statement* (B) : The contact pressure distribution under a rigid footing in cohesionless soil, uniform throughout the width of the footing.**

(a) Both the statements (A) and (B) are correct
(b) Statement (A) is correct but (B) is wrong
(c) Statement (A) is wrong but (B) is correct
(d) Both the statements (A) and (B) are wrong

[MPSC 2015 Paper-II Mains]

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Ans. (b) : Contact pressure distribution under a rigid footing in cohesionless soil is as shown.



Rigid foundation sand | Rigid foundation of saturated soil

1432. Minimum depth of foundation calculated by Rankine's formula for the design of retaining wall is _____.
Where, q_0 = Safe bearing capacity of soil
γ = Density of soil
ϕ = Angle of repose

(a) $\frac{\gamma}{q_0}\left(\frac{1-\sin\phi}{1+\sin\phi}\right)^2$ (b) $\frac{q_0}{\gamma}\left(\frac{1-\sin\phi}{1+\sin\phi}\right)^2$

(c) $\frac{\gamma}{q_0}\left(\frac{1+\sin\phi}{1-\sin\phi}\right)^2$ (d) $\frac{q_0}{\gamma}\left(\frac{1+\sin\phi}{1-\sin\phi}\right)^2$

LMRC AM 2019
Maharashtra PSC Mains 2018 Paper-1
APTRASCO 2017
TSGENCO_2015
Maharashtra PSC AE (Mains) Paper–I 2012, 2015

Ans. (b) : Minimum depth of foundation as given by Rankine is

$$D_f = \frac{q_0}{\gamma}\left(\frac{1-\sin\phi}{1+\sin\phi}\right)^2$$

1433. What does not fit into the classification of pile based on function?
(a) Bearing pile (b) Friction pile
(c) Compaction pile (d) Steel pile

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (d) : Classification of pile
1. Based on material and composition
   (a) Timber piles
   (b) Steel piles
   (c) Concrete piles
   (d) Composite piles
2. Based on mode of transfer of load
   (a) End bearing piles
   (b) Friction piles
   (c) Combined end bearing and friction piles
3. Based on method of installation
   (a) Driven piles
   (b) Bored and cast in situ piles
   (c) Driven and cast in situ piles
   (d) Jack piles
   (e) Screw piles
4. Based on function or action
   (a) Load bearing piles
   (b) Tension or uplift piles
   (c) Compaction piles
   (d) Anchor piles
   (e) Fender piles
   (f) Sheet piles
   (g) Batter piles

1434. *Statement* A : Terzaghi's bearing capacity theory assumes strip foundation in the analysis.
***Statement* B : Terzaghi's theory does not consider development of shear resistance in the soil mass above founding level-**
(a) Both the statements A and B are true
(b) Statement A is true but B is false
(c) Statement A is false but B is true
(d) Both the statements A and B are false

MPSC (Mains) 2017 Paper–II

Ans. (a) : Both the statement A and B assumptions are correct w.r.t. Terzaghi's theory.

1435. *Statement* A : Plate load test is a short duration test and is not suitable in cohesive soils.
***Statement* B : Plate load test does not record the total settlement of the test plate in clayey soils.**
(a) Both the statements A and B are true but B is not the correct explanation of A
(b) Statement A is true but B is false
(c) Statement A is false but B is true
(d) Both the statements A and B are true and B is the correct explanation of A

MPSC (Mains) 2017 Paper–II

Ans. (d) : Plate load test is a field test, which is performed to determine the ultimate bearing capacity of soil and settlement of foundation under the loads for clayey and sandy soils.

1436. Following are the statements about the major differences between Terzaghi's analysis "T" and Meyerhoff's analysis ('M') of bearing capacity-
(i) 'T' is for homogeneous and isotropic soils but ('M') accounts for non-isotropy
(ii) In 'T', the failure surfaces form upto founding level but in 'M', they are extended upto ground level
(iii) In 'T', the angle of wedge formed beneath the foundation is assumed to be equal to the angle of internal friction of the soil but in 'M', it varies
(iv) In 'T' the load acting on the foundation is concentric and vertical but in 'M', it is assumed as eccentric

Ascertain the correctness of the above statements and write the correct code.
(a) Statement (i) is the only correct statement
(b) Statement (i) and (ii) are correct
(c) Statement (ii) and (iii) are correct
(d) Statement (i) and (iv) are correct

MPSC (Mains) 2017 Paper–II

Ans. (*) : The difference between Terzaghi's and Meyerhoff's approaches is that the latter consider the shearing resistance of the soil above the base of the foundation, while the former ignores it. Meyerhof allows the failure zone to extend up to the ground surface.

1437. Which is an example of cased cast-in-situ concrete pile?
(a) Raymond pile (b) Watson pile
(c) Reynold pile (d) Boston pile

Maharashtra PSC (Mains) 2017 Paper–I

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Ans. (a) : Different types of cast-in-situ piles are given as :
i) Simplex pile
ii) Franki pile
iii) Vibro pile
iv) Mac-Arthur pile
v) Raymond pile
vi) Vibro expanded pile

1438. Raft foundation are generally preferred to when the area required for individual footing, is more than-
(a) 25% to total area (b) 30% of total area
(c) 40% of total area (d) 50% of total area

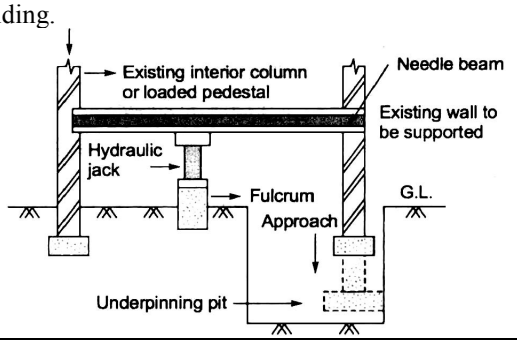
OPSC AE -2016 (II)
OPSC AEE 2015 PAPER–I

Ans. (d) : Raft foundation are generally preferred when area required for individual footing is more than 50% of total area.

1439. The arrangement of supporting an existing structure by providing supports underneath, is known as-
(a) Shoring (b) Underpinning
(c) Jacking (d) Piling

OPSC AE -2016 (II)

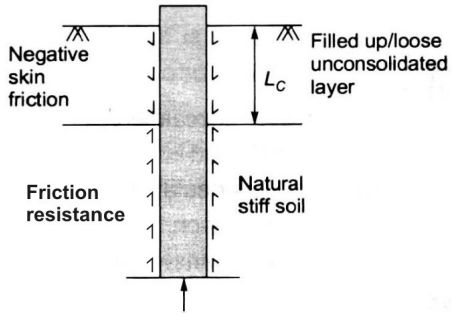
Ans. (b) : Underpinning – a solid foundation laid below ground level to support or strength on a building.



1440. Select the correct statement-
(a) Both negative skin friction and skin frictional resistance are caused by relative settlement of soil
(b) Both negative skin friction and skin frictional resistance are caused by relative settlement of pile
(c) Negative skin friction is caused by relative settlement of soil and skin friction is caused by relative settlement of pile
(d) Negative skin friction is caused by relative settlement of pile and skin frictional resistance is caused by relative settlement of soil

OPSC AE -2016 (II)
OPSC AEE 2015 PAPER–I

Ans. (c) :
- Negative skin friction or 'downward drag' phenomenon, which occurs when a portion of soil layer surrounding a pile settles more than the pile.
- When a compressive load (P) is applied at the top of pile, the pile will tend to move vertically downward relative to the soil. Due to this, shear or friction develops between soil and surface of shaft.



1441. The static cone penetration test and standard penetration test are performed on a soil at a certain depth. The value of static cone penetration test is 8 MPa and the N value is 20. The soil met with at that depth is
(a) sandy silt
(b) silty clay
(c) sand and gravel mixture
(d) medium dense sand

BPSC AE 2019 Paper (V) Section-I

Ans. (a) :

1442. Deflection of a sheet pile in a braced cut
(a) increases from top to bottom
(b) decreases from top to bottom
(c) increases from top and then decreases
(d) decreases from top and then increases

BPSC AE 2019 Paper (V) Section-I

Ans. (a) : Braced cuts are the excavations that are laterally supported. The vertical side of the excavations are supported by a sheeting and bracing.
Deflection of sheet pile in a braced cut increase with an increase in the depth of excavation.

1443. For reinforced concrete structures supported by isolated footing resting on sand, the maximum permissible settlement as per IS : 1904-1986 recommendation is

TNPSC AE 2018

(a) 75 mm (b) 60 mm
(c) 40 mm (d) 50 mm

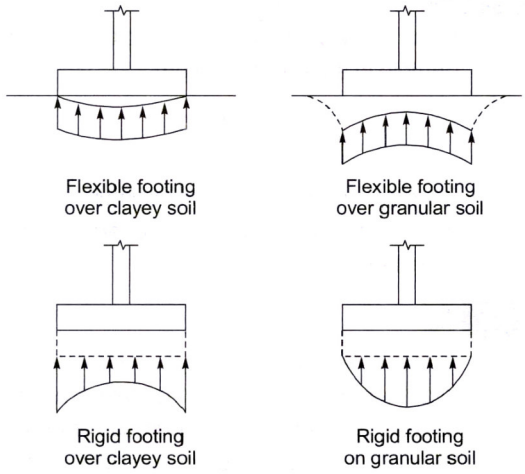
Ans. (c) : Permissible settlement in mm
Isolated footing on clay - 65 mm
Isolated footing on sand - 40 mm
Raft on clay – 65 to100 mm
Raft on sand – 40 to 65 mm

1444. Soil pressure distribution below a rigid footing on the surface of a cohesive soil is
(a) Minimum at the centre and maximum at edges
(b) Maximum at the centre and minimum at edges
(c) Uniform throughout
(d) Maximum at one end and minimum at other end

TNPSC AE 2018

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Ans. (a) : Soil pressure distribution



1445. A group of nine piles 12 m long and 250 mm in diameter is arranged in a square form in clay having undrained shear strength of 30 kN/m^2,. Neglecting bearing at top of the piles and taking adhesion factor as 0.9, ultimate capacity of all piles in individual action is :

(a) 2045 kN (b) 2290 kN
(c) 2545 kN (d) 2690 kN

[HPPSC AE 2016]

Ans. (b) : Ultimate capacity of all piles system
$= n \times (\propto C_u \times \text{perimeter} \times \text{length})$
$= 9 \times 0.9 \times 30 \times \pi \times 0.25 \times 12$
$= 2290$ kN

1446. Two footings, one circular and the other strip, are founded on the surface of a purely Cohesionless soil. Diameter of the circular footing and width of strip footing is same. Then, the ratio of ultimate bearing capacity of circular to strip footing is :

(a) 0.75 (b) 0.60
(c) 1.20 (d) 1.33

[HPPSC AE 2016]

Ans. (b) : For strip footing,
$(q_u) = C N_c + q N_q + 0.5 B\gamma N_\gamma$
Soil is cohesionless $\therefore$ C = 0
Footing on ground surface, $\therefore$ q = 0
$(q_u)_{strip} = 0.5 \gamma B N\gamma$
For circular footing, $(q_u) = 1.3 C N_c + q N_q + 0.3 \gamma B N_\gamma$
$(q_u)_{circular} = 0.3 \gamma B N_\gamma$

$$\frac{(q_u)_{Circular}}{(q_v)_{strip}} = \frac{0.3\gamma BN_\gamma}{0.5\gamma BN_\gamma} = 0.6$$

1447. The ultimate load cannot be clearly recognized in case of :

(a) Local shear failure
(b) General shear failure
(c) Punching shear failure
(d) None of these

Karnataka PSC AE 2015 Paper - II

Ans. (c) : In punching shear failure, no heaving of soil adjacent to footing occurs. Hence ultimate bearing capacity is not well defined for punching shear failure.

1448. In plate load test, the excavated width of test pit is ..... times the width of test plate.

(a) 5 (b) 4
(c) 3 (d) 2

Karnataka PSC AE 2015 Paper - II

Ans. (a) :
- In plate load test, a test pit of width of size equal to five times the width of plate and depth equal to the depth of the foundation (D_f) is to be excavated.
- Test plates are usually square or circular in shape.
- Size of plate varying from 300-750 mm is used and minimum thickness recommended is 25 mm.

1449. Suggest a most suitable foundation when the loads are heavy and clay soil is soft with a basement floor.

(a) Combined footing (b) Isolated footing
(c) Raft footing (d) Wall footing

Karnataka PSC AE 2015 Paper - II

Ans. (c) : When the allowable soil pressure is low, or the building loads are heavy, the use of spread footing would cover more than one-half of area and it proves more economical to use mat or raft foundation.

1450. The process of recording the resistance to the driving of bearing tubes as a quick site exploration is :

(a) Trenching (b) Jacking
(c) Sounding (d) Dredging

Karnataka PSC AE 2015 Paper - II

Ans. (c)
- A recording the resistance to the driving of bearing tubes as a quick site exploration is sounding.
- An undisturbed sample is that in which the natural structure and properties remain preserved.

1451. Which one of the following states of field compaction of sand deposit truly represents the corrected standard penetration test value N (Corrected) = 12 ?

(a) Loose (b) Medium dense
(c) Dense (d) Very dense

Karnataka PSC AE 2017 Paper - II
ESE 2005

Ans. (b) : We can corrected the standard penetration test number N within the relative density of sands.

Soil density description	Relative density(%)	Standard penetration test No. (N)
Very loose	0-15	0-4
Loose	16-35	5-10
Mediumm dense	36-65	11-30
Dense	66-85	51-50
Very dense	86-100	>50

$\therefore$ For standard penetration test No. (N) - 12; soil is described as medium dense.

1452. The ultimate bearing capacity of a soil is 310 kN/m^2, the depth of foundation is 0.80 m and γ is 20 kN/m^3. If the F.O.S. is 3, what will be the net safe bearing pressure ?

(a) 98.0 kN/m^2 (b) 98.86 kN/m^2
(c) 100.98 kN/m^2 (d) 100.86 kN/m^2

JPSC AE (Pre) 2019

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Ans. (a) : Net safe bearing capacity,

$$q_{ns} = \frac{q_u - \gamma D_f}{Fos}$$

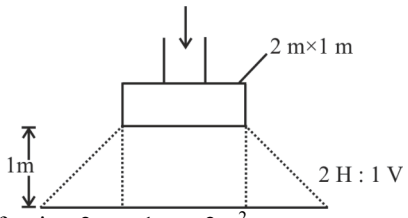
$$= \frac{310 - 20 \times 0.80}{3} = \frac{294}{3} = 98 kN/m^2$$

1453. A footing of 2m x 1m exerts a uniform pressure of 150 kN/m^2 on the soil. Assuming a load dispersion of 2 vertical to 1 horizontal, the average vertical stress (kN/m^2) at 1.0 m below the footing is -

(a) 75 (b) 80
(c) 50 (d) 100

RPSC AE 2018

Ans. (c) :



Area of footing 2m × 1m = 2m^2
Pressure at soil by footing = 150 kN/m^2
Total pressure act = 150 × 2 = 300 kN
Area, 3 × 2 = 6 m^2

Average vertical stress $= \frac{300}{6} = 50$ kN/m^2

1454. The load carrying capacity of an individual friction pile is 200 kN. What is the total load capacity of a group of 9 such piles with a group efficiency factor of 0.8?

(a) 1800 kN (b) 1640 kN
(c) 1440 kN (d) 900 kN

GPSC AE Class (1&2) Paper - 2 2017

Ans. (c) : Load carrying capacity of group pile = No. of piles × Group efficiency factor × individual pile capacity = 9 × 0.8 × 200
= 1440 kN.

1455. The under-reamed piles are connected by a reinforced beam known as :

(a) plinth beam (b) grade beam
(c) plate beam (d) arch beams

GPSC AE Class (1&2) Paper - 2 2017

Ans. (b) : The under-reamed piles are connected by a reinforced concrete beam known as Capping beam or grade beam under reamed piles are bared cost in situ concrete piles having one or more bulbs formed by enlarging the bore hole.

1456. A raft foundation of 6 m × 9 m is placed at a depth of 3 m in a cohesive soil having c = 120 kN/m^2. The net ultimate bearing capacity of the soil using Terzaghi's theory will be :

(a) 820 kN/m^2 (b) 1020 kN/m^2
(c) 1220 kN/m^2 (d) 1420 kN/m^2

GPSC AE Class (1&2) Paper - 2 2017
ESE 1999

Ans. (a) : $q_{ultimate} = S_c.CN_C$
where shape factor

$$S_C = 1 + \frac{0.3B}{L}$$

$$S_C = 1 + 0.3\frac{6}{9} = 1.2$$

$q_{ultimate} = 1.2 \times 120 \times 5.8 = 820.8$ kN/m^2

1457. A gross bearing capacity of a 2.0m wide strip footing at a depth of 1.5 m is 450 KN/m^2. If γ = 20 KN/m^3, what is the net bearing capacity ?

(a) 430 KN/m^2 (b) 410 KN/m^2
(c) 400 KN/m^2 (d) 420 KN/m^2

Rajasthan JEn (Degree) 2016

Ans. (d) : $q_{net} = q_{gross} - \gamma D_F$
$= 450 - 20 \times 1.5$
$= 420$ KN/m^2

1458. Batter Piles are designed to carry which type of load?

(a) Uplift Load
(b) Lateral Load
(c) Load due to negative skin friction
(d) Buckling load

GPSC AE (CLASS 1 & 2) 2019

Ans : (b) : Batter pile- It is a type of pile provided when vertical pile can not resist horizontal load effectively. It may be grouped with vertical pile to increase its lateral capacity generally for dry ground inclination angle should be 30º for water logged area it should be limited to 15º with the horizontal.

1459. As per IS 1904-1986, all foundations shall extend to a depth of at least_____below natural ground level

(a) 500 mm (b) 1000 mm
(c) 15000 mm (d) 2000 mm

GPSC AE (CLASS 1 & 2) 2019

Ans : (a) : IS 1904-1986, all foundations shall extend to a depth of at least 50 cm below natural ground level or rock such other weather resisting natural ground removal of the top soil may be all that is required.

1460. With respect to the site investigation for a foundation, significant depth is defined as

(a) the depth upto which the stress increment due to superimposed load can produce significant settlement
(b) the depth at which the ground water is encountered
(c) the depth at which a rock strata is obtained
(d) the depth at which coefficient of consolidation equals coefficient of compressibility

GPSC AE (CLASS 1 & 2) 2019

Ans : (a) : The depth upto which the live load or super impose load affect the stress & settlement below the soil is called as significant depth

- depth of exploration is equal to the depth of significant depth (S_d)
- S_d generally taken as 20% of vertical load of vertical stress
  $S_d \rightarrow 1.5$ B for square footing
  $S_d \rightarrow 3.0$ B for strip footing

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1461. For shallow foundations if P is the load, γ is the unit weight of soil and ϕ is angle of repose, which of the following is equal to total depth of foundation according to Rankine's formula?

(a) $\frac{P}{\gamma}\left(\frac{1+\sin\phi}{1-\sin\phi}\right)$ (b) $\frac{P}{\gamma}\left(\frac{1-\sin\phi}{1+\sin\phi}\right)$

(c) $\frac{P}{\gamma}\left(\frac{1-\sin\phi}{1+\sin\phi}\right)^2$ (d) $\frac{P}{\gamma}\left(\frac{1+\sin\phi}{1-\sin\phi}\right)^2$

Kerala PSC AE 2015

Ans. (c) : Minimum depth of foundation as given by Rankine's formula–

$$d_{min} = \frac{P}{\gamma}\left(\frac{1-\sin\phi}{1+\sin\phi}\right)^2$$

where P = load acting at the base
γ = unit weight of soil
ϕ = angle of repose

1462. A wooden pile is being driven with a drop hammer weighing 18 kN and having a free fall of 1 m. The penetration in the last blow is 5 mm. Determine the load carrying capacity of pile according to the Engineering News formula

(a) 110 kN (b) 90 kN
(c) 110 kN (d) 180 kN

Kerala PSC AE 2015

Ans. (a) : According to engineering news formula, load carrying capacity of pile is given as

$$Q_u = \frac{WH}{6(S+C)}$$

where H = height of free fall = 1 m
W = weight of hammer = 18 kN
6 = factor of safety used
S = penetration of pile per hammer blow = 5 mm
C = 25 mm

$$Q_u = \frac{18\times1000\,\text{kN mm}}{6(25+5)\,\text{mm}} = 100\,\text{kN}$$

1463. The ultimate bearing capacity of a surface strip footing on clay, according to Terzaghi's theory in:

(a) 5.7C (b) 5.14C
(c) q_u B (d) 9C

Where, C = unit cohesion,
q_u = unconfined compression strength and
B = width of footing

KPSC AE 2016

Ans. (a) : According to Terzaghi theory, the ultimate bearing capacity of soil is given as–

$$q_u = CN_c + \gamma DN_q + 0.5\gamma BN\gamma \quad \text{..... (i)}$$

For pure cohesive soils $\phi_u = 0$ and Terzaghi's bearing capaicty factors are $N_c = 5.7$ (rough footing)

$N_a = 1$, $N\gamma = 0$ and $C = C_u$.

By substituting these values in equation (i), the ultimate capacity of strip footing for pure clays is given by –

$$q_u = 5.7\,C_u + \gamma D$$

The ultimate bearing capacity of strip footings for clays at surface (i.e., D = 0) is, therefore, given by–

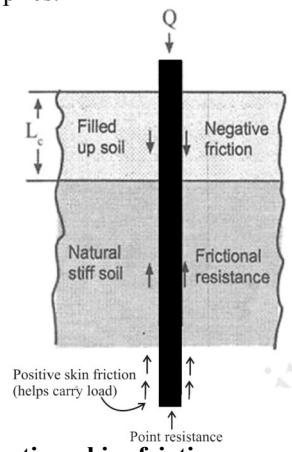
$$q_u = 5.7\,C_u$$

1464. Negative skin friction on a pile under vertical compressive load acts:

(a) Downwards and increases the load carrying capacity of the pile
(b) Downwards and reduces the load carrying capacity of the pile
(c) Upwards and increases the load carrying capacity of the pile
(d) Downwards and maintain the same load carrying capacity of the pile

DSSB AE 2019
KPSC AE 2016
BPSC AE 2012 Paper–V

Ans. (b) : Negative skin friction is usually a downward shear drag acting on a pile or pile group because of downward movement of surrounding solid relative to the piles.



Effect of negative skin friction on piles and pile groups are –

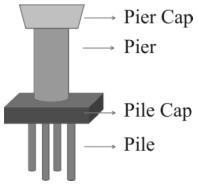
- Negative skin friction reduces the load bearing capacity of the pile.
- For piles in compressible soils were pile capacity is contributed by both point resistance and shaft adhesion, the problem of negative skin friction should be considered a settlement problem.
- In bearing piles where the settlement of the pile is negligible, Negative skin friction becomes a pile capacity problem.

1465. The type of foundation suitable for under water structures is

(a) Cast in situ concrete piles
(b) Pier foundation
(c) Continuous footing
(d) Stepped foundation

KPSC AE 2016

Ans. (b) : A pier foundation is a collection of large diameter cylindrical columns to support the superstructure and transfer large super-imposed loads to the firm strata below.



A caisson foundation also called as pier foundation is watertight retaining structure used as a bridge pier, in

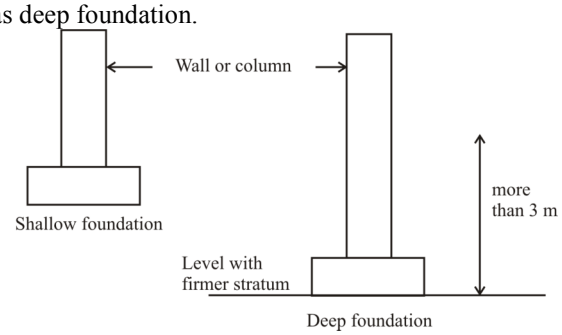
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the construction of a concrete dam, or for the repair of ships
Caisson foundation is most often used in the construction of bridge piers & other structure that require foundation beneath rivers & other bodies of water. This is because caissons can be floated to the job site and sunk into place.

1466. A foundation is considered as shallow if its depth is:
(a) Equal to or less than its width
(b) Less than 1 meter
(c) Greater than its width
(d) Greater than 1 meter

Arunachal Pradesh Poly. Tech. Lect. 2019
Nagaland PSC Paper-II 2017
KPSC AE 2016
RPSC AE (GWD) 2014

Ans. (a) : A foundation is the element of a structure which connects it to the ground, and transfers loads from the structure to the ground.
Foundations are classified as shallow and deep foundations.
Shallow Foundation – The depth of shallow foundation is generally about 3 meters or the depth of foundation is less than the footing width is considered as shallow foundation.
Deep Foundation – The depth of deep foundation is generally more than 3 meters or the depth of foundation is more than the footing width is considered as deep foundation.



1467. Terzaghi's equation of ultimate bearing capacity for a strip footing may be used for square footing resting on pure clay soil with the correction factor
(a) 0.5 (b) 0.8
(c) 1.0 (d) 1.3

TNPSC AE 2019

Ans. (d) Terzaghi for strip foundation
$q_u = CN_c + qN_q + 0.5B\gamma N\gamma$
for square $\rightarrow q_u = 1.3\ CN_c + qN_q + 0.4B\gamma N\gamma$
this (1.3) is correction factor.

1468. For $\phi = 0$, the Terzahgi bearing capacity factors are
(a) $N_c = 1, N_q = 5.7, N_\gamma = 0$
(b) $N_c = 0, N_q = 5.7, N_\gamma = 1$
(c) $N_c = 5.7, N_q = 1, N_\gamma = 0$
(d) $N_c = 1, N_q = 0, N_\gamma = 5.7$

TNPSC AE 2019

Ans. (c) As per terzahgi $N_c = 5.7, N_q = 1\ N_\gamma = 0$

1469. While driving a large number of piles in loose sand
(a) it is advantageous to follow a sequence of pile driving such that the inner piles are driven first and then proceed outwards
(b) it is advantages to follow a sequence of pile driving such that the piles……. periphery are driven first and inner piles are driven later
(c) it is advantageous to follow a sequence of pile driving such that…….inner and outer piles are driven
(d) driving of piles can be done in any random series

TNPSC AE 2019

Ans. (b) While pr driving a pile in group it is good to follow sequence by 1st inner pile.
First then move outwardly so as no error.

1470. Efficiency of pile group is defined as
(a) $\dfrac{\text{load carried by the large pile in the group}}{\text{load carried by the smallest pile in the group}}$
(b) $\dfrac{\text{Maximum load carried by a pile in the group}}{\text{Minimum load caried by a pile in the group}}$
(c) $\dfrac{\text{Minimum load carried by a pile in the group}}{\text{Maximum load carried by a pile in the group}}$
(d) $\dfrac{\text{Average load carried by a pile in the group}}{\text{load carried by a single pile}}$

TNPSC AE 2019

Ans. (d) Efficiency of Pile group (η)
$$\eta = \frac{\text{Maximum (ultimate) load carried by a pile group}}{\text{wt. load carried by a Single pile}}$$

1471. Permissible settlement is relatively higher for
(a) isolated footing on clays
(b) rafts on clay
(c) rafts on sand
(d) isolated footing on sands

TNPSC AE 2019

Ans. (b) Permissible, settlement on sand
(i) Raft footing in Sand $\rightarrow$ 65mm Raft footing in clay $\rightarrow$ (65 to 100)mm
(ii) For isolated footing in clay$\rightarrow$40mm
For isolated footing in sand $\rightarrow$ 25mm

1472. The contact pressure of flexible footing on non-cohesive soils is
(a) more in the centre thanat the edges
(b) less in the centre than at the edges
(c) uniform thorugh out
(d) none of these

GUJRAT PSC AE - II 2017

Ans : (a) For flexible footing on non cohesive soil settlement is maximum of center of footing and minimum at the edges which for bowl. But contact pressure is distributed uniformly along the settlement line or deflected line.

1473. Area of footing of an axially loaded column subjected to working load of 1000 kN and safe bearing capacity of soil 250 kN/m^2, is
(a) 4.0 m^2 (b) 4.4 m^2

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(c) 6.0 m^2 (d) 5.5 m^2

Maharashtra PSC AE (Mains) 2012 Paper–I

Ans : (b) total load = column load + working load

$= 1000 \times \frac{10}{100} + 1000$

$= 100 + 1000$

$= 1100$ kN

Area of footing $= \frac{1100 \text{ kN}}{250 \text{ kN/m}^2} = 4.4 \text{ m}^2$

1474. The in-plane shear deformation effect by which concentrated forces tangential to the surface of plate get distributed over the entire section perpendicular to the load over a finite length of the plate along the direction of the load is called :

(a) Point of contraflexure (b) Shear force
(c) Angle of repose (d) Shear lag

CIL MT 2017

Ans. (d) The in-plane shear deformation effect by which concentrated forces tangential to the surface of plate get distributed over the entire section perpendicular to the load over a finite length of the plate along the direction of the load is called shear lag.

1475. The net intensity of loading which the foundation will carry without undergoing settlement in excess or the permissible value for the structure under consideration but not exceeding net safe bearing capacity is termed as :

(a) Net loading intensity
(b) Ultimate bearing capacity
(c) Safe bearing capacity
(d) Allowable bearing capacity

CIL MT 2017
UPRVUNL AE 2016
(RRB SSE (shift-II, III), 02.09.2015)

Ans. (d) The net intensity of loading which the foundation will carry without undergoing settlement in excess or the permissible value for the structure under consideration but not exceeding net safe bearing capacity is termed as allowable bearing capacity.

Ultimate bearing capacity (q_u) : Ultimate bearing capacity is defined as the minimum gross pressure at the base of the foundation at which the soil to just fails in shear.

Safe bearing capacity (q_u) : It is that gross pressure at the base of footing which can be applied safety without risk of shear failure.

1476. What does the plate load test help in estimating?

(a) Lateral movement of foundation
(b) Saturation in foundation
(c) Depth of foundation
(d) Settlement of foundation

CIL MT 2017

Ans. (d) : The plate load test help in estimating settlement of foundation.

1477. Which of the following combination of factors is used for determining allowable capacity of a foundation?

(a) Tensile failure, compression failure
(b) Settlement, shear failure
(c) Settlement, bond failure
(d) Settlement, tensile failure

NPCC AE 2017
CIL MT 2017

Ans. (b) : Settlement, shear failure combination of factors is used for determining allowable capacity of a foundation.

1478. Which of the following is NOT a recognized failure type of soil as support beneath the foundation, depending upon the deformations associated with the load and the extent of development of failure surface?

(a) bending (b) General shear
(c) Punching shear (d) Local shear

CIL MT 2017

Ans. (a) : Mode of shear failure in shallow foundation:
- General shear failure
- Local shear failure
- punching shear failure

1479. Which of the following is a reason for using pile caps over a group of piles?

(a) For spreading vertical and horizontal loads to all piles
(b) For enhancing the load bearing capacity of each pile
(c) For protecting piles from displacing laterally
(d) For protecting piles from rising up

CIL MT 2017
Chandigarh AE 2017

Ans. (a) : For spreading vertical and horizontal loads to all piles is a reason for using pile caps over a group piles.

1480. In shallow foundation design the correction factor for water table correction in case of water table permanently remaining at or below a depth of (depth + width) of the foundation is taken as :

(a) 0.25 (b) 1.00
(c) 0.50 (d) 0.75

CIL MT 2017

Ans. (b) : In shallow foundation design, the correction factors for water table correction in case of water table permanently remaining at or below a depth of (depth + width) of the foundation is taken as 1.00.

1481. What will be the load carrying capacity (kN) of a group of 6 piles, if the individual load carrying capacity is 150 kN and group efficiency is 75%.

(a) 542 (b) 621
(c) 675 (d) 500

CIL MT 2017

Ans. (c) : Here, Q_{up} (ultimate load carrying capacity of single pile)

$Q_{up} = 150$ kN

η (efficiency) = 0.75 = 75%

We know,

$\frac{Q_{ug}}{n.Q_{up}} = \eta$

$Q_{ug} = \eta \times (n \times Q_{up}) = 0.75 \times 6 \times 150$

$= 675$ kN

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1482. While designing a pile, negative skin friction is considered when the pile is constructed in :

(a) Dense fine sand (b) Fill material
(c) Black cotton soil (d) Dense coarse sand

GUJRAT PSC AE - II 2017
CIL MT 2017

Ans. (b) : While designing a pile, negative skin friction is considered when the pile is constructed in fill material.
In case of filled up soil, soil settles downward relatively to pile thus imparts drag in downward direction which is known as negative skin friction. It should be considered in design.

1483. Which of the following is NOT an advantage of well foundation?

(a) It can resist scouring better.
(b) Possible to ensure its resting on suitable bearing stratum of uniform nature
(c) It can withstand large lateral loads.
(d) Sinking of the well causes vibrations that are good for adjacent structures.

APPSC Poly. Tech. Lect. 2020

Ans. (d) : Advantage of well foundation :
– It can resist scouring better
– Possible to ensure its resting on suitable bearing stratum of uniform nature.
– It can withstand large lateral loads.

1484. According to the Indian standard, the maximum settlement allowed for isolated foundation of a reinforced concrete structure constructed on plastic clay is limited to:

(a) 25mm (b) 50mm
(c) 75mm (d) 100mm

APPSC Poly. Tech. Lect. 2020

Ans. (c) : According to the indian standard, the maximum settlement allowed for isolated foundation of a reinforced concrete structure constructed on plastic clay is limited to 75mm.

1485. Which of the following is NOT an advantage of driven concrete precast piles?

(a) Quality of concrete is never ensured
(b) Suitable for aggressive environment
(c) High displacement piles
(d) Suitable for underwater installations

APPSC Poly. Tech. Lect. 2020

Ans. (a) : Quality of concrete is rules ensured is not an advantage of driven concrete precast piles.

1486. Which of the following is a WRONGLY stated design for machine foundations?

(a) The settlements should be within prescribed limits.
(b) The center of gravity of the machine should preferably lie in the same vertical line as the center of gravity of the foundation system.
(c) There must be resonance
(d) The dynamic amplitudes of the machine-foundation-soil system must be within the prescribed limits under service conditions

APPSC Poly. Tech. Lect. 2020

Ans. (c) : There must be resonance is a wrongly stated design criteria for machine foundation.

1487. A 30 cm diameter friction pile is embedded 10 m into a homogenous consolidated deposit. Unit adhesion developed between clay and pile shaft is 4 t/m^2 and the adhesion factor is 0.7. If the factor of safety to be adopted is 3.0, then what would be safe load carrying capacity of the friction pile?

(a) 26.4t (b) 13.3t
(c) 13.3t (d) 8.8t

Gujarat PSC AE (N.W.R.) 2020

Ans. (b) : For friction pile
$q_u = \alpha Fs.As$
a = adhesion factor = 0.7
$A_S = \pi dl = \pi \times 0.3 \times 10 = 9.42\ m^2$
FOS = 3
$q_u = 0.7 \times 4 \times 9.42$
$q_u = 26.38$ t

safe load = $\frac{q_u}{FOS} = \frac{26.38}{3} = 8.8t$

1488. Minimum centre to centre spacing of friction piles of diameter (D) is-

(a) 2D (b) 3D
(c) 4D (d) 5D

Gujarat PSC AE (N.W.R.) 2020
ESE 1998

Ans. (b) : Minimum centre to centre spacing of friction piles of diameter (D) is 3D.

1489. The load carrying capacity of an individual friction pile is 100 kN. What is the total load carrying capacity of a group of 9 such piles with a group efficiency factor of 0.8?

(a) 1125 kN (b) 900 kN
(c) 720 kN (d) 680 kN

Gujarat PSC AE (N.W.R.) 2020

Ans. (c) : Efficiency factor = 0.8
capacity of single pile = 100 kN
Number of piles = 9
Load capacity of the group of pile
Q_{ug} = Efficiency × capacity of single pile × Number of piles
$= 0.8 \times 100 \times 9 = 720$ kN.

1490. Which of the following piles should be used when weak foundation soil is to be compacted?

(a) Sand Piles (b) Friction Piles
(c) Pedestal Piles (d) End bearing Piles

Gujarat PSC AE (N.W.R.) 2020

Ans. (a) : Sand Piles should be used when weak foundation soil is to be compacted.

1491. Arrange the bearing capacities of shallow foundation in descending order (from highest to lowest)–

(i) Ultimate Bearing Capacity
(ii) Net Ultimate Bearing Capacity
(iii) Net Safe Bearing Capacity
(iv) Gross Safe Bearing Capacity

(a) (i), (ii), (iv), (iii) (b) (i), (ii), (iii), (iv)
(c) (ii), (i), (iii), (iv) (d) (ii), (i), (iv), (iii)

Gujarat PSC AE (N.W.R.) 2020

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Ans. (a) : Bearing capacities of shallow foundation in descending order (from highest to lowest)–
Ultimate Bearing Capacity > Net Ultimate Bearing Capacity > Gross Safe Bearing Capacity > Net Safe Bearing Capacity.

1492. The efficiency of a pile group
(a) will be always less than 100%
(b) will be always more than 100%
(c) may be less than 100% or more than 100%
(d) will be more than 100% in cohesive soil and less than 100% in cohesionless soil

ASSAM PSC CCE (PRE) 2015

Ans : (c) : Pile group efficiency can be less than 100% or more than 100%.

Group efficiency $(\eta) = \frac{Q_{g(u)}}{\sum Q_u}$

where $Q_{g(u)}$ = ultimate load carrying capacity of group pile.
Q_u = ultimate load bearing capacity of each pile without group effect.

1493. Which of the following statements is not a limitation in a 'Plate Load Test' ?
(a) In case of cohesive soils this test does not give the ultimate settlement
(b) The test results reflect only the character of the soil located within a depth less than twice the width of the bearing plate
(c) The bearing capacity of sands and gravels increases with the size of the footing
(d) In this test a rigid plate is loaded at the foundation level and the settlements are determined corresponding to each load in settlement

Rajasthan AE (Nagar Nigam) 2016

Ans. (d) : Limitation in a 'Plate Load Test'–
- In case of cohesive soils this test does not give the ultimate settlement.
- The test results reflect only the character of the soil located within a depth less than twice the width of the bearing plate.
- The bearing capacity of sands and gravels increases with the size of the footing.

1494. The net pressure the soil can carry without exceeding settlement, (the settlement varying in the range of 25 mm to 40 mm for individual footings) is called_____.
(a) Net safe bearing capacity
(b) Net allowable bearing pressure
(c) Safe bearing pressure
(d) Ultimate bearing capacity

LMRC AM 2019

Ans. (c) Safe bearing pressure is the net pressure the soil can carry without exceeding settlement.

1495. Newmark's chart is used in foundation engineering to determine :
(a) earth pressure
(b) seepage loss
(c) permeability of soils
(d) stresses in soil due to surface loading

KPSC AE 2020

Ans. (d) : Newmark's chart is used in foundation engineering to determines stresses in soil due to surface loading.

1496. The full fixity at the column may be valid only for columns supported by:
(a) an individual footing on rock
(b) a thick pile cap
(c) a raft foundation
(d) all of the above

KPSC AE 2020

Ans. (d) : The full fixity at the column may be valid for columns supported by
- An individual footing on rock.
- A thick pile cap
- A raft foundation

1497. Caissons are used for structures such as
(a) Bridge piers (b) 10 storey building
(c) River abutments (d) Both (a) and (c)

Nagaland PSC Paper-II 2017

Ans. (d) : Caissons are used for structure such as –
(1) Bridge piers
(2) River abutment
(3) Dams

1498. Floating caissons are used where
(a) Loads are not very heavy
(b) Loads are very heavy
(c) Bearing stratum is at a shallow depth
(d) Both (a) and (c)

Nagaland PSC Paper-II 2017

Ans. (d) : Floating caissons are used where
(1) Loads are not very heavy
(2) Bearing stratum is at a shallow depth.

1499. In case of isolated footings, offset provided in concrete bed on all sides is :
(a) 15 cm minimum (b) 10 cm minimum
(c) 5 cm minimum (d) 10 cm maximum

Nagaland PSC Paper-II 2017

Ans. (a) : In case of isolated footings, offset provided in concrete bed on all sides in general by 15 cm minimum.

1500. Inverted arch footing is more suitable for
(a) Residential buildings
(b) Bridges
(c) Single storey buildings
(d) Light wave

Nagaland PSC Paper-II 2017

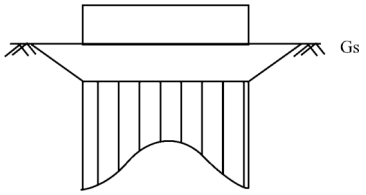
Ans. (b) : Inverted arch footing is more suitable for bridges, because, with the advent of reinforced cement concrete construction practices, inverted arch footing is rarely done for building these days.

1501. The contact pressure of rigid footing on cohesive soils is
(a) More in the centre than at the edges
(b) Less in the centre than at the edges
(c) Uniform throughout
(d) None of these

Nagaland PSC Paper-II 2017

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Ans. (b) :



→ For rigid footing on cohesive soils, settlement is uniform but contact pressure varies.
→ At edges contract pressure is maximum and at centre it is minimum which forms inverted bowl shape as shown in above figure.

1502. Sheet piles are
(a) Non-load bearing (b) Load bearing
(c) Both of the above (d) None of the above

Nagaland PSC Paper-II 2017

Ans. (c) : Sheet pipes are :
(1) Non-load bearing
(2) Load bearing

1503. Example of cased pile is
(a) Sewage pile (b) Vibro-pile
(c) Raymond pile (d) Both (a) and (c)

Nagaland PSC Paper-II 2017

Ans. (d) : Sewage pile and Raymond pile both the example of cased pile

1504. Pressure piles are
(a) Uncased (b) Cased
(c) Precast (d) All of the above

Nagaland PSC Paper-II 2017

Ans. (a) : Pressure piles are the uncased pile.

1505. Local shear failure occurs at strains
(a) More than 10% (b) Less than 10%
(c) Between 5% to 10% (d) More than 90%

Nagaland PSC Paper-II 2017

Ans. (d) : Local shear failure occurs at more than 90% vertical strain.

1506. Which type of foundation is preferable on soil of poor bearing capacity ?
(a) Raft (b) Stepped
(c) Grillage (d) None of the above

Nagaland PSC Paper-II 2017

Ans. (a) : Raft foundation is suitable for soil, which have poor bearing capacity.

1507. Raft foundation will be better solution for
(a) Highly compressible soil
(b) High settlement under individual footing
(c) Both of the above
(d) None of the above

Nagaland PSC Paper-II 2017

Ans. (c) : Raft foundation will be better solution for
→ highly compressible soil
→ high settlement under individual footing

1508. Piles under marine structures and wave action are spread to a
(a) Maximum of 5 diameter apart
(b) Minimum of 5 diameter apart
(c) Maximum of 3 diameter apart
(d) Minimum of 3 diameter apart

Nagaland PSC Paper-II 2017

Ans. (c) : As per IS 2911 (part I) – 1979

Type of pile	min. spacing
→ Friction pile	3D
→ point bearing pile	2.5D
→ pile in loose sand or fill deposit	2D

D → dia of shaft of pile

1509. Piers in pier foundation are built by
(a) Excavation (b) Hammer blows
(c) Vibro equipments (d) Hydro forces

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Ans. (a) : Piers in pier foundation, most commonly made of concrete, masonary or treated timber and installed into prepared holes or shafts. Piers can serve a similar purpose to piles, But are not installed by hammer, pier shaft have be excavated with width more than 1.8 m & Depth more than 30 m, capable of supporting high rise building.

1510. For each storey of a building, the boring should be for a depth of about
(a) 3 m (b) 1 m
(c) 1.5 m (d) 6 m

Nagaland PSC Paper-II 2017

Ans. (a) : For each storey of building, the Depth of exploration should be 3 meter.

1511. Minimum depth of embedment below scour level for piers should not less than
(a) 1 m (b) 2 m
(c) 0.5 m (d) 10 m

Nagaland PSC Paper-II 2017

Ans. (b) : Minimum Depth of embedment below scour level for piers should not be less than 2m .

1512. Factor of safety for pier foundation on rock bed is generally taken as
(a) 5 to 8 (b) 1.5 to 2
(c) 2 to 3 (d) 3 to 5

Nagaland PSC Paper-II 2017

Ans. (c) : Factor of safety for pier foundation on rock bed is generally taken as 2 to 3.

1513. According to IS code, the total settlement of isolated footings for cohesive soil should be :
(a) 30 mm (b) 40 mm
(c) 50 mm (d) 65 mm

Nagaland PSC Paper-II 2017

Ans. (d) : According to IS code, the total settlement of Isolated footing for cohesive soil should be 65mm.

Type of foundation	permissible total settlement	permissible differential settlement
Isolated foundation on clay	65 mm	40 mm
Isolated foundation on sand	40 mm	25 mm
Raft foundation on clay	65-100 mm	40 mm
Raft foundation on sand	40-65 mm	25 mm

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1514. A grillage foundation is essentially a

(a) Shallow foundation (b) Deep foundation
(c) Spread foundation (d) Pile foundation

DMRC AM 2017
Nagaland PSC Paper-II 2017

Ans. (c) : A Grillage foundation is essentially a spread foundation

1515. The type of foundation suitable for under water structures is :

(a) Cast in situ concrete piles
(b) Continuous footing
(c) Pier foundation
(d) Stepped foundation

Nagaland PSC Paper-II 2017

Ans. (a) : Cast in situ concrete piles foundation suitable for under water structure.

1516. The foundation suitable for multi story building is

(a) Isolated column footing
(b) Mono-tube piles
(c) Under reamed pile with many bulbs
(d) Sand piles

Nagaland PSC Paper-II 2017

Ans. (c) : For super high rise buildings one can consider raft foundation, pile foundation, pile assisted raft foundation, under reamed pile foundation etc.

1517. Cast in situ concrete piles are not suitable for

(a) Under water purposes
(b) With in city
(c) Light weight casing shell
(d) On river banks

Nagaland PSC Paper-II 2017

Ans. (b) : Cast in situ concrete piles are not suitable for with in city.
They are suitable in soil of poor drainage qualities since cast-in-situ piles do no significantly disturb of all the materials used in the construction.

1518. Maximum bearing capacity can be expected from :

(a) Laminated rocks (b) Compact coarse sand
(c) Granite rocks (d) Soft rocks

Nagaland PSC Paper-II 2017

Ans. (c) : Among Laminated rock, compacted coarse sand, granite rock & soft, rock, maximum bearing capacity of Granite will be maximum.

1519. Plate loading test is for:

(a) Short duration (b) Long duration
(c) Medium duration (d) Very long duration

DMRC AM 2017

Ans. (a) : Plate loading test is for short duration.

1520. The skin frictional resistance of a pile driven in sand does not depend on

(a) Lateral earth pressure coefficient
(b) Angle of friction between pile and soil
(c) Pile material
(d) Total stress analysis

ESE 2019

Ans. (d) Skin frictional resistance of a pile driven in sand does not depend on total stress analysis. Skin frictional resistance of pile in sands

$\left[Q_s = As.K.\sigma_{va} \tan\delta\right]$

It is depends on lateral earth pressure coefficient (k) and the angle of friction between pile and soil (δ).

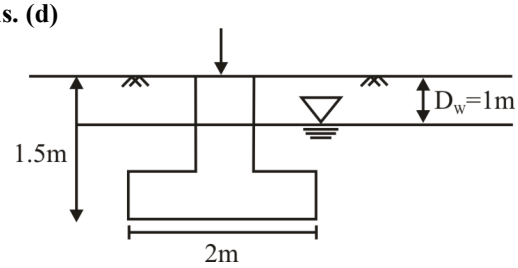
In case of sandy soils, effective stress analysis is generally used and the total stress analysis is used in clayey soil.

1521. A strip footing 2 m in width, with its base at a depth of 1.5 m below ground surface, rests on a saturated clay soil with γ_{sat} = 20 kN/m^3, c_u = 40 kN/m^2, ϕ_u = 0, C' = 10 kN/m^2 and ϕ' = 20^0. The natural water table is at 1 m depth below ground level. As per IS: 6403-1981, the ultimate bearing capacity of this footing will be (taking the relevant N_c as 5.14)

(a) 327 kN/m^2 (b) 285 kN/m^2
(c) 253 kN/m^2 (d) 231 kN/m^2

ESE 2019

Ans. (d)



Strip Footing
Width 'B' = 2m
Depth 'D' = 1.5m
γ_{sat} = 20 kN/m^3
C_u = 40 kN/m^2
ϕ_u = 0
C' = 10 kN/m^2
ϕ' = 20^0
As per 156403 : 1981

The value of N_c = 5.14 for ϕ_u = 0 Condition.

For ϕ_u = 0, $[q_u = C_u N_c + q']$ Ultimate Bearing Capacity

$q_u = 40 \times 5.14 + [\gamma \times 1 + \gamma' \times 0.5]$
$q_u = 205.6 + [20 \times 1 + (20–10)\ 0.5]$
$q_u = 205.6 + [20 + 5]$
$q_u = 205.6 + 25$
$q_u = 230.6$ kN/m^2

$\boxed{q_u \simeq 231 kN/m^2}$

γ is taken equal γ_{sat} and γ_w = 10 kN/m^3

So, $[\gamma' = \gamma_{sat} - \gamma_w]$

1522. Consider the following statements as suggestive of the bearing capacity of soil:

1. The maximum net loading intensity at which neither the soil fails in shear nor is there excessive settlement detrimental to the structure.
2. The maximum net pressure which the soil can carry without shear failure.
3. The net ultimate bearing capacity of the soil divided by a factor of safety.

Which of the above statements is/are ***correct?***

(a) 1, 2 and 3 (b) 1 only
(c) 2 only (d) 3 only

ESE 2017

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Ans. (b)

- The bearing capacity of soil is defined as the capacity of the soil to bear the loads coming from the foundation. The pressure which the soil can easily withstand against load is called allowable bearing pressure.
- By neglecting the overburden pressure from ultimate bearing capacity we will get net ultimate bearing capacity.

$$\left[q_{nu} = q_u - \gamma D_f\right]$$

D_f = Depth of foundation
γ = Unit weight of soil.

- Net ultimate bearing capacity is divided by certain factor of safety will give the net safe bearing capacity.

$$\left[q_{ns} = q_{nu}/f\right]$$

F = Factor of safety

1523. Consider the following statements:
For a rigid footing placed at the ground surface on sand, the contact pressure

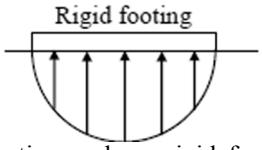
1. is maximum at the edges
2. is zero at the edges
3. distribution is parabolic
4. is uniform throughout the base of the footing.

Which of the above statements are ***correct?***

(a) 1 and 3 only (b) 1 and 4 only
(c) 2 and 3 only (d) 2 and 4 only

ESE 2017

Ans. (c)



Pressure distribution under a rigid footing resting on sand.

1524. In a plate load test on a soil, at a particular magnitude of the settlement, it was observed that the bearing pressure beneath the footing is 100 kN/m^2 and the perimeter shear is 25 kN/m^2. Correspondingly, the load capacity of a 2m square footing at the same settlement will be

(a) 200 kN (b) 300 kN
(c) 400 kN (d) 600 kN

ESE 2017

Ans. (d)

Bearing pressure 'm' = 100 kN/m^2
Peripheral shear, n = 25kN/m
Given area A = 2 × 2 = 4m^2
Perimeter P = 4 × 2 = 8m
Q = Am + Pn
Q = 4 × 100 + 8 × 25
$\boxed{Q = 600kN}$

1525. Consider the following statements:

1. According to Terzaghi, a foundation is shallow if its depth is equal to or less than its width.
2. Spread footing, strap footing and raft footing are type of shallow foundations.
3. Combined footing may be trapezoidal if the two columns carry unequal loads; and rectangular if both columns carry equal loads.
4. For water tanks, providing raft foundations will avoid unequal settlements?

Which of the statements are correct?

(a) 1, 2, 3 and 4 (b) 1, 2 and 3 only
(c) 1, 2 and 4 only (d) 3 and 4 only

ESE 2017

Ans. (a)

According to Terzaghi, a foundation is shallow if its depth is equal to or less than its width $\left[D \le B\right]$

- Types of shallow foundations are–
  - → Combined Footing
  - → Isolated Footing
  - → Spread Footing
  - → Strap Footing
  - → Raft Footing
- Combined footing may be trapezoidal if two columns carry unequal loads in order to make uniform stress distribution and uniform settlement.
- Water tanks foundations carry heavy loads with low and high safety level is required for its construction hence its foundation is designed as raft foundation.

1526. Which of the following is not considered in the design of the isolated footings?

(a) Bending moment (b) Shear
(c) Punching stress (d) Torsion

ISRO Scientist/Engineer 2020

Ans. (d) : In the design of isolated footing Bending moment, Shear, Punching stress are considered.

1527. The cyclic plate load test in soils can be adopted to determine the

(a) Consolidation characteristics of soils
(b) Dynamic shear modulus of the soil
(c) Permeability characteristics of the soil
(d) None of the above

ISRO Scientist/Engineer 2019

Ans : (b) :

• The cyclic plate load test in soils can be adopted to determine the dynamic shear modulus of the soil.

Cyclic plate load test is similar to the plate bearing test for evalution of the allowable bearing capacity of soil for foundation. The plate used for tests in the field are usually made of steel and are 25mm thick and 150mm to 762 mm in diameter.

The nature of the plot of 'q' versus settlement 's' obtained from cyclic plate load test.

$$\left[q = \frac{\text{Load on the plate, Q}}{\text{Area of the plate, A}}\right]$$

The number of cycles of loading of the plate may be needed to replicate the elastic condition in the soil under footing.

1528. Piles used to resist uplift forces are known as

(a) Bearing piles (b) Friction piles
(c) Anchor piles (d) Batter piles

ISRO Scientist/Engineer 2019

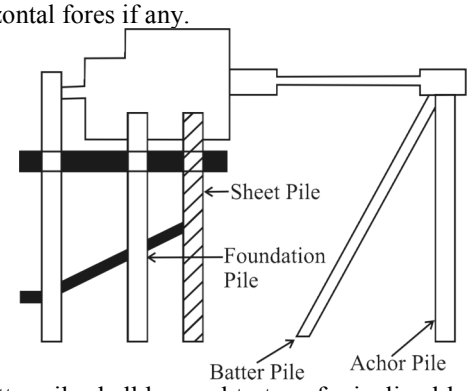
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Ans : (c) :

- Pile foundation that is constracted to resist the uplift forces can be called as anchor piles or tension piles. In those areas where there are chances of extraction of piles from the ground the uplift piles will work well.
- Reasons for the development of the uplift forces are the

→ Seismic forces
→ Overturring Moments
→ Hydrostatic pressure

The anchor piles use sheet piling walls to resist the horizontal fores if any.



• Batter pile shall be used to transfer inclined load and horizontal load. The load on batter pile is generally considered to be axial.

1529. When the water table is under the base of the footing at a depth equal to half of the width of the footing, the bearing capacity of the soil is reduced to its

(a) One-fourth (b) One-half
(c) Three-fourth (d) None of the above

ISRO Scientist/Engineer 2019
(UKPSC AE (Paper-I) 2007)

Ans : (c) :

- When the water table is under the base of the footing at a depth equal to half of the width of the footing, the bearing capacity of the soil is reduced to its three-fourth.
- Bearing capacity of soil depends upon type of soil shape and size of footing and is independent of rate of loading.
- Rise of water table up to ground surface reduce the net ultimate bearing capacity of soil approximately by 50%.

1530. Fender piles are

(a) Used to function as retaining walls
(b) Used to protect concrete deck or other water front structures from the abrasion or impact
(c) Driven at an inclination to resist large horizontal inclined forces
(d) Driven in granular soil with the aim of increasing the bearing capacity of the soil

ISRO Scientist/Engineer 2017

Ans : (b) : Fender piles are used to protect concrete deck or other water front structures from the abrasion or impact.

1531. A square footing 2m × 2m is built in a homogenous bed of sand of density 1.9t/m^3 and having an angle of shearing resistance of 38^0. The depth of base footing is 0.8m below the ground surface. What is the safe load according to Terzaghi analysis which can be carried by the footing with a factor of safety of 3 against complete shear failure? Assume φ = 38^0; N_q = 65, N_γ = 80

(a) 298t (b) 928t
(c) 892t (d) 829t

ISRO Scientist/Engineer 2014

Ans : (a)

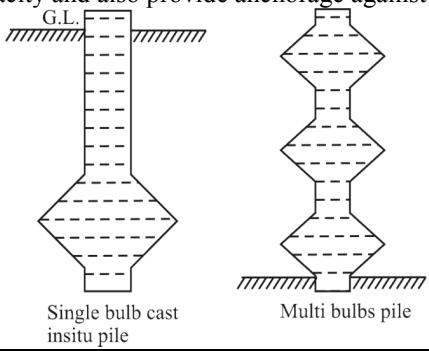
1532. The pile provided with one or more bulbs in its vertical shaft, is generally known as

(a) Under reamed pile (b) Friction-pile
(c) Bearing-pile (d) Sheet-pile

ISRO Scientist/Engineer 2015

Ans : (a) : • Under-reamed pile foundations are bored cast in situ concrete piles with one or more bulbs or under reams. The bulb is formed by enlarging the stem of the pile near the bottom in double conical shape.

- The minimum length of under-ream pile's in deep deposits of expansive soils, should be 3.5m below the ground levels.
- Diameter of pile is 20 to 50cm and bulb diameter is 2-3 times of diameter of pile.
- For black cotton soil the bulb is increase bearing capacity and also provide anchorage against uplift.

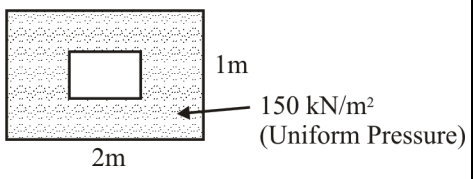


1533. A footing 2m x 1m exerts a uniform pressure of 150 kN/m^2 on the soil. Assuming a load dispersion of 2 vertical to 1 horizontal, the average vertical stress in (kN/m^2) at 1.0 m below the footing is

(a) 50 (b) 75
(c) 80 (d) 100

ISRO Scientist/Engineer 2013

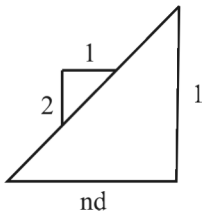
Ans : (a)



Average vertical stress

$$\left[\sigma_z = \frac{P\times(BL)}{(B+2nd)\times(L+2nd)}\right] \begin{matrix} B=\text{Bridh} \\ L=\text{Length} \end{matrix} \text{------(i)}$$

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$P = 150\ N/m^2$

$B = 2m,\ L = 1m$

$$nd = \frac{1}{2}$$

Put the value in equation (i)

$$\sigma_z = \frac{150 \times 2 \times 1}{\left(2 + 2 \times \frac{1}{2}\right)\left(1 + 2 \times \frac{1}{2}\right)}$$

$$\sigma_z = \frac{300}{6}$$

$$\boxed{\sigma_z = 50 kN/m^2}$$

1534. A cyclic pile load test performed to determine the pile

(a) Ultimate load capacity under repetition
(b) Skin frictional resistance and base resistance separately
(c) Skin frictional resistance
(d) Tip resistance

APTRASCO 2017

Ans : (b) The vertical cyclic pile load test is specially carried out when it is required to separate the pile load into skin friction and point bearing.
The load shall be applied in increment of about 20% of the estimated safe load.

1535. The settlement of a footing resting on a sandy soil is

(a) $D_f = \dfrac{D_p}{\left(\dfrac{B_p}{B_f}\right)}$

(b) $D_f = \dfrac{D_p}{\left(\dfrac{B_p}{B_f}\right)^2}$

(c) $D_f = \dfrac{D_p}{\left(\dfrac{B_p[B_f + 30.48]}{B_f[B_p + 30.48]}\right)}$

(d) $D_f = \dfrac{D_p}{\left(\dfrac{B_p[B_f + 30.48]}{B_f[B_p + 30.48]}\right)^2}$

Where D_f and D_p are the depth of foundation and depth of the plate respectively. B_f and B_p are the width of the foundation and width of the plate respectively in cm.

APTRASCO 2017

Ans : (d) Settlement of a footing resisting on a sandy soil–

$$D_f = \frac{D_P}{\left(\dfrac{B_P[B_f + 30.48]}{B_f[B_P + 30.48]}\right)}$$

On clayey soil–

$$D_f = \frac{D_P}{\left(\dfrac{B_P}{B_f}\right)}$$

where, D_f = Depth of foundation
D_p = Depth of plate
B_f = Width of foundation
B_p = Width of plate

1536. A square pile of 25 cm × 25 cm in size and 10 m long is driven in a uniform clay of large depth. The clay has an unconfined compression strength of 100 kPa. Determine the safe frictional resistance of the pile, assuming a factor of safety is 3. Also assume the adhesion factor is 0.6.

(a) 100 kN (b) 50 kN
(c) 300 kN (d) 625 kN

APTRANSCO 2017

Ans : (a) : Given,

$$q_u = 100 kPa$$

$$\text{Cohesion } (C) = \frac{q_u}{2} = 50 kPa$$

Friction resistance of piles $= \propto .C.A$

$\propto = 0.6$

Total surface Area

$$(A) = (0.25 + 0.25 + 0.25 + 0.25) \times 10 m^2$$

$$= 10 m^2$$

Friction resistance of pile = 0.6 × 50 × 10
= 300 kN

$$\text{Coefficient of resistance} = \frac{300 kN}{3}$$

= 100 kN

1537. A square footing of size 1 m, transmits a uniform pressure of 400 kPa to the underlying sandy soil. Determine the vertical stress at a depth of 1 metre below the foundation using the 2 : 1 method.

(a) 400 kPa (b) 200 kPa
(c) 100 kPa (d) 80 kPa

APTRANSCO 2017

Ans : (c) : Vertical stress at depth below 1 m of foundation

$$D_{SZ} = Q/(B + Z) \times (B + Z)$$

where Z = 1 m
B = 1 m
Q = 400 kPa

$$D_{SZ} = \frac{400}{(1+1) \times (1+1)}$$

$$\boxed{D_{SZ} = 100 kPa}$$

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1538. Depth of soil investigation (boring) required for construction of 10 m wide footing of a transmission tower is

(a) 5 m (b) 10 m
(c) 15 m (d) 100 m

APTRANSCO 2017

Ans : (c)

1539. The sheet piles used to protect water front structure from the impacts of ships and vessels, are called:

(a) Tension pile (b) Load bearing pile
(c) Anchor pile (d) Fender pile

WBPSC Poly Lect. 2019

Ans. (d) : The sheet piles used to protect water front structure from the impacts of ships and vessels, are called fender pile

1540. A pile is driven with a drop hammer of weight 1.8t and free fall of 1 m. If the average penetration in last few flows is 5 mm, then safe load carrying capacity of the pile as per Engineering News formula, is:

(a) 10 t (b) 15 t
(c) 25 t (d) 60 t

WBPSC Poly Lect. 2019

Ans. (a) : w = 1.8 t
h = 1 m = 100 cm
s = 5 mm = 0.5 cm
As per engineering news formula:

$$Q_{safe} = \frac{w.h}{fos(s+c)}$$

C = 2.5 cm for drop hammer
fos = 6

$$Q_{safe} = \frac{1.8\times100}{6(2.5+0.5)}$$

Q_{safe} = 10 t

1541. What is the maximum settlement allowed for isolated foundations for RCC structures on sand?

(a) 40 mm (b) 50 mm
(c) 75 mm (d) 00 mm

OPSC AE Paper- II 2019

Ans. (b) Allowable maximum settlement of building for RCC structure

Types of foundation	Sand and hard clay	Plastic clay
Isolated foundation	50 mm	75 mm
Raft foundation	75 mm	100 mm

1542. If L and B are the length and breadth of a footing, e the eccentricity along the length and P and Q are the axial force and bearing capacity of the soil, then, to avoid tension

(a) $BL = \frac{P}{Q}\left(1+\frac{6e}{L}\right)$ (b) $BL = \frac{Q}{P}\left(1+\frac{6e}{L}\right)$

(c) $BL = \frac{P}{Q}\left(1-\frac{6e}{L}\right)$ (d) $BL = \frac{P}{Q}\left(1-\frac{3e}{L}\right)$

OPSC AE Paper- II 2019

Ans. (c)
L = Length
B = Breadth
P = Axial force
Q = Bearing capacity

$$BL = \frac{P}{Q}\left(1-\frac{6e}{L}\right)$$

1543. A 30 cm square bearing plate settles by 8 mm in the plate load test on cohesionless soil when the intensity of loading is 180 kN/m^2. The settlement of a shallow foundation of 1.5 m square under the same intensity of loading will be nearly

(a) 30 mm (b) 26 mm
(c) 22 mm (d) 18 mm

ESE 2020

Ans. (c) : Given,
B_P = 0.3m, B_F = 1.5m, S_P = 8mm, S_F = ?
For cohesionless soil,

$$\frac{S_F}{S_P} = \left[\frac{B_F(B_P+0.3)}{B_P(B_F+0.3)}\right]^2$$

$$\frac{S_F}{8} = \left[\frac{1.5(0.3+0.3)}{0.3(1.5+0.3)}\right]^2$$

S_F = 22.22 mm
or, $\boxed{S_F \simeq 22mm}$

1544. Which of the following two criteria are used to determine the allowable bearing capacity of a foundation?

(a) tensile failure and compressive failure
(b) settlement and tensile failure
(c) bond failure and shear failure
(d) shear failure and settlement

PSPCL AE 2012

Ans. (d) : The allowable bearing capacity of foundation is the maximum soil pressure without any shear failure or settlement failure, the above two condition need to be satisfy for any foundation under soil.

1545. The distance 'D' between centers of piles with top diameter 'd' should not be less than (from practical consideration)

(a) 2d (b) 3d
(c) 4d (d) 5d

Maharashtra PSC Mains 2018 Paper-2

Ans. (b) : The distance 'D' between centers of piles with top diameter 'd' should not be less than 3d.

1546. Match List-I and List-II and select the correct answer using the codes given below:

List I (Construction type)	List II (Suitable Cofferdam Type)
A. Cut-off trench of a dam to be constructed across flowing river	1. Cellular sheet pile cofferdam
B. Shallow foundation of a bridge pier	2. Embankment type cofferdam

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C. **Sequential repetition of underwater foundation work** — 3. **Single wall sheet pile cofferdam**

D. **Control of groundwater to prevent entry into deep excavation** — 4. **Floating steel cylinder cofferdam**

	A	B	C	D
(a)	4	3	2	1
(b)	2	1	4	3
(c)	2	3	1	4
(d)	3	4	2	1

Maharashtra PSC (Mains) 2018 Paper-2

Ans. (c) :

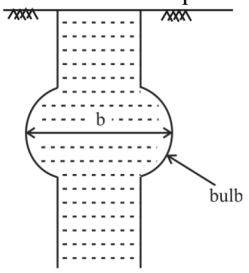
Construction type	Suitable Cofferdam Type
A. Cut-off trench of a dam to be constructed across flowing river	Embankment type cofferdam
B. Shallow foundation of a bridge pier	Single wall sheet pile cofferdam
C. Sequential repetition of underwater foundation work	Cellular sheet pile cofferdam
D. Control of groundwater to prevent entry into deep excavation	Floating steel cylinder cofferdam

1547. Under reamed piles are usually

(a) Precast Piles (b) Driven Piles
(c) Bore Piles (d) Bore or Driven Piles

GPSC AE March 2018
GPSC AE January 2018

Ans. (c) : Under reamed pile is a special type of bored pile with one or more bulb in the pile shaft as figure



This bulb is called under ream with diameter 2-3 time the diameter of pile shaft.

1548. According to Terzaghi's theory, the ultimate bearing capacity at ground surface for a purely cohesive soil and for a smooth base of a strip footing is (Note: C is the unit of cohesive soil).

(a) 2.57 C (b) 5.14 C
(c) 5.7 C (d) 6.2 C

GPSC AE March 2018
OPSC AE -2016 (II)
OPSC AEE 2015 PAPER–I
WBPSC AE 2014

Ans. (b) : Terzaghi's theory for strip footing.

$q_u = CN_c + \gamma D_f N_q + 0.5\,\gamma B N_\gamma$

For purely cohesive soil and for smooth base, $N_c = 5.14$, $N_q = 1$, $N_\gamma = 0$, $D_f = 0$.

$\boxed{q_u = 5.14\,C}$

1549. If the ratio of width of foundation to the width of plate is 5 times, the ratio of corresponding settlements in a clayey soil is

(a) 1:1 (b) 1:5
(c) 15:1 (d) 5:1

GPSC AE March 2018

Ans. (d) : Given data

$$\frac{B_f}{B_p} = 5$$

Then $\boxed{\frac{S_f}{S_p} = \frac{B_f}{B_p}}$ for clayey soil

$$\frac{S_f}{S_p} = \frac{5}{1}$$

1550. Group efficiency of a friction pile in a clay is

(a) Exactly 100% (b) Greater than 100%
(c) Less than 100% (d) Almost 100%

GPSC AE March 2018

Ans. (c) : The efficiency of pile group is defined as,

$$\boxed{\eta_g = \frac{Q_{ug}}{N.Q_u} \times 100}$$

N = no. of piles in pile group
Q_{ug} = ultimate capacity of pile group
Q_u = ultimate load capacity of individual piles.
$\eta_g > 100\%$ for piles in sand
$\eta_g < 100$ for piles in clay.

1551. In a plate load test, the size of plate recommended for gravely and dense sandy soil is

(a) 30 cm square (b) 60 cm square
(c) 75 cm square (d) 90 cm square

GPSC AE March 2018

Ans. (a) : In plate load test, the size of plate recommended for gravely and dense sandy soil is 30 cm square.

Plate → Mild steel

$$S_f = S_p \left[\frac{B_f(B_p + 0.3)}{B_p(B_f + 0.3)}\right]^2$$

S_f = settlement of footing in m
S_p = settlement of plate in m
B_p = width of test plate in m
B_f = width of footing in m

1552. In the plate load test for finding the bearing capacity of soil, the size of square bearing plate should be

(a) less than 300 mm
(b) 750 mm to 1.0 meter
(c) 300 mm to 750 mm
(d) more than 1 meter

GPSC Poly. Tech. Lect. 2016

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Ans. (c) : The plate load test for the bearing capacity of soil are plate to be used is square in section of minimum size 30 cm × 30 cm and maximum size 75 cm × 75 cm. Plate thickness should be sufficient to provide rigidity and is ≃ 25 mm (⩽ 10 mm).

1553. To resist heavy lateral loads which type of piles are used

(a) Friction and bearing piles
(b) Under reamed piles
(c) Displacement piles
(d) Batter piles

GPSC Poly. Tech. Lect. 2016

Ans. (d) : Batter or raked have been used, especially in offshore structures, bridge, and towers to resist large lateral loads generated due to winds, tidal waves, soil pressure etc. The performance of batter piles were found poor for seismic resistance, resulting in bad reputation.

1554. The maximum total settlement for raft foundation on clayey soils should be limited to

(a) 25 mm (b) 25 to 40 mm
(c) 40 to 65 mm (d) 65 to 100 mm

Punjab PSC Sub divisional Engineer 2014

Ans. (d) : The maximum total settlement for raft foundation on clayey soils limits to 65 to 100 mm.

1555. The type of foundation recommended for heavily loaded steel stanchions in soils with poor bearing capacity is

(a) pile foundation (b) well foundation
(c) grillage foundation (d) raft foundation

GWSSB AAE 2015

Ans. (c) A grillage foundation is a special type of isolated footing, generally provided from heavily loaded steel stanchions. It is a type of shallow foundation.
It is used in the location where safe bearing capacity of soil is very poor.

1556. The net ultimate bearing capacity of a purely cohesive soil

(a) depends on the width of the footing and is independent of the depth of the footing
(b) depends on the width as well as the depth of the footing
(c) depends on the depth, but is independent of the width of the footing
(d) is independent of both the width and the depth of the footing

GOA PSC Assistant Prof. 2020

Ans. (d) : The net ultimate bearing capacity of a purely cohesive soil is independent of both the width and the depth of the footing.

1557. Dolphin is a type of which one of the following?

(a) Pile foundation (b) Isolated footing
(c) Raft foundation (d) Caisson

JPSC AE 2009
Ans. (d) : Dolphin is a man made marine structure that extends above the water level and is not connected to shore. It is a type of caisson.

1558. Ratio of bearing capacity of double Under Reamed (UR) pile to that of single UR pile is nearly-

(a) 2 (b) 1.5
(c) 1.2 (d) 1.7

JPSC AE 2009
ESE 1999

Ans. (b) : According to IS 2911 : part III 1973, the ratio of bearing capacity for double under reamed pile to that of single under reamed pile is 1.45 to 1.55 and for sandy and clayey soil is 1.5.

1559. Minimum center-to-center spacing of friction piles of diameter D as per BIS code is-

(a) 1.5 D (b) 2 D
(c) 2.5 D (d) 3 D

GPSC AE Dec 2018
WBPSC AE 2016, 2014
OPSC AEE 2015 PAPER–I
JPSC AE 2009

Ans. (d) : As per BIS - IS : 2911 (Part 1) 1979 recommands a minimum spacing of 2.5 times the shaft diameter for point bearing piles and 3 time the shaft diameter for friction piles.

1560. In using the data from a plate bearing test for determining the modulus of subgrade reaction, the value of settlement to be used is :

(a) 1.25 mm (b) 2.50 mm
(c) 3.75 mm (d) 1.75 mm

KPTCL AE 2016

Ans. (a) : If P is the pressure sustained in kg/cm^2 by the rigid plate of diameter 75 cm at a deflection $\Delta = 0.125$ cm the modulus of subgrade reaction K is given by

$$K = \frac{P}{\Delta} = \frac{P}{0.125} \text{ kg/cm}^3$$

Hence the correct answer A.

1561. Bearing capacity of soil is not influenced by

(a) shape and depth of footing
(b) position of water table
(c) overcoming load on footing
(d) type of soil

Maharashtra PSC Mains 2012 Paper -2

Ans. (c) : Bearing capacity of soil is influenced by:
(1) Shape and depth of footing
(2) Position of water table
(3) Type of soil.

1562. When the allowable soil pressure is low or building loads are heavy, suitable type of foundation is

(a) Strap footing (b) Raft foundation
(c) Spread footing (d) Combined footing

Maharashtra PSC Mains 2012 Paper -2

Ans. (b) : Raft foundation is a thick concrete slab reinforced with steel which covers the entire contact area of the structure like a thick floor. Raft foundation is required where soils have low bearing capacity and have to support heavy structural loads.

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1563. The allowable load on a pile from pile load test is calculated as

(a) 50% load corresponding to a settlement of 10% pile diameter

(b) $\frac{2}{3}$ of load corresponding to a settlement of 12 mm

(c) 50% load corresponding to a settlement of 25 mm

(d) lesser of (a) and (b)

Maharashtra PSC Mains 2012 Paper -2

Ans. (d) :

$$\text{Allowable load on pile} = \min^{m} \begin{cases} \text{50\% load corresponding to a settlement of 10\% pile diameter} \\ \text{or} \\ \text{50\% load corresponding to a settlement of 25 mm} \end{cases}$$

1564. Pneumatic caissons are preferred in situations where the soil flow into the excavated area is ..... than it can be removed.

(a) slower (b) faster

(c) initially faster (d) initially slower

Maharashtra PSC Mains 2012 Paper -2

Ans. (b) : A Pneumatic caisson is a type of water tight enclosed structure used during construction in water or wet areas. In these place the soil flow into the excavated area is faster than it can be removed.

1565. The value of bearing capacity factor N_c for piles as per Meyerhof is taken as :

(a) 5.14 (b) 5.70

(c) 9.0 (d) 6.2

KPTCL AE 2016

OPSC Civil Services Pre 2011

Ans. (c) : Value of bearing capacity factor N_C for piles as per meyerhoff's is 9.0.

1566. The ultimate bearing capacity of a purely cohesive soil at ground surface for a based footing as given by Terzaghi, is (C_u = undrained strength of the soil, D_f = depth of foundation and γ = unit weight of soil).

(a) $5.7\ C_u$ (b) $5.14\ C_u$

(c) $5.14\ C_u + \gamma D_f$ (d) $5.7\ C_u + \gamma D_f$

OPSC Civil Services Pre 2011

Ans. (a) :

Ultimate bearing capacity = $5.7\ Cu + \gamma\ D_F Nq + \frac{1}{2} N\gamma B\gamma$

At ground $D_F = 0$

purely cohesive $N\gamma = 0$

$qu = 5.7\ C_u$

1567. Which type of failure occurs in deep footing in case of soils of low compressibility?

(a) Bending failure

(b) Punching shear failure

(c) Local shear failure

(d) General shear failure

Rajasthan AE Nagar Nigam Shift-3, 2016

Ans. (b) : Punching shear failure occurs in deep footing in case of soils of low compressibility.

Local shear failure:

- Failure is not sudden with well defined load
- occurs in highly compressive soil associated with large settlement.

1568. Ultimate bearing capacity of soil is

(a) Total load on the bearing area

(b) Net load on the bearing area

(c) Load at which soil is fully compacted

(d) Load at which soils fails

Sikkim PSC AE 2017

Ans. (a) : The bearing capacity of soil is the max. avg. contact pressure between the foundation and the soil which should not produce shear failure in the soil.

1569. The standard penetration test is most frequently used to measure the

(a) shear strength of soft clays

(b) undrained strength of fissured clays

(c) relative density of granular soils

(d) consistency of clay

TANGEDCO AE 2018

Ans. (c) : The standard penetration test is most frequently used to measure the relative density of granular soils.

1570. The allowable bearing pressure is the maximum pressure that can be applied on the soil by satisfying the requirement of :

(a) Settlement

(b) Shear

(c) Both shear failure and settlement

(d) None of the above

TRB Poly. Lect. Tech. 2017

Ans. (c) : The allowable bearing pressure is the maximum pressure that can be applied on the soil by satisfying the requirement of both shear failure and settlement.

1571. The total vertical settlement of a rigid foundation under a pressure 'q' is taken to be – ——— times the total vertical settlement at the centre of the flexible foundation. Choose the correct answer from those listed below.

(a) 1.4 (b) 1.2

(c) 0.8 (d) 0.6

TRB Poly. Lect. Tech. 2017

Ans. (a) The total vertical settlement of a rigid foundation under a pressure 'q' is taken to be 1.4 times the total vertical settlement at the centre of the flexible foundation.

1572. When the standard penetration test is carried out in a sandy soil below the water table, the water level in the borehole is always maintained :

(a) Slightly above the ground water level

(b) Slightly below the ground water level

(c) At a depth of ground water level

(d) None of the above

TRB Poly. Lect. Tech. 2017

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Ans. (a) : When the standard penetration test carried out in a sandy soil below the water table, the water level in the borehole is always maintained slightly above the ground water level. Standard penetration test can be used to determine :
(i) Relative density of sand
(ii) Angle of internal friction (ϕ)
(iii) Unconfined compressive strength of clays
(iv) Ultimate bearing capacity on the basis of shear criteria.
(v) Allowable bearing pressure on the basis of settlement criteria.
(vi) Load carrying capacity of pile

1573. In which mode of shear failure, all the three zones of failure develops fully
(a) local shear failure
(b) punching shear failure
(c) general shear failure
(d) vertical shear

TRB Poly. Tech. Lect. 2012

Ans. (c) : General shear failure : in this mode a slight downward movement of the footing develop fully plastic zones and a sudden failure taken place with considerable bulging of the ground surface adjacent to the footing.

1574. A 0.3 m square bearing plate settles by 10 mm in the plate load test cohesion less soil, when the intensity of load is 200 kN/m^2. Estimate the settlement of a shallow foundation of 2 m square under the same intensity loading
(a) 30.25 mm (b) 3.31 mm
(c) 17.39 mm (d) 5.75 mm

TRB Poly. Tech. Lect. 2012

Ans. (a) : Since load intensity is same in both the plate & footing. So,

$$\frac{S_F}{S_P} = \left[\frac{B_F}{B_P}\left(\frac{B_P + 0.3}{B_f + 0.3}\right)\right]^2$$

$$\Rightarrow S_F = 10 \times \left[\frac{2}{0.3}\left(\frac{0.3+0.3}{2+0.3}\right)\right]^2$$

$$= 10 \times [3.0245]$$

$$\boxed{S_F = 30.245\,\text{mm}}$$

1575. Example of spread foundation is:
(a) Cantilever footing
(b) Inverted arch footing
(c) Pier foundation
(d) 1 and 2 both

UJVNL AE 2012

Ans. (a) A cantilever footing is a component of a building's foundation. It is a type of combined footing, consisting of two or more coloumn footing connected by a concrete beam.

1576. The causes of failure of foundation of a building are:
(a) Unequal settlement of soil
(b) Withdrawl of sub soil moisture
(c) Lateral escape of the supporting material
(d) All of the above

UJVNL AE 2012

Ans. (d) : The causes of failure of foundation of a building are:
1. Unequal settlement of soil
2. withdrawl of sub soil moisture
3. Lateral escape of the supporting material.

1577. Black Cotton soil is unsuitable for foundations because of its:
(a) Low bearing capacity
(b) Uncertain permeability
(c) Cohesive particles
(d) Property to undergo a volumetric change due to variation of moisture constant

UJVNL AE 2012

Ans. (d) : Black cotton soil is unsuitable for foundations because of its property to undergo a volumetric change due to variation of moisture constant.

1578. The determination of ultimate bearing capacity on eccentrically loaded square footing depends upon the concept of useful
(a) square (b) width
(c) triangle (d) circle

APPSC AEE 2012

Ans. (b) : Ultimate bearing capacity (square footing)

$$= CN_c + qN_q + \frac{1}{2}\gamma\, BN_\gamma$$

$$= CN_c + \gamma D_f N_q + \frac{1}{2}\gamma\, BN_\gamma$$

depends upon width of footing

1579. A plate load test is useful to estimate
(a) Bearing capacity of foundation
(b) Settlement of foundation
(c) Both (1) and (2)
(d) None of these

Chandigarh AE 2017
NPCC AE 2017/ LMRC AM 2015
ASSAM PSC CCE (PRE) 2015
APPSC AEE 2012 / TSGENCO_2015
CGPSC AE Shift - 2 2014/ WBPSC AE 2007

Ans. (c) : A plate load test is used to
- Bearing capacity of foundation
- Settlement of foundation

taking empirical formula as relation b/w plates and foundation

1580. Pick up the incorrect statement on dynamic pile driving formulae from the following :
(a) In Hiley's formulae a number of constants are involved, which are difficult to determine
(b) The formulae do not take into account the reduced bearing capacity of pile when in group
(c) Dynamic formulae are best suited to coarse grained soils for which shear strength is independent of rate of loading
(d) Law of impact used for determining energy loss is strictly valid for piles, subjected to restraining influence of the surrounding soil

TRB Poly. Lect. Tech. 2017

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Ans. (d) : Correct statement are :
- In Hiley's formulae a number of constants are involved, which are difficult to determine.
- The formulae do not take into account the reduced bearing capacity of pile when in group.
- Dynamic formulae are best suited to coarse grained soils for which shear strength in independent of rate of loading.

1581. Relationship between settlement (S_f) of a foundation width (B_f) and settlement (S_p) of test plate width (B_p), as per as Terzaghi and peck is concerned, is :

(a) $\frac{S_f}{S_P} = \left[\frac{B_P(B_P+0.3}{B_f(B_f+0.3)}\right]^2$

(b) $\frac{S_f}{S_P} = \frac{B_P^2}{B_f^2}$

(c) $\frac{S_f}{S_P} = \left[\frac{B_f(B_P+0.3}{B_p(B_f+0.3)}\right]^2$

(d) $\frac{S_f}{S_P} = \left[\frac{B_P+0.3}{B_f+0.3}\right]^2$

TRB Poly. Lect. Tech. 2017

Ans. (c) : $\frac{S_f}{S_P} = \left[\frac{B_f(B_P+0.3)}{B_p(B_f+0.3)}\right]^2$

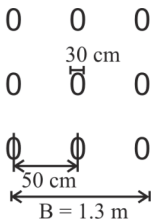
B_f = width of foundation
B_p = width of test plate
S_f = settlement of foundation
S_p = settlement of plate

1582. A group of 9 piles each 10 m long and 30 cm in diameter is placed in a 3 by 3 array and spaced 50 cm c/c are driven into a soft clay of unconfined compressive strength of 70 kN/sq m. Take the value of m = 0.6 for shear mobilization around each pile, neglect bearing at the tip of piles. If the factor of safely is 1.5, then what is the allowable load on the group of pile?

(a) 135 π (b) 810π
(c) 540π (d) 270π

UPPCL AE 2015

Ans. (c) : L = 10 m, m = 0.6, UCS = 70 kN/m^2
FOS = 1.5



$Q_u = Q_{pu} + Qf$

Q_u n α C_u A_s

$= 9 \times 0.6 \times \frac{70}{2} \times 0.3 \times 10 \times \pi$

$Q_u = 567\ \pi$

$\therefore (Q_u)_{Allowable} = 567\pi$

1583. Piles that are used in locations where increase in the end bearing area is desired are called :

(a) Western button bottom piles
(b) Swage piles
(c) Raymond piles
(d) Union metal monotube piles

SJVNL ET 2019

Ans. (a) : Western button bottom piles are used in locations where increase in the end bearing area is desired.

1584. In general, if the density of a soil remains more or less the same, the standard penetration resistance value (N) changes with overburden pressure. If the over burden pressure increases, the standard penetration value :

(a) decreases
(b) increases
(c) remains unchanged
(d) is independent of the overburden pressure

UPPCL AE 2015

Ans. (b) : If the density of soil remains more or less the same, the standard penetration resistance value (N) changes with over burden pressure. If over burden pressure increase then standard penetration value increases.

$$N = 0.77 \log\left(\frac{2000}{\bar{\sigma}_0}\right)$$

1585. According to Schmertmann, the static cone resistance given by CPT is X times that of Standard Penetration Test (SPT) for sandy gravel and gravel, where X is :

(a) 5 (b) 3.5
(c) 2 (d) 6

UPPCL AE 2015

Ans. (d) :

1586. The maximum intensity of loading that the foundation will safely carry without the risk of shear failure of soil irrespective of any settlement that may occur is defined as the :

(a) Allowable bearing capacity
(b) Net loading intensity
(c) Ultimate bearing capacity
(d) Safe bearing capacity

OPSC AE Paper- II 2019
UPRVUNL AE 2015
UPPCL AE 2015

Ans. (d) : The maximum intensity of loading that foundation will safely carry without risk of shear failure of soil irrespective of any settlement.
- The maximum pressure which the soil can carry safely without risk of shear failure.

$$q_s = q_{ns} + \bar{\sigma}$$
$$\downarrow$$
$$q_s = q_{ns} + \gamma' D_f$$
$$\downarrow$$
$$q_s = \frac{q_{nu}}{F} + \gamma' D_f$$

- The order of factor of safety is 2 to 3 (most commonly adopted 2.5).

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1587. If the water table below a footing is just at the base of the footing, then while using the generalized shear failure equation for estimating the bearing capacity of the soil, the value of water table correction shall be taken as:

(a) 0 (b) 0.5
(c) 1 (d) 0.25

UPPCL AE 2015

Ans. (b) : Water table correction–

$$C_w = \frac{1}{0.5 + 0.5\left[\frac{D_w - D_f}{B}\right]}$$ for settlement criteria

$$R_w = \frac{1}{2}\left(1 + \frac{D_w}{B}\right)$$

D_w = depth of water flow footing base

$$R_w = \frac{1}{2}\left(1 + \frac{0}{B}\right)$$

$$\boxed{R_w = 0.5}$$

$$Q_u = CNc + qNq + \frac{1}{2}R_w \gamma BN\gamma$$

1588. A wooden pile is being driven with a drop hammer weighing 20 kN and having a free fall of 2 m. The penetration in the last blow is 5 mm. If Engineering News formula is used, then what would be the estimated load carrying capacity (kN) of the pile?

(a) 222 (b) 125
(c) 111 (d) 333

UPPCL AE 2015

Ans. (a) : Engineering News Formula–

$$Q = \frac{wh}{F(S + C)}$$

2.5 (drop hammer)
S = measured penetration per blow = 5 mm
h = 2 m
w = 20 kN
F = 6 (FOS)

$$Q = \frac{(20\text{kN})(2\text{m})}{6(2.5 + 0.5\text{cm})} = \frac{(20 \times 1000\text{N})(200\text{cm})}{(3 \times 6)\text{cm}}$$

Q = 222 kN

1589. Which of the following is not a cause of differential settlement in soil foundations?

(a) Varying water regime on a construction site
(b) Overlap of stress distribution in soil from adjoining structures
(c) Consolidation including secondary compression
(d) Unequal expansion of the soil due to excavation for footings

UPRVUNL AE 2016

Ans.: (c) Consolidation including secondary compression is not a cause of differential settlement in soil foundations.

1590. Which of the following is not a method to determine the ultimate load capacity of a pile?

(a) Single tangent
(b) Double tangent
(c) Secant
(d) Vander Veen's method

UPRVUNL AE 2016

Ans.: (c) The method of single tangent, double tangent and Vander Veen's is appropriate to determine the ultimate load capacity of a pile

1591. In a raft foundation, the ratio of intensity of soil pressure 'P' and the average settlement 'S' is known as

(a) Modulus of subgrade reaction
(b) Reaction coefficient
(c) Settlement coefficient
(d) Compressibility coefficient

GPSC AE June 2019

Ans. (a) : In raft foundation, the ratio of intensity of soil pressure 'P' and average settlement 'S' is known as Modulus of subgrade reaction.

Modulus of subgrade reaction = $\frac{P}{S}$

1592. A raft foundation is to be constructed on a sandy soil. As per I.S. code, the maximum differential settlement and maximum limiting settlement range are respectively

(a) 40 mm, 65 mm to 100 mm
(b) 40 mm, 40 mm to 65 mm
(c) 25 mm, 65 mm to 100 mm
(d) 25 mm, 40 mm to 65 mm

GPSC AE June 2019

Ans. (d) : As per IS code, when a raft foundation is to be constructed on sandy soil, the maximum differential settlement and maximum limiting settlement range are 25 mm and 40 mm to 65 mm.

1593. In case of foundations on black cotton soils, the most suitable method to increase the bearing capacity of soils is to

(a) Increase the depth of foundation
(b) Drain the soil
(c) Compact the soil
(d) Replace the poor soil

GPSC AE June 2019

Ans. (d) : In case of foundations on black cotton soils, the most suitable method to increase the bearing capacity of soil is to replace the poor soil.

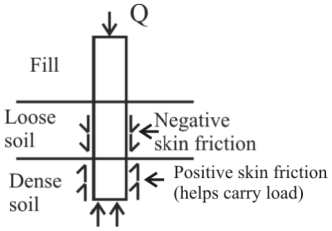
1594. Which one of the following is responsible for development of negative skin friction on a pile?

(a) Settlement of pile more than that of surrounding soil
(b) Settlement of pile less than that of surrounding soil
(c) Downward movement of surrounding soil, irrespective of pile movement
(d) Downward movement of pile irrespective of settlement of soil

GPSC AE Dec 2018

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Ans. (b) : Settlement of pile less than that of surrounding soil is responsible for development of negative skin friction.



1595. Consider the following statements:

1. **Triangulation is another system of increasing the ground control points**
2. **Triangulation is used for filling in the minute details of an area**
3. **Triangulation system of quadrilaterals is most suitable for railways.**

Which of the above statements is correct?

(a) 1 only (b) 2 only
(c) 3 only (d) 3 and 1 only

GPSC AE Dec 2018

Ans. (c) : Triangulation system of quadrilaterals is most suitable for railways.

1596. The net ultimate bearing capacity of a purely cohesive soil

(a) Depends on the width of the footing and is independent of the depth of the footing
(b) Is independent of both the width and the depth of the footing
(c) Depends on the depth, but is independent of the width of the footing
(d) Depends on the width of well as the depth of the footing

GPSC AE Dec 2018

Ans. (b) : $q_{nu} = CN_c$
q_{nu} is independent of depth and width of footing.

1597. If the weight of excavated soil is equal to the superimposed load, then the foundation is known as

(a) Dolphin type foundation
(b) Will foundation
(c) Floating foundation
(d) Balanced foundation

GPSC AE Dec 2018

Ans. (c) : If the weight of excavated soil is equal to the superimposed load, then the foundation is known as Floating foundation.

1598. Well foundation is recommended for a bridge where the standard penetration N values = 22 for the soil constituting the foundation. When earthquake forces are included in the design, the permissible increase in the allowable bearing pressure as per IS 1893 : 2002 can be :

(a) 25% (b) 50%
(c) 40% (d) zero

BHEL ET 2019

Ans. (a) :

1599. Match the items in List 1 (Name of field exploration) with those in List 2 (Soil properties). and select the correction option. options.

List 1	List 2
A. Cyclic Pile load test	**1. Modulus of subgrade reaction**
B. Plate load test	**2. Relative density and strength**
C. Pressure meter test	**3. Skin friction and point bearing resistance**
D. Standard penetration test	**4. Elastic constants**

(a) A–3, B–2, C–1, D–4
(b) A–3, B–1, C–4, D–2
(c) A–2, B–4, C–3, D–1
(d) A–1, B–3, C–4, D–2

BHEL ET 2019

Ans. (b) :

	Name of field exploration	Soil properties
A.	Cyclic pile load test	Skin friction and point bearing resistance
B.	Plate load test	Modulus of subgrade reaction
C.	Pressure meter test	Elastic constant
D.	Standard penetration test	Relative density and strength

1600. A raft foundation is proposed on a clay soil. The permissible differential settlement (D) and limiting maximum settlement (M) as IS code are :

(a) D = 40 mm; M = 65 to 100 mm
(b) D = 40 mm; M = 40 to 65 mm
(c) D = 25 mm; M = 40 to 65 mm
(d) D = 25 mm; M = 65 to 100 mm

BHEL ET 2019

Ans. (a) : A raft foundation is proposed on a clay soil. The permissible differential settlement (D) and limiting maximum settlement (M) as per IS code are 40 mm and 65 to 100 mm respectively.

1601. Match the Items in List 1 (Requirement condition) with those in List 2 (Foundation type) and choose the best answer using the codes.

List 1	List 2
A. When columns are very close to the property line.	1. Floating rafts
B. To transfer the moment between two adjacent footings.	2. Under-reamed piles
C. To restrict settlement of soft clays/silts.	3. Strap Footing
D. To restrict damages due to volume changes of swelling soils.	4. Combined Footing

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(a) A–3, B–1, C–4, D–2
(b) A–2, B–1, C–4, D–3
(c) A–1, B–3, C–2, D–4
(d) A–4, B–3, C–1, D–2

BHEL ET 2019

Ans. (d) : A–4, B–3, C–1, D–2

1602. A shallow foundation square in shape is placed in a strain for which $\phi = 0$; the Meyerhof's bearing capacity shape factor for the foundation is
(a) 1 (b) 0.6
(c) 1.2 (d) 1.1

GPSC Assistant Prof. 2016

Ans. (a) : Shape factor value for-

$S_e = (1+0.2\, k_\gamma \frac{B}{L})$	Any ϕ
$S_q = S_\gamma = (1+0.1\, k_\gamma \frac{B}{L})$	$\phi > 1.0$
$S_q = S_\gamma = 1$	$\phi = 0$

$k_\gamma = \tan^2\left(45+\frac{\phi}{2}\right)$

ϕ = Angle of resultant measured from vertical without sign
B → width of footing
L → length of footing
D → Depth of footing

1603. A pile 0.5 m in diameter is driven in a clay deposit, having an undrained shear strength of 50 kN/m^2 at the base of the pile. The base capacity of the pile is given by
(a) 88.36 kN (b) 9.82 kN
(c) 55.96 kN (d) 78. 54 kN

GPSC Assistant Prof. 2016

Ans. (a) : Pile capacity in clay

$$Q_u = \underbrace{CN_C A_b}_{\text{Due to end bearing of pile}} + \underbrace{\alpha C_u A_s}_{\text{friction capacity of pile}}$$

But here in question not provided detail for friction pile capacity calculation (like type of pile, undrain cohesion in pile length etc.)
$Q_u = CN_C A_b$
C = undrain shear strength at base = 50 kN/m^2
N_C =q (standard result)

$$A_b = \frac{\pi}{4}d^2$$

$$\therefore\ Q_u = 50\times 9\times\frac{\pi}{4}(0.5)^2$$

$= 450 \times 0.1963$
$Q_u = 88.36$ kN

1604. The action of negative skin friction on the pile is
(a) increase working load on pile
(b) reduce settlement of pile
(c) reduce the ultimate load on pile
(d) reduce the allowable load on pile

GPSC Assistant Prof. 2016

Ans. (d) : The action of Negative skin friction on the pile is reduced the allowable load on the pile.

1605. Black cotton soil is unsuitable for foundation because of its:
(a) Low bearing capacity
(b) Uncertain permeability
(c) Cohesive particle
(d) Property to undergo a volumetric change due to variation of moisture content

PTCUL AE 2016

Ans. (d) : Black cotton soil has sufficient moisture due to which change in moisture attains change in volume.

Direction : Match list I with list II in Q. No. 53 to 63 and select the correct answer using code given below the two list in each question.

1606.

List I	List II
A. Friction pile	1. Stiff clay
B. Batter pile	2. Loose granular soil
C. Tension pile	3. Lateral load
D. Compaction pile	4. Uplift load

Code:

	A	B	C	D
(a)	1	3	4	2
(b)	3	2	1	4
(c)	3	1	4	2
(d)	1	3	2	4

PTCUL AE 2016

Ans. (a) : 1. Friction pile used in stiff clay.
2. Batter pile used in lateral load.
3. Tension pile used in uplift load.
4. Compaction pile used in loose granular soil.

1607. A square pile of section 30 cm × 30 cm and length of 10 m penetrates a deposit of clay having cohesion 'C' = 5 kN/m^2 and mobilizing factor 'α' = 0.8. Load carried by pile by skin friction only?
(a) 192 kN (b) 48 kN
(c) 37.7 kN (d) 60 kN

UJVNL AE 2016
ESE 2004

Ans. (b) : Square pile = 30 × 30 cm^2
Length of pile = 10 m
Cohesion 'C' = 5 kN/m^2
Mobility factor ($\propto$) = 0.8
Load carried by pile by skin friction only
$= C \propto \times$ surface of pile
$= C \times 0.8 \times (4DL)$
$= 5 \times 0.8 \times 4 \times 0.3 \times 10$
$= 48$ kN

1608. The ultimate bearing capacity and unit weight of soil are 300 kN/m^2 and 20 kN/m^3 respectively. The depth of foundation is 1 m, factor of safety is 2.5, the net safe bearing capacity is
(a) 100 kN/m^2 (b) 80 kN/m^2
(c) 112 kN/m^2 (d) 100.5 kN/m^2

UJVNL AE 2016
(UKPSC AE 2012 Paper–I)

Ans. (c) : Ultimate bearing capacity = 300 kN/m^2
Unit weight = 20 kN/m^3
Depth = D_f = 1 m
FOS = 2.5

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$$\text{Net safe bearing} = \frac{\text{Ultimate bearing capacity} - \gamma D_f}{\text{FOS}}$$
$$= \frac{300 - 20\times 1}{2.5}$$
$$= 112 \text{ kN/m}^2$$

1609. In using the data from a plate bearing test for determining the modulus of subgrade reaction, the value of settlement to be used is:
(a) 1.25 mm (b) 2.50 mm
(c) 3.75 mm (d) None of the above

UJVNL AE 2016

Ans. (a) : In using the data from a plate bearing test for determining the modulus of sub-grade reaction, the value of settlement to be used is 1.25 mm.

1610. Negative skin friction in a soil is considered when the pile is constructed through a
(a) Fill material
(b) Dense coarse sand
(c) Over consolidated stiff clay
(d) None of the above

UJVNL AE 2016

Ans. (a) : Negative skin friction is a downward shear drag acting on the pile surface due to relative downward movement of soil strata surrounding the pile.

1611. A square pile of 25 cm × 25 cm in size and 10 m long is driven in a uniform clay of large depth. The clay has an unconfined compression strength of 100 kPa. Determine the safe frictional resistance of the pile, assuming a factor of safety is 3. Also assume the adhesion factor is 0.6.
(a) 100 kN (b) 50 kN
(c) 300 kN (d) 625 kN

AP TRANSCO AEE 2017

Ans. (a) :
L = 10 m
q_u = 100 kPa
FOS = 3
$\alpha = 0.6$
$Q_f = \alpha\, Cu_{avg} \times As$
$= 0.6\times\frac{100}{2}\times 4\times 0.25\times 10$
= 300 kN
$\therefore (Q_f)_{safe} = \frac{Q_f}{fos} = \frac{300}{3} = 100$ kN

1612. Depth of soil investigation (boring) required for construction of 10 m wide footing of a transmission tower is
(a) 5 m (b) 10 m
(c) 15 m (d) 100 m

AP TRANSCO AEE 2017

Ans. (c) : width = 10 m, As per thumb rule,
(1) The depth of exploration should be 1.5 to 2 times width (lower dimension) of the footing.
(2) Pile foundation, 10 to 30 meters or more or at least 1.5 times width,
Depth = 15 m

1613. The following assumption is not made, for design of a strap footing:-
(a) The strap is perfectly rigid
(b) The strap is not subjected to any direct soil pressure
(c) The soil pressure varies linearly
(d) The interior footing is centrally loaded

UPPCL AE 2014

Ans. (c) The soil pressure linear variation is not an assumption made during design of a strap footing.

1614. The maximum depth of foundation, according to the Rankine's formula will be:- (Given that q = 180 KN/m², γ = 20KN/m² and Φ = 30°)
(a) 0.8 m (b) 1 m
(c) 2 m (d) 0.5 m

UPPCL AE 2014

Ans. (b) : $D_f = \frac{q}{\gamma}\left(\frac{1-\sin\phi}{1+\sin\phi}\right)^2$
$= \frac{180}{20}\times\left(\frac{1}{3}\right)^2 = 1$
$D_f = 1$ m

1615. The load carrying capacity of a board pile in sand in comparison to a driven pile will be :-
(a) 5/4 to 3/4 (b) 3/4 to 2/3
(c) more than 5/4 (d) 2/3 to 1/2

UPPCL AE 2014

Ans. (d) : The load carrying capacity of a board pile in sand in comparison to a driven pile will be2/3 to 1/2.

1616. Some facts about pile foundation is given below. Find the correct combination:-
i. Loads are heavy
ii. Loads are light
iii. Soil stratum near ground surface is week
iv. Soil stratum near ground surface is strong
(a) (ii) and (iv) (b) (i) and (iv)
(c) (i) and (iii) (d) (ii) and (iii)

UPPCL AE 2014

Ans. (c) : Pile foundation is used when loads are heavy and soil stratum near the ground surface is weak.

1617. Negative skin friction on piles
(a) Is developed due to relative settlement of the soil
(b) Develops in hard clays
(c) Increases the pile capacity
(d) Develops due to driving of piles

GPSC AE January 2018

Ans. (a) : Negative skin friction on piles
• Is developed due to relative settlement of the soil
• when recently dumped soil is present.

1618. According to Vesic, deep foundations in sand having D_f/B ratio greater than 10 are likely to fail in
(a) General shear failure mode
(b) Local shear failure mode
(c) Punching shear failure mode
(d) Local bending failure mode

GPSC AE January 2018

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Ans. (c) : Deep foundations in sand having D_f/B ratio greater than 10 are likely to fail in punching shear failure mode, according to Vesic.

1619. Multi-under-reamed piles are generally provided in

(a) Salty soil (b) Sandy soil
(c) Expansive soil (d) Gravity sandy soil

TSPSC Deputy Engineer 2015
UPRVUNL AE 2015

Ans. (c) : Multi-under reamed piles which has more than two bulbs, it is cast in-situ. Expansive soil have more expanding property so multi under reamed piles are used.

1620. Match List I (Field test) with List II (Property) and select correct answer using the given lists.

List I	List II
A. Pumping test	**a. Bearing capacity**
B. Plate load test	**b. Load carrying capacity**
C. Pile load test	**c. Permeability**

(a) A-a, B-b, C-c (b) A-c, B-a, C-b
(c) A-b, B-c, C-a (d) A-c, B-b, C-a

UPRVUNL AE 2015

Ans. (b) :

Pumping test	Permeability
Plate load test	Bearing capacity
Pile load test	Load carrying capacity

1621. The most comprehensive of the pile driving formula is

(a) Engineering new formula
(b) Hiley's formula
(c) Static capacity formula
(d) Danish formula

MPPSC AE 2014

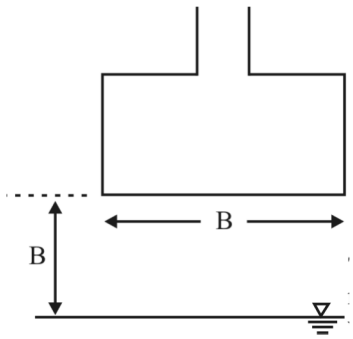
Ans. (b) : Hiley's formula is most comprehensive of the pile driving formula.

1622. The bearing capacity of soil supporting a footing of size 3m × 3m will not be affected by the presence of water table located at a depth below the base of footing of

(a) 1.0m (b) 1.5m
(c) 3.0m (d) 6.0m

MPPSC AE 2014

Ans. (c) : If depth of water table, below base of footing is greater than depth equal to width of footing then there is no effect of water table



Here, footing is 3m × 3m

So, $\boxed{B = 3m}$

1623. The immediate settlement of a rigid footing is about --------- times the maximum settlement of an equal flexible footing

(a) 0.90 (b) 0.80
(c) 0.70 (d) 0.60

LMRC AM 2018
UPPCL AE 2014 / MPPSC AE 2014

Ans. (b) : immediate settlement of a rigid footing is about 0.80 times the maximum settlement of an equal flexible footing

1624. The Permissible settlement is the maximum in the case of

(a) Isolated footing on clay
(b) Isolated footing on sand
(c) raft on clay
(d) raft on sand

UPPCL AE 2014 / MPPSC AE 2014

Ans. (c) :
⇒ IS code guideline for permissible settlement –
For isolated footing on clay = 65 mm
For isolated footing in sand = 40 mm
For raft footing on clay = 65-100 mm
For raft footing on sand = 40-65 mm
⇒ IS code guideline for permissible differential settlement –
For isolated footing on clay = 40 mm
For isolated footing in sand = 25 mm
So, the Permissible settlement is the maximum in the case of raft on clay.

1625. Two footings, one circular and the other square, are founded in a pure clay. The diameter of the circular footing is the same as the side of the square footing. The ratio of their net ultimate bearing capacities

(a) is unity
(b) is 1.3
(c) is 1/1.3
(d) cannot be determined with the provided data

MPPSC AE 2014

Ans. (a) : Net ultimate capacity formula :–

q_{nu} (circular) = $1.3\,CN_C + q\,(N_q - 1) + 0.3\,B\gamma N_\gamma$

q_{nu} (square) = $1.3\,CN_C + q\,(N_q - 1) + 0.4B\gamma N_\gamma$

For pure clay

$(\phi = 0)$, $N_C = 5.7$, $N_q = 1$ $N_\gamma = 0$

$\therefore$ q_{nu} (circular) = $1.3 \times 5.7 \times C + q\,(1-1) + 0.3B\gamma \times O$

$= (1.3 \times 5.7 \times C)$

q_{nu} (square) = $1.3 \times 5.7 \times C + O$

$= 1.3 \times 5.7 \times C$

$$\boxed{\frac{q_{nu}\,(\text{circular})}{q_{nu}\,(\text{square})} = 1}$$

i.e unity

1626. If the ultimate bearing capacity of a 1 m wide strip footing resting on the ground surface is 250 kN/m^2, what will be the allowable bearing pressure of a 3 m × 3 m square resting on the same surface with factor of safety 2.5? Use

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Terzaghi's bearing capacity theory and assume the soil is cohesion less.

(a) 240 kN/m^2 (b) 250 kN/m^2
(c) 600 kN/m^2 (d) 440 kN/m^2

UPPCL AE 2018

Ans. (a) : Given,
strip footing = 1 m wide
ultimate bearding capacity = 250 kN/m^2
square footing = 3 m × 3 m
FOS = 2.5, C = 0
Allowable bearing pressure = 240 kN/m^2

1627. Which of the following of soil types is responsible for negative skin friction in piles?

(a) Over consolidated clay (b) Dense sand
(c) Soft clay (d) Stiff clay

UPPCL AE 2018

Ans. (c) Soft clay is responsible for negative skin friction in piles.
Negative skin friction is developed when settlement of soil is more than settlement of pile (when soil moves down relative to the pile) it occurs when.
1. The pile is driven through a recently deposited layer.
2. Lowering of ground water table, this causes increase in settlement of soil with the resultant downward drag (negative skin friction) on the pile.

1628. The Piles used to resist large horizontal or inclined loads are called :

(a) Compaction pile (b) Batter pile
(c) Tension pile (d) Fender pile

UPPCL AE 2019

Ans. (b) The pile used to resist large horizontal or inclined loads are called batter pile.
Compaction pile : Compaction pile one of group of piles, driven inepter, to compact a surface layer of loose granular soil to increase is bearing capacity.

1629. The load carrying capacity of a friction pile in a clayey strata is 180 kN. If a group of 7 piles are designed with an efficiency factor of 0.75, then the total load carrying capacity (kN) of the group will be:

(a) 1260 (b) 945
(c) 1680 (d) 1200

LMRC AM 2015

Ans (b) Load capacity of the group of pile
Q_{ug} = Efficiency × capacity of single pile × number of piles
= 0.75 × 180 × 7
= 945 kN

1630. Downward drag acting on a pile, due to the downward movement of the surrounding compressible soil relative to pile, is called:

(a) Positive skin friction
(b) Negative skin friction
(c) Settlement
(d) Sunken support

LMRC AM 2018

Ans. (b) : Downward drag acting on a pile, due to the downward movement of the surrounding compressible soil relative to pile, is called negative skin friction.

1631. According to Feld's rule, the load capacity of each pile is reduced by _____ on account of the effect of the nearest pile in each diagonal or straight row of which the pile in question is a member.

(a) $\frac{1}{16}$ (b) $\frac{1}{4}$
(c) $\frac{1}{12}$ (d) $\frac{1}{8}$

LMRC AM 2018

Ans. (a) :

1632. If an angular disortion of $\frac{1}{300}$ is allowed between columns 7.5 m apart, what is the corresponding value of differential settlement ?

(a) 12.5 mm (b) 50 mm
(c) 25 mm (d) 40 mm

CIL MT 2020

Ans. (c) :
Differential settlement = $\frac{7.5\text{m}}{300} = \frac{7500}{300} = 25\text{mm}$

1633 According to IS 8009, Part1(1976), the recommended value of rigidity factor for computing the settlement of a rigid footing is:

(a) 0.8 (b) 1.2
(c) 0.5 (d) 1.5

CIL MT 2020
Chandigarh AE 2017

Ans. (a) According to IS 8009, Part I (1976), the recommended value of rigidity factor for computing the settlement of a rigid footing is 0.8

1634. Match the items under List 1 (Field situation) with those in List 2 (Type of pile foundation) and select the answer using the codes given in the lists.

List 1	List 2
M. To resist lateral loads	1. Sheet piles
N. Top retain soil filling	2. Compaction piles
O. To density loose soils	3. Point bearing piles
P. To transfer heavy loads to strong stratum below a weak stratum.	4. Batter piles

(a) M- 1; N- 3; O- 2 ; P- 4
(b) M- 4; N- 3; O- 2 ; P- 1
(c) M- 3; N- 1; O- 4 ; P- 2
(d) M- 4; N- 1; O- 2 ; P- 3

CIL MT 2020

Ans. (d) :

1635. Identify the factor that does not affect the bearing capacity of a purely cohesive soil.

(a) Density of soil
(b) Depth of footing
(c) Width of footing
(d) Angle of shearing resistance of soil

CIL MT 2020

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Ans. (c) : Width of footing

Terzaghi general bearing capacity equation for a strip footing

$$q_u = CN_C + qN_q + \frac{1}{2}\gamma BN_\gamma$$

for purely cohesive soil

$N_q = 1$

$N_\gamma = 0$

So,

$$q_u = CN_C + q = CN_C + \gamma D_f$$

This expression is independent of base width of foundation, but it increases with depth below the ground level i.e. as 'D_f' increase 'q_u' also increases.

1636. The following statements pertain to the occurrence of negative skin friction in piles.
S1: It is developed when the pile is driven through a recently deposited clay layer.
S2: It is developed when the pile is driven through a layer of dense sand.
S3: It is developed due to a sudden drawdown of the water table.
Which of the above statement(s) is/are incorrect?

(a) S3 only (b) S2 and S3
(c) S1 and S3 (d) S2 only

CIL MT 2020

Ans. (c) : Negative skin friction is developed when settlement of soil is more than the settlement of pile (when soil moves down relative to the pile). it occurs when.
(1) Pile is driven through a recently deposited layer.
(2) Lowering of ground water table, this causes increase in settlement of soil with the resultant downward drag (negative skin friction) on the pile.

1637. In pile foundations, the negative skin friction can be developed by the :
(a) upward movement of the surrounding compressible solid relative to the pile
(b) decrease in effective stress causing consolidation of the soil
(c) increase in the ground water level
(d) lowering of the ground water level

DMRC AM 2020

Ans. (d) : Negative skin friction developed because
- Soil settle more than the pile.
- lowering of ground water table.

1638 What is the mode of failure for the condition of 'footing on very sand underlain by loose sand or soft clay'?
(a) General shear failure
(b) Local shear failure
(c) Normal failure
(d) Punching shear failure

DMRC AM 2020

Ans. (d) The mode of failure for the condition of 'footing on very dense sand underlain by loose sand or soft clay is punching shear future.

1639. Piles, which are used to protect waterfront structures against the impact from ships and other floating objects, are called :
(a) fender piles (b) anchor piles
(c) friction piles (d) end-bearing piles
(e) compaction piles

DMRC AM 2020
CGPSC AE 2017
TRB Poly. Tech. Lect. 2012

Ans. (a) Piles, which are used to protect waterfront structures against the impact from ships and other floating objects, are called fender piles.
End bearing piles : Used to transfer load through the piles tip to suitable bearing stratum, passing soft soil or water.
Friction piles : Used to transfer loads to a depth in a frictional material by means of skin friction along the surface area of the pile.
Friction piles are also called as "Floating piles", as they do not reach the hard stratum.

1640. As per the National Building Code of India 2016, the safe bearing capacity of a cohesion less fine loose dry sand is :
(a) 440 kN/m^2 (b) 300 kN/m^2
(c) 100 kN/m^2 (d) 500 kN/m^2
(e) 550 kN/m^2

CGPSC AE 2017

Ans. (c) As per the National Building Code of India 2016, the safe bearing capacity of a cohesion less fine loose dry sand is 100 kN/m^2.

1641. As per IS 8009 Part I : 1976, the influence factor for rigid square footing is :
(a) 0.82 (b) 1.00
(c) 1.56 (d) 2.20
(e) 5.62

DMRC AM 2020
CGPSC AE 2017

Ans. (a) As per IS 8009 part I : 1976, the influence factor for rigid square footing is 0.82.

1642. As per IS: 2974 Part I, the size of the foundation block (in plan) should be larger than bed plate of the machine with a minimum all-round clearance of :
(a) 70 cm (b) 110 cm
(c) 15 cm (d) 25 cm
(e) 10 cm

DSSSB AE 2019
CGPSC AE 2017

Ans. (c) As per IS : 297 part I, the size of the foundation block (in plan) should be larger than the bed plate of the machine with a minimum all round clearance of 15 cm.

1643. Dynamic formulae are best suited for :
(a) silt (b) coarse grained soils
(c) clayey soil (d) fine grained soil
(e) alluvial soil

CGPSC AE 2017

Ans. (b) Dynamic formula are best suited for coarse grained soil.

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1644. A wooden pile is being driven with a drop hammer weighing 20 kN and having a free fall of 1.0 m. The penetration in the last blow is 5 mm. Determine the load bearing capacity of the pile according to the Engineering News formula.

(a) 220 kN (b) 500.04 kN
(c) 765.20 kN (d) 111.11 kN
(e) 52.67 kN

CGPSC AE 2017

Ans. (d) Given, W = 20 kN
h = 1 m = 100 cm
S = 5 mm = 0.5 cm
As per Engineering news formula

$$Q_{allowable} = \frac{Wh}{FOS(S+C)}$$

C = 2.5 cm for drop hammer
FOS = 6

$$Q_{allowable} = \frac{20\times100}{6(0.5+2.5)} = 111.11 \text{ kN}$$

1645. .....failure is a mode of failure in a very deep footing in dense sand.

(a) Punching shear (b) General shear
(c) Local shear (d) Buckling
(e) Twisting

CGPSC AE 2017

Ans. (a) Punching shear failure is a made of failure is a very deep footing in dense sand.
Local shear failure :
- Failure is not sudden with well defined load
- Occurs in highly compressible soil associated with large settlement.

General shear failure : General shear failure, usually associated with medium to dense or stiff soils of relatively low compressibility.

1646. The ultimate bearing capacity of a soil, is:

(a) Load at which soil consolidates
(b) Load at which soil fails
(c) Total load on the bearing area
(d) Safe load on the bearing area

OPSC AEE 2015 PAPER–I

Ans : (b) The ultimate bearing capacity of a soil, is Load at which soil fails.

1647. Ultimate bearing capacity (in kN/sq.m) of a square footing on surface of a saturated clay having unconfined compressive strength of 60 kN/sq.m, using Skemption equation will be :

(a) 150 (b) 120
(c) 100 (d) 125
(e) 180

CGPSC AE Shift - 2 2014

Ans. (e) : $q_{nu} = C_u N_c$

where $C_u = \frac{60}{2} = 30 kN/m2$

$N_c = 6(1 + 0.2D_f/b)$
for $D_f = 0$, we get $N_c = 6$
$q_{nu} = 30 \times 6 = 180 \text{ kN/m}^2$

1648. If the load carrying capacity of an individual friction pile is 250 kN. What is the total load carrying capacity of a group (in kN) of 8 such piles with group efficiency factor of 0.75 ?

(a) 2000 (b) 1600
(c) 2400 (d) 1500
(e) 1875

CGPSC AE Shift - 2 2014

Ans. (d) : Total load carrying capacity = efficiency factor × load carrying capacity × No. of piles
= 0.75 × 8 × 250
= 1500

1649. Which type of foundation is used for the construction of school building on black cotton soil -

(a) Well foundation
(b) Stepped foundation
(c) Mat foundation
(d) Cantilever foundation

Rajasthan VP ITI 2018

Ans. (c) The foundation is suitable for black cotton/expansive soils bearing capacity is less for suitability of spread footing and wall footings. It is also recommended and or economical when the area of footings becomes greater than 50% area of the building.

1650. The modulus of subgrade reaction is obtained from load-deformation curve corresponding to the settlement of -

(a) 0.125 cm (b) 0.25 cm
(c) 0.375 cm (d) 0.5 cm

Rajasthan VP ITI 2018

Ans. (a) The modulus of sub-grade reaction is obtained from load-deformation curve corresponding to the settlement of 0.125 cm.

1651. The group efficiency of a pile group for closely spaced piles -

(a) will always be less than 100%
(b) will always be more than 100%
(c) often greater than 100% for cohesion-less soils
(d) often greater than 100% for cohesive soils

Rajasthan VP ITI 2018

Ans. (c) :

1652. For a pile load test in compression, the total weight to be arranged should be :

(a) at least 50% greater than the anticipated maximum test load
(b) at least 100% greater than the anticipated maximum test load
(c) at least 10% greater than the anticipated maximum test load
(d) equal to the anticipated maximum test load

UPRVUNL AE 2014

Ans. (c) : For a pile load test in compression, the total weight to be arranged should be at least 10% greater than the anticipated maximum test load.

1653. In context of a Strip Spread footing, which of the following expressions is correct for Length (L) and Width (W)?

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(a) L/W < 10 (b) L/W ≥ 10
(c) L = W (d) L/W < 5

UPRVUNL AE 2014

Ans. (b) : For strip footing = $\frac{L}{W} \geq 10$

1654. In the Plate Loading Test, the ratio of settlement of a shallow foundation to that of a test plate is :
(a) greater than 4
(b) less than or equal to 4
(c) greater than 8
(d) less than or equal to 8

UPRVUNL AE 2014

Ans. (b) : In the Plate Loading Test, the ratio of settlement of a shallow foundation to that of a test plate is less than or equal to 4.

1655. In case of an end bearing pile, the neutral planes will be located close to :
(a) the surface of compressible stratum
(b) the base of compressible stratum
(c) the middle of compressible stratum
(d) the surface of tensile stratum

UPRVUNL AE 2014

Ans. (b) : In case of an end bearing pile, the neutral planes will be located close to the base of compressible stratum.

1656. On which of the following factors does the allowable bearing pressure for a foundation depend?
(a) Allowable settlement
(b) Ultimate bearing capacity of soil
(c) Void ratio
(d) Both allowable settlement and ultimate bearing capacity of soil

UPRVUNL AE 2014

Ans. (d) : The allowable bearing pressure for a foundation depends upon the both allowable settlement and ultimate bearing capacity of soil.

1657. For determining the bearing capacity of soil by plate loading test, the size of square bearing plate should be :
(a) less than 300 mm
(b) between 300 mm and 750 mm
(c) between 750 mm and 1 m
(d) greater than 1 m

Rajasthan JEn (Degree) 2016
UPRVUNL AE 2014

Ans. (b) : For determining the bearing capacity of soil by plate loading test, the size of square bearing plate should be between 300 mm and 750 mm.

1658. In case of continuous footings, load intensity is given in terms of :
(a) load per unit area
(b) load per unit volume
(c) load per unit length
(d) load per unit equivalent area

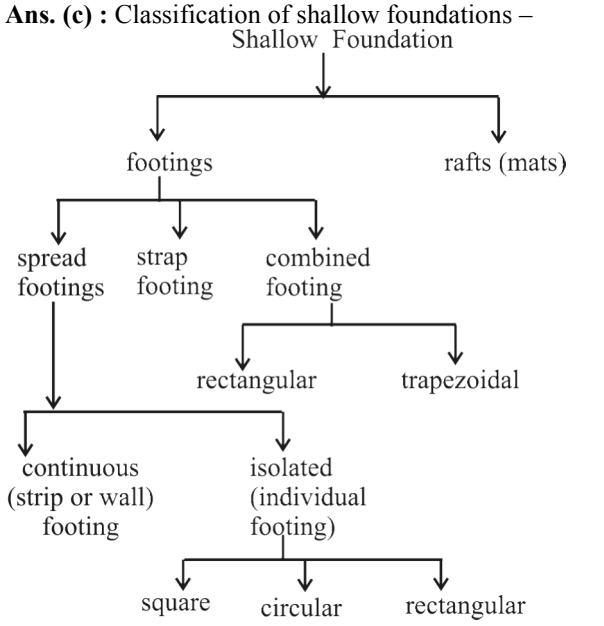
UPRVUNL AE 2014

Ans. (c) : In case of continuous footings, load intensity is given in terms of load per unit length.

1659. Which of the following is not a type of shallow foundation?
(a) Spread Footing
(b) Combined Footing
(c) Discrete Footing
(d) Continuous Footing

UPRVUNL AE 2014

Ans. (c) : Classification of shallow foundations –



1660. The primary purpose of ---------- is to obtain a load settlement curve of a soil at a particular depth.
(a) plate load test
(b) cone penetration test
(c) standard penetration test
(d) settlement test

UPRVUNL AE 2014

Ans. (a) : The primary purpose of plate load test is to obtain a load settlement curve of a soil at a particular depth..

1661. The ultimate settlement of a soil is directly proportional to:
(a) Depth of the compressible soil strata
(b) Compressive index
(c) Void ratio
(d) Both (a) and (b)

OPSC AEE 2015 PAPER–I

Ans : (d) The ultimate settlement of a soil is directly proportional to depth of the compressible soil strata, compressive index.

1662. In the Standard Penetration Test what is the driving depth of the sample for which the number of hammer blows are counted?
(a) 100 mm (b) 150 mm
(c) 200 mm (d) 300 mm

OPSC AE Paper- II 2019

Ans. (d) : In the Standard Penetration Test 300 mm depth of the sample is driving for which the number of hammer blows are counted.

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1663. The contact pressure for cohesionless soil at the centre is

(a) sinusoidal (b) maximum
(c) exponential (d) minimum

BPSC AE Aug. 2019 Paper-V

Ans. (b) : Contact pr. at center for cohesionless soil is minimum for flexible pavement and maximum for rigid pavement.

1664. Rise of water table in cohesionless soils upto ground surface reduces the net ultimate bearing capacity approximately by

(a) 25% (b) 33%
(c) 50% (d) 75%

Gujarat PSC AE (N.W.R.) 2020
GPSC AE June 2019
ISRO Scientist/Engineer 2017
UPPCL AE 2014
OPSC Civil Services Pre 2011

Ans. (c) : According to IS 1904-1978 recommends, for non-cohesive soils the values net ultimate bearing capacity should be reduced by 50% if water table is above or near base of footing.

1665. The ultimate bearing capacity of a 700-mm diameter plate as determined from field load tests is 300 kN/sq.m. The ultimate bearing capacity (kN/sq.m) of a circular footing with a diameter of 2.1 m in a sandy soil will be:

(a) 100 (b) 150
(c) 300 (d) 900

UPRVUNL AE 2016

Ans.: (d) q_{uf} for circular footing = ?
q_{up} for circular plate = 300 kN/m^2
B_p = width/diameter of plate = 700mm
B_f = width/ diameter of footing = 2.1 m
for sandy soil

$$\frac{q_{uf}}{q_{up}} = \frac{B_f}{B_p}$$

$$q_{uf} = \frac{2.1\text{m}}{0.7\text{m}} \times 300\,\text{kN/m}^2$$

$$q_{uf} = 900\,\text{kN/m}^2$$

1666. Foundation of a structure resting on sand settles by 20 mm. Differential settlement is expected be 30% of total settlement. If the column spacing for the structure is 6 m, then the angular distortion will be :

(a) 1/1000 (b) 1/100
(c) 1/50 (d) 1/500

BHEL ET 2019

Ans. (a) :

1667. A continuous wall footing 1.5 m wide supports a load of 596 kN/m. The unit weight of the soil beneath the foundation is 18.6 kN/m^3. The soil has cohesion of 14 kPa and an angel of internal friction of 25° with Terzaghi bearing capacity factors, N_c = 25.1 and N_γ = 9.7. If the footing is placed very near the ground surface, the factor of safety against bearing capacity failure is most nearly :

(a) 0.8 (b) 1.22
(c) 1.82 (d) 2.82

HPPSC Poly. Lect. 2016

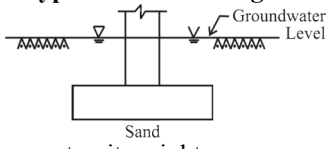
Ans : (b) $q_u = CN_C + \gamma D_f N_q + \frac{1}{2} B \times \gamma\, N_\gamma$

$$= 14 \times 25.1 + 0 + \frac{1}{2} \times 1.5 \times 18.6 \times 9.7$$

$$= 486.7$$

$$\text{FOS} = \frac{596}{486.7} = 1.22$$

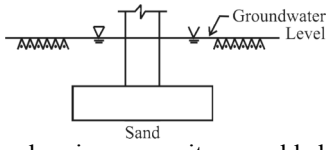
1668. A bridge footing is to be constructed in sand. The groundwater level is at the ground surface. The ultimate bearing capacity would be based on what type of soil unit weight?



(a) Buoyant unit weight
(b) Saturated unit weight
(c) Dry unit weight
(d) Total unit weight

HPPSC Poly. Lect. 2016

Ans : (a)



The ultimate bearing capacity would be based on buoyant unit weight type of soil.

1669. In the design of strap footing, which of the following assumption is not made?

(a) Soil pressure varies linearly
(b) The interior footing is centrally loaded
(c) The strap is weightless
(d) The strap is perfectly rigid

IRCON AE 2017

Ans. (a) : Assumption in design of strap footing–
i.) Strap must be rigid to control rotation of exterior
ii.) A strap footing is commonly used to footing connect an eccentrically loaded column at centre to an interior column footing having weightless strap.
iii.) Soil pr. is not linear.

1670. The method used for improvement of bearing capacity of cohesive soil in the field is:

A. Per compression
B. Compaction piles
C. Stone column

(a) Only B and C (b) Only A and C
(c) Only A (d) Only A and B

IRCON AE 2017

Ans. (b) : The method used for improvement at bearing capacity of cohesive soil in the field is stone column (It helps to limit the amount and consequences of future liquefaction by densify the soil through vibration and introducing stone into the soil) and compaction piles are also used.

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1671. Which of the following is a type of cased cast in situ concrete piles?

(a) Franki pile (b) Simplex piles
(c) Vibro piles (d) Swage piles

IRCON AE 2017

Ans. (d) : A pipe pile, having a thin wall, the bottom of pile is closed with a precast point. It is type of cased-cast-in-situ concrete pile are widely used in case of hard soils.

1672. According to IS code, maximum settlement permitted in case of raft foundation on clay soil is:

(a) 25 mm (b) 60 mm
(c) 40 mm (d) 70 mm

IRCON AE 2017

Ans. (b) : As per IS code recommendations:
Maximum settlement for raft foundation on clay = 65 mm.
Maximum settlement for raft foundation on sand = 40 mm.

1673. Negative skin friction on a pile develops when:

(a) Ground water table rises
(b) The soil surrounding it settles more than the pile
(c) Ground water table decreases
(d) The soil in which it is driven sandy soil

IRCON AE 2017
WBPSC AE (PHE) 1998

Ans. (b) : Negative skin friction is usually a downward shear drag acting on a pile group because of downward movement of surrounding soil relatives to the piles. The shear drag movements are anticipated to occur when a penetrates into compressible soil layer that can consolidate as soil settles more than the pile.

1674 According to IS code, in plate load test following load arrangements are used:

A. Gravity loading platform method
B. Reaction truss method

(a) Neither A nor B (b) Only A
(c) Only B (d) Both A and B

IRCON AE 2017

Ans. (d) : Plate load test is a field test for determining the ultimate bearing capacity of soil. The load is transmitted to the column by given two methods–
i. By gravity loading platform method
ii. By loading truss method.

1675. Height of fall in standard penetration test for a mass of 63.5 kg is:

(a) 750 mm (b) 650 mm
(c) 775 mm (d) 700 mm

IRCON AE 2017

Ans. (a) : Standard Penetration Test (STP) is an in situ dynamic penetration test designed to provide information on the soil properties.
Weight of hammer = 65 kg.
Free fall height of hammer = 750 mm.

1676. The pressure sustained per unit deformation of subgrade at specified deformation level using specified plate size is termed as:

(a) CBR
(b) Safe bearing capacity
(c) Modulus of subgrade reaction
(d) Tangent modulus

NPCC AE 2017

Ans. (c) : Reaction P = kΔ (pressure on soil)
k = modulus of sub grade reaction
Δ = deflection
It is based on plate load test.

1677. The Ultimate bearing capacity as per Terzhaghis Analysis is given by :

(a) $q_f = 0.75\,\gamma\,BN_\gamma + \gamma\,D_f N_q$
(b) $q_f = CN_c + 0.5\,\gamma\,BN_\gamma + N_q$
(c) $q_f = CN_c + 0.5\,\gamma\,BN_\gamma + \gamma\,D_f N_q$
(d) $q_f = CN_c + 0.35\,\gamma\,B + \gamma\,D_f N_q$

GPHC AE 2018

Ans : (c)
Terzaghi's ltinate bearing capacity formula :

$$q_f = CN_c + \gamma D_f N_q + 0.5\gamma BN_\gamma$$

C → Unit cohesion
γ → Unit weight of soil
D → Depth of foundation
B → Width of foundation
N_C, N_q, N_γ → Bearing capacity factor.

1678. Terzaghi's bearing capacity factors are function of

(a) C and ϕ
(b) only ϕ
(c) ϕ and depth of foundation
(d) ϕ depth and width of foundation

WBPSC AE 2016

Ans. (b) : Terzaghi's bearing capacity factors are function of only ϕ.

1679. The bearing capacity of a strip footing resting on a saturated clay is 120 kN/m^2. The bearing capacity of a circular footing with diameter same as the width of strip footing resting on same soil will be

(a) more than 120 kN/m^2
(b) equal to 120 kN/m^2
(c) less than kN/m^2
(d) Any one of the above

WBPSC AE 2016

Ans. (a) :

1680. Pile foundations are normally used

(a) In soft clayey soils
(b) In heavy-loaded situations
(c) In loose sandy soils
(d) When required bearing area is not available

WBPSC AE 2012

Ans. (b) : Pile foundations are normally used in heavy-loaded situations.

1681. A 30 cm square bearing plate settles by 1.5 cm in a plate loading test on a cohesive soil when the intensity of loading is 2 kg/cm^2. The settlement of a prototype shallow footing 1.0 m square under the same intensity of loading is

(a) 1.5 cm (b) 3 cm

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(c) 5 cm (d) 10 cm

WBPSC AE 2014

Ans. (c) : Given,

$S_p = 1.5$ cm, $B_P = 0.3$ m, $B_f = 1$m

$$\frac{S_f}{S_p} = \frac{B_f}{B_p} \quad \text{(For clay soil)}$$

$$S_f = 1.5 \times \frac{1}{0.3} = 5$$

$S_p = 5$ cm

1682. Deep foundations are normally defined as those that have depth to width ratio

(a) Greater than 4 (b) Less than 4

(c) Equal to 2 (d) Greater than 2

WBPSC AE 2012

Ans. (d) : Deep foundations are normally defined as those that have depth to width ratio Greater than 2.

$$\frac{D}{B} > 2$$

1683. The additional pile driven to increase the capacity of supporting loads on vertical piles is known as

(a) sinking pile (b) raymond pile

(c) eccentric pile (d) simplex pile

WBPSC AE 2007

Ans. (b) : The additional pile driven to increase the capacity of supporting loads on vertical piles is known as raymond pile.

1684. The net ultimate bearing capacity of a strip footing resting on clay compared to a square footing of same size is

(a) more (b) less

(c) equal (d) cannot be predicted

WBPSC AE 2007

Ans. (a) : The net ultimate bearing capacity of a strip footing resting on clay compared to a square footing of same size is more.

$q_{nu} = q_u - \gamma.D_f$

Where, q_{nu} = net ultimate bearing capacity.

q_u = ultimate bearing capacity.

D_f = Depth of footing.

1685. The capacity of a group of piles driven in clay is

(a) equal to single pile capacity

(b) less than that of pile capacity

(c) greater than that of pile capacity

(d) multiplied by number of piles

WBPSC AE 2007

Ans. (d) : The capacity of a group of piles driven in clay is multiplied by number of piles.

1686. The load carrying capacity of a pile, driven with fall of 1 m by 1800 kg. drop hammer, penetrating 3 mm in the last blow is

(a) 100 t (b) 50 t

(c) 20 t (d) 10 t

WBPSC AE 2003

Ans. (d) :

1687. The maximum settlement of raft foundation on sand should be limited to the following values:

(a) 20 to 40 mm (b) 40 to 65 mm

(c) 65 to 80 mm (d) 80 to 10 mm

WBPSC AE 2001

Ans. (b) : The maximum settlement of raft foundation on sand should be limited to the following values 40 to 65 mm.

1688. The pile tip and shaft resistances can be assessed separately from load test by

(a) equilibrium method

(b) constant rate of penetration method

(c) Housed method

(d) Cyclic method

WBPSC AE (PHE) 1998

Ans. (d) : The pile tip and shaft resistances can be assessed separately from load test by cyclic method.

1689. Two footing A (2m × 2m) and B (4m × 4m) are placed on a saturated cohesive soil and subjected to the same intensity of pressure. The immediate settlement of B will be

(a) same as A

(b) twice the settlement of A

(c) four times the settlement of A

(d) very small compared to A

WBPSC AE (PHE) 1998

Ans. (b) :

1690. For footings on piles the thickness of concrete tops of piles shall not be less than

(a) 150 mm (b) 200 mm

(c) 250 mm (d) 300 mm

WBPSC AE (PHE) 1998

Ans. (d) : For footings on piles the thickness of concrete tops of piles shall not be less than 300 mm.

1691. The ultimate bearing capacity of a shallow foundation laid on a cohesion-less soil medium was estimated as 200 kN/sqm, when the water table was far below. All other conditions remaining the same, the ultimate bearing capacity of the foundation, when the water table was risen to ground level, is:

(a) 400 kN/sqm (b) 200 kN/sqm

(c) 100 kN/sqm (d) 50 kN/sqm

TSPSC Deputy Engineer 2015

Ans. (c) : when the water table was far below. All other conditions remaining the same, the ultimate bearing capacity of the foundation reduce to 50%.

So, ultimate bearing capacity of foundation

$= 200/2 = 100$ kN/m^2

1692. The ultimate load carrying capacity of an end bearing type of bored cast-in-situ circular pile is estimated as 100 kN All other parameters remaining the same, the capacity of the Pile if the diameter is doubled, is:

(a) 100 kN (b) 200 kN

(c) 300 kN (d) 400 kN

TSPSC Deputy Engineer 2015

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Ans. (d) : Ultimate load bearing capacity
$= N_C \times C \times$ pile tip area

$$(q) = N_C \times C \times \frac{\pi}{4} \times d^2$$

$$q \propto d^2$$

$$\frac{q_1}{d_1^2} = \frac{q_2}{d_2^2}$$

$$\frac{100}{d^2} = \frac{q_2}{4d^2} \Rightarrow q_2 = 400 \text{ kN}$$

1693. Which of the following statements about Dynamic Formulae is correct?
(a) Applicable for the bored cast in situ piles only.
(b) Overestimate the capacity in sensitive clays.
(c) Underestimate the capacity wherein localized momentary liquefaction takes place while driving the pile.
(d) Set value is average of the first few blows.

TSPSC AEE 2017

Ans. (c) : The correct statements about Dynamic Formulae is underestimate the capacity wherein localized momentary liquefaction takes place while driving the pile.

1694. As per IS : 3955–1967, tilt of a well foundation should generally be limited to
(a) 1 in 20 (b) 1 in 40
(c) 1 in 60 (d) 1 in 100

TSPSC AEE 2017

Ans. (c) : As per IS : 3955–1967, tilt of a well foundation should generally be limited to 1 in 60.

1695. The load carrying capacity of an end bearing pile is found to be 400 kN. All other parameters remaining same, what will be capacity if its diameter is halved.
(a) 100 kN (b) 500 kN
(c) 1000 kN (d) 2000 kN

TSPSC AEE 2017

Ans. (a) : Ultimate bearing capacity (q) = $N_c \times C \times$ pile tip area.

$$q = N_c \times C \times \frac{\pi}{4} \times d^2$$

$$q \propto d^2$$

$$\frac{q_1}{d_1^2} = \frac{q_2}{d_2^2}$$

$$\frac{400}{d^2} = \frac{q_2}{(d/2)^2} \Rightarrow q_2 = 100 \text{ kN}$$

1696. Which type of caissons are preferred in emergency conditions where time for sinking of caissons is not available?
(a) Box caisssons
(b) Pneumatic caisssons
(c) Open caissons
(d) Box, Pneumatic and Open caissons

TSPSC AEE 2017

Ans. (a) : Which type of caissons are preferred in emergency conditions where time for sinking of caissons is not available box caisssons.

1697. The permissible settlement is the maximum in the case of
(a) Isolated footing on clay
(b) Raft on clay
(c) Isolated footing of sand
(d) Raft on sand

Chandigarh AE 2017

Ans. (b) : The permissible settlement is the maximum in the case of raft on clay.

1698. The adhesion factor for drilled piers on clay is usually taken as
(a) 1.0 (b) 0.80
(c) 0.60 (d) 0.40

Chandigarh AE 2017

Ans. (d) : The adhesion factor for drilled piers on clay is usually taken as 0.40.

1699. An infinite fill is to be placed with a granular soil (bulk unit weight = 20 kN/m^3) on a saturated clayey silt deposit (undrained shear strength = 25 kPa). Assuming undrained general shear failure and bearing capacity factor 5.7. The maximum height (in m) of the fill at the point of failure is
(a) 7.125 (b) 5.0
(c) 0.7125 (d) 14.250

APTRASCO 2017

Ans : (a) : Maximum height of filter at the considered failure h-meter

$$CN_C = \gamma_b \times h$$

$$25 \times 5.7 = 20h$$

$$h = \frac{25 \times 5.7}{20}$$

$$\boxed{h = 7.125 \text{ m}}$$

11. Soil Stabilization

1700. Bitumen stabilization is suitable for
(a) Clayey soil (b) Sandy soil
(c) Peaty soil (d) Silty soil

WBPSC AE (PHE) 1998

Ans. (b) : Bitumen stabilization is suitable for sandy soil.

1701. The most effective and economical method to stabilize Black Cotton Soils is
(a) Cement stabilization
(b) Lime stabilization with suitable additives
(c) Thermal stabilization
(d) None of the given answers

APPSC AEE (Mains) 2016

Ans. (b) : Highly unstable plastic and swelling clayey soil such as black cotton soils may be stabilized by hydrated lime.
Normal requirement is 4% to 6% of the soil.

1702. Best suited material for stabilization of black cotton soils is :
(a) Sodium silicate (b) Calcium chloride
(c) Hydrated lime (d) Bitumen

Rajasthan AE (Nagar Nigam) 2016

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Ans. (c) Highly unstable plastic and swelling clayey soils such as black cotton soils may be stabilized by hydrated lime (Calcium hydroxide).

1703. Which type of stabilization will be the best for black cotton soil

(a) Mechanical stabilization
(b) Soil-bitumen stabilization
(c) Soil-cement stabilization
(d) Soil-lime stabilization

RPSC AE (GWD) 2014

Ans : (d)

- Highly unstable, plastic and swelling clayey soil such as "black cotton soils" may be stabilized by hydrated lime (calcium hydroxide).
- Normal dose requirement is 4% to 6% of the soil weight.

1704. Lime stabilization is very effective in treating -

(a) Silty soil (b) Sandy soil
(c) Plastic clayey soil (d) Non-plastic soil

GPSC AE (CLASS 1 & 2) 2019
RPSC VPITI 2012

Ans : (c)

- Black cotton soil is used to high amount of clay mineral montmorillonite soil.
- Plastic clayey soil such as "black cotton soils" may be stabilized by hydrated lime (calcium hydroxide).

1705. A wet cohesive soil is most effectively stabilized by the addition of :

(a) Lime (b) Bitumen
(c) Fly Ash (d) Cement

Punjab PSC SDE 2016
[UPPSC State Eng. A.E. 2008]

Ans : (a) Wet cohesive soils are most effectively stabilized by the addition of lime because lime reduces the plasticity of wet soil quickly.

1706. Lime stabilization method is particularly suitable and has been found successful in stabilizing :

(a) Clayey soil with expansive nature in water logged areas
(b) Sandy desertic area
(c) Soil having liquid limit less than 30%
(d) Wind blown soil deposits

[UPPSC State Eng. A.E. 2003]

Ans : (a) Lime stablization: Lime has been widely used as a modifier for clay soil or as a binder.
When clay soils with high plasticity are treated with lime, the plasticity index is decreased and the soil becomes flexible and easy to pulverize having less affinity with water.

1707. Hydrated lime $Ca(OH)_2$ can be effectively used for stabilization of-

(a) Plastic equilibrium (b) Radial shear
(c) Elastic equilibrium (d) Linear shear

MPPSC AE 2017

Ans. (c) : Lime is used for stabilization of plastic clayey soils. Ca^{2+} ions migrate to the surface of the clay and displace water and other ions. It reduces the plasticity index of the soil and stabilizes the soil.

1708. For clays of moderate to high plasticity, which of the following agent is the best for improving soil properties?

(a) Cement (b) Fly ash
(c) Lime (d) Slag

(LMRC AE 2017 I–Shift)

Ans : (c)

- Lime its better workability early stiffening. good strength and resistance.
- Highly unstable, plastic and swelling clayey soil.
- Normal dose is 4% to 6% of the soil weight.
- Generally 5% of soil is the requirement of cement for sandy soil and 15% for clays soil. This requirement may be reduced.

1709. If water table is high in stiff clay, which of the following stabilizing method should be followed?

(a) Bore hole be kept full of water to a level higher than ground water table
(b) Bore hole be kept full of drilling mud to a level higher than ground water table
(c) Bore hole be cased with pipe casing
(d) Nothing needs to be done

[HPPSC AE 2016]

Ans. (d) : Stiff clay are stable and they do not require any stabilization.

1710. In the analysis of sediment transport the soil is assumed as

(a) Coherent (b) Incoherent
(c) Cohesive (d) Medium texture

TNPSC AE 2019

Ans. (b) For sediment to transport the soil should be inchorent.

1711. Mud-Pumping is

(a) Ejection of soil slurry
(b) Failure of soil sub grade
(c) Failure of R.C. Slab
(d) Cracks in pavement

TNPSC AE 2019

Ans. (a) The process of ejecting out the soil slurry through joints and cracks under the action of wheel loads is known as mud-pumping.

1712. Minimum thickness of a layer of fine sand required to cutoff capillary rise of water completely should be

(a) 40 cm (b) 52 cm
(c) 64 cm (d) 76 cm

Gujarat PSC AE 2017

Ans : (d) Minimum thickness of a layer of fine sand required to cutoff capillary rise of water completely should be 76 cm.

1713. With the increase in value of plasticity index of soil, the quantity of lime required for stabilization,

(a) Increases
(b) Decreases
(c) Remains unaffected
(d) first increases and then decreases

Rajasthan AE (Nagar Nigam) 2016

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Ans. (a) : The plasticity index of the soil should be greater than 10.
With increase in value of plasticity index of soil, the quantity of lime required for stabilization is increase.

1714. Mechanical stabilization requires
(a) Addition of chemical to soils
(b) Mixing of two or more types of natural soils
(c) Mixing of lime with soils
(d) Addition of cementing material to soils

GPSC AE Dec 2018
WBPSC AE 2014

Ans. (b) : Mechanical stabilization requires mixing of two or more types of natural soils.

1715. Lime stabilization of clayey soil generally leads to
(a) decrease in shrinkage limit
(b) decrease in plastic limit
(c) increase in liquid limit
(d) flocculation of particles

MPPSC AE 2014

Ans. (d) : Lime stabilization of clayey soil is generally leads to flocculation of particles.

1716. When bitumen is added to soil mainly it
(a) increases cohesion
(b) accelerates cohesion
(c) acts as water proofing agent
(d) fills void

WBPSC AE 2014

Ans. (a) : When bitumen is added to soil mainly it increases cohesion.

1717. The bearing capacity of a soil cannot be increased by
(a) chemical treatment (b) grouting
(c) compacting (d) moistening the soil

Chandigarh SDE 2017

Ans. (d) : The bearing capacity of a soil can be increased by chemical treatment, grouting, compacting.

1718. For clays of moderate to high plasticity, which of the following agent is the best for improving soil properties?
(a) Cement (b) Fly ash
(c) Lime (d) Slag

LMRC AM 2015

Ans (c) Lime has been widely used as a modifier for clayey soil or as a blinder.

12. Soil Exploration

1719. The observations of a Standard Penetration Test are reported as 10/15/20. Assuming correction for overburden is not required and correction for dialatancy only is require, the corrected N-value is:
(a) 45 (b) 35
(c) 25 (d) 15

TSPSC Deputy Engineer 2015

Ans. (c) : $N_R = 15 + 20 = 35$
Record value

$$\text{Correct value, } N_c = 15 + \frac{1}{2}(N_R - 15)$$
$$= 15 + \frac{1}{2}(35 - 15)$$
$$= 25$$

1720. In a standard penetration test, the blows are recorded as 1/20/30. If correction factor for overburden is 0.90, the standard penetration resistance, N-value corrected for overburden and dilatancy is
(a) 57 (b) 32
(c) 37 (d) 30

TSPSC AEE 2017

Ans. (d) : $N_R = 20+30 = 50$
$N_R = 50 \times 0.9 = 45$

$$N_C = 15 + \frac{1}{2}(N_R - 15)$$
$$= 15 + \frac{1}{2}(45 - 15)$$
$$N_C = 30$$

1721. Penetration resistance 'N' in a Standard Penetration Test (SPT) is the total number of blows for
(a) the first and second 15 cm of penetration
(b) the second 15 cm of penetration
(c) any 30 cm of penetration
(d) the second 30 cm of penetration

UKPSC(Urban engineering)-2011

Ans. (d) : Penetration resistance 'N' standard penetration test is the total number of blow for the second 30 cm of penetration.

1722. In the Standard Penetration Test, the split spoon sampler is penetrated into soil stratum by giving blows from a drop weight whose weight (in Kg) and free fall (in cm) are respectively
(a) 65 and 75 (b) 75 and 65
(c) 65 and 30 (d) 30 and 65

GPSC AE (CLASS 1 & 2) 2019
UJVNL AE 2016

Ans : (a) : In SPT → Split spoon sampler is used
weight (w) = 65 kg
Height of free fall Hammer = 75 cm
Standard penetration = 300 mm

1723. The standard penetration test is useful to measure:-
(a) Consistency of clays
(b) Shear strength of soft clays
(c) Shear strength of sands
(d) Shear strength of stiff clay

UPRVUNL AE 2014
UPPCL AE 2014

Ans. (c) : The standard penetration test is useful to measure shear strength of sands.

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1724. The standard penetration test value in a deposit of fully saturated fine silty sand was 43. The corrected SPT value for dilatancy

(a) 29 (b) 33
(c) 39 (d) 45

GPSC Assistant Prof. 2016

Ans. (a) : For saturated fine sand and silt, when $(N_1 > 15)$

$$N_2 = \frac{1}{2}(N_1 - 15) + 15 \qquad N_1 = 43 \text{ (given)}$$

$$\therefore N_2 = \frac{1}{2}(43-15) + 15 = 29$$

1725. Standard Penetration Test (SPT) gave an average count of 35 in fine sand below water table, then what is the correct SPT count?

(a) 15 (b) 20
(c) 25 (d) 35

GPSC AE Dec 2018

Ans. (c) : Given, avg. SPT count, N' = 35

$$N'' = 15 + \frac{(N' - 15)}{2}$$

$$= 15 + \frac{(35-15)}{2}$$

Correct SPT count, N'' = 25

1726. The range of N values for a very loose sand is

(a) 0 to 4 (b) 4 to 10
(c) 10 to 30 (d) 30 to 50

GPSC AE March 2018

Ans. (a) : SPT value = N – value

N value	Compactness
0 - 4	Very loose
4 - 10	Loose
10 - 30	Medium
30 - 50	Dense

1727. In a standard penetration test (SPT) for soil, the SPT number is

(a) Penetration depth of sampler in the soil for any given number of blows
(b) Number of blows required for 30 cms of penetration of sampler in the soil
(c) Number of blows required for 20 cms of penetration of sampler in the soil
(d) None of the above

ISRO Scientist/Engineer 2019

Ans : (b) : (SPT) Standard Penetration test for soil, the SPT number is 'N' number of blows required for 30 cm of penetration of sampler in the soil. (IS: 2131 - 1981). In very fine silt, saturated sand an apparent increase in resistance occurs. Terzaghi and Reck have recommended the use of an equivalent penetration resistance -N_c in place of actually observed value of N. When N is greater than 15, then,

$N \rightarrow$ SPT Number.

1728. The observe N value from a standard penetration test conducted on a saturated sandy soil is 30; the N-value corrected for dilatancy may be taken as :

(a) 15 (b) 20
(c) 23 (d) 29

GPSC AE Class (1&2) Paper - 2 2017

Ans. (c) : Given, Record value N = 30

N-Value of corrected dilatancy

$$N_1 = 15 + \frac{1}{2}(N - 15)$$

$$N_1 = 15 + \frac{1}{2}(30-15) = 15 + 7.5$$

$$= 22.5 \simeq 23$$

1729. When the observed value of N exceeds 15, the corrected penetration number N_c as per Terzaghi and Peck recommendation in the silty fine sands will be

(a) $15 + \frac{1}{2}(N_R + 15)$ (b) $15 - \frac{1}{2}(N_R - 15)$
(c) $15 + \frac{1}{2}(N_R - 15)$ (d) $15 - \frac{1}{2}(N_R + 15)$

Where :

N = Penetration number, and
N_R = Recorded value

DMRC AM 2020
ESE 2020
Gujarat PSC AE (N.W.R.) 2020
RPSC AE 2018
MPPSC AE 2014

Ans. (c) : As per Terzaghi and Peck, 1948 corrected penetration number,

$$N_C = 15 + \frac{1}{2}(N_R - 15)$$

where,

N_R = recorded value
N_C = corrected penetration value

Two corrections, which are usually applied, on the observed N-value are–

(i) over burden correction
(ii) dilatancy correction

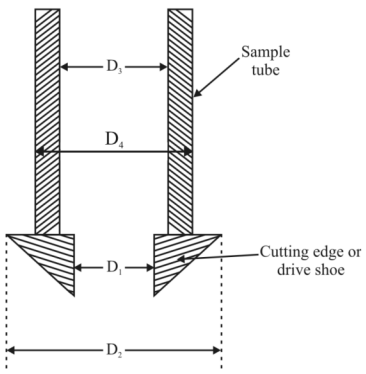
As overburden pressure is not known and fine sand is below water table so we will only apply dilatancy correction.

1730. The area ratio of thin wall sampler should not normally exceed more than :

(a) 8% (b) 16%
(c) 24% (d) 32%

TRB Poly. Tech. Lect. 2012
UPPSC State Eng. A.E. 2007(II)
UPPSC State Eng. A.E. 2003

Ans : (a)



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Where,

D_1 = inside diameter of cutting edge

D_2 = outside diameter of cutting edge

D_3 = inside diameter of sampling tube

D_4 = outside diameter of sampling tube

Area ratio,

$$A_r\% = \frac{D_2^2 - D_1^2}{D_1^2} \times 100\%$$

- Should be as low as possible
- For soft sensitive soil, > 25%
- Preferably, ≤ 10%

1731. In order to minimize sampling disturbance the area ratio A_r, $\left(A_r = \frac{D_o^2 - D_i^2}{D_i^2}\right)$ should be:

(a) zero
(b) as high as possible
(c) as low as possible
(d) equal to unity

UPPSC AE 12.04.2016 Paper–I

Ans : (d) To minimize sampling disturbance the area ratio. Ar, should be equal to unity.

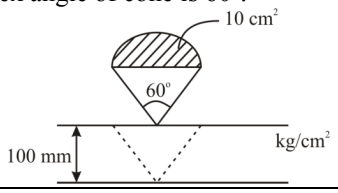
1732. The cone penetration test (CPT) has four main advantages over the usual combination of boring, sampling and standard penetration testing; which of the following is not an advantage?

(a) It provides a continuous or virtually continuous record of ground conditions
(b) It avoids the disturbance of the ground associated with boring and sampling, particularly that which occurs with the Standard Penetration Test (SPT)
(c) It is an expensive as conventional boring method
(d) It is faster by a factor of about 10

(UP RVNL AE 2016)

Ans.: (c) Cone penetration Test (CPT) :

- Also known as the Dutch cone for CPT gives a continuous record of variation of both cone resistance and friction resistance with depth.
- CPT however does not yield any sample.
- Suitable for soft clay, silts and fine to medium sand deposits.
- It is as expensive as conventional boring method.
- The vertex angle of cone is 60°.



1733. Undisturbed samples are obtained by:

(a) Direct excavations
(b) Thin walled samplers
(c) Thick walled samplers
(d) None of the above

Uttrakhand AE 2013)

Ans : (b) undisturbed samples are obtained by thin walled samplers.

1734. As per IS : 4453 the clear working space at the bottom of a soil exploration pit should be–

(a) 1.0 m × 1.0 m (b) 1.2 m × 1.0 m
(c) 1.2 m × 1.2 m (d) 0.9 m × 1.2 m

(UKPSC AE (Paper-I) 2007)

Ans : (c) IS : 4453 the clear working space at the bottom of a soil exploration pit should be 1.2 m × 1.2 m.

1735. Significant depth of exploration in soil is given by :

(a) 1.0 q (b) 0.5 q
(c) 0.4 q (d) 0.2 q

q = Intensity of loading

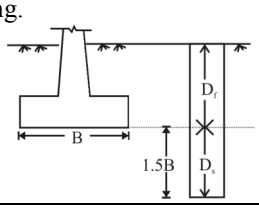
[UPPSC State Eng. A.E. 2007(II)]

Ans : (d) The significant depth of exploration is taken as the depth of which the vertical stress is 20% of the load intensity

$D_s = 0.2\ q$

q = load intensity

If width of footing is B, the depth of exploration should be about 1.5 times the width of footing (B) from the bottom of footing.



1736. Consider the following statements-

1. **Dynamic cone penetration test for site investigation is based on the principle that elastic shock waves travel in different materials at different velocities.**
2. **Electrical resistivity method of subsurface investigation is capable of detecting only the strata having different electrical resistivity.**
3. **In-situ vane shear test is useful for determining the shear strength of very soft soil and sensitive clays and is unsuitable for sandy soil.**

Which of these statements is/are correct?

(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 2 alone

MPPSC AE 2017

Ans. (b) : Dynamic Cone Penetration Test (DCPT) is based on shear resistance provided to cone by the soil and not an elastic shock waves.

1737. During a sampling operation, the drive sampler is advanced 600 mm and the length of the sample recovered is 525 mm. What is the recovery ratio of the sample?

(a) 0.125 (b) 0.140
(c) 0.143 (d) 0.875

MPPSC AE 2017
ESE 2006

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Ans. (d) :

$$\text{Recovery ratio} = \frac{\text{Length of sample recovered}}{\text{Length of drive sample}}$$

$$= \frac{525}{600} = 0.875$$

1738. For sampling natural sands and other soft and wet soils satisfactorily, the most suitable soil sampler is

(a) open drive thin-walled tube sampler
(b) standard spilt-spoon sampler
(c) stationary piston sampler
(d) rotary sampler

BPSC AE 2019 Paper (V) Section-I
TNPSC AE 2018

Ans. (c) : Depending upon the mode of operation, the sampler are divided into four types-
(i) **Open drive sampler–**Used for sampling of most soft or moderately stiff cohesive soil.
(ii) **Piston sampler–**Used for sampling natural sand and soft and wet soil.
(iii) **Rotary sampler–**Used for sampling in firm to hard cohesive soil and particularly in rocks.
(iv) **Split-spoon sampler–**Used for sampling cohesion less soil, for which undisturbed samples cannot be simply obtained.

1739. The maximum force per unit area that needs to be placed over an expansive soil to prevent value increase is known as

(a) Pore water pressure (b) Swelling pressure
(c) Safe bearing capacity (d) Plastic limit

APPSC AEE (Mains) 2016

Ans. (b) : Swelling pressure $= \frac{\text{Maximum force}}{\text{Area}}$, that needs to be placed over an expansive soils to prevent value increase.

1740. Study the statements listed below :
1. Area ratio should be low
2. Cutting edge should be thick
3. Outside clearance should be low
Which of these statements is/are correct for a good quality soil samples ?

(a) 1 and 2 (b) 1 and 3
(c) 2 and 3 (d) Only 1

HPPSC AE 2016
ESE 2016

Ans. (b) : To obtain good quality sample the area ratio of sampler should be as low as possible. The inside clearance should lie between 1 to 3 percent and outside not be much greater than inside clearance. The wall of sampler should be smooth and cutting edge should be thin.

1741. Auger boring is suited for which of the following exploratory strata?

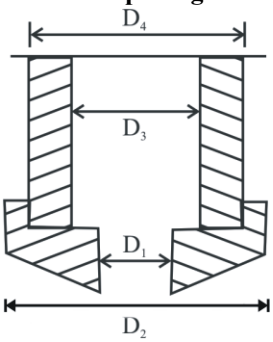
(a) Partly saturated sands, silts and medium to stiff cohesive soils
(b) All types of soils and rocks except fissured rocks
(c) Practically all types of soils except hard and cemented soil or rock
(d) All types of soils and rocks except loose sands and soft sticky clays

GPSC AE (CLASS 1 & 2) 2019

Ans : (a) : Boring is done for soil depth greater than 3m or when difficult ground water condition are wet.

- Auger Boring:- done when soil sample are distributed and tell how for ground water table i.e., saturated soil i.e., clay or other not Rock
- Wash boring:- are done when ground water table on soil done for all soil except (Rock)
- Precussion drilling when hole is made upon hard rock or soft
- Rotatory drilling Primarily meant for Rock all other soil also.

1742. The given figure relates to design features of samplers affecting soil disturbance. The area ratio of the soil sample is given by

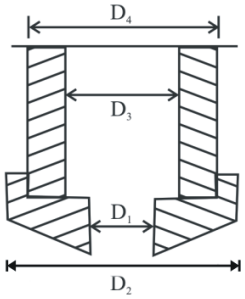


(a) $\frac{D_2^2 - D_3^2}{D_4^2}$ (b) $\frac{D_2^2 - D_1^2}{D_1^2}$

(c) $\frac{D_2^2 - D_4^2}{D_4^2}$ (d) $\frac{D_3^2 - D_1^2}{D_1^2}$

TNPSC AE 2019

Ans. (b)



$$\text{Area Ratio\% } A_r = \frac{A_2^2 - A_1^2}{A_1^2} = \frac{D_2^2 - D_1^2}{D_1^2}$$

- Area ratio should be less because it make cutting edge is sharp
- It should be less than 10% but not ≠ 20%

1743. In which of the following samplings is care take to ensure that at least the composition of the soil as it exists in nature remains unaltered?

(a) Undisturbed samples
(b) Disturbed representative
(c) Disturbed non-representative samples
(d) Intact samples

APPSC Poly. Tech. Lect. 2020

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Ans. (b) : disturbed representation sampling is care taken to ensure that at least the composition of the soil as it exists in nature remains unaltered.

1744. A sampling tube has an inner diameter of 80 mm and outer diameter of 88 mm. The area ratio is

(a) 0.21 (b) 0.12
(c) -0.21 (d) -0.12

ISRO Scientist/Engineer 2014

Ans : (a) Given, Inner diameter, $D_1 = 80$ mm
Outer diameter, $D_2 = 88$ mm

$$\text{area ratio} = \frac{D_2^2 - D_1^2}{D_1^2} = \frac{(88)^2 - (80)^2}{(80)^2} = \frac{7744 - 6400}{6400}$$

$$= \frac{1344}{6400} = 0.21$$

1745. Which one of the following methods is not the category of Geophysical methods of sub-surface investigation ?

(a) Electrical resistivity method
(b) Electric logging
(c) Gamma-ray logging
(d) Electrical response surveying

ESE 2020

Ans. (d) : Geophysical investigations involve simple methods of study made on the surface with the aim of ascertaining subsurface details.
Geophysical methods of sub-surface investigation are–

- **Electric resistivity method–** It is based on the difference in the electrical conductivity or the electrical resistivity of different soil.
- **Electric Logging/well logging–** This technique can be used in geotechnical investigations to assess the variation with depth of geologic materials and associated fluids.
- **Gamma ray logging–** This is a method of measuring naturally occurring gamma radiation to characterize the rock or sediment in a borehole or drill hole.
- **Seismic Refraction method–** This is a geophysical method used for investigating subsurface ground conditions by utilising surface-sourced seismic waves. This method is based on measurements of the time interval between initiation of a seismic (elastic) wave and its arrival at detectors.

1746. A solid sampler has inner and outer radius of 25 mm and 30 mm respectively. The area–ratio of the soil sampler is :

(a) 14% (b) 24%
(c) 34% (d) 44%

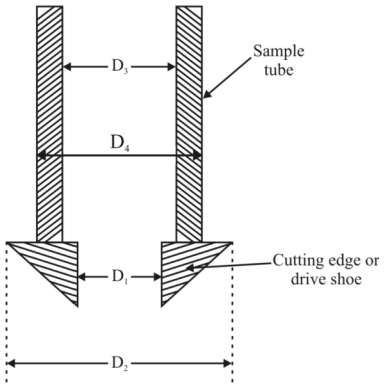
UPPSC State Eng. A.E. 2008, 2003

Ans : (d) Given, $D_1 = 25$ mm and $D_2 = 30$ mm
Area ratio,

$$A_r\% = \frac{D_2^{\,2} - D_1^{\,2}}{D_1^{\,2}} \times 100\%$$

$$A_r\% = \frac{30^2 - 25^2}{25^2} \times 100\%$$

$$A_r\% = 44\%$$



Where,
D_1 = inside diameter of cutting edge
D_2 = outside diameter of cutting edge
D_3 = inside diameter of sampling tube
D_4 = outside diameter of sampling tube

1747. Samples of high fissures soils can be obtained from

(a) assessable explorations
(b) open-drive sampling
(c) split-spoon sampling
(d) thin-walled sampling

TANGEDCO AE 2018

Ans. (a) : Samples of high fissures soils can be obtained from assessable explorations.

1748. The degree of disturbance for a soil samples collected from the sampler is usually expressed by :

(a) Void ratio (b) Area ratio
(c) Recovery ratio (d) Consolidation ratio

TRB Poly. Lect. Tech. 2017

Ans. (b/c) : The degree of disturbance for a soil sample collected from the sample is usually expressed by Area ratio or recovery ratio.

1749. The area ratio of core cutter sampler having inner diameter 150mm and outer diameter 165 mm respectively is :

(a) 17.35% (b) 12.8%
(c) 112% (d) 21%

TRB Poly. Lect. Tech. 2017

Ans. (d) : Given, Inner diameter (D_i) = 150 mm
outer diameter (D_0) = 165 mm

$$\text{Area ratio } (A_r) = \frac{D_0^2 - D_i^2}{D_i^2} \times 100$$

$$= \frac{165^2 - 150^2}{150^2} \times 100$$

Area ratio = 21%

1750. The cone penetration test (CPT) has four main advantages over the usual combination of boring, sampling and standard penetration testing; which of the following is not an advantage?

(a) It provides a continuous or virtually continuous record of ground conditions
(b) It avoids the disturbance of the ground associated with boring and sampling, particularly that which occurs with the Standard Penetration Test (SPT).

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(c) It is as expensive as conventional boring method
(d) It is faster by a factor of about 10

UPRVUNL AE 2016

Ans.: (c) Advantage of cone penetration test (CPT)
- It provides a continuous or virtually continuous record of ground conditions.
- It avoids the disturbance of the ground associated with boring and sampling particularly that which occurs with the standard penetration test (SPT)
- It is faster by a factor of about 10

1751. The most commonly used sampler for obtaining a disturbed sample of soil is
(a) Split spoon sampler
(b) Open drive sampler
(c) Piston Sampler
(d) Thin wall shell by tube sampler

GPSC AE June 2019

Ans. (a) : The most commonly used sampler for obtaining a disturbed sample of soil is split spoon sampler. Sample in which natural structure of soil of disturbed either portly or fully during sampling called disturbed sample.

1752. The number of blows observed in a Standard Penetration Test (SPT) for different penetration depths are given as follows

Penetration of sampler	Number of blows
0-150 mm	6 mm
150-30 mm	8 mm
300-450 mm	10 mm

The Standard Penetration Number (N) will be
(a) 8 (b) 14
(c) 18 (d) 24

GPSC AE June 2019
ISRO Scientist/Engineer 2014
JPSC AE 2009

Ans. (c) : In a standard penetration test (SPT)

Penetration of sampler	Number of blows
0–150 mm	$N_1 = 6$
150–300 mm	$N_2 = 8$
300–450 mm	$N_3 = 10$

Observed N-value = N_2+N_3
N-Value = 18

1753. The observed N value from a standard penetration test conducted on a saturated sandy soil stratum is 33. The corrected N value for dilatancy can be estimated as :
(a) 24 (b) 15
(c) 48 (d) 17

BHEL ET 2019

Ans. (a) : Given, $N_1 = 33$
The corrected N_2 value for dilatancy

$$N_2 = 15+\frac{1}{2}(N_2-15)$$

$$= 15+\frac{1}{2}(33-15)$$

$$= 15+\frac{1}{2}\times 18 = 24$$

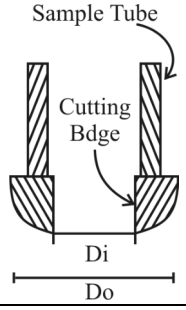
1754. The degree of disturbance of the sample collected by the sampler is expressed by a term called "area ratio". If the outer diameter and inner diameter of the sampler are D_0 and D_i respectively, then the area ratio is given by
(a) $D_i^2 - D_0^2 / D_i^2$ (b) $D_0^2 - D_i^i / D_i^2$
(c) $D_0^2 - D_i^2 / D_0^2$ (d) $D_i^2 - D_o^2 / D_o^2$

GPSC AE June 2019

Ans. (b) :

$$\text{Area of Ratio} = \frac{D_0^2 - D_i^2}{D_i^2}\times 100$$

Area ratio should not be greater than 20% for stiff formations and 10% for soft sensitive clays.



1755. Match List I (Type of exploration) with List II (Soil profile) and select the correct answer using the codes given below the lists:

List I description	List II property of fluid
A. Diamond Core drilling	1. Medium strong cohesive soils
B. Open pit excavation	2. Soft cohesive soils and cohesionless soils, for large depth
C. Uncashed wash boring	3. Rocky formation
D. Cased boring	4. Exploration upto a shallow depth below ground level

	A	B	C	D
(a)	3	4	1	2
(b)	4	3	2	1
(c)	3	1	4	2
(d)	2	3	4	1

GPSC AE Dec 2018

Ans. (a) : Diamond Core drilling → Rocky formation
Open pit excavation → Exploration upto a shallow depth below ground level
Uncashed wash boring → Medium strong cohesive soils
Cased boring → Soft cohesive soils and cohesionless soils, for large depth

1756. A stand pipe in the grout hole is absolutely essential during grouting operation of soil when
(a) There are many surface cracks
(b) Rock is very weak
(c) Drainage is towards the hole
(d) All of the above

GPSC AE January 2018

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Ans. (d) : A stand pipe in the grout hole is absolutely essential during grouting operation of soil when,
• There are many surface cracks
• Rock is very weak
• Drainage is towards the hole.

1757. Cone Penetration Tests (CPT) are best suited for :
(a) loose to medium dense silts and sands
(b) highly dense silts and sands
(c) any type of silts and sands
(d) weathered rocks

UPRVUNL AE 2014

Ans. (a) : Cone Penetration Tests (CPT) are best suited for loose to medium dense silts and sands.

1758. The Standard Penetration Test (SPT) is widely used as a simple and economic means of obtaining which of the following:
(a) A measurement of Soil compressibility expressed in terms of a compression index
(b) A direct measurement of the undrained shear strength
(c) An indirect indication of the relative density of cohesionless soil
(d) A direct measurement of the angle of internal friction

HPPSC Poly. Lect. 2016

Ans : (c) The main purpose of SPT test is to provide an indication of the relative density of granular deposits such as sands and gravels from which it is vertically impossible to obtain undisturbed samples.

1759. In a Standard Penetration Test, at every 0.15 m penetration the recorded blow counts were 8, 10 and 7. Determine the N value for 60% energy efficiency.
(a) 10 (b) 15
(c) 28 (d) 42

UPPCL AE 2018

Ans. (c) : For N value we take sum of last 300 mm blow :

$N = 17 = 10 + 7$

For 60% energy efficiency :

$$N = \frac{17}{0.6} \simeq 28$$

13. Stability Analysis of Slopes

1760. The Swedish circle method of slope stability assumes that soil is :
(a) Purely cohesive (b) C-ϕ soil
(c) Purely granular (d) Well graded gravels

UPPCL AE 2019

Ans. (a) The Swedish circle method of slope stability assumes that soil is purely cohesive.

1761. The factor of safety with respect to friction for a soil for stability :
(a) $F_s = \frac{C + \sigma \tan\phi}{C_m + \sigma \tan\phi_m}$ (b) $F_s = \frac{C}{C_m}$
(c) $F_s = \frac{\tan\phi}{\tan\phi_m}$ (d) $F_s = \frac{C + \tan\phi}{C_m + \tan\phi_m}$

Where F_s is the factor of safety, C_m and ϕ_m are mobilized cohesion and mobilized angle of internal friction and C and ϕ are the cohesion and angle of internal friction. σ is normal stress.

APTRASCO 2017

Ans : (c) : Safety in terms of friction for soil stability $= (F_S) = \frac{\tan\phi}{\tan\phi_m}$

where C_m and ϕ_m are the movable cohesion and moving internal friction angle respectively and C and ϕ are the cube and inward friction angle respectively.

1762. Method of slices for determining stability of slope was first suggested by :
(a) Taylor (b) Bishop
(c) Fellenius (d) Terzaghi

[UPPSC State Eng. A.E. 2011 Ist Paper]

Ans : (c) The method of slices for determination of stability of the slope was first suggested by Fellenius in 1926.
According to this method,

$$F.O.S = \frac{C\gamma\theta + \mu \sum N}{\sum T}$$

where, C = cohesion of soil
μ = friction coefficient of soil
N = Normal component of weight of soil
T = Tangential component of weight of soil

1763. An excavation is made with a vertical face in a clay soil which has C_u = 50 kN/m², γ_t = 18 kN/m³ and S_n = 0.261. The maximum depth of a stable excavation will be nearly
(a) 10.6 m (b) 12.4 m
(c) 14.2 m (d) 16.0 m

ESE 2019

Ans. (a) Given,
Excavation is clay soil
$C_u = 50$ kN/m²
$\gamma_t = 18$ kN/m³
$S_n = 0.261$
For maximum depth of a stable excavation
Factor of safety [$F_s = 1$]

Taylor's Stability number, $S_n = \frac{C}{F_s \gamma H}$

$$\left[S_n = \frac{C}{\gamma H}\right]$$

$\therefore$ Critical height 'H_c' $= \frac{C}{S_n \gamma}$ $(\gamma = \gamma_t)$

$$H_c = \frac{50}{0.261 \times 18}$$

$$\boxed{H_c = 10.6\text{m}}$$

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1764. A cut is to be made in clay for which the cohesion is 30 kN/sq.m and negligible friction angle. If the density of the soil is 20 kN/cum and the factor of safety for the design is 1.5, the maximum depth for a cutting of side slope 1.5:1 is (Assume stability number as 0.17).

(a) 6.28 (b) 5.88
(c) 3.36 (d) 4.88
(e) 5.3

CGPSC AE Shift - 2 2014

Ans. (b) : Given, cohesion, C = 30 kN/sq.m
Unit weight, $\gamma = 20\ kN/m^3$
stability number $S_n = 0.17$
$F_C = 1.5$

$$S_n = \frac{C}{\gamma F_C . H}$$

$$H = \frac{30}{20 \times 1.5 \times 0.17}$$

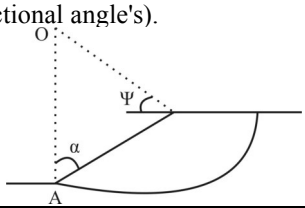
$$H = 5.88m$$

1765. For stability analysis of slopes of purely cohesive soils, the critical centre is taken to lie at the intersection of

(a) The perpendicular bisector of the slope and the locus of the centre
(b) The perpendicular drawn at the one-third slope from the toe and the locus of the centre
(c) The perpendicular drawn at the two-third slope from the toe and the locus of the centre
(d) Directional angles

ESE 2018

Ans. (d) ■ Fellenius proposed on empirical procedure to find the centre of the most critical circle in purely cohesive soil. The centre 'O' for the toe faivere case can be located at the intersection of the two lines drawn from the ends A & B of the slope at angles 'α' and 'Ψ' (directional angle's).



1766. Factor of safety with respect to cohesion gives the same concept as that of factor of safety w.r.t.:

(a) height
(b) average shear strength
(c) friction
(d) None

JPSC AE (Pre) 2019

Ans. (c) :
1. When the factor of safety with respect to cohesion is unity.
2. When the factor of safety with respect to friction is unity. We can have any combination of F_C and F between these two extremes cited above to give the same mobilized shearing resistance of 39.6 kN/m^2.

$$F = \frac{\text{shearing strength}}{\text{shear stress}}, \quad f_c = \frac{C}{C_m}$$

1767. The factor of safety of an infinite slope in a sand deposit is found to be 1.732. The angle of shearing resistance of the sand in 30 degree. The average slope of the sand deposit is given by

(a) sin-1 (0.333) (b) cos-1 (0.252)
(c) tan-1 (0.333) (d) cot-1 (0.621)

BPSC AE 2019 Paper (V) Section-I

Ans. (c) : Given, FOS = 1.732, $\phi = 30^0$

We know that, $FOS = \frac{\tan\phi}{\tan\beta}$

$$1.732 = \frac{\tan 30^0}{\tan\beta}$$

$$\tan\beta = 0.333$$

$$\boxed{\beta = \tan^{-1}(0.333)}$$

1768. Which one of the following pairs is not correctly matched?

(a) Critical void ratio – Rapid draw down
(b) Swedish arc – Stability of slopes
(c) Critical height – Stability number
(d) Base failure – Soft clay

MPPSC AE 2017

Ans. (a) The Swedish method of slope stability analysis for C - ϕ soil is often referred to as Swedish method of slices. The critical void ratio is defined as the void ratio at which failure will occur at zero volume change.

1769. A 10 m deep canal is constructed in purely cohesive soil having $C = 0.2\ kg/cm^2$, $\phi = 0^o$, $G = 2.5$, $e = 0.5$. The stability number is 0.1. In a canal running in full condition, the factor of safety w.r.t. cohesion against failure of side slopes will be-

(a) 1.0 (b) 1.5
(c) 2.0 (d) 2.5

MPSC (Mains) 2017 Paper–II

Ans. (a) : We know that,

$$S_n = \frac{C}{\gamma H F_C}$$

Give data,
$S_n = 0.1$, $C = 0.2\ kg/cm^2$, $\gamma = ?$, $H = 10m$

$$0.1 = \frac{0.2 kg/cm^2}{\gamma \times 10 \times F_C}$$

$$\gamma = \left(\frac{G+e}{1+e}\right)\gamma_w = \frac{3}{1.5}\gamma_w = 2\gamma_w$$

$$0.1 = \frac{200 g/cm^2}{2 g/cm^2 \times 1000 cm \times F_C}$$

$$F_C = \frac{200}{2 \times 1000 \times 0.1} = 1$$

$$\boxed{F_C = 1}$$

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1770. A slope is to be constructed at an angle of 30^0 to the horizontal from a soil having the properties, C = 15 kN/m², γ = 19 kN/m³. Taylor's stability number is 0.046. If a factor of safety (with respect of cohesion) of 1.5 is required, then the safe height of the slope will be

(a) 25.7 m. (b) 17.2 m.
(c) 12.8 m. (d) 11.4 m.

MPPSC AE 2016

Ans : (d) $\beta = 30^0$, $C_u = 15$ kN/m², $\gamma = 19$ kN/m³
$S_n = 0.046$, $F_c = 1.5$, H = ?

$$S_n = \frac{C_u}{F_c \gamma H}$$

$$0.046 = \frac{15}{1.5 \times 19 \times H}$$

$$H = \frac{15}{1.5 \times 19 \times 0.046} = 11.44 \text{ m}$$

$$H_c = 11.44 \text{ m}$$

1771. Taylor's stability number N is given by :

(a) $N = \frac{\gamma H}{C_m}$ (b) $N = \frac{C_m}{\gamma H}$
(c) $N = \frac{C_m H}{\gamma}$ (d) $N = \frac{\gamma C_m}{H}$

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (b) Taylor's stability number N is given as

$$N = \frac{C_m}{\gamma H F_c}$$

C = cohesion
F_c = factor of safety
γ = density of soil
and H = height of slope

1772. If an infinite slope of clay at a depth 5 m has cohesion of 1 t/m² and unit weight of 2 t/m³, then the stability number will be :

(a) 0.1 (b) 0.2
(c) 0.3 (d) 0.4

[UPPSC State Eng. A.E. 2003]

Ans : (a) Given
C = 1 t/m²; γ = 2 t/m³; h = 5m
stability number,

$$S_\eta = \frac{C}{\gamma h} = \frac{1}{2 \times 5} = 0.1$$

1773. For a clay slope of height 10m, the stability number (S_n) is 0.05, γ = 2.0 t/m³, C = 2.5 t/m². The critical height of the slope of the soil is :

(a) 4.0 m (b) 12.5 m
(c) 25.0 m (d) 15.0 m

UKPSC AE (Urban) 2011
UPPSC State Eng. A.E. 2011 Ist Paper

Ans : (c) Given, z = 10 m, Stability No. (S_n) = 0.05
Unit weight of soil, γ = 2.0 t/m³, cohesion, C = 2.5 t/m²

$$S_n = \frac{C}{\gamma H_C}$$

$$0.05 = \frac{2.5}{2.0 H_c}$$

$$H = \frac{2.5}{0.05 \times 2} = 25 \text{m}$$

1774. An infinite slope is inclined at angle, i and has its angle of internal friction, ϕ, the stability number, S_n is :

(a) $\frac{\cos^2 i}{\tan i - \tan \phi}$ (b) $\frac{\sin^2 i}{\tan i - \tan \phi}$
(c) $(\tan i - \tan \phi)\cos^2 i$ (d) $(\tan i - \tan \phi)\sin^2 i$

[UPPSC State Eng. A.E. 2011 Ist Paper]

Ans : (c) For C - ϕ soils taylor's stability no. (S_n) is defined as

$S_n = (\tan i - \tan \phi) \cos^2 i$

where, i = Angle of inclination of slope
ϕ = Angle of internal friction.

1775. A deep cut of 7 m has to be made in a clay with unit weight of 16 kN /m³ and a cohesion of 25 kN/m². The stability number is 0.178. The factor of safety will be :

(a) 1.00 (b) 1.25
(c) 1.50 (d) 2.00

[UPPSC State Eng. A.E. 2008]

Ans : (b)
Given,
γ = 16 kN/m³
C = 25 kN/m²
S_n = 0.178
H = 7 m

$$S_n = \frac{C}{\gamma H F_c}$$ S_n = Taylor's stability number

$$F_c = \frac{C}{\gamma H S_n}$$ F_c = factor of safety

$$F_c = \frac{25}{16 \times 0.178 \times 7} = 1.25$$

1776. In a clayey soil having 50 kN/m² as unit cohesion and 18 kN/m³ as unit weight, an excavation is made with a vertical face. Taking Taylor's stability number as 0.261, what is the maximum depth of excavation so that the vertical face remains stable?

(a) 5.30 m (b) 7.06 m
(c) 10.6 m (d) 12.4 m

ESE 2018

Ans. (c)
Maximum depth upto which vertical face remain stable is critical height. (H_C)
At H = H_C

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$F_c = 1$,

$S_n = \dfrac{C}{F_c \gamma H_c}$ (or) $S_n = \dfrac{C}{\gamma . H_c}$

$C = 50\ kN/m^2$

$\gamma = 18\ kN/m^3$ $H_c = ?$

$S_n = 0.261$ $F_c = 1$

$\therefore H_c = \dfrac{C}{S_n . \gamma} = \dfrac{50}{0.261 \times 18} = 10.64m$

1777. If the failure of a finite slope occurs in which the failure surface passes below the toe to known as

(a) Slope failure (b) Face failure
(c) Base failure (d) Toe failure

OPSC AE Paper- II 2019

Ans. (c) : In case of finite slope, three types of failure of surfaces occur, which has failure surface as circular arc.

(i) Base failure–In base failure, rupture surface passes below the toe and it occurs when soil mass below the toe is relatively weak of soft.

(ii) Face failure or slope failure–It is when failure surface intersects above the toe of slope. This occurs when soil close to the toe is quite strong but in upper part soil is weak, and slope angle B is very high.

(iii) Toe failure–Failure surface passes through the toe and it occurs in steep slope. Soil mass above the base and below the base is homogeneous. This is the most common mode of failure.

1778. The factors leading to the failure of slopes are classified as

(a) Increase in the shear stress
(b) Decrease in the shear strength of the soil
(c) No shear stress in the failure plain
(d) Both (a) and (b)

OPSC AE Paper- II 2019

Ans. (d) : The factors leading to the failure of slopes are classified as :

(i) Increase in the shear stress
(ii) Decrease in the shear strength of the soil.

1779. Taylor's stability number curves are used for the analysis of stability of slopes. The angle of shearing resistance used in the chart is the :

(a) effective angle (b) apparent angle
(c) mobilized angle (d) weighed angle

Karnataka PSC AE Paper-II-2017, 2015

Ans. (c) : Taylor's stability number curves are based on mobilized angle of shearing resistance.

• Stability Number = $S_n = \dfrac{C_u}{F_e \gamma H}$

1780. With respect to C–ϕ soil in an infinite slope, identify if the following two statements are True or False :

I. The stable slope angle can be greater than ϕ

II. The factor of safety of the slope does not depend on the height of soil in the slope.

(a) Both statements are false
(b) I is true but II is false
(c) I is false but II is true
(d) Both statements are true

UKPSC AE 2012 Paper–I

Ans : (b) C-ϕ soil–

(i) C-ϕ soil in an infinite slope is stable slope angle can be greater than (ϕ).

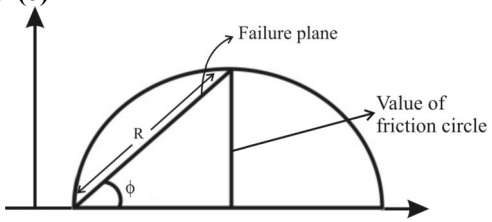
(ii) C-ϕ soil are also referred to as shear strength parameters of the soil. C and ϕ are not inherent properties of soil. These are in-fact related to the type of test and the condition under which these are measured.

1781. If R is the radius of failure and ϕ is the angle of internal friction of a soil, the value of friction circle is :

(a) R tan ϕ (b) R sin ϕ
(c) R cos ϕ (d) $\dfrac{R}{\phi}$

UPPCL AE 2014
(UKPSC AE (Paper-I) 2007)

Ans : (b)



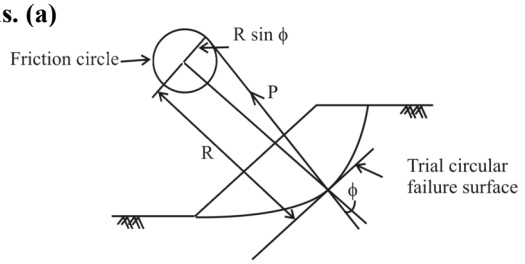
friction circle = R sin ϕ

1782. The radius of friction circle used in slope stability analysis with circular failure surface with radius, r, in soil having shear strength parameter can ϕ, is

(a) r sin ϕ (b) r cos ϕ
(c) r tan ϕ (d) r cot ϕ

CIL MT 2020
WBPSC Poly Lect. 2019

Ans. (a)



• The friction circle method of slope analysis is a convenient approach for both graphical and mathematical solution.

• The radius of friction circle = r sin ϕ

1783. The maximum angle of inclination of the plane (with horizontal) at which a body remains in equilibrium under the action of friction only is termed as

(a) Angle of wedge
(b) Angle of equilibrium
(c) Angle of repose
(d) All the given answers

APPSC AEE MAINS 2016

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Ans. (c) : Angle of repose– The maximum angle of inclination of the plane (with horizontal) at which a body remains in equilibrium under the action of friction only is termed as angle of repose.

1784. An embankment has a slope of 30° which was constructed with soil having C = 30 kN/m², f = 20° and g = 15 kN/m³. The height of embankment is 20m. Using Taylor's stability no. $\frac{1}{40}$, the factor of safety with respect to cohesion is

(a) 0.25 (b) 2
(c) 4 (d) 1.5

Maharashtra PSC Mains 2018 Paper-2

Ans. (c) : Given, unit weight, $\gamma = 15\ kN/m^3$
Cohesion, $C = 30\ kN/m^2$
height, $H = 20\ m$

Stability Number, $S_n = \frac{1}{40}$

$$S_n = \frac{C}{F_c \gamma H}$$

$$\frac{1}{40} = \frac{30}{F_c \times 15 \times 20}$$

$F_c = 4$

1785. Taylor's stability number is given by :

(a) $\frac{F_c}{C\gamma H}$ (b) $\frac{\gamma H}{CF_c}$
(c) $\frac{C}{F_c \gamma H}$ (d) $\frac{H}{CF_c\gamma}$

WBPSC AE 2016
Rajasthan AE Nagar Nigam Shift-3, 2016
OPSC AE -2016 (II)
OPSC AEE 2015 PAPER–I
OPSC Civil Services Pre 2011

Ans. (c) : Taylor's stability number

$$S_n = \frac{C}{F_c \gamma H}$$

C = cohesion
F_c = factor of safety
H = height

1786. Inclination of a clay slope

(a) can be greater than the angle of shearing resistance
(b) cannot be greater than the angle of shearing resistance
(c) cannot be greater than the angle of repose
(d) cannot be greater than 45 degree

TANGEDCO AE 2018

Ans. (a) : Inclination of a clay slope can be greater than the angle of shearing resistance.

1787. Total stress method of stability analysis may be applied to find the factor of safety in the case of newly cut slope in

(a) fissured over-consolidated saturated clay
(b) non-fissured over-consolidated saturated clay
(c) normally consolidated saturated clay
(d) partially saturated expansive clay

TANGEDCO AE 2018

Ans. (c) : Total stress method of stability analysis may be applied in case of newly slope in normally consolidated clay.

1788. In stability analysis of slopes, factor of safety with respect to height is given by

(a) $F_H = \frac{H_C}{H}$ (b) $F_H = \frac{H}{H_C}$
(c) $F_H = \frac{2H}{H_C}$ (d) $F_H = \frac{H_C}{2H}$

TRB Poly. Tech. Lect. 2012

Ans. (a) : In stability analysis of slopes, factor of safety with respect to height is given by.

$$F_H = \frac{H_C}{H}$$

H_c = critical height
H = actual height

1789. Stability analysis by Swedish method of slices gave following values per meter run of 10 m high embankment (i) total shearing force = 480 kN (ii) total normal force = 1950 kN (iii) total neutral force = 250 kN (iv) length of arc = 22 m, if C = 24 kN/m², ϕ = 6°, then the factor of safety with respect to shear strength is

(a) 1.57 (b) 1.67
(c) 1.47 (d) 1.75

TRB Poly. Tech. Lect. 2012

Ans. (c) : Given, $C = 20\ kN/m^2$
$\phi = 6^\circ$, N = 1950 kN
U = 250 kN
T = 480 kN

Factor of safety with respect of hear strength

$$Fs = \frac{C'(\gamma\theta) + \Sigma(N - U)\tan\phi}{\Sigma T}$$

$$= \frac{24(22) + (1950 - 250)\tan 6^\circ}{480} = 1.47$$

Note : In question U (total neutral force) is not given, so we can't calculate Fs.

1790. For a base failure of a finite slope the depth factor

(a) $D_f = 1$
(b) $D_f < 1$
(c) $D_f > 1$
(d) None of the above

TRB Poly. Lect. Tech. 2017

Ans. (c) The ratio of the total depth (H + D) to depth H is called the depth factor D_f, for toe failure $D_f = 1$ for base failure $D_f > 1$. Planer failure surface may commonly occur is a soil deposit or embankment with a specific plane of weakness.

1791. A temporary cutting 10 m deep is to be made in a clay of unit weight 20 kN/cum and average cohesion of 25 kN/sq m. Compute the available factor of safety for a 30° angle, if a factor of safety 1.25 is desired for stable slope, there is a

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rock stratum at 15 m below the ground level and the Taylor's stability number is 0.16.

(a) 1.74 (b) 1.52
(c) 0.78 (d) 0.85

UPPCL AE 2015

Ans. (c) : Given,

$S_n = 0.16$
$\gamma = 20\ kN/m^3$
$C = 25\ kN/m^3$
$H = 10\ m$
$F_c = ?$

$$S_n = \frac{C}{F_c.\gamma.H}$$

$$\text{Factor of safety } (F_c) = \frac{25}{0.16\times 20\times 10}$$

$F_c = 0.781$

1792. A canal of 4 m deep has side slopes of 1 : 1. The properties of the soil are $C = 15\ kN/m^2$, $\phi = 15^0$, e = 0.76 and G = 2.7. Taylor's stability number for that sudden drawdown = 0.136. The factor of safety with respect to cohesion in the case of sudden drawdown will be

(a) 0.64 (b) 1.43
(c) 2.22 (d) 3.01

ESE 2020

Ans. (b) : Given,

$H = 4m$, $C = 15\ kN/m^2$, $e = 0.76$, $G = 2.70$
$S_n = 0.136$

we know that,

$$\text{Saturated unit weight } (\gamma_{sat}) = \left(\frac{G+e}{1+e}\right)\gamma_w$$

$$= \left(\frac{2.7+0.76}{1+0.76}\right)\times 9.81$$

$$\gamma_{sat} = 19.28\ kN/m^3$$

For sudden draw down,

Taylor's stability number,

$$S_n = \frac{C}{FOS.\gamma_{sat}.H}$$

$$0.136 = \frac{15}{FOS\times 19.28\times 4}$$

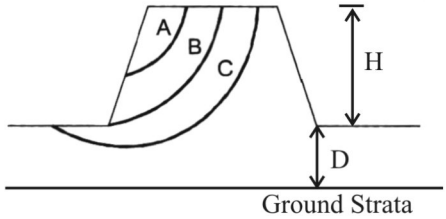
$$\boxed{FOS = 1.43}$$

1793. Failure of a finite slope along a surface that intersects the slope above the toe:

(a) Compound failure (b) Base failure
(c) Slope failure (d) Toe failure

Kerala PSC AE 2015

Ans. (c) :



A → Slope failure
B → Toe failure
C → Base failure

$$\text{Depth factor} = \frac{H+D}{H}$$

If Depth factor = 1 – Toe failure
Depth factor > 1 – Base failure

Note :

(i) for soil with $\phi < 3^0$ the critical slip circle invariable through the toe.
(ii) TOE failure if $\beta > 53^0$
(iii) When there is a stiff layer at the base of the slope the slip circle will be tangential to it.

1794. A plane inclined at an angle 'ϕ' to the horizontal at which the soil is expected to stay in the absence of any lateral support is known as the

(i) natural slope line
(ii) repose line
(iii) ϕ line
(iv) phreatic line

(a) (i) and (ii) (b) (ii) and (iv)
(c) (i), (ii) and (iii) (d) (i), (ii), (iii) and (iv)

GPSC AE June 2019

Ans. (c) : A plane inclined at an angle 'ϕ' to the horizontal at which the soil is expected to stay in the absence of ay lateral support is known as the natural slope line or repose line or ϕ-line.

1795. During stability analysis, the term mobilised shear strength means:-

(a) Shear strength
(b) Applied shear stress
(c) Maximum shear stress
(d) Minimum shear stress

UPPCL AE 2014

Ans. (b) : During stability analysis, the term mobilised shear strength means applied shear stress.

1796. The following assumption is not made for the friction circle method of slope stability analysis:-

(a) The resultant passes through the centre of friction circle.
(b) Friction is fully mobilised.
(c) The resultant is tangential to the friction circle.
(d) Total stress analysis is applicable.

UPPCL AE 2014

Ans. (d) :

1797. Taylor's stability chart are based on the total stresses using:-

(a) Method of slices
(b) Culmann's method
(c) $\phi_u = 0$ Analysis
(d) Friction circle method

UPPCL AE 2014

Ans. (d) : Taylor's stability chart are based on the total stresses using Friction circle method.

1798. Bishop's method of stability analysis

(a) Is more conservative
(b) Is not preferred

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(c) Neglects the effect of forces acting on the sides of the slices
(d) Assumes the slip surface as an arc of a circle

GPSC AE January 2018

Ans. (d) : Bishop's method of stability analysis assumes the slip surface as an arc of a circle and resultant interslice shear force is zero.

1799. To determine stability of earth dam slopes immediately after construction the pore pressures are generally estimated using
(a) Bishop's method (b) Hilf's method
(c) Culmann's method (d) Kirchoff's method

GPSC AE January 2018

Ans. (b) : Pore pressure of earth dam are generally estimated using Hilf's method.
Hilf's formula for evaluating pore pressure is

$$U = \frac{Pa.\Delta}{V_a + 0.2V_w = \Delta}$$

U = induced pore pressure in kg/cm^2.
Pa = absolute atmospheric pressure in kg/cm^2.
Δ = embankment compression in percent of original total embankment volume.
V_a = Volume of free air in the voids of soil directly after compaction.
V_w = Volume of pure water.

1800. Identify the incorrect statement the stability of a slope is decreased by
(a) removal of a part of slope by excavation
(b) Shock caused by an earthquake
(c) Pore water pressure in the soil
(d) Providing a berm at the toe

MPPSC AE 2014

Ans. (d) : The stability of a slope is increased by providing a berm at the toe

1801. If an infinite slope of clay at a depth 5 m has cohesion of 1 t/m^2 and unit weight 2 t/m^3 the stability number will be:
(a) 0.1 (b) 0.2
(c) 0.3 (d) 0.4

Haryana PSC AE 2017

Ans. (a) : Given,
C = 1 t/m^2
H = 5 m
γ = 2 t/m^3
Taylor's stability No.

$$S_n = \frac{C}{\gamma H} = \frac{1}{2\times 5} = 0.1$$

S_n = 0.1

1802. The cohesion and the density of a soil are 2 t/m^2 and 2 t/m^3 respectively. For a factor of safety of 2 and stability number 0.1, the safe height of slope is
(a) 2.5 m (b) 10 m
(c) 5 m (d) 50 m

GPSC AE March 2018

Ans. (c) : S_n = 0.1, C = 2 t/m^2, ρ = 2t/m^3, FOS = 2

$$\gamma = \rho.g = \left(2\times\frac{9.81}{10}\right) t/m^3$$

Sn = stability number = $\frac{C}{F_c.\gamma.H}$

$$\boxed{H = \frac{C}{S_n F_c \gamma}}$$

$$H = \frac{2}{0.1\times 2\times\left(2\times\frac{9.81}{10}\right)}$$

$\boxed{H = 5\ m}$

1803. Failure of the stability of slopes generally occurs along:
(a) Slip plane
(b) A horizontal surfaces
(c) A curved surfaces
(d) All the surfaces
(e) None of these options

Haryana PSC AE Paper–II 2018

Ans. (c) : Types of slope failure
1. Translational failure
2. Rotational failure
3. Wedge failure
4. Compound failure

In Rotational failure the material fails along a curved surface.

1804. The Limit Equilibrium method used commonly in stability analysis of slopes :
(a) considers stress-strain relationship of soil
(b) does not concentrate on the equilibrium and yield conditions and considers the stress-strain relationship of soil
(c) concentrates on the equilibrium and yield conditions and considers the stress-strain relationship of soil
(d) concentrates on the equilibrium and yield conditions but does not consider the stress-strain relationship of soil

UPRVUNL AE 2014

Ans. (d) : The limit equilibrium method used commonly in stability analysis of slopes.
Concentrates on the equilibrium and yield conditions but does not consider the stress strain relationship of soil.

1805. An infinite slope of clay having cohesion of 10 kN/m^2 and unit weight of 20 kN/m^3. If the depth of clay is 5m, determine the stability number for the slope.
(a) 0.4 (b) 0.1
(c) 0.2 (d) 0.3

BHEL ET 2019

Ans. (b) : Given value
C = 10 kN/m^2
γ = 20 kN/m^3
h_c = 5 m
By using formula –

Stability number for the slope $(S_n) = \frac{C}{\gamma h_c}$

$$S_n = \frac{10}{20\times 5} = 0.1$$

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1806. In stability analysis of infinite slope, for cohesionless soil, the factor of safety (F) against sliding is given by (where τ_f is shear strength and τ is shear stress) :

(a) $\frac{\tau_f}{\tau}$ (b) $\frac{\tau}{\tau_f}$

(c) $\tau f \times \tau$ (d) $\tau f + \tau$

LMRC AM 2018

Ans. (a) :

factor of safety against sliding = $\frac{\text{shear strength}}{\text{shear stress}}$

$$FOS = \frac{\tau_f}{\tau}$$

1807. A slope of infinite extent is made in dense sand layer at an angle of 30° to horizontal. The factor of safety of the slope against shear failure, if the angle of internal friction of sand is 36° is :

(a) 1.000 (b) 1.258

(c) 1.500 (d) 1.558

HPPSC Poly. Lect. 2016

Ans : (b) $FOS = \frac{\tan\phi}{\tan i}$ for cohession less soil.

$$FOS = \frac{\tan 36^\circ}{\tan 30^\circ} = 1.258$$

FOS = 1.258

1808. A slope is to be constructed at an angle of 30° to the horizontal for a soil with C = 15 kN/m², ϕ = 22.5°, γ = 19 kN/m³. Taylor's stability number is 0.046. If a factor of safety (with respect to cohesion) of 1.5 is required, then the safe height of the slope will be

(a) 25.8 m (b) 19.1 m

(c) 17.2 m (d) 11.5 m

WBPSC AE 2016

Ans. (d) : Given,

$C = 15\ kN/m^2$

$\gamma = 19\ kN/m^3$

$S_n = 0.046$

$F_c = 1.5$

Taylor stability No :

$$S_n = \frac{C}{F_c.\gamma.H}$$

$$H = \frac{15}{1.5 \times 19 \times 0.046}$$

$H = 11.5\ m$

1809. If a cohesive soil specimen is subjected to a vertical compressive load, the inclination of the cracks to the horizontal is

(a) 90° (b) 45°

(c) 22.5° (d) 0°

WBPSC AE 2014

Ans. (b) :

1810. For the upstream face of an earthen dam, most adverse condition for stability of slope is

(a) Sudden drawdown (b) Steady seepage

(c) During construction (d) Sloughing of slope

WBPSC AE 2012

Ans. (a) : For the upstream face of an earthen dam, most adverse condition for stability of slope is sudden drawdown.

1811. A 45° slope of 15 m high in a soil with an undrained strength of 115 KN/m² and γ = 20.2 KN/m² stability number - 0.181, then the factor of safety of the slope is

(a) 2.10 (b) 1.50

(c) 1.83 (d) 2.81

WBPSC AE 2003

Ans. (a) : Given, undrained strength (C_u) = 115 kN/m²

Unit weight (γ) = 20.2 kN/m³

Height (H) = 15 m

Stability no (S_n) = 0.181

$$S_n = \frac{C_u}{F_c.\gamma.H}$$

$$F_c = \frac{115}{0.181 \times 20.2 \times 15} = 2.1$$

1812. For a clay slope of height 10m, stabiliyn number is 0.05, γ = 2 kN/m³ C = 2.5 kN/m², the critical height of the slope of the soils is

(a) 4 m (b) 12.5 m

(c) 25 m (d) 15 m

WBPSC AE 2003

Ans. (c) : Given,

$S_n = 0.05$

$\gamma = 2\ kN/m^3$

$C = 2.5\ kN/m^2$

Critical height of slope $S_n = \frac{C}{\gamma.H}$

$$H = \frac{2.5}{0.05 \times 2} = 25\ m$$

1813. The factor of safety of a slope of given inclination of 6 m high constructed using a soil with c = 60 kPa, y = 20 kN/cum and its Taylor's stability number = 0.20, will be:

(a) 2.50 (b) 5.00

(c) 1.00 (d) 2.00

TSPSC Deputy Engineer 2015

Ans. (a) : Given,

Height, H = 6 m

cohesion, C = 60 kPa

Unit weight (γ) = 20 kN/m³

Taylor's stability no. (S_n) = 0.2

$$S_n = \frac{C}{\gamma.H.F_c} = \frac{60}{20 \times 6 \times F_c}$$

Factor of safety, $F_c = \frac{60}{20 \times 6 \times 0.2} = 2.5$

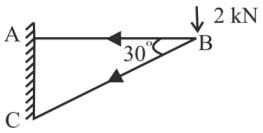
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03.

Engineering Mechanics

1. Forces and Force Systems

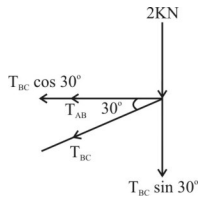
1. The load shared by the member BC of the structure shown in figure below is



(a) $2\sqrt{3}$kN (b) $3\sqrt{2}$kN
(c) 4 kN (d) 3 kN

TNPSC AE 2012

Ans. (c) :



$T_{BC} \sin 30^o = -2$ kN

$T_{BC} = \dfrac{-2}{\sin 30^\circ} = 4\,\text{kN}\,(\text{Compressive})$

2. The algebraic sum of moments of two unlike parallel forces about any point in their plane will be

(a) Zero
(b) Constant
(c) Always anticlockwise
(d) Always clockwise

APPSC AEE 2016

Ans. (b) : The algebraic sum of moments of two unlike parallel forces about any point in their plane will be constant.

3. A force equal in magnitude, opposite in direction and collinear with the resultant is known as :

(a) Resolution of force
(b) Equilibrant force
(c) Reaction
(d) Non collinear force

LMRC AM 2018

Ans. (b) : A force equal in magnitude, opposite in direction and collinear with the resultant is known as equilibrant force.

4. A force of magnitude 5 N moves through a distance of 4mm in a direction, inclined at 60° to the direction of force. The magnitude of the work done by the force is _____.

(a) $10\sqrt{3}$ N.mm (b) 0 N.mm
(c) 10 N.mm (d) 20 N.mm

TSPSC Deputy Engineer 2015

Ans. (c) : $W = Fd \cos\theta$

$W = 5 \times 4 \times \cos 60°$

$= 20 \times \dfrac{1}{2} = 10$ Nmm

5. A point subjected to a number of forces will be in equilibrium if

(a) sum of the resolved parts in any two directions at right angles are both zero
(b) the algebraic sum of the forces is zero
(c) two resolved parts in any two directions at right angles are equal
(d) the algebraic sum of the moments of the forces about the point is zero

WBPSC AE 2007

Ans. (a) : Sum of the resolved parts in any two directions at right angles are both zero.

6. Consider below statements,

A. According to principle of transmissibility, "force can be applied at any point on the object in its line of action, which creates same moment about the considered point"

B. According to Varignon's theorem "Moment of resultant force of a system of force acting on a body about a point is inversely proportional to moment of all the forces acting on same object about the same point."

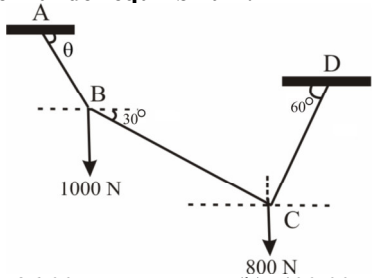
Identify the correct statement's

(a) Statement B is true A is false
(b) Both statements are false
(c) Both statements are true
(d) Statement A is true B is false

UPPCL AE 2019

Ans. (d)

7. Two forces are acting at point B and C in a system of cable as shown in figure calculate the tension force developed in cable CD to keep the system under equilibrium.

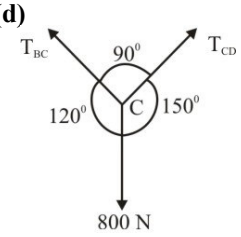


(a) 526.00 N (b) 400.00 N
(c) 725.64 N (d) 692.82 N

UPPCL AE 2019

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Ans. (d)



Force in CD = ?

From Lami's theorem

$$\frac{T_{CD}}{Sin120^0} = \frac{800}{Sin90^0}$$

$$T_{CD} = \frac{800 \times Sin120^0}{Sin90^0}$$

$$\boxed{T_{CD} = 692.82N}$$

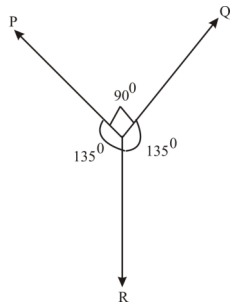
8. If three forces P, Q and R keep a body in equilibrium, the angle between P & Q is 90º and R is acting vertically downward, then magnitude of forces R and Q respectively in terms of P will be

(a) $P/\sqrt{2}, \sqrt{2}P$ (b) $\sqrt{2}P, P/\sqrt{2}$

(c) $P, \sqrt{2}P$ (d) $\sqrt{2}P, P$

APPSC AEE 2016

Ans. (d) :



Using Lami's theorem

$$\frac{R}{\sin 90^\circ} = \frac{Q}{\sin 135^\circ} = \frac{P}{\sin 135^\circ}$$

$$R = \frac{P}{\sin 135^\circ} = \frac{P}{1/\sqrt{2}} = \sqrt{2}P$$

$Q = P$ $R = \sqrt{2}P$

9. When two concurrent forces of 20 kg and 15 kg act at right angles on a particle, then their resultant will be equal to

(a) 35 kg (b) 25 kg

(c) 5 kg (d) $20\sqrt{15}$ kg

Punjab PSC SDE 2016

Ans. (b) : $R^2 = F_1^2 + F_2^2 + 2F_1F_2\cos\theta$

$$R^2 = 15^2 + 20^2 + 2 \times 15 \times 20 \times \cos 90^\circ$$

(As they act at right angle)

$$R^2 = 15^2 + 20^2 + 0$$

$$R^2 = 225 + 400$$

$$R = \sqrt{625}$$

$R = 25$ kg

10. If the system of forces reduces to a single resultant, then the moment of this resultant about any point on the line of action of the resultant will be

(a) Equal to the resultant

(b) More than the resultant

(c) Less than the resultant

(d) Zero

Punjab PSC SDE 2016

Ans. (d) : As the line of action is same for resultant about any point, the moment will be zero.

11. A smooth cylinder lying on its convex surface remains

(a) in stable equilibrium

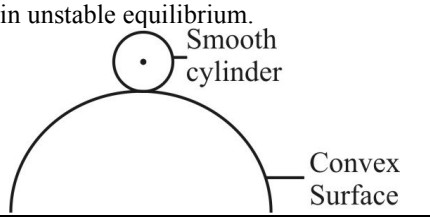
(b) in unstable equilibrium

(c) in neutral equilibrium

(d) out of equilibrium

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (b) A smooth cylinder lying on its convex surface remains in unstable equilibrium.



12. The forces which meet at one point and have their lines of action in different planes are called.

(a) coplanar non-concurrent forces

(b) non-coplanar concurrent forces

(c) non-coplanar non-concurrent forces

(d) intersection forces

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Ans. (b) : The line of action don't lie in the same plane implies - non-coplanar & meet at one point implies-concurrent force.

13. If the given forces P_1, P_2, P_3 and P_4 are such that the force polygon does not close, then the system will-

(a) be in equilibrium

(b) always reduce to a resultant force

(c) always reduce to a couple

(d) None of the above

JPSC AE 2009

Ans. (b) :

Polygon law of forces– If a number of co-planer forces acting at a point such that they can be represented in magnitude and direction by the sides of a polygon taken in the same order, then their resultant represented in magnitude and direction by the closing side of the polygon taken in opposite order.

■ The force polygon must be closed for equilibrium.

■ If the force polygon is open the system will always reduce to a resultant force.

14. Lami's theorem states that if

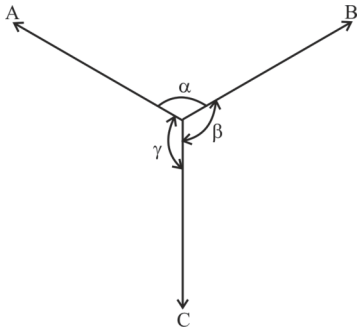
(a) Three forces acting at a point are in equilibrium; they can be respresented by the three sides of a triangle

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(b) The three forces acting at a point can be represented in magnitude and direction by the sides of a triangle; the forces are in equilibrium
(c) Three forces acting at a point are in equilibrium; each force is proportional to the sine of the angle between the other two forces
(d) None of these

Punjab PSC SDE 2016

Ans. (c) : Lami's theorem states if three coplanar forces acting at a point be in equilibrium, then each force is proportional to the sine of angle between the other two forces.



Mathematically,

$$\frac{A}{\sin\beta} = \frac{B}{\sin\gamma} = \frac{C}{\sin\alpha}$$

Where, A, B, C, are three forces and α, β, γ are the angles.

15. Newton is that force which produces an accleration of 1 m/sec^2 in a mass of 1 kg and is equal to
(a) 981 dynes (b) 10^3 dynes
(c) 10^5 dynes (d) 322×10^3 dynes

Punjab PSC SDE 2016

Ans. (c) : F = ma
$= 1 \times 1 = 1N$
We know that,
$\boxed{1N = 10^5 \text{ dynes}}$

16. Static equilibrium implies that the
(a) forces have zero resultant
(b) forces have zero resultant and zero moment
(c) forces are stationary
(d) forces are equal

TRB Poly. Tech. Lect. 2012

Ans. (b) : A body said to be static equilibrium if–
■ Algebric sum of all the external forces acting on the body zero. it means
$\Sigma F_x = 0, \Sigma F_y = 0, \Sigma F_z = 0$
■ Algebric sum of all the moments of external forces about a point is equal to zero. it means
$\Sigma M_x = 0, \Sigma M_y = 0, \Sigma M_z = 0$

17. If the given forces P_1, P_2, P_3 and P_4 which are coplanar and concurrent are such that the force does not close, then the system will-
(a) be in equilibrium
(b) always reduced to a resultant force
(c) always reduced to a couple
(d) always reduced to a couple and always be in equilibrium

JPSC AE 2009

Ans. (b) : If the given forces P_1, P_2, P_3 and P_4 which are co-planer and concurrent are such that the forces doesn't close, then the system will be always reduced to a resultant force.

18. The angles between two vectors to make their resultant a minimum and maximum respectively, are
(a) 0º and 90º (b) 180º and 90º
(c) 90º and 180º (d) 180º and 0º

OPSC Civil Services Pre 2011

Ans. (d) : The resultant of two vector is minimum when both vectors are equal and in opposite direction, i.e. the angle between the vector is 180^0. Resultant is maximum when vectors are parallel to each other i.e. angle between vectors is zero.

19. Match the list-I with List-II and select the correct answer using the codes given below the lists:

List- I	List-II
A. Coplanar force	1. Lines of action of all forces lie in the same plane, but do not pass through a common point
B. Concurrent forces	2. Lines of actions of all forces lie in the same plane and pass through a common point
C. Concurrent Coplanar forces	3. Lines of actions of all forces lie in the same plane.
D. Collinear forces	4. Lines of actions of all forces pass through a common point
	5. Lines of actions of all forces lie along same line.

Codes:
(a) A – 5, B – 4, C – 1, D – 3
(b) A – 4, B – 3, C – 2, D – 1
(c) A – 4, B – 5, C – 2, D – 1
(d) A – 3, B – 4, C – 2, D – 5

OPSC Civil Services Pre 2011

Ans. (d) : A. Coplanar forces – Lines of actions of all forces lies in the same plane.
B. Concurrent forces – Lines of actions of all forces pass through a common point.
C. Concurrent coplanar forces – Lines of actions of all forces lie in the same plane and pass through a common point.
Collinear forces – Lines of actions of all forces lie along same line.

20. The resultant of 2 forces Q and 2Q acting at a point P at an angle 90° to each other is 40R N. The value of Q is:
(a) 8 RN (b) $8\sqrt{5}$ RN
(c) $40\sqrt{5}$ RN (d) 20 RN

WBPSC Poly Lect. 2019

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Ans. (b) : Given $\theta = 90^o$

So, Resultant $R = \sqrt{Q_1^2 + Q_2^2}$

$$40R = \sqrt{Q^2 + (2Q)^2}$$

$$40R = \sqrt{5Q^2}$$

$$40R = \sqrt{5}\,Q$$

$$\frac{40R \times \sqrt{5}}{\sqrt{5} \times \sqrt{5}} = Q$$

$$Q = 8\sqrt{5}\,RN$$

21. If two concurrent forces P and Q having an angle α between them, then the direction of their resultant defined by angle θ with reference to the directions of P is given by tan θ which is equal to

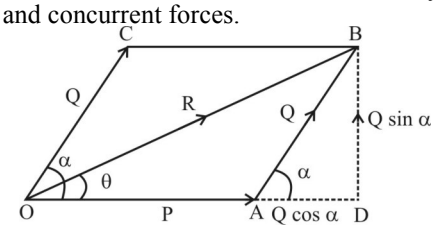
(a) $\frac{Q\sin\alpha}{P+Q\cos\alpha}$ (b) $\frac{P\sin\alpha}{Q+P\cos\alpha}$

(c) $\frac{P\sin\alpha}{Q+Q\cos\alpha}$ (d) $\frac{Q\sin\alpha}{P+P\cos\alpha}$

APPSC AEE MAINS 2016

Ans. (a) : Law of parallelogram of forces–

It states that if two forces acting simultaneously at a point be represented in magnitude and direction by the two adjacent sides of a parallelogram, their resultant may be represented in magnitude and direction by the diagonal of the parallelogram, which passes through their point of intersection. this law is valid only for co-planer and concurrent forces.



in ΔOBD : -

$OD = OA + AD = P + Q\cos\alpha$

$DB = Q\sin\alpha$

$$\tan\theta = \frac{BD}{OD} = \frac{Q\sin\alpha}{P+Q\cos\alpha}$$

22. If two equal forces of magnitude P act at an angle θ, their resultant, will be

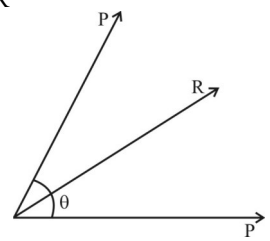
(a) P cos θ/2 (b) 2P sin θ/2

(c) P tan θ/2 (d) 2P cos θ/2

ISRO Scientist/Engineer 2015

Ans : (d) : • Resultant is defined as a single force which produces the same effect as the system of forces produce.

• Resultant 'R'



$$R = \sqrt{P^2 + P^2 + 2P^2\cos\theta}$$

$$R = \sqrt{2P^2 + 2P^2\cos\theta}$$

$$R = P\left(\sqrt{2(1+\cos\theta)}\right)$$

$$R = P\sqrt{2\left(2\cos^2\frac{\theta}{2}\right)} \qquad \left\{1+\cos\theta = 2\cos^2\frac{\theta}{2}\right\}$$

$$R = P\sqrt{4\cos^2\frac{\theta}{2}}$$

$$\boxed{R = 2P\cos\frac{\theta}{2}}$$

23. If three coplanar, concurrent forces are acting at a point are in equilibrium, of which two of them are collinear, then what is the magnitude of third force which is acting at an angle θ with other two forces?

(a) zero

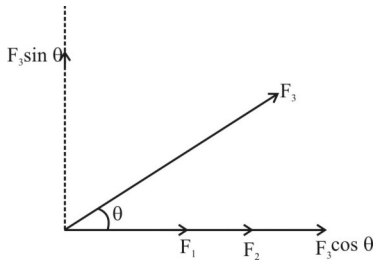
(b) algebraic

(c) vector sum of other two forces

(d) none of the above

Kerala PSC AE 2015

Ans. (a) :



Now from condition of equilibrium

If : $\Sigma V = 0$

$\Rightarrow F_3 \sin\theta = 0$

$\Rightarrow F_3 = 0$

24. Non coplanar concurrent forces are those forces which

(a) Meet at one point, but their lines of action do not lie on the same plane

(b) Do not meet at one point and their lines of action do not lie on the same plane

(c) Meet at one point and their lines of action also lie on the same plane

(d) Do not meet at one point, but their lines of action lie on the same plane

Haryana PSC AE 2018 Paper–I

ISRO Scientist/Engineer 2017

Ans : (a)

- **Non-Coplanar Concurrent forces–**
  Forces whose line of action do not lie on the same plane, but they meet at one point.
- **Non-Coplanar Non-Concurrent forces–**
  Forces whose line of action do not lie on the same plane and they do not meet at one point.
- **Coplanar Concurrent forces–**
  Forces which meet at one point and lines of action also lie on the same plane are known as coplanar concurrent forces.

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25. The resultant of two forces each equal to P and acting at right angles is

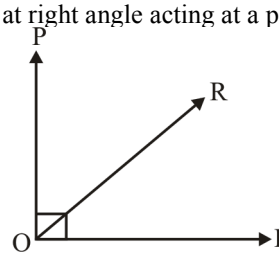
(a) $P/\sqrt{2}$ (b) $P/2$

(c) $P/2\sqrt{2}$ (d) $\sqrt{2}\,P$

ISRO Scientist/Engineer 2017

Ans : (d)

Two force 'P' at right angle acting at a point 'O'.



According to Parallelogram law of forces if two forces acting at a point are represented in magnitude and direction by the two sides of parallelogram, then the resultant of these two force is represented in magnitude and direction by the diagonal of parallelogram passing through the same point.

So,

$$R = \sqrt{P^2 + P^2 + 2P.P\cos 90^0} \qquad \because [\theta = 90^0]$$

$$R = \sqrt{2P^2 + 0}$$

$$\boxed{R = \sqrt{2}\,P}$$ Resultant.

26. If the angle between two equal forces is zero degree, their resultant will be _____.

(a) 0.2 P (b) Zero

(c) 2P (d) 0.5P

LMRC AM 2019

Ans. (c) Given,

$F_1 = F_2 = P$

$\theta = 0^o$

$\because$ Resultant force (R) = $\sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$

Resultant force (R)= $\sqrt{P^2 + P^2 + 2P^2}$ =2P

27. The force system that is applied to open a bottle cap is called :

(a) Collinear forces

(b) Like parallel forces

(c) Unlike parallel forces

(d) Couple

TNPSC AE 2015

Ans. (d) The force system that is applied to open a bottle cap is called couple.

28. The resultant of two collinear forces P and Q, which are acting in opposite direction is :

(a) P + Q (b) P – Q

(c) $\sqrt{P^2 + Q^2}$ (d) $\sqrt{P^2 - Q^2}$

TNPSC AE 2015

Ans. (b) : Resultant of two collinear forces P and Q–

(1) When two forces are in opposite direction.

R = P – Q

(2) When two forces are in same direction

R = P+ Q

29. For a given system of coplanar concurrent forces, if $\Sigma F_x = -20$ N and $\Sigma F_y = -20$N, then :

(a) $R = -20\sqrt{2}$ N and $\alpha = 45^0$ with east

(b) $R = 20\sqrt{2}$ N and $\alpha = 135^0$ with east

(c) $R = 20\sqrt{2}$ N and $\alpha = 225^0$ with east

(d) $R = -20\sqrt{2}$ N and $\alpha = 315^0$ with east

TNPSC AE 2015

Ans. (c) : Given,

$$\Sigma F_x = -20N$$

$$\Sigma F_y = -20N$$

$$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$$

$$R = \sqrt{(-20)^2 + (-20)^2}$$

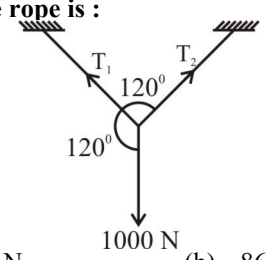
$$R = 20\sqrt{2}N$$

$$\tan\alpha = \frac{\Sigma F_y}{\Sigma F_x}$$

$$\tan\alpha - \frac{-20}{-20} = 1$$

$$\alpha = 45^0,\ 225^0$$

30. For the force system shown below, the tension T_1 in the rope is :

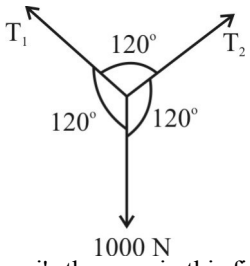


(a) 500 N (b) 866 N

(c) 1000 N (d) 1732 N

TNPSC AE 2015

Ans. (c)



Now apply the Lami's theorem in this fig.

$$\frac{T_1}{\sin 120^o} = \frac{1000}{\sin 120^o} = \frac{T_2}{\sin 120^o}$$

$$\frac{T_1}{\sin 120^o} = \frac{1000}{\sin 120^o}$$

$$\boxed{T_1 = 1000N}$$

31. The two forces of 9 Newtons and 12 Newtons which are acting at right angles to each other will have a resultant of

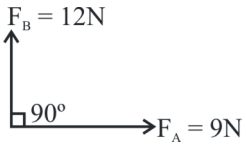
(a) 8 Newtons (b) 10 Newtons

(c) 15 Newtons (d) 20 Newtons

ISRO Scientist/Engineer 2018

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Ans : (c)



Force $F_A = 9$ Newtons
Force $F_B = 12$ Newtons

Resultant $'R' = \sqrt{(F_A)^2 + (F_B)^2 + 2F_AF_B\cos\theta}$

$R = \sqrt{(9)^2 + (12)^2 + 2\times 9\times 12\cos 90^o}$

$R = \sqrt{81+144} \quad \because \{\cos 90^o = 0\}$

$R = \sqrt{225}$

$R = 15$ Newtons

32. If a body is acted upon by a number of coplaner non-concurrent forces, it may:
(a) Rotate about itself without moving
(b) Move in any one direction
(c) Move in any one direction rotating about itself
(d) Be completely at rest
(e) All of the above options

Haryana PSC AE 2018 Paper–I

Ans. (e) : All of the above options

33. One kgf/cm^2 when converted into SI units is
(a) 0.0981 MPa (b) 0.981 Pa
(c) 10^4 Pa (d) 1 Pa

APPSC (PHE) AE Paper-2 2012

Ans. (a) : In SI units –
1 kgf/cm^2 = 0.0981 MPa

34. One Newton force is equivalent to:
(a) 1 kg wt (b) 9.81 kg wt
(c) 1/9.81 kg wt (d) 981 dyne

UJVNL AE 2012

Ans. (c) : 1 Newton force is equivalent to 1/9.81 kg. wt.

35. A system of coplanar forces is in equilibrium when
(a) Force polygon closes
(b) Funicular polygon closes
(c) Both force and funicular polygon closes
(d) All of the forces are coplanar concurrent

Punjab PSC SDE 2016

Ans. (c) :

36. The forces, whose line of action lies on the same line are known as
(a) Coplanar forces (b) Concurrent forces
(c) Collinear forces (d) None of these

Punjab PSC SDE 2016

Ans. (c) : The forces, whose line of action lies on the same line are known as Collinear forces.
• The forces, whose lines of action lie on same plane are termed as coplanar forces.
The forces, which meet at one point are known as Concurrent forces.

37. According to the law of moments, if a number of coplanar forces acting on a particle are in equilibrium, then
(a) Their lines of action are at equal distances
(b) The algebraic sum of their moments about any point in their plane is zero
(c) The algebraic sum of their moments about any point is equal to the moment of their resultant force about the same point
(d) Their algebraic sum is zero

GOA PSC Assistant Prof. 2020

Ans. (c) : According to law of moments, if a number of co-planer forces acting on a particle are in equilibrium, the algebric sum of their moments about any points equal to moment of their resultant force about same point

38. The forces, whose line of action lie on the same line, are known as
(a) Collinear forces
(b) None of three mentioned here
(c) Coplanar forces
(d) Concurrent forces

LMRC AM 2018
GWSSB AAE 2015

Ans. (a) : The forces whose line of action lie on the same line are known as collinear force.

39. Which is the correct statement as per Law of polygon forces?
(a) If any number of forces acting at point can be represented by the sides of a polygon taken in order, then the forces are in equilibrium
(b) If any number of forces acting at point can be represented in direction and magnitude by the sides of polygon, then the forces are in the equilibrium
(c) If a polygon representing forces acting at a point is closed then forces are in equilibrium
(d) If any number of forces acting at point can be represented in direction and magnitude by the sides of polygon taken in order, then the forces are in equilibrium

ISRO Scientist/Engineer 2015

Ans : (d) : If any number of forces acting at point can be represented in direction and magnitude by the sides of polygon taken in order, then the forces are in equilibrium.

40. The resultant of two perpendicular forces each equal to P/2 will be equal to
(a) $\frac{P}{\sqrt{2}}$ (b) $\sqrt{2}P$
(c) 2P (d) $\sqrt{2}P$

APPSC AEE MAINS 2016

Ans. (a) : Given,
$\theta = 90^o$

$P' = Q = \frac{P}{2}$

$R = \sqrt{P'^2 + Q^2 + 2P'Q\cos\theta}$

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$$= \sqrt{\left(\frac{P}{2}\right)^2 + \left(\frac{P}{2}\right)^2 + 2.\left(\frac{P}{2}\right).\left(\frac{P}{2}\right).\cos 90^o}$$

$$= \sqrt{\frac{P^2}{4} + \frac{P^2}{4}} = \sqrt{\frac{2P^2}{4}} = \sqrt{\frac{P^2}{2}}$$

$$R = \frac{P}{\sqrt{2}}$$

41. If a body is in equilibrium under several forces, vector diagram should remain :

(a) closed (b) opened
(c) rectangular (d) linear

PSPCL AE 2012

Ans. (a) : When a single force on a body act and its is maintained by the second force of an equal of opposite sign then the body will be in equilibrium state these the vector diagram drawn by connecting finishing point to the starting point will be closed one diagram.

42. The force polygon represeting a set of forces in equilibrium is a :

(a) triangle (b) open polygon
(c) closed polygon (d) parallelogram

Punjab MCP SDO 2015

Ans. (c) :

43. If forces of 1N, 2N, 3N, 4N, 5N and 6N act in order along the sides of a regular hexagon, their resultant is :

(a) 0 (b) 6N
(c) 12 N (d) 21 N

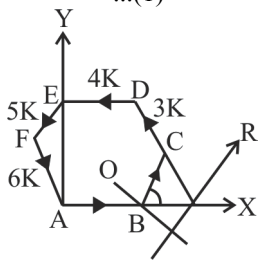
Punjab MCP SDO 2015

Ans. (b) : Let us consider a Hexagon ABCDEF with center O. Let us consider AB and AE as the axes. If K is proportionality constant, then the force along the sides AB, BC, CD, DE, EF and FA are the magnitudes K, 2K, 3K, 4K, 5K, 6K. Let R is resultant

$R\cos\theta = K + 2K\cos 60^o + 3K\cos 120^o - 4K - 2K\cos 60^o - 6K\cos 120^o$

$$= K + 2K\frac{1}{2} + 3K\left(-\frac{1}{2}\right) - 4K - 5K\left(\frac{1}{2}\right) - 6K\left(-\frac{1}{2}\right)$$

$= -3K$ ...(1)



Resolving along AY

$R\sin\theta$

$= 2K\sin 60 + 3K\sin 120^o - 5K\sin 60^o - 6K\sin 120^o - 5K\sin 60^o - 6K\sin 60^o$

$= 6K\sin 60^o$

$= -3\sqrt{3}.K$ ...(2)

$R^2 = 9K^2 + 27K^2$

$= 36K^2$

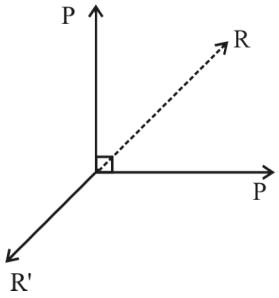
$\boxed{R = 6K}$

44. If two forces each equal to p act at right angles, their effect may be neutralized by a third force acting along their bisector in opposite direction whose magnitude is equal to

(a) $\frac{P}{2}$ (b) 2P
(c) $P\sqrt{2}$ (d) 3P

TRB Poly. Tech. Lect. 2012

Ans. (c) :



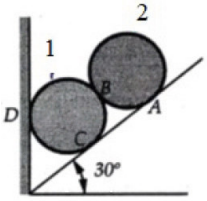
$\because R = \sqrt{P^2 + P^2 + 2P^2\cos 90^0}$

$R = \sqrt{2P^2} = \sqrt{2}P$

$\because$ R' is equilibrant force which brings a body into mechanical equilibrium, magnitude of equilibrant force is equal to resultant force (R) but opposite in direction so that.

$\therefore R = R' = \sqrt{2}P$

45. Two identical rollers each weighing 200 N are placed in a trough as shown in figure, assuming all contact surfaces are smooth, find the reaction developed at contact surface A to keep the system under equilibrium.

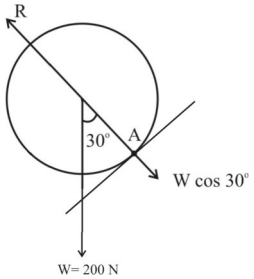


(a) 100.000 N (b) 173.205 N
(c) 230.94 N (d) 288.675 N

UPPCL AE 2019

Ans. (b)

FBD of body -2:-



$\because$ R = w cos 30° = 200 × cos 30° = 173.205 N

46. The type of force system that consists of forces whose lines of action do not meet at one single point is known as :

(a) Concurrent force system
(b) Coplanar force system

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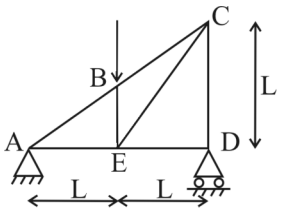
(c) Non concurrent force system
(d) Non coplanar force system

UPPCL AE 2019

Ans. (c) The type of force system that consists of forces whose lines of action do not meet at one single point is known as non concurrent force system.

- **Concurrent force system :** The forces, which meet at one point, are known as concurrent forces.
- **Coplanar forces system :** The forces whose line of action lie on the same plane are known as coplanar forces.

47. The forces in the members BE, CD, ED for the truss shown in are :

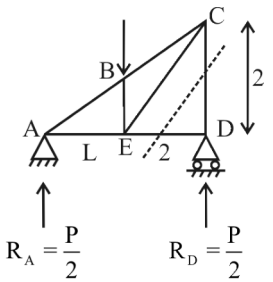


(a) P, $\frac{P}{2}$, 0 (b) $\frac{P}{2}$, P, 0

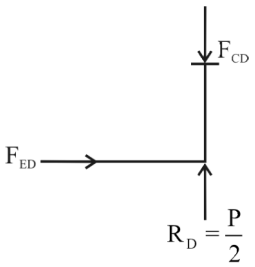
(c) P, P, P (d) $\frac{P}{2}$, $\frac{P}{2}$, 0

HPPSC Poly. Lect. 2016

Ans : (a)



At joint D



$F_{CD} = \frac{P}{2}$

$F_{CD} = \frac{P}{2}$

$F_{ED} = 0$, $F_{BE} = P$

48. The resultant of two forces acting at right angles is 5 kg_f and if they act at an angle of 60°, it is $\sqrt{37}$ kg_f. The magnitude of the forces are:

(a) 2 kg_f, 3 kg_f (b) 5 kg_f, 3 kg_f
(c) 4 kg_f, 5 kg_f (d) 3 kg_f, 4 kg_f
(e) 6 kg_f, 5 kg_f

Haryana PSC AE 2018 Paper–I

Ans. (d) Case-I

$\because R^2 = P^2 + Q^2 + 2PQ\cos\theta$

When $\theta = 90^0$, R = 5 kg_f

$P^2 + Q^2 = 25$ ..... (i)

Case - II

When $\theta = 60^0$, R = $\sqrt{37} kg_f$

$\therefore P^2 + Q^2 + 2PQ\cos 60 = \left(\sqrt{37}\right)^2$

$\therefore P^2 + Q^2 + PQ = 37$ ..... (ii)

From equation (i) and (ii)

$PQ = 12$ ..... (iii)

Solve the equation (i) and (iii) you will get

P = 3 kg_f and Q = 4 kg_f

49. The eccentric force induces an internal..... which is opposite that caused by the external load.

(a) Force (b) Shear force
(c) Resilience (d) Moment

AP (VSS-1) PART - A 2017

Ans : (d) When an eccentric force is applied on a body, always an internal moment induces in it which opposes that external eccentric force.
When the applied force is axial or concentric, then no any internal moment induces.

50. A single force and a couple acting in the same plane upon a rigid body

(a) Balance each other
(b) Cannot balance each other
(c) Produce moment of a couple
(d) Produce a couple

GOA PSC Assistant Prof. 2020
J & K PSC Pre 2013

Ans. (b) : A single force and a couple acting in the same plane upon a rigid body cannot balance each other because force and couple both are distinct quantity.

2. Friction

51. Which of the following is not a type of friction?

(a) Static friction (b) Ladder friction
(c) Fluid friction (d) Rolling friction

LMRC AM 2018

Ans. (b) : Ladder friction is not a type of friction.
Classification of friction :

- Dry friction – (a) Static friction
  (b) Dynamic friction – Ex. sliding friction, rolling friction
- Fluid friction – Ex. Lubricated friction, Skin friction, Viscous friction.
- Internal friction.

52. A block of weight 60 kg is resting on horizontal surface as shown in figure, calculate the maximum frictional force that develops between surfaces of contact, take coefficient of static friction as 0.3.

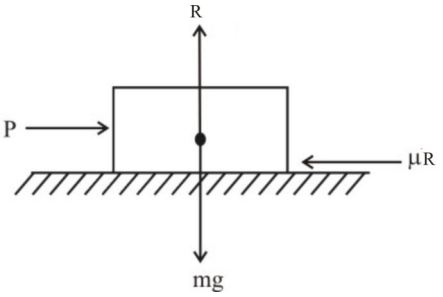


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(a) 176.58 N (b) 180.00 N
(c) 18.00 N (d) 155.64 N

UPPCL AE 2019

Ans. (a) :
weight of block = 60 kg
coefficient of static friction = 0.3



$\sum Fx = 0$
$P = \mu R$
$P = \mu mg$
$P = 0.3 \times 60 \times 9.81 = 176.58$ N

53. Kinetic friction may be defined as:
(a) Friction force acting when the body is just about to move
(b) Friction force acting when the body is in motion
(c) Angle between normal reaction and resultant of normal reaction and limiting friction
(d) Ratio of limiting friction and normal reaction
(e) None of the above options

Haryana PSC AE 2018 Paper–I

Ans. (b) : Static friction:-The friction between two surface of bodies in rest condition is called static friction.
Dynamic/kinetic friction:-The friction force when surface in contact are in relative motion is called Dynamic/kinetic friction.

54. The coefficient of friction does not depend on :
(1) area of interface of two mating surfaces
(2) roughness of two mating surfaces
(3) the time of contact
Out of these statements :
(a) (1), (2) and (3) are correct
(b) (1) and (2) are correct
(c) (2) and (3) are correct
(d) (1) and (3) are correct

HPPSC Poly. Lect. 2016

Ans : (d) Coefficient of friction depends only on the nature of the surface. It does not depends on any other factors including relative speed, surface area of contact.
Note : By HPPSC Poly. Lect. option (a) is given.

55. Friction acting on a body which is just on the point or verge of sliding is:
(a) Limiting friction (b) Static friction
(c) Dynamical friction (d) Dry friction

IRCON AE 2017

Ans. (a) : The maximum friction that can be generated between two static surfaces in contact with each other. Once a force applied to the two surfaces exceeds the limiting friction motion will occur.

56. The coefficient of friction between two surfaces is the constant of proportionality between the applied force and normal reaction
(a) At the instant of impending motion
(b) At an instant after the motion takes place
(c) At the instant of application of force
(d) At the instant when the body is at rest

GOA PSC Assistant Prof. 2020

Ans. (a) : The coefficient of friction between two surfaces is the constant of proportionality between the applied force and normal reaction at the instant of impending motion.

57. On a ladder resisting on a smooth ground and leaning against a rough vertical wall, the force of friction acts:
(a) Towards the wall at its upper end
(b) Away from the wall at its upper end
(c) Upwards at its upper end
(d) Downwards at its lower end

OPSC Civil Services Pre 2011

Ans. (c) : On a ladder resisting on a smooth ground and leaning against a rough vertical wall the force of friction acts upwards at its upper end because ladder will try to move away from the wall.

58. Static friction is always
(a) Greater than the dynamic friction
(b) Less than the dynamic friction
(c) Equal to the dynamic friction
(d) None of these

Punjab PSC SDE 2016

Ans. (a) : Static friction i.e. friction experienced by the body at rest is always greater than kinetic friction experienced by body at motion.

59. A mass of 10 kg rests on a horizontal plane. The plane is gradually inclined until at an angle $\theta = 14^0$ with the horizontal, the mass just begins to slide. What is the co-efficient of static friction between the block and the surface ?
(a) 0.15 (b) 0.42
(c) 0.31 (d) 0.25

BHEL ET 2019

Ans. (d) : Given, mass = 10 kg, $\theta = 14^0$

$\mu = \tan\theta = \tan 14^0 = 0.25$

60. The coefficient of friction depends on
(a) nature of surface (b) area of surface
(c) shape of surface (d) all of these

GPSC Assistant Prof. 2016

Ans. (a) : The coefficient of friction depends on nature of the surface and independent of shape, area of the surface.

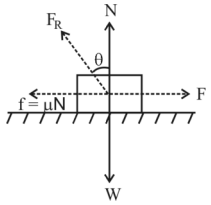
61. The angle of friction is
(a) The ratio of friction and the normal reaction
(b) The force of friction when the body is in motion
(c) The angle between the normal reaction and the resultant of normal reaction and limiting friction

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(d) The force of friction at which the body is just about to move

ISRO Scientist/Engineer 2018

Ans : (c) : Angle of friction is defined as the angle made by the resultant of frictional force and the normal reaction with the normal reaction.



$$\tan\theta = \frac{f}{N} = \frac{\mu N}{N}$$

$$\boxed{\tan\theta = \mu}$$

62. The ratio of Static Friction to Dynamic friction is always

(a) Equal to 1 (b) Less than 1
(c) Greater than 1 (d) None

ISRO Scientist/Engineer 2013

Ans : (c)

- When the body is in rest condition then the friction force exerted by the body is called "static friction".
- When the body is in moving condition then the friction force exerted by the body is called "Dynamic friction".
- The ratio of static friction to Dynamic friction is always greater than '1'.

63. The ratio of limiting frictional force to normal reaction is known as _____.

(a) Coefficient of expansion
(b) Coefficient of restitution
(c) Coefficient of friction
(d) Coefficient of resistance

LMRC AM 2019

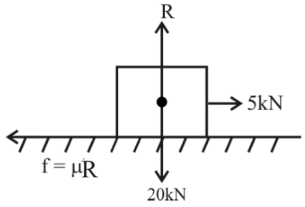
Ans. (c) The ratio of limiting frictional force to normal reaction is known as coefficient of friction.

64. A block of weight 20 kN just begins to move along a horizontal surface on application of 5 kN horizontal force. The coefficient of friction between block and surface is:

(a) 0.10 (b) 0.20
(c) 0.25 (d) 0.50

Kerala PSC AE 2015

Ans. (c) :



$f = \mu R = 5$ kN

$\mu \times 20 = 5$

$$\mu = \frac{5}{20} = 0.25$$

65. When a body resting on an inclined surface just begins to slide down the inclined plane under the action of its own weight, then the angle of inclination of the inclined plane is

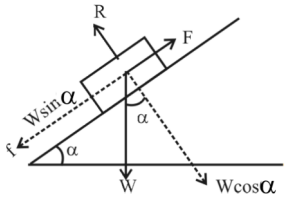
(a) Greater than the angle of repose
(b) Less than the angle of repose
(c) Equal to the angle of repose
(d) Vertical
(e) Horizontal

CGPSC AE 2014 Shift–I

Ans. (a) Angle of repose (α_R) : It is the angle of inclined plane at which block will be just about to slide down or in impending motion.

i.e –

$\mu = \tan\theta = \tan\alpha_R = \tan\alpha$



Note :

If $\alpha < \theta \rightarrow$ Rest
$\alpha > \theta \rightarrow$ motion
$\alpha = \theta \rightarrow$ impending motion

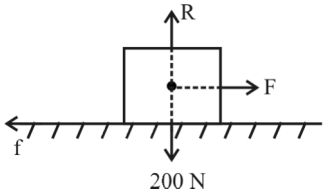
Where, θ = Friction Angle
α = Angle of inclined surface form horizontal.

66. A body having a weight of 200 N is placed on a rough horizontal plane. If coefficient of friction between the body and the horizontal plane is 0.3, the horizontal force required to just slide the body on the plane is

(a) 120 N (b) 200 N
(c) 60 N (d) 666.67 N

ASSAM PSC CCE (PRE) 2015

Ans : (c)



$F = \mu R$ .....(i)

Weight of the body (W) = R (Reaction force)

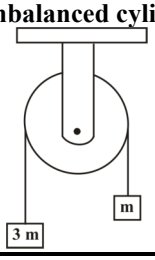
$W = 200$ N

From Equation (i)

$F = 0.3 \times 200$

$F = 60$ N

67. What is the minimum coefficient (μ) of friction between the rope and the fixed shaft which will prevent the unbalanced cylinder from moving?

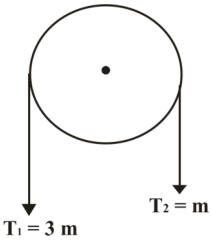


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(a) $\mu = 0.333$ (b) $\mu = 0.350$
(c) $\mu = 0.253$ (d) $\mu = 0.372$

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (b) :



$$\frac{T_1}{T_2} = e^{\mu\alpha}$$

θ is angle of contact between rope and cylinder

$\theta = \pi$ radian

$$\frac{3m}{m} = e^{\mu\pi}$$

$$3 = \left(e^{\pi}\right)^{\mu}$$

$$3 = (23.14)^{\mu}$$

$$\mu = \frac{\log 3}{\log(23.14)} = 0.3497$$

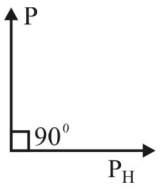
$\therefore \ \mu \approx 0.35$

68. The horizontal component of a force P acting towards north is :

(a) 0 (b) P
(c) 2P (d) ∞

TNPSC AE 2015

Ans. (a) :



$P_H = P\cos\theta$

$= P\cos 90^0$

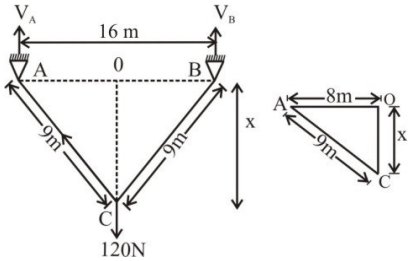
$= 0$

69. A light cable 18 m long, is supported at two ends at the same level. The supports are 16 m apart. The cable supports 120 N load dividing the distance into two equal parts. Find the tension in cable.

(a) 60.00N (b) 156.34 N
(c) 116.42 N (d) 130.97 N

UPPCL AE 2019

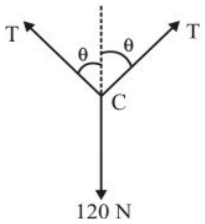
Ans. (d) :



In, ΔAOC

$9^2 = 8^2 + x^2$

$x = 4.123$ m and $\cos\theta = \frac{4.123}{9}$



at C; $\Sigma F_y = 0$

$2T\cos\theta = 120$

$$2 \times T \times \frac{4.123}{9} = 120$$

$$T = \frac{120 \times 9}{4.123 \times 2} = 130.97 \text{ N}$$

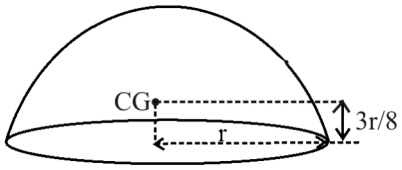
3. Moment of Inertia & Centre of Gravity

70. Position of C.G. of hemisphere of radius 16 mm measured form the base along the vertical radius lies at distance.

(a) 6 mm (b) 4 mm
(c) $\frac{48}{3\pi}$ mm (d) $\frac{24}{3\pi}$ mm

UP Jal Nigam AE 2016

Ans. (a) :



Here r = 16 mm

Distance of C.G. from base $= \frac{3 \times 16}{8} = 6\,\text{mm}$

71. The unit of moment of inertia of an area is:

(a) Kg/m (b) Kg/sq.m
(c) m^4 (d) m^3

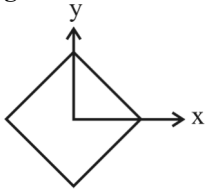
ISRO Scientist/Engineer 2020

Ans. (c) : $I = \frac{bd^3}{12}$ [For rectangle section]

Unit $\Rightarrow$ mm × $(mm)^3$

$\Rightarrow$ mm^4 or m^4

72. A square section of side 'a' is oriented as shown in the figure. Determine the section modulus of the following section?



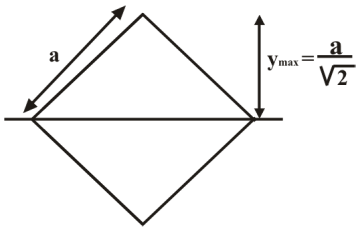
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(a) $\frac{a^4}{12\sqrt{2}}$ (b) $\frac{a^3}{12\sqrt{2}}$

(c) $\frac{a^4}{6\sqrt{2}}$ (d) $\frac{a^3}{6\sqrt{2}}$

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (d) :



The section modulus is

$$Z=\frac{I_G}{y_{max}}=\frac{\left[\frac{a.a^3}{12}\right]}{\left[\frac{a}{\sqrt{2}}\right]}=\frac{a^3}{6\sqrt{2}}$$

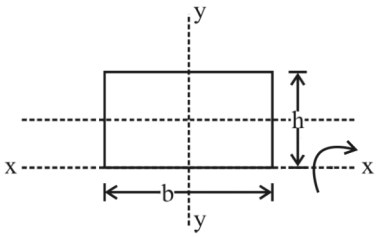
73. The moment of inertia of a rectangular section base b and height h about the base of rectangle is:

(a) $\frac{bh^3}{12}$ (b) $\frac{bh^3}{3}$

(c) $\frac{bh^3}{2}$ (d) $\frac{bh^3}{6}$

PTCUL AE 2016

Ans. (b) : Moment of inertia $(I_{xx}) = \frac{bh^3}{12}+bh\left(\frac{h}{2}\right)^2$



$$I_{xx}=\frac{bh^3}{12}+\frac{bh^3}{4}$$

$$=bh^3\left(\frac{1+3}{12}\right)=\frac{4bh^3}{12}$$

$$=\frac{bh^3}{3}$$

74. When the diameter of circular section is doubled, its radius of gyration is

(a) Reduced to half

(b) Doubled

(c) Increased by 8 times

(d) Increased by 3 times

UJVNL AE 2016

Ans. (b) : Radius of gyration (k) $=\sqrt{\frac{I_{min}}{A}}$

$$k=\sqrt{\frac{\frac{\pi}{64}d^4}{\frac{\pi}{4}d^2}}=d\sqrt{\frac{1}{16}}=\frac{d}{4}$$

So, $k \propto d$

Then, if diameter is doubled the radius of gyration also gets doubled.

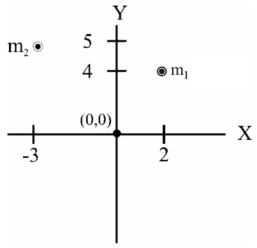
75. Two particles of mass m_1 = 2 kg and m_2 = 3 kg are located at coordinates (2m, 4m) and (-3m, 5m), respectively, in the xy - plane. Find the location of their centre of mass.

(a) – 1.2 m, 4.2 m (b) –1 m, 4.6 m

(c) – 0.5 m, 4.8 m (d) – 1.5 m, 4.2 m

UPPCL AE 2019

Ans. (b) :



For given masses, the centre of mass at x-axis

$$\bar{x}=\frac{m_1x_1+m_2x_2}{m_1+m_2}$$

$$=\frac{2\times2+3\times(-3)}{2+3}=-1\text{ m}$$

and, the centre of mass at y-axis

$$\bar{y}=\frac{m_1y_1+m_2y_2}{m_1+m_2}$$

$$=\frac{2\times4+3\times5}{2+3}=4.6\text{ m}$$

So, the point of centre of masses = (-1m, 4.6m)

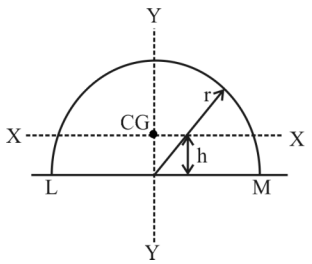
76. Moment of inertia of a semicircle about its XX-axis is given by

(a) 0.22 r^3 (b) 0.11 r^4

(c) 0.14 r^4 (d) 0.2 r^4

APPSC (PHE) AE Paper-2 2012

Ans. (b) :



Let,

r = radius of semi-circle

LM = base of the semi-circle or lamina

Moment of inertia of a circular lamina about its diameter

$$I_G=\frac{\pi d^4}{64}=\frac{\pi r^4}{4}$$

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Moment of inertia of the semi-circle about LM

$I_{LM} = = \frac{I_G}{2} = \frac{\pi r^4}{8}$

Let XX be centroidal axis parallel to the base LM and Let h be the distance between the axis XX and base LM

$h = \frac{4r}{3\pi}$

Applying parallel axis theorem

$I_{LM} = I_{XX} + Ah^2$

$\frac{\pi r^4}{8} = I_{XX} + \frac{\pi r^2}{2} \times \left(\frac{4r}{3\pi}\right)^2$

$I_{XX} = \frac{\pi r^4}{8} - \frac{8\pi r^4}{9\pi^2}$

$I_{XX} = 0.11\ r^4$

77. The diameter of kernel of a circular section of diameter 'd' is

(a) d/2 (b) d/3

(c) $d/\sqrt{2}$ (d) d/4

APPSC (PHE) AE Paper-2 2012

Ans. (d) :

The diameter of kernel of a circular section = $\frac{d}{4}$

78. Moment of inertia of a circle about XX plane :

(a) $\pi d^4/64$ (b) $bd^4/64$

(c) $bd^3/12$ (d) $bh^4/12$

(e) $bh^4/16$

CGPSC AE 2017

Ans. (a)

Moment of Inertia of circle about XX plane= $\frac{\pi d^4}{64}$

79. Which of the following cases gives the least moment of inertia, in case of a square ?

(a) M.I. about the base side

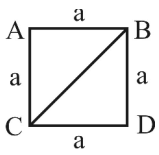
(b) M.I. about the top side

(c) M.I. about the diagonal

(d) M.I. about one vertical side

TNPSC AE 2015

Ans. (c) :



Moment of inertia about the base side (I_{CD}) = $\frac{a^4}{3}$

Moment of inertia about the top side (I_{AB}) = $\frac{a^4}{3}$

Moment of inertia about the vertical side (I_{AC} or I_{BD})

= $\frac{a^4}{3}$

Moment of inertia about the diagonal side (I_{CB}) = $\frac{a^4}{12}$

80. Polar moment of inertia of a rectangular section having width 'a' and depth 'b' is given by _____.

(a) $\frac{ab \times (a^2 + b^2)}{10}$ (b) $\frac{ab \times (a^2 + b^2)}{12}$

(c) $\frac{ab \times (a^2 + b^2)}{20}$ (d) $\frac{ab \times (a^2 + b^2)}{16}$

LMRC AM 2019

Ans. (b) Polar moment of Inertia is equal to the sum of moment of Inertia about x-axis and y-axis.

$I_P = I_{xx} + I_{yy}$

$= \frac{ab^3}{12} + \frac{ba^3}{12}$

$= \frac{ab(a^2 + b^2)}{12}$

81. The product moment of inertia of a rectangular section of size b × d about its horizontal centroidal axis is equal to :

(a) $\frac{b^2d^2}{4}$ (b) $\frac{bd^2}{6}$

(c) 0 (d) $\frac{b^2d^2}{6}$

KPSC AE 2020

Ans. (c) The product moment of inertia of a rectangular section is zero because x and y are symmetry axis.

82. The point through which the whole weight of the body acts, irrespective of its position, is known as

(a) Moment of inertia

(b) Centre of mass

(c) Centre of percussion

(d) Centre of gravity

ISRO Scientist/Engineer 2017

Ans : (d)

- **Centre of Gravity**–It is defined as a point through which the whole weight of the body is assumed to be act.

OR

The point from which the object will balance.

- **Centre of mass**–It is the point in a body at which the entire mass may be assumed to be concentrated.
- **Centre of Percussion**–It is the point on an extended massive object attached to a pivot where a perpendicular impact will produce no reactive shock at the pivot.

83. The moment of inertia of a hollow circular section whose external diameter is 8 cm and internal diameter is 6 cm, about centroidal axis is

(a) 437.5 cm^4 (b) 337.45 cm^4

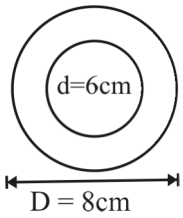
(c) 237.5 cm^4 (d) 137.45 cm^4

Punjab PSC SDE 2016

ISRO Scientist/Engineer 2015

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Ans : (d) : Moment of inertia 'I_G' of hollow circular section is



$$I_G = I_{xx} = I_{yy} = \pi\frac{(D^4 - d^4)}{64}$$

$$= \pi\frac{\left((8)^4 - (6)^4\right)}{64}$$

$$= \frac{\pi(4096 - 1296)}{64}$$

$$= \frac{\pi}{64} \times 2800 = 137.446$$

$$\therefore I_G \approx 137.45 \text{ cm}^4$$

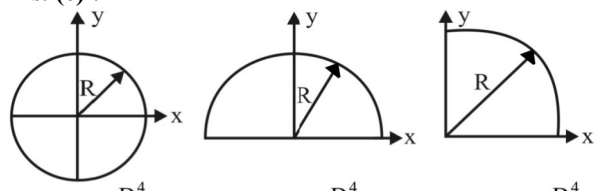
84. The moment of inertia of a semi-circle of diameter D about its base diameter is :

(a) $\frac{\pi}{32}D^4$ (b) $\frac{D^4\pi}{64}$

(c) $\frac{\pi D^4}{128}$ (d) $\frac{\pi D^4}{256}$

TNPSC AE 2015

Ans. (c) :



$I_x = I_y = \frac{\pi D^4}{64}$ $I_X = \frac{\pi D^4}{128}$ $I_x = I_y = \frac{\pi D^4}{256}$

$J_0 = \frac{\pi D^4}{32}$ $J_0 = \frac{\pi D^4}{128}$

85. The square root of the ratio of moment of inertia of the cross section to its cross-sectional area is called

(a) Second moment of area
(b) Slenderness ratio
(c) Section modulus
(d) Radius of gyration

Gujarat PSC AE 2017

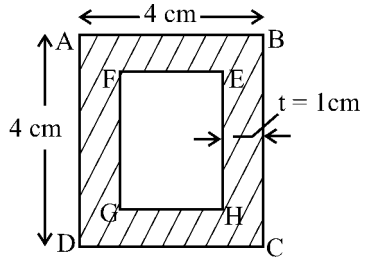
Ans : (d) The square root of the ratio of moment of inertia of the cross section to its cross-sectional area is called radius of gyration.

86. A rectangular hollow section with internal dimensions 4 cm by 4 cm and thickness of 1 cm will have its moment of inertia about centroidal X-X axis

(a) 21.33 cm^4 (b) 1.33 cm^4
(c) 256 cm^4 (d) 20 cm^4
(e) 2.0 cm^4

CGPSC AE 2014 Shift–I

Ans. (d) :



dimension of internal section (EFGH) = 2 cm × 2 cm
MOI of hollow
Section = MOI of ABCD - MOI of EFGH

$$= \frac{4(4)^3}{12} - 2\frac{(2)^3}{12}$$

$$= \frac{2\times 2^3}{12}\left[2\times 2^3 - 1\right]$$

$$= \frac{16}{12}[16 - 1]$$

$$= \frac{16}{12}\times 15$$

$$\boxed{I_{XX} = 20\text{cm}^4}$$

87. The distance from the center of gravity of a cutting to that of the tipped material is called as _____.

(a) Free haul distance (b) Over-haul
(c) Average haul distance (d) Haul

NPCC AE 2017

Ans. (c) : The average distance from centre of gravity of cutting or trench (using plumb bob) to the point of material tipping is known as average haul distance.

88. The C.G. of a quadrant of a circle lies along its central radius at a distance of

(a) 0.3 R (b) 0.44 R
(c) 5 R (d) 0.6 R

Punjab PSC SDE 2016

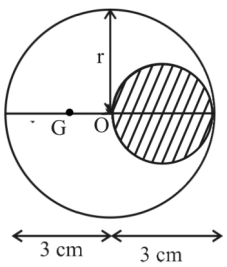
Ans. (d) : The center of gravity of a quadrant of a circle lies along its central radius at a distance of 0.6 R. Where R is radius.

89. From a circular plate of diameter 6 cm is cut out a circle whose diameter is equal to the radius of the plate. The CG of the remainder from the centre of circular plate is at a distance of

(a) 2.0 cm (b) 1.5 cm
(c) 1.0 cm (d) 0.5 cm

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (d) :



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Distance of CG of remainder from 'O'

$$\bar{x} = \frac{A_1x_1 - A_2x_2}{A_1 - A_2} = \frac{\pi r^2 \times 0 - \pi\left(\frac{r}{2}\right)^2 \times \frac{r}{2}}{\pi r^2 - \pi\left(\frac{r}{2}\right)^2}$$

$$= -\frac{\frac{\pi r^3}{8}}{\frac{3}{4}\pi r^2} = -\frac{r}{6} = -\frac{3}{6} = -0.5 \text{ cm}$$

Note : Negative sign (–) is indicates that the CG of remainder is opposite of the cutting side.

90. The point, through which the whole weight of the body acts, irrespective of its position, is known as

(a) centre of percussion (b) moment of inertia
(c) centre of mass (d) centre of gravity

Arunachal Pradesh Poly. Tech. Lect. 2019
GPSC R&B AE 2016

Ans. (d) : Center of gravity is the point of the body at which whole weight of body acts.

91. The circular area is cut into four equal parts. The quarter - circular area is separated. Now centroid distance of quarter - circular area from base and vertical straight line is .....

(a) 2 r/π (b) 2 r/3π
(c) 4 r/3π (d) 4 π/3r

GPSC R&B AE 2016

Ans. (c) : For circular section –

$$\boxed{x = \frac{2r\sin\alpha}{3\alpha}}$$

For quarter circular area

$$x = \frac{2r\sin 90^\circ}{3\frac{\pi}{2}}$$

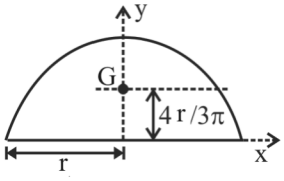
$$\boxed{x = \frac{4r}{3\pi}}$$

92. The centre of gravity of a semicircle of radius r from the diameter is-

(a) $\frac{3r}{2\pi}$ (b) $\frac{4r}{3\pi}$
(c) $\frac{2r}{3\pi}$ (d) $\frac{3r}{4\pi}$

JPSC AE 2009
UJVNL AE 2012

Ans. (b) : The centre of gravity of semi circle is located at a distance of $\frac{4r}{3\pi}$ from the base or diameter of semi circle.



93. The center of gravity of a hemisphere of radius r from its base measured along the vertical radius is

(a) $\frac{4r}{3}$ (b) $\frac{3r}{2}$
(c) $\frac{3r}{8}$ (d) $\frac{4r}{7}$

Punjab PSC SDE 2016

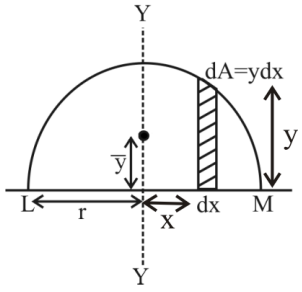
Ans. (c) : The center of gravity of a hemisphere lies at a distance of 3r/8 from its base, measured along the vertical radius.
Note-The center of gravity of a semi-circle lies at a distance of 4r/3π from its base, measured along the vertical radius.

94. The C.G. of a semicircular lamina lies at a distance of

(a) $\frac{3r}{8}$ from base diameter
(b) $\frac{4r}{3\pi}$ from base diameter
(c) $\frac{8r}{3}$ from base diameter
(d) none of these

TNPSC AE 2012

Ans. (b) :



$$\bar{y} = \frac{\int \bar{y}'\,dA}{A} \quad (\bar{y}' = y/2)$$

$$= \frac{2\int_0^r \frac{y}{2} \times y\,dx}{\pi r^2/2} = \frac{2}{\pi r^2}\int_0^r \left(r^2 - x^2\right)dx$$

(For circle $x^2 + y^2 = r^2$)

$$= \frac{2}{\pi r^2}\left[r^2x - \frac{x^3}{3}\right] = \frac{2}{\pi r^2}\left[r^3 - \frac{r^3}{3}\right]$$

$$= \frac{2}{\pi r^2} \times \frac{2r^3}{3} = \frac{4r}{3\pi}$$

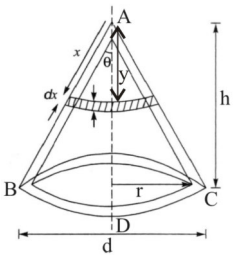
95. The centre of gravity of a right circular hollow cone of diameter *d* and height h lies at a vertical distance _____ from the base.

(a) $\frac{h}{2}$ (b) $\frac{h}{3}$
(c) $\frac{h}{4}$ (d) $\frac{h}{6}$

Kerala PSC AE 2015

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Ans. (b) :



$$dm = \frac{M}{\pi r\sqrt{h^2 + r^2}} \times 2\pi \times \sin\theta . dx$$

$$dm = \frac{M}{r\sqrt{h^2 + r^2}} \times 2x \frac{r}{\sqrt{h^2 + r^2}} \times dx$$

$$dm = \frac{2M}{h^2 + r^2} \times xdx$$

We consider a strip ring of mass *dm* as an element

$$y_c = \frac{1}{M}\int dm\, y$$

$$= \frac{1}{M}\int \frac{2M}{\left(h^2 + r^2\right)}(xdx) \times \left[x \frac{h}{\sqrt{h^2 + r^2}}\right] \quad \because \cos\theta = \frac{h}{\sqrt{h^2 + r^2}}$$

$$= \frac{2h}{\left(h^2 + r^2\right)^{3/2}} \int_0^{\sqrt{h^2 + r^2}} x^2 dx$$

$$= \frac{2h}{\left(h^2 + r^2\right)^{3/2}} \left[\frac{x^3}{3}\right]_0^{\sqrt{h^2 + r^2}}$$

$$y_c = \frac{2h}{\left(h^2 + r^2\right)^{3/2}} \times \frac{\left(h^2 + r^2\right)^{3/2}}{3} = \frac{2h}{3}$$

y_c = Distance of the CG of the hollow cone from A distance of the CG of the hollow cone from the base = $h - \frac{2h}{3} = \frac{h}{3}$

96. A force equal to product of mass times acceleration and acting in direction opposite to acceleration is called

(a) Damping (b) Friction
(c) Inertia force (d) Equilibrant force

GPSC AE January 2018

Ans. (c) : Inertia force is equal to the accelerating force in magnitude and opposite in direction.

97. The unit of measurement of inertia of mass is

(a) Kg/sq metre (b) Kg/ metre
(c) Kg-sq metre (d) Cubic metre

ISRO Scientist/Engineer 2019

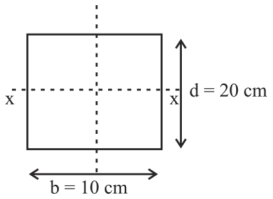
Ans : (c) : $I = mk^2$
m = mass
k = radius of gyration
I = moment of inertia
$I = kg\ m^2$

98. A rectangular section has dimensions of 10 cm × 20 cm. The ratio of the moment of inertia about x-axis passing through its centroid to the moment of inertia about y-axis passing through its centroid is equal to

(a) 8 (b) 4 (c) 6 (d) 2

GPSC AE March 2018

Ans. (b) :



$$I_{xx} = \frac{bd^3}{12} = \frac{10 \times 20^3}{12}\ cm^4$$

$I_{xx} = 6666.67\ cm^4$

$$I_{yy} = \frac{db^3}{12} = \frac{20 \times 10^3}{12}\ cm^4$$

$I_{yy} = 1666.67\ cm^4$

$$\Rightarrow \quad \frac{I_{xx}}{I_{yy}} = \frac{6666.67}{1666.67} = 4$$

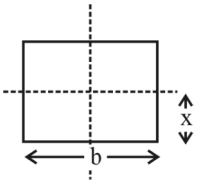
99. The moment of inertia of a rectangular section about the base is

(a) twice the moment of inertia about the centroidal axis
(b) three times the moment of inertia about the centroidal axis
(c) four times the moment of inertia about the centroidal axis
(d) six times the moment of inertia about the centroidal axis

GPSC AE March 2018

Ans. (c) : Moment of inertia of rectangular section about centroidal axis

$$I_c = \frac{bd^3}{12}$$



About base axis-

$I_b = I_c + Ax^2$

$$I_b = \frac{bd^3}{12} + bd\left(\frac{d}{2}\right)^2$$

$$I_b = \frac{bd^3}{12} + \frac{bd^3}{4}$$

$$I_b = \frac{bd^3}{3}$$

$$I_b = 4\left(\frac{bd^3}{12}\right)$$

$$\boxed{I_b = 4I_c}$$

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100. The square root of ratio of moment of inertia of cross section to its cross sectional area is called.

(a) second moment of area
(b) radius of gyration
(c) section modulus
(d) slenderness ratio

GPSC Poly. Tech. Lect. 2016

Ans. (b) : The square root of ratio of moment of inertia of cross section to its cross sectional area is called as radius of gyration.

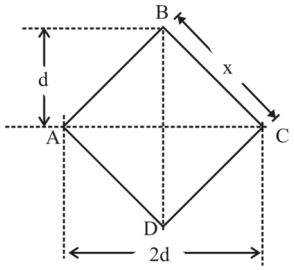
$$\text{Radius of gyration} = \sqrt{\frac{I}{A}}$$

101. The moment of Inertia of a square of side 'x' about its diagonal is .....

(a) $x^4/16$ (b) $x^3/12$
(c) $x^4/12$ (d) $x^3/8$

GPSC R&B AE 2016
JPSC AE 2009

Ans. (c) :



Moment of inertia about diagonal or AC –

$$= \frac{2\times(2d)\times d^3}{12} = \frac{d^4}{3} = \frac{1}{3}\left(\frac{x}{\sqrt{2}}\right)^4 = \frac{x^4}{12}$$

102. Modified moment of inertia of sections with a single web, is equal to moment of inertia of the section about Y-Y axis at the point of maximum bending moment and is multiplied by the ratio of :

(a) area of compression flange at the minimum bending moment to the corresponding area at the point of maximum bending moment
(b) area of tension flange at the minimum bending moment of the corresponding area at the point of maximum bending moment
(c) total area of flanges at the maximum bending moment to the corresponding area at the point of maximum bending moment
(d) none of the other three options

KPTCL AE 2016

Ans. (c) : Modification moment of inertia of section with a single web, is equal to moment of inertia of section about Y-Y axis at the point of maximum bending moment and is multiple by the ratio of total area of flange at the maximum bending moment to the corresponding area at the point of maximum bending moment.

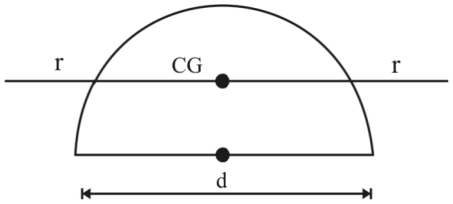
103. The moment of inertia of a circular lamina of a diameter d, about an axis perpendicular to the plane of the lamina and passing through its centre, is:

(a) $\frac{\pi d^4}{12}$ (b) $\frac{\pi d^4}{16}$
(c) $\frac{\pi d^4}{32}$ (d) $\frac{\pi d^4}{64}$

OPSC Civil Services Pre 2011

Ans. (c) : The moment of inertia of a circular lamina of a diameter d, about an axis perpendicular to the plane of lamina and passing through its centre is $\frac{\pi d^4}{32}$.

104. The moment of inertia of a semicircular section about its centre of gravity parallel to rr axis is given by :



(a) $0.0032\ d^4$ (b) $0.0069\ d^4$
(c) $0.0089\ d^4$ (d) $0.0016\ d^4$

TRB Poly. Lect. Tech. 2017

Ans. (b) : $I_{rr} = \frac{\pi r^4}{8} - \frac{8\pi r^4}{9\pi^2} = 0.11\ r^4 = 0.0069\ d^4$

105. The sum of the moment of inertias of two mutually perpendicular arbitrary axis is

(a) varies directly as the square of smaller value
(b) zero
(c) always constant
(d) different for different orientations

TRB Poly. Tech. Lect. 2012

Ans. (d) : The sum of the moment of inertias of two mutually perpendicular arbitrary Axis is different for different orientations.

106. The moment of inertia of a triangle of base b and height h about the base is

(a) $\frac{bh^3}{12}$ (b) $\frac{bh^3}{36}$
(c) $\frac{bh^3}{6}$ (d) $\frac{bh^3}{18}$

BPSC AE Aug. 2019 Paper-V
J&K PSC Pre 2013

Ans. (a) • Moment of inertia about axis passing through centroid parallel to base $I_c = \frac{bh^3}{36}$

• Moment of inertia about the axis passing through its base $I_b = \frac{bh^3}{12}$

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107. The condition for identifying the two perpendicular axes X and Y of a section as principal axes is :

(a) product moment of Inertia (I_{XY}) is zero

(b) product of moment of Inertia (I_X. I_Y) is zero

(c) sum of squares of moment of Inertia about the axes $\left(I_X^2 + I_Y^2\right)$ is zero

(d) moment of inertia about the axes are equal ($I_X = I_Y$)

BHEL ET 2019

Ans. (a) : If two I_X and I_Y are moment of Inertia of axes X and Y of a section then product moment of inertia (I_{XY}) is zero. Moment of inertia about a principle axis is called the principle moment of inertia.

108. The density at any point of a slender rod of length L varies with the first power of the distance of the point from one end of the rod. Locate the mass centre.

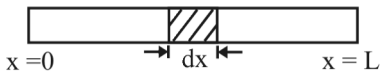
(a) $\frac{2L}{3}$ (b) $\frac{3L}{5}$

(c) $\frac{L}{4}$ (d) $\frac{L}{2}$

BHEL ET 2019

Ans. (a) :

$\rho = \rho_0 \frac{x}{L}$



$dm = \rho.Adx$

Position of centre of mass $(X_{cm}) = \frac{\int xdm}{\int dm}$

$= \frac{\int x.\rho Adx}{\int \rho.Adx}$

$= \frac{\int_0^L x.\rho_0 \frac{x}{L}dx}{\int_0^L \rho_0 \frac{x}{L}dx} = \frac{\int_0^L x^2 dx}{\int_0^L xdx}$

$= \frac{\left[\frac{x^3}{3}\right]_0^L}{\left[\frac{x^2}{2}\right]_0^L} = \frac{\frac{L^3}{3}}{\frac{L^2}{2}} = \frac{2L}{3}$

$X_{cm} = \frac{2L}{3}$

109. The moment of inertia of a rectangular section of width B and depth D about an axis passing through C.G. and parallel to its width is

(a) $\frac{BD^3}{6}$ (b) $\frac{BD^2}{6}$

(c) $\frac{BD^3}{12}$ (d) $\frac{BD^2}{12}$

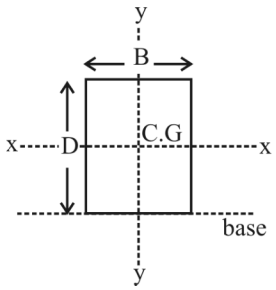
GPSC Assistant Prof. 2016

Ans. (c) :

$I_{xx} = \frac{BD^3}{12}$

$I_{Base} = \frac{BD^3}{3}$

$I_{yy} = \frac{DB^3}{12}$



4. Kinematics & Kinetics of Rigid Body

110. A particle is dropped from the top of a tower of 60 m high and another is projected upwards from the foot of the tower to meet the first particle at a height of 15.9m. The velocity of the second particle is:

(a) 16 m/sec (b) 18 m/sec

(c) 20 m/sec (d) 22 m/sec

OPSC Civil Services Pre 2011

Ans. (c) : Distance traveled by 1st particle when they meet = 60 – 15.9

= 44.1 m

By equation $S = ut + \frac{1}{2}gt^2$

for 1st particle

$44.1 = 0 \times t + \frac{1}{2} \times 9.8 \times t^2$

$t = 2.99 \simeq 3$ sec.

For second particle

$S = ut - \frac{1}{2}gt^2$

$15.9 = u \times 3 - \frac{1}{2} \times 9.8 \times 3^2$

$u = 20$ m/sec

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111. The velocity ratio of a first system of pulleys with four movable pulleys:

(a) 16 (b) 8
(c) 15 (d) 4

JUVNL AE 2016

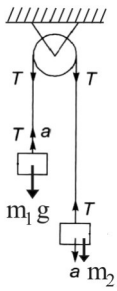
Ans. (a) : First system of pulley (VR) = 2^n
n = number of moving pulley
n = 4
$= 2^4 = 16$

112. Two bodies of masses 5.5 kg and 4.3 kg are hung to the ends of rope, passing over a smooth frictionless pulley. With what acceleration the heavier mass comes down?

(a) 9.8 m/s^2 (b) 4.73 m/s^2
(c) 80 m/s^2 (d) 1.2 m/s^2

Kerala PSC AE 2015

Ans. (d) :



Since the pulley is frictionless, the tension in the rope will remain same at each point.

$m_1a = T - m_1g$

$T = m_1a + m_1g$ ..... (i)

$m_2a = m_2g - T$

$T = m_2g - m_2a$ .....(ii)

Equating equation (i) and (ii)

$m_1a + m_1g = m_2g - m_2a$

$$a = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)g$$

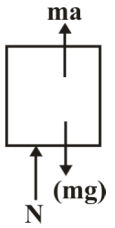
$$= \left(\frac{5.5 - 4.3}{5.5 + 4.3}\right) \times 9.8 = 1.2 \text{ m/s}^2$$

113. A body exerts a force of 1 kN on the floor of the lift which moves upward with a retardation of 1.5 m/s^2. What is the mass of the body, in kg, carried in the lift?

(a) 120.33 (b) 101.94
(c) 88.42 (d) 77.32

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : FBD of mass in the lift



N = Normal reaction between lift and mass

$\sum F_y = 0$

(N - mg) = m(-a) (-a = Retardation)

$1 \times 10^3 = m(g - a)$

$1 \times 10^3 = m(9.81 - 1.5)$

$$m = \frac{1 \times 10^3}{(9.81 - 1.5)} = 120.33 \text{ kg}$$

114. When a particle moves with a uniform velocity along a circular path, then the particle has

(a) tangential acceleration only
(b) centripetal acceleration only
(c) both tangential and centripetal acceleration
(d) none of the mentioned

GOA PSC Assistant Prof. 2020

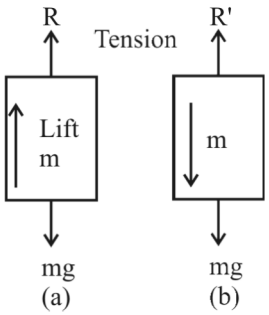
Ans. (b) : When a particle moves with a uniform velocity, then $\left(\frac{dv}{dt}\right)$ will be zero. In other words, there will be no tangential acceleration, but the particle will have only normal or radial or centripetal acceleration.

115. If the tension in a cable supporting a lift moving upwards is twice the tension when the lift is moving downwards, the acceleration of the lift, is

(a) g/2 (b) g/3
(c) g/4 (d) g/5

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (b) : Consider lift carrying some mass (m) and moving with uniform acceleration (a)



R = Tension in the cable

When lift is moving upward

R– mg = ma $\Rightarrow$ R = m(g+a) .....(i)

When lift is moving downward

mg-R' = ma $\Rightarrow$ R' = m(g-a) .....(ii)

From the question

R = 2 R'

m (g+a) = 2 m (g–a)

g+a = 2g–2a

3a = 2g–g

$$a = \frac{g}{3}$$

g = acceleration due to gravity

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116. If a body moves in such a way that its velocity increase by equal amount in equal intervals of time, it is said to the moving with

(a) a uniform retardation
(b) a uniform acceleration
(c) a variable acceleration
(d) a variable retardation

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (b) : Acceleration is the rate of change of velocity of on object with respect to time.

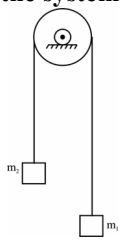
$$\text{Acceleration} = \frac{\Delta v}{\Delta t}, \quad \text{if}\left(\frac{\Delta v}{\Delta t}\right) \text{is constant}$$

Then, acceleration is uniform

$+ \rightarrow$ Acceleration

$- \rightarrow$ Retardation

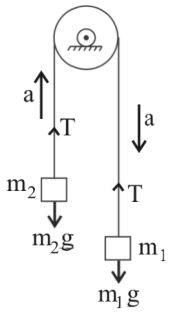
117. Two bodies of masses m_1 and m_2 are connected by a light inextensible string passing over a small smooth fixed pulley; $m_1 > m_2$. What is the acceleration of the system?



(a) $g(m_1 + m_2)/(m_1 - m_2)$
(b) $g(2m_1 + m_2)/(m_1 - 2m_2)$
(c) $g(m_1 + 2m_2)/(2m_1 - m_2)$
(d) $g(m_1 - m_2)/(m_1 + m_2)$

JPSC AE 2009

Ans. (d) :



By force equilibrium equation $\Sigma fy = 0$

$$m_1 g - m_2 g = (m_1 + m_2)a$$

$$a = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) g$$

118. If a particle starting from rest performs rectilinear motion and its velocity is given by $v = 2t + 9t^2$ m/s, what will be its position and acceleration at t = 3 sec?

(a) 90 m and 56 m/s^2
(b) 76 m and 52 m/s^2
(c) 92 m and 46 m/s^2
(d) 82 m and 36 m/s^2

LMRC AM 2019

Ans. (a) For position,

$$S_{t=3} = \int_0^3 V dt = \int_0^3 2t + 9t^2$$

$$S = \left[\frac{2t^2}{2} + \frac{9t^3}{3}\right]_0^3$$

$$= \left[t^2 + 3t^3\right]_0^3$$

$$= 9 + 3 \times 3^3$$

$$= 9 + 81$$

$$S = 90\,\text{m}$$

$$a = \frac{dV}{dt} = \frac{d(2t + 9t^2)}{dt}$$

$$(a)_{t=3} = 2 + 9 \times 2t$$

$$= 2 + 18 \times 3$$

$$a = 56\ \text{m/s}^2$$

5. Work, Power and Energy

119. A body falls from rest. Find the velocity of the body at the instant it has fallen through a height of h metres.

(a) gh (b) 2gh
(c) $\sqrt{2}$ gh (d) $\sqrt{2gh}$

ASSAM PSC CCE (PRE) 2015

Ans : (d) : According to energy conservation law– kinetic energy = potential energy.

$$\frac{1}{2}mv^2 = mgh$$

$$v^2 = 2gh$$

$$\boxed{v = \sqrt{2gh}}$$

120. For a conservative force, the work done is independent of

(a) path (b) time
(c) distance (d) All of the above

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (d) : A conservative force is a force whose work done is independent of path, time, distance and depends only on the initial and final position.

121. In conservation of energy principle, normally energy due to motion of 'm' mass at velocity 'v' in straight line is indicated by:

(a) mv^2 (b) mv
(c) $\frac{1}{2}mv^2$ (d) m^2v

GPSC R&B AE 2016

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Ans. (c) : The Conservation of energy principle of physics according to which the energy of interacting bodies or particles in a closed system remains constant. The first kind of energy recognized is

$$\text{Kinetic energy} = \frac{1}{2}mv^2$$

122. Which of the following is a scalar quantity?

(a) Momentum (b) Impulse
(c) Energy (d) Torque

ASSAM PSC CCE (PRE) 2015

Ans : (c) Energy is a scalar quantity.

123. Dimensions of Torque are :

(a) ML^2T^2 (b) ML^2T^{-3}
(c) ML^2T^{-2} (d) ML^2T^3

OPSC Civil Services Pre 2011

Ans. (c)
Torque (T) = Moment of Inertia x angular acceleration
Dimension of angular acceleration = $[M^0L^0T^{-2}]$
Dimension of moment of Inertia = $[ML^2T^0]$
Dimension of torque = $[ML^2T^{-2}]$

124. One megapascal is equal to

(a) 10^6 N/m^2 (b) 1 kgf/m^2
(c) 1 N/mm^2 (d) 1 N/m^2

ASSAM PSC CCE (PRE) 2015

Ans : (a, c) :

125. The potential energy of a system is given by the expression PE = $10x^3 + 8x^2 - 9x$. Determine the stable equilibrium position ?

(a) x = –0.678 m (b) x = –0.876 m
(c) x = 0.3425 m (d) x = 0.423 m

CIL MT 2020

Ans. (c) : Given, PE = U_x

$PE = 10x^3 + 8x^2 - 9x$

For equilibrium condition $\frac{dU_x}{dx} = 0$

hence, $\frac{dU_x}{dx} = 30x^2 + 16x - 9$

a= 30, b = 16, c = –9

$$x_{1,2} = \frac{-16 \pm \sqrt{16^2 - (4 \times 30 \times -9)}}{2 \times 30}$$

$$= \frac{-16 \pm \sqrt{256 + 1080}}{60}$$

Take (+) sign –

$$x = \frac{-16 + \sqrt{1336}}{60} = \frac{-16 + 36.55}{60}$$

$$= \frac{20.55}{60} = 0.3425\,m$$

Take (–) sign–

$$x = \frac{-16 - \sqrt{1336}}{60}$$

$$x = \frac{-60 - 36.55}{60}$$

$x = -0.8758$

6. Impulse, Momentum & Collision

126. Which of the following do not have same dimensions?

(a) Kinetic energy and Potential energy
(b) Torque and energy
(c) Impulse and Momentum
(d) Angular momentum and Moment of force

TSGENCO_2015

Ans. (d) :

(i) Kinetic energy = $\frac{1}{2}mV^2 = [ML^2T^{-2}]$

Potential energy = $mgh = [ML^2T^{-2}]$

(ii) Torque = $[ML^2T^{-2}]$

Energy = $[ML^2T^{-2}]$

(iii) Impulse = Force × time = $[MLT^{-1}]$

Momentum = mass × velocity = $[MLT^{-1}]$

(iv) Angular momentum = mvr = $[ML^2T^{-1}]$

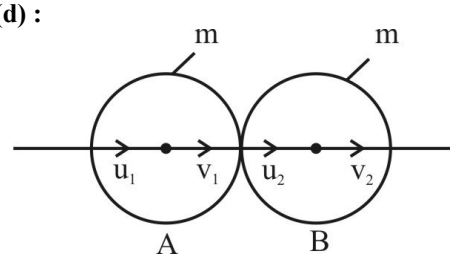
moment of force = force × distance = $[ML^2T^{-2}]$

127. Two perfectly elastic spheres of equal masses moving in the same direction with their velocities in the ratio 2 : 1 have an impact. After the impact, they will

(a) move in the same direction with the same velocity ratio
(b) have a velocity ratio of 1 : 2
(c) move in opposite direction with a velocity ratio of 1 : 2
(d) move in the same direction with a velocity ratio of 1 : 2

WBPSC AE 2007

Ans. (d) :



given $m_1 = m_2$, $u_1 : u_2 = 2 : 1$
by momentum conservation law–
$m_1v_1 + m_2v_2 = m_1u_1 + m_2u_2$
$v_1 + v_2 = 2 + 1 = 3$ ..... (i)
by Newton's Experimental Law–
$v_1 - v_2 = -e(u_1 - u_2)$ [e = 1 Perfectly elastic]
$v_1 - v_2 = -1(2 - 1)$
$v_1 - v_2 = -1$ ..... (ii)
from equation (i) & (ii)
$v_1 = 1$ $v_2 = 2$

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128. Consider the below statements :

A. Change in impulse can be achieved by changing the magnitude of force applied.

B. Change in impulse can be achieved by changing the time interval in which that force acts.

Identify the correct statement/s.

(a) Both statements are false
(b) Statement B is true A is false
(c) Both statements are true
(d) Statement A is true B is false

UPPCL AE 2019

Ans. (c)
Impulse I = F × dt
∵ I = f(F, dt)
So if any one changes, impulse will be change.

129. The maximum number of unknown forces that can be determined in a concurrent force system under equilibrium is

(a) zero (b) 2
(c) 3 (d) 6

TRB Poly. Tech. Lect. 2012

Ans. (b) : In case of system of concurrent coplanar forces, as we have two equilibrium equations, two unknown can be determined.

130. The principle of impulse and momentum is given by :

(a) Initial momentum + Impulse = Final momentum
(b) Initial momentum + Final momentum = Impulse created
(c) $\frac{\text{Initial momentum}}{\text{Final momentum}}$ = Impulse created
(d) Final momentum = Impulse × Initial momentum

TRB Poly. Lect. Tech. 2017

Ans. (a) : The Principle of impulse momentum states that the change in momentum of an object equals to the impulse applied it.
Final momentum – initial momentum = impulse
Final momentum = Initial momentum + Impulse

131. When a large force acts for a very short amount of time and causes a considerable change in a particle's momentum it is called_____.

(a) Impulsive force (b) Gravity force
(c) Inertia force (d) Resultant force

LMRC AM 2019

Ans. (a) In Impulsive force large force acts for a very short amount of time and causes a considerable change in a particle's momentum.

132. The product of mass and velocity of a moving a body is called:

(a) Moment (b) Power
(c) Momentum (d) Impulse
(e) Torque

Haryana PSC AE 2018 Paper–I

Ans. (c) : Momentum = Mass × Velocity (m × v)
Moment = Force × Perpendicular Distance
Power = Rate of doing work
Torque = Tendency of force to rotate the body on which it is applied.

$$T = F \times r \sin\theta$$

θ = Angle b/w F and r
r = Distance measured from axis of rotation where linear force is applied.

133. A bullet weighing 200 g is fired horizontally with a velocity of 25 m/sec from a gun carried on a carriage which together with the gun weights 100 kg. The velocity of recoil of the gun will be

(a) 0.01 m/sec (b) 0.05 m/sec
(c) 1.00 m/s (d) 1.5 m/sec

Arunachal Pradesh Poly. Tech. Lect. 2019

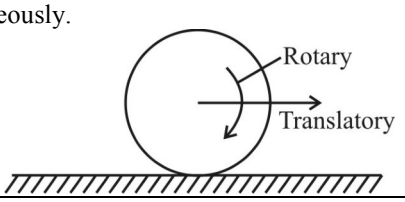
Ans. (b) : Momentum conservation law–
$m_1v_1 = m_2v_2$
$100v_1 = 25 \times 0.2$
$v_1 = \frac{0.2}{4} \Rightarrow v_1 = 0.05$ m/sec

134. The motion of the wheel of a bicycle is known as:

(a) Rotary
(b) Curvilinear
(c) Translatory
(d) Rotary and Translatory

UJVNL AE 2012

Ans. (d) : The wheel of a bicycle is rotating about its axis as well as moving forward direction simultaneously.



135. The dimension of linear momentum in MLT system is:

(a) MLT (b) MLT^{-1}
(c) MT (d) LT^{-1}

OPSC Civil Services Pre 2011

Ans. (b) : Linear momentum = mass × velocity
= m × v
$= [M^1L^0T^0][M^0L^1T^{-1}]$
Dimension of Linear Momentum = $[MLT^{-1}]$

136. For perfectly elastic bodies, the value of coefficient of restitution is

(a) Zero (b) 0.5
(c) 1.0 (d) Between 0 and 1

ISRO Scientist/Engineer 2015

Ans : (c) : Coefficient of Restitution - It is the ratio of the final velocities and initial velocities of two objects after they collide with each other. It's value lies between 0 to 1.

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$\left[e = \frac{v_2 - v_1}{u_1 - u_2} \right]$

Where v = Final velocity

u = Initial velocity

- In perfectly elastic collision, the coefficient of restitution [e = 1]. In it no energy is lost and relative separation velocity equals the relative approach velocity of particles.
- Plastic Impact, [e = 0]

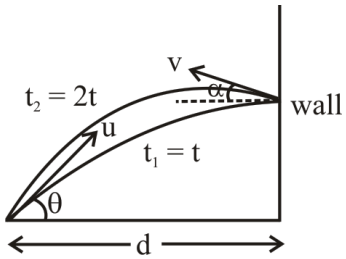
137. A glass ball is shot to hit a wall from a point on a smooth floor. If the ball returns back to the point of the projection in twice the time taken in reaching the wall, the coefficient of restitution between the glass and the wall is:

(a) 0.25 (b) 0.33

(c) 0.40 (d) 0.50

OPSC Civil Services Pre 2011

Ans. (d) :



Velocity along horizontal does not change in case of projectile motion.

$u \cos\theta = v \cos\alpha$

for collision with wall

$ev \cos\alpha = eu \cos\theta$

time taken by the ball of collide using wall

$t(u \cos\theta) = d$ ..... (1)

time taken by ball to come back to point

$2t(eu \cos\theta) = d$ ..... (2)

from equation (1) and (2)

$t u \cos\theta = 2t\, eu \cos\theta$

$e = \frac{1}{2} = 0.50$

Where, e = coefficient of restitution

7. Simple Harmonic Motion & Projectile Motion

138. If f is frequency and T is the time period, then the Angular frequency of a Simple Harmonic Motion is

(a) 2π (b) $2\pi f$

(c) $2\pi T$ (d) $1/T$

GPSC AE June 2019

Ans. (b) : Angular frequency of a simple Harmonic Motion $= 2\pi f$

$f = \frac{1}{T}$ where $T = \frac{2\pi}{\omega}$

1389. A particle moves along a circular path with constant speed. What is the nature of its acceleration?

(a) its is zero

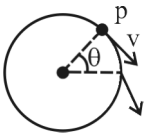
(b) It is Uniform

(c) Its direction changes

(d) its magnitude changes

RRB SSE (Bilaspur) Yellow Paper : 21.12.2014

Ans : (c) when a particle moves along a circular path with constant speed, its angular velocity does not change w.r.t. time but its direction always changes.



140. For undamped free vibrations, the natural frequency is given in usual notation by

(a) $\sqrt{\frac{m}{k}}$ (b) $\sqrt{\frac{k}{m}}$

(c) $\sqrt{\frac{2k}{m}}$ (d) $\sqrt{\frac{k}{2m}}$

GPSC AE January 2018

Ans. (b) : For undamped free vibrations,

Natural frequency $(f) = \sqrt{\frac{k}{m}}$

where–

k = stiffness, m = mass

Damped natural frequency $= f\sqrt{1 - R_D}$

R_D = Damping ratio

141. To double the period of oscillation of a simple pendulum

(a) The mass of its bob should be doubled

(b) The mass of its bob should be quadrupled

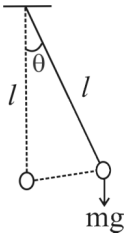
(c) Its length should be quadrupled

(d) Its length should be doubled

ISRO Scientist/Engineer 2015

Ans : (c) : Simple pendulum

- For pendulums, the period get large as the length of pendulum increases.
- If the time of oscillation is to be doubled the length should be quadrupled



$T = 2\pi\sqrt{\frac{l}{g}}$ ..... (i)

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$2T = 2\pi\sqrt{\frac{l'}{g}}$ ..... (ii)

Divide (i) and (ii)

$$\frac{T}{2T} = \frac{2\pi\sqrt{\frac{l}{g}}}{2\pi\sqrt{\frac{l'}{g}}}$$

$$\frac{1}{2} = \sqrt{\frac{l}{g} \times \frac{g}{l'}}$$

$$\frac{1}{4} = \frac{l}{l'}$$

$$\boxed{l' = 4l}$$

142. A satellite moves in its orbit around the earth due to:

(a) Compressive force (b) Centripetal force
(c) Centrifugal force (d) Tensile force
(e) None of these options

Haryana PSC AE 2018 Paper–I

Ans. (b) : Centripetal force:-A force acting on a moving body at an angle to the direction of motion, tending to make the body follow a circular or curved path. The force of gravity acting on a satellite in Orbit is an example of a centripetal force; the friction of the tyres of a car making a turn similarly provides centripetal force on the wheel.

143. A body is moving with a velocity of 10 m/sec. The time required to stop the body within a distance of 5m is equal to

(a) 3 seconds (b) 5 seconds
(c) 1 second (d) 0.5 second

Punjab PSC SDE 2016

Ans. (c) : $v^2 = u^2 + 2as$

where v = 0 (final velocity)
(a = (-ve) because vehicle is de-accelerating)
s = distance
u = initial velocity

$u^2 = -2as$

$$-a = \frac{10^2}{2 \times 5}$$

$a = -10\ m/sec^2$

$v = u - at$

$u = at$

$10 = 10 \times t$

$$t = \frac{10}{10} = 1 sec$$

144. A mass m is suspended from a massless spring of spring constant k. The spring is cut in half and the same mass is suspended from one of the halves. If the frequency of oscillation in the first case is f, then what is the frequency in the second case?

(a) $\sqrt{2}f$ (b) f (c) f/2 (d) $f/\sqrt{2}$

JPSC AE 2009

Ans. (a) : Simple harmonic motion frequency in a spring.

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

m = suspended body mass,
k = stiffness of spring
If the length of spring is cut into halves, the stiffness of new springs become 2k.

New frequency $f' = \frac{1}{2\pi}\sqrt{\frac{2k}{m}}$

$$f' = \frac{1}{2\pi}\sqrt{\frac{k}{m}}\sqrt{2} = \sqrt{2}f$$

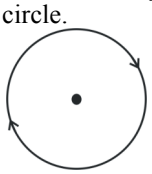
145. When a body moves around a fixed axis it has

(a) A rotary motion
(b) A circular motion
(c) A translatory motion
(d) A rotary and translatory motion

ISRO Scientist/Engineer 2018

Ans : (b)

- A body moves around a fixed axis it has circular motion. A special form of curvilinear motion, which is the motion when a body moves along a circumference of a circle.



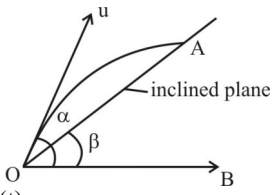
- A motion in which all particles in the object moves around a imaginary axis is called rotary motion **Eg :** Rotation of earth around its own axis.
- When an object moves in such a way that every point of the object moves the same longitudinal distance in the same time then it is said to be in translatory motion.

146. The expression for time of flight (t) of a projectile on an upward inclined plane is _____ . (Where u = Velocity of projection. α = Angle of projection, and β = Inclination of the plane with the horizontal.)

(a) $t = 2u\cos\beta / g\sin(\alpha + \beta)$
(b) $t = 2u\sin(\alpha + \beta) / g\cos\beta$
(c) $t = 2u\cos\beta / g\sin(\alpha - \beta)$
(d) $t = 2u\sin(\alpha - \beta) / g\cos\beta$

UPPCL AE 2019

Ans. (d)



Time of flight (t)

$$t = \frac{2u\sin(\alpha - \beta)}{g\cos\beta}$$

Range : $(OA) = \frac{2u^2\sin(\alpha - \beta)}{g\cos\beta}$

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8. Principle of Virtual Work & Simple Machines

147. A load of 500 kg was lifted through a distance of 13 cm by an effort of 25 kg which moved through a distance of 650 cm. The mechanical advantage of the lifting machine is

(a) 15 (b) 18
(c) 20 (d) 26

ISRO Scientist/Engineer 2015

Ans : (c) : Mechanical advantage $= \frac{W}{P} = \frac{\text{Load}}{\text{Effort}}$

$= \frac{500}{25} = 20$

148. If a man has a mass of 80 kg on earth, then compute his weight.

(a) 700.28 N (b) 784.8 N
(c) 282 N (d) 327.13N

DSSSB AE 2019

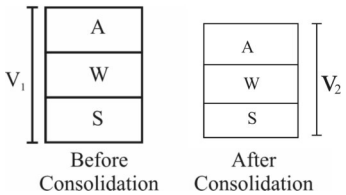
Ans. (b) mass = 80 kg.
W = mg
= 80 × 9.81 N
= 784.8 N

149. The compression resulting from a long term static load and consequent escape of pore water is known as

(a) Consolidation (b) Compaction
(c) Swelling (d) None of these

ISRO Scientist/Engineer 2018

Ans : (a) Consolidation is the process in which reduction in volume takes place by expulsion of pore water under long-term static loads.
When stress is applied to a soil that causes the soil particles to pack together more tightly. When this occurs in a soil that is saturated with water, water will be squeezed or escape out of the soil.



The consolidation of soil usually occurs in three stages :
⇒ Initial Consolidation
⇒ Primary Consolidation
⇒ Secondary Consolidation

150. According to principle of virtual work, if a body is in equilibrium, then the algebraic sum of the virtual work done by all the forces and couple moments acting on the body is :

(a) zero
(b) doubled
(c) maximum
(d) Measured with non-zero displacement

UPPCL AE 2019

Ans. (a) According to principle of virtual work, if a body is in equilibrium, then the algebraic sum of the virtual work done by all the forces and couple moments acting on the body is zero.

151. Virtual work means

1. **work done by real forces due to virtual displacement**
2. **work done by virtual forces during real displacement**

(a) Both 1 and 2 are correct
(b) Both 1 and 2 are wrong
(c) 1 is correct and 2 is wrong
(d) 2 is correct and 1 is wrong

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : Virtual work = virtual force × real displacement (or) virtual displacement × real force.

152. According to the principle of virtual work, if a system of rigid bodies is in equilibrium under the action of a set of forces, then the work done by those forces during a small virtual displacement of the system must be equal to :

(a) Infinite (b) Zero
(c) Unity (d) A non-zero constant
(e) always changing

CGPSC AE 2014 Shift–I

Ans. (b) : If a body/system is in equilibrium under the application of several forces, then the sum of all virtual works done by all forces due to virtual displacement of their point of application will be equal to zero & it is called principal of virtual work.

153. A weighing scale in a stationary lift shows a reading of 90 kg for a man standing on it. Subsequently, the lift starts descending at an acceleration of g/5. The revised reading on the scale would be

(a) 108 kg (b) 90 kg
(c) 72 kg (d) None of these

GESCOM (KARNATAKA) AE 2011

Ans : (c) : Weight of man on lift coming down ward.

$w' = w(g - a)$

$w' = 90 \times \left(g - \frac{g}{5}\right)$

$w' = 90 \times \frac{4g}{5}$

w' = 72 g
m'g = 72 g
m' = 72 kg

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9. Miscellaneous

154. From a circular sheet of paper having radius 50cm, a sector of 40% area is removed in the ratio of the radius to height of the cone is :

(a) 4/3 (b) 5/4

(c) 3/4 (d) 7/8

ISRO Scientist/Engineer 2019

Ans : (c) : Radius of circular sheet of paper = R

Radius of cone = r

Height of cone = H

Perimeter of base of cone = $0.6 \times 2\pi R$

$2\pi r = 0.6 \times 2\pi R$

$r = 0.6\ R$

Now, height of cone $H = \sqrt{R^2 - r^2}$

$$H = r\sqrt{\left(\frac{R}{r}\right)^2 - 1}$$

$$\frac{r}{H} = \frac{1}{\sqrt{\left(\frac{1}{0.6}\right)^2 - 1}}$$

$$\frac{r}{H} = \frac{1}{1.33}$$

$$\frac{r}{H} = 0.75$$

$$\frac{r}{H} = \frac{3}{4}$$

155. The primary mode of force transmission in HSFG bolts is _____:

(a) friction (b) shearing

(c) bearing (d) tension

UPPCL AE 2018

Ans. (a) In HSFG bolts primary mode of force transmission is friction.

156. Atmometer is used to measure

(a) evaporation

(b) transpiration

(c) evapotranspiration

(d) all of the above

GUJRAT PSC AE - II 2017

Ans : (a) An atmometer or evaporimeter is a scientific instrument used to measure the rate of water.

Evaporation from a wet surface to the atmosphere. Atmometer are mainly used by formers and growers to measure evapotranspiration (ET) rates of crops at any field location.

157. The relation between intensity of wind pressure P and velocity V is taken as

(a) $P \propto V$ (b) $P \propto V^{1/2}$

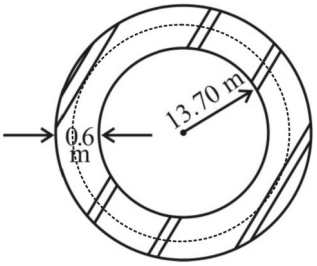
(c) $P \propto 1/V$ (d) $P \propto V^2$

Gujarat PSC AE 2017

Ans : (d) pressure intensity $P = 0.0026 \times V^2$

So, $\boxed{P \propto V^2}$

158. The centre line length of the following fig is :



(a) 26.20 πm (b) 28 πm

(c) 28.60 πm (d) 196 πm

TNPSC AE 2015

Ans. (b) Centre line length or middle circle length

$= 2\pi r$

$= 2\pi \times 14$ (where $r = 13.7 + \frac{0.6}{2} = 14m$)

$= 28\pi$

159. The acceptable value of copper (in mg/litre) as recommended by the Ministry of Urban Development in India is:

(a) 0.05 (b) 0.10–0.15

(c) 0.05–0.095 (d) 0.15–0.165

LMRC AM 2019

Ans. (a) The acceptable value of copper (in mg/litre) as recommended by the Ministry of urban development in India is 0.05.

160. As per Pappus theorem, if straight line having length '*l*', parallel to axis of rotation, at distance 'r' from rotating axis, then curved surface area generated is .....

(a) $2\pi r^2 l$ (b) $\pi r l$

(c) $2\pi r l^2$ (d) $2\pi r l$

GPSC R&B AE 2016

Ans. (d) : Pappus theorem for surface area–

A torus is the solid revolution obtained by rotating a circle about an external coplanar axis.

If l : length of straight line

r : distance from rotating axis

Curved surface area $\boxed{(A) = (2\pi r l)}$

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04.

Strength of Materials (Solid Mechanics)

1. Properties of Materials, Stress and Strain

1. If a shaft of diameter d and length ℓ has been loaded axially, then ratio of change in diameter to the original diameter is called

(a) longitudinal strain (b) shear strain
(c) volumetric strain (d) lateral strain

ASSAM PSC CCE (PRE) 2015

Ans : (d) Ratio of change of lateral dimension (Δd) to the original lateral dimension (d) in the lateral direction of applied load is termed as lateral strain.

$$\text{Lateral strain} = \frac{\Delta d}{d}$$

2. Young's Modulus is the ratio of the normal stress to the:

(a) longitudinal stress within proportional limit
(b) longitudinal stress as yield point
(c) normal strain within elastic limit
(d) reciprocal of normal strain within elastic limit

JUVNL AE 2016

Ans. (c) : Young's modulus of elasticity (E) $= \frac{\text{Normal stress}}{\text{Normal strain}}$ within elastic limit.

$$\boxed{E = \frac{\sigma}{e}}$$

3. Which of the following material has a property of malleability?

(a) Bick (b) Cement
(c) Concrete (d) Copper

GPHC AE 2018

Ans : (d) Copper material has a property of malleability.

4. Poisson Ratio of steel in elastic range is :

(a) 0.5 (b) 0.4
(c) 0.3 (d) 0.2

GPHC AE 2018

Ans : (c)

For engineering material, the value of poisson's ratio range between 0 to 0.5.

Material		Poisson's Ratio
Aluminium	→	0.33
Cast Iron	→	0.2-0.27
Steel	→	0.25-0.33

5. The stress corresponding to the maximum load in stress-strain curve is called as

(a) Breaking stress (b) True stress
(c) Ultimate stress (d) Yield stress

Rajasthan Nagar Nigam AE Shift-2, 2016

Ans. (c) : A stress-strain curve for a material gives the relationship b/w stress and strain. It is obtained by g gradually Ultimate stress.

6. If a material is heated up, its Elastic modulus

(a) decreases (b) increases
(c) remains constant (d) None of the above

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : As the material is heated up, it becomes soft. It undergoes more strain for a given stress

$\left(\because E = \frac{\sigma}{\in}\right)$ The modulus of elasticity decreases.

7. Identify the correct expression among the following:

(a) Young's modulus = Strain/Stress
(b) Lateral strain = Poisson's ratio × Longitudnal strain
(c) Young's modulus = Strain × Stress
(d) Lateral strain = Poisson's ratio / Longitudnal Strain

UPRVUNL AE 2014

Ans. (b) : Young's modulus = $\frac{\text{stress}}{\text{strain}}$

$$\text{Poisson's ratio} = \frac{\text{Lateral strain}}{\text{Longitudinal strain}}$$

i.e Lateral strain = Poissons's ration × Longitidinal strain

8. The ratio of change of dimension of the body to the original dimension is known as

(a) Strain (b) Stress
(c) Force (d) Pressure

GPSC AE January 2018

Ans. (a) :

$$\text{Strain} = \frac{\text{Change in dimension of body}}{\text{Original dimension of body}}$$

$$\boxed{\in = \frac{\Delta D}{D}} \qquad \boxed{\sigma = \in E}$$

9. The stress induced in a body, when subjected to two equal and opposite forces which are acting tangentially across the section is known as

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(a) Compressive stress (b) Tensile stress
(c) Shear stress (d) Bending stress

GPSC AE January 2018

Ans. (c) : Shear stress is the stress acts parallel to the surface. It can cause one surface to slide over other. It is the stress occurred across the cross section.

10. The ratio of axial deformation to the original length of the body is known as

(a) Laterial strain (b) Linear stress
(c) Linear strain (d) Poisson's ratio

GPSC AE January 2018

Ans. (c) :

$$\text{Linear strain} = \frac{\text{Axial Deformation}}{\text{Original length}}$$

$$\in = \frac{\Delta l}{l}$$

11. If longitudinal strain is tensile, the lateral strain will be .....

(a) tensile (b) compressive
(c) zero (d) not determined

GPSC AE January 2018

Ans. (b) : As the volume of any body cannot be changed but shape may be changed.

$V = \frac{\pi}{4} d^2 L$ Volume of cylindrical body

If L increase, d decreases (lateral strain ↓)

12. If Poisson's ratio of a material is 0.5, then the elastic modulus for the material is

(a) Three times its shear modulus
(b) Four times its shear modulus
(c) Equal to its shear modulus
(d) Indeterminate

UPRVUNL AE 2015
ESE 2014

Ans. (a) : $E = 2G(1+\mu)$ Where $\mu = 0.5$
$E = 2G(1 + 0.5)$ E = young modulus
$\boxed{E = 3G}$ G = Shear Modulus

13. A member which is subjected to reversible tensile or compressive stresses may fail at stresses lower than the ultimate stresses of the material. This property of metal is called

(a) Plasticity of the metal
(b) Workability of the metal
(c) Fatigue of the metal
(d) Creep of the metal

MPPSC AE 2014

Ans. (c) : A member which is subjected to reversible tensile or compressive stresses may fail at stresses lower than the ultimate stresses of the material, this property of metal is called fatigue of the metal.

14. Pressure meter test is used for the determination of _____.

(a) Poisson's ratio (b) shear modulus
(c) bulk modulus (d) Young's modulus

UPPCL AE 2018

Ans. (b) Pressure meter test is used for the determination of shear modulus.

15. In stress – strain curve of structural steel with high carbon content, which of the following zone is not prominent?

(a) Elastic zone (b) Plastic zone
(c) Yield zone (d) Strain hardening

UPPCL AE 2019

Ans. (c) The yield zone of the following is not considered in the stress - strain curve of structural steel with high carbon content because the yield zone becomes almost negligible if its increase the amount of carbon in the steel and steel starts to act like brittle material.

16. During tensile-testing of a specimen using a UTM, the parameters actually measured include.

(a) True stress and true strain
(b) Poisson's ratio and Young's modulus
(c) Engineering stress and Engineering strain
(d) Load and elongation

APPSC (PHE) AE Paper-2 2012

Ans. (d) : From the Universal testing machine (UTM) during tensile testing of a specimen load and elongation measured.

17. To measure_____strain, strain rosettes are used

(a) Linear (b) Shear
(c) Volumetric (d) Any of the above

APPSC (PHE) AE Paper-2 2012

Ans. (a) : Strain rosettes method is used to measure Linear strain.

18. A localised compressive stress at the area of contact between two members is known as

(a) Shear (b) Crushing
(c) Bending (d) Tensile

APPSC (PHE) AE Paper-2 2012

Ans. (b) : A localised compressive stress at the area of contact between two members is known as crushing.

19. The work done per unit volume in elongating a body by a uniaxial force is

(a) stress/strain (b) stress × strain
(c) $\frac{1}{2}$ stress × strain (d) None of the above

APPSC (PHE) AE Paper-2 2012

Ans. (c) :

$$\frac{\text{Work done}}{\text{volume}} = \frac{1}{2}\text{stress} \times \text{strain} = \frac{1}{2}\sigma.\delta$$

20. If a rigidly connected bar of steel and copper is cooled, the copper bar will be subjected to

(a) compression (b) shear
(c) tension (d) None of these

APPSC AEE (Civil/Mechanical) 2012

Ans. (c) If a rigidly connected bar of steel and copper is cooled, the copper bar will be subjected to tension.

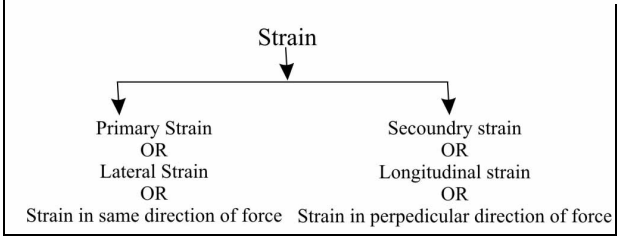
21. Strain in a direction at right angles to the direction of applied force is known as

(a) shear strain (b) lateral strain
(c) longitudinal strain (d) volumetric strain

APPSC AEE (Civil/Mechanical) 2012

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Ans. (b)



22. Modulus of Rigidity G can be expressed in usual notations as

(a) $\frac{E}{2(1+\mu)}$ (b) $\frac{E}{2(1+2\mu)}$

(c) $\frac{2E}{2(1+\mu)}$ (d) $\frac{2E}{2(1+\mu)}$

TNPSC AE 2019, GPSC AE January 2018
GPSC R&B AE 2016, PSPCL AE 2012, ESE 2006

Ans. (a) : $E = 2G(1 + \mu)$

$$\boxed{G = \frac{E}{2(1+\mu)}}$$

G = modulus of rigidity
E = Young's modulus of elasticity
μ = Poisson's ratio

23. Which of the following expression gives the correct relationship between Young's modulus (E), Bulk modulus (K) and Poisson's ratio (μ)?

(a) $E = 3K(1-2\mu)$ (b) $K = 3E(1-2\mu)$

(c) $E = 3K(1-\mu)$ (d) $K = 3E(1-\mu)$

GPSC AE March 201 /January 2018
RPSC AE 2018 / BPSC AE (Pre) 2018
OPSC AEE 2015 PAPER–I
UPPCL AE 2015 / LMRC AM 2015
UPRVUNL AE 2014 / TNPSC AE 2012

Ans. (a) :

$E = 3K(1-2\mu)$

and

$E = 2G(1+\mu)$

$$E = \frac{9KG}{3K+G}$$

$$\mu = \frac{3K-2G}{6K+2G}$$

E = young's modulus
G = Shear modulus
K = bulk modulus
μ = Poisson ratio

24. If a material has identical elastic properties in all directions, it is said to be-

(a) Homogenous (b) Isotropic
(c) Elastic (d) Orthotropic

APPSC AEE (CIVIL/MECHANICAL) 2019
GPSC AE March 2018/CGPSC AE 2014 Shift–I
OPSC AEE 2015 PAPER–I
Punjab PSC SDE 2016
APPSC (PHE) AE Paper-2 2012 / ESE 2012

Ans. (b) :

- **Isotropic–** Elastic properties are same in each and every direction.
- **Homogenous–** Materials having uniform composition throughout.
- **Anisotropic –** Elastic properties are not same in any direction, Ex- wood.
- **Orthotropic –** Elastic properties are same in all direction other than that in perpendicular direction eg- cooled Rolled steel.

25. Bulk modulus is defined as the ratio of

(a) Direct stress to direct strain
(b) Direct stress to shear strain
(c) Direct stress to lateral strain
(d) Direct stress to volumetric strain

SJVNL ET 2019, GPSC AE January 2018
UKPSC AE 2013 PAPER–I, TNPSC AE 2012
BPSC AE Paper V 2001

Ans : (d) Bulk modulus- It is defined as ratio of direct stress to the volumetric strain.

$$K = \frac{\sigma}{\varepsilon_V}$$

Young's modulus (E):- It is ratio of direct stress to linear strain

$$E = \frac{\sigma}{\varepsilon} = \frac{F/A}{\Delta l/l} \ (N/mm^2)$$

⇒ **Modulus of Rigidity (G)** :- It is ratio of shear stress to shear strain

$$G = \frac{q}{\phi}$$

$$\boxed{E = 2G(1+\mu)}$$

$$\boxed{E = 3K(1-2\mu)}$$

$$\boxed{\mu = \frac{9KG}{3K+G}}$$

26. Strain energy (U) caused by bending is given by the relation

(a) $U = \int \frac{M}{2EI} dx$ (b) $U = \int \frac{M^2}{2EI} dx$

(c) $U = \int \frac{M^2}{EI} dx$ (d) $U = \int \frac{M^2}{3EI} dx$

OPSC AEE 2019 Paper–I,
Haryana PSC AE 2018 Paper–I
Vizag steel MT 2017 / IRCON AE 2017
BPSC Poly. Tech. Lect. 2016
APPSC AEE 2012/ APPSC (PHE) AE Paper-2 2012

Ans. (b) : Strain energy (u) caused by bending is given by the relation

$$U = \int \frac{M^2}{2EI} dx$$

27. What will be the unit of Strain energy Density in SI system-

(a) J/m^3 (b) J/m^3s
(c) $Inlb/in^3$ (d) Js/m^3

Sikkim PSC AE 2017

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Ans. (a) : The strain energy density (u) is expressed in units obtained by dividing units of energy to the unit of volume, thus, if SI metric units are used, the strain energy density is expressed in J/m^3 or kJ/m^3 and MJ/m^3.

28. Modulus of rigidity is the ratio of

(a) Linear stress to linear strain
(b) Linear stress to lateral stress
(c) Volumetric strain to linear strain
(d) Shear stress to shear strain

SJVNL ET 2019, ISRO Scientist/Engineer 2019
Arunachal Pradesh Poly. Tech. Lect. 2019
APPSC AEE (CIVIL/MECHANICAL) 2019
Haryana PSC AE 2018 Paper–I
GPSC AE January 2018 / GPSC R&B AE 2016
Rajasthan AE Nagar Nigam Shift-3, 2016

Ans : (d) : The modulus of rigidity, also known as shear modulus, is defined as a material property with a value equal to the shear stress divided by the shear strain.

$[\tau = \gamma \times G]$ G = Modulus of rigidity

$\left[G = \frac{\tau}{\gamma}\right]$ γ = shear strain
τ = shear stress

29. If the depth of a rectangular section is reduced to half, strain energy stored in the beam becomes :

(a) $\frac{1}{4}$ times (b) $\frac{1}{8}$ times
(c) 4 times (d) 8 times

OPSC AEE 2019 Paper–I

Ans. (d) : Strain energy stored in the beam, $U \propto \frac{1}{I}$

$$\frac{U_2}{U_1} = \frac{I_1}{I_2} = \left(\frac{d}{\frac{d}{2}}\right)^3 = 8$$

$\because$ b = constant

30. If σ and E for a body of volume 2×10^5 mm^3 are 10 N/mm^2 and 1×10^5 N/mm^2, Resilience of the body is :

(a) 10 N mm (b) 20 N mm
(c) 100 N mm (d) 200 N mm

TNPSC AE 2015

Ans. (c) : Given that,

$V = 2 \times 10^5$ mm^3
$\sigma = 10$ N/mm^2
$E = 1 \times 10^5$ N/mm^2

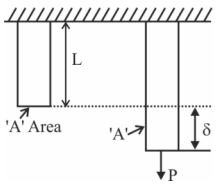
$$\text{Resilience} = \frac{\sigma^2}{2E} \times V = \frac{(10)^2}{2 \times 1 \times 10^5} \times 2 \times 10^5 = 100 \text{ Nmm}$$

31. A rod of uniform cross-section A and length L is deformed by δ, when subjected to a normal force P. The Young's Modulus E of the material is,

(a) $E = \frac{P\delta}{A.L}$ (b) $E = \frac{A\delta}{P.L}$
(c) $E = \frac{P.L}{A\delta}$ (d) $E = \frac{A.L}{P\delta}$

Haryana PSC AE 2018 Paper–I
ISRO Scientist/Engineer 2013

Ans : (c)



Elongation $'\delta' = \frac{PL}{AE}$

$\left[E = \frac{PL}{A\delta}\right]$ Young's Modulus

■ $P = \left(\frac{AE}{L}\right) \times \delta = k\delta$

k = Stiffness of the prismatic bar = $\frac{AE}{L}$

■ $\delta = \left(\frac{L}{AE}\right) P = f \times p$

f = flexibility of the bar = $\frac{L}{AE}$

32. A prismatic beam of uniform flexural rigidity EI is simply supported over a span, L. If a moment, M is applied at one support, the resulting bending strain energy is :

(a) $\frac{M^2L}{4EI}$ (b) $\frac{ML^2}{EI}$
(c) $\frac{M^2L}{6EI}$ (d) $\frac{M^2L}{2EI}$

Haryana PSC AE 2018 Paper–I
Rajasthan JEn (Degree) 2016

Ans. (d) :

1. Strain energy due to flexure,

$U_i = \int \frac{M_x^2 dx}{2EI}$ or $U = \frac{M^2L}{2EI}$

2. Strain energy due to axial loading

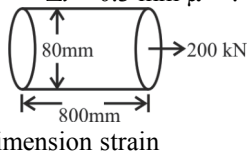
$U_i = \int \frac{P^2 dx}{2AE}$ or $U = \frac{P^2L}{2AE}$

33. A bar of 80 mm diameter and 800 mm length is subjected to an axial load of 200 kN. It elongates by 0.30 mm and the diameter decreases by 0.01 mm. What is the Poisson's ratio of the material of the bar?

(a) 0.25 (b) 0.33
(c) 0.5 (d) 0.75

GPSC AE (CLASS 1 & 2) 2019

Ans : (b) : Given as, d = 80 mm
$\Delta d = 0.01$ mm
$l = 800$ mm
$\Delta l = 0.3$ mm $\mu = ?$



$$\mu = \frac{\text{Lateral dimension strain}}{\text{Longitudinal dimension strain}}$$

$$= \frac{0.01 \times 800}{0.3 \times 80} = \frac{1}{3} = \boxed{0.33}$$

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34. A bar of 40 mm diameter and 400 mm length is subjected to an axial load of 100 kN. It elongates by 0.150 mm and the diameter decreases by 0.005 mm. What is the Poisson's ratio of the material of the bar?

(a) 0.25 (b) 0.33
(c) 0.5 (d) 0.75

ESE 2006

Ans : (b) : Given as
d = 40 mm
$\Delta d = 0.005$ mm
$l = 400$ mm
$\Delta l = 0.150$ mm

$$\mu = \frac{\text{Lateral dimension strain}}{\text{Longitudinal dimension strain}}$$

$$= \frac{\frac{\Delta d}{d}}{\frac{\Delta l}{l}} = \frac{\frac{0.005}{40}}{\frac{0.150}{400}} = 0.33$$

35. A bar made of a material which has modulus of elasticity 2×10^5 MPa and poisson's ratio 0.25. Which one of the following corresponds to shear modulus of the material?

(a) 0.5×10^5 MPa (b) 0.2×10^5 MPa
(c) 0.8×10^5 MPa (d) 0.4×10^5 MPa

Rajasthan JEn (Degree) 2016

Ans. (c) : Given,
Modulus of elasticity (E) = 2×10^5 MPa
Poisson's ratio (μ) = 0.25
G = ?
By using formula
$E = 2G(1+\mu)$

$$G = \frac{E}{2(1+\mu)}$$

$$G = \frac{2\times10^5}{2(1+0.25)}$$

$$\boxed{G = 0.8\times10^5 \text{ MPa}}$$

36. The ratio of lateral strain to linear strain is known as :

(a) Modulus of elasticity (b) Modulus of rigidity
(c) Poisson's ratio (d) Elastic limit

SJVNL ET 2019, GPSC AE January 2018
Nagaland PSC CTSE 2017 Paper–I
GPSC R & B AE 2016, TSGENCO_2015
APPSC (PHE) AE Paper-2 2012
BPSC AE Paper V 2001 /ESE 2010

Ans. (c) : Poisson's Ratio (μ)

$$\boxed{\mu = \frac{-(\text{Lateral strain})}{\text{Longitudinal strain}}}$$

37. In a particular material, if the modulus of rigidity is equal to the bulk modulus, the Poisson Ratio would be :

(a) 1/8 (b) 1/4
(c) 1/2 (d) 1

GPSC AE Class (1&2) Paper - 2 2017
APPSC AEE 2012, WBPSC AE 2007

Ans. (a) We know that-
$E = 2G(1+\mu)$ (E = modulus of elasticity)
$E = 3K(1-2\mu)$
Given, G = K

$$\frac{E}{2(1+\mu)} = \frac{E}{3(1-2\mu)} = 3-6\mu = 2+2\mu$$

$$\mu = \frac{1}{8}$$

38. A circular bar of diameter 50 mm is subjected to pull of 50 kN. Take poission's ratio = 0.33 and Young's Modulus of Elasticity = 7.07×10^4 N /mm^2. Value of shear modulus is:

(a) 2.65×10^4 N/mm^2 (b) 3.6×10^4 N/mm^2
(c) 6.93×10^4 N/mm^2 (d) 9.36×10^4 N/mm^2

APPSC Poly. Tech. Lect. 2020

Ans. (a) : Given

$$E = 7.07\times10^4 \text{ N/mm}^2$$
$$\mu = 0.33$$
$$E = 2G(1+\mu)$$
$$G = \frac{E}{2(1+\mu)} = \frac{7.07\times10^4}{2(1+0.33)} = 2.65\times10^4 \text{ N/mm}^2$$

39. The strain energy of a pin-jointed structure can be expressed as :

(a) $\Sigma F^2L/2AE$ (b) $\Sigma F^2L/AE$
(c) $\Sigma L^2F/2AE$ (d) $\Sigma FL^2/AE$

APPSC Poly. Tech. Lect. 2020

Ans. (a) : The strain energy of a pin-jointed structure can be expressed as $\sum F^2L/2AE$

40. The Poisson's ratio of a material is 0.3 and Young's modulus is 200 GPa. Its Rigidity Modulus is:

(a) 77 GPa (b) 51 GPa
(c) 125 GPa (d) 333 GPa

Kerala PSC AE 2015

Ans. (a) : $G = \frac{E}{2(1+\mu)} = \frac{200}{2(1+0.3)} = 77$ GPa

41. The ratio of modulus of rigidity and modulus of elasticity (G/E) for any elastic isotropic material is :

(a) less than $\frac{1}{2}$ (b) less than $\frac{1}{3}$
(c) more than $\frac{1}{3}$ (d) both (a) and (c)

[HPPSC AE 2016]

Ans. (d) :
Poisson's ratio for elastic isotropic material is between 0 to 0.5 relation b/w E and G
$E = 2G(1+\mu)$
for $\mu = 0$; $\frac{G}{E} = \frac{1}{2} = 0.5$
for $\mu = 0.5$; $\frac{G}{E} = \frac{1}{3}$
so, $\boxed{\frac{1}{3} \le \frac{G}{E} \le \frac{1}{2}}$

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42. The ratio of linear stress to linear strain is called

(a) Modulus of elasticity (b) Bulk modulus
(c) Rigidity modulus (d) Poisson's ratio

TNPSC AE 2019
BPSC AE Aug. 2019 Paper-V
UPRVUNL AE 2016

Ans. (a) $E = \frac{\sigma}{\varepsilon}$
σ = Linear stress,
ε = Linear strain,
E = Modulus of elasticity.

43. For a material, the modulus of Rigidity is 100 GPa and Poisson's ratio is 0.25. The value of elastic modulus in GPa is

(a) 100 (b) 150
(c) 200 (d) 250

TNPSC AE 2019

Ans. (d) $E = 2G(1 + \mu)$
$E = 2 \times 100 (1 + 0.25) = 250$ GPa

44. The loss of stress with the time at constant strain is called

(a) relaxation (b) creep
(c) shrinkage (d) ductility

Gujarat PSC AE 2017

Ans : (a) Relaxation : The loss of stress with the time at constant strain.
Creep : A certain additional strain developed in material due to axially loaded for a long time.

45. Young's modulus of a perfectly rigid body is

(a) Unity (b) Negative
(c) Zero (d) Infinity
(e) A non-zero, finite constant

CGPSC AE 2014 Shift–I

Ans. (d) Perfectly rigid body : Since a perfectly rigid body would not yield any strain, irrespective of amount of stress, its young's modulus (E) is infinite.

$$E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{0}$$

46. A glass rod having an elastic modulus of 90 GPa and Poisson's ratio of 0.2 will have its bulk modulus (in GPa)

(a) 50 (b) 108
(c) 270 (d) 91
(e) 100

CGPSC AE 2014 Shift–I

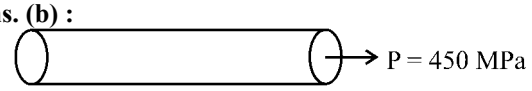
Ans. (a) : $E = 90$ GPa $= 90 \times 10^3$ MPa
$\mu = 0.2$
$E = 3K(1-2\mu)$
$90 \times 10^3 = 3K[1-2(0.2)]$
$K = \frac{90 \times 10^3}{3 \times 0.6} = \frac{90 \times 10^4}{18}$
$K = 5 \times 10^4$ MPa
$K = 50\,GPa$

47. A circular silicon carbide rod exhibiting an elastic modulus of 450 GPa under a uniaxial stress of 450 MPa. The strain energy density stored in this specimen is

(a) 225 Nmm/mm^3 (b) 0.225 Nmm/mm^3
(c) 2.25 Nmm/mm^3 (d) 22.5 Nmm/mm^3
(e) 225 kNmm/mm^3

CGPSC AE 2014 Shift–I

Ans. (b) :



$E = 450$ GPa $= 450 \times 10^3$ MPa
$\sigma = 450$ MPa

$$\text{Strain energy density} = \frac{1}{2}(\text{stess} \times \text{strain})$$
$$= \frac{1}{2}\sigma \times \varepsilon$$
$$= \frac{1}{2} \times \sigma \times \left(\frac{\sigma}{E}\right) = \frac{1}{2}\frac{\sigma^2}{E}$$
$$= \frac{1}{2} \times \frac{(450\text{MPa})^2}{450 \times 10^3}$$
$$= \frac{1}{2} \times \frac{450 \times 450}{450 \times 10^3}$$

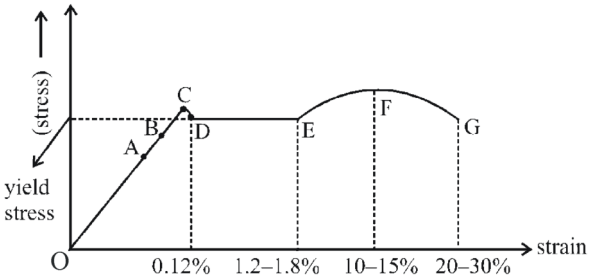
$U = 0.225$ Nmm/mm^3

48. In a stress-strain plot, the following points appear : a) Limit of proportionally, b) Elastic limit, c) Yield point; d) Ultimate strength point e) point of rupture. The correct sequence of occurrence of these points is

(a) a-b-c-d-e (b) e-d-c-b-a
(c) a-c-b-d-e (d) b-a-c-d-e
(e) a-e-d-b-c

CGPSC AE 2014 Shift–I

Ans. (a) :



OA = Linear curve
A = Proportional limit
B = Elastic limit
C = Upper yield point
D = Lower yield point
DE = Plastic Region
EF = Strain hardening region
FG = Necking region
F = Ultimate stress point
G = Fracture point

49. Modulus of Rigidity is

(a) axial stress divided by axial strain
(b) shear stress divided by shear strain
(c) increase or decrease in volume divided by original volume
(d) direct stress divided by volumetric strain

TNPSC AE 2012

Ans. (b) : Modulus of Rigidity $= \frac{\text{shear stress}}{\text{shear strain}}$

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50. The effect of external forces or the nature of internal forces developed in a body may be

(a) tension (b) compression
(c) shear (d) all of these

TNPSC AE 2012

Ans. (d) Effect of external force or the nature of internal forces developed in a body may be tension, shear and compression or any combination of these three.

51. Poisson's ratio may vary from

(a) 0.52 to 0.60 (b) 0.55 to 0.60
(c) 0.25 to 0.40 (d) none of these

TNPSC AE 2012

Ans. (d) Poisson's ratio = –1 to 0.5 → in general
Poisson's ratio = 0 to 0.5 → for engineering material

52. The unit for modulus of resilience is :

(a) Joule (b) Joules/mm
(c) Joules/mm^2 (d) Joules/mm^3

TNPSC AE 2015

Ans. (d) : (Modulus of resilience) $= \frac{\sigma^2}{2E}$

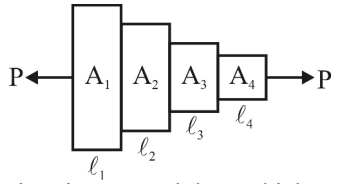
$$\text{Unit} = \frac{\left(N/m^2\right)^2}{N/m^2}$$
$$= \frac{N}{m^2} \times \frac{m}{m}$$
$$= \frac{N-m}{m^3}$$
$$\frac{\text{Joule}}{m^3} = \text{Joule/mm}^3$$

53. In case of a stepped bar of a material subjected to an axial load, the total elongation is proportional to :

(a) P/E
(b) $\left[\ell_1 + \ell_2 + ... + \ell_n\right]$
(c) $\left[\frac{\ell}{A_1} + \frac{\ell}{A_2} + ... + \frac{\ell}{A_n}\right]$
(d) $\left[\frac{\ell_1}{A_1} + \frac{\ell_2}{A_2} + ... + \frac{\ell_n}{A_n}\right]$

TNPSC AE 2015

Ans. (d) :



Total elongation in stepped bar which made by one metal.

$$(\Delta\ell) = \frac{P\ell_1}{A_1E} + \frac{P\ell_2}{A_2E} + ... \frac{P\ell_n}{A_n.E}$$

E is constant for all strip of one metal

$$\Delta\ell \propto \left[\frac{\ell_1}{A_1} + \frac{\ell_2}{A_2} + ... \frac{\ell_n}{A_n}\right]$$

54. Under compression loading, what would be area under the stress-strain curve represent?

(a) Energy required to cause failure
(b) Elasticity of the material
(c) Breaking stress of the material
(d) Fatigue

CIL MT 2017
ASSAM PSC CCE (PRE) 2015

Ans. (a) : Under compression loading, the area under the stress-strain curve represents energy required to cause failure.

55. Which type of cross-section can attain yield moment but not the plastic moment before failure by plate buckling?

(a) Plastic section (b) Compact section
(c) Semi-compact section
(d) Slender section

Gujarat PSC AE (N.W.R.) 2020

Ans. (c) Semi-compact section can attain yield moment but not the plastic moment before failure by plate buckling.

56. A uniform bar of length L, cross sectional area A and material density ρ is suspended vertically from one end. Axial elongation of the bar will be

(a) $\frac{\rho g L^2}{2E}$ (b) $\frac{\rho g L}{2E}$
(c) $\frac{\rho g L}{AE}$ (d) $\frac{\rho g L^2}{AE}$

ASSAM PSC CCE (PRE) 2015

Ans : (a) :
Let self weight of rod = W

Elongation due to self load $(\Delta\ell) = \frac{WL}{2AE}$

Weight (W) = mass. (m) × gravitational acceleration (g)
Mass (m) = Density (ρ) × Volume (V)

$$m = \rho \times AL$$
$$W = mg$$
$$W = \rho \times AL \times g$$
$$\Delta\ell = \frac{\rho ALg \times L}{2AE}$$
$$\boxed{\Delta\ell = \frac{\rho g L^2}{2E}}$$

57. A torsion test gives the shear modulus of a specimen as 12 × 10^4 N/mm^2. When the same sample was subjected to a tensile test, its elastic modulus was found to be 3 × 10^5 N/mm^2. Poisson's ratio of the material would be

(a) 0.35 (b) 0.16
(c) 0.75 (d) 0.25

ASSAM PSC CCE (PRE) 2015

Ans : (d) : Given that :

Shear modulus $G = 12 \times 10^4$ N/mm^2
$= 1.2 \times 10^5$ N/mm^2

young's modulus $E = 3 \times 10^5$ N/mm^2

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$$E = 2G(1+\mu)$$
$$3\times10^5 = 1.2\times10^5 \times 2(1+\mu)$$
$$\mu = \frac{(3-2.4)\times10^5}{2.4\times10^5}$$
$$\mu = \frac{0.6}{2.4}$$
$$\boxed{\mu = 0.25}$$

58. What will be the relation between E (Young's modulus) and K (Bulk modulus) when μ (Poisson's ratio) is 0.25?
(a) E = K (b) E = 2K
(c) E = 1.5 K (d) E = K = 0

ASSAM PSC CCE (PRE) 2015

Ans : (c) : Given that,
$\mu = 0.25$
$E = 3K(1 - 2\mu)$
$E = 3K(1 - 2\times 0.25)$
$\boxed{E = 1.5K}$

59. The process of tempering is applied to steel in hardening process for improving
(a) ductility (b) strength
(c) roughness (d) All of the above

Maharashtra PSC AE (Mains) 2012 Paper–I

Ans : (d) Tempering–It is a process of improving characteristics of a metal specially steel, by heating it to a high temperature (below melting point), then cooling it in air. This process has effect of toughening by lessening brittleness and reducing internal stresses in metal.

60. A plastic member
(a) Does not regain its original shape after removal of load producing deformation
(b) Regains its original shape after removal of load producing deformation
(c) Regains its original shape after removal of load producing deformation when heat it applied to it
(d) None of the above

GESCOM (KARNATAKA) AE 2011 2011

Ans : (a) Plasticity–Property of material by virtue of which it does not regain it's shape after removal of load.
Elasticity–Property of material in which it regain it's shape after removal of load.

61. Proof Resilience is the
(a) Maximum energy stored at elastic limit of a material
(b) Minimum energy stored at elastic limit of a material
(c) Average energy stored at elastic limit of a material
(d) None of the above

GESCOM (KARNATAKA) AE 2011 2011

Ans : (a) Proof resilience–It is the maximum energy which is stored in material upto the elastic limit. It is calculated as area under stress strain curve upto elastic limit.

62. Poisson's ratio of concrete is frequently taken as:
(a) 0.15 – 0.25 (b) 0.5 – 0.75
(c) 0.05 – 0.15 (d) none of above

KPSC AE 2020

Ans. (a) : Poisson's ratio of concrete is frequently taken as 0.15 – 0.25.

63. Which of the following statements is incorrect ?
(a) Stress is directly proportional to strain within elastic limit
(b) The stress is force per unit area
(c) Hook's law hold good up to the breaking point
(d) The ratio of linear stress to linear strain is called Young's Modulus.

Nagaland PSC CTSE 2017 Paper–I

Ans. (c) : Hooke's law hold's good up to the elastic limit (proportional limit).

64. When a bar is subjected to a change of temperature and its deformation is prevented. Which of the following stresses is induced ?
(a) Thermal stress (b) Shear stress
(c) Tensile stress (d) Compressive stress

Nagaland PSC CTSE 2017 Paper–I

Ans. (a) : Thermal stresses : Thermal stress is mechanical stress created by any change in temperature of a material - these stresses can lead to fracturing or plastic deformation depending on the other variable of heating, which induce material type and constraints.

65. The unit of modulus of elasticity is same as those of :
(a) Stress, strain and pressure
(b) Stress, force and modulus of rigidity
(c) Stress, pressure and modulus of rigidity
(d) Strain, force & pressure

Nagaland PSC CTSE 2017 Paper–I
APPSC (PHE) AE Paper-2 2012

Ans. (c) : Stress, pressure, modulus of rigidity & modulus of elasticity, they All have same unit. i.e. MPa or N/mm^2 or N/m^2

66. When a body is subjected to two equal and opposite pulls, as a result of which the body tends to extend its length, the stress and strain induced is :
(a) Compressive stress, tensile strain
(b) Tensile stress, compressive strain
(c) Compressive stress, compressive strain
(d) Tensile stress, tensile strain

Nagaland PSC CTSE 2017 Paper–I

Ans. (d) :



When a body is subjected to two equal and opposite pulls, as a result of which the body tends to extend its length, the stress induced tensile stress while strain induced tensile strain.

67. According to Robert Hooke, stress is directly proportional to strain within :
(a) Proportional limit (b) Elastic limit
(c) Yield point (d) Ultimate stress

TSGENCO_2015, RPSC ACF 2011
RPSC GWD A. En. 2014
RPSC POLY. TECH. LECT. 2011

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Ans : (a) Proportionality limit– It is limit within which stress is directly proportional to strain. Hooke's law is valid within this limit.
Hooke's law- Stress $\propto$ Strain
Elastic limit- Within elastic limit if a member is stressed, then on removal of applied stress, it regains its original shape.

68. The strength at which steel fails under repeated load applications is known as:
(a) Impact strength (b) Tensile strength
(c) Yield strength (d) Fatigue strength
[UPPSC State Eng. A.E. 2003 & 2008]
[UPPSC AE 12.04.2016 Paper–I]

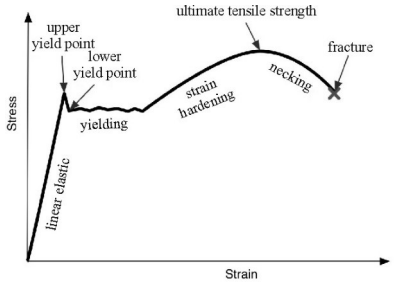
Ans : (d) Due to cyclic or reverse cyclic loading if total accumulated strain energy exceeds the toughness, then fracture failure occur and strength corresponding to fatigue is known as fatigue strength.
The strength of material against impact loading is known as impact strength (toughness).
The maximum resistance offered by material against tension is known as tensile strength whereas yield strength is stress corresponding to yield point.

69. The property by which a metal resists impact load is called
(a) Ductility (b) Toughness
(c) Elasticity (d) Malleability
[UPPSC State Eng. A.E. 2004(I) & 2007(I)]

Ans : (b) Toughness : Toughness is the resistance to the impact loading if a metal is tough then it is able to store large strain energy before fracture.
Ductility : Ductility of a metal is that property due to which a metal can be drawn into wire of thin section.
Elasticity : It is that property of material by which the original dimensions can be recovered after unloading.
Malleability : It is that property of a metal due to which a piece of metal can be converted into a thin sheet.

70. The maximum stress that can be applied to a material for an infinite number of cycles of repeated stress without causing failure is called
(a) elastic limit (b) proportional limit
(c) ultimate strength (d) endurance limit
Punjab PSC SDE 2016
[UPPSC State Eng. A.E. 2004 (I)]

Ans : (d) Endurance Limit : It is the stress level below which even large number of stress cycle can not produce fatigue failure.



Proportional Limit – Proportional limit determines the greatest stress that is directly proportional to strain.
Elastic limit– Maximum stress or force per unit area within a solid material that can arise before the permanent deformation, when stresses upto the elastic limit are removed, the material resumes its original size and shape, stresses beyond the elastic limit cause a material to yield or flow.
Ultimate Strength– Ultimate strength of a material in tension, compression or shear, respectively is the maximum tensile, compressive or shear stress that the material can sustain, calculated on the basis of the ultimate load and the original or unstrained dimensions.

71. The modulus of elasticity of rolled structural mild steel is about
(a) 2×10^5MPa (b) 2×10^6MPa
(c) 2×10^7MPa (d) 2×10^8MPa
[UPPSC State Eng. A.E. 2007(I)]

Ans : (a) The modulus of elasticity for steel,
$E_s = 2 \times 10^5$ N/mm^2
$E_s = 2 \times 10^5$ MPa

72. Poisson's ratio is involving
(a) Elastic Modulus (b) Stresses
(c) Strains (d) None of these
OPSC AEE 2015 PAPER–I
RPSC POLY. TECH. LECT. 2014

Ans : (c) The ratio of lateral strain to the longitudinal strain of a material when it is subjected to longitudinal stress, is known as poisson's ratio.

$$\mu = \frac{-\text{lateral strain}}{\text{longitudinal strain}}$$

Its value varies between 0.25 to 0.34 for most of the metals.
For engineering material it is between 0 to 0.5.

73. The value of Poisson's ratio of the materials lies between
(a) 1 and 2 (b) 0 and 1/2
(c) 0 and 1 (d) 2 and 3
UJVNL AE 2016
APPSC AEE 2012
RPSC VPITI 2012

Ans : (b) Poisson's ratio varies between 0 to 0.5. For cork it is zero and for natural rubber it is 0.5. for metals it varies between 0.25 to 0.34.

74. The property of a material by which it returns partially or completely to its initial shape after unloading is called
(BPSC AE Paper V 2006)
OR
The value of stress up to which a member regains its original shape or size after load removal is called
(a) Elastic limit (b) Proportional limit
(c) Yield stress (d) Plastic limit
RPSC POLY. TECH. LECT. 2014

Ans : (a) Elastic limit- Within elastic limit if a member is stressed, then on removal of external load, it regains its original shape.
Proportionality limit– It is limit within which stress is directly proportional to strain. Hooke's law is valid within this limit.
Hooke's law → Stress $\propto$ Strain

75. A rectangular bar having cross section A, length L, modulus of elasticity E and Poisson's ratio 1/m, is subjected to axial pull of P. The volumetric strain will be given by

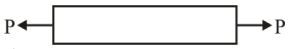
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(a) $\frac{PL}{AE}\left(1-\frac{2}{m}\right)$ (b) $\frac{P}{AE}\left(1-\frac{2}{m}\right)$

(c) $\frac{PL}{AE}\left(1-\frac{1}{m}\right)$ (d) $\frac{P}{AE}\left(1-\frac{1}{m}\right)$

RPSC POLY. TECH. LECT. 2014

Ans : (b)



Given Area = A

Elasticity = E; poisson's ratio = $\frac{1}{m}$

Volumetric strain

$\varepsilon_v = \varepsilon_x + \varepsilon_y + \varepsilon_z$

$\varepsilon_v = \frac{P}{AE} - \frac{P}{mAE} - \frac{P}{mAE}$

$\boxed{\varepsilon_v = \frac{P}{AE}\left(1-\frac{2}{m}\right)}$

76. A metal bar of length 100 mm is inserted between two rigid supports and its temperature is increased by 10^0C. If the coefficient of thermal expansion is 8×10^{-6} per ^{0}C and the young's modulus is 1.5×10^5 MPa, the stress in the bar is :

(a) Zero (b) 12 MPa

(c) 24 MPa (d) 2400 MPa

RPSC POLY. TECH. LECT. 2014

Ans : (b) Given,

Change in temperature (ΔT) = 10^oC

Elasticity modulus (E) = 1.5×10^5 MPa

Coefficient of thermal expansion (α) = $8 \times 10^{-6}/^oC$

Thermal stress in bar = E α ΔT

= $1.5 \times 10^5 \times 8 \times 10^{-6} \times 10 = 12$ MPa

77. Choose the correct statements from the following-

(1) Hooke's law is valid upto the limit of proportionality.

(2) Hooke's law is valid up to elastic limit

(3) Limit of proportionality is always less than the elastic limit

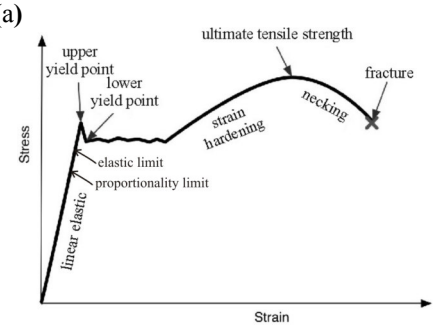
(4) Limit of proportionality is either equal to or less than the elastic limit

(a) 1 and 4 (b) 2 and 4

(c) 1 and 3 (d) 2 and 3

BPSC AE 2012 Paper–V

Ans : (a)



78. A copper rod of square cross section is fixed between two rigid supports and over which a steel rod of square cross-section is simply placed. If the temperature of the whole assembly is raise T^0C, the stresses in steel and copper respectively are

(a) tensile and compressive

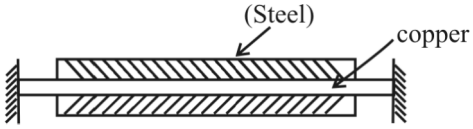
(b) zero and compressive

(c) compressive and tensile

(d) compressive and zero

BPSC AE 2012 Paper–V

Ans : (b) Copper rod is rigidly fixed between supports and steel rod is simply placed over it. When temperature is increased copper will try to expand, but as it is rigidly placed, compressive stresses will develop in it. In steel rod, as they are free to expand, there will be no stress induced in them.



79. The stress developed in bar which is not free to expand due to increase in temperature, is given as

(a) $p = \alpha tE$ (b) $p = \frac{\alpha}{tE}$

(c) $p = \frac{t}{\alpha E}$ (d) $p = \frac{E}{\alpha t}$

TNPSC AE 2018

RPSC AE 2013

Ans : (a) change in length due to stress $\Delta l = \frac{pl}{E}$

change in length due to temperature change $\Delta l = \alpha\, t l$

$\therefore \frac{pl}{E} = \alpha t.l \Rightarrow \boxed{p = \alpha tE}$

80. The endurance limit is found by testing specimen of standard size, usually 7.5 mm, in a standard rotating beam test. What will be the actual endurance limit of parts with size greater than that used in the test

(a) Less than the value determined with standard test

(b) More than the value determined with standard test

(c) Same as the value determined with standard test

(d) Less than the value determined with standard test upto certain size then it will increasing with the increasing size

[RPSC AE 2013]

Ans : (c) For a given material endurance limit is same irrespective of size of component.

81. When a prismatic steel bar is subjected to axially applied tensile force the total tension in the bar is called

(a) Axial resultant (b) Reactive force

(c) Stress resultant (d) Normal stress

(BPSC AE Paper V 2006)

Ans : (a) When prismatic bar is subjected to axially applied tensile force the total tension in bar is called axial resultant.

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82. **The value of Poisson's ratio of cork is approximately equal to :**

(a) 1 (b) 0.50
(c) 0.1 (d) 0

(BPSC AE Paper V 2006)

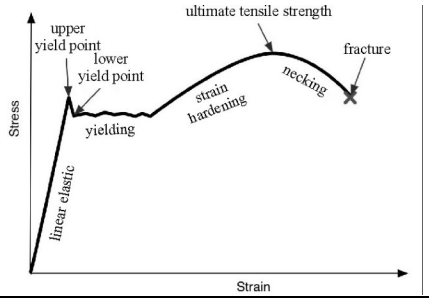
Ans : (d) Poisson's ratio varies between 0 to 0.5. For cork it is zero and for natural rubber it is 0.5. for metals it varies between 0.25 to 0.34.

83. **Among the materials mentioned below the pronounced yield point is the characteristic of**

(a) Cast-iron (b) Structural steel
(c) Concrete (d) Stone

(BPSC AE Paper V 2006)

Ans : (b) Stone, concrete and cast-iron are brittle in nature and they do not have defined yield point. Only structural steel among given possesses ductility and has pronounced yield point.



84. **A bar of length L and cross–sectional area A is subjected to gradually applied load w. The strain energy stored in the bar is :**

(a) $\frac{wL^2}{2AE}$ (b) $\frac{wL^2}{AE}$
(c) $\frac{w^2L}{AE}$ (d) $\frac{w^2L}{2AE}$

UKPSC AE (Paper-I) 2007
ESE 2001, 2003

Ans : (d) Strain energy for gradually applied load = $\frac{w^2L}{2AE}$

Strain energy for suddenly applied load = $\frac{W^2L}{AE}$

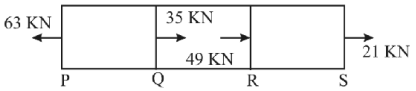
85. **When a nut is tightened on the bolt the nature of stress in the bolt is :**

(a) compressive (b) tensile
(c) bending (d) shearing

(UKPSC AE (Paper-I) 2007)

Ans : (b) When nut is tightened on bolt, tensile stresses are induced in the bolt.

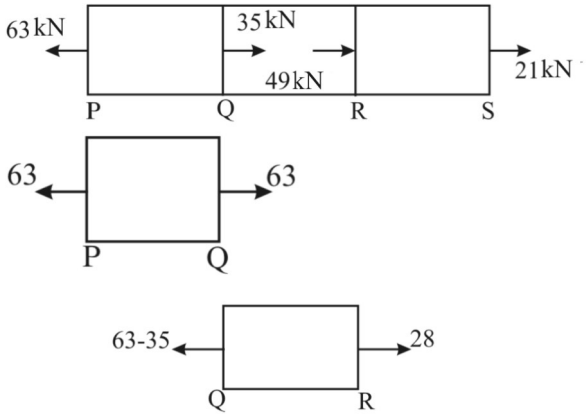
86. **A bar having a cross-sectional area of 700 mm2 is subjected to axial loads at the positions indicated. The value of stress in the segment QR is**



(a) 40 MPa (b) 50 MPa
(c) 70 MPa (d) 120 MPa

TRB Poly. Lect. Tech. 2017
(UPRVUNL AE 2015)

Ans : (a)



Force in QR = 63 – 35 = 28 kN (tensile)
Area, A = 700 mm^2

Stress = $\frac{P}{A} = \frac{28 \times 1000}{700} = 40 MPa$

87. **The number of elastic constants for a completely anisotropic elastic material which follows Hooke's law is?**

(a) 3 (b) 4
(c) 21 (d) 25

(UPRVUNL AE 2015)
UPRVUNL AE 2015
APPSC (PHE) AE Paper-2 2012

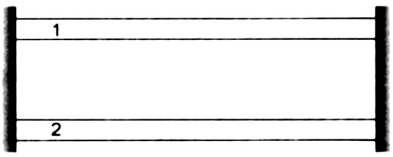
Ans : (c) Independent elastic constant for different type of material-
(i) Isotropic → 2, ex. steel
(ii) orthotropic → 9, ex. wood, ply, etc
(iii) Anisotropic → 21, ex. wood, biological tissue etc.

88. **Two roads of different materials having coefficient of linear expansion α_1, α_2 and Young's module E_1, E_2 respectively are fixed between two massive walls. The rods are heated such that they undergo the same increase in temperature. There is no bending of the rods. If $\alpha_1 : \alpha_2 = 2 : 3$, the thermal stresses are equal, what is the ratio of E_1 to E_2?**

(a) 2 : 3 (b) 1 : 1
(c) 3 : 2 (d) 4 : 9

Maharashtra PSC (Mains) 2015 Paper–I

Ans. (c) :



Stress generated due to increase in temperature may be given as,

$$\sigma = \varepsilon \times E = (\alpha \Delta T) E$$

$$\sigma_1 = \alpha_1 \Delta T_1 E_1$$

and $\sigma_2 = \alpha_2 \Delta T_2 E_2$

Given, $\sigma_1 = \sigma_2$

and $\Delta T_1 = \Delta T_2$

$$\alpha_1 \Delta T_1 E_1 = \alpha_2 \Delta T_2 E_2$$

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$$\frac{\alpha_1}{\alpha_2} = \frac{E_2}{E_1}$$

$$\Rightarrow \quad \frac{2}{3} = \frac{E_2}{E_1}$$

$$\Rightarrow \quad \frac{E_1}{E_2} = \frac{3}{2}$$

89. In an experiment it is found that the bulk modulus of a material is equal to its shear modulus then the Poisson's ratio is-

(a) 0.125 (b) 0.250
(c) 0.375 (d) 0.500

Maharashtra PSC (Mains) 2015 Paper–I
ESE 1998

Ans. (a) : $K = \frac{E}{3(1-2\mu)}$

and $G = \frac{E}{2(1+\mu)}$

Given, $K = G$

$$\frac{E}{3(1-2\mu)} = \frac{E}{2(1+\mu)}$$

$$3(1-2\mu) = 2(1+\mu)$$

$$3 - 6\mu = 2 + 2\mu$$

$$1 = 8\mu$$

$$\mu = \frac{1}{8} = 0.125$$

90. The stress developed due to external force in an elastic material-

(a) Depends on elastic constant
(b) Does not depend on elastic constant
(c) Depends partially on elastic constant
(d) Depends on limit of proportionality

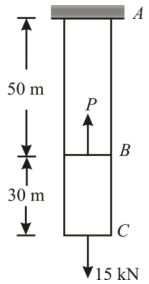
Maharashtra PSC (Mains) 2017 Paper–I

Ans. (b) : Stress in a elastic body is developed to resist the deformation produced due to external load.

$$\sigma = \frac{P}{A}$$

It does not depend upon elastic constant.

91. A steel bar *ABC* of uniform cross-section 100 mm^2 is suspended vertically and loaded as shown in the figure. If the lower end of bar C does not move when loads are applied (neglect self-weight), then the value of force *P* is- (E_s = 200 kN/mm^2)



(a) 24 kN (b) 42 kN
(c) 36 kN (d) 15 kN

Maharashtra PSC (Mains) 2017 Paper–I

Ans. (a) :

For C to remain in its position

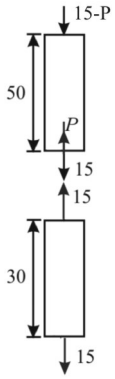
$$\Delta_{AB} + \Delta_{BC} = 0$$

$$\therefore \frac{(15-P)\times 50}{AE} + \frac{15\times 30}{AE} = 0$$

$$\frac{(15-P)\times 50}{AE} = \frac{-15\times 30}{AE}$$

$$15(50+30) = 50P$$

$$P = \frac{15\times 80}{50} = 24\,kN$$

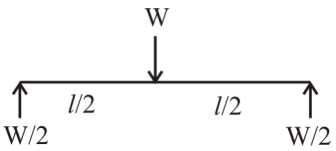


92. The strain energy stored in a simply supported beam of span *'L'* and flexural rigidity *EI* due to central concentrated load *W* is given by-

(a) $\frac{W^2L^2}{48EI}$ (b) $\frac{W^2L^3}{48EI}$
(c) $\frac{W^2L^2}{96EI}$ (d) $\frac{W^2L^3}{96EI}$

OPSC AEE 2019, 2015 Paper–I
WBPSC AE 2012

Ans. (d) :



The deflection under the load,

$$\delta = \frac{WL^3}{48EI}$$

Strain Energy due to load,

$$U = \frac{W}{2}\times\frac{WL^3}{48EI} = \frac{W^2L^3}{96EI}$$

93. Every material obeys the Hooke's law within its-

(a) Elastic limit
(b) Plastic limit
(c) Limit of proportionality
(d) None of these

OPSC AEE 2015 PAPER–I

Ans. (c) : The limit of proportionality is slightly less than the limit of elasticity in stress-strain graph

$$\sigma \propto \varepsilon$$

Elastic Limit : It can be defined as the point in the graph upto which material comes back to its original shape when loads are removed.

94. A bar of brass is enclosed in a steel tube and is rigidly fastened at both the ends. If the coefficient of thermal expansion of brass is more than that of steel, when temperature rises, the nature of stresses developed are-

(a) tensile in steel tube and compressive in brass bar
(b) compressive in steel tube and tensile in brass bar
(c) tensile in both steel tube and brass bar
(d) compressive in both steel tube and brass bar

BPSC AE Pre 2018

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Ans. (a) : If a composite bar of Steel & Brass is heated, brass tries to expand more than the steel because the coefficient of thermal expansion of brass is more than steel. Brass brings a tensile stress on the composite bar. On the other hand, steel tries to withhold the expansion of brass there by introducing a compressive stress on the composite bar

95. In case of brittle materials, the ratio of ultimate compressive stress to ultimate tensile stress is-

(a) equal to 1 (b) more than 1
(c) less than 1
(d) may be anything. No definite relation exists

BPSC AE Pre 2018

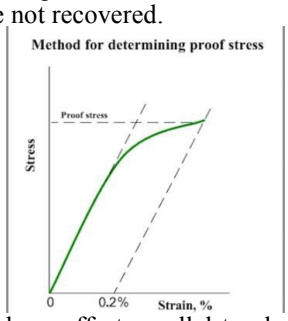
Ans. (b) : Compressive strength for brittle materials (concrete) is generally higher than their tensile strength. So, their ratio will be more than 1.

96. 0.2 percent proof stress means-

(a) stress corresponding to 0.2 percent strain
(b) 0.2 percent of ultimate stress
(c) stress at which if unloading is made, there will be 0.2 percent permanent strain
(d) None of the above

BPSC AE Pre 2018

Ans. (c) : 0.2 percent proof stress is an indicator of yield stress of steel. Like all elastic materials, a steel body stretches itself when subjected to stress. Upon release of stress the body regains it's original dimensions. However, there exists a value of stress, beyond which, upon release of the stress, original dimensions are not recovered.



It is a stress where offset parallel to elastic line drawn from 0.20% strain intersects in stress-strain curve

97. Hooke's law holds good upto

(a) Upper yield point (b) Elastic limit
(c) Lower yield point (d) Plastic limit

SJVNL ET 2019
TNPSC AE 2018

Ans. (b) : Hook's law holds good upto limit of proportionality. It states that stress is proportional to strain within the limit of proportionality.
If a member is strained upto elastic limit then on unloading the member it regains its original shape. Elastic limit is greater than limit of proportionality. No option is correct but the closest one is elastic limit.

98. Yield strength is :

(a) stress required to produce certain arbitrary plastic deformation
(b) stress required to produce certain arbitrary elastic deformation
(c) stress required to cause fracture
(d) stress required to cause fatigue

[HPPSC 2016-II]

Ans. (a) : Yield strength is stress at which a specific amount of deformation is produced which is permanent (plastic deformation).

99. Match List-I (Elastic constant) with List-II (Definition) and select the correct answer using the codes given below the list :

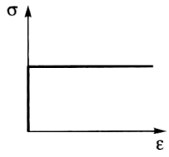
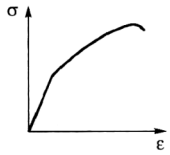
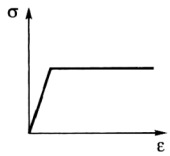
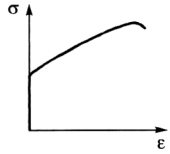
List - I	List-II
A. Young's modulus	1. The ratio of lateral strain to linear strain within elastic limit
B. Poisson's ratio	2. The ratio of stress to strain within elastic limit
C. Bulk modulus	3. The ratio of shear stress to shear strain within elastic limit
D. Rigidity modulus	4. The ratio of direct stress to corresponding volumetric strain

	A	B	C	D
(a)	3	1	4	2
(b)	2	1	4	3
(c)	2	4	1	3
(d)	3	4	1	2

[HPPSC 2016-II]
ESE 2000

Ans. (b) :
$\Rightarrow$ Young's modulus of elasticity is ratio of longitudinal stress to longitudinal strain within elastic limit. Poisson's ratio is ratio of lateral strain to longitudinal strain within elastic limit. Bulk modulus of elasticity is ratio of direct stress to corresponding volumetric strain. Rigidity modulus is ratio of shear stress to shear strain within elastic limit.

100. Choose the correct combination for the given Table:

Stress-strain Graph		Material
P.		I. Elastic-Plastic
Q.		II. Rigid Plastic
R.		III. Elastic-Perfectly Plastic
S.		IV. Perfectly Plastic

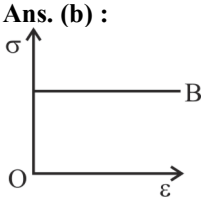
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Codes:

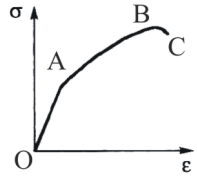
	P	Q	R	S
(a)	IV	III	II	I
(b)	IV	I	III	II
(c)	I	II	III	IV
(d)	III	I	IV	II

[HPPSC AE 2016]

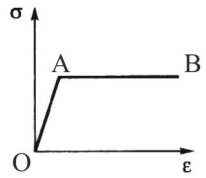
Ans. (b) :



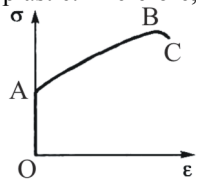
for perfectly plastic, strain increases at a particular value without further increase in stress



Initially stress is proportional to strain, therefore elastic. After point A it shows plastic nature.



Initially upto A it is elastic, but after A it is perfectly plastic. Therefore, elastic perfectly plastic.



Initially no strain for stress increase, hence rigid, after that it shows plastic nature.

101. Which quantity will not be zero for a plane strain problem in *x-y* plane?

(a) Shear strain in *x-z* plane
(b) Normal strain in *x* direction
(c) Normal stress in *z* direction
(d) Shear stress in *y-z* plane

[HPPSC AE 2016]

Ans. (c) : For plane strain condition, $\varepsilon_z = 0$

i.e $\dfrac{\sigma_z}{E} - \dfrac{\mu}{E}(\sigma_x + \sigma_y) = 0$

$\Rightarrow \quad \sigma_z = \mu(\sigma_x + \sigma_y)$

Hence $\sigma_z \neq 0$

102. If E, G, K and μ represent the elastic modulus, shear modulus, bulk modulus and Poisson's ratio respectively of a linear elastic, isotropic and homogeneous material, and if you need to expresses the stress-strain relationships completely for this material, at least:

(a) All the four must be known
(b) E, G and μ must be known
(c) E, K and μ must be known
(d) Any two of the four must be known

[HPPSC AE 2016]

Ans. (d) : For linear elastic, isotropic and homogeneous material, $E = 2G(1 + \mu) = 3K(1 - 2\mu)$.
For above equation we have three unknown for any relation between E,G,K and μ, there fore we require any two to find the remaining one.

103. Ductility depends on :
(i) Temperature of the structure
(ii) Size of the structure
(iii) Applied loading time
Which of the above is/are true?

(a) (i) and (iii) (b) (i) and (ii)
(c) (i) only (d) All of these

[HPPSC AE 2016]

Ans. (b) : Ductility is property of material due to which it can be drawn into wires. Ductility depends upon temperature and size of structure. It does not depend upon loading time.

104. The maximum value of Poisson's ratio for an elastic material is :

(a) 0.25 (b) 0.5
(c) 0.75 (d) 1.0

UPPCL AE 2018
Karnataka PSC AE 2015 Paper - II

Ans. (b) : We know that

$G = \dfrac{E}{2(1+\mu)}$ and $K = \dfrac{E}{3(1-2\mu)}$,

poisson's ratio $(\mu) = \dfrac{|\text{lateral strain}|}{\text{longitudinal strain}}$

$E = 3K(1-2\mu)$

for E to be positive

$1 - 2\mu > 0$

$\mu < \dfrac{1}{2}$

Under uniaxial loading

$\boxed{0 \le \mu \le 0.5}$

$\mu = 0$ for cork
$\mu = 0.5$ for perfectly plastic body (Rubber)
$\mu = 0.25$ to 0.42 for Elastic materials
$\mu = 0.1$ to 0.2 for concrete
$\mu = 0.286$ mild steel
μ is greater for ductile materials than for brittle materials.

105. If the temperature of a rod of length L is increased by T^0C, whose coefficient of linear expansion is α and Young's modulus is E, then free expansion of the rod due to increase in temperature is given by :

(a) αTEL (b) αTL
(c) αTE (d) None of the above

Karnataka PSC AE 2015 Paper - II

Ans. (b) : Thermal expansion is where an object expands and becomes larger due to change in its Temperature.

$\dfrac{\Delta L}{L} = \alpha_L \Delta T$

$\boxed{\Delta L = L\alpha_L \Delta T}$ where

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ΔL = change in length
L = original length
ΔT = change in Temperature
α_L = Linear coefficient of Thermal expansion.

106. Volumetric strain is defined as which of the following :

(a) $\frac{\delta V}{V}$ (b) $\frac{V}{\delta V}$

(c) $\frac{m}{\delta V}$ (d) $\frac{\delta V}{m}$

Karnataka PSC AE 2017 Paper - II

Ans. (a) : Volumetric strain of a deformed body is defined as the ratio of change in volume of the body to its original volume.

$$\varepsilon_v = \frac{\delta V}{V} = \left(\varepsilon_x + \varepsilon_y + \varepsilon_z\right)$$

107. Which one of the following expression represents strain energy density ?

(a) Stress x Strain (b) $\frac{1}{2}$ Stress x Strain

(c) $\frac{1}{3}$ Stress x Strain (d) $\frac{1}{4}$ Stress x Strain

Rajasthan JEN (Degree) Shift - II 2016

Ans. (b) $u = \frac{U}{V} = \frac{1}{2}\sigma \times \in$

$$= \frac{1}{2}\sigma\left(\frac{\sigma}{E}\right) = \frac{1}{2}\frac{\sigma^2}{E}$$

108. For a given material the Poisson's ratio is 0.25, the ratio of Young's modulus and modulus of rigidity is :

(a) 2.5 (b) 0.4

(c) 2.75 (d) 3.0

JPSC AE (Pre) 2019
ESE 1997

Ans. (a) : Given, $\mu = 0.25$

Since, $E = 2G(1+\mu)$

$$\therefore \frac{E}{G} = 2(1+\mu) = 2\times(1+0.25) = 2.5$$

109. A body having similar properties throughout its volume is said to be :

(a) Homogenous (b) Isotropic

(c) Continuous (d) Anisotropic

GPSC AE Class (1&2) Paper - 2 2017

Ans. (a) : A body having similar property in its volume is called Homogenous section.

110. Stress is :

(a) a measure of deformation of material
(b) resistance to forces resulting in elongation
(c) internal force exerted by neighbouring particles over each other
(d) resistance to bending and torsion

GPSC AE Class (1&2) Paper - 2 2017

Ans. (c) : Stress is the internal force exerted on particles that push each other which oppose the change in dimension (strain).

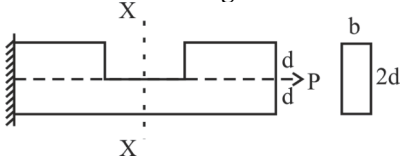
111. Which one is correct about prying force ?

(a) Additional torsional force
(b) Additional compressive force
(c) Additional tension force
(d) Additional bending moment

Rajasthan JEn (Degree) 2016

Ans. (c) The prying force is nothing but the additional tensile force which are developed in the bolts. When connection is subjected to tension force the connected parts can leads to deformation that increase the tension applied to the bolt, so that the additional force is generated in the bolts due to joint plates.

112. Determine the maximum tensile stress at the section XX in the figure below:

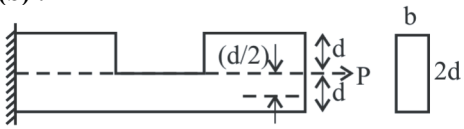


(a) 2P/bd (b) 4P/bd

(c) 6P/bd (d) 8P/bd

GPSC AE (CLASS 1 & 2) 2019

Ans : (b) :



$\sigma_{Tensile}$ = Due to direct action of P + Due to bending

$$= \frac{P}{bd} + \frac{M}{I_{xy}}y = \frac{P}{bd} + \frac{P\,d/2}{\frac{b\times d^3}{12}} \times \left(d/2\right)$$

$$= \frac{P}{bd} + \frac{P\left(1/2\right)}{\frac{bd}{12}}\left(1/2\right) = \frac{P}{bd} + \frac{3p}{bd}$$

$$= \boxed{\frac{4P}{bd}}$$

113. The principle of superposition is not applicable when

(i) The material does not obey Hooke's law
(ii) The effect of temperature changes is taken into consideration
(iii) The structure is being analysed for the effect of support settlement

Which of these statement(s) is/are correct?

(a) (i) only (b) (i) and (ii)

(c) (ii) and (iii) (d) (i), (ii) and (iii)

GPSC AE (CLASS 1 & 2) 2019

Ans : (a) The Principle of superposition says that when a number of load are acting on a body, the resulting strain, according to the principle of superposition, will be the algebraic sum of strain caused by individual loads.

114. Assertion (A): Normal stress of one nature (compressive or tensile) acting along one of the three orthogonal axes of a member will produce strains of the same nature in its direction and strains of opposite nature along the other two directions.

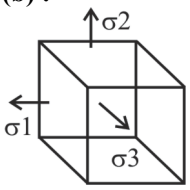
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Reason (R): Sum of the strains along the three orthogonal axes equals volumetric strains. Of these statements:

(a) both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

GPSC AE (CLASS 1 & 2) 2019

Ans : (b) :



$$\varepsilon_x = \frac{\sigma_1}{E} - \frac{\mu\sigma_2}{E} - \frac{\mu\sigma_3}{E}$$

$$\varepsilon_y = \frac{\sigma_2}{E} - \frac{\mu\sigma_1}{E} - \frac{\mu\sigma_3}{E}$$

$$\varepsilon_z = \frac{\sigma_3}{E} - \frac{\mu\sigma_2}{E} - \frac{\mu\sigma_1}{E}$$

$$\varepsilon_v = \varepsilon_x + \varepsilon_y + \varepsilon_z$$

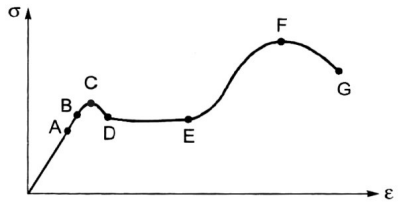
$$\frac{\Delta V}{V} = \frac{\sigma_1}{E} - \frac{\mu\sigma_2}{E} - \frac{\mu\sigma_3}{E} + \frac{\sigma_2}{E} - \frac{\mu\sigma_1}{E} - \frac{\mu\sigma_3}{E} + \frac{\sigma_3}{E} - \frac{\mu\sigma_2}{E} - \frac{\mu\sigma_1}{E}$$

115. During strain hardening:

(a) Material undergoes changes in atomic and crystalline structures
(b) Increased resistance to further deformation
(c) Stress strain diagram has positive slope
(d) All of the above

Kerala PSC AE 2015

Ans. (d) :



A = Proportional limit
B = Elastic limit
C = Upper yielding
D = Lower yielding
F = Ultimate stress point
G = Fracture point
EF Region is strain hardening where strain increases by large amount with small increase in stress value.

116. A steel rod of length 20 m at 30°C is heated upto 40°C. What is the temperature stress devloped if the expansion is prevented?
Given, $\alpha = 12 \times 10^{-6}$ per °C, $E = 2 \times 10^5$ N/mm²

(a) 2.4 N/mm² (b) 24 N/mm²
(c) 240 N/mm² (d) 0.24 N/mm²

Kerala PSC AE 2015

Ans. (b) : $\sigma = \varepsilon E$

$$\varepsilon = \frac{\Delta l}{l} = \frac{L\alpha\Delta T}{L}$$

$$\sigma = \alpha E \Delta T$$

$$= 12 \times 10^{-6} \times 2 \times 10^5 (40 - 30)$$

$$\sigma = 24 \text{ N/mm}^2$$

117. The stress at which a material fractures under large number of reversal of stresses is called:

(a) Residual stress (b) Creep
(c) Endurance limit (d) Ultimate stress

KPSC AE 2016, UJVNL AE 2016
RPSC POLY. TECH. LECT. 2014
ESE 1996

Ans. (c) : Endurance limit – Endurance limit is the stress level below which a specimen can withstand cyclic stress (reversal of stress) indefinitely without exhibiting fatigue failure.
Residual stress – Residual stresses are stresses that remain in a solid material after the original cause of the stresses has been removed.
Creep – creep is the tendency of a solid material to move slowly and deform permanently under the influence of persistent mechanical stress.
Ultimate stress – Ultimate stress is the maximum value of stress that a material can resist.

118. Hooke's law is valid:

(a) Only above elastic limit
(b) Only within elastic limit
(c) Only within plastic limit
(d) Till breaking of Substance

KPSC AE 2016

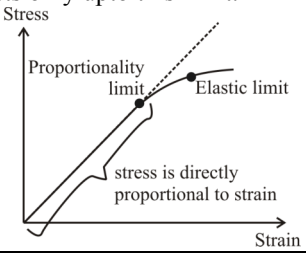
Ans. (b) : Hooke's law states that for homogenous isotropic material, stress is directly proportional to the strain, within limit of proportionality.

$$\sigma \propto \varepsilon$$

$$\sigma = E\varepsilon$$

Where,
- σ is the stress
- E is the modulus of elasticity also known as Young's modulus
- ε is the strain

But, truly Hooke's law is valid upto limit of proportionality as the linear relationship between stress and strain exists only upto this limit.



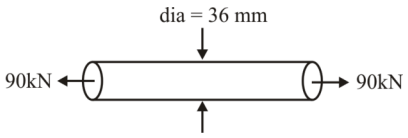
119. A bar specimen of 36 mm diameter is subjected to a pull of 90 kN during a tension test. The extension on a gauge length of 200 mm is measured to be 0.089 mm and the change in diameter to be 0.0046 mm. The Poisson's ratio will be

(a) 0.287 (b) 0.265
(c) 0.253 (d) 0.241

ESE 2019

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Ans. (a)



Diameter 'D' = 36mm
Pull 'P' = 90 kN
ℓ = 200mm
$\Delta\ell$ = 0.089
ΔD = 0.0046mm

Poisson's ratio 'μ' = $-\dfrac{\text{Lateral strain}}{\text{Longitudinal strain}}$

$$\mu = -\frac{\left(\frac{\delta D}{D}\right)}{\left(\frac{\delta \ell}{\ell}\right)} \Rightarrow \mu = -\frac{\left(\frac{-0.0046}{36}\right)}{\left(\frac{0.089}{200}\right)} \Rightarrow \boxed{\mu = 0.287}$$

120. A steel rod 15 m long is at a temperature of 15^0C. The values of $\alpha = 12\times 10^{-6}$ /^{0}C and E = 200 GN/m^2 are adopted. When the temperature is raised to 65^0C, what is the free expansion of the length; and if this expansion of the rod is fully prevented, what is the temperature stress produced?

(a) 5 mm and 120 MN/m^2
(b) 9 mm and 120 MN/m^2
(c) 5 mm and 150 MN/m^2
(d) 9 mm and 150 MN/m^2

ESE 2019

Ans. (b)

Length (L) = 15m
Temperature difference $\Delta t = 65-15 = 50^0$C
$\alpha = 12\times 10^{-6}/^0$C
E = 200×10^3MPa
Free expansion = $\ell\alpha\Delta T$
= 15000mm $\times$ 12 $\times$ $10^{-6}\times(50)$
[Free expansion = 9mm]
Temperature stress
σ = E $\alpha\Delta t$
$\sigma = (200\times10^3)\,(12\times10^{-6})\times(50)$
σ = 120MPa (compressive)

$\boxed{\text{Temperature stress '}\sigma\text{'} = 120\text{MN/m}^2}$

121. For a material, the modulus of rigidity is 100 GPa and the modulus of elasticity is 250 GPa. The value of the Poisson's ratio is

(a) 0.20 (b) 0.25
(c) 0.30 (d) 0.35

ESE 2018

Ans. (b) : Modulus of rigidity 'G' = 100 GPa
Modulus of elasticity 'E' = 250 GPa
Now, relation between G, E and μ are–

$E = 2G(1+\mu)$
$250 = 2\times 100(1+\mu)$
$1+\mu = \dfrac{250}{200} = 1.25$

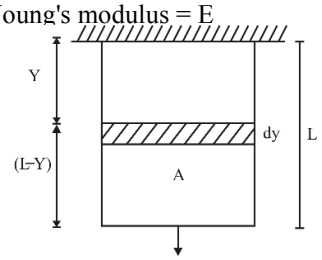
$\boxed{\mu = 0.25}$ Poisson's ratio.

122. The deformation of a vertically held bar of length *L* and cross-section *A* is due to its self-weight only. If Young's modulus is *E* and the unit weight of the bar is γ, the elongation d*L* is

(a) $\dfrac{\gamma L^3}{2E}$ (b) $\dfrac{EL^2}{2\gamma}$
(c) $\dfrac{\gamma L^2}{2E}$ (d) $\dfrac{\gamma L^2}{2AE}$

ESE 2018

Ans. (c) : Unit weight of bar = γ
Young's modulus = E



Total elongation (dL) = $\dfrac{PL}{AE}$

$$\Rightarrow dL = \int_0^L \frac{\gamma A(L-y)dy}{AE}$$

by Integrating we get,

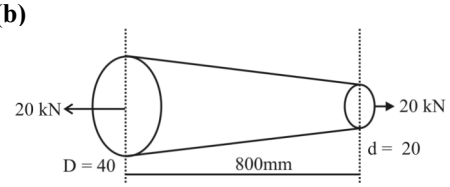
$$\boxed{dL = \frac{\gamma L^2}{2E}}$$

123. A mild steel bar, circular in cross-section, tapers from 40 mm diameter to 20 mm diameter over its length of 800 mm. It is subjected to an axial pull of 20 kN. E = 2×10^5 N/mm^2. The increase in the length of the rod will be

(a) $\dfrac{1}{10\pi}$mm (b) $\dfrac{2}{5\pi}$mm
(c) $\dfrac{4}{5\pi}$mm (d) $\dfrac{1}{5\pi}$mm

ESE 2017

Ans. (b)



Axial pull 'P' = 20 kN
E = 2×10^5 N/mm^2
For a tapering circular bar, change in length due to axial force.

$$\delta\ell = \frac{4P\ell}{\pi E.D.d}$$

$$\delta\ell = \frac{4\left(20\times10^3\right)\times800}{\pi\times2\times10^5\times40\times20} = \frac{64\times10^6}{160\times10^6\pi}$$

$$\boxed{\delta\ell = \frac{2}{5\pi}\text{mm}}$$

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124. In mild steel specimens subjected to tensile test cycle, the elastic limit in tension is raised and the elastic limit in compression is lowered. This is called

(a) Annealing effect (b) Bauschinger effect
(c) Strain rate effect (d) Fatigue effect

ESE 2017

Ans. (b) : The Bauschinger effect refers to a property of materials where the material's stress/strain characteristic change as a result of the microscopic stress distribution of the material. For example, an increase in tensile yield strength occurs at the expenses of compressive yield strength.
These effect is normally associated with conditions where the yield strength of a metal decreases when the direction of strain is changed.

125. A solid uniform metal bar of diameter D mm and length *l* mm hangs vertically from its upper end. The weight density of the material is ρ N/mm³ and its modulus of elasticity is E N/mm². The total extension of the rod due to its own weight would be

(a) $\frac{\rho\ell^2}{2E}$ (b) $\frac{\rho\ell}{2E}$
(c) $\frac{\rho\ell}{4E}$ (d) $\frac{\rho\ell^2}{4E}$

Maharashtra PSC (Mains) 2019 Paper–I
ESE 2017

Ans. (a) : Total extension of the rod due to its own weight is–

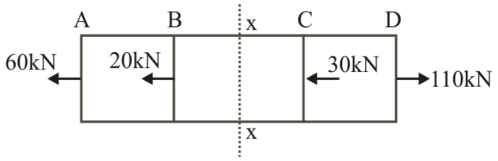
$$\left[\delta\ell = \frac{W\ell}{2AE}\right]$$

$$\rho = \frac{W}{V}$$

$$W = \rho A\ell$$

$$\delta\ell = \frac{(\rho A\ell)\ell}{2EA} \qquad \boxed{\delta\ell = \frac{\rho\ell^2}{2E}}$$

126. What is the stress at the section x- x for the bar ABCD with uniform cross-section 1000 mm² ?

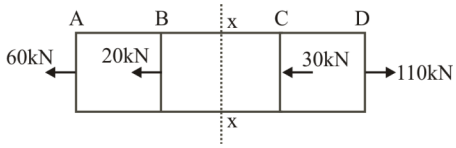


(a) 20 N/mm² (Tensile)
(b) 30 N/mm² (Compressive)
(c) 80 N/mm² (Tensile)
(d) 50 N/mm² (Compressive)

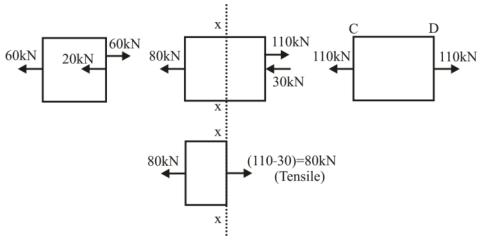
ESE 2017

Ans. (c)

■ Cross-section 'A' = 1000mm².



⇒ The stress at section x – x for a given cross-section.

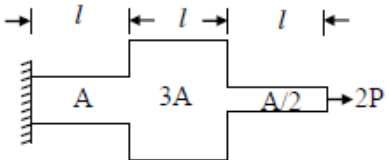


$$\text{Stress } '\sigma' = \frac{P}{A}$$

$$\sigma = \frac{80\times10^3}{1000}$$

$$\boxed{\sigma = 80\text{N/mm}^2\text{(Tensile)}}$$

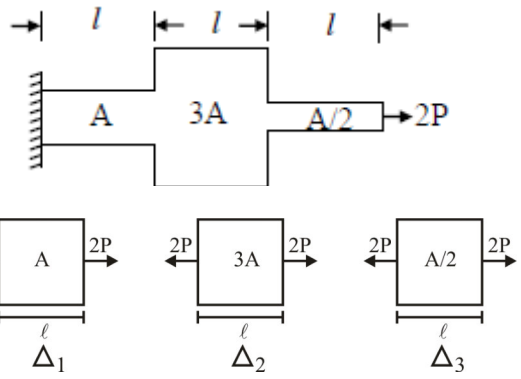
127. The total elongation of the structural element (fixed at one end, free at the other end, and of varying cross-section) as shown in the figure, when subjected to load 2P at the free end is



(a) $6.66\frac{P\ell}{AE}$ (b) $5.55\frac{P\ell}{AE}$
(c) $4.44\frac{P\ell}{AE}$ (d) $3.33\frac{P\ell}{AE}$

ESE 2017

Ans. (a)



Elongation $'\Delta_{total}' = \Delta_1 + \Delta_2 + \Delta_3$

$$\Delta_{total} = \frac{2P\ell}{AE} + \frac{2P\ell}{3AE} + \frac{2P\ell}{\frac{A}{2}E}$$

$$\Delta_{total} = \frac{2P\ell}{AE} + \frac{2P\ell}{3AE} + \frac{4P\ell}{AE}$$

$$\Delta_{total} = \frac{6P\ell}{AE} + \frac{2P\ell}{3AE}$$

$$\Delta_{total} = \frac{6P\ell}{AE} + 0.666\frac{P\ell}{AE}$$

$$\boxed{\Delta_{total} = 6.66\frac{P\ell}{AE}}$$

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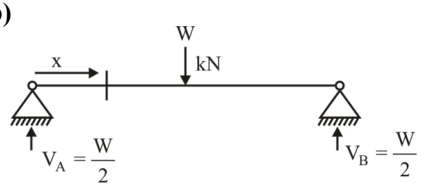
128. A simply supported beam of span *l* and flexural rigidity EI carries a unit load at its mid-span. The strain energy at this condition in the beam due to bending is

(a) $\frac{\ell^3}{48EI}$ (b) $\frac{\ell^3}{96EI}$

(c) $\frac{\ell^3}{192EI}$ (d) $\frac{\ell^3}{16EI}$

ESE 2017

Ans. (b)



The bending moment at a section is given by

$$M_x = V_A \times x = \frac{W}{2}x$$

Strain energy due to bending is given by

$$U_b = \int_o^{\ell} \frac{(Mx)^2\,dx}{2EI} = 2\int_o^{\ell/2}\left(\frac{w}{2}x\right)^2 \frac{dx}{2EI}$$

$$U_b = 2x\int_o^{\ell/2}\left[\frac{W}{2}x\right]^2 \frac{dx}{2EI}$$

$$U_b = 2\left[\frac{W^2}{4}\left[\frac{x^3}{3}\right]_0^{\ell/2}\right]\frac{1}{2EI}$$

$$U_b = \frac{W^2}{12EI}\left[x^3\right]_0^{\ell/2}$$

$$\left[U_b = \frac{W^2\ell^3}{96EI}\right] \quad \text{When W = 1,} \quad \boxed{U_b = \frac{\ell^3}{96EI}}$$

129. Ability of a material to absorb energy within the elastic range:

(a) Toughness (b) Elasticity

(c) Stiffness (d) Resilience

DSSSB AE 2019, GPSC AE March 2018
APPSC AEE MAINS 2016, Kerala PSC AE 2015

Ans. (d) : Toughness: It is the measure of how much energy the material can absorb before its breaking.
Resilience: How much energy can be stored in material within elastic range.
Proof resilience- How much maximum energy can be store by material at elastic range.
Stiffness- Required force for unit deflection.

130. The bulk modulus of elasticity of a material is twice its modulus of rigidity. The Poission's ratio of the material is:

(a) 1/7 (b) 2/7

(c) 3/7 (d) 4/7

ISRO Scientist/Engineer 2020
APPSC AEE MAINS 2016, ESE 2002

Ans. (b) : K = 2G

$$\frac{1}{m} = \mu = \frac{3K-2G}{6K+2G} = \frac{3K-K}{6K+K} = \frac{2}{7}$$

131. A steel rod of 30 mm diameter and 3 m length is subjected to an axial pull of 50 kN. If E = 200 × 10^9 pa, the elongation of the rod will be:

(a) 2.225 mm (b) 1.062 mm

(c) 0.525 mm (d) 3.152 mm

ISRO Scientist/Engineer 2020

Ans. (b) : Given that
d = 30 mm
l = 3m = 3 × 10^3 mm
P = 50 kN = 50 × 10^3 N

$$\Delta\ell = \frac{P\ell}{AE} = \frac{50\times10^3\times3\times10^3}{\frac{\pi}{4}\times(30)^2\times2\times10^5}$$

Δl = 1.062 mm

132. The stress in a body, when suddenly loaded is _____ the stress when same load is applied gradually

(a) Equal to (b) One half

(c) Twice (d) Four times

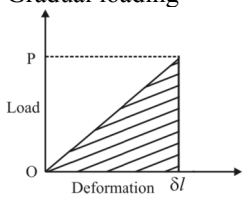
ISRO Scientist/Engineer 2019

Ans : (c)

- The Stress in a body, when suddenly loaded is twice the stress when same load is applied gradually.
- Stress due to sudden loading–
  When the load is applied suddenly the value of the load is P throughout the deformation.

Now,

- Stress due to Gradual loading–



Work done on the bar = Area of load

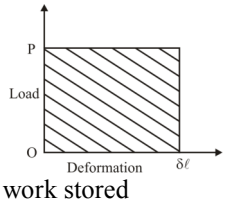
$$\left[\text{Work done} = \frac{1}{2}\times P\times\delta\ell\right]$$

Work done = work stored

$$\therefore \frac{1}{2}P\delta\ell = \frac{1}{2}\sigma\times A\times\delta\ell$$

$$P = \sigma\times A$$

$$\left[\sigma = \frac{P}{A}\right]$$



Work done = work stored

$$\therefore P\delta\ell = \frac{1}{2}\times\sigma\times A\times\delta\ell$$

$$P = \frac{1}{2}\sigma A$$

$$\left[\sigma = \frac{2P}{A}\right]$$

so, $\sigma_{sudden} = 2 \times \sigma_{gradual}$

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133. The unit of strain is

(a) N-mm (b) N/mm
(c) N/Sq.mm (d) No unit

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Ans : (d) : • The normal strain is a dimensionless quantity or unit of strain is no unit, since it is a ratio of two lengths, but it is common in practice to state it in terms of a ratio of length units i.e.meters per meter (m/m).

• Strain is a deformation of a solid due to stress

$$\left[\varepsilon = \frac{d\ell}{\ell_o}\right]$$

$$\left[\frac{m}{m} = \text{Dimensionless}\right]$$

134. A steel bar of 8 mm is heated from 10°C to 25°C and the bar is free to expand. The bar will induce

(a) Tensile stress (b) No stress
(c) Compressive stress (d) Shear stress

ISRO Scientist/Engineer 2019

Ans : (b)

- If the body is allowed to expand or contract freely, no stress will be set up in the body.
- An increase the temperature produces an increase in length and a decrease in temperature results in a decrease in length.
- When the free expansion or contraction is partially or completely prevented then there will be internal resisting force induced in the body i.e. thermal stress will be developed in the body.
- Thermal strain ε_T is proportional to the temperature change ΔT.

$$\left[\varepsilon_T = \alpha \Delta T\right]$$

$$\left[dL = \alpha.\Delta T.L \quad \varepsilon \neq 0, \sigma = 0\right]$$

135. Two bars of different materials and same size are subjected to the same tensile force. If the bars have unit elongation in the ratio of 3:5, then the ratio of modulus of elasticity of the two materials will be

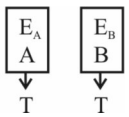
(a) 2:5 (b) 3:5
(c) 5:3 (d) 3:4

ISRO Scientist/Engineer 2017

Ans : (c)

Bar A $= E_A T$

Bar B $= E_B T$



Elongation $\left[\Delta = \frac{PL}{AE}\right]$

Elongation ratio i.e. $\frac{\Delta_A}{\Delta_B} = \frac{3}{5}$

So, $\Delta_A = \frac{PL}{AE_A}$ $\Delta_B = \frac{PL}{AE_B}$

$$\Rightarrow \frac{\Delta_A}{\Delta_B} = \frac{3}{5} = \frac{PL}{AE_A} \Big/ \frac{PL}{AE_B} \quad \because \left\{\begin{matrix}\text{Same size and} \\ \text{same tensile force}\end{matrix}\right\}$$

$$\Rightarrow \frac{\frac{1}{E_A}}{\frac{1}{E_B}} = \frac{3}{5}$$

$$\therefore \frac{E_B}{E_A} = \frac{3}{5} \quad \text{and} \quad \boxed{\frac{E_A}{E_B} = \frac{5}{3}}$$

136. Modular ratio of the two materials is the ratio of

(a) Linear stress to linear strain
(b) Shear stress to shear strain
(c) Their modulus of elasticity's
(d) Their modulus of rigidities

ISRO Scientist/Engineer 2017

Ans : (c) : Modular ratio of the two materials is the ratio of their young's modulus of elasticity's.

$$\left[m = \frac{E_1}{E_2}\right]$$

137. The young's modulus of a material is 150 GPa and Poisson's ratio is 0.25, the modulus of rigidity of the material is

(a) 30 GPa (b) 50 GPa
(c) 60 GPa (d) 100 GPa

ISRO Scientist/Engineer 2017

Ans : (c)

E = 150 GPa

$\mu = 0.25$

Modulus of rigidity 'G' and E, μ relationship

$$\left[E = 2G(1+\mu)\right]$$

So, $$\left[G = \frac{E}{2(1+\mu)}\right] = \frac{150}{2(1+0.25)} = \frac{150}{2.5}$$

$$\boxed{G = 60 GPa}$$

138. The property of a material by which it can be beaten or rolled into thin plates, is called

(a) Malleability (b) Ductility
(c) Plasticity (d) Elasticity

ISRO Scientist/Engineer 2015

Ans : (a) : Malleability : It is that property of a metal due to which a piece of metal can be converted into a thin sheet.

Ductility : Ductility of a metal is that property due to which a metal can be drawn into wire of thin section.

Elasticity : It is that property of material by which the original dimension can be recovered after unloading.

139. When a body is elastically stretched, the value of Poisson's ratio always remains

(a) Greater than one (b) Less than one
(c) Equal to one (d) None of these

ISRO Scientist/Engineer 2018

Ans : (b) : Poisson's ratio is the ratio of transverse contraction strain to longitudinal extension strain in the direction of stretching force.

Poisson's ratio of elastically stretched body always remains less than one.

$$\left[\mu = -\frac{\text{Lateral Strain}}{\text{Longitudinal Strain}}\right]$$

$$\boxed{-1 < \mu < 0.5}$$

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140. A bar of copper and bar of steel form a composite system, which is heated to a temp of 40^0C. The stress in steel bar is

(a) Tensile (b) Compressive
(c) Zero (d) Shear

ISRO Scientist/Engineer 2014

Ans : (a) If a composite bar of Steel & Copper is heated, copper tries to expand more than the steel because the coefficient of thermal expansion of copper is more than steel. Copper brings a tensile stress on the composite bar.

On the other hand, steel tries to withhold the expansion of copper there by introducing a compressive stress on the composite bar.

141. A standard measure of ductility of a material is

(a) Percent elongation in length
(b) Percent increase in the Area
(c) Percent decrease in the length
(d) Percent decrease in length & increase in Area

ISRO Scientist/Engineer 2014

Ans : (a) Ductility is a measure of length. If the length of material during testing elongates more than 5% of its original length then that material is ductile material, otherwise brittle.

142. A rod of length L = 500 mm and cross sectional area A= 60mm^2 is made of an elasto-plastic material having a modulus of elasticity E = 200 GPa in its elastic range and yield point σy = 300 MPa. The rod is subjected to an axial load until it is stretched 7 mm and the load is then removed. What is the resulting permanent strain ?

(a) 6.25 mm (b) 6.50 mm
(c) 6.00 mm (d) 5.95 mm

ISRO Scientist/Engineer 2014

Ans : (a)

$\sigma = E.\varepsilon$, given $\sigma_y = 300$ MPa; $E = 200 \times 10^3$ MPa

Strain, $\varepsilon = \dfrac{300}{200\times10^3} = 1.5\times10^{-3}$

Now strain $= \dfrac{\Delta\ell}{\ell} \Rightarrow \dfrac{\Delta\ell}{500} = 1.5\times10^{-3}$

$\Delta\ell = 0.75$mm

Remaining strain = 7 – 0.75
= 6.25 mm

143. The change in the volume per Unit volume occurs when normal stresses are applied on an element of isotropic material is called as

(a) Bulk modulus of material
(b) Shearing Strain of material
(c) Dillatation of material
(d) Shearing stress of material

ISRO Scientist/Engineer 2014

Ans : (c) Volumetric strain is also known as Dillatation of material.

144. A bar 40 mm in diameter and subjected to a tensile force of 40,000 kg$_f$. Undergoes elongation resulting in decrease in diameter considering the properties of the material as E = 2×10^5 N/mm^2 and Poisson's ratio as 0. 3 the modulus of rigidity will be :

(a) 76923.07 N/mm^2 (b) 20×10^4 kg/cm^2
(c) 56898.50 N/mm^2 (d) 3×10^5 Kg/cm^2

ISRO Scientist/Engineer 2013

Ans : (a) $E = 2 \times 10^5$ N/mm^2

$\mu = 0.3$

Relation between young modulus 'E', Poisson's ration 'μ' and modulus of rigidity. 'G' will be

$$\left[E = 2G(1+\mu)\right]$$

So,

$2 \times 10^5 = 2G\ (1+0.3)$

$2G = \dfrac{20\times10^4}{1.3}$

$2G = 153{,}846.154$

$$\boxed{G = 76{,}923.077\ N/mm^2}$$

145. If Poisson's ratio of a material is 0.3, then the ratio of Young's Modulus to bulk modulus is

(a) 0.6 (b) 0.8
(c) 1.2 (d) 1.4

APTRANSCO 2017
AP TRANSCO AEE 2017
ESE 2013

Ans : (c) : $E = 3K(1-2\mu)$,

where, E = Young's modulus
K = Bulk modulus

$\dfrac{E}{K} = 3(1-2\times0.3)$

$$\boxed{\frac{E}{K} = 1.2}$$

146. The property of material due to which it can be drawn into a thin wire is called:

(a) malleability (b) ductility
(c) strength (d) softness

WBPSC Poly Lect. 2019

Ans. (b) : The property of material due to which it can be drawn into a thin wire is called ductility.
Malleability : it is a physical property of metals that defines their ability to be hammered, pressed or rolled into thin sheet without breaking.

147. The ultimate tensile strength of structural mild steel is about :

(a) 30 N/mm^2 (b) 260 N/mm^2
(c) 420 N/mm^2 (d) 840 N/mm^2

OPSC AEE 2019 Paper–I

Ans. (c) : The yield strength of mild steel is around 250 N/mm^2 and its ultimate strength is around 420 N/mm^2.

148. If the Young's modulus of elasticity of a material is twice its modulus of rigidity, then the Poisson's ratio of the material is :

(a) Zero (b) 0.5
(c) – 0.5 (d) – 11

OPSC AEE 2019 Paper–I

Ans. (a)

$E = 2G\ (1+\mu)$

given, $E = 2G$

$2G = 2G\ (1+\mu)$

$1 = 1 + \mu$

$\Rightarrow \mu = 0$

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149. Given that for an element in a body of homogeneous isotropic material subjected to plane stress; ε_x, ε_y and ε_z are normal strains in x, y and z directions respectively and μ is the Poisson's ratio, the magnitude of unit volume change of the element is given by :

(a) $\varepsilon_x + \varepsilon_y + \varepsilon_z$ (b) $\varepsilon_x - \mu(\varepsilon_y + \varepsilon_z)$

(c) $\mu(\varepsilon_x + \varepsilon_y + \varepsilon_z)$ (d) $\mu(\varepsilon_y + \varepsilon_z) - \varepsilon_x$

OPSC AEE 2019 Paper–I, ESE 1995

Ans. (a)

$$\varepsilon_V = \frac{\delta v}{v} = \varepsilon_x + \varepsilon_y + \varepsilon_z$$

$$\Rightarrow \quad \frac{\delta v}{V} = \varepsilon_x + \varepsilon_y + \varepsilon_z$$

150. The phenomenon of decreased resistance of a material to reversal of stress is called

(a) Creep (b) Fatigue

(c) Resilience (d) Plasticity

OPSC AEE 2019 Paper–I

Ans. (b) : The phenomenon of decreased resistance of a material to reversal of stress is called fatigue.
Some sort of microscopic cracks will begin to appear on the material when the loads on the material cross the certain limit which leads to the fatigue in that material.

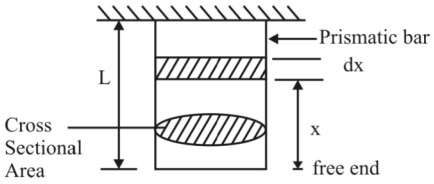
151. What is the total elongation of a prismatic bar of length (L) and cross-sectional area (A) and Young's Modulus of its material (E) hangs vertically under its own weight (W)?

(a) $\frac{WL}{2AE}$ (b) $\frac{WL}{8AE}$

(c) $\frac{WL}{6AE}$ (d) $\frac{WL}{4AE}$

OPSC AEE 2019 Paper–I

Ans. (a)



Where, W_x = weight of the portion below the section = γAx

The extension of the entire bar can be obtained by integrating the equation,

$$\int_O^L \delta dx = \delta L \int_O^L \frac{W_x d_x}{AE}$$

$$\delta L = \int_O^L \frac{\gamma AX\, dx}{AE}$$

$$\delta L = \frac{\gamma}{E}\int_O^L x.dx$$

$$\delta L = \frac{\gamma}{E}\left[\frac{x^2}{2}\right]_O^L$$

$$\delta L = \frac{\gamma L^2}{2E}$$

if W is the total weight of the bar, then

$$\gamma = \frac{W}{AL}$$

$$\boxed{\delta L = \frac{WL}{2AE}}$$

The deformation of the bar under its own weight is equal to the Half of the deformation, if the body is subjected to the direct load equal to the weight of the body.

152. Strain energy per unit volume that a material can absorb without exceeding its proportional limit is called :

(a) Strain hardening
(b) Shear modulus of material
(c) Bulk modulus of material
(d) Modulus of resilience

OPSC AE Paper- I 2019

Ans. (d) : The modulus of resilience is maximum strain energy that can be absorbed per unit volume without creating a permanent distortion (within the elastic limit) Maximum strain energy which a body can stores upto the elastic limit is called proof resilience.

$$\text{Modulus of resilience} = \frac{\text{Proof resilience}}{\text{Unit volume of the body}}$$

i.e. the strain energy stored below the elastic limit called resilience.

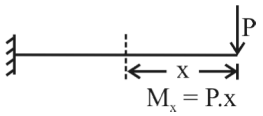
153. If a cantilever beam of length L is subjected to a point load P acting in the downward direction at its free end and the flexural rigidity (EI) of the beam is constant, the elastic strain energy due to bending would be :

(a) $\frac{PL}{6EI}$ (b) $\frac{P^2L^3}{6EI}$

(c) $\frac{P^2L^2}{3EI}$ (d) $\frac{P^2L^3}{3EI}$

ESE 2020
HPPSC Poly. Lect. 2016
APPSC AEE MAINS 2016
WBPSC AE 2014, 2012

Ans : (b)



Strain energy is given by

$$U = \int_0^L \frac{(M_x)^2}{2EI}dx$$

$$= \int_0^L \frac{(Px)^2}{2EI}dx$$

$$= \int_0^L \frac{P^2x^2}{2EI}dx$$

$$= \frac{P^2}{2EI}\int_0^L x^2dx = \frac{P^2}{2EI} \times \left[\frac{x^3}{3}\right]_0^L$$

$$\boxed{U = \frac{P^2L^3}{6EI}}$$

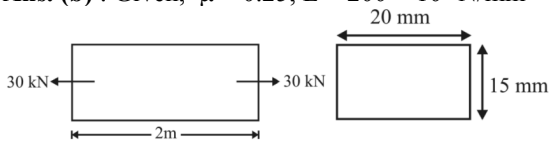
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154. A steel bar 2 m long, 20 mm wide and 15mm thick is subjected to a tensile load of 30 kN. If Poisson's ratio is 0.25 and Young's modulus is 200 GPa, an increase in volume will be :

(a) 160 mm^3 (b) 150 mm^3
(c) 140 mm^3 (d) 130 mm^3

ESE 2020

Ans. (b) : Given, $\mu = 0.25$, $E = 200 \times 10^3$ N/mm^2



$$\text{Axial stress } (\sigma_x) = \frac{P}{A} = \frac{30\times10^3}{20\times15} = 100\,\text{MPa}$$

$$(\sigma_x, \sigma_z) = 0$$

Now,

$$\text{Volumetric strain } (\varepsilon_v) = \frac{\sigma_x + \sigma_y + \sigma_z}{E}(1-2\mu)$$

$$\varepsilon_v = \frac{100}{200\times10^3}(1-2\times0.25)$$

$$\boxed{\varepsilon_v = 2.5\times10^{-4}}$$

We know that,

$$\Delta V = \varepsilon_v \times V$$
$$= 2.5 \times 10^{-4} \text{ x } (2000 \times 20 \times 15)$$
$$\boxed{\Delta V = 150 mm^3}$$

155. When a round bar material with diameter of 37.5 mm, length of 2.4 m, Young's modulus of 110 GN/m^2 and shear modulus of 42 GN/m^2 is stretched for 2.5 mm, its Bulk modulus will be nearly :

(a) 104 GN/m^2 (b) 96 GN/m^2
(c) 84 GN/m^2 (d) 76 GN/m^2

ESE 2020

Ans. (b) : Given,
$d = 37.5$ mm, $L = 2400$ mm, $E = 110 \times 10^3$ N/mm^2
$G = 42 \times 10^3$ N/mm^2, $\delta L = 2.5$ mm
we know that,
$E = 2G(1 + \mu)$
$110 \times 10^3 = 2 \times 42 \times 10^3 (1 + \mu)$

$$\boxed{\mu = 0.31}$$

Now,
$E = 3K(1-2\mu)$
$110 \times 10^3 = 3K(1-2 \times 0.31)$

$$\boxed{K = 96.49\times10^3\,\text{MPa} \text{ or } 96.49\,\text{GPa} \simeq 96\,\text{GPa}}$$

156. U_1 and U_2 are the strain energies stored in a prismatic bar due to axial tensile forces P_1 and P_2 respectively. The strain energy U stored in the same bar due to combined action of P_1 and P_2 will be

(a) $U = U_1 + U_2$ (b) $U = U_1 \times U_2$
(c) $U < U_1 + U_2$ (d) $U > U_1 + U_2$

Maharashtra PSC AE (Mains) 2012 Paper–I

Ans : (d) For unit volume,
for P_1 force –

$$U_1 = \frac{1}{2}\text{ stress} \times \text{strain}$$
$$= \frac{1}{2}\times\frac{P_1}{A}\times\varepsilon_1$$

for P_2 force –

$$U_2 = \frac{1}{2}\times\frac{P_2}{A}\times\varepsilon_2$$

for combined force $(P_1 + P_2)$

$$U = \frac{1}{2}\times\frac{(P_1+P_2)}{A}\times(\varepsilon_1+\varepsilon_2)$$

$$U = \frac{1}{2A}\times(P_1\varepsilon_1 + P_1\varepsilon_2 + P_2\varepsilon_1 + P_2\varepsilon_2)$$

$$U_1 + U_2 + \frac{1}{2A}\times(P_1\varepsilon_2 + P_2\varepsilon_1)$$

$$U > U_1 + U_2$$

157. A mild steel bar is in two parts having equal length. The area of cross-section of part-1 is double that of part-2. If the bar carries an axial load P, then the ratio of elongation in part-1 to that in part-2 will be

(a) 2 (b) 4
(c) 1/2 (d) 1/4

Gujarat PSC AE 2017

Ans : (a)

$\because$ Elongation $\propto \frac{L}{A}$

$$\frac{\text{Ratio of elongation in 1}^{st}\text{ section}}{\text{Ratio of elongation in 2}^{nd}\text{ section}} = \frac{L_1}{L_2}\times\frac{A_1}{A_2}$$

since $L_1 = L_2$ and $2A_1 = A_2$

$$\text{Ratio} = 1\times\frac{2}{1} = 2$$

158. A prismatic bar has a cross-section of 25 mm × 50 mm and a length of 2000 mm. Under an axial load of 100 kN, the measured elongation of the bar is 2 mm. The tensile stress and % strain in the bar are :

(a) 80 MPa and 0.1%
(b) 0.08 GPa and 1.0%
(c) 80 N/mm^2 and 1%
(d) 800 MPa ad 0.01%
(e) 8 kN/mm^2 and 0.001%

CGPSC AE 2014 Shift–I

Ans. (a) :
Cross-sectional area = 25mm × 50 mm
Length (L)=2000 mm
P(Load) = 100 kN
$\delta = 2$ mm

$$\delta = \frac{PL}{AE}$$

$$2 = \frac{100\times10^3\times2000}{25\times50\times E}$$

$$E = 8\times10^4\text{ N/mm}^2$$

$$\varepsilon = \frac{\delta}{L} = \frac{2}{2000}$$

$$\boxed{\varepsilon = \frac{2}{2000}}$$

$$\%\,\varepsilon = \frac{2}{2000}\times100$$

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$\boxed{\%\ \varepsilon = 0.1\%}$

$\sigma = E\varepsilon = 8\times10^4 \times \frac{2}{2000}$

$\sigma = 80\ \text{N/mm}^2$

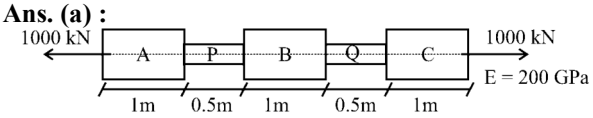
$\boxed{\sigma = 80\,\text{MPa}}$

159. A metal bar of variable cross section as shown and length is subjected to 1000 kN of force. The sections are of 1 m long section is 10000 sq. mm and that of 500 mm long section is 5000 sq. mm. Assuming the elastic modulus of 200 GPa, the total elongation of the bar will be :

(a) 2.5 mm (b) 0.025 mm
(c) 0.25 mm (d) 25 mm
(e) 1.5 m

CGPSC AE 2014 Shift–I

Ans. (a) :



A,B,C → Block have the same dimension
& P & Q → have same dimension
Cross-sectional area of (A,B,C) = 10,000 mm^2
Cross-sectional area of (P,Q) = 5000 mm^2

$\delta_T = \delta_A + \delta_P + \delta_B + \delta_Q + \delta_C$

$= \delta_A + \delta_A + \delta_A + \delta_P + \delta_P \quad \left[\because \delta_A = \delta_B = \delta_C,\ \delta_P = \delta_Q\right]$

$= 3\delta_A + 2\delta_P$

$= 3\left[\frac{1000\times10^3\times1000}{10,000\times200\times10^3}\right] + 2\left[\frac{1000\times10^3\times500}{5000\times200\times10^3}\right]$

$= 3\,[1/2] + 2\times\frac{1}{2}$

$= 1.5 + 1$

$\boxed{\delta_T = 2.5}$ mm

160. While calculating the elastic modulus, the stress difference between two points under consideration is 40 MPa and the strain difference is 2000 $\times 10^{-6}$. The elastic modulus of this material in GPa is :

(a) 20000 (b) 0.02
(c) 2000000 (d) 20
(e) 2000

CGPSC AE 2014 Shift–I

Ans. (d) :

$\Delta\sigma = 40\,\text{MPa}$

$\Delta\varepsilon = 2000\times10^{-6}$

$E = \frac{\Delta\sigma}{\Delta\in} = \frac{40\,\text{MPa}}{2000\times10^{-6}} = \frac{40\times10^6\,\text{MPa}}{2000}$

$= \frac{40\times10^3}{2000}\,\text{GPa} = 20\ \text{GPa}$

161. A circular steel bar of 2 cm diameter carries a tensile load of 30 kN. If E = 2×10^7 N/mm^2 the elongation in the 30 cm length of bar will be

(a) 0.0143 mm (b) 0.1430 mm
(c) 1.43 mm (d) 14.30 mm

TNPSC AE 2012

Ans. (b) : d = 2cm, f = 30 kN $\quad e = \frac{\delta\ell}{\ell}$

$\ell = 30\text{cm} \quad E\times e = P$

$A = \frac{\pi}{4}d^2 = \frac{\pi}{4}(2)^2 = 3.14\text{cm}^2 \quad e = P/E$

$E = 2\times10^7\ \text{N/mm}^2$

$P = F/A$

$P = \frac{30\times10^3}{3.14}\,\text{N/cm}^2$

$\frac{P}{E} = \frac{\delta\ell}{\ell}$

$\delta\ell = \frac{P\times\ell}{E} = \frac{30\times10^3\times30}{3.14\times2\times10^7}$

$P = \frac{9}{628}$

$\delta\ell = 0.01433\text{cm}$

$\boxed{\delta\ell = 0.1433\text{mm}}$

162. When a rectangular bar of length *l*, breadth b and thickness *t* is subjected to a pull of *p*, then volumetric strain is

(a) *e(1–2m)* (b) *e(2m–1)*
(c) *e(1–2/m)* (d) *e(2/m–1)*

TNPSC AE 2012

Ans. (c) : $V = \ell\times b\times t$

$\delta V = \delta\ell\times b\times t + \delta b\times\ell\times t + \delta t\times b\times\ell$

Divided by V

$\frac{\delta V}{V} = \frac{\delta\ell}{\ell} + \frac{\delta b}{b} + \frac{\delta t}{t}$

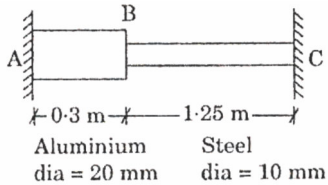
ℓ increase and t and b decrease

Then $e = \frac{\delta\ell}{\ell}, \frac{e}{m} = \frac{-\delta b}{b}, \frac{e}{m} = \frac{-\delta t}{t}$

$\frac{\delta V}{V} = e - \frac{e}{m} - \frac{e}{m} \quad \frac{1}{m} = \frac{\text{Lateral strain}}{\text{Longitudinal strain}}$

$\boxed{\frac{\delta V}{V} = e\left(1 - \frac{2}{m}\right)}$

163. At room temperature the rods are shown in figure. When temperature is raised, what is the nature of stresses developed in each rod, if $E_{Al} < E_S$ and $\alpha_{Al} > \alpha_S$?



Supports are unyielding

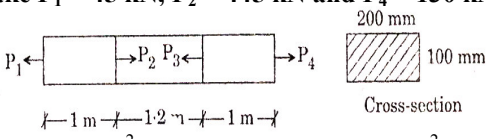
(a) Compressive in aluminium and steel
(b) Tensile in aluminium and steel
(c) Compressive in aluminium and tensile in steel
(d) Tensile in aluminium and compressive in steel

Maharashtra PSC AE (Mains) 2012 Paper–I

Ans : (a) If temperature is increased and member is restrained, the force produced is compressive. If temperature is decreased the force produced is tensile.

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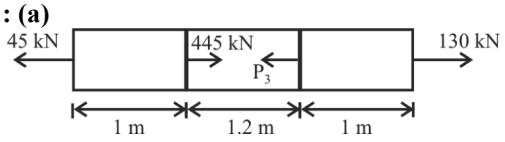
164. Calculate the maximum stress acting on the cross-section of following element:
Take P_1 = 45 kN, P_2 = 445 kN and P_4 = 130 kN.



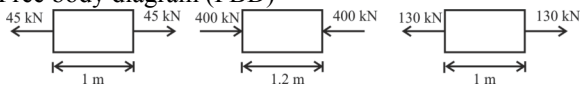
(a) 20 N/mm^2 (b) 22.5 N/mm^2
(c) 28.75 N/mm^2 (d) 6.5 N/mm^2

Maharashtra PSC AE (Mains) 2012 Paper–I

Ans : (a)



For equilibrium condition = $\Sigma H = 0$
$-45 - P_3 + 445 + 130 = 0$
$P_3 = 530$ kN
Free body diagram (FBD)



Maximum force (P) = 400 kN = 400 × 1000 N
Cross - sectional Area (A) = (200×100) mm^2

$$\text{Maximum stress (P)} = \frac{F}{A} = \frac{400\times1000}{200\times100}$$

P = 20 N/mm^2

165. A bar of length 'l' and cross sectional area (b×d) is subjected to an axial force P. Young's modulus of the material is E, Poisson ratio = 1/m. What is the change in volume ?

(a) $dv = \frac{Pl}{E}\left[1-\frac{2}{m}\right]$ (b) $dv = \frac{Pl}{E}\left[m-\frac{2}{m}\right]$

(c) $dv = \frac{m}{Pl}\left[1-\frac{2}{m}\right]$ (d) $dv = \frac{Pl}{E}\left[E-\frac{2}{m}\right]$

Rajasthan AE (Nagar Nigam) 2016

Ans. (a) : Volumetric strain

$$\frac{\delta v}{v} = \frac{1}{E}\left(1-\frac{2}{m}\right)\left(\sigma_x+\sigma_y+\sigma_z\right)$$

$\sigma_y = \sigma_z = 0$

$$\frac{\delta v}{v} = \frac{1}{E}\left(1-\frac{2}{m}\right)\sigma_x \qquad \{v = b\times d\times\ell\}$$

$$\delta v = \frac{1}{E}\left(1-\frac{2}{m}\right)\times\left(\frac{P}{b\times d}\right)\times\left(b\times d\times\ell\right)$$

$$\boxed{\delta v = \frac{P\ell}{E}\left(1-\frac{2}{m}\right)}$$

166. A steel bar having cross sectional area of 100 π mm^2 is subjected to a tensile force of 50 kN. Determine the tensile stress.

(a) 175.25 MPa (b) 150 MPa
(c) 159.15 MPa (d) 200.05 MPa

DSSSB AE 2019

Ans. (c) Tensile stress $= \frac{\text{Tensile force}}{\text{Cross sectional Area}}$

$$= \left[\frac{50\times1000\text{N}}{100\times\pi\times10^{-6}\text{m}^2}\right] = \frac{7\times50\times1000}{100\times22}\frac{\text{N}}{\text{mm}^2}$$

= 159.15 MPa

167. The stress in a steel bar is given as 250 kN/m^2 and length of the bar is 40m. Calculate the shortening of the bar if E = 2.14×10^5 N/mm^2.

(a) 0.250mm (b) 0.305mm
(c) 0.160mm (d) 0.0467mm

DSSSB AE 2019

Ans. (d) $E = \frac{\sigma}{\varepsilon}$

$\varepsilon = \frac{\sigma}{E}$

$\frac{\Delta\ell}{\ell} = \frac{\sigma}{E}$

$$\Delta\ell = \frac{\sigma}{E}\times\ell = \frac{250\times10^{-3}\times40\times1000}{2.14\times10^5} = 0.0467\text{mm}$$

168. A tapered circular section is of diameter 10 mm gradually increasing to 40 mm over an axial length of 200 mm. It is subjected to a pull P. Let the elongation be Δ_1. Another circular section made up of the same material of uniform diameter 20 mm is subjected to same pull P. If its elongation Δ_2. Which is of the following statement true :

(a) $\Delta_1 > \Delta_2$ (b) $\Delta_1 < \Delta_2$
(c) $\Delta_1 = \Delta_2$ (d) $\Delta_1 = 0.5\,\Delta_2$

APTRASCO 2017

Ans : (c) : For tapered circular section,

$$\Delta_1 = \frac{4P\ell}{\pi D_1 D_2 E} = \frac{P\times4\times200}{\pi\times10\times40\times E}$$

$$\Delta_1 = \frac{2P}{\pi E} \qquad \text{.....(i)}$$

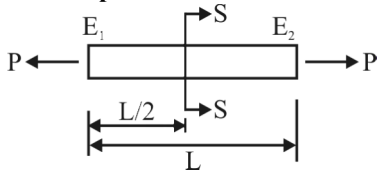
For circular section,

$$\Delta_2 = \frac{P\ell}{AE} = \frac{P\times200}{\frac{\pi}{4}D^2E} = \frac{P\times200\times4}{\pi\times20^2E}$$

$$\Delta_2 = \frac{2P}{\pi E} \qquad \text{.....(ii)}$$

Equation (i) = Equation (ii)
$\therefore\ \Delta_1 = \Delta_2$

169. A rod of length L having uniform cross-sectional area A is subjected to a tensile force P as shown in the figure below. If the Young's modulus of the material varies linearly from E_1 to E_2 along the length of the rod, the normal stress developed at the section SS is:



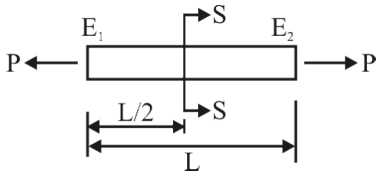
(a) $\frac{P}{A}$ (b) $\frac{P(E_1-E_2)}{A(E_1+E_2)}$

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(c) $\frac{PE_2}{AE_1}$ (d) $\frac{PE_1}{AE_2}$

PTCUL AE 2016

Ans. (a) :



Normal stress developed at section SS = $\frac{P}{A}$

170. Plastic response of a material to compressive force is known as

(a) Elasticity (b) Ductility
(c) Plasticity (d) Malleability

APPSC AEE MAINS 2016

Ans. (d) : Malleability– It is a special case of ductility which permits materials to be rolled or hammered into thin sheets. A malleable material should be plastic but it is not essential to be so strong. The malleable materials commonly used in engineering practice (In order of diminishing malleability) are lead, soft steel, wrought iron, copper and aluminium.

171. A prismatic bar 1 m long and 4 sq. cm in cross sectional area is compressed by a force of 80 kN. If E = 200 kN/sq.mm, the total strain energy stored in the bar is equal to

(a) 40 kN-mm (b) 0.05 kN-mm
(c) 400 kN-mm (d) 80 kN-mm

APPSC AEE MAINS 2016

Ans. (a) : Strain energy stored in prismatic bar

$$= \frac{P^2L}{2AE} = \frac{80\times10^3\times80\times10^3\times1\times1000}{2\times400\times200\times1000}$$
$$= 40000\text{kN/mm} = 40\text{kN-mm}$$

172. The phenomenon of the decreased resistance of a material to repeated stresses is called

(a) fatigue (b) creep
(c) freezing (d) melting

APPSC AEE MAINS 2016

Ans. (a) : Fatigue– When a material is subjected to repeated stress, it fails at stresses below the yield point stresses. Such type of failure of a material is known as fatigue.

173. Creep of a material is

(a) Continued deformation with time under sustained loading
(b) Disappearance of deformation on removal of load
(c) Not being ductile
(d) To become brittle

Chandigarh SDE 2017/Haryana PSC AE 2017
APTRANCO AEE 2017, APPSC AEE MAINS 2016
Maharashtra PSC (Mains) 2015 Paper–I
CGPSC AE 2014 Shift–I, ESE 2002

Ans. (a) : Creep– When a part is subjected to a constant stress at high temp. for a long period of time, it will undergo a slow and permanent deformation called creep. This property is considered in designing internal combustion engines, boilers and turbines.

174. Let the strain produced in length and diameter of cylindrical rod be 0.02 and 0.005 respectively. Then the volumetric strain is given by

(a) 0.03 (b) 0.025
(c) 0.015 (d) 0.01

APPSC AEE MAINS 2016

Ans. (d) : let, ϵ_v = Volumetric strain for cylindrical rod.

ϵ_L = Longitudinal strain = 0.02

ϵ_d = Dimatrical strain = –0.005

$$\epsilon_v = \epsilon_L + 2\epsilon_d$$
$$= 0.02 + \left[2\times(-0.005)\right] = 0.02 - 0.01$$
$$\epsilon_v = 0.01$$

175. A material with large deformation at failure is termed

(a) Brittle (b) Elastic
(c) Ductile (d) Elasto-plastic

APPSC AEE MAINS 2016
ESE 2011, 2003

Ans. (c) : Ductility– It is property of a material enabling it to be drawn into wire with the application of a tensile force.

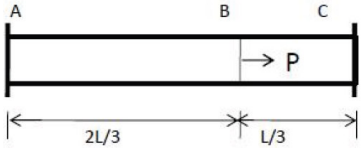
176. The diameter of a tapering rod varies from 'D' to 'D/2' in length of 'L' m. If it is subjected to an axial tension of 'P' the change in length is

(a) $4PL/(\pi ED^2)$ (b) $8PL/(\pi ED^2)$
(c) $2PL/(\pi ED^2)$ (d) None of the given answers

APPSC AEE MAINS 2016

Ans. (b) : $\delta = \frac{4PL}{\pi E\frac{D}{2}D} = \frac{8PL}{\pi ED^2}$

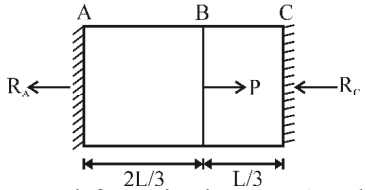
177. The ratio of loads shared by parts 'AB' and 'BC' of the bar shown below is



(a) 1 : 1 (b) 2 : 1
(c) 3 : 1 (d) 1 : 2

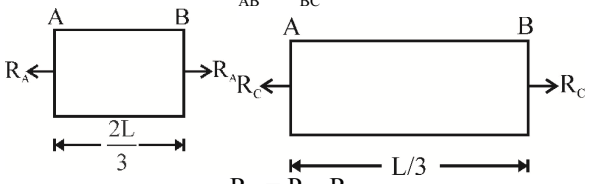
APPSC AEE MAINS 2016

Ans. (d) :



Rigid section, so deformation between A and C is zero.

$$\delta_{AB} + \delta_{BC} = 0$$



$$R_C = P - R_A$$

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$$\frac{R_A(2L/3)}{AE}-\frac{R_C(L/3)}{AE}=0$$

$$\frac{R_A\times 2L}{3AE}=\frac{R_C\times L}{3AE}\Rightarrow\frac{R_A}{R_C}=\frac{1}{2}$$

178. Two bars of same area and length but of different materials are subjected to same tensile force. If the bars have their axial elongation in the ratio of 4:6, then the ratio of modulus of elasticity of the two materials would be

(a) 4 : 6 (b) 6 : 4

(c) $2:\sqrt{6}$ (d) $\sqrt{6}:2$

APPSC AEE MAINS 2016

Ans. (b) : $\delta\ell_1=\frac{PL}{AE_1}$

$$\delta\ell_2=\frac{PL}{AE_2}$$

$$\frac{\delta\ell_1}{\delta\ell_2}=\frac{\frac{PL}{AE_1}}{\frac{PL}{AE_2}}\Rightarrow\frac{4}{6}=\frac{E_2}{E_1}$$

$$\Rightarrow\frac{E_1}{E_2}=6:4$$

179. A brass tube has enclosed a steel bar and they have equal cross-sectional area. The Young's modulus of elasticity is 200 GPa and 100 GPa for steel and brass respectively. Then the ratio of stress developed in the steel bar to that in the brass tube under compression is

(a) 0.5 (b) 1

(c) 1.5 (d) 2

APPSC AEE MAINS 2016

Ans. (d) : By using compatibility equation

$$\frac{\sigma_S}{E_S}=\frac{\sigma_B}{E_B}$$

$$\frac{\sigma_S}{200}=\frac{\sigma_B}{100}$$

$$\frac{\sigma_S}{\sigma_B}=2$$

180. The strain energy stored in a solid circular shaft, under pure torque, per unit its volume (take max. shear stress = q and modulus of rigidity = C) is expressed as

(a) q^2/C (b) $q^2/(4C)$

(c) $q^3/(4C)$ (d) $q^4/(4C)$

APPSC AEE MAINS 2016

Ans. (d) : Strain energy stored in a solid circular shaft under pure torque

$$U=\frac{T^2L}{2CJ}=\frac{q^2}{4C}\times\text{Volume}$$

Strain energy per unit volume under pure torque = $\frac{q^2}{4C}$

■ Strain energy in a prismatic bar under pure axial loading.

$$U=\frac{P^2L}{2AE}=\frac{\sigma^2}{2E}\times\text{Volume}$$

$$=\frac{1}{2}\times\sigma\times\in\times\text{Volume}$$

■ Strain energy in a tapered bar under purely axial load

$$U=\frac{1}{2}P.\delta,\left(\text{where }\delta=\frac{4PL}{\pi Ed_1d_2}\right)$$

■ Strain energy in a prismatic bar under self weight.

$$U=\frac{W^2L}{6AE}=\frac{\gamma^2AL^3}{6E}\quad(\text{Where, }W=\gamma AL)$$

■ Strain energy in a conical bar under self weight

$$U=\frac{W^2L}{6AE}=\frac{\gamma^2AL^3}{90E}$$

$$\left(\text{Where, }W=\frac{\gamma AL}{3}\right)$$

■ Strain energy in a prismatic bar under moment

$$U=\frac{M^2L}{2EI_{N.A}}$$

181. The principle of super position is valid only for

(a) Any elastic materials

(b) All metals

(c) Linear elastic materials

(d) Homogeneous materials

APPSC AEE MAINS 2016

Ans. (c) : Principle of super position is valid only for linear elastic materials. This principle says that when a number of load are acting on a body, the resulting strain, according to the principle of super position, will be the algebaric sum of strains caused by individual loads.

182. The value of Poisson's ratio for which bulk modulus of a material will be equal to its Young's modulus

(a) 0.33 (b) 0.15

(c) 0.45 (d) 0.25

APPSC AEE MAINS 2016

Ans. (a) : $E = 3K(1-2\mu)$

Given as, $K = E$

$E = 3E(1-2\mu)$

$$\frac{1}{3}=(1-2\mu)$$

$$\frac{1-3}{3}=-2\mu$$

$\mu = 0.33$

183. A steel bar is heated from 20°C to 35°C and it is free to expand. Then bar will have

(a) No stress (b) Tensile stress

(c) Compressive stress (d) Shear stress

APPSC AEE MAINS 2016

Ans. (a) : If a material expands or contract freely due to heating or cooling then no stress will develop in material.

184. If a solid steel sphere of 10,000 cm^3 subjected to a uniform hydrostatic pressure of 700 × 10^4 kg(f)/sq.m, then decrease in volume of the sphere for $1/K = 6\times10^{-11}$ $[kg(f)/sq.m]^{-1}$

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(a) 2.4 cm^3 (b) 8.4 cm^3
(c) 4.2 cm^3 (d) 1.2 cm^3

APPSC AEE MAINS 2016

Ans. (c) : $V = 10000\ cm^3$

$p = 700 \times 10^4\ kg(f)/m^2$

$$\frac{1}{K} = 6\times10^{-11}\left[(kg(f)/sq.m)\right]^{-1}$$

$$K = \frac{p}{\frac{\Delta V}{V}} \Rightarrow \Delta V = \frac{pV}{K}$$

$= 700 \times 10^4 \times 10^4 \times 6 \times 10^{-11}$

$= 700 \times 6 \times 10^{-3}$

$= 4.2cm^{3ss}$

185. If temperature changes from t_0 to t in a simple bar with both end constrained and having modulus of elasticity E and thermal coefficient α, the thermal stress due to the temperature rise is

(a) $\frac{E(t-t_0)}{\alpha}$ (b) $\alpha\frac{(t-t_0)}{E}$

(c) $E\alpha(t-t_0)$ (d) $\frac{E\alpha}{(t-t_0)}$

APPSC AEE MAINS 2016

Ans. (c) : when a bar of length (l) is subjected to an increase in temperature t_0 to t, then the increase in length when the bar is free to expand is given by-

$\delta\ell = \ell\alpha\left(t - t_0\right)$

α = coeff. of linear expansion

If the ends of the bar are fixed to rigid supports so that its expansion is prevented, then compressive thermal strain induced in the bar,

$$\varepsilon = \frac{\delta\ell}{\ell}$$

$= \alpha(t - t_0)$

And thermal stress (σ_T) = εE

$= E\alpha(t - t_0)$

186. A circular solid uniform metal of cross section area A and length L is hanging vertically from its upper end. If the bar is having weight of W and modulus of rigidity E, the total elongation of the bar due to its own weight will be

(a) $\frac{2WL}{AE}$ (b) $\frac{WL}{AE}$

(c) $\frac{WL}{2AE}$ (d) $\frac{WL}{3AE}$

Kerala PSC AE 2015

APPSC AEE MAINS 2016

Ans. (c) : Deflection of bar (δl) due to self weight.

(a) For Prismatic bar–

$$\delta\ell = \frac{WL}{2AE} = \frac{\gamma L^2}{2E}$$

(b) For Conical bar

$$\delta\ell = \frac{\gamma L^2}{6E} = \frac{1}{3}\times \text{Deflection of Prismatic bar}$$

Here, γ = specific weight

L = Length of bar

E = Young's modulus

Deflection (δl) of tapered bar

$$\delta\ell = \frac{4PL}{\pi E D_1 D_2}$$

Here,

P = Force Applied

D_1, D_2 = Diameter of big and small end respectively

L=Length of bar

(c) For Deflection of rectangular tapering bar

$$\delta\ell = \frac{PL\log_e\left(\frac{B_2}{B_1}\right)}{E.t\left(B_1 - B_2\right)}$$

Here,

t = thickness

P = Force applied

E = Young's modulus

B_2 = width of big taper end

B_1 = width of small taper end

187. Volumetric strain of a sphere of diameter increasing from d_1 to d_2 is equal to

(a) $\frac{d_2 - d_1}{d_1}$ (b) $3\frac{(d_2 - d_1)}{d_1}$

(c) $3\frac{d_2 + d_1}{2d_1}$ (d) $\frac{d_2 + d_1}{2d_1}$

APPSC AEE MAINS 2016

Ans. (b) : Volumetric strain of a sphere of diameter increasing from d_1 to d_2 is equal to

$$= \frac{3\left(d_2 - d_1\right)}{d_1}$$

188. If a circular rod having Poisson's ratio μ is subjected to an axial pull and if the strain is e, then the lateral strain is equal to

(a) $-\mu e$ (b) μe

(c) $-e/\mu$ (d) e/μ

APPSC AEE MAINS 2016

Ans. (a) : We know that Poisson's ratio

$$= \frac{-\text{Lateral strain}}{\text{Longitudinal strain}}$$

$$\mu = \frac{-e_x}{e}$$

$\therefore e_x = -\mu e$

189. For linear elastic systems displacement function varies :

(a) Linearly (b) quadratically

(c) cubically (d) quartically

PSPCL AE 2012

Ans. (a) : In linear elastic system, displacement function varies linearly, as strain varies linearly

$$\varepsilon = \frac{\Delta\ell}{\ell}$$

So, $\Delta\ell \propto \varepsilon$

190. Compared to mild steel, cast iron has :

(1) high compressive strength

(2) high tensile strength

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(3) low compressive strength
(4) low tensile strength

The correct answer is :

(a) 1 and 4 (b) 1 and 2
(c) 2 and 3 (d) 3 and 4

PSPCL AE 2012

Ans. (a) : Cast iron has high compressive strength and low tensile strength as compared to mild steel.

- Carbon content in cast iron has range 2 to 4%, where as mild steel has carbon content range less than 0.25% which make mild steel more ductile and less hard.
- Melting point of cost Iron is approx 1200^0C and mild steel is 1530^0C.

191. Isotropic hardening is associated with

(a) dilatation of yield surface
(b) translation of yield surface
(c) metals under cyclic loading
(d) Bauschinger effect

GPSC AE March 2018

Ans. (a) : Isotropic hardening:-It is where the yield surface remains the same shape but expands with increasing stress.

192. A pull of 20 t is suddenly applied to a rod of cross-sectional area 40 cm^2. The stress produced in the rod is equal to

(a) 0.5 t/cm^2 (b) 1.0 t/cm^2
(c) 2.0 t/cm^2 (d) 4 t/cm^2

GPSC AE March 2018

Ans. (b) : Given as,
P = 20t
A = 40 cm^2
For gradual load

Stress $(\sigma) = \frac{P}{A} = \frac{20t}{40} = 0.5t/cm^2$

For suddenly applied load = 2 × stress for gradually applied load = 2 × 0.5 = 1.0 t/cm^2

193. Let the strains produced in length and diameter of the cylindrical rod be α and β respectively. Then the volumetric strain is given by

(a) α + 2β (b) α + β
(c) α – β (d) α – 2β

GPSC AE March 2018

Ans. (a) : Volume of cylinder

$$V = \frac{\pi}{4} d^2 l$$

$$\frac{\partial V}{V} = \frac{\partial l}{l} + \frac{2\partial d}{d}$$

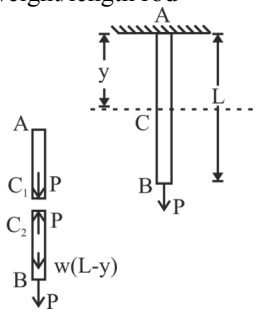
$$\in_v = \alpha + 2\beta$$

194. A rod of length L is hanging vertically and carries a load P at the bottom. If the weight per unit length of the rod be w, then the tensile force in the rod at a distance y from the support is given by

(a) P (b) P – wy
(c) P + wy (d) P + w (L – y)

GPSC AE March 2018

Ans. (d) : w = weight/length rod



So, at point C:
Total tensile force

$$= P + w(L - y)$$

195. What is tenacity ?

(a) ultimate strength in tension
(b) ultimate strength in compression
(c) ultimate shear stress
(d) ultimate impact strength

GPSC AE March 2018

Ans. (a) : The maximum stress that a material can withstand while being stretched or pulled, known as ultimate tensile strength or tenacity.

196. If the Poisson's ratio of a material is 0.25, the ratio of Modulus of Rigidity to Young's Modulus will be

(a) 2 (b) 0.4
(c) 2.5 (d) 4

GPSC AE March 2018

Ans. (b) : Young's modulus E = 2G(1 + μ)

$$\frac{G}{E} = \frac{1}{2(1+\mu)}$$

$$\frac{G}{E} = \frac{1}{2(1+0.25)}$$

$$\frac{G}{E} = \frac{1}{2(1.25)}$$

$$\frac{G}{E} = 0.4$$

Where, G = Modulus of rigidity

197. A rod of length L and uniform cross-section area A is rigidly fixed at its top and is hanging. At any section which is at a distance x from the lower end, the stress due to its self-weight is proportional

(a) x^2 (b) 1/x
(c) x (d) $1/x^2$

GPSC AE March 2018

Ans. (c) : $E = \frac{\text{stress}}{\text{strain}}$

stress = E × strain

stress due to self weight = $\frac{Edy}{l}$

stress due to self weight = $\frac{P}{A} = \frac{\rho(x)A}{A}$

stress due to self weight = x

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198. A member with a cross-section of A and length L is subjected to a force of P. If Young's Modulus is E, then linear strain will be

(a) PL/AE (b) PA/LE
(c) P/AE (d) PE/AL

GPSC AE March 2018

Ans. (c) : Young's modulus of elasticity

$$E = \frac{\text{Stress}}{\text{Strain}} = \frac{PL}{A\Delta L}$$

$$\boxed{\Delta L = \frac{PL}{AE}}$$

$$\boxed{\frac{\Delta L}{L} = e = \frac{P}{AE}}$$

199. How much (in %) fracture strain is commonly adopted as border between brittle and ductile materials?

(a) 5 (b) 4 (c) 7 (d) 3.5

Vizag steel MT 2017

Ans. (a) : Generally 5% fracture strain is adopted as border between brittle and ductile material.

200. If each dimension of a triangular bar is tripled then elongation due to self-weight will be how much times initial value?

(a) Remains same (b) 9 times
(c) 27 times (d) 1.732 times

Vizag steel MT 2017

Ans. (b) : Elongation of triangular bar due to its self weight is 'Δ'.

and $\Delta \propto L^2$, L = length of bar

Now,

$$\frac{\Delta_1}{\Delta_2} = \left(\frac{L_1}{L_2}\right)^2 = \left(\frac{L}{3L}\right)^2$$

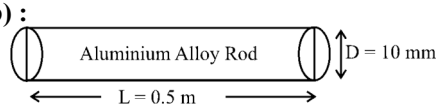
i.e., $\boxed{\Delta_2 = 9\Delta}$ i.e. 9 times.

201. A rod made of Aluminium alloy (E = 72 GPa) has length 0.5 m and diameter 10 mm. The tensile stiffness (N/m) of this rod is

(a) $18\pi \times 10^5$ (b) $36\pi \times 10^5$
(c) $9\pi \times 10^5$ (d) $12\pi \times 10^5$

CGPSC AE 2014 Shift–I

Ans. (b) :



Given,

D (diameter) = 10 mm , L (Length) = 0.5 m

E = 72 GPa = 72×10^3 MPa

$$\text{Stiffness (k) of a elastic body} = \frac{\text{Applied force (P)}}{\text{Deformation } (\delta)}$$

$$k = \frac{P}{\delta} = \frac{P}{PL/AE} = \frac{AE}{L}$$

$$k = \frac{\frac{\pi}{4}(10)^2 \times 72 \times 10^3}{0.5}$$

$$\boxed{k = 36\pi \times 10^5 \,(N/m)}$$

202. If the bulk modulus K, and modulus of rigidity, G are given then that will be the Poisson's ratio.

(a) $\frac{3K-2G}{6K+2G}$ (b) $\frac{3K+4G}{6K-4G}$
(c) $\frac{3K-4G}{6K+4G}$ (d) $\frac{3K+2G}{6K-2G}$

HPPSC Poly. Lect. 2016, GPSC Poly. Tech. Lect. 2016 ESE 2016, 1995, Kerala PSC AE 2015, JPSC AE 2009

Ans. (a) : Relation between E, G, K and μ

- $E = 3K(1 - 2\mu)$
- $E = 2G(1 + \mu)$
- $E = \frac{9KG}{3K+G}$
- $\mu = \frac{3K-2G}{6K+2G}$

203. If a metal bar fixed at both ends is cooled by reducing the temperature by 30°C, the nature of the stress developed in the bar will be

(a) Compressive (b) Shear
(c) Tensile (d) None of the above

GPSC Poly. Tech. Lect. 2016

Ans. (c) : The bar is fixed at both ends, and cooling the bar at 30°C. The bar will shrink because of change in temperature this will produce a reaction force in both ends and trying to opposed the action of shrinking so those force will produce tensile stress in the body.

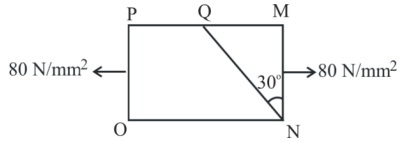
204. A mild steel bar of uniform cross-section is subjected to a uni-axial tensile stress of 80 N/mm^2. Consider a plane NQ making an angle of 30° with MN. What will be the normal stress and the tangential stress on the plane MQ?



(a) $20\sqrt{3}$ N/mm^2 and 80 N/mm^2
(b) 40 N/mm^2 and 80 N/mm^2
(c) 60 N/mm^2 and $20\sqrt{3}$ N/mm^2
(d) 40 N/mm^2 and 60 N/mm^2

UPSC JWM 2017

Ans. (c)



Normal stress –

$$\sigma_n = \frac{80}{2} + \frac{80}{2}\cos 60^0 = 40 + 40 \times \frac{1}{2}$$

$\sigma_n = 60$ N/mm^2

Tangential stress –

$$\sigma_t = \frac{80}{2}\sin 60^0 = 40 \sin 60^0 = 40 \times \frac{\sqrt{3}}{2} = 20\sqrt{3} \text{ N/mm}^2$$

205. A material has a Young's Modulus of 1.25 ×10^5 N/mm^2 and a Poisson's ratio of 0.25. The Modulus of Rigidity is-

(a) 0.65 ×10^5 N/mm^2 (b) 0.60 ×10^5 N/mm^2

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(c) 0.50×10^5 N/mm^2 (d) 0.45×10^5 N/mm^2

UPSC JWM 2017

Ans. (c) : Young's Modulus (E) $= 1.25 \times 10^5$ N/mm^2
$\mu = 0.25$
$G \rightarrow$ Modulus of rigidity
Relation between E, G and μ
$E = 2G(1+\mu)$
$1.25 \times 10^5 = 2G(1+0.25)$
$1.25 \times 10^5 = 2.5\,G$
$G = 0.50 \times 10^5$ N/mm^2

206. The ratio of Young's modulus of high tensile steel to that of mild steel is about :
(a) 0.5 (b) 1.0
(c) 1.5 (d) 2.0

Punjab MCP SDO 2015

Ans. (b) :

$$\frac{\text{Young's modulus of high tensile steel}}{\text{Young's modulus of mild steel}} = 1$$

207. For an isotropic, homogeneous and elastic material obeying Hooke's law, number of independent elastic constant is
(a) 2 (b) 3
(c) 9 (d) 1

Punjab MCP SDO 2015
UPRVUNL AE 2014
RPSC POLY. TECH. LECT. 2011
ESE 2000

Ans : (a) For an isotropic homogenous and elastic material obeying Hooke's law, only E and μ are independent elastic coefficients. other coefficients can be derived from E and μ.
For anisotropic and orthotropic material number of independent elastic constants are 21 and 9 respectively.

208. In a strained body, three principal stresses at a point are denoted by σ_1, σ_2, σ_3 such that $\sigma_1 > \sigma_2 > \sigma_3$. If σ_0 denoted yield stress, then according to the maximum shear stress theory
(a) $\sigma_1 - \sigma_2 = \sigma_0$ (b) $\sigma_1 - \sigma_3 = \sigma_0$
(c) $\sigma_2 - \sigma_3 = \sigma_0$ (d) $\sigma_1 + \sigma_2 = \sigma_0$

Punjab PSC Sub divisional Engineer 2014
ESE 1997

Ans. (b) : Maximum shear stress theory is used only in the ductile material.

$$\tau_{max} = \frac{\sigma_0}{2} \text{ (for simple tension)}$$

$$\tau_{max} = \frac{\sigma_1 - \sigma_3}{2}$$

The maximum shear stress theory predicts that failure will occur whenever.

$$\tau_{max} = \frac{\sigma_1 - \sigma_3}{2} \le \frac{\sigma_0}{2} \quad \text{or} \quad \sigma_1 - \sigma_3 \le \sigma_0$$

209. The diagram showing the variation of axial load along the span is called
(a) shear force diagram
(b) bending moment diagram
(c) thrust diagram
(d) influence line diagram

Punjab PSC Sub divisional Engineer 2014

Ans. (c) : The diagram showing the variation of axial load along the span is called thrust diagram.

210. Total energy possessed by a system of moving bodies
(a) is constant at every instant
(b) varies from point to point
(c) is maximum in the start and minimum at the end.
(d) is minimum in the start and maximum at the end

GPSC R&B AE 2016

Ans. (a) : The mechanical energy of an object can be the result of its motion (K.E.) or the result of its stored energy (potential energy) of position.
Mechanical Energy = Kinetic Energy + Potential
always constant Energy

211. If the Young's modulus E is equal to bulk modulus K. What would be the Poisson's ratio?
(a) 1/4 (b) 1/2
(c) 1/3 (d) 3/4

JPSC AE 2009, ESE 2005

Ans. (c) : Relation between E, G and μ
$E = 3K(1 - 2\mu)$
$E = 3E(1 - 2\mu)$ $(\because E = K)$
$\mu = \frac{1}{3}$

212. For linear elastic system, the top of displacement for strain energy is given by-
(a) linear (b) quadratic
(c) cubic (d) quartic

JPSC AE 2009

Ans. (b) : For a perfectly elastic material –
External work done = Internal work done.
Internal work done is the strain energy stored.

$$U = \frac{1}{2} PL \times \Delta = \frac{\sigma}{2} \in V$$

$$U = \frac{\sigma^2}{2E} V \text{ (This is quadratic equation)}$$

$$\boxed{U = \frac{P^2 L}{2AE}}$$

213. If a composite bar of steel and copper is heated, then the copper will be under-
(a) tension (b) compression
(c) shear (d) torsion

OPSC AEE 2015 PAPER–I
UTTRAKHAND AE 2013
RPSC POLY. TECH. LECT. 2011
OPSC Civil Services Pre 2011
RPSC ACF - 2011, JPSC AE 2009

Ans. (b) :

Thermal linear coefficient of expansion (α)	Substance
13.2×10^{-6}/°C	Steel
16.5×10^{-6}/°C	Copper

Composite bar of steel and copper have same expansion, But $\alpha_{copper} > \alpha_{steel}$. So, more expansion of copper will occurred which will be restricted. So copper will be under compression.

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214. Red short iron cracks when bent due to the presence of :

(a) sulphur (b) carbon
(c) Phosphorus (d) Silicon

KPTCL AE 2016

Ans. (a) : Red short iron cracks when bent due to the presence of sulphur.

215. The property of a material by which it can be drawn to a smaller section due to tension is called:

(a) Plasticity (b) Ductility
(c) Elasticity (d) Malleability

OPSC Civil Services Pre 2011

Ans. (b) Plasticity : It is the property of a material by virtue of which it undergoes permanent deformation.

Ductility : Ductility is a measure of a material ability to undergo significant plastic deformation before rupture or breaking. It is also the ability of a material by which it can be drawn into wires.

Elasticity : The ability of an object or material to resume its original shape after being stretched or compressed.

Malleability : The state of being malleable or capable of being shaped as by hammering or pressing.

216. A solid cube is subjected to equal normal forces on all its faces. The volumetric strain will be x-times the linear strain in any of the three axes when:

(a) x = 1 (b) x = 2
(c) x = 3 (d) x = 4

OPSC Civil Services Pre 2011

Ans. (c) : $\frac{dV}{V} = 3\left[\frac{\sigma}{E}\left(1-\frac{2}{m}\right)\right]$

It is 3 times because cube is subjected to 3 actually perpendicular stresses.

217. Which of the following is the correct relation between the modulus of rigidity (C), Young's modulus of elasticity (E) and bulk modulus(K)?

(a) $E = \frac{3CK}{9K+C}$ (b) $C = \frac{9EK}{3K+E}$

(c) $K = \frac{9EC}{9C+E}$ (d) $E = \frac{9CK}{3K+C}$

Maharashtra PSC (Mains) 2017 Paper–I
Rajasthan Nagar Nigam AE Shift-2, 2016
ESE 2013
APPSC AEE (Civil/Mechanical) 2012

Ans. (d) : Young's modulus (E) can be expressed using Bulk modulus (K) and Poisson's ratio (μ) as-

$E = 3K(1 - 2\mu)$ ...(1)

Similarly, Young's modulus (E) can also expressed using rigidity modulus (C) and Poisson's ratio (μ) as-

$E = 2C(1 + \mu)$ ...(2)

Combining the above two equation and solving then to eliminate Poisson's ratio we can get relationship b/w Young's modulus (E), Bulk modulus (K) and Modulus of rigidity (C) as-

$$E = \frac{9KC}{3K+C}$$

218. The rectangular bar is subjected to triaxial stress f_x, f_y, f_z. If algebraic sum of all these stresses is zero, what is the change in volume?

(a) decrease in volume
(b) No change in volume
(c) Increase in volume
(d) depends on the size proportions

Rajasthan AE Nagar Nigam Shift-3, 2016

Ans. (b) : Normal strain $e = \frac{\Delta V}{V_o} = f_x + f_y + f_z$

$\frac{\Delta V}{V_o} = 0$

So, $\Delta V = 0$

219. The stress is 'S' when a body is subjected to a gradual load W on it. If the load is applied suddenly, the stress shall be

(a) 2 S (b) S
(c) 1.5 S (d) 4 S

Sikkim PSC AE 2017

Ans. (a) : In case of sudden applied load,

Stress $(\sigma) = 2\left(\frac{W}{A}\right) = 2$ S

S ⇒ Stress due to gradual load.

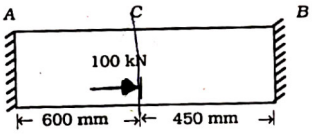
220. The stress at which extension of a material takes place more quickly as compare to the increase in load is called

(a) Elastic point (b) Yielding point
(c) Plastic point (d) Breaking point

Sikkim PSC AE 2017

Ans. (b) : The stress at which extension of a material takes place more quickly as compare to the increase in load is called Yielding point.

221. In the bar shown below the force in the part AC is

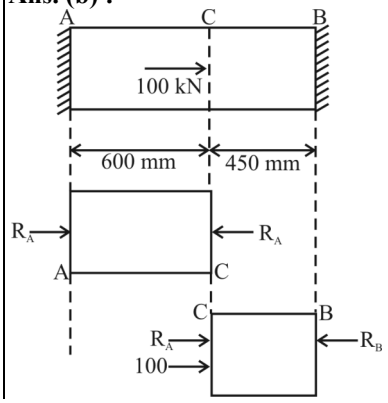


(a) $\frac{600}{7}$ kN (b) $\frac{300}{7}$ kN

(c) $\frac{450}{7}$ kN (d) 100 kN

TRB Poly. Tech. Lect. 2012

Ans. (b) :



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From Compatibility eq.

$\delta_{AC} + \delta_{BC} = 0$

$$\frac{R_A(600)}{AE} + \frac{(R_A + 100)450}{AE} = 0$$

$\Rightarrow 600\ R_A + 450\ R_A + 450{,}00 = 0$

$1050\ R_A = -450{,}00$

$$R_A = \frac{-450{,}00}{1050}$$

$$R_A = \frac{300}{7}\text{ kN (Compressive)}$$

222. A bar of cross-sectional area 2 cm^2 and length 100 cm fixed to rigid supports at both ends at a temperature of 25°C. The bar is subjected to an increase in temperature of 30°C. If the coefficient of linear thermal expansion of the material is 10×10^{-6}/°C and Young's modulus 2×10^5 N/mm^2 the stress in the bar is

(a) 10 N/mm^2 comp (b) 20 N/mm^2 comp
(c) 50 N/mm^2 comp (d) 50 N/mm^2 tension

TRB Poly. Tech. Lect. 2012

Ans. (a) : Given,

$\alpha = 10 \times 10^{-6}$ /0C

$E = 2 \times 10^5$ N/mm^2

$t_1 = 25°$, $t_2 = 30°$

$\therefore \Delta t = t_2 - t_1 = 5°C$

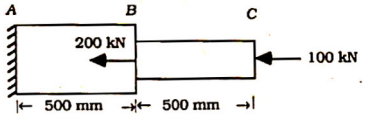
$\therefore$ Stress $= E\alpha\Delta t = 2 \times 10^5 \times 10 \times 10^{-6} \times 5$

$= 10\text{ N/mm}^2\text{ (compressive)}$

↓

As fixed at both ends

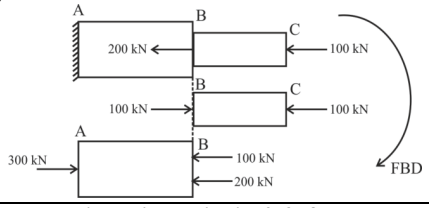
223. The portion AB is subjected to a force equal to



(a) 200 kN (b) 300 kN
(c) 100 kN (d) 400 kN

TRB Poly. Tech. Lect. 2012

Ans. (b) :



224. If the Poisson's ratio is 0.3 for a material, the ratio of Young modulus to shear modulus is

(a) 1.3 (b) 2.6
(c) 3.9 (d) 5.2

TRB Poly. Tech. Lect. 2012

Ans. (b) : $\mu = 0.3$ (given)

We know,

$$E = 2G(1 + \mu)$$

$$\frac{E}{G} = 2(1 + \mu) = 2(1 + 0.3) = 2(1.3)$$

$$\boxed{\frac{E}{G} = 2.6}$$

225. The maximum permissible load on a steel bar of 20 mm dia when the factor of safety is 1.8 and the ultimate stress is 495 MPa is

(a) 863.9 kN (b) 86.39 kN
(c) 8.639 kN (d) 0.86 kN

TRB Poly. Tech. Lect. 2012

Ans. (b) : given

d = 20 mm

F.O.S. = 1.8

Ultimate stress = 495 MPa

$$\text{The max permissible load} = \frac{495}{1.8} \times \frac{\pi}{4}(20)^2$$

$$= \boxed{86.39\text{ kN}}$$

226. If the ratio of Young's modulus to bulk modulus is 1.8, the Poisson's ratio is

(a) 0.3 (b) 0.25
(c) 0.2 (d) 0.275

TRB Poly. Tech. Lect. 2012

Ans. (c) : Given as,

$$\frac{E}{K} = 1.8$$

We know, $E = 3K(1 - 2\mu)$

$$\frac{E}{K} = 3(1 - 2\mu)$$

$1.8 = 3(1 - 2\mu)$

$\Rightarrow \boxed{\mu = 0.2}$

227. For a beam carrying a UDL, the strain energy will be maximum in case the beam is

(a) cantilever
(b) simply supported
(c) propped cantilever
(d) fixed at both ends

TRB Poly. Tech. Lect. 2012

Ans. (a) : Strain energy $(U) = \int \frac{M^2}{2EI} dx$, here M is Maximum bending moment will depend upon the value of 'M'.

Here M is maximum cantilever beam

$$M = \frac{wl^2}{2} \rightarrow \text{cantilever}$$

$$= \frac{wl^2}{8} \rightarrow \text{propped cantilever}$$

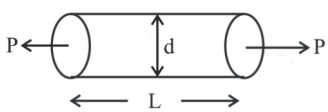
$$= \frac{wl^2}{12} \rightarrow \text{fixed beam}$$

228. Two uniform steel rods A and B of same length having diameters d and 2d are subjected to tensile forces P and 2P respectively. Then the strain energy in both rods will be

(a) equal (b) rod A has more
(c) rod B has more (d) zero in both

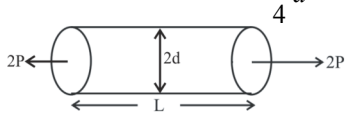
TRB Poly. Tech. Lect. 2012

Ans. (a) :



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$$U_A = \frac{1}{2}(P)\times(\Delta_A) = \frac{1}{2E}P\times\frac{(PL)}{\frac{\pi}{4}d^2}$$



$$U_B = \frac{1}{2}(2P)\times(\Delta_B) = \frac{1}{2E}2P\times\frac{(2P\times L)}{\frac{\pi}{4}(2d)^2}$$

$$= \frac{1}{2E}(P)\times\frac{PL\times 2\times 2}{\frac{\pi}{4}d^2\times 2\times 2} = \frac{1}{2E}(P)\times\frac{PL}{\frac{\pi}{4}d^2}$$

$\therefore$ $\boxed{U_A = U_B}$

Same strain energy stored in both bars.

229. The ratio of lateral strain to axial strain of a homogeneous material is known as

(a) Yield ratio (b) Hooke's ratio
(c) Poisson's ratio (d) Plastic ratio

UKPSC AE (Urban) 2011

Ans. (c) : The ratio of lateral strain to axial strain of homogeneous material is known as poison's ratio

$$\text{Poisson's ratio } (\mu) = \left|\frac{\text{Lateral strain}}{\text{Axial strain}}\right|$$

230. In a bar of length L, diameter D and cross-sectional area A, the strain produced is S under the action of load W. The Young's modulus of elasticity for the wire is

(a) W/DLS (b) WL/DS
(c) W/AS (d) SDA/WL

Punjab PSC SDE 2016

Ans. (c) : Deformation of bar–

$$\Delta = \frac{WL}{AE}$$

where, Δ = change in length, S = strain, L = length, W = load, A = area of cross-section, E = modulus of elasticity.

$$S = \frac{\Delta}{L} = \frac{W}{AE}$$

$$\boxed{E = \frac{W}{AS}}$$

231. A bar of 4 cm diameter is subjected to an axial load of 4T. The extension of the bar over a gauge length of 20 cm is 0.03. The decrease in diameter is 0.0018. The Poisson's ratio is

(a) 0.25 (b) 0.30
(c) 0.33 (d) 0.35

Punjab PSC SDE 2016

Ans. (b) :

$$\mu = \frac{\text{Lateral strain}}{\text{Longitudinal strain}}$$

$$= \frac{\text{Strain due to change in diameter}}{\text{Strain due to Change in length}}$$

$$\mu = \frac{0.0018/4}{0.03/20} = 0.3$$

232. The ability of the material to deform permanently under compression without breaking is called

(a) Malleability (b) Ductility
(c) Softness (d) Hardness

Punjab PSC SDE 2016

Ans. (a) : The ability of the material to deform under compression is called Malleability
The ability of the material to deform under tension is called ductility.

233. Which of the following materials is more elastic?

(a) Steel (b) Plastic
(c) Wood (d) Rubber

Punjab PSC SDE 2016
APPSC (PHE) AE Paper-2 2012
UJVNL AE 2012

Ans. (a) : Elasticity is the property of material by virtue of which it regains its original dimension on unloading. Therefore, steel is more elastic than rubber and wood.

234. When the strain in a material increases with time under sustained constant stress, the phenomenon is known as

(a) strain hardening (b) hysteresis
(c) creep 0 (d) visco-elasticity

Arunachal Pradesh Poly. Tech. Lect. 2019
ESE 2008

Ans. (c) When the strain in a material increases with time under sustained constant stress, the phenomenon is known as creep.
The phenomenon of creep, which is the increase in strain with time at constant applied stress and stress relaxation, which is decrease in stress at constant strain.

235. For which of the following materials, the poisson's ratio is expected to be the least?

(a) Steel (b) Cast iron
(c) Concrete (d) Copper

Arunachal Pradesh Poly. Tech. Lect. 2019

Ans. (c) : Poisson ratio $(\mu) = -\frac{\text{lateral strain}}{\text{longitudinal strain}}$

concrete – 0.1 to 0.2
cost iron – 0.21 to 0.26
steel – 0.3
copper – 0.35

236. The ratio of young's modulus to the bulk modulus is:

(a) More than one third (b) Less than one third
(c) Less than one half (d) None of the above

UJVNL AE 2012

Ans. (a) : The ratio of young's modulus to the bulk modulus is more than one third.

237. Modulus of elasticity is:

(a) Compressive stress / tensile strain
(b) Tensile stress / Compressive stress
(c) Compressive stress / Compressive strain
(d) Tensile stress / Compressive or tensile strain

UJVNL AE 2012

Ans. (c) : Modulus of elasticity $= \frac{\text{compressive stress}}{\text{compressive strain}}$

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238. The unit of stress in S.I. units is

(a) Kg/cm^2 (b) Kg/m^2
(c) N/m^2 (d) N/cm^2

UJVNL AE 2012

Ans. (c) : The force of resistance per unit area, offered by a body against deformation is known as stress.
Units of stress :

In M.K.S. $\Rightarrow \frac{Kgf}{cm^2}$

In S.I. $\Rightarrow \frac{N}{m^2}$

239. The ratio of Young's modulus to modulus of rigidity for a material having poisson's ratio 0.2 is:

(a) 12/5 (b) 5/12
(c) 5/14 (d) 14/5

UPPCL AE 2015
UJVNL AE 2012

Ans. (a) : $E = G(1+\mu)$ E = young's modulus
$E = G(1+0.2)$ G = Modulus of rigidity
$E = 2G(1.2)$ μ = Poisson ratio

$\frac{E}{G} = 2.4 = \frac{24}{10}$

$\frac{E}{G} = \frac{12}{5}$

240. With a percentage increase of carbon in steel, decreases

(a) Hardness (b) Ductility
(c) Brittleness (d) Strength

APPSC AEE 2012

Ans. (b) : With a percentage increase of carbon in steel, ductility decreases because of increment in hardness.

241. Shape factor may be defined as the ratio of.....

(a) Elastic Modulus to Plastic Modulus
(b) Plastic Modulus to Rigidity Modulus
(c) Elastic Modulus to Lateral Strain
(d) Plastic Modulus to Elastic Modulus

SJVNL ET 2019

Ans. (d) :

Shape factor = $\frac{\text{Plastic modulus}}{\text{elastic modulus}}$

242. 1 N/mm^2 is equal to :

(a) 1 Pa (b) 1 MPa
(c) 1 GPa (d) 1 KPa

SJVNL ET 2019

Ans. (b) : 1 N/mm^2 = 1 MPa = 1×10^6 N/m^2

243. What is the Poisson's ratio of Ice?

(a) 0.29 (b) 0.37
(c) 0.33 (d) 0.48

SJVNL ET 2019

Ans. (c) : Poisson's ratio of ice is 0.33

244. A member subjected to a rise in temperature develops thermal stresses

(a) when it is allowed to expand
(b) when it is not allowed to expand
(c) irrespective of above two conditions
(d) when thermal strain develops

BPSC AE Aug. 2019 Paper-V

Ans. (b) Thermal stresses tends to increase the length of body or member. If the body is not allowed to expand they develops thermal stresses, on increment of temperature. If it is allowed to expand free then there will be no thermal stress.

245. Subjected to axial load, the ductile material fails due to

(a) axial stress (b) ductility
(c) axial strain (d) shear stress

BPSC AE Aug. 2019 Paper-V

Ans. (d) : Since ductile materials are weak in shear ductile material failure occurs due to principle shear stress.

246. Poisson's ratio is defined as

(a) stress/strain
(b) lateral strain/axial strain
(c) (–) lateral strain/axial strain
(d) normal strain/shear strain

BPSC AE Aug. 2019 Paper-V
UKPSC AE 2012 Paper–I)

Ans. (c) : Poisson's ratio is defined as the ratio of change in the width per unit width, to the change in its length per unit length, as a result of strain.

$$\mu = \frac{\text{lateral strain}}{\text{axial strain}} = \frac{\Delta d/d}{\Delta l/l}$$

247. In the bulk modulus of Brass is 110 GPa and its Poisson's ratio is 0.30, then the elastic modulus (GPa) of this material is

(a) 33 (b) 367
(c) 222 (d) 132

UPRVUNL AE 2016

Ans.: (d) Bulk modulus (k) · 110 GPa
(μ) Poisson's ratio · 0.30
(E) = Elastic modulus = ?

$E = 3k(1-2\mu) = 3\times110(1-2\times0.30)$

$E = 3\times110\times0.40$

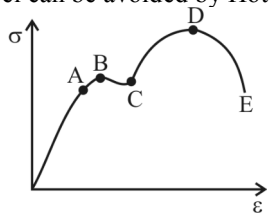
$E = 132$ GPa

248. Upper yielding point in a stress-strain curve of structural steel can be avoided by

(a) Cold working (b) Hot working
(c) Quenching (d) Galvanizing

GPSC AE June 2019

Ans. (b) : Upper yielding point in a stress-strain curve of structural steel can be avoided by Hot working.



B = Upper Yield point

249. Strain energy stored in a member is given by

(a) 0.5 × stress × volume
(b) 0.5 × strain × volume
(c) 0.5 × stress × strain × volume
(d) 0.5 × stress × strain

OPSC AEE 2019 Paper–I, GPSC AE June 2019
Punjab PSC Sub divisional Engineer 2014

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Ans. (c) : Strain energy stored in a member given by

$$\boxed{U = \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}}$$

or $\boxed{U = \frac{1}{2}\frac{\sigma^2 V}{E}}$ $\boxed{\sigma = \varepsilon E}$

250. For a given material if the Young's modulus is 200 GN/m^2 and modulus of rigidity is 80 GN/m^2, then the Poisson's ratio will be

(a) 0.40 (b) 0.50
(c) 0.25 (d) 0.30

GPSC AE June 2019

Ans. (c) : E = 200 GN/m^2
G = 80 GN/m^2
E = 2G (1 + μ)
200 = 2 × 80 (1 + μ)

$$\boxed{\mu = \frac{1}{4}} = 0.25$$

251. The material of a rubber balloon has Poisson's ratio of 0.5. If uniform pressure is applied to blow the balloon, the volumetric strain of the material will be

(a) 0.25 (b) 0.50
(c) 0.20 (d) 0.00

GPSC AE June 2019

Ans. (d) : E = 3k (1–2μ)

$$K = \frac{\sigma}{\varepsilon_v}$$

$$E = 3\left(\frac{\sigma}{\varepsilon_v}\right)(1-2\mu)$$

$$\varepsilon_v = 3 \times \frac{\sigma}{E}(1 - 2 \times 0.5)$$

$$\boxed{\varepsilon_v = 0}$$

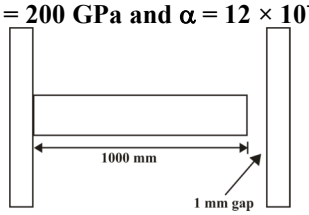
252. Crystals those are developed equally in all direction are known as

(a) Isotropant (b) Equant
(c) Prismatic (d) Deadritic

GPSC AE June 2019

Ans. (b) : A crystal or crystalline solid is a solid material whose constituents are arranged in a highly ordered microscopic structure, forming a crystal lattice that extends in all direction. Crystals these are developed equally in all directions are known as equant.

253. A 1m long rod is fixed at one end. There is a rigid wall at a distance 1 mm from the free end of the rod as depicted in the figure. What is the thermal stress generated in the rod if its temperature is increased by 100°C?
Take E = 200 GPa and α = 12 × 10^{-6}/°C



(a) 40 MPa (b) 80 MPa
(c) 120 MPa (d) 240 MPa

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : Free expansion = $l\,\alpha\,\Delta T$
= (1000) (12 × 10^{-6}) (100) = 1.2 mm
Expansion prevented = 1.2 - 1 = 0.2 mm

$$0.2 = \frac{Pl}{AE}$$

$$\left(\because \sigma = \frac{P}{A}\right)$$

$$\frac{2}{10} = \frac{(\sigma)(1000)}{200 \times 10^3}$$

σ = 40 MPa

254. Find out the Lame constants (λ and μ) for an isotropic material having modulus of elasticity (E) and Poisson's ratio (ν) as 200 GPa and 0.2, respectively.

(a) 80 GPa, 80 GPa
(b) 35.71 GPa, 166.6 GPa
(c) 55.55 GPa, 83.33 GPa
(d) 73.33 GPa, 66.66 GPa

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (c) : E = 200 GPa
Poisson's ratio ν = 0.2

$$\text{Lame constant } (\lambda) = \frac{E\,\nu}{(1+\nu)(1-2\nu)}$$

$$\lambda = \frac{200 \times 0.2}{(1+0.2)(1-2(0.2))} = 55.55\,\text{GPa}$$

$$\mu = \frac{2E\,\nu}{(1+\nu)(1-\nu)} = \frac{2 \times 200 \times 0.2}{(1+0.2)(1-0.2)} = 83.33\ \text{GPa}$$

255. Consider the state of stress at any point as σ_{xx} = 250 MPa, σ_{zz} = 250 MPa. The Young's modulus and Poisson's ratio of the material is considered as 2 GPa and 0.18 respectively. Determine the ε_{zz} at the point.

(a) -0.125 (b) 0.103
(c) -0.103 (d) 0.125

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (b) : σ_{xx} = 250 MPa = 250 × 10^6 Pa
σ_{yy} = 0
σ_{zz} = 250 MPa = 250 × 10^6 Pa
E = 2 × 10^9 Pa
μ = 0.18

$$\varepsilon_{zz} = \frac{\sigma_{zz}}{E} - \mu\frac{\sigma_{yy}}{E} - \mu\frac{\sigma_{xx}}{E}$$

$$= \frac{\sigma_{zz}}{E} - \mu\frac{\sigma_{xx}}{E} = \frac{\sigma_{xx}}{E}(1-\mu)$$

$$(\because \sigma_{xx} = \sigma_{zz})$$

$$= \frac{250 \times 10^6}{2 \times 10^9}(1 - 0.18) = 0.1025 \simeq 0.103$$

256. What is the number of non-zero strain components for a plane stress problem?

(a) 6 (b) 4
(c) 3 (d) 2

APPSC AEE (CIVIL/MECHANICAL) 2019

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Ans. (c) : In a plane stress (2D) problem the number of non-zero strain components are $\in_x$, $\in_y$ and ϕ_{xy}
Totally three numbers.

The 2D tensor is $\begin{bmatrix} \in_x & \phi_{xy} \\ \phi_{xy} & \in_y \end{bmatrix}$

257. If for a given material, E = 2G (E is modulus of elasticity, G is the shear modulus), then the bulk modulus K will be

(a) $\frac{E}{2}$ (b) $\frac{E}{3}$

(c) E (d) $\frac{E}{4}$

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (b) : E = 2G (given)
We have E = 2G (1 + μ)
2G = 2G (1 + μ)
μ = 0
and E = 3K (1 - 2μ)
E = 3K [1 - 2(0)]
E = 3K

$\Rightarrow \quad K = \frac{E}{3}$

258. In case of a torsional problem the assumption- "Plane sections perpendicular to longitudinal axis before deformation remain plane and perpendicular to the longitudinal axis after deformation" holds true for a shaft having

(a) circular cross-section
(b) elliptical cross-section
(c) circular and elliptical cross-section
(d) any cross-section

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : In case of a torsional problem the assumption-
"Plane sections perpendicular to longitudinal axis before deformation remain plane and perpendicular to the longitudinal axis after deformation" holds true for a shaft having circular cross-section only. It is not valid for other shape of cross-section.

259. A cylindrical rod with length L, cross-sectional area A and Young's modulus E is rigidly fixed at its upper end and hangs vertically. The elongation of the rod due to its self weight W is

(a) $\frac{WL}{3AE}$ (b) $\frac{WL}{2AE}$

(c) $\frac{2WL}{3AE}$ (d) $\frac{WL}{AE}$

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (b) : Self weight = W = weight density × volume

$$= \gamma \times A \times L \Rightarrow \gamma = \frac{W}{AL}$$

and self weight elongation $\delta l_{sw} = \frac{\gamma L^2}{2E}$

$$= \frac{\frac{W}{AL}.L^2}{2E} = \frac{WL}{2AE}$$

260. A metal sphere of diameter D is subjected to a uniform increase in temperature ΔT. E, ν and α are the Young's modulus, Poisson's ratio and Coefficient of thermal expansion respectively. If the ball is free to expand, the hydrostatic stress developed within the ball due to temperature change is

(a) 0 (b) $\frac{\alpha \Delta T E}{1-2\nu}$

(c) $-\frac{\alpha \Delta T E}{1-2\nu}$ (d) $\frac{\alpha \Delta T E}{3(1-2\nu)}$

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : The hydrostatic stress developed within the ball due to temperature change is zero as the ball is free to expand.

261. A mild steel flat of width 100 mm and thickness 12 mm is bent into an arc of a circle of radius 10 m by applying a pure moment M. If Young's modulus E = 200 GPa, then the magnitude of M is

(a) 72 Nm (b) 144 Nm
(c) 216 Nm (d) 288 Nm

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (d) : Radius of curvature (R) = 10,000 mm
Modulus of elasticity (E) = 200 × 10^3 MPa
Thickness (t) = 12 mm
From bending equation

$$\frac{M}{I} = \frac{E}{R}$$

$$\frac{M}{\left[\frac{100 \times 12^3}{12}\right]} = \frac{200 \times 10^3}{10,000}$$

M = 288000 N-mm
= 288 N-m

262. The rate of increase of stress is large in case of:

(a) Bonded beams (b) Unbonded beams
(c) Tensioned beams (d) Anchorage beams

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (a) : The rate of increase of stress is large in case of bonded beams.

263. A concrete beam is post-tensioned by a cable carrying an initial stress of 1000 N/mm^2, the slip of jacking end was observed to be 5 mm, modulus of steel is 210 kN/mm^2 and span of beam is 30 m ; what is % of loss of stress due to anchorage?

(a) 3.5% (b) 2.5%
(c) 1.5% (d) 4.0%

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (a) : Loss in stress = $\frac{\Delta L}{L} \times E$

$$= \frac{5 \times 10^{-3}}{30} \times 210 \times 10^3 = 35 \text{ N/mm}^2$$

% Loss = $\frac{36}{1000} \times 100 = 3.5\%$

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264. A steel bar of 5 mm is heated from 15° to 40°C and it is free to expand. The bar will induce .....

(a) No stress (b) Shear stress
(c) Tensile stress (d) Compressive stress

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (a) : If steel bar is free to expand then the bar will induce no stress.

265. Type of stress that causes dilation is known as

(a) Thermal stress (b) Shear stress
(c) Deviatoric stress (d) Hydrostatic stress

GPSC AE Dec 2018

Ans. (d) : Dilation is the change in volume. Hydrostatic stress causes dilation.

266. The shearing stress in a piece of structural steel is 100 MPa. If the elastic modulus is 200 GPa and the Poisson's ratio is 0.25, then the shearing strain γ would be

(a) 0.00125 rad (b) 0.8 rad
(c) 1.25 (d) 800

GPSC AE Dec 2018

Ans. (a) :

$$G = \frac{E}{2(1+\mu)} = \frac{200}{2\times 1.25} = 80GPa$$

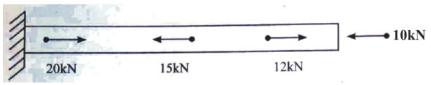
$\tau = G\gamma$

$$\gamma = \frac{\tau}{G} = \frac{100\times 10^6}{80\times 10^9} \qquad \left[\gamma = \frac{\theta.R}{\ell}\right]$$

$\gamma = 1.25 \times 10^{-3}$

$\gamma = 0.00125$ rad.

267. The largest internal force in the bar shown in the following figure is



(a) 20 kN (b) 13 kN
(c) 10 kN (d) 7 kN

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Ans. (b) : Largest internal force = 15 + 10 – 12 = 13 kN

268. Modulus of toughness is the area of the stress-strain diagram upto

(a) Rupture Point (b) Yield Point
(c) Limit of Proportionality (d) Ultimate Point

GPSC AE Dec 2018

Ans. (a) : Modulus of toughness is the area of the stress-strain diagram upto Rupture Point.

269. The Bulk modulus (K) of a material having Young's Modulus (E) = 200 GPa and Modulus of Rigidity (G) = 80 GPa is

(a) 233.3 GPa (b) 133.3 GPa
(c) 250 GPa (d) 160 GPa

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Ans. (b) :

$$E = \frac{9KG}{3K+G}$$

$$200 = \frac{9\times K\times 80}{3K+80}$$

K = 133.33 GPa

270. Two bars A and B are of equal length but B has an area half that of A and bar A has young's modulus double that of B. When a load 'P' is applied to the two bars, the ratio of deformation between A and B is

(a) $\frac{1}{2}$ (b) 1
(c) 2 (d) $\frac{1}{4}$

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Ans. (d)

Bar 'A'	Bar 'B'
$\ell_1 = \ell$	$\ell_2 = \ell$
$A_1 = A$	$A_2 = A/2$
$E_1 = 2E$	$E_2 = E$
$P_1 = P$	$P_2 = P$
$\delta\ell_1 = \frac{P_1\ell_1}{A_1E_1}$	$\delta\ell_2 = \frac{P_2\ell_2}{A_1E_2}$
$\delta\ell_1 = \frac{P\ell}{2AE}$	
$\frac{\delta\ell_1}{\delta\ell_2} = \frac{1}{4}$	$\delta\ell_2 = \frac{2P\ell}{AE}$

271. The elongation of beam of length 'l' and cross-sectional area 'A' subjected to a load 'P' is δl. If the modulus of elasticity is halved, the new elongation will be

(a) $\frac{\delta\ell}{2}$ (b) $2(\delta\ell)$
(c) $\delta\ell$ (d) $\sqrt{2}\delta\ell$

APPSC AEE (Civil/Mechanical) 2012

Ans. (b) Case 1st :-

Elongation of beam

$$\delta\ell_1 = \frac{P\ell}{AE}$$

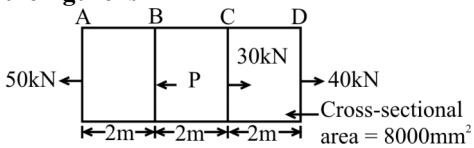
Case 2nd :-

Modulus of elasticity is halved

$$S\ell_2 = \frac{2P\ell}{AE}$$

$$\boxed{\delta\ell_2 = 2\times\delta\ell_1}$$

272. The force P for equilibrium of the bar shown in the figure is



(a) 60kN (b) 40kN
(c) 120kN (d) 20kN

APPSC AEE (Civil/Mechanical) 2012

Ans. (d) For equilibrium total horizontal force is equal to zero.

$$\sum H = 0$$

$\therefore$ $50 + P - 30 - 40 = 0$

$$\boxed{P = 20\,kN}$$

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273. The relationship between Young's modulus and shear modulus when $\frac{1}{m} = 0$, is

(a) $E = 2G$ (b) $E = 3G$
(c) $E = 2G+1$ (d) $C = 2E$

APPSC AEE (Civil/Mechanical) 2012

Ans. (a)

$$E = 2G\left(1+\frac{1}{m}\right)$$

$$E = 3K\left(1-\frac{2}{m}\right)$$

If $\frac{1}{m} = 0$ then

$E = 2G$, $E = 3K$

274. The force required to punch a 10mm diameter hole in a mild steel plate 10mm thick, if the shear strength of mild steel is 360MPa is

(a) 9πkN (b) 36πkN
(c) 18πkN (d) 2.25kN

APPSC AEE (Civil/Mechanical) 2012

Ans. (b) Hole dia = 10mm = 10×10^{-3}m
Thickness of plate = 10mm = 10×10^{-3} m
Shear strength of mild steel = 360 MPa.
Force required to punch = $\pi dt \times \tau$

$$F = \pi \times 10 \times 10^{-3} \times 10 \times 10^{-3} \times 360 \times 10^{6} \text{ N}$$

$$\boxed{F = 36\pi \text{ kN}}$$

275. The ratio between tensile stress and tensile strain or compresive stress and compressive strain is termed as

(a) modulus of rigidity
(b) modulus of elasticity
(c) bulk modulus
(d) modulus of subgrade reaction

APPSC AEE (Civil/Mechanical) 2012

Ans. (b) According to Hook's law
stress $\propto$ strain (up to proportional limit)

$\sigma \alpha e$

$$\frac{\sigma}{e} = E$$

E = Modulus of Elasticity.
where stress and strain both are tensile or compressive nature.

276. The elongation of a conical bar due to its self weight is

(a) $\frac{\gamma \ell^2}{6E}$ (b) $\frac{\gamma \ell^2}{2E}$
(c) $\frac{\gamma \ell^2}{2E}$ (d) $\frac{\gamma \ell^2}{E}$

Where γ = unit weight of the material.

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Ans. (a) Elongation of conical bar due to self weight.

$$\Delta = \frac{\gamma \ell^2}{6E} = \frac{1}{3} \times \text{Deflection of prismatic bar}$$

γ = Specific weight
l = length of bar
E = Modulus of Elasticity.

277. The strain due to a temperature change in a simple bar is

(a) αt (b) α / t
(c) t / α (d) $\alpha + 1$

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Ans. (a) Thermal stress and strain :-

$$\sigma_{th} = E \alpha T$$

$$\Delta = L \alpha T$$

$$\text{strain} = \frac{L\alpha T}{L} = \alpha T$$

Δ = Change in lenth
T = Temperatore Change
α = coefficient of thermal expansion.
σ = thermal stress

278. The ratio of total elongation of a bar of uniform cross-section produced under its own weight to the elongation produced by an external load equal to the weight of the bar is

(a) 1 (b) 2
(c) $\frac{1}{2}$ (d) $\frac{1}{4}$

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Ans. (c) Axial elongation (Δ_1) of bar due to external load $(\Delta_1) = \frac{PL}{AE}$

Axial elongation (Δ_2) of bar due to self weight

$$\Delta_2 = \frac{PL}{2AE}$$

$$\boxed{\frac{\Delta_2}{\Delta_1} = \frac{1}{2}}$$

279. The ratio between the change in volume and original volume of the body is called

(a) tensile strain
(b) compressive strain
(c) volumetric strain
(d) shear strain

APPSC AEE (Civil/Mechanical) 2012

Ans. (c) Volumetric strain $= \frac{\text{change in volume}}{\text{original volume}}$

$$\boxed{e_v = \frac{\Delta V}{V}}$$

280. A 10 cm gauge length is marked on a tensile test specimen which has a working area of 1.0 cm^2. A force of 10 kN is applied to the specimen and with that load still applied, the gauge length is found to be 10.1 cm. The modulus of the material is

(a) 10 MPa (b) 100 MPa
(c) 1 GPa (d) 10 Gpa

GPSC Assistant Prof. 2016

Ans. (d) : L = 10 cm $\Rightarrow \delta L = 0.1$ cm = 1 mm
L + δL = 10.1 cm
A = 1.0 cm^2 = 100 mm^2
P = 10 kN
E = ?

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$E = \frac{\sigma}{E}$

$\sigma = \frac{P}{A} = \frac{10 \times 10^3}{100}$

$\Rightarrow \sigma = 100\ N/mm^2$

OR $\boxed{\sigma = 100\ MPa}$

$\varepsilon = \frac{\delta L}{L} = \frac{1}{10 \times 10} = 10^{-2}$

$E = \frac{\sigma}{\varepsilon} = \frac{100}{10^{-2}} = 10^4\ MPa = 10\,GPa$

E = 10 GPa

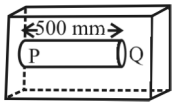
281. In a simple bending theory, one of the assumption is that the material of the beam is isotropic. This assumption means that the

(a) normal stress remains constant in all directions
(b) normal stress varies linearly in the material
(c) elastic constants are same in all the directions
(d) elastic constants varies linearly in the material

GPSC Assistant Prof. 2016

Ans. (c) In a simple bending theory, one of the assumption is that the material of the beam is isotropic. This assumption means that the elastic constants are same in all the directions.

282. In a circuit box, a copper rod at room temperature of 25°C has a gap of 0.3 mm between the end of the rod Q and the rigid wall. If temperature o the rod increases to 70°C, then what is the compressive stress acting on the rod PQ? (Assume Young's modulus = 100 GPa & $\alpha = 15 \times 10^{-6}$/°C



(a) 6 MPa (b) 7.5 MPa
(c) 8.2 MPa (d) None of the above

PTCUL AE 2016

Ans. (b) : $L = L_0(1 + \alpha\Delta T)$, $\alpha = 15 \times 10^{-6}$/°C

$L = 500(1 + 15 \times 10^{-6} \times 45)$

$L = 500.3375$ mm

Compressive stress

$= \varepsilon E = \left(\frac{0.3375 - 0.3000}{500}\right) \times 100 \times 10^9\ Pa$

= 7.5 MPa

283. Strain resisters are used to:

(a) Measure shear strain
(b) Measure volumetric strain
(c) Measure linear strain
(d) Relieve strain

PTCUL AE 2016

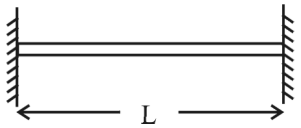
Ans. (c) : Strain resisters are used to measure linear strain.

284. A steel rod of length 'L' and diameter 'd' fixed at both ends, is uniformly heated to a temperature rise of ΔT. The modulus of elasticity of material is 'E' and Thermal expansion coefficient is 'α'. Thermal stress in rod is:

(a) zero (b) $E\alpha\Delta T$
(c) $\alpha\Delta T$ (d) $E\alpha\Delta TL$

UJVNL AE 2016

Ans. (b) :



Length = L
Diameter = d
Temperature increase = ΔT
Modulus of elasticity = E
Expansion coefficient = α
$\Delta L = L\alpha\Delta T$

$E = \frac{stress}{strain}$

$E = \frac{stress}{\frac{L\alpha\Delta T}{L}}$

$E = \frac{stress}{\alpha\Delta T}$

stress = E α ΔT (Thermal stress in rod)

285. A steel rod of c/s area 100 mm^2 and 1 m long is subjected to a tensile force of 40 kN. What is the total elongation of the rod? If modulus of elasticity of steel is 200 GPa.

(a) 0.5 mm (b) 0.7 mm
(c) 1.2 mm (d) 2.0 mm

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (d) : $A = 100\ mm^2$

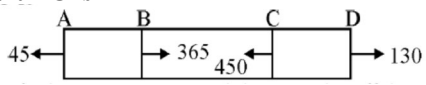
$l = 1\ m = 1000\ mm$

$E = 200\ GPa = 200 \times 10^3\ N/mm^2$

$P = 40\ kN = 40 \times 10^3\ N$

$\delta l = \frac{Pl}{AE} = \frac{40 \times 10^3 \times 1000}{100 \times 200 \times 10^3} = 2\ mm$

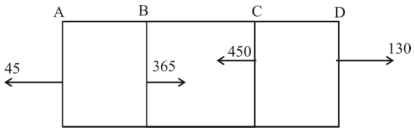
286. A member ABCD is subjected to a force system as shown in the figure. The resistive force in the part BC is



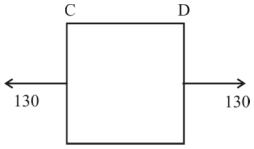
(a) 365 (compressive) (b) 450 (tensile)
(c) 85 (compressive) (d) 320 (compressive)

AP TRANSCO AEE 2017
AP TRANSCO 2017
ESE 2014

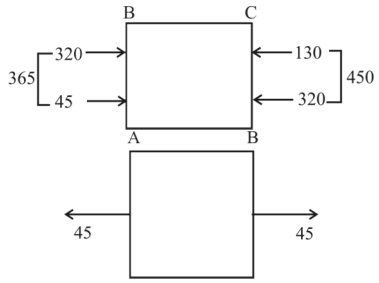
Ans. (d)



Find resistive force in BC part, break all the portion and applied load from one side maintain equilibrium condition.



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That's why resistive force in BC portion = 320 compressive

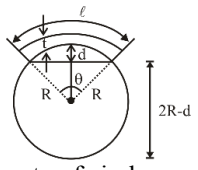
287. A long rod of uniform rectangular section with thickness *t*, originally straight, is bent into the form of a circular arch with displacement d at the mid-point of span *l*. The displacement *d* may be regarded as small as compared to the length *l*. The longitudinal surface strain is

(a) $\frac{2td}{l^2}$ (b) $\frac{4td}{l^2}$

(c) $\frac{8td}{l^2}$ (d) $\frac{16td}{l^2}$

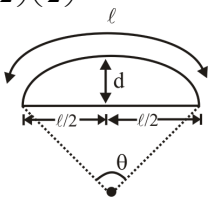
ESE 2018

Ans. (b)



$\therefore$ From the property of circle

$(2R-d) \times d = \left(\frac{\ell}{2}\right)\left(\frac{\ell}{2}\right)$



$2Rd - d^2 = \frac{\ell^2}{4}$

If d is very small, then d^2 will be neglected.

$d = \frac{\ell^2}{8R}$ or $R = \frac{\ell^2}{8d}$

Bending Equation is–

$\left[\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}\right]$

$\therefore \frac{\sigma}{y} = \frac{E}{R} \Rightarrow \frac{\sigma}{\frac{t}{2}} = \frac{E}{\frac{\ell^2}{8d}}$

$\therefore \frac{\sigma}{E} = \frac{\frac{t}{2}}{\frac{\ell^2}{8d}} \Rightarrow \frac{\sigma}{E} = \frac{4dt}{\ell^2}$

The longitudinal surface strain $\left[\frac{\sigma}{E}\right] = \frac{4dt}{\ell^2}$

288. A metal bar of 10 mm diameter when subjected to a pull of 23.5 kN gave an elongation of 0.3 mm on a gauge length of 200 mm. The Young's modulus of elasticity of the metal will nearly be

(a) 200 kN/mm^2 (b) 300 kN/mm^2

(c) 360 kN/mm^2 (d) 400 kN/mm^2

AP TRANSCO AEE 2017

APTRANSCO 2017, ESE 2014

Ans. (a) : Given,

P = 23.5 kN, Diameter = 10 mm

Elongation δl = 0.3 mm

Gauge length l = 200 mm

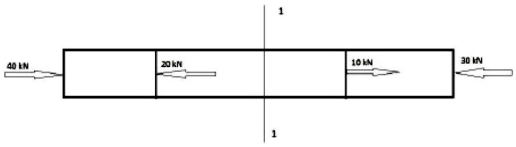
$\delta l = \frac{Pl}{AE}$

$E = \frac{Pl}{A.\delta l} = \frac{23.5 \times 200}{\frac{\pi}{4} \times (10)^2 \times 0.3}$

E = 199.47 kN/mm^2

E = 200 kN/mm^2

289. A bar of uniform cross section of 400mm^2 is loaded as shown in figure below. The stress at section 1-1 is –

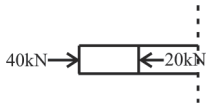


(a) 125 N/mm^2 (b) 50 N/mm^2

(c) 100 N/mm^2 (d) 200 N/mm^2

UPPCL AE 2014

Ans. (b) : Stress = $\frac{\text{Force at x-x}}{\text{Area}}$



$= \frac{20 \times 1000}{400} N/mm^2$

= 50 N/mm^2.

290. A solid cube is subjected to equal normal forces on all its faces. Volumetric strain will be how many times of the linear strain.

(a) 4 (b) 3

(c) 1 (d) 2

UPPCL AE 2014

Ans. (b) : $\varepsilon_v = \frac{(\sigma_x + \sigma_y + \sigma_z)}{E}[1 - 2\mu]$

$\varepsilon_x = \frac{\sigma_x}{E}(1 - 2\mu)$

Now,

$\sigma_x = \sigma_y = \sigma_z$ [given]

$\frac{\varepsilon_v}{\varepsilon_x} = \frac{3\sigma(1-2\mu)/E}{\sigma(1-2\mu)/E} = 3$

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291. What is the correct sequence of the following metals in the decreasing order of their poisson's ratio –
(1) Aluminum (2) Cast iron (3) Steel
(a) 2-1-3 (b) 3-1-2
(c) 1-2-3 (d) 1-3-2

UPPCL AE 2014

Ans. (d) : $H_{Al} > H_{steel} > H_{cast\ iron}$

292. The stress due to suddenly applied load is _____ times that of gradually applied load.
(a) 2 (b) 3
(c) 4 (d) 5

APPSC (PHE) AE Paper-2 2012

Ans. (a) : Let,

σ = Stress due to gradually applied load = $\frac{P}{A}$

$\sigma_{imp.}$ = Impact stress due to suddenly applied load

$$\therefore\ \sigma_{imp} = \sigma \times \left[1+\sqrt{1+\frac{2h}{\delta_{static}}}\right]$$

where ;

δ_{static} = static deflection

for suddenly applied load; h = 0

$$\sigma_{imp} = \sigma \times \left[1+\sqrt{1+0}\right] = \sigma \times 2$$

$$\sigma_{imp} = 2\sigma$$

Therefore, the stress due to suddenly applied load is 2 times that of gradually applied load.

293. The ratio of the deformation of a bar due to its own weight, to the deformation due to axial load equal to its weight, is
(a) 1 (b) 1/2
(c) 2 (d) 4

APPSC (PHE) AE Paper-2 2012

Ans. (b) : We know that -

Deformation of a bar due to its own weight

$$(\Delta_1) = \frac{WL}{2AE}$$

Deformation due to axial load equal to its weight

$$(\Delta_2) = \frac{WL/4AE}{WL/AE} = \frac{1}{2}$$

294. Young's modulus of elasticity and Poisson's ratio of a material are 1.25×10^5 MPa and 0.34 respectively. The modulus of rigidity of the material is
(a) 0.4025×10^5MPa (b) 0.4664×10^5MPa
(c) 0.8375×10^5MPa (d) 0.9469×10^5MPa

APPSC (PHE) AE Paper-2 2012

Ans. (b) :

Given : Young's Modulus of elasticity (E)

$= 1.25 \times 10^5$ MPa

Poisson's ratio (μ) = 0.34

Let, Modulus of rigidity = G

$E = 2G(1+\mu)$

$$\Rightarrow G = \frac{E}{2\times(1+\mu)} = \frac{1.25\times10^5\ MPa}{2\times(1+0.34)} = \frac{1.25\times10^5}{2\times1.34}$$

$G = 0.4664 \times 10^5$ MPa

295. Which one of the following features improves the fatigue strength of a metallic material?
(a) Increasing the temperature
(b) Searching the surface
(c) Over stressing
(d) Under stressing

APPSC (PHE) AE Paper-2 2012

Ans. (d) : When we applying the stress at the surface of metallic material, then the fatigue strength of material improves.

296. A tensile force of P is applied on a compound bar having two members X and Y. Then the load shared by the member X is
(a) $\frac{PA_XE_X}{A_XE_X + A_YE_Y}$ (b) $\frac{PA_YE_Y}{A_XE_X + A_YE_Y}$
(c) P/2 (d) $\frac{PE_X}{E_Y}$

APPSC AEE 2016
APPSC AEE MAINS 2016

Ans. (a) : Tensile force P is applied on a compound bar having two members X and Y then

load shared by $X = \frac{PAxEx}{AxEx + AyEy}$

load shared by $Y = \frac{PAyEy}{AyEy + AyEy}$

297. Relation between the elastic constants E and C for poisson's ratio of 0.25 is
(a) E=10C (b) E=5C
(c) E=2.5C (d) E=0.25C

APPSC AEE 2016

Ans. (c) : Relation between E and C with μ is

$E = 2C(1+\mu)$

$E = 2C(1+.25)$

$E = 2.5C$

298. A steel bar of 500mm length is under tensile stress of 100 N/mm^2. If the modulus of elasticity is 2×10^{11} N/m^2, then the total elongation of the bar will be
(a) 0.25 mm (b) 2.50 mm
(c) 5.00 mm (d) 0.50 mm

APPSC AEE 2016

Ans. (a) : Strain = $\frac{\text{stress}}{\text{modulus of elasticity}}$

$$\frac{\Delta l}{l} = \frac{\sigma}{E}$$

$$\Delta l = \frac{\sigma}{E}\times l = \frac{100}{2\times10^{11}\times10^{-6}}\times500$$

Total elongation (Δl) = .25mm

299. A square steel rod 20 mm × 20 mm in section is to carry an axial load of 100 kN. Calculate the stress.
(a) 2,50,000 kN/m^2 (b) 1,50,000 kN/m^2
(c) 3,50,000 kN/m^2 (d) 4,50,000 kN/m^2
(e) 6,50,000 kN/m^2

CGPSC AE 2017

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Ans. (a)

$$\text{Stress} = \frac{\text{Load}}{\text{Area}} = \frac{100}{20\times 20\times (100)^{-2}}$$

$$= \frac{100}{(20\times 20)\times (1000)^{-2}}$$

Stress = 250000 kN/m^2

300. The stress in a steel bar is given as 25,0000 kN/m^2 and length of the bar is 50 m. Calculate the shortening of the bar if E = 2.14×10^8 N/mm^2

(a) 0.160 mm (b) 0.0584 mm
(c) 0.250 mm (d) 0.305 mm
(e) 1.2.2 mm

CGPSC AE 2017

Ans. (b) Stress (σ) = 250000 kN/m^2 = 250 N/mm^2

length of bar (l) = 50 m

E = 2.14 × 10^8 N/mm^2

$$\delta l = \frac{\sigma}{E} l \qquad \left[\delta l = \frac{Pl}{AE} = \frac{\sigma l}{E}\right]$$

$$= \frac{250}{2.14}\times\frac{50000}{10^8} = \frac{125}{2140} = 0.0584\,\text{mm}$$

301. A Steel bar having cross sectional area of 100π mm^2 is subjected to a tensile force of 40 kN. Determine the tensile stress.

(a) 127.23 MPa (b) 150 MPa
(c) 175.25 MPa (d) 200.05 MPa
(e) 160 MPa

CGPSC AE 2017

Ans. (a)

Cross sectional area = 100 π mm^2

Tensile force (P) = 40 kN

$$\text{Tensile stress} = \frac{P}{A} = \frac{40\times 1000}{100\pi} = 127.32\ \text{MPa}$$

302. The principle of virtual work can be applied to elastic system by considering the virtual work of:

(a) Internal forces only
(b) External forces only
(c) Internal as well as external forces
(d) All of the above options
(e) None of the above options

Haryana PSC AE 2018 Paper–I

Ans. (c) : Internal as well as external forces.

303. The Poisson's ratio for cast-iron varies from:

(a) 0.25 to 0.33 (b) 0.31 to 0.34
(c) 0.23 to 0.27 (d) 0.32 to 0.42
(e) 0.42 to 0.45

Haryana PSC AE 2018 Paper–I

Ans. (c) :

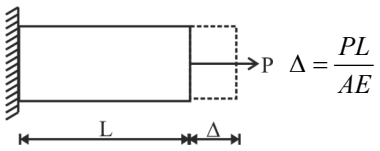
Poisson's Ratio	Material
0.02 - 0.03	Glass
0.1 - 0.2	Concrete
0.2 - 0.27	Cast Iron
0.27 - 0.30	Steel
0.3	Wrought Iron
0.33	Aluminium
0.33 - 0.36	Copper
0.34	Brass and Bronze
0.40	Nylon
0.45 - 0.50	Rubber

304. When a mild steel bar of length *l*, width b and thickness t is subjected to a pull of P, its:

(a) Length, width and thickness increases
(b) Length, width and thickness decreases
(c) Length decreases, width and thickness increases
(d) Length increases, width and thickness decreases
(e) None of these options

Haryana PSC AE 2018 Paper–I

Ans. (d) :



$\Delta = \frac{PL}{AE}$

$\rightarrow \Delta \alpha L \qquad \therefore L = Length$

$\rightarrow \Delta \alpha \frac{1}{A} \qquad b = Width$

$\rightarrow \Delta \alpha \frac{1}{b\times t} \qquad t = thickness$

305. The ratio of elongations of a conical bar due to its self-weight and that of a prismatic bar of the same length is:

(a) 1/2 (b) 1/3
(c) 1/4 (d) 1/5
(e) 1/6

Haryana PSC AE 2018 Paper–I
UPPCL AE 2014

Ans. (b) : Elongations of a conical bar due to its self-weight (Δ) $= \frac{\gamma L^2}{6E}$

Elongations of a Prismatic bar due to its self-weight (Δ) $= \frac{\gamma L^2}{2E}$

$$\text{Ratio} = \frac{\frac{\gamma L^2}{6E}}{\frac{\gamma L^2}{2E}} = \frac{2}{6} = \frac{1}{3}$$

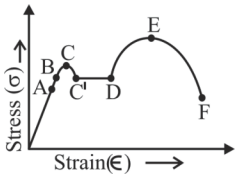
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306. The stress at which extension of a material takes place more quickly as compared to the increase in load is called:

(a) Elastic point (b) Plastic point
(c) Breaking point (d) Yielding point
(e) None of these options

Haryana PSC AE 2018 Paper–I

Ans. (d) :



307. A round steel bar is of length 40 cm consists of two equal portions of 20 cm each having diameters of 10 cm and 8 cm respectively. Take E as 2×10^6 kg/cm^2. If the rod is subjected to a tensile load of 10 tonnes, the elongation in cm will be given by:

(a) $\frac{1}{10\pi}\left(\frac{1}{25}+\frac{1}{16}\right)$ (b) $\frac{2}{10\pi}\left(\frac{1}{25}+\frac{1}{16}\right)$

(c) $\frac{3}{10\pi}\left(\frac{1}{25}+\frac{1}{16}\right)$ (d) $\frac{4}{10\pi}\left(\frac{1}{25}+\frac{1}{16}\right)$

Haryana PSC AE 2017
ESE 1997

Ans. (a) :

$$\Delta\ell = \frac{PL}{AE} = \frac{P_1L_1}{A_1E_1} + \frac{P_2L_2}{A_2E_2}$$

$P_1 = P_2 = 10\times10^3\,kg = P = 10^4\,kg$

$L_1 = L_2 = 20$ cm $= L$

$E_1 = E_2 = 2 \times 10^6$ kg/cm$^2 = E$

$$\therefore \quad \Delta\ell = \frac{PL}{E}\left[\frac{1}{A_1}+\frac{1}{A_2}\right]$$

$$= \frac{10^4\times20}{2\times10^6}\left[\frac{1}{\frac{\pi}{4}\times10^2}+\frac{1}{\frac{\pi}{4}\times8^2}\right]$$

$$= \frac{4\times10^4\times20}{\pi\times2\times10^6}\left[\frac{1}{100}+\frac{1}{64}\right] = \frac{1}{10\pi}\left(\frac{1}{25}+\frac{1}{16}\right)$$

hence a is right answer.

308. A bar of diameter 30 mm is subjected to a tensile load such that the measured extension on a gauge length of 200 mm is 0.09 mm and the change in diameter is 0.0045 mm. The Poisson's ratio will be:

(a) 1/4 (b) 1/3
(c) 1/5 (d) 1/6

Haryana PSC AE 2017, ESE 1995

Ans. (b) : Poisson Ratio : $\frac{\text{Lateral strain}}{\text{Longitudinal strain}}$

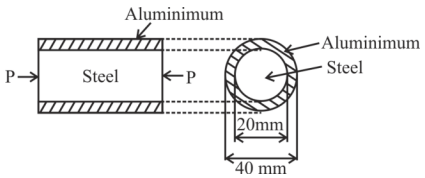
$$= \frac{-\left(\frac{0.0045}{30}\right)}{\left(\frac{0.09}{200}\right)} = \frac{1}{3}$$

309. A tube of aluminium of 40 mm external diameter and 20 mm internal diameter is snugly fitted on a solid steel rod of 20 mm diameter. The composite bar is subjected to an axial compressive force P. If the stress on steel bar is 70 N/mm^2, the stress in the aluminium tube and corresponding value of P will be: (E for steel: 2×10^5 N/mm^2 and E for aluminium 7×10^4 N/mm^2)

(a) 24.5 N/mm^2, 45.08 kN
(b) 36.5 N/mm^2, 60.10 kN
(c) 54.5 N/mm^2, 73.10 kN
(d) 73.80 N/mm^2, 92.60 kN

ISRO Scientist/Engineer 2020

Ans. (a) :



Given– Dia of steel bar = 20 mm
External dia of aluminium = 40 mm
Internal dia of aluminium = 20 mm
Stress on steel = 70 N/mm^2
P_{Steel} – Force in steel
P_{Al} – Force in aluminium
Area of aluminium bar – 942.477 mm^2
Area of steel bar – 314.159 mm^2
Now– $P = P_{Steel} + P_{Al}$ .....(i)
We know $\delta_{Steel} = \delta_{Aluminium}$

$$\left(\frac{PL}{AE}\right)_{Steel} = \left(\frac{PL}{AE}\right)_{AL}$$

$$\frac{\sigma_{steel}}{E_{steel}} = \frac{\sigma_{Al}}{E_{Al}}$$

$$\sigma_{Al} = \frac{70\times7\times10^4}{2\times10^5} = 24.5\,N/mm^2$$

$$P_{Al} = \sigma_{Al}\times A_{Al} = 24.5\times942.477 = 23.09\,kN$$

$$P_{Steel} = \sigma_{Steel}\times A_{Steel} = 70\times314.159 = 21.991\,kN$$

Hence $P = P_{Steel} + P_{Al} \Rightarrow 45.08$ kN

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310. The approximate value of the thermal coefficient of expansion of concrete is :

(a) 9×10^{-6} per ^0C (b) 10×10^{-6} per ^0C

(c) 9×10^{-7} per ^0F (d) 9×10^{-6} per ^0F

UP Jal Nigam AE 2016

Ans. (b) The approximate value of the thermal coefficient of expansion of concrete is 10×10^{-6} per ^0C.

311. Total Number of elastic constant and independent elastic constant for material are :

(a) 5, 3 (b) 4, 3

(c) 4, 4 (d) 4, 2

UP Jal Nigam AE 2016

Ans. (d) : Total number of elastic constant = 4

Total Independence elastic constant for material = 2

Material	**No. of independent elastic constant**
(1) isotropic	2 (E and μ)
(2) Orthotropic	9
(3) Anisotropic or non-isotropic	21

312. What would be the value of Bulk modulus for a material which has Young's modulus = 1.2×10^5 N/mm^2 and modulus of Rigidity = 4.8×10^4 N/mm^2 ?

(a) 4×10^4 N/mm^2 (b) 8×10^4 N/mm^2

(c) 2×10^4 N/mm^2 (d) 6×10^4 N/mm^2

UPRVUNL AE 2014

Ans. (b) : Bulk Modulus K = ?

Young's modulus (E) = 1.2×10^5 N/mm^2

(G) Modulus of rigidity = 4.8×10^5 N/mm^2

$$E = \frac{9KG}{3K+G}$$

$$1.2\times10^5 = \frac{9\times K\times 4.8\times10^4}{3K+4.8\times10^4}$$

$(3K + 4.8 \times 10^4)\ 1.2 \times 10^5 = 4.8 \times 10^4 \times 9 \times K$

$3.6 \times 10^5\ K + 5.76 \times 10^9 = 43.2 \times 10^4\ K$

$43.2 \times 10^4\ K - 36 \times 10^4\ K = 5.76 \times 10^9$

$7.2 \times 10^4\ K = 5.76 \times 10^9$

$$K = \frac{57.6\times10^4}{7.2}$$

$K = 8\times10^4\ N/mm^2$

313. Which of the following statement are true?

(i) Two-dimensional stresses applied to a thin plate in its own plane represent the plane stress condition.

(ii) Under plane stress condition , the strain in the direction perpendicular to the plane is zero.

(iii) Normal and shear stresses may occur simultaneously on a plane.

Choose the correct answer from the options given below.

(a) Only i (b) i and ii

(c) ii and iii (d) i and iii

UPRVUNL AE 2014

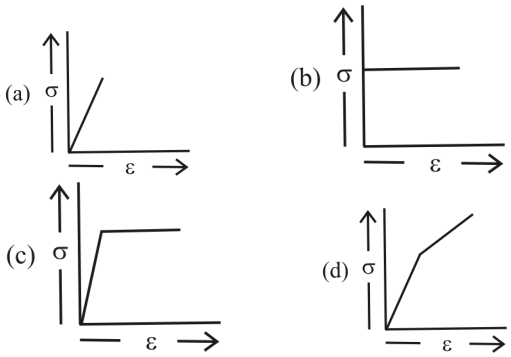
Ans. (d) : (i) Two-dimensional stresses applied to a thin plate in its own plane represent the plane stress condition.

(ii) Under plane stress condition, the strain in the direction perpendicular to the plane is lateral strain

(iii) Normal and shear stress as may occur simultaneously on a plane.

314. Which of the following is the correct representation of an idealized stress-strain curve for a perfectly plastic material?

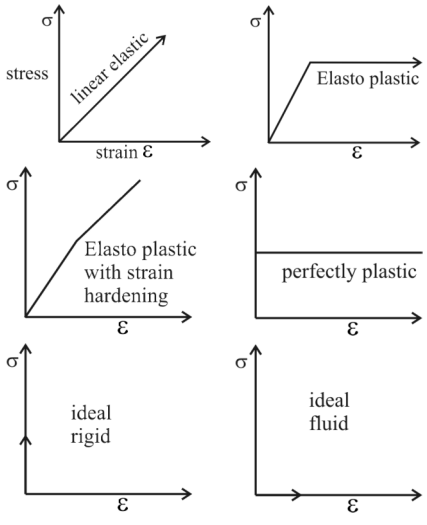
(where σ = stress, ε = strain)



UPRVUNL AE 2014

ESE 1995

Ans. (b) : Behaviour of various material



σ = stress, ε = strain

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315. The value of Poisson's ratio to be zero illustrates that :

(a) The material is rigid.

(b) The material is perfectly plastic.

(c) There is no longitudinal strain in the material.

(d) There is no longitudinal strain in the material is infinite.

UPRVUNL AE 2014

Ans. (a) : Poisson's ratio (μ) = 0; for rigid material

$$\text{Poisson's ratio} = \frac{\text{Latera strain}}{\text{Linear strain}}$$

under uniaxial loading

$\mu = 0$ for cork

$\mu = 0.5$ for rubber

$\mu = 0.25$ to 0.42 for metals

$\mu = 0.1$ to 0.2 for concrete

$0 \le \mu \le 0.5$

μ is greater for ductile metals than for brittle metals.

316. A rod 20mm × 20 mm cross-section is carrying an axial tensile load 40 kN.what is the value of tensile stress devloped in this rod?

(a) 10 MPa (b) 100MPa

(c) 1000MPa (d) 1MPa

UPRVUNL AE 2014

Ans. (b) : Given,

(P) axial tensile load = 40 kN

$P = 40 \times 10^3$ N

Area = 20 mm × 20 mm

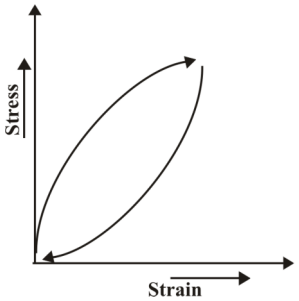
$$\text{tensile stress } (\sigma) = \frac{P}{A}$$

$$= \frac{40000}{20 \times 20}$$

$= 100$ N/mm^2

$= 100$ MPa

317. The following diagram is a stress-strain diagram of any material. Which kind of material is it?

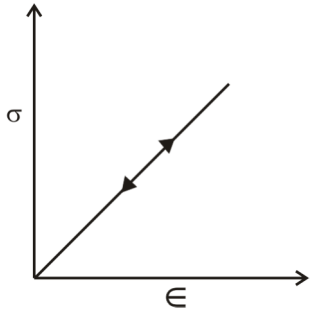


(a) Plastic (b) Linear Elastic

(c) Non-linear Elastic (d) Visco-elastic

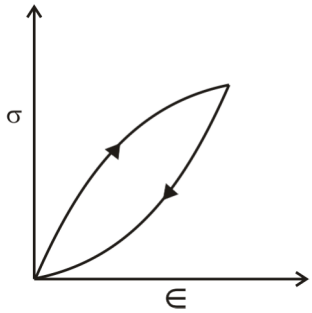
APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (d) : Purely elastic material



Loading and unloading to energy loss.

Viscoelastic material



Loading and unloading the area inside the curve, hysteresis loop is energy loss.

318. A metal sphere of diameter D is subjected to a uniform increase in temperature ΔT. E, ν and α are the Young's modulus, Poisson's ratio and Coefficient of thermal expansion respectively. If the ball is free to expand, the hydrostatic stress developed within the ball due to temperature change is

(a) 0 (b) $\dfrac{\alpha \Delta T E}{1-2\nu}$

(c) $-\dfrac{\alpha \Delta T E}{1-2\nu}$ (d) $\dfrac{\alpha \Delta T E}{3(1-2\nu)}$

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : The hydrostatic stress developed within the ball due to temperature change is zero as the ball is free to expand.

319. Shear strain γ_{yz} is given by :

(a) $\dfrac{\partial u}{\partial x} + \dfrac{\partial v}{\partial y}$ (b) $\dfrac{\partial w}{\partial y} + \dfrac{\partial v}{\partial z}$

(c) $\dfrac{\partial u}{\partial z} + \dfrac{\partial w}{\partial x}$ (d) $\dfrac{\partial v}{\partial x} + \dfrac{\partial u}{\partial y}$

where u, v and w are the displacement vectors along x, y and z axis respectively.

HPPSC Poly. Lect. 2016

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Ans : (b) Shear strain $\gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}$

where u, v, w are the displacement vectors along x, y and z axis respectively

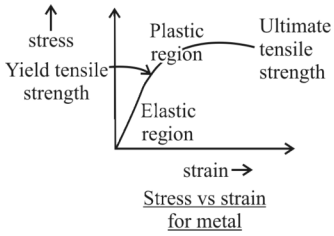
320. The region in the stress strain curve between elastic limit and point of rupture is:

(a) Elastic limit (b) Plastic limit

(c) Elastic range (d) Yield strength

IRCON AE 2017

Ans. (b) : When the stress is sufficient to permanently deform that metal, it is called plastic deformation, that will occur even if the load is removed.



Stress vs strain for metal

321. A rod of length 1 m, area 300 sq.mm and self weight 6 kN is suspended vertically and is firmly fixed at the top. Modulus of elasticity is 200 GPa. What is the total elongation of the bar due to its self weight?

(a) 1 mm (b) 0.1 mm

(c) 0.05 mm (d) 0.5 mm

TSPSC AEE 2017

Ans. (c) : Total elongation of the bar due to its self weight

$$= \frac{wL}{2AE} = \frac{6 \times 1000 \times 1000}{2 \times 300 \times 200 \times 10^3} = 0.05 \text{ mm}$$

322. What will be the strain induced in a steel rod of length 200mm that is subjected to a tensile load of 100kN when the extension of the rod is equal to 0.4 mm?

(a) 0.02 (b) 0.0002

(c) 0.002 (d) 0.2

LMRC AM 2019

Ans. (c) Given,

L = 200 mm, ΔL = 0.4 mm

P = 100 kN

$$\text{Strain } (\varepsilon) = \frac{\Delta L}{L} = \frac{0.4}{200} = 0.002$$

323. A load of 100 kN suddenly acts on a bar of 100 cm^2 area of cross section and of length 300 mm. Maximum stress developed in the bar is

(a) 10 MPa (b) 20 MPa

(c) 30 MPa (d) 40 MPa

APTRASCO 2017

Ans : (b) A load of 100 kN suddenly acts on a bar of 100 cm^2 area of cross section and of length 300 mm.

$$\sigma = 2 \times \frac{P}{A} = 2 \times \frac{100 \times 10^3}{100 \times 10^2} = 20 \text{ N/mm}^2$$

$$\sigma = 20 \text{ MPa}$$

324. If all the dimensions of a prismatic bar are increases in the proportion n:1, the proportion with which the maximum stress produced in the bar by its own weight will change by

(a) $1:n^2$ (b) 1:n

(c) $\sqrt{n}:1$ (d) n:1

MPPSC AE 2014

Ans. (d) : Let prismatic Beam a Rectangular section (we can assume any shape)

$$\therefore \text{Stress} = \frac{\text{Force}}{\text{c/s area}} = \frac{\text{Self w.t}}{\text{c/s area}} = \frac{\text{density x volume}}{\text{c/s area}}$$

$$= \frac{\text{density} \times (\text{c/s Area}) \times \text{length}}{(\text{c/s Area})}$$

$$\boxed{\sigma_{max} \propto L}$$

Hence, if dimensions increase by (n:1), then stress too increase by (n:1).

325. A metal bar of length 200 m is inserted between 2 rigid supports and its temperature is increased by 30^0C. If the coefficient of thermal expansion is 15×10^{-6} per ^{0}C and the Young's modulus is 2×10^5 MPa, then the stress in the bar is :

(a) 90,000 MPa (b) 60 MPa

(c) 90 MPa (d) 6000 MPa

DMRC AM 2020

Ans. (c) : Length ℓ = 200 mm, $\alpha = 15 \times 10^{-6}$ per ^{0}C

$E = 2 \times 10^5$ MPa, $\Delta T = 30^0$C

Stress, $\sigma = E\alpha\Delta T = 2 \times 10^5 \times 15 \times 10^{-6} \times 30^0 \text{C}$

$= 90 \text{ N/mm}^2 = 90 \text{ MPa}$

326. The coefficient of thermal expansion (in degree / degree C) of structural steel, irrespective of its grade can be assumed as :

(a) 10×10^{-6} (b) 10.5×10^{-6}

(c) 10.75×10^{-6} (d) 13.25×10^{-6}

(e) 12×10^{-6}

CGPSC AE 2014 Shift–I

Ans. (e) : Coefficient of thermal expansion (α) = 12×10^{-6} ^{0}C

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2. Bending Moment and Shear Force

327. The bending moment is maximum on a section where shearing force:
(a) Is maximum (b) Is minimum
(c) Is equal (d) Changes sign
(e) None of these options

Haryana PSC AE 2018 Paper–I

Ans. (d) : (a) If shear force changes sign at a point then bending moment at that point will be maximum.
(b) If bending moment changes sign, then curvature also changes sign at that point, then such a point is called a point of contraflexure or point of inflection.

328. If M is the external moment which rotates the near end of a prismatic beam without translation, the far end being fixed, then the moment induced at the far end is
(a) Zero
(b) M/2 in the same direction as M
(c) M/2 in the opposite direction as M
(d) None of these

Punjab PSC SDE 2016
Haryana PSC AE 2010

Ans. (b) : The moment induced at far end is half the moment which rotates the near end of a prismatic beam without translation.

329. In a continuous bending moment diagram, the point where it changes sign is called:
(a) Point of inflection
(b) Point of contraflexure
(c) Point of virtual hinge
(d) All of the above options
(e) None of these options

Haryana PSC AE 2018 Paper–I

Ans. (d) If shear force changes sign at a point, then B.M. at that point will be maximum.
If Bending moment changes sign, then curvature also changes sign at that point then such a points is called a point of contraflexure or point of inflection or point of virtual hinge.

330. For a beam, Shear span is defined as the zone where:
(a) Shear force is constant
(b) Shear force is zero
(c) Bending moment is constant
(d) Bending moment is zero

CIL MT 2020
Rajasthan AE Nagar Nigam Shift-3, 2016
(UKPSC AE (Paper–I) 2007
ESE 2010

Ans. (a) : For a beam, shear span is defined as the zone where shear force is constant.

331. For a beam subjected to point loads, the shear force between point loads
(a) is constant (b) varies linearly
(c) has parabolic variation (d) always remain zero

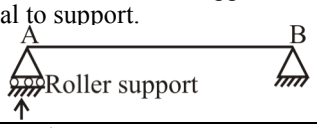
APPSC AEE 2016

Ans. (a) : When a beam is subjected to points load shear force between point loads is constant.
• Any beam segment where no loads are applied the shear force remains constant (Horizontal line).

332. The reaction at a roller support of beam will be
(a) Tangential to support (b) Normal to support
(c) Unknown in direction (d) Inclined to support

APPSC AEE 2016

Ans. (b) : Reaction at Roller support of beam will be always Normal to support.



333. At a hinge in a beam
(a) Shear force is zero
(b) Bending moment is maximum
(c) Bending moment is zero
(d) Bending moment changes sign

APPSC AEE 2016

Ans. (c) : At a hinge in a beam bending moment is zero.

334. The bending moment for a beam with equal overhangs and carrying equal point loads each of P at the free ends, the shear force for the beam portion between the supports is
(a) Zero
(b) equal to P
(c) Linearly varying from –P to +P
(d) Linearly varying from 0 to P

APPSC AEE 2016

Ans. (a) : • If load applied at overhang portion of beam than shear force is zero between supports of beam portion.
• If loading is applied at beam and a beam segment between loading is present then at that segment shear force is constant.

335. The point of contraflexure is also called as
(a) the point of inflexion (b) a virtual hinge
(c) Either of the above (d) None of the above

APPSC (PHE) AE Paper-2 2012

Ans. (c): The point of contraflexure is also called as the point of inflection or virtual hinge. Because at the point of contraflexure bending moment changes sign and bending moment is zero.

336. The flexural rigidity of a beam is
(a) EI (b) E/I
(c) I/E (d) E/I^2

APPSC (PHE) AE Paper-2 2012

Ans. (a) : The flexural rigidity of a beam is the product of the elastic constant (E) and Moment of inertia of the beam. i.e, flexural rigidity of a beam = EI

337. In a fixed beam, at the fixed ends
(a) slope is zero and deflection is maximum
(b) slope is maximum and deflection is is zero
(c) both slope and deflection are maximum
(d) slope and deflection are zero

APPSC AEE (Civil/Mechanical) 2012

Ans. (d) In a fixed beam, at the fixed ends slope and deflection are zero.

338. If a fixed beam is subject to a point load at mid span, total number of points of contra-flexure are
(a) 1 (b) 2
(c) 3 (d) zero

APPSC AEE (Civil/Mechanical) 2012

Ans. (b) In a fixed beam is subjected to a point load at mid span total number of points of contra-flexure are two.

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339. When there is a sudden increase or decrease in shear force diagram between any location, it indicates that the location has
(a) a point load
(b) an acting moment
(c) no load
(d) maximum bending moment

GPSC Assistant Prof. 2016

Ans. (a) : When there is a sudden increase or decrease in shear force diagram between any location, it indicates that the location has a point load.

340. Rate of change of shear force is equal to
(a) Bending moment
(b) Intensity of loading
(c) Maximum deflection
(d) Slope

APPSC AEE (Civil/Mechanical) 2012

Ans. (b) (i) Rate of change of Shear force is equal to intensity of loading

$$\frac{ds}{dx} = W$$

W = load per unit length.

(ii) Rate of change of bending moment along the length of beam is equal to shear force.

$$\frac{dM}{dx} = S_x$$

341. At the point of contraflexure
(a) bending moment changes sign
(b) bending moment is maximum
(c) shear force is maximum
(d) None of the above

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : The point at which bending moment changes its sign (from +ve or -ve and vice-versa) is known as point of contra flexure.

342. Consider the following:
1. Bending moment is a moment about the longitudinal axis of a beam.
2. A structural component cannot have axial force and shear force together.
(a) Only 1 is correct
(b) Only 2 is correct
(c) Both 1 and 2 are correct
(d) Both 1 and 2 are incorrect

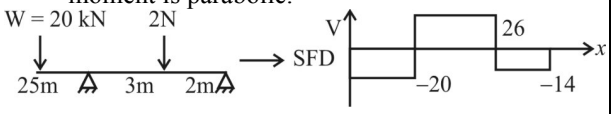
APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (d) : • Bending moment is the moment about neutral axis, but not about longitudinal axis.
• Structural components can have axial forces, shear forces together when inclined loads are acting, on a beam.

343. Between the point loads the variation of shear is
(a) Linear (b) Constant
(c) Parabolic (d) Cubic

GPSC AE June 2019

Ans. (b) :
(i) Between the points load the variation of shear force is constant.
(ii) Between the point load the variation of bending moment is linear.
(iii) Between the udl load the variation of shear force is linear.
(iv) Between the udl load the variation of bending moment is parabolic.



344. If the shape of a cable is a parabola, it implies that it is subjected to
(a) Point loads
(b) Uniformly distributed load
(c) Uniformly varying load
(d) Uniformly varying load with point loads

GPSC AE June 2019

Ans. (b) : If the shape of a cable is a parabola, it implies that it is subjected to a uniformly distributed load.

345. In case of pure bending, the beam will bend into an arc of a/an
(a) parabola (b) hyperbola
(c) circle (d) ellipse

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (c) : If a beam is subjected to pure bending the elastic curve is a circular arc with constant radius.

346. Which type of support prevents the movement of the support in each of the three directions?
(a) Roller support
(b) Ball support
(c) Fixed support
(d) Ball and socket support

SJVNL ET 2019

Ans. (d) : Ball and socket support prevents the movement of the support in each of the three directions.

347. The bending moment on a section is maximum where shear force is
(a) zero (b) minimum
(c) maximum (d) changing sign

BPSC AE Aug. 2019 Paper-V
UJVNL AE 2012

Ans. (d) : The maximum bending moment occurs in a beam, when the shear force at the section is zero. At a point of contraflexure bending moment is zero or bending moment changes sign.

348. The point where bending moment changes direction is a beam carrying load is called as:
(a) The point of contraflexure
(b) Point of zero stress
(c) Point of zero deflection
(d) Non-yielding support point

UPRVUNL AE 2016

Ans.: (a) If bending changes sign at a section then curvature will also change at that section. Such point is called point of contraflexure.

349. When shear force at a point is zero, then bending moment at that point is
(a) Zero (b) Minimum
(c) Maximum (d) Infinity

TSGENCO AE 2015

Ans. (c) :

$$\frac{dm}{dx} = V$$

when shear force is equal to zero, then bending moment is maximum.

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350. The maximum bending moment due to a single point moving load in a simply supported beam occurs at
(a) mid span
(b) quarter span
(c) a section under the load
(d) anywhere in the beam

UKPSC AE (Urban) 2011

Ans. (c) : The shear force should change sign at point where the single point load is applied, Thus maximum moment will be act at this point. If the load moves, the location of maximum moment moves with it.

351. If the shear force diagram of a simply supported beam is parabolic, then the load on the beam is
(a) uniformly distributed load
(b) concentrated load at mid span
(c) external moment acting at mid span
(d) linearly varying distributed load

ESE 2003

Ans. (d) :

$$\frac{dv}{dx} = -w$$

if the shear force diagram is parabolic then its shape or load will be linearly varying distributed.

352. The maximum bending moment in a beam occurs where
(a) Shear force is minimum
(b) Shear force is maximum
(c) Shear force changes sign
(d) Deflection is zero

TSGENCO AE 2015

Ans. (c) :

$$\frac{dm}{dx} = V$$

Bending moment is maximum where shear force change sign.

353. The point where the bending moment is zero after changing its sign is known as
(a) point of failure
(b) point of contraflexure
(c) point of maximum shear force
(d) point of maximum deflection

Rajasthan Nagar Nigam AE Shift-2, 2016

Ans. (b) : Point of contraflexure/Inflection is that point where bending moment changes, its sign. It is the point at which the bending moment curve intersects with zero line.

354. A simply supported beam is subjected to uniformly varying load. Here at the point of zero shear force the bending moment will be
(a) one third of maximum bending moment
(b) zero bending moment
(c) maximum bending moment
(d) maximum or minimum depending on span

Rajasthan Nagar Nigam AE Shift-2, 2016

Ans. (c) : The bending end of simply supported and roller supported beam is taken zero because the ends of simply supported beam resist the motion of the beam in two direction i.e. horizontal and vertical direction, while the end of roller supported beam prevents the beam in moving in only one direction.

355. A prismatic beam of length *l* and fixed at both ends carries a uniformly distributed load. The distance of points of contraflexure from either end is
(a) 0.207 *l* (b) 0.211 *l*
(c) 0.277 *l* (d) 0.25 *l*

Punjab PSC Sub divisional Engineer 2014

Ans. (b) : A prismatic beam of length *l* and fixed at both ends carries a uniformly distributed load. The distance of points of contraflexure from either end is 0.211 *l*.

356. A cantilever beam is the one which is supported with
(a) both ends on rollers
(b) one end fixed and the other free
(c) one end hinge and other on rollers
(d) one end fixed and the other on rollers

GWSSB AAE 2015

Ans. (b) : Cantilever beam has got fixed support at one end while the other end is free to move.

357. The bending moment in a beam will be maximum when
(a) the S.F. is zero
(b) None of three mentioned here
(c) the S.F. is uniform
(d) the S.F. is maximum

GWSSB AAE 2015

Ans. (a) The maximum bending moment occurs in a beam, when the shear force that section is zero or changes the sigh because at point of contraflexure the bending moment is zero.
If bending moment changes sigh at a section that curvature will also change sigh at that section such a point is called point of contraflexure or point of inflection.

358. A uniformly distributed load longer than the span of girder moves from left to right. The maximum bending moment at the center of the span occurs when the uniformly distributed load occupies
(a) whole span of girder
(b) right half span of girder
(c) left half span of girder
(d) none of these

GOA PSC Assistant Prof. 2020

Ans. (a) : For the maximum B.M. at a section or absolute maximum B.M. the moving UDL should be divided by the section in the same ratio in which the section divide the span. If length of UDL is more than the span, than for the same ratio it will cover entire span.

359. For no torsion, the plane of bending should-
(a) be parallel to one of the principal axes
(b) pass through shear center of the section
(c) pass through neutral axis of the section
(d) pass through center of gravity of the section

JPSC AE 2009

Ans. (b) : For no torsion, the beams or member is subjected to pure bending means the shear force in the particular beam is zero and no torsional and axial loads. So, the plane of bending is passed through shear center of section.

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360. A beam of uniform strength refers to which one of the follows?

(a) A beam in which extreme fibre stresses are same at all cross-sections along the length of the beam

(b) A beam in which the moment of inertia about the axis of bending is constant at all cross-section of the beam

(c) A beam in which the distribution of bending stress across the depth of cross-section is uniform at all cross-sections of the beam

(d) A beam in which the bending stress is uniform at the maximum bending moment cross-section

JPSC AE 2009

Ans. (c) : A beam of uniform strength have uniform cross section throughout their length. When they are loaded, there is a variation in bending moment from section to section along length. The stress in extreme outer fibre also varies from section to section along their length.

361. In fixed beam (fixed supports at both ends) having 'l' span with 'w' as uniformly distributed loads, acting in downward direction, then fixed end bending moment at supports is .....

(a) $\frac{wl^2}{12}$ (b) $\frac{wl^2}{8}$

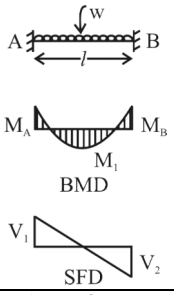
(c) $\frac{wl^3}{12}$ (d) $\frac{wl^8}{12}$

GPSC R&B AE 2016

Ans. (a) : $M_A = M_B = \boxed{\frac{-wl^2}{12}}$

Moment at center–

$$M_1 = \frac{wl^2}{24}$$



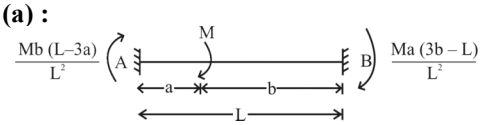
362. A fixed beam AB of span L is subjected to a clockwise moment M at a distance 'a' from end A. Fixed end moment at end A will be:

(a) $\frac{M}{L^2}(L-a)(L-3a)$ (b) $\frac{M}{L^2}a(2L-3a)$

(c) $\frac{M}{L^2}a(L-a)$ (d) $\frac{M}{L^2}(L-a)(2L-a)$

Maharashtra PSC Mains 2018 Paper-1
APPSC AEE 2012

Ans. (a) :



$$M_{AB} = \frac{Mb(L-3a)}{L^2} \quad (\text{as } a+b=L)$$
$$= \frac{M(L-a)(L-3a)}{L^2}$$

363. A moment M is applied at the propped end of a propped cantilever beam of span *l* and flexural rigidity EI. The moment at fixed end is

OPSC Civil Services Pre 2011

OR

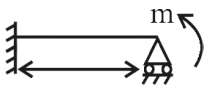
If a moment is applied to the hinged end of a prismatic propped cantilever, then the moment at the fixed end will be

(a) $\frac{M}{2}$ (b) $\frac{M}{3}$

(c) $\frac{M}{4}$ (d) M

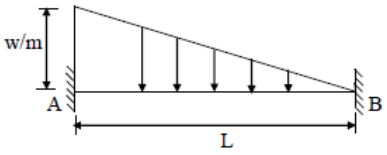
GPSC Poly. Tech. Lect. 2016
TRB Poly. Tech. Lect. 2012

Ans. (a) :



When the member is fixed at one end and a moment is applied at the other end which is simply supported or hinged, the moment induced the fixed end is one half of the applied moment. The induced moment at the fixed end is in the same direction as the applied moment.

364. A fixed beam is loaded as in figure. The fixed end moment at support *A* is



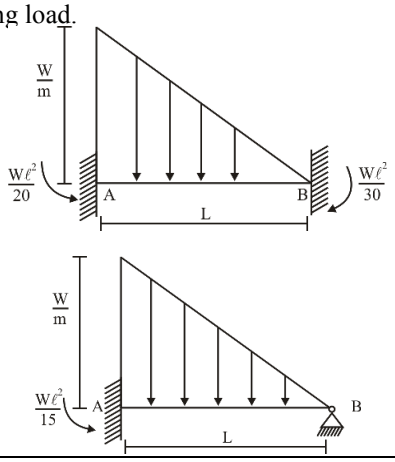
(a) $\frac{wL^2}{30}$ (b) $\frac{wL^2}{20}$

(c) $\frac{wL^2}{10}$ (d) $\frac{wL^2}{8}$

ESE 2018
GPSC R&B AE 2016

Ans. (b)

- Fixed end moment for beam loaded with uniformly varying load.



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365. The angle between the axis of a beam and normal to the transverse plane in beams subjected to pure bending moment is:

(a) 90°
(b) 0°
(c) Close to but not equal to 0°
(d) Close to but not equal to 90°

Haryana PSC AE 2017

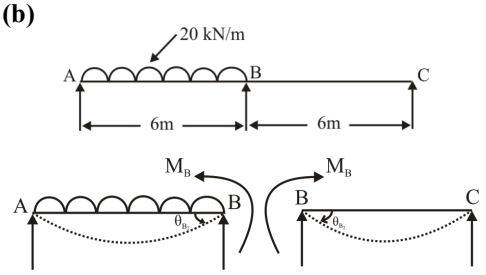
Ans. (b) : The angle between the axis of a beam and normal to the transverse plane in beams subjected to pure bending moment is 0°.

366. Each span of a two-span continuous beam of uniform flexural rigidity is 6 m. All three supports are simple supports. It carries a uniformly distributed load of 20 kN/m over the left span only. The moment at the middle support is

(a) 90 kNm Sagging (b) 45 kNm Hogging
(c) 90 kNm Hogging (d) 45 kNm Sagging

ESE 1998, 2018

Ans. (b)



$\because \quad \theta_{B_1} = \theta_{B_2}$

$$\frac{W\ell^3}{24EI} + \frac{M_B\ell}{3EI} = \frac{-M_B\ell}{3EI}$$

(negative sign due to clockwise direction of θ_{B_2})

$$M_B = \frac{-W\ell^2}{16}$$

M_B = – 45 KN-m -------- (negative sign show hogging)

OR

Using the theorem of three moments

$$M_A L_1 + 2M_B(L_1 + L_2) + M_c L_2 + \frac{6A_1\bar{x}_1}{L_1} + \frac{6A_2\bar{x}_2}{L_2} = 0$$

$\because M_A = 0, \; M_C = 0, \; A_2 = 0$

then

$$2 \times M_B(6+6) + 0 = \frac{-6 \times \frac{2}{3} \times \frac{20 \times 6^2}{8} \times 6 \times 3}{6} = -5 \times 6^3$$

$M_B = -45\,\text{kNm}$ or $M_B = 45\,\text{kNm}$ (Hogging)

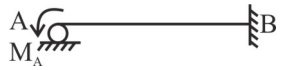
367. A propped cantilever beam, AB, fixed at B is subjected to the action of a couple of moment M_A at the end A. What is the value of the reactive moment M_B will be induced at the built-in end B?

(a) $M_B = \frac{M_A}{4}$ (b) $M_B = M_A$

(c) $M_B = \frac{M_A}{8}$ (d) $M_B = \frac{M_A}{2}$

OPSC AEE 2019 Paper–I

Ans. (d)



Deflection due to M_A

$$\delta_1 = \frac{M_A L^2}{2EI}$$

Deflection due to propped reaction (R_A)

$$\delta_2 = \frac{R_A L^3}{3EI}$$

At equilibrium $\delta_1 = \delta_2$

$$\frac{R_A L^3}{3EI} = \frac{M_A L^2}{2EI}$$

$$R_A = \frac{3M_A}{2L}$$

B.M. at B = $M_B = R_A.L - M_A$

$$M_B = \frac{3M_A \times L}{2L} - M_A$$

$$= \frac{3M_A}{2} - M_A$$

$$M_B = \frac{M_A}{2}$$

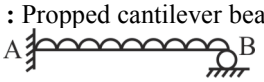
368. A propped cantilever beam of span 'L' is loaded with UDL of intensity W/unit length, all through the span. The BM at the fixed end is

(a) $\frac{WL^2}{8}$ (b) $\frac{WL^2}{2}$

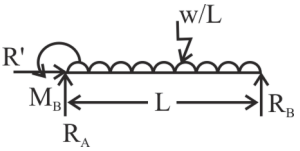
(c) $\frac{WL^2}{12}$ (d) $\frac{WL^2}{24}$

TNPSC AE 2012

Ans. (a) : Propped cantilever beam with UDL is



FBD ⇒



At propped end–

δ_{B_1} due to UDL $= \frac{wL^4}{8EI}$

δ_{B_2} due to support $= \frac{R_B L^3}{3EI}$

$$\delta_{net} = 0 \Rightarrow \delta_{B_1} + \delta_{B_2} = 0$$

$$\delta_{B_1} = -\delta_{B_2} \Rightarrow \frac{wL^4}{8EI} = \frac{R_B L^3}{3EI}$$

$$\therefore \quad R_B = \frac{3wL}{8}$$

B.M. at B = M_B

$$= R_B \times L - wL \times \frac{L}{2} = \frac{3wL^2}{8} - \frac{wL^2}{2} = \frac{-wL^2}{8}$$

Resistive or Reactive BM $= -M_B = \frac{wL^2}{8}$

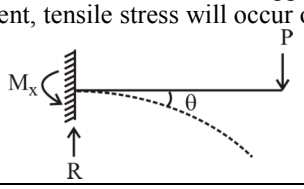
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369. The maximum tensile stress in a cantilever beam with concentrated load acting downwards on the span is caused at-

(a) Top fibre at mid span
(b) Bottom fibre at mid span
(c) Bottom fibre at support
(d) Top fibre at support

OPSC AEE 2015 PAPER-I

Ans. (d) : Maximum moment will be at support. Hence maximum tensile stress will occur at support. Since it is hogging moment, tensile stress will occur on top.

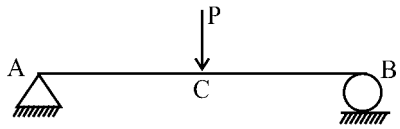


370. A simply supported beam with rectangular cross section is subjected to a central concentrated load. If the width and depth of the beam are doubled, while retaining the same elastic properties, then the deflection at the centre of the beam w.r.t. the original deflection will be reduced to

(a) 50% (b) 25%
(c) 75% (d) 12.50%
(e) 6.25%

OPSC AEE 2019, 2015 Paper–I
CGPSC AE 2014 Shift–I
UTTRAKHAND AE 2013, RPSC ACF 2011

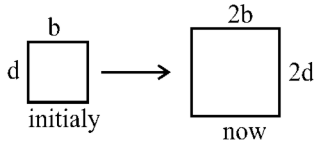
Ans. (e) :



$$\Delta_{\text{midspan}} = \frac{PL^3}{48EI}$$

$P, L, E \rightarrow$ constant

$$\therefore \Delta \propto \frac{1}{I} \propto \frac{1}{bd^3}$$



$$\therefore \frac{\Delta_1}{\Delta_2} = \frac{1/bd^3}{1/2b \times (2d)^3}$$

$$= \frac{2b \times 8d^3}{bd^3}$$

$$\frac{\Delta_1}{\Delta_2} = 16 \Rightarrow \Delta_1 = 16\Delta_2$$

$$\%\Delta_2 = \frac{\Delta_2}{\Delta_1} \times 100$$

$$= \frac{\Delta_2}{16\Delta_2} \times 100$$

$$\boxed{\%\Delta_2 = 6.25\%}$$

371. The theorem of three moments for the solution of continuous beams is formulated by

(a) Young (b) Maxwell
(c) Williot-Mohr (d) Clapeyron

TNPSC AE 2019

Ans. (d) Theory of three-moments equation is also known Clapeyron.

372. A single concentrated load w rolling over the beam of span L will cause the maximum bending moment and shear force on a section x at a distance x from left support. When the load is on the section, its maximum bending moment will be

(a) $\frac{wxL^2}{(L-x)}$ (b) $\frac{wx(L-x)}{L}$

(c) $\frac{wx^2(L-x)}{L^2}$ (d) $\frac{wL(L-x)}{L}$

Maharashtra PSC AE (Mains) 2012 Paper–I

Ans : (b) According to question, if W is act 'x' distance from left end of beam.

Reaction for R_A and R_B

(i) $\Sigma V = 0$

$R_A + R_B = w$ .....(i)

(ii) Moment about A

$R_B \times L = w \times x$

$$R_B = \frac{wx}{L}$$

from eq^n (i)

$$R_A = w - R_B$$

$$R_A = w - \frac{wx}{L} = \frac{w(L-x)}{L}$$

Bending moment at section x

$$M_X = R_A \times x$$

$$M_X = \frac{w(L-x)}{L} \times x$$

$$\boxed{M_X = \frac{w \times x(L-x)}{L}}$$

373. A prismatic bar when subjected to pure bending assumes the shape of-

(a) Catenary (b) Cubic Parabola
(c) Quadratic Parabola (d) Arc of circle

RPSC AE 2018

Ans. (d) A prismatic bar when subjected to pure bending assumes the shape of arc of circle.

374. Where a member is built into a masonry wall which develops only partial restraint, the member shall be designed to resist a negative moment at the face of support is _____, Where W is total design load L is the effective span.

(a) WL/8 (b) WL/12
(c) WL/16 (d) WL/24

KPSC AE 2020

Ans. (d) : Where a member is built into a masonry wall which develops only partial restraint, the member shall be designed to resist a negative moment at the face of support is WL/24.

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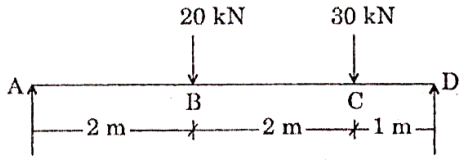
375. When a beam is loaded transversely the maximum bending compressive stress develops on

(a) top fibre
(b) bottom fibre
(c) neutral axis
(d) the section passing through the c.g.

ASSAM PSC CCE (PRE) 2015

Ans : (a) When a beam is loaded transversely the maximum bending compressive stress develops on top fibre.

376. Which part of the beam is subjected to pure bending in the following figure?



(a) AB
(b) BC
(c) CD
(d) No part of beam is subjected to pure bending

Maharashtra PSC AE (Mains) 2012 Paper–I

Ans : (d) Draw for SFD

(i) To find reaction

(a) $\Sigma V = 0$

$R_A + R_D = 20 + 30 = 50$ kN ..... (i)

(b) $\Sigma M = 0$

$R_A \times 5 - 20 \times 3 - 30 \times 1 = 0$

$$R_A = \frac{90}{5} = 18 \text{ kN}$$

From equation (i)

$R_D = 50 - R_A = 50 - 18 = 32$ kN

Calculate SF in beam

at D

$F_D = +32$ kN

at C

$F_C = +32 - 30 = 2$ kN

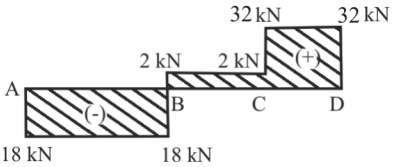
at B

$F_B = 2 - 20 = -18$ kN

at A

$F_A = -18 + 18 = 0$ kN

SFD –



No part of the beam is free from shear force according to figure.

377. The maximum bending moment due to a train of wheel loads on a simply supported girder

(a) always occurs at centre of span
(b) always occurs under a wheel load
(c) never occurs under a wheel load
(d) None of the above

Gujarat PSC AE 2017

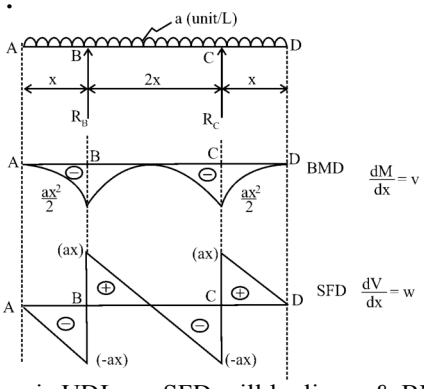
Ans : (b) The maximum bending moment due to a train of wheel loads on a simply supported girder always occurs under a wheel load.

378. In the adjacent beam loaded uniformly at an intensity of a/unit length, the maximum bending moment is :

(a) $ax^2/2$ (b) $ax^2/4$
(c) $ax^2/8$ (d) $2ax^2$
(e) $4ax^2$

CGPSC AE 2014 Shift–I

Ans. (a) :



As loading is UDL, so SFD will be linear & BMD will be parabolic in shape Due to symmetry

$$R_B = R_C = \frac{a \times (4x)}{2} = 2ax$$

from BMD, It is clear Max. B.M. at B & C

$M_B - M_A$ = Area under SFD between A & B (M_A=0)

$$M_B = \frac{1}{2}(x)(-ax) = \frac{ax^2}{2} \text{ (Hogging B.M.)}$$

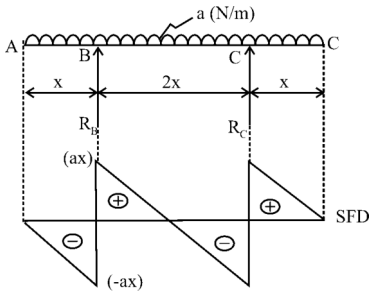
$$M_C = \frac{ax^2}{2} \text{ (Hogging B.M.)}$$

379. In the adjacent beam, the maximum shear stress is

(a) ax (b) 2ax
(c) ax/2 (d) ax/4
(e) 4ax

CGPSC AE 2014 Shift–I

Ans. (a) :



Due to symmetry $R_B = R_C = \frac{a(4x)}{2} = 2ax$

From SFD, it is clear $\boxed{S.F_{max} = ax}$

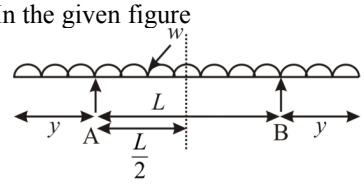
380. A steel beam is carrying just its self load. It needs to be so supported on two columns (span length, L) with equal overhangs (y) on both the sides such that the bending moment at the mid-span is zero. The ratio of L/y should be:

(a) 0.25 (b) 0.5
(c) 1.00 (d) 2.00

UPRVUNL AE 2016

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Ans.: (d) In the given figure



Reaction at support

$$R_A = R_B = \left(\frac{L+2y}{2}\right)w$$

Moment at mid span = 0

$$-\frac{w}{2}\left(\frac{L}{2}+y\right)^2 + R_A \times \frac{L}{2} = 0$$

$$-\frac{w}{2}\left(\frac{L}{2}+y\right)^2 + \frac{w}{2}(L+2y)\times\frac{L}{2} = 0$$

By solving–

$$y = \frac{L}{2}$$

Then, $\frac{L}{y} = \frac{L}{L/2} = 2$

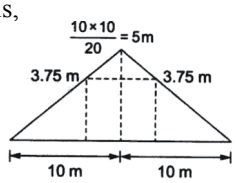
381. The absolute maximum bending moment in a simply supported beam of span 20 m due to moving u.d.l. of 4 kN/m spanning over 5 m is

(a) 87.5 kNm at the support/
(b) 87.5 kNm near the mid-point
(c) 3.5 kNm at the mid-point
(d) 87.5 kNm at the mid-point

BPSC AE 2019 Paper (V) Section-I

Ans. (d) : Given, $l = 20$ m, $w = 4$ kN/m

UDL for absolute maximum bending moment in simply supported beam is,



$$M_{max} = \left(5\times 3.75 + \frac{1}{2}\times 1.25\times 5\right)\times 4$$

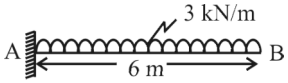
M_{max} = 87.5 kNm (at mid point)

382. A cantilever beam of span 6 metre with left end fixed carries a uniformly distributed load of 3kN/m throughout its length. The magnitude of maximum shear force is

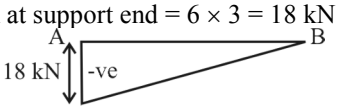
(a) –24 kN-m (b) –12 kN
(c) –24 kN (d) –18 kN

TNPSC AE 2012

Ans. (d)



Reaction at support end = 6 × 3 = 18 kN



Maximum shear force at A = –18 kN

383. The point where the bending moment is zero is called as

(a) Point of contraflexure (b) Yield point
(c) Plastic hinge (d) Limit of elasticity

TNPSC AE 2019
Arunachal Pradesh Poly. Tech. Lect. 2019
UPPCL AE 2018, Nagaland PSC CTSE 2017 Paper–I
***UPRVUNL AE 2016,* CGPSC AE 2014 Shift–I**
RPSC POLY. TECH. LECT. 2014
APPSC AEE (Civil/Mechanical) 2012, PSPCL AE 2012

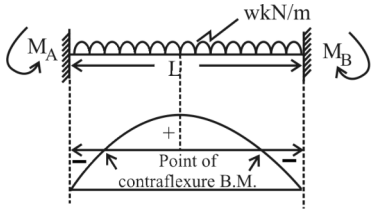
Ans : (a) Point of contraflexure:- It is a point where bending moment changes its sign. At point of contraflexure bending moment is zero but the support ends are not counted as point of contraflexure.

384. The number of points of contra flexure for a fixed beam carrying uniformly distributed load over the whole span is

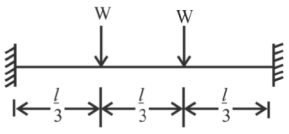
(a) 6 (b) 2
(c) 3 (d) 4

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Ans. (b) The number of points of contraflexure for a fixed beam carrying uniformly distributed load over the whole span is 2.



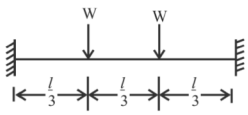
385. Fixed end moment of end A of the fixed beam shown in fig.



(a) $\frac{0.5WL}{9}$ (b) $\frac{WL}{9}$
(c) $\frac{6WL}{27}$ (d) $\frac{4WL}{9}$

TNPSC AE 2019

Ans. (c)





$$M_1 = \frac{W\times l/3\left(\frac{2l}{3}\right)^2}{l^2} \qquad M_2 = \frac{W\times 2l/3\times\left(\frac{l}{3}\right)^2}{l^2}$$

$$M_1 = W\times\frac{l}{3}\times\frac{4l^2}{9\times l^2} \qquad M_2 = W\times\frac{2l^2}{27l^2}\times l$$

$$M_A = M_1 + M_2 = \frac{4Wl}{27} + \frac{2Wl}{27} = \frac{6Wl}{27}$$

386. A propped cantilever of span 'L' is subjected to a concentrated load at mid-span. If M_p is plastic moment capacity of the beam, then the value of the collapse load will be

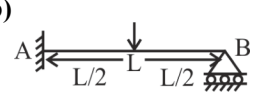
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(a) $\frac{12M_P}{L}$ (b) $\frac{6M_P}{L}$

(c) $\frac{4M_P}{L}$ (d) $\frac{8M_P}{L}$

TNPSC AE 2019

Ans. (b)



$W = \frac{6M_P}{L}$

387. A simply supported beam of span '*l*' carries a point load 'W' at the centre of the beam. The bending moment diagram will be a

(a) Parabola with maximum ordinate at the centre of the beam

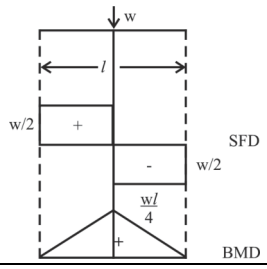
(b) Parabola with maximum ordinate at one end of the beam

(c) Triangle load with maximum ordinate at the centre of the beam

(d) Triangle with maximum ordinate at one end of the beam

TNPSC AE 2019

Ans. (c)



388. The shear force and bending moment are zero at the free end of a cantilever, if it carries a

(a) Point load at the free end

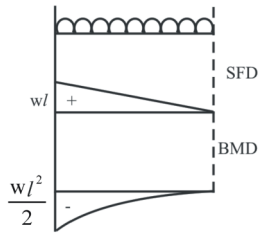
(b) Uniformly distributed load throughout the span

(c) Uniformly verying load with maximum ordinate at the free end

(d) Couple at the free end

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Ans. (b) We get shear force and bending moment value zero at free end if we provide an UDL, throughout the span.



389. A series of concentrated load traverses a simply supported beam. For maximum B.M. at a given section to occur, the load system must be placed such that

(a) The heaviest load is placed over the section

(b) The average loading on the two sides of the section is same i.e when the section divides the load in the same ratio it divides the span.

(c) Maximum number of loads will cover the span

(d) The load is placed over the centre of the span and the rest of the load system acts accordingly

TNPSC AE 2019

Ans. (b) Maximum BM at section occurs when the average loading to the section's rights side and the left side are equal. It means the section divide according to the ratio of load distributed in the section.

390. The point of contraflexure occurs in

(a) Cantilever beams

(b) Simply supported beams

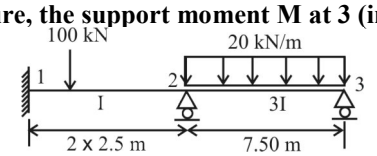
(c) Overhanging beams

(d) Fixed beams

Gujarat PSC AE 2017

Ans : (c & d) The point of contraflexure occurs in overhanging beam.
The point of contraflexure also occur in Indeterminate Beam.
both (c) and (d) are correct.

391. In the continuous beam shown in the following figure, the support moment M at 3 (in kN-m) is



(a) 93.8 (b) 46.9

(c) 62.5 (d) -93.5

(e) 0.0

CGPSC AE 2014 Shift–I

Ans. (e) : Since point (3) is roller support & we know, at hinge/roller support moment is always zero.

392. Where maximum bending moment occurs in a beam the shear force at that location is

(a) Zero

(b) Maximum

(c) Acts in a vertically upward direction

(d) Acts in a vertically downwards direction

(e) Has a non-zero non-negative value

CGPSC AE 2014 Shift–I

Ans. (a) : Since $\frac{dM}{dx} = S_{xx}$

For B.M. to be maximum

$$\frac{dM}{dx} = 0$$

$\therefore \boxed{S_{xx} = 0}$

393. For the conjugate beam method to be applicable, which of the following rules is not followed

(a) The original beam must be statically determinate

(b) The conjugate beam and original beam must be of same length

(c) The conjugate beam must be in static equilibrium

(d) No other loading should be applied to the conjugate beam other than the elastic weight

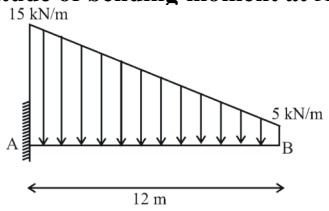
(e) The maximum indeterminacy can be upto 5 degree

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Ans. (e) : Conjugate beam method can not be apply to statically indeterminate real beam because indeterminate real beam gives unstable conjugate beam.

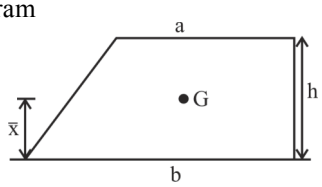
394. The beam is loaded as shown in the figure. The magnitude of bending moment at A is



(a) 120 kN-m (b) 600 kN-m
(c) 240 kN-m (d) 180 kN-m
(e) 60 kN-m

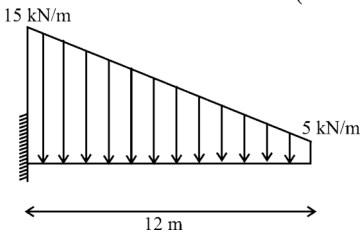
CGPSC AE 2014 Shift–I

Ans. (b) :
Concept :
Total BM = Area of loading diagram × Centroid of loading diagram



Centroid of trapezoidal from bottom $(\bar{x}) = \frac{2a+b}{3(a+b)} \times h$

$= \frac{2\times5+15}{3(5+15)} \times 12$



$(B.M.)_A = \frac{1}{2}(5+15)\times12\times\left[\frac{2\times5+15}{3(5+15)}\times12\right]$

= 600 kN-m

395. Continuous beams are
(a) stronger and much stiffer than simple beams
(b) weaker and less stiffer than simple beams
(c) subjected to excessive shear strain
(d) with standing double the maximum bending moment on simple beams

TNPSC AE 2012

Ans. (a) Continuous beams are stronger and much stiffer than simple beams.

396. A beam supported on more than two supports is called
(a) simply supported beam
(b) fixed beam
(c) cantilever beam
(d) continuous beam

TNPSC AE 2012

Ans. (d) A beam supported on more than two supports is called continuous beam.

397. A simple support offers
(a) only vertical reaction
(b) only a horizontal reaction
(c) a vertical reaction and a horizontal reaction
(d) a horizontal reaction and a moment reaction

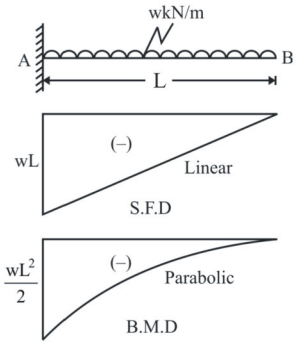
TNPSC AE 2012

Ans. (a) A simple support offers only vertical reaction.

398. Variation of bending moment under a UDL, in a beam is :
(a) Straight line variation
(b) Parabolic variation
(c) Cubic variation
(d) Zero i.e. horizontal straight line

TNPSC AE 2015

Ans. (b) : Bending Moment under UDL is parabolic variation.
Shear force under UDL is linear variation.

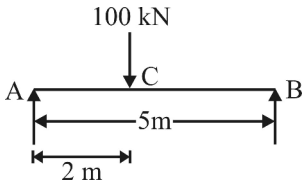


If the UDL is applied to the full length of the cantilever beam its shear force diagram will be triangular (linear) and bending moment diagram will be parabolic.

399. If a simply supported beam of 5 m span carries a point load of 100 kN at 2 m from LHS, the maximum bending moment on it is :
(a) 62.5 kN m (b) 120 kN m
(c) 125 kNm (d) 312.5 kN m

TNPSC AE 2015

Ans. (b) :



$R_A + R_B = 100\,kN$

Taking moment about point A.

$100 \times 2 - R_B \times 5 = 0$

$R_B = 40$ kN

$R_A = 60$ kN

Maximum Bending moment $(BM_c) = 60 \times 2$

= 120 kN-m

400. Under a beam loading condition, if the bending moment is constant over a certain span length, then the shear force would be :
(a) decreasing
(b) increasing
(c) constant and a non-zero value
(d) zero

CIL MT 2017

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Ans. (d) : Under a beam loading condition, if the bending moment is constant over a certain span length, then the shear force would be zero, because –

$$\frac{dM_x}{dx} = S_{xx}$$

if M_x = Constant (C)

then, $\frac{d(C)}{dx} = 0$

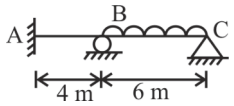
So, $S_{xx} = 0$

401. A two span continuous beam ABC with end A fixed and end C hinged is having AB = 4m and BC = 6m, $I_{AB} : I_{BC} = 1 : 2$. It is subjected to udl (uniformly distributed load) of 10 kN/m over the entire right span. The moment at the end C is :

(a) 25 kNm hogging (b) 11.25 kNm hogging
(c) 11.25 kNm sagging (d) 0

APPSC Poly. Tech. Lect. 2020

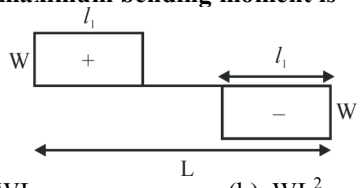
Ans. (d) :



Given point C is hinged, as we know that B.M about a hinge is zero

$BM_C = 0$

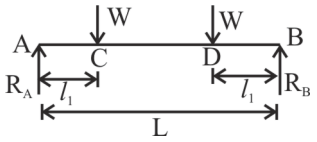
402. The shear force diagram for a simply supported beam of span L is given in the figure. The maximum bending moment is–



(a) WL (b) WL^2
(c) Wl_1 (d) $Wl_1/2$

Gujarat PSC AE (N.W.R.) 2020

Ans. (c) :



Let X-X section from B in part BC

$M_{xx} = W_x + W(x - l_1)$

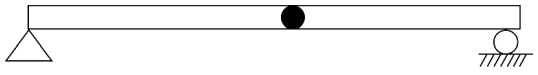
at $x = l_1$

$M_D = M_C = M_{max}$

$M_{max} = M_x l_1 - 0$

$\therefore \quad \boxed{M_{max} = Wl_1}$

403. A prismatic beam is shown in the figure given below:



Consider the following statements:

(i) The structure is unstable
(ii) The bending moment is zero at support and internal hinge
(iii) It is a mechanism
(iv) It is statically indeterminate

Which of these statements are correct?

(a) i, ii and iii (b) i, ii, iii and iv
(c) i and ii (d) iii and iv

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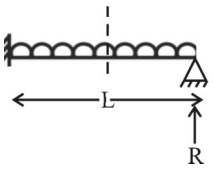
Ans. (a) : Correct statement are :
(i) The structure is unstable.
(ii) The bending moment is zero at support and at internal hinge
(iii) It is a mechanism.

404. A propped cantilever beam of span L is subjected to Uniformly Distributed Load of w kN/m throughout the span. What is the distance of Point of Contraflexure from the Fixed end?

(a) 0.175L (b) 0.25L
(c) 0.33L (d) 0.375L

Gujarat PSC AE (N.W.R.) 2020

Ans. (b) :



$$\frac{wL^4}{8EI} = \frac{RL^3}{3EI} \Rightarrow R = \frac{3WL}{8}$$

$$\frac{wx^2}{2} - \frac{3wl}{8}x = 0$$

$4x^2 = 3Lx$

$x = \frac{3L}{4}(\text{from free end})$

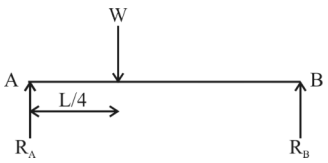
$\frac{L}{4}$ from fixed end

405. A simply supported beam of span L, width B and depth D is subjected to a rolling concentrated load of magnitude W. The maximum flexural stress developed at the section L/4 distance from the end support is

(a) $(3WL)/(4BD^2)$ (b) $(4WL)/(3BD^2)$
(c) $(9WL)/(8BD^2)$ (d) $(8WL)/(9BD^2)$

Gujarat PSC AE (N.W.R.) 2020

Ans. (c) :



$R_A + R_B = W$

$\Sigma M_A = 0; \; W \times \frac{L}{4} = R_B \times L$

$R_B = \frac{W}{4}, \; R_A = \frac{3W}{4}$

now at L/4, $\sigma = \frac{My}{I} = \frac{\left(\frac{3W}{4} \times \frac{L}{4}\right) \times \left(\frac{D}{2}\right)}{\frac{BD^3}{12}}$

$\sigma_{flexure} = \frac{9WL}{8BD^2}$

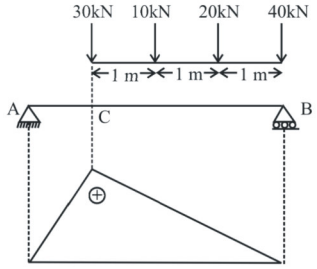
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406. A train of four wheel loads 30 kN, 10 kN, 20 kN and 40 kN (distance between each sequential wheel is 1 m) is passing in the given order leading with 30 kN, from right to left on a simply supported girder of 10 m span. Which of these wheel loads should be placed at the section 2m from the left side support, to produce Maximum Bending Moment at that section?

(a) 10 kN (b) 20 kN
(c) 30 kN (d) 40 kN

Gujarat PSC AE (N.W.R.) 2020

Ans. (c) :



ILD for BM at C

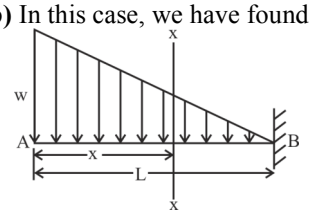
Certainly, BM will be more, when more load will be at C.
But as load is moving from right to left.
When 40 kN will be at C, 30 kN will be end of the beam So, maximum BM will be obtained when 30 kN will be at C.

407. A cantilever beam of span L is subjected to uniformly varying load having value of w kN/m at the free end and 0 kN/m at the fixed end. The magnitude of slope of shear force diagram at the fixed end is.....the magnitude of the slope of shear force diagram at the free end.

(a) same as (b) less than
(c) w times more than (d) wL/2 times more than

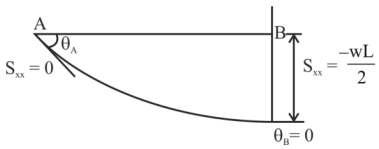
Gujarat PSC AE (N.W.R.) 2020

Ans. (b) In this case, we have found



$$S_{xx} = \frac{-w}{2L}x^2 - \frac{w}{L}\left(Lx - x^2\right)$$ (Parabolic equation)

SFD $\Rightarrow$



The magnitude of slope of shear force diagram at the fixed end is less than the magnitude of the slope of shear force diagram at the free end
Hence,

$$(\text{slope})_{\text{fixed end}} < (\text{slope})_{\text{free end}}$$

408. The point of contra flexure is the point

(a) where the bending moment changes sign
(b) where bending moment is maximum
(c) where bending moment is minimum
(d) where shear force is zero

APPSC (PHE) AE Paper-2 2012
UKPSC AE (Urban) 2011

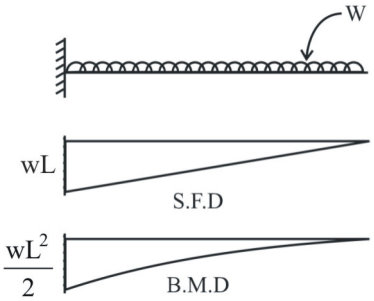
Ans. (a) : In bending beam, a point of contra flexure if it is a location where bending moment is zero (change its sign)

409. The shear force diagram for a cantilever beam carrying uniformly distributed load per unit length will be a/an

(a) rectangle (b) isosceles triangle
(c) right-angled triangle (d) parabola

ASSAM PSC CCE (PRE) 2015

Ans : (c)

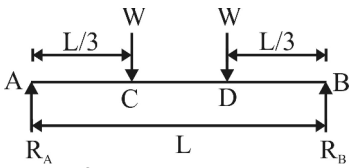


410. A simply supported beam carries two equal concentrated loads W at distances L/3 from either support. The maximum bending moment in the beam will be

(a) $\frac{WL}{2}$ (b) $\frac{WL}{3}$
(c) $\frac{WL}{6}$ (d) $\frac{WL}{8}$

ASSAM PSC CCE (PRE) 2015

Ans : (b) :



$R_A + R_B = 2W$
$R_A = R_B = W$
Maximum bending moment developed at C and D

$$BM_C = W \times \frac{L}{3}$$

$$\boxed{BM_C = BM_D = \frac{WL}{3}}$$

411. When will the bending moment of a section of a beam be maximum?

(a) When the shear force along the section of the beam is zero
(b) When the shear force along the section of the beam is non zero
(c) When the stress on the beam is zero
(d) None of these

GESCOM (KARNATAKA) AE 2011 2011

Ans : (a) When the shear force along the section of the beam is zero, bending moment of that section will be maximum.

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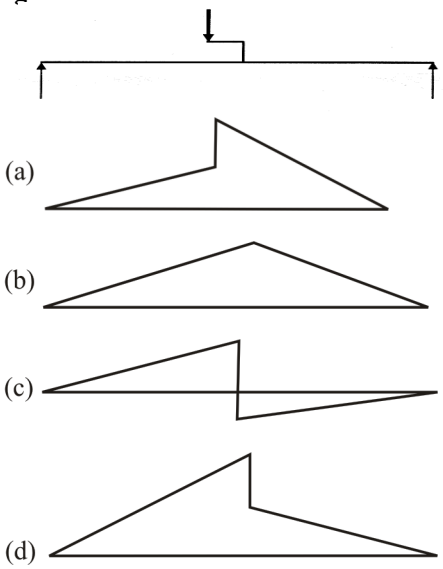
412. If shear force diagram of a beam is an ascending or descending straight line, bending moment diagram will be_____.

(a) Zero
(b) Same as shear force diagram
(c) Horizontal line
(d) Parabolic curve

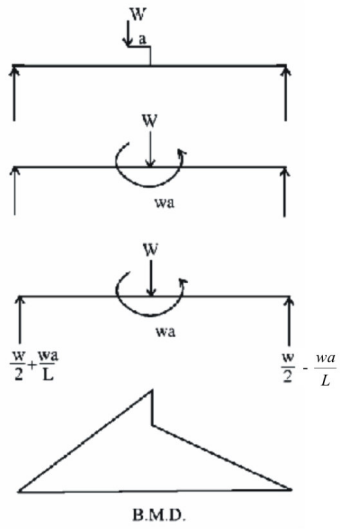
LMRC AM 2019

Ans. (d) If shear force diagram of a beam is an ascending or descending straight line, bending moment diagram will be parabolic curve

413. Select the bending moment diagram for the simply supported beam loaded as shown in the figure :



KPSC AE 2020



Ans. (d)

414. In a simple bending theory, one of the assumption is that the plane sections before bending remain plain after bending. This assumption means that :

(a) Stress is uniform throughout the beam
(b) Strain is uniform throughout the beam
(c) Stress is portional to the distance from the neutral axis.
(d) Strain is proportional to the distance from the neutral axis

Nagaland PSC CTSE 2017 Paper–I

Or

The assumption that the cross section plane before bending remain plane even after bending means

(a) the neutral axis lies at mid height
(b) the bending moment will be resisted by the central core of the section
(c) the stresses in the fibres are proportional to their distances from the neutral axis
(d) the strain in the fibres is proportional to their distances from the neutral axis

GPSC AE March 2018
GPSC Assistant Prof. 2016

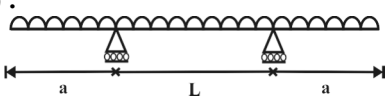
Ans. (d) : In Bending theory, one of the Assumption is that, the plane section before bending remain plane After Bending this Assumption means that –
→ Strain at any point on the x-section is directly proportional to its distance from its neutral Axis, it mean strain diagram is linear.

415. A simply supported beam with equal overhang on both sides carry Uniformly Distributed Load on entire length. What would be the percentage overhang on each side so that maximum bending moment is minimum?

(a) 17.7% (b) 20.7%
(c) 25.2% (d) 33.3%

APPSC Poly. Tech. Lect. 2020

Ans. (b) :

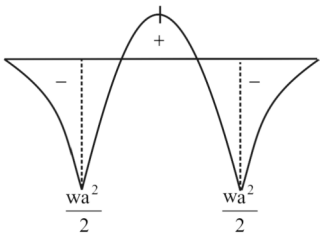


Now for maximum BM to be negative

$$\frac{wL^2}{8}-\frac{wa^2}{2}=\frac{wa^2}{2}$$

or $$\frac{L^2}{8}=a^2$$

$$a=\frac{L}{2\sqrt{2}}$$

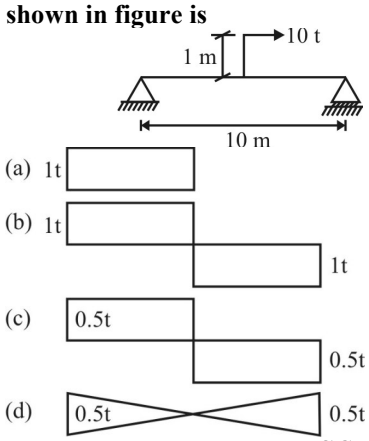


$$\%\ \text{overhang}=\frac{a}{L+2a}\times 100$$

$$=\frac{\frac{L}{2\sqrt{2}}}{L+\frac{L}{\sqrt{2}}}\times 100=20.71\%$$

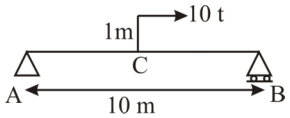
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416. The shear force diagram (SFD) for the beam shown in figure is



UKPSC AE 2012 Paper–I

Ans : (a)



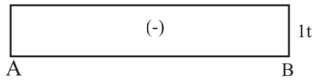
$R_A + R_B = 0$

$\Sigma M_B = 0$

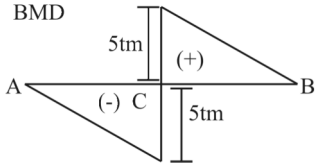
$\therefore \quad R_A \times 10 + 10t \times 1 = 0$

$\boxed{R_A = -1t}$; $\boxed{R_B = 1t}$

SFD



BMD



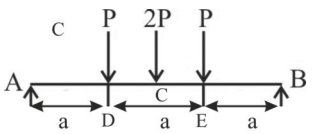
417. A simply supported beam of length '3a' supports three loads separated by distance 'a'. The middle load is 2P, while the loads at distance 'a' from each support are P each. What will be the maximum shear force in the beam?

(a) 2P (b) 4P

(c) 6P (d) Zero

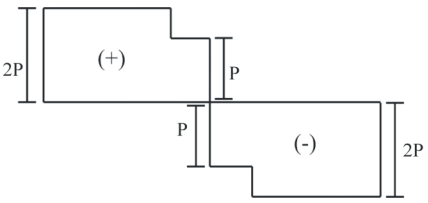
(LMRC AE 2017 I–Shift)

Ans : (a)



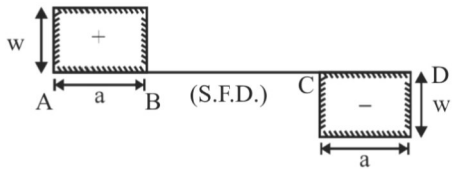
$R_A + R_B = 4P$, $R_A = R_B = 2P$

SFD



Therefore maximum shear force is 2P.

418. The shear force diagram for a simply supported beam of span L is shown in the figure. The maximum bending moment in the beam is :

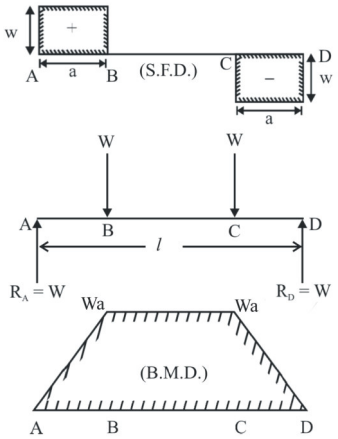


(a) $\frac{wL}{2}$ (b) $w\left(\frac{L}{2}-a\right)$

(c) wa (d) w (L–a)

(UKPSC AE (Paper–I) 2007)

Ans : (c)



BM at B = W × a

BM at C = W × (l – a) – w (l – 2a)

= Wa

$\therefore$ $\boxed{\text{maximum BM} = Wa}$

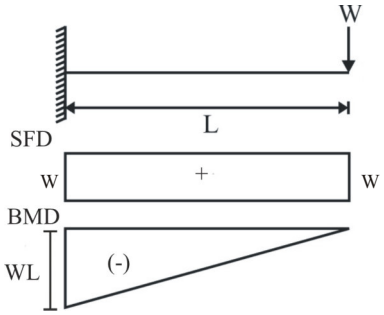
419. The bending moment diagram for a cantilever whose free end is subjected to a point load will be of shape–

(a) Triangle (b) Parabola

(c) Rectangle (d) Cubic Parabola

UKPSC AE 2013 Paper–I

Ans : (a)



420. The maximum bending moment (M) caused by a concentrated load (W) at the mid span of a simply supported beam is

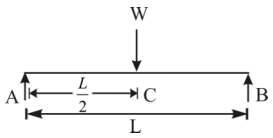
(a) M = (WL/2) (b) M= (WL/8)

(c) M = (WL/4) (d) M = (WL/12)

RPSC ACF 2011

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Ans : (c)



$R_A = R_B = \frac{W}{2}$

$BM_C = R_A \times \frac{L}{2} = \frac{W}{2} \times \frac{L}{2} = \frac{WL}{4}$

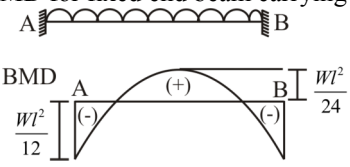
$$M_{max} = \frac{WL}{4}$$

421. A prismatic beam fixed at both ends carries a uniformly distributed load. The ratio of bending moment at supports to the bending moment of mid span

(a) 0.5 (b) 1.0
(c) 1.5 (d) 2.0

RPSC POLY. TECH. LECT. 2011

Ans : (d) BMD for fixed end beam carrying UDL is–



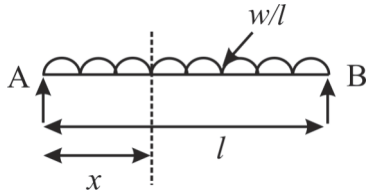
Ratio of BM at support to BM at center $= \frac{\frac{w\ell^2}{12}}{\frac{w\ell^2}{24}} = \frac{24}{12} = 2$

422. The variation of the bending moment in the segment of a beam where the load is uniformly distributed is

(a) Zero (b) Linear
(c) Parabolic (d) Cubic

OPSC AEE 2015 PAPER-I
RPSC POLY. TECH. LECT. 2014

Ans : (c)



$R_A = R_B = \frac{Wl}{2}$

Moment at distance x from support A $= \frac{Wl}{2}x - \frac{Wx.x}{2}$

$$M_x = \frac{Wlx}{2} - \frac{Wx^2}{2}$$

As it is equation of 2 degree therefore variation of BM is parabolic in nature.

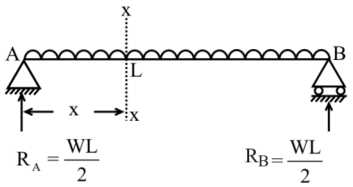
423. Under uniformly distributed load, the shear varies

(a) constant (b) linearly
(c) parabolically (d) cubically

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (b) Considering a simply supported beam subjected to UDL over entire span

B.M. at a section x–x



M_x (x from A) $= R_A x - \frac{wx^2}{2}$

$M_x = \frac{wL}{2}x - \frac{wx^2}{2}$ (Parabolic)

shear force at a section x–x

S_x (x from A) $= +R_A - Wx$

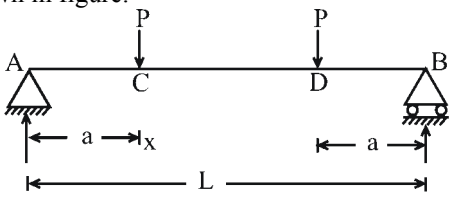
$S_x = \frac{Wl}{2} - Wx$ (Linear)

424. Between point loads, the shear force varies

(a) constant (b) linearly
(c) parabolically (d) cubically

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (a) Considering a simply supported beam loaded as shown in figure.



By symmetry

$R_A = R_B = P$

Portion AC :

S_x (x from A) $= + R_A$

$S_x = +P$ (constant)

Portion CD :

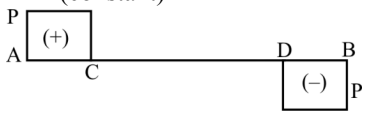
$S_x(x \text{ from } A) = +R_A - P$

$S_x = P\text{-}P = 0$ (constant)

Portion BD :

S_x (x from B) $= - R_B$

$S_x = -P$ (constant)



Between two point loads i.e, in portion C D shear force is constant which is equal to zero.

425. The relation between Shear force and Bending moment in a beam is:

(a) $V = \frac{d^2M}{dx^2}$ (b) $V = \frac{dM}{dx}$

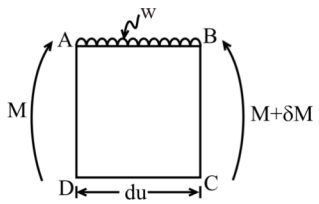
(c) $V = \int m dx$ (d) $V = \frac{d^3M}{dx^3}$

Where V = Shear force
M = Bending moment

[UPPSC State Eng. A.E. 2007 (I)]
[UPPSC State Eng. A.E. 2007 (II)]

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Ans : (b)



The relation between shear force and Bending moment is such that

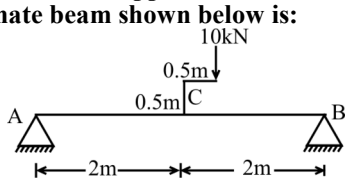
$$\frac{dM}{dx} = SF = V$$

$$V = \frac{dM}{dx}$$

V = shear force

M = bending moment

426. The reaction at support 'B' of the statically determinate beam shown below is:

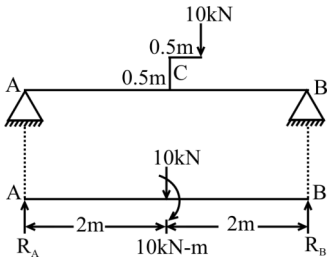


(a) 3.75kN (b) 4.25kN

(c) 5.75kN (d) 6.25kN

[UPPSC State Eng. A.E. 2011]

Ans : (d)



$\Sigma F_y = 0;\ R_A + R_B = 10$ .....(i)

$\Sigma M_B = 0;$ $5\ R_A \times 4 + 5 - 10 \times 2 = 0$

$R_A \times 4 = 15$

$$R_A = \frac{15}{4} = 3.75KN(\uparrow)$$

from equation (i) $R_B = 10 - R_A = 10 - 3.75$

427. The relation between transverse load and shear in a beam is

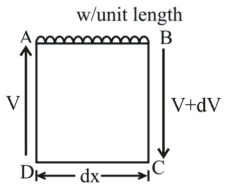
(a) $W = \frac{dV}{dx}$ (b) $\frac{dW}{dx} = V$

(c) WL=V (d) $\frac{d^2W}{dx^2} = V$

Where symbols have their usual meaning.

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (a)



For equilibrium

$\Sigma f_y = 0$

$V + w_x \times dx - V + dV = 0$

$$\boxed{\frac{dV}{dx} = w_x}$$

V – Shear force

w – Intensity of distributed load

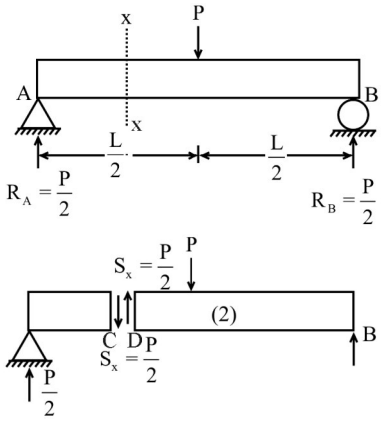
428. Shear force at any section is the algebraic sum of the following on either side of member.

(a) Axial force (b) Transverse force

(c) Moments (d) None of the above

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (b) Shear force at a section is resultant transverse force either from the left or from the right of the cross section.



$$S_x = +\frac{P}{2}$$

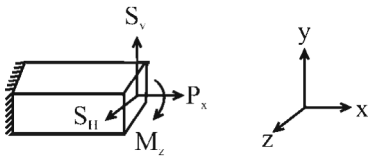
429. A beam is capable of resisting

(a) axial force only (b) shear force only

(c) bending moment only (d) All the above

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (d)



P_x = Axial force

S_v = Vertical shear force

S_H= Horizontal shear force

M_Z =Bending moment

So a beam is capable for resisting axial froce, shear force and bending moment.

430. In a fixed beam carrying a uniformly distributed load W over the whole span, two points of contraflexure occur. They are equidistant from the centre of the span and this distance is

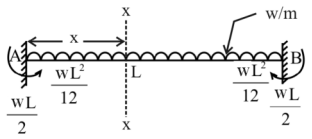
(a) $\frac{L}{3}$ (b) $\frac{L}{\sqrt{3}}$

(c) $\frac{L}{2\sqrt{3}}$ (d) $\frac{2L}{\sqrt{3}}$

RPSC AE 2013

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Ans : (c)



At any section at a distance x from the left end A the BM is given by

$$M_x = R_A.x - \frac{wx^2}{2} - \frac{wL^2}{12}$$

$$M_x = \frac{wL}{2}x - \frac{wx^2}{2} - \frac{wL^2}{12}$$

for points of contraflexure,

$$M_x = 0$$

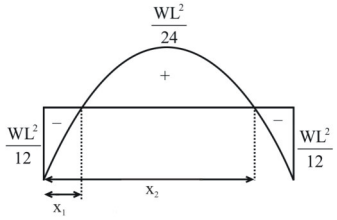
$$\frac{wL}{2}x - \frac{wx^2}{2} - \frac{wL^2}{12} = 0$$

$$Lx - x^2 - \frac{L^2}{6} = 0$$

$$\Rightarrow x^2 - Lx + \frac{L^2}{6} = 0$$

$$x_1 = \frac{L}{2} + \frac{L}{2\sqrt{3}}$$

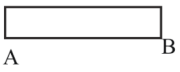
$$x_2 = \frac{L}{2} + \frac{L}{2\sqrt{3}}$$

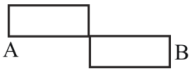


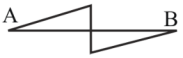
So, from the center distance of left point of contraflexural

$$= \frac{L}{2} - \left(\frac{L}{2} - \frac{L}{2\sqrt{3}}\right) = \frac{L}{2\sqrt{3}}$$

431. Shape of the shear force Diagram for a simply supported beam (AB) subjected to point load W at the centre of span shall be

(a) 

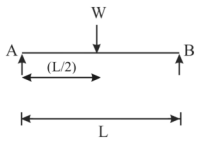
(b) 

(c) 

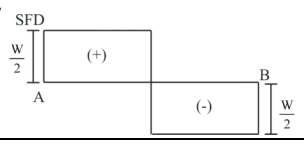
(d) 

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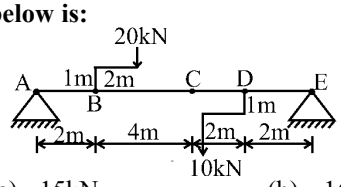
Ans : (b)



$$R_A = R_B = \frac{W}{2}$$



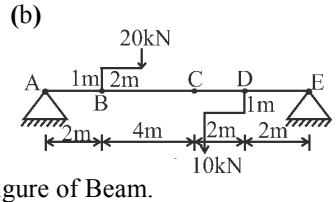
432. The reaction at support A for the beam shown below is:



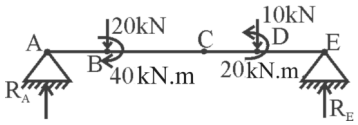
(a) 15kN (b) 16kN
(c) 17kN (d) 18kN

[UPPSC AE 12.04.2016 Paper–I]

Ans : (b)



final figure of Beam.



$$\Sigma Fy = 0$$

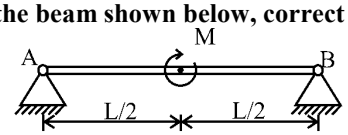
$$R_A + R_E = 20+10 = 30 \text{ kN}$$

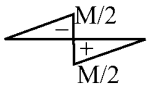
$$\Sigma M_E = 0$$

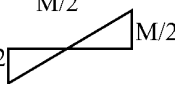
$$R_A = \frac{10\times 2 + 20 + 20\times 8 - 40}{10}$$

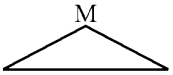
$$\boxed{R_A = 16 \text{ kN}}$$

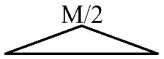
433. For the beam shown below, correct BMD is:



(a) 

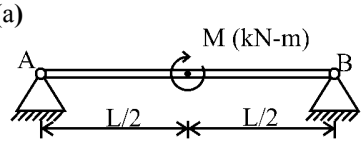
(b) 

(c) 

(d) 

[UPPSC AE 12.04.2016 Paper–I]

Ans : (a)



$R_A + R_B = 0$

Moment about A

$R_B \times L = M$

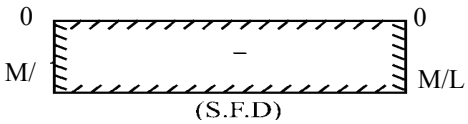
$R_B = M/L, \quad R_A = -M/L$

For Shear force diagram –

$(S.F)_B = +M/L$

$(S.F)_A = +M/L - M/L = 0$

$(S.F)_{mid} = +M/L$



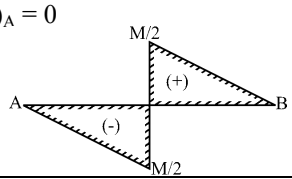
For Bending moment diagram (B.M.D)-

$(B.M)_B = 0$

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$(B.M)_{mid} = +\frac{M}{L}\times\frac{L}{2} - M = -M/2$

$(B.M)_A = 0$

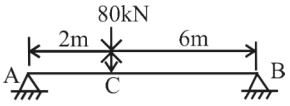


434. A simply supported beam is of span 8m and carries a point load of 80 kN at 2 m from left support. The net BM in the beam is.... kNm.

(a) 120 (b) 100
(c) 102.67 (d) 80

(BPSC AE Paper V 2001)

Ans : (a)



$R_A + R_B = 80$ kN .....(i)

$\sum M_A = 0;\ -80\times 2 + R_B \times 8 = 0$

$R_B = \frac{80\times 2}{8}$

$R_B = 20$kN

From equation (i)

$R_A = 60$kN

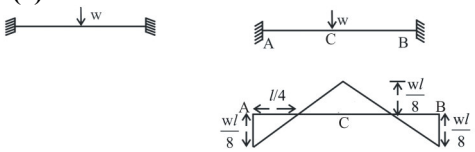
$M_c = 60\times 2 = 120$ kNm

435. The maximum positive Bending Moment in a fixed beam of span 8 m and subjected to a central point load of 120 kN is (in kN-m)

(a) 240 (b) 120
(c) 80 (d) None of these

(BPSC AE Paper V 2001)

Ans : (b)



$\therefore$ Maximum positive BM $= \frac{wl}{8}$

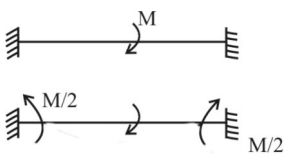
$= \frac{120\times 8}{8} = 120$ kNm

436. A fixed beam of length L is subjected to a bending moment M at the centre. The fixing couple at the ends are

(a) M (b) 3/4M
(c) M/2 (d) M/4

(BPSC AE Paper V 2006)

Ans : (c) If bending moment M is applied at centre of fixed beam, then fixing couple at ends will be $\frac{M}{2}$

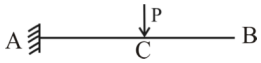


437. A cantilever beam of span 'L' subjected to a concentrated load 'P' at mid span. The bending moment at free end is equal to

(a) PL/2 (b) PL/4
(c) PL (d) 0

(BPSC AE Paper V 2006)

Ans : (d)



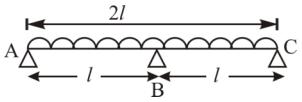
BM at any distance between C to B will be zero as no load is acting between them.

438. A continuous beam 8m long simply supported over two spans 4m long each carries a UDL of 10 kN/m run over its entire length. The support moment at the central support is

(a) 20kN-m (b) 30kN-m
(c) 40kN-m (d) 45kN-m

(BPSC AE Paper V 2006)

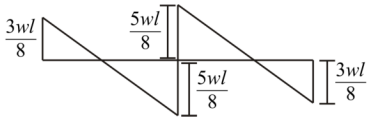
Ans : (a)

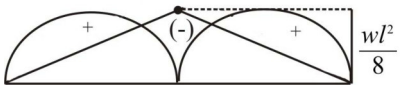


$R_A = R_C = \frac{3}{8}wl$

$R_B = \frac{5}{4}wl$

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BMD

$M_B = \frac{w\ell^2}{8} = \frac{10\times 4^2}{8} = 20\text{kN}-\text{m}$

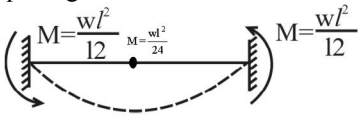
439. The support moment in a fixed beam carrying uniformly distributed load w on span *l* is:

(a) $\frac{w\ell^2}{8}$ (b) $\frac{w\ell^2}{4}$

(c) $\frac{w\ell^2}{12}$ (d) $\frac{w\ell}{12}$

[UPPSC State Eng. A.E. 2007 (II)]

Ans : (c) Equating the areas of the fixed and free BMD



$\frac{2}{3}\times l\times\frac{wl^2}{8} = M_a \times l$

$M_a = \frac{w\ell^2}{12}$

$M_a = \frac{wl^2}{12}$

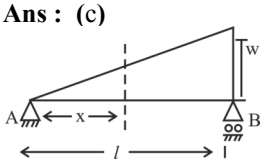
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440. Nature of bending moment of simply supported beam loaded with triangular load is

(a) Triangular (b) Square parabola
(c) Cubic parabola (d) None of these

(BPSC AE Paper V 2006)

Ans : (c)



$$\Sigma M_B = 0 \Rightarrow R_A \times l - \frac{1}{2} \times w \times l \times \frac{l}{3} = 0$$

$$\boxed{R_A = \frac{wl}{6}}, \quad \boxed{R_B = \frac{2wl}{6}}$$

Moment at distance x from end A

$$M_X = R_A.x - \frac{1}{2} \times x \times \frac{wx}{l} \times \frac{x}{3}$$

$$\boxed{M_n = \frac{wl.x}{6} - \frac{1}{6}.\frac{wx^3}{l}}$$

∴ Variation of BM is cubic parabola.

441. Where a member is built into a masonry wall which develops only partial restraint, the member shall be designed to resist a moment at the face of the support of

(a) WL/12 (b) WL/10
(c) WL/48 (d) WL/24

WhereW = Total design load and
L = Effective span

(BPSC AE Paper V 2006)

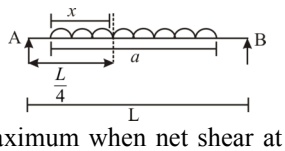
Ans : (d) If a member is built into masonry wall which develops only partial restraint, the member shall be designed to resist negative bending moment of $\frac{WL}{24}$ or $\frac{w\ell^2}{24}$.

442. The maximum bending moment at the left quarter point of a simply supported beam due to crossing of UDL of length shorter than span in the direction left to right, would occur when the load has just crossed the section by

(a) one-fourth of its span
(b) half of its span
(c) three-fourth of its span
(d) its full length

(JPSC Combined AE Paper V 2013)

Ans : (c)

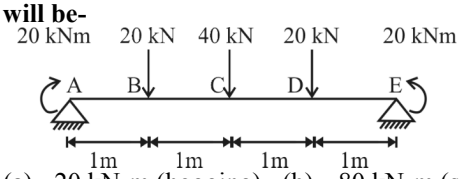


BM will be maximum when net shear at left quarter is zero

$$\therefore \quad \frac{x}{L/4} = \frac{(a-x)}{\frac{3L}{4}} \Rightarrow \boxed{x = \frac{a}{4}}$$

∴ Maximum BM will occur when $\frac{3}{4}$ of UDL crosses the quarter.

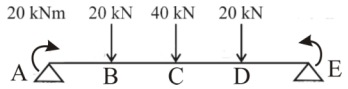
443. A simply supported beam is loaded as shown in the figure below. The bending moment at C will be-



(a) 20 kN-m (hogging) (b) 80 kN-m (sagging)
(c) 60 kN-m (sagging) (d) 40 kN-m (sagging)

(JPSC Combined AE Paper V 2013)

Ans : (b)



$\Sigma M_A = 0$

$\therefore \quad 20 + 20 + 80 + 60 - 20 - R_E \times 4 = 0$

$\therefore \quad \boxed{R_E = 40kN}; \quad \boxed{R_A = 40kN}$

$M_C = -20 \times 1 + 20 + 40 \times 2$

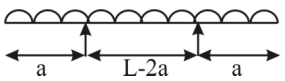
$= 80kN - m$

444. A beam of length 10m, carrying a uniformly distributed load of 10 kN/m over the entire length, rests on two simple supports symmetrically. In order to have maximum sagging bending moment developed in the beam least possible, the distance between the supports must be

(a) 2.07 m (b) 4.14 m
(c) 5.86 m (d) 7.93 m

(JPSC Combined AE Paper V 2013)

Ans : (c)



For maximum hogging moment, length of each overhang must be equal to 0.207 L

∴ Distance between supports = L – 2a
= L – 2 × 0.207L
= 10 × (1 – 0.414) = 5.86 m

• For maximum sagging moment, distance between supports must equal to L i.e., 10 m.

445. For the simply supported beam of span L, the shape of the shear force diagram will be rectangle of length L if the beam is subjected to-

(a) a point load at mid-span
(b) uniformly distributed load over the full span
(c) a clockwise moment at any point within the span
(d) equal magnitude hogging moments at supports

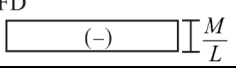
(JPSC Combined AE Paper V 2013)

Ans : (c)



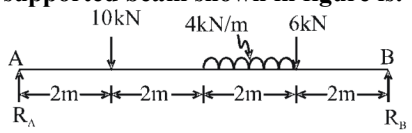
$R_A \times L + M = 0 \Rightarrow \boxed{R_A = -\frac{M}{L}} \quad \therefore \boxed{R_B = \frac{+M}{L}}$

SFD



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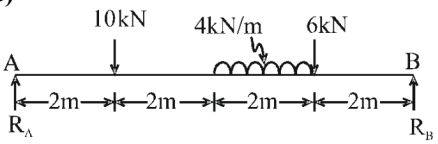
446. The ratio of reactions R_A and R_B in the simply supported beam shown in figure is:



(a) 0.50 (b) 0.75
(c) 1.00 (d) 1.50

[UPPSC State Eng. A.E. 2007 (I)]

Ans : (c)



$\Sigma F_Y = 0$

$R_A + R_B = 10+6+4\times2 = 24\text{kN}$ .....(i)

$\Sigma M_B = 0$

$R_A\times8 - 10\times6 - 4\times2\times3 - 6\times2 = 0$

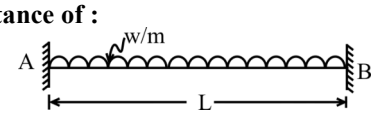
$$R_A = \frac{96}{8} = 12\text{kN}$$

from eq (i),

$R_B = 12$ kN

$$\therefore \quad \frac{R_A}{R_B} = \frac{12}{12} = 1$$

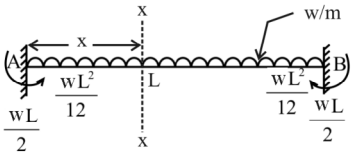
447. For the fixed beam shown in figure, the point of contraflexure nearer to end A is at a distance of :



(a) 0.112 L from end A
(b) 0.212 L from end A
(c) 0.412 L from end A
(d) 0.500 L from end A

[UPPSC State Eng. A.E. 2008]

Ans : (b)



At any section at a distance x from the left end A the BM is given by

$$M_x = R_A.x - \frac{wx^2}{2} - \frac{wL^2}{12}$$

$$M_x = \frac{wL}{2}x - \frac{wx^2}{2} - \frac{wL^2}{12}$$

for points of contraflexure,

$M_x = 0$

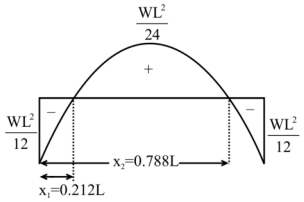
$$\frac{wL}{2}x - \frac{wx^2}{2} - \frac{wL^2}{12} = 0$$

$$Lx - x^2 - \frac{L^2}{6} = 0$$

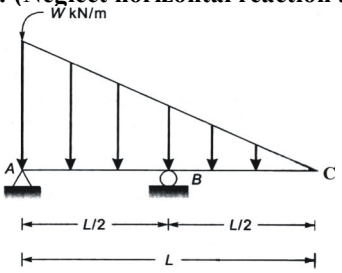
$$\Rightarrow x^2 - Lx + \frac{L^2}{6} = 0$$

$$x_1 = \frac{L}{2} - \frac{L}{2\sqrt{3}} = 0.212\text{ L from end A}$$

$$x_2 = \frac{L}{2} + \frac{L}{2\sqrt{3}} = 0.788\text{ L from end A}$$



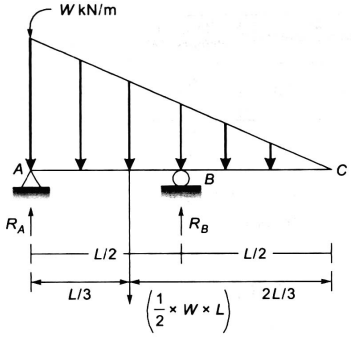
448. A beam *ABC* is supported and loaded as shown in the figure. Find the support reactions at *A* and *B*. (Neglect horizontal reaction at *A*)-



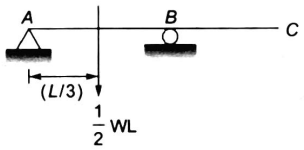
(a) $\frac{WL}{3}, \frac{WL}{3}$ (b) $\frac{WL}{3}, \frac{WL}{6}$
(c) $\frac{WL}{6}, \frac{WL}{3}$ (d) $\frac{WL}{6}, \frac{WL}{6}$

Maharashtra PSC (Mains) 2017 Paper–I

Ans. (c) :



Moment at point A -



$$\Sigma M_A = 0$$

$$R_B \times \frac{L}{2} - \frac{1}{2}\times WL \times \frac{L}{3} = 0$$

$$R_B = \frac{WL}{3}$$

$$\Sigma F_V = 0$$

$$R_A + R_B = \frac{1}{2}WL$$

$$R_A = \frac{WL}{2} - \frac{WL}{3} = \frac{WL}{6}$$

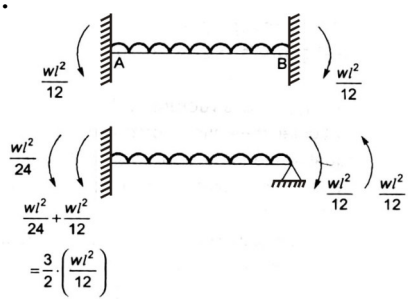
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449. A beam AB is fixed at both ends and carries a uniformly distributed load of intensity W per unit length run over its entire length. Due to some construction defects, the end B is now reduced to a simple support. The percentage increase in bending moment at A is-

(a) 25 (b) 50
(c) 75 (d) 100

OPSC AEE 2015 PAPER-I

Ans. (b) :



% age of increase in Bending Moment of A :

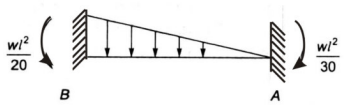
$$= \frac{\frac{3}{2}\cdot\left(\frac{wl^2}{12}\right) - \frac{wl^2}{12}}{\frac{wl^2}{12}} \times 100 = 50\%$$

450. A fixed beam AB is subjected to a triangular load varying from zero at end A to W per unit length at end B. The ratio of fixed end moment at B to A will be-

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) $\frac{2}{3}$ (d) $\frac{3}{2}$

OPSC AEE 2015 PAPER-I

Ans. (d) :



$$\frac{M_{BA}}{M_{AB}} = \frac{\frac{Wl^2}{20}}{\frac{Wl^2}{30}} = \frac{3}{2}$$

451. Maximum bending moment in a beam occurs where

(a) deflection is zero
(b) shear force is maximum
(c) shear force is minimum
(d) shear force changes sign

Rajasthan VP ITI 2018
RPSC POLY. TECH. LECT. 2011
***RPSC ACF 2011,* OPSC AEE 2015 PAPER-I**

Ans : (d) Shear force $S_{xx} = \frac{dM}{dx}$

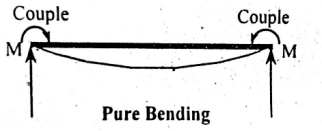
for maximum bending moment $\frac{dM}{dx}$ should zero, i.e. shear force must be zero or we can say shear force changes sign.

452. A portion of beam between two sections is said to be in pure bending, when there is-

(a) constant bending moment and constant shear force
(b) constant bending moment and zero shear force
(c) zero bending moment and constant shear force
(d) zero bending moment and zero shear force

BPSC AE Pre 2018

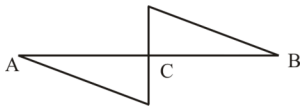
Ans. (b) :



Pure bending– Pure bending (Theory of simple bending) is a condition of stress where a bendiing moment is applied to a beam without the simultaneous presence of axial, shear or torsional forces.

Pure bending occurs only under a constant bending moment (M). So the shear force (S_{xx}), which is equal to dM/dx is equal to zero.

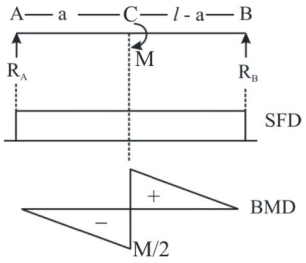
453. If the BMD for a simply supported beam is as shown below. the load on the beam will be



(a) a concentrated load at C
(b) equal and opposite couples at the ends A and B
(c) a uniformly distributed load acting on the entire span
(d) concentrated couple at C

APPSC AEE (Mains) 2016

Ans. (d) In bending moment diagram-sudden jump indicates couple and straight line means-constant shear force-no load.



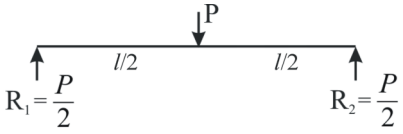
454. The ratio of maximum deflection to maximum flexural stress in a simply supported beam of span *l* and depth *d* subjected to a concentrated load at mid-span is

(a) $\frac{l^2}{(6Ed)}$ (b) $\frac{l^2}{(8Ed)}$
(c) $\frac{l^2}{(16Ed)}$ (d) $\frac{l^2}{(60Ed)}$

APPSC AEE (Mains) 2016

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Ans. (a) :



$$\sigma_{max} = \frac{M_{max} y_{max}}{I}$$

We know, max deflection

$$\therefore \quad \delta_{max} = \frac{PL^3}{48EI}$$

$$M_{max} = \frac{P}{2} \times \frac{l}{2} = \frac{Pl}{4}$$

$$y_{max} = \frac{d}{2}$$

$$\sigma_{max} = \frac{My}{I} = \frac{Pl}{4} \times \frac{d}{2I} = \frac{Pld}{8I}$$

$$\frac{\delta_{max}}{\sigma_{max}} = \frac{Pl^3}{48EI} \times \frac{8I}{Pld} = \frac{l^2}{6Ed}$$

455. A cantilever carries a uniformly distributed load from fixed end to the mid-span in the first case and a UDL of same intensity from mid-span to the free end in the second case. The ratio of maximum deflections in the two cases is-

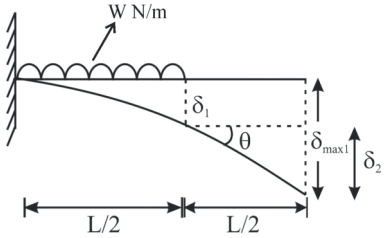
(a) $\frac{1}{3}$ (b) $\frac{3}{21}$

(c) $\frac{5}{24}$ (d) $\frac{7}{41}$

BPSC AE Pre 2018

Ans. (d) : Case – I–

As per given condition



From above figure

$$\delta_{max1} = \delta_1 + \delta_2$$

$$= \frac{wl^4}{8EI} + \theta l$$

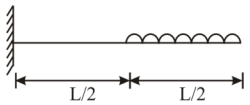
$$= \frac{wl^4}{8EI} + \frac{wl^3}{6EI} \times l$$

$$= \frac{w\left(\frac{L}{2}\right)^4}{8EI} + \frac{w\left(\frac{L}{2}\right)^3}{6EI} \times \frac{L}{2} \quad \ldots\ldots \left(l = \frac{L}{2}\right)$$

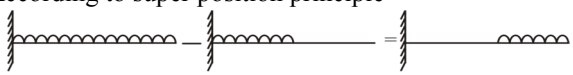
$$\boxed{\delta_{max1} = \frac{7}{384} \cdot \frac{wL^4}{EI}}$$

Case – II–

As per given condition



According to super position principle



$$\frac{wl^4}{8EI} - \frac{7}{384} \cdot \frac{wL^4}{EI} = \delta_{max2}$$

$$\delta_{max2} = \frac{wL^4}{8EI} - \frac{7wL^4}{384EI}$$

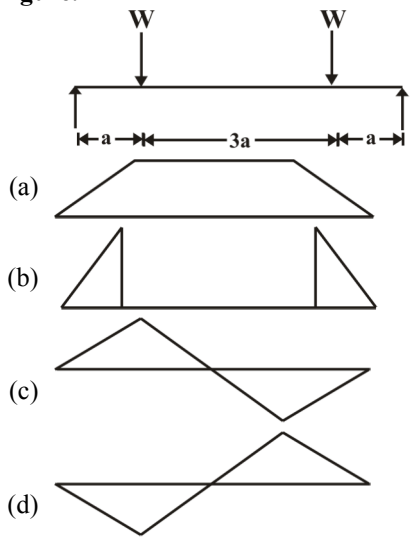
$$\delta_{max2} = \frac{41}{384} \frac{wL^4}{EI}$$

Now, ratio of maximum deflection in case 1 and case 2 is,

$$\frac{\delta_{max1}}{\delta_{max2}} = \frac{\frac{7}{384} \frac{wL^4}{EI}}{\frac{41}{384} \frac{wL^4}{EI}}$$

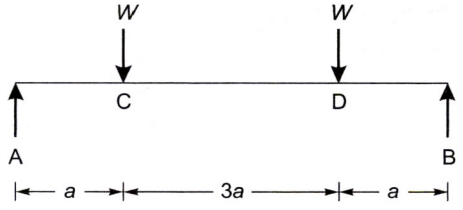
$$\boxed{\frac{\delta_{max1}}{\delta_{max2}} = \frac{7}{41}}$$

456. The bending moment diagram for the case shown in figure below will be as shown in figure.



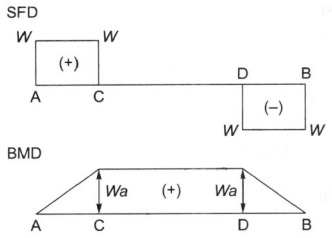
TNPSC AE 2018

Ans. (a) :



$R_A = R_B = W$, symmetrically loaded beam

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457. The variation of BM in the portion of a beam carrying uniformly varying load is

TNPSC AE 2018

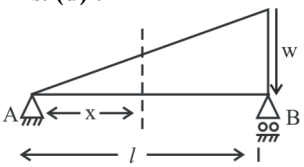
Or

The shape of the bending moment diagram over the length of a beam, carrying a uniformly increasing load is always:

(a) Constant (b) Linear
(c) Parabola (d) Cubic parabola

ISRO Scientist/Engineer 2020

Ans. (d) :



$$\sum M_B = 0 \Rightarrow R_A \times l - \frac{1}{2} \times w \times l \times \frac{l}{3} = 0$$

$$\boxed{R_A = \frac{wl}{6}}, \quad \boxed{R_B = \frac{2wl}{6}}$$

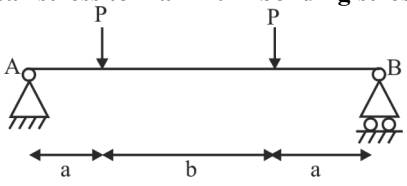
Moment at distance x from end A

$$M_X = R_A.x - \frac{1}{2} \times x \times \frac{wx}{l} \times \frac{x}{3}$$

$$M_x = \frac{wl.x}{6} - \frac{1}{6}.\frac{wx^3}{l} \qquad \text{(Cubic equation)}$$

∴ Variation of BM is cubic parabola.

458. For a rectangular beam with cross - section having width b and depth d and loaded as shown in figure, choose the ratio of maximum shear stress to maximum bending stress.

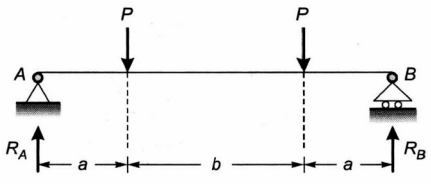


(a) $\frac{d}{4a}$ (b) $\frac{d}{2a}$

(c) $\frac{b}{4a}$ (d) $\frac{b}{2a}$

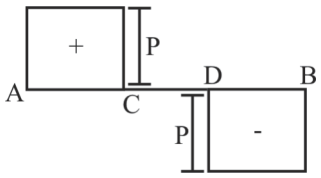
[HPPSC 2016-II]

Ans. (a) :



As beam is symmetrically loaded;

$$R_A = R_B = \frac{P+P}{2} = P$$



Maximum SF = P

Maximum shear stress = $\frac{3}{2} \times \frac{P}{bd}$

Maximum bending moment = P.a

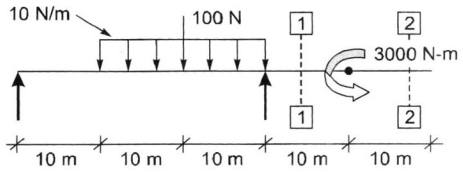
Maximum bending stress

$$= \frac{P.a \times \frac{d}{2}}{\frac{bd^3}{12}} = \frac{6Pa}{bd^2}$$

∴ Ratio of maximum shear stress to bending stress is

$$= \frac{\frac{3}{2}\frac{P}{bd}}{\frac{6Pa}{bd^2}} = \frac{d}{4a}$$

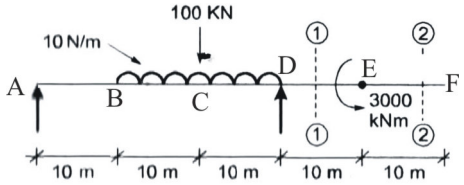
459. For the beam shown in figure, bending moment at sections 1-1 and 2-2 respectively are :



(a) +3000 N-m, –3000 N-m
(b) –3000 N-m, 0
(c) –3000 N-m, +3000 N-m
(d) +3000 N-m

[HPPSC AE 2016]

Ans. (d) :



Taking section 2-2 between E and F at distance x from F. $M_{2\text{-}2} = 0$; No force is acting.

Taking section 1-1 between D and E at distance x from F.

$M_{1\text{-}2}$ = +3000 KNm + 0

+ve as moment is sagging in nature

460. Consider the following statements :

A. The intersection of the first and the last ray in funicular diagram is on the line of action of the resultant.

B. Intersection of first and last ray in polar diagram gives the magnitude of the resultant.

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Which of the above statements is/are correct ? Select the code for the correct answer from the options given below :

(a) A only (b) B only
(c) Both A and B (d) Neither A nor B

Karnataka PSC AE 2017 Paper - II

Ans. (a) : Funicular polygon: A graphical method of finding the reactive forces and resultants of a system of forces, constructing bending moment diagrams, determining the rational shapes of arched and suspended systems and solving other problems of the statics of two dimensional systems.

Polar diagram: A diagram in which a point of origin is surrounded by a curve whose radius at any given point is proportional to the magnitude of resultant property measured in the direction of that point

Hence statement A is correct and statement B is incorrect.

461. An overhanging beam on both sides is acted upon by a uniformly distributed load (kN/m) throughout. Overhanging on each side is L_1 m and clear span is L m. For maximum positive moment equal to the negative moment, what is the ratio of L_1 to L ?

(a) 0.7071 (b) 0.3535
(c) 0.5 (d) 0.5773

JPSC AE (Pre) 2019

Ans. (b) : As in question when max. negative and max. positive BM are equal, then

$$\frac{wL_1^2}{2} = \frac{wL^2}{8} - \frac{wL_1^2}{2}$$

$$L_1 \quad \frac{L}{2\sqrt{2}} \quad 0.3535L$$

462. A simply supported beam of length 10 m is acted upon by a clockwise moment of 100 kNm at the beam, the value of shear force at left support is :

(a) 20 kN (b) 10 kN
(c) Zero (d) 1000 kN

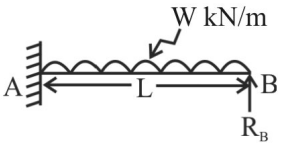
JPSC AE (Pre) 2019

Ans. (b) :

$$R \times 10 = -100 \quad R \quad 10kN$$

shear force at left support = 10kN

463. In the propped cantilever beam carrying a uniformly distributed load of w kN/m, shown in the following figure, the reaction at the support B is-



(a) $\frac{5}{8}wL$ (b) $\frac{3}{8}wL$
(c) $\frac{1}{2}wL$ (d) $\frac{3}{4}wL$

RPSC AE 2018

Ans. (b) : Deflection of free end due to UDL;

$$\delta_B = \frac{wL^4}{8EI}$$

Deflection of free end due to point load (R_B) at free end:

$$\delta_B = \frac{R_B L^3}{3EI}$$

net deflection is zero

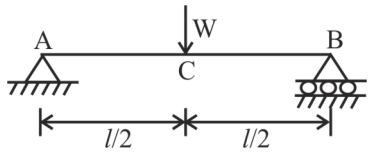
So, $$\frac{wL^4}{8EI} = \frac{R_B L^3}{3EI} \quad \Rightarrow \quad \boxed{R_B = \frac{3}{8}wL}$$

464. If a simply supported beam is loaded with point load W at the centre then what is the ratio of bending moment at the support to the bending moment at the centre?

(a) 0.6 (b) 0
(c) 1 (d) 2

GPSC AE Class (1&2) Paper - 2 2017

Ans. (b) :



In this case–

$$B.M_A = B.M_B = 0$$

and at mid point C, B.M. is maximum

$$BM_C = \frac{wl}{2}$$

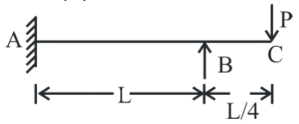
So, $$\frac{(BM)_A}{(BM)_C} = \frac{0}{\frac{wl}{2}} = 0$$

465. In a propped cantilever beam ABC, Fixed at end A and propped at B. AB = L and overhang BC = L/4. A point load "P" is applied at C will generate moment at A :

(a) PL/10 (b) PL/4
(c) PL/2 (d) PL/8

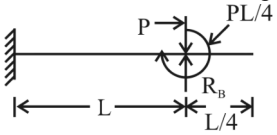
Rajasthan JEn (Degree) 2016

Ans. (d) :



Total deflection at point B = 0

Total deflection in downward due to point load P = Total deflection in upward due to Reaction R_B



$$\frac{R_B L^3}{3EI} = \frac{PL^3}{3EI} + \frac{(PL/4)L^2}{2EI}$$

$$R_B = \frac{11}{8}P$$

Bending moment at point A.

$$= -P \times \frac{5}{4}L + \frac{11}{8}P \times L = \frac{PL}{8}$$

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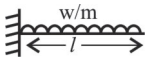
466. A beam carries uniformly distributed load throughout its length. In which of the following configuration will the strain energy be maximum?

(a) Cantilever
(b) Simply supported beam
(c) Propped cantilever
(d) Fixed

GPSC AE (CLASS 1 & 2) 2019

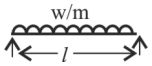
Ans : (a) :

- Cantilever beam



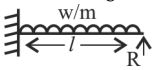
$$M_{max} = wl\frac{l}{2}$$

- Simple support beam



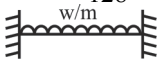
$$M_{max} = \frac{wl^2}{8}$$

- Propped cantilever



$$M_{max} = \frac{9}{128}wl^2$$

- Fixed Beam



$$M_{max} = \frac{wl^2}{12}$$

Among all, cantilever beam has maximum bending moment. So, in this beam strain energy will be maximum because

$$\text{strain energy} = \int_0^l \frac{M_x^2.dx}{2EI}$$

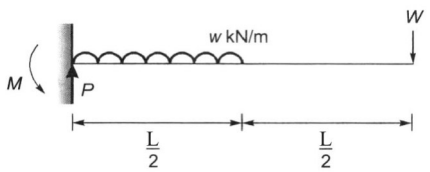
467. A cantilever beam fixed at left end carries a *udl w/*unit length over the left half portion and a point load *W* at the free end. If *L* is the length of beam, the bending moment at fixed end is:

(a) $\frac{WL}{2}+\frac{wl}{4}$ (b) $\frac{wl}{2}+\frac{WL^2}{4}$

(c) $wL+\frac{wL^2}{8}$ (d) $WL+\frac{wL^2}{8}$

Kerala PSC AE 2015

Ans. (d) :



$$P = w\times\frac{L}{2}+W \qquad (\Sigma V = 0)$$

$$M = w\times\frac{L}{2}\times\frac{L}{4}+W\times L \qquad (\Sigma M_A = 0)$$

$$= \frac{wL^2}{8}+WL$$

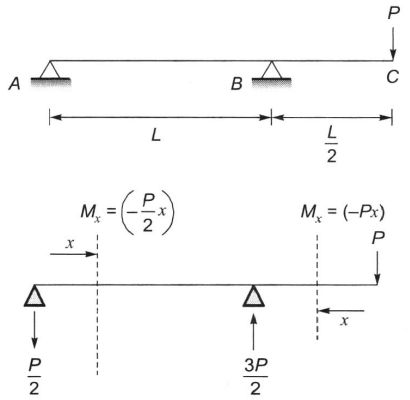
468. A beam *ABC*, is simply supported at A and B and *BC* is overhanging. *AB* = *L* and *BC* = $\frac{L}{2}$ and it carries a point load *P* at *C*. The deflection at *C* is:

(a) $\frac{PL}{24EI}$ (b) $\frac{PL^3}{8EI}$

(c) $\frac{PL^3}{48EI}$ (d) $\frac{PL^2}{16EI}$

Kerala PSC AE 2015

Ans. (b) : Method 1



$$R_A = \frac{P}{2},\ R_B = \frac{3P}{2},\ \delta_C = \frac{\partial U}{\partial P}$$

$$U = \int \frac{M^2 dx}{2EI}$$

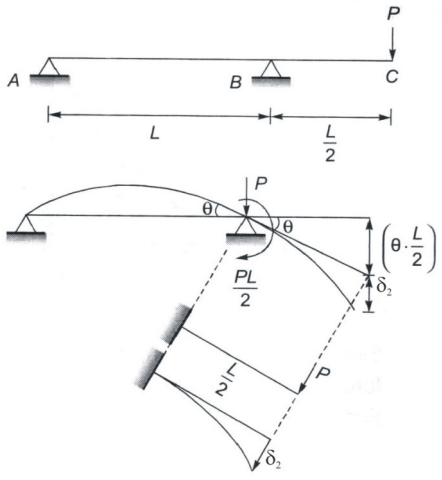
$$= \int_0^L \frac{\left(-\frac{Px}{2}\right)^2 dx}{2EI} + \int_0^{L/2} \frac{(-Px)^2 dx}{2EI}$$

$$= \frac{P^2}{8EI}\int_0^L x^2\,dx + \frac{P^2}{2EI}\int_0^{L/2} x^2\,dx$$

$$= \frac{P^2L^3}{24EI}+\frac{P^2L^3}{48EI} = \frac{3P^2L^3}{48EI}$$

$$U = \frac{P^2L^3}{16EI},\ \frac{\partial U}{\partial P} = \frac{2PL^3}{16EI} = \frac{PL^3}{8EI}$$

Method 2: Using standard result



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$$\theta = \frac{\left(\frac{PL}{2}\right)\cdot L}{3EI} = \frac{PL^2}{6EI}$$

$$\delta_2 = \frac{P\cdot\left(\frac{L}{2}\right)^3}{3EI} = \frac{PL^3}{24EI}$$

$$\delta_1 = \theta\cdot\frac{L}{2} = \frac{PL^2}{6EI}\times\frac{L}{2} = \frac{PL^3}{12EI}$$

$\therefore$ Deflection at C:

$$\delta_C = \frac{PL^3}{12EI} + \frac{PL^3}{24EI} = \frac{2PL^3 + PL^3}{24EI} = \frac{PL^3}{8EI}$$

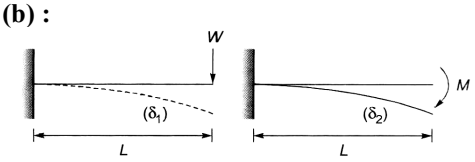
469. For a cantilever beam of length *L*, what bending moment at free end would produce a deflection equal to that produced by a concentrated load *W* at free end?

(a) WL (b) $\frac{2}{3}WL$

(c) $\frac{2}{3}W$ (d) $\frac{WL}{EI}$

Kerala PSC AE 2015

Ans. (b) :



M = ? for $\delta_1 = \delta_2$

$$\delta_1 = \frac{WL^3}{3EI} \text{ and } \delta_2 = \frac{ML^2}{2EI}$$

$$\frac{WL^3}{3EI} = \frac{ML^2}{2EI}$$

$$M = \frac{2}{3}WL$$

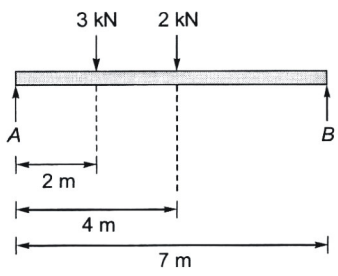
470. A simply supported beam of span 7 m has a point load of 3 kN at a distance of 2 m from left end A and a point load of 2 kN at a distance of 4 m from left end A. What will be the support reaction at B?

(a) 3 kN (b) 2.5 kN

(c) 2 kN (d) $\frac{19}{7}$ kN

Kerala PSC AE 2015

Ans. (c) :



$\Sigma M_A = 0$

$3 \times 2 + 2 \times 4 - R_B \times 7 = 0$

$6 + 8 = R_B \times 7$

$R_B = 2$ kN

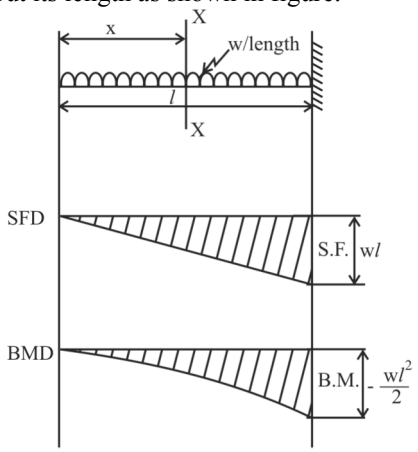
471. The bending moment diagram for a cantilever carrying uniformly distributed load will be:

(a) A triangle (b) A parabola

(c) A cubic parabola (d) A rectangle

KPSC AE 2016

Ans. (b) : Consider a cantilever beam of length *l* subjected to uniformly distributed load w N/m throughout its length as shown in figure.



Hence bending moment diagram for cantilever carrying uniformly distributed load will be parabola.

472. A beam with uniform strength refers to which of the following?

(a) Where moment of inertia remains same all along the section

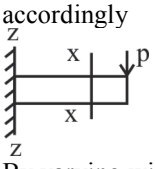
(b) Where bending stress is uniform at maximum bending moment of the section

(c) If the loading passes through the shear centre, the beam doesn't twist

(d) If the beam has two axes of symmetry, the shear center coincides with the centroid

GPSC AE (CLASS 1 & 2) 2019

Ans : (b) : An uniform strength beam is when bending stress (σ_b) is uniform at maximum bending moment & used economically by varying width (b) & depth (d) accordingly



By varying width:-

$$\boxed{B_x = \frac{xb}{L}}$$

By varying depth:-

$$d_x = \sqrt{\frac{x}{L}}\times d$$

473. A timber beam, 100 mm wide and 150 mm deep, supports a UDL over a span of 2 m. If the safe stresses are not to exceed 28 MPa in bending and 2 MPa in shear, the maximum load that the beam can support is

(a) 16 kN/m (b) 20 kN/m

(c) 24 kN/m (d) 28 kN/m

ESE 2019

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Ans. (b)



$F_{max} = 28MPa$

$\tau_{max} = 2MPa$

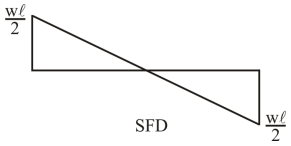
For rectangular section,

$$\tau_{max} = \frac{3}{2}\left[\tau_{avg}\right] = \frac{3}{2}\left[\frac{F}{bd}\right]$$

$$2 = \frac{3}{2}\left[\frac{F}{100\times150}\right]$$

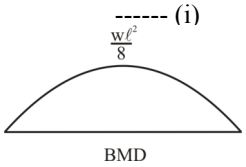
$$\frac{F}{15000} = \frac{4}{3} \Rightarrow F = \frac{4\times15000}{3}$$

[F=20kN]



$$F = \frac{w\ell}{2} \qquad 20 = \frac{w\times2}{2}$$

w = 20kN/m ------ (i)



$$F_{max} = \frac{M}{Z} - \Rightarrow 28 = \frac{M}{\left[\frac{100\times150^2}{6}\right]}$$

M = 10.5 kN–m

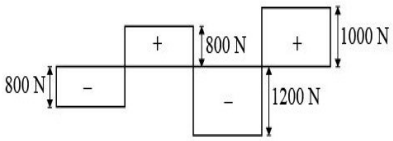
$$\frac{W\ell^2}{8} = 10.5 \Rightarrow \frac{W\times(2)^2}{8} = 10.5$$

[w = 21kN/m] ------ (ii)

use minimum of (i) and (ii)

Maximum Safe Load 'w' = 20kN/m

474. The shear-force diagram of a beam is shown in the figure

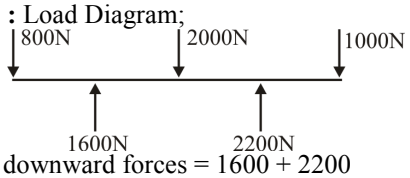


The total of the vertically downward loads on the beam is

(a) 2600 N (b) 3000 N
(c) 3400 N (d) 3800 N

ESE 2019

Ans. (d) : Load Diagram;



Vertical downward forces = 1600 + 2200

Loads = 3800N

475. The change in shearing force between two points on the beam is equal to the area of

(a) Loading diagram between the two points
(b) Shear force diagram between the two points
(c) Bending moment diagram between the two points
(d) M/EI diagram between the two points

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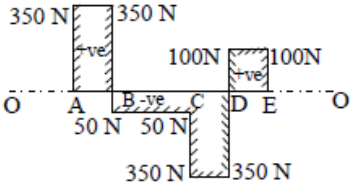
Ans. (a) : Change in shear force between any two points equal to area of loading diagram.

$$\frac{dF_{xx}}{dx} = w$$

$$dF_{xx} = w.dx$$

where F_{xx} = shear force
w = load per length

476.

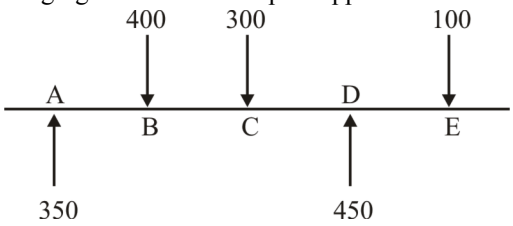


The shear force diagram of a single overhanging beam is shown in figure. One simple support is at end *A*. The 'total' downward load acting on the beam is

(a) 800 N (b) 600 N
(c) 400 N (d) 200 N

ESE 2018

Ans. (a) : For the given shear force diagram of a single overhanging beam at one simple support at end A.



Now, total downward load = 400 + 300 + 100 = 800 N

477. The maximum bending moment at a given section, in which a train of wheel loads moves, occurs when the average load on the left segment is

1. Equal to the average load on the right segment.
2. More than the average load on the right segment
3. Less than the average load on the right segment.

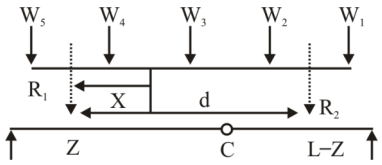
Select the *correct* answer using the codes given below.

(a) 1, 2 and 3 (b) 1 only
(c) 2 only (d) 3 only

ESE 2017

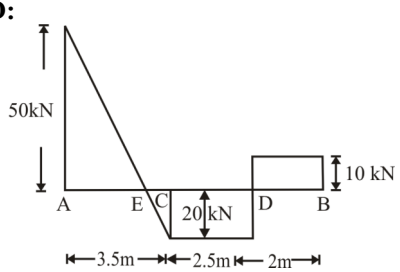
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Ans. (b)



- For train of wheel loads moving on the given span, the maximum bending moment at a given section occurs when average load on the left segment = Average load on the right segment.

478. Consider the following statements with regards to the shear force diagram for the beam ABCD:



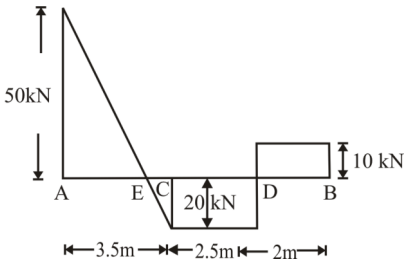
1. The beam ABCD is an overhanging beam having supports at A and D only.
2. The beam carries a point load of 20 kN at C.
3. The beam carries a concentrated load of 10 kN at the end B.
4. The beam is an overhanging beam having supports at C and D only.
5. The beam carries a uniformly distributed load of 70 kN over the left hand portion AC only.

Which of the above statements are *correct*?

(a) 1, 2 and 3 only (b) 1, 3 and 5 only
(c) 2, 3 and 4 only (d) 2, 4 and 5 only

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Ans. (b)



$$W = \frac{dV}{dx}$$

At A, w = 50 kN

Between A and C, $W = \frac{(-20)-(50)}{3.5}$

$\boxed{W = -20\text{kN/m}}$

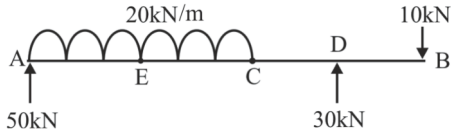
Uniformly distributed load of 20 kN/m

Between C and D; $W = \frac{20-20}{2.5} = 0$

At D, W = – (–20) + 10 = + 30 kN

Between D and B; $W = \frac{10-10}{2} = 0$

At B, $\boxed{W = -10\text{kN}}$



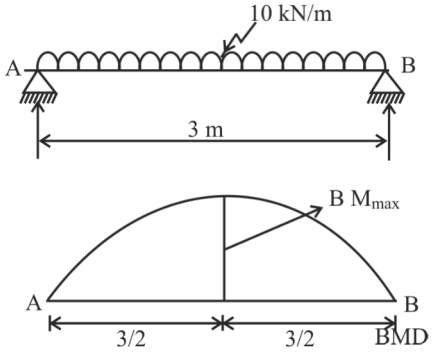
- The beam carries a point load of 20kN at C is wrong. There must be a sudden jump of 20kN at C if there is point load.

479. A simply supported beam carrying a uniformly distributed load of 10 kN/m per unit over the whole span of 3m. What will be the maximum bending moment of the beam?

(a) 11.25 kN-m (b) 9 kN-m
(c) 15.17 kN-m (d) 8.34 kN-m

DSSSB AE 2019

Ans. (a)



Maximum bending moment $BM_{max} = \frac{wl^2}{8}$

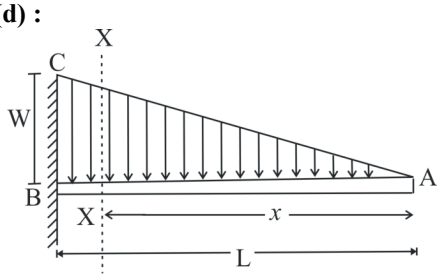
$$= \frac{10(3)^2}{8} = 11.25 \text{ kN-m}$$

480. The maximum bending moment of a cantilever beam of length L and carrying a gradually varying load from zero at free end to *w* per unit length at the fixed end is _____.

(a) $wL/2$ (b) wL
(c) $wL^2/2$ (d) $wL^2/6$

ISRO Scientist/Engineer 2019

Ans : (d) :



At section, X–X–

$$M_x = \frac{-wx}{L} \times \frac{1}{2} \times x \times \frac{x}{3} = \frac{-wx^3}{6L}$$

The bending moment variation is cubic

at x = 0, $M_x = 0$

at x = L, $M_x = \frac{-wL^2}{6}$

$$\left[\text{i.e. Maximum Bending Moment}(M_{max}) = \frac{wL^2}{6}\right]$$

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481. A simply supported beam of span L carries a total W which is uniformly distributed. The maximum bending moment M is

(a) WL/2 (b) WL/4

(c) WL/12 (d) WL/8

ISRO Scientist/Engineer 2015

Ans : (d) : For simply supported beam with uniformly distributed load,

Maximum bending moment $= \frac{wL^2}{8}$

$\because$ Total load W = wL

So, $BM_{max} = \frac{WL}{8}$

482. The shear force on a simply supported beam is proportional to

(a) Displacement of the neutral axis

(b) Sum of the moments divided by the effective depth

(c) Sum of transverse forces

(d) Algebraic sum of the transverse forces of the section

ISRO Scientist/Engineer 2018

Haryana PSC AE (Paper–I) 2018, 2017

Maharashtra PSC Mains 2018 Paper-1

Ans : (d) : The shearing force at the section is defined as the algebraic sum of transverse components of all forces acting on the beam, either from left or from the right of the section.

Shear force is the force in the beam acting perpendicular to its longitudinal axis.

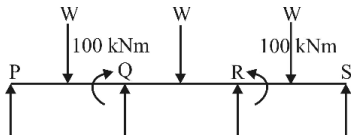
483. A continuous beam PQRS is supported over three equal spans PQ = QR = RS. Each span carries a concentrated load W at the centre. If the support moment at Q is –100 kNm, what is the support moment at R

(a) –100 kNm (b) +100 kNm

(c) –50 kNm (d) +50 kNm

APTRASCO 2017

Ans : (b) :



If moment at support Q is –100 kNm, then to balance this moment, +100 kNm will act at support R.

so, support moment at R = +100 kNm.

$\boxed{M_R = +100\,kNm}$ (because at P and S, M = 0)

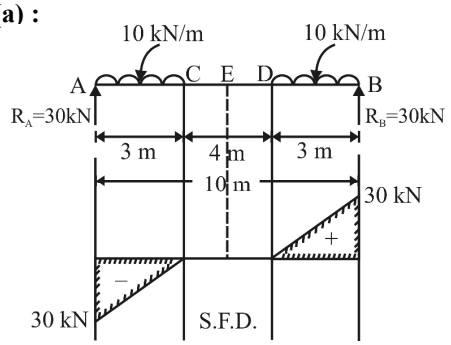
484. A beam of 10 m long is simply supported at the ends. It carries a UDL of 10 kN/m upto a distance of 3 m from either end. The shear force at the centre of the beam is

(a) 0.0 (b) 30 kN

(c) 10.0 kN (d) 50.0 kN

APTRASCO 2017

Ans : (a) :



$R_A = R_B = \frac{30+30}{2} = 30\,kN$

S.F at center of beam is zero.

485. Which of the following loads should be applied on a simply supported beam, so that the shear force is constant throughout its span?

(a) U.D.L. over the entire span

(b) Two concentrated loads equally spaced in the span

(c) A central concentrated load and a U.D.L. over the entire span

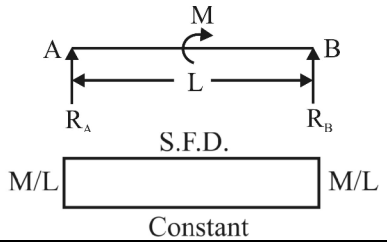
(d) A couple anywhere in the span

APTRANSCO 2017

AP TRANSCO AEE 2017

ESE 2004

Ans : (d) : If a couple is applied anywhere in simply supported beam then S.F will be constant throughout the beam.



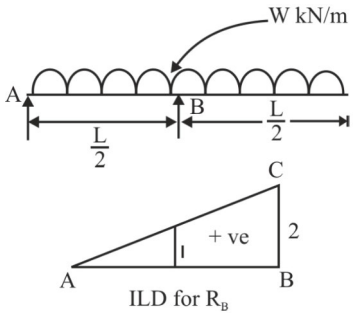
486. A simply supported beam of length L over hanged to the right by $\frac{L}{2}$ at support B. The maximum reaction at support B will be when the uniformly distributed load is loaded :

(a) Only in AB (b) Only in BC

(c) Entire span (d) None of these

OPSC AEE 2019 Paper–I

Ans. (c)



To get the max reaction at B, place the UDL over entire span.

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Max. reaction at B = Intensity of UDL × Area of ILD under the UDL

$$= W\left[\frac{1}{2}\times L\times 2\right] = W\,L \text{ kN}$$

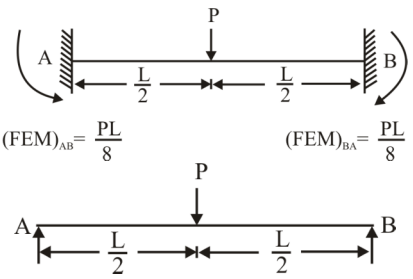
487. Due to some point load anywhere on a fixed beam, the maximum free bending moment is M. The sum of fixed end moment is

(a) M (b) 1.5 M

(c) 2.0 M (d) 3.0 M

OPSC AEE 2019 Paper–I

Ans. (a)



The maximum free bending moment (M)

$$= \frac{PL}{4}$$

The sum of fixed end moment $= \frac{PL}{8} + \frac{PL}{8}$

$$= \frac{PL}{4}$$

$$= M$$

488. Choose the correct statement :

(a) At the point of contraflexure, the shear force changes sign.

(b) At the point of contraflexure, the bending moment is not zero.

(c) At the point of contraflexure, the bending moment is maximum.

(d) At the point of contraflexure, the gradient of elastic line changes sign.

OPSC AEE 2019 Paper–I

Ans. (d) : In a bending beam, a point is known as a point of contra flexure. It is a location where bending moment is zero (changes its sign). In a bending moment diagram, it is the point at which the bending moment curve intersects with the zero line.

So that, at the point of contraflexure, the gradient of elastic line changes sign.

489. A simply supported beam of span L carries over its full span of load varying linearly from zero at either end to W/unit length at midspan. The maximum bending moment occurs at :

(a) Quarter points and is equal to $\frac{WL^2}{8}$

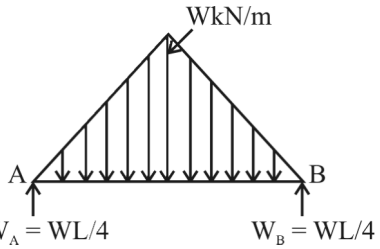
(b) Quarter points and is equal to $\frac{WL^2}{12}$

(c) Midspan and is equal to $\frac{WL^2}{8}$

(d) Midspan and is equal to $\frac{WL^2}{12}$

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Ans. (d)



$$M_{max} = M_{midpoint}$$

$$= R_A\left(\frac{L}{2}\right) - \left(\frac{1}{2}.W\frac{L}{2}\right)\left(\frac{L}{2}.\frac{1}{3}\right)$$

$$= \frac{WL}{4}\left(\frac{L}{2}\right) - \frac{WL^2}{24} = \frac{WL^2}{12}$$

490. If the area under the shear force diagram for a beam between the two points C and D is 'K", then the difference between the moments at the two points C and D will be equal to :

(a) K (b) 2K

(c) $\frac{K}{2}$ (d) K^2

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Ans. (a)

$$\frac{dM_x}{dx} = S_{xx}$$

$$\int_D^C dM_x = \int_D^C S_{xx}.dx = \text{Area under the shear force diagram}$$

$$\boxed{M_C - M_D = K}$$

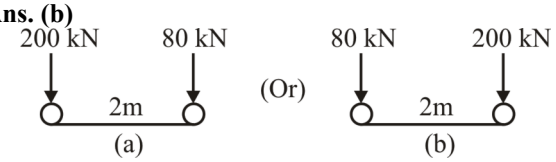
491. Two wheel loads 200 kN and 80 kN spaced at 2 m apart move on the span of girder of 16 m. If any wheel load can lead the other, the maximum bending moment that can occur at a section of 6 m from the left end will be :

(a) 1050 kNm (b) 990 kNm

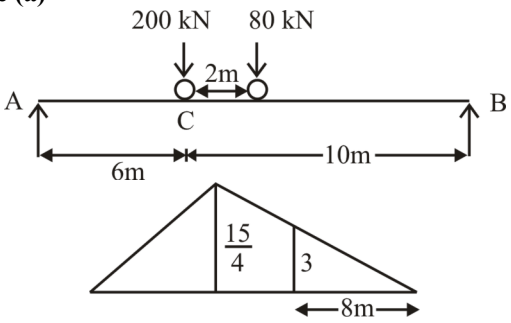
(c) 870 kNm (d) 750 kNm

ESE 2020

Ans. (b)



Case (a)

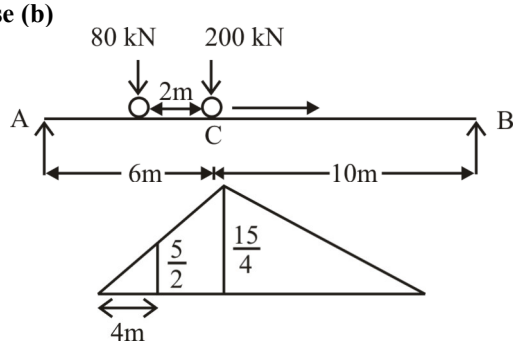


Max^m B.M. at C $= 200\times\frac{15}{4} + 80\times 3$

$= 990$ kN-m ..... (ii)

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Case (b)



$$\text{Max}^{m}\ \text{B.M. at C} = 200\times\frac{15}{4}+80\times\frac{5}{2}$$

$$= 950\ \text{kN–m} \ldots\ldots\ldots\ldots\ldots \text{(i)}$$

from eq^{n} (i) & (ii), we can say that maximum bending moment occur at section 6m from left end will be 990 kN-m.

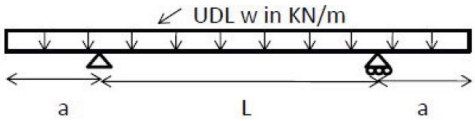
492. The number of contra flexure points that occur in a cantilever beam subjected uniformly distributed load is

(a) 3 (b) 2
(c) 1 (d) 0

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Ans. (d) : The number of contra flexure points that occur in a cantilever beam subjected uniformly distributed load is zero.

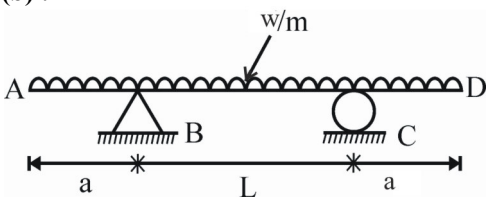
493. A beam with equal overhangs is carrying UDL as shown in the following figure. The bending moment at the centre of the beam will be zero for the condition



(a) a = L/3 (b) a = L/2
(c) a = 3L/2 (d) a = L

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Ans. (b) :



$$R_A + R_C = w(L+2a) \qquad \ldots\ldots(i)$$

$$\sum M_B = 0$$

$$R_C\times L + w\times a\times\frac{a}{2} = w\times L\times\frac{L}{2} + wa\times\left(L+\frac{a}{2}\right)$$

$$R_C\times L = \frac{wL^2}{2}+\frac{waL}{2}+\frac{wa^2}{2}-\frac{wa^2}{2}$$

$$\therefore \qquad R_C = \frac{w}{2}(L+2a)$$

from equation (i)

$$R_B = \frac{w}{2}(L+2a)$$

S.F.in $AB(x \text{ from } A)$

$$S.F_x = -wx$$

$$\text{S.F. at } A(x=0) = 0$$

$$\text{S.F. at } B(x=a) = -wa$$

S.F. in BC(x from A)

$$S.F_x = R_B - w(x+a)$$

$$= \frac{w}{2}(L+2a) - w(x+a)$$

$$S.F_x = \frac{wL}{2} + wa - wx - wa$$

$$S.F_x = \frac{wL}{2} - wx$$

$$\text{S.F. at } B(x=a) = \frac{wL}{2} - wa$$

$$S.F_x \text{ at } C(x=L+a) = \frac{wL}{2} - w(L+a)$$

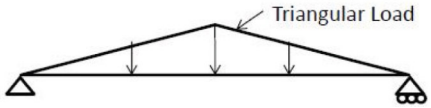
$$= \frac{wL}{2} - wa$$

So, if we put, $a = \frac{L}{2}$

Then, at the center (C)

$$SF_C = \frac{wL}{2} - \frac{wL}{2} = 0$$

494. A simply supported beam is carrying a linearly varying load from zero at either ends to the maximum value at the mid span. Then the shape of shear force diagram (SFD) is



(a) Rectangle
(b) Triangular
(c) Second Degree Parabola
(d) Cubic

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Ans. (c) : In this case shape of shear force diagram (SFD) is second degree parabola and bending moment diagram (BMD) is cubic.

495. Shear force at a section of a horizontal beam is the

(a) algebraic sum of the vertical forces to any one side of the section
(b) algebraic sum of the vertical forces on both sides of the section
(c) algebraic sum of the moments of all forces on any one side of the section
(d) algebraic sum of the moments of all forces on both sides of the section

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Ans. (a) : Shear force at a section of a horizontal beam is the algebraic sum of the vertical force to any one side of the section.

496. For UDL (uniformly distributed load) on a beam, the bending moment diagram (BMD) shows

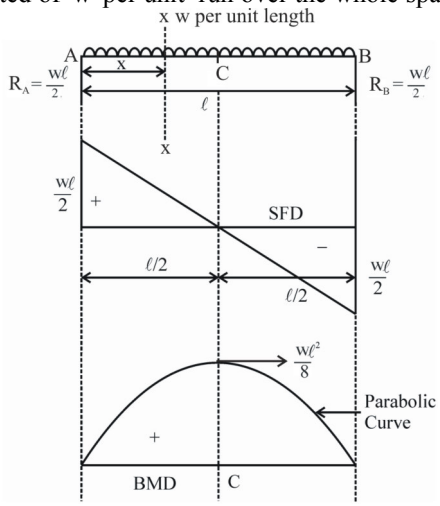
(a) Linear variation
(b) Parabolic (curve) variation

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(c) Rectangle or constant value
(d) Trapezoidal variation

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Ans. (b) : Simply supported beam carrying a uniformly distributed of 'w' per unit run over the whole span.



$$S.F_x = \frac{+w\ell}{2} - wx \quad \text{(Linear equation)}$$

$$BM_x = \frac{wlx}{2} - \frac{wx^2}{2} \text{ (parabolic equation)}$$

497. Bending moment is maximum at a section of a beam where
(a) Shear force changes from –ve to +ve
(b) Shear force is zero
(c) Shear force changes from +ve to –ve
(d) All the given answers

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Ans. (d) : Bending moment is maximum at a section of a beam where-
: Shear force changes from –ve to +ve.
: Shear force is zero.
: Shear force changes from +ve to –ve

498. Which of the following is giving correct relation between load (w), shear force (F) and bending moment (M)
(a) $M = \frac{dF}{dx}$ (b) $F = \frac{dM}{dx}$
(c) $w = \frac{dM}{dx}$ (d) $M = \frac{dw}{dx}$

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Ans. (b) : Correct relation between load(w) shear force (F) and bending moment (M).

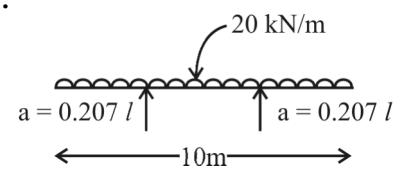
$$\boxed{F = \frac{dM}{dx}}, \quad w = \frac{dF}{dx}$$

and $$w = \frac{d^2M}{dx^2}$$

499. A beam of length 10 m carries a uniformly distributed load of 20 kN/m over its entire length and rests on two simple supports. In order that the maximum BM produced in the beam is the least possible, the supports must be placed from the ends at a distance of
(a) 5.86 m (b) 4.14 m
(c) 2.93 m (d) 2.07 m

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ESE 1998

Ans. (d) :

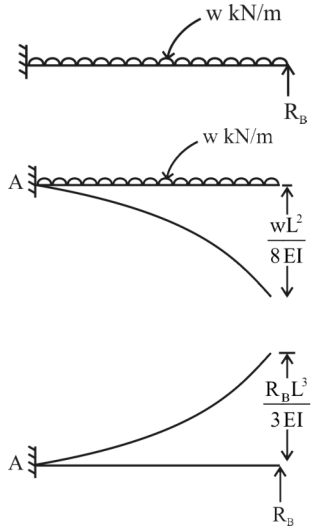


Maximum bending moment in beam is least possible, when overhang = 0.207 l
$\therefore$ a = 0.207 l = 0.207 × 10 = 2.07 m

500. A beam of span l is fixed at one end and simply supported at other end. It carries uniformly distribution load of w per unit run over the whole span. The reaction (R) at the simply supported end is:
(a) $R = \frac{3}{8}wl$ (b) $R = \frac{5}{8}wl$
(c) $R = \frac{1}{2}wl$ (d) $R = \frac{1}{3}wl$

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Ans. (a) :



Upward deflection = Downward deflection

$$\frac{R_BL^3}{3EI} = \frac{wL^2}{8EI}$$

$$R_B = \frac{3wL}{8}$$

501. Maximum shear force for three equal spans of beam/slab occur at
(a) inner side of end support
(b) inner side of support next to end support
(c) outer side of support next to end support
(d) outer side of end support

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Ans. (c) :

Shear force = (shear coefficient) × design load

= (Shear coefficient) × wl

since all spans are of same length maximum shear force occurs on the support having maximum shear coefficient.

As per table 13 of IS 456 (cl 22.51 to 22.52) shear force coefficient is maximum at outer side of support next to end support = 0.6.

502. The expression EI(d^4y/dx^4) at any section for a beam is equal to

(a) load intensity at the section
(b) S. F. at the section
(c) B. M. at the section
(d) the slope at that section

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Ans. (a) :

Load intensity $\boxed{w = \frac{d^2M}{dx^2}}$

$\because \quad \frac{d\theta}{dx} = \frac{M}{EI}$

so, $\frac{d^2y}{dx^2} = \frac{M}{EI}$

Hence,

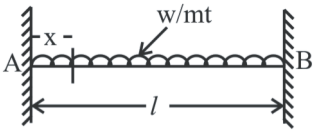
$$\boxed{w = \left(\frac{d^4y}{dx^4}\right)EI}$$

503. A prismatic beam of length L and fixed at both ends carries a uniformly distributed load. The distance between points of contra flexure is

(a) 0.207 L (b) 0.211 L
(c) 0.277 L (d) 0.25 L

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Ans. (*) :



At any section at a distance x from the left end A the BM is given by

$$M_x = R_A.x - \frac{wx^2}{2} - \frac{wL^2}{12}$$

$$M_x = \frac{wL}{2}x - \frac{wx^2}{2} - \frac{wL^2}{12}$$

for points of contraflexure,

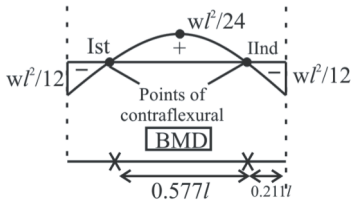
$M_x = 0$

$$\frac{wL}{2}x - \frac{wx^2}{2} - \frac{wL^2}{12} = 0$$

$$Lx - x^2 - \frac{L^2}{6} = 0$$

$$\Rightarrow x^2 - Lx + \frac{L^2}{6} = 0$$

$$x = \frac{L}{2} \pm \frac{L}{2\sqrt{3}}$$



$\therefore$ distance of point of contraflexure from each ends (x_1) = 0.211 l and 2nd contraflexure point from point A (x_2) = 0.788 l.

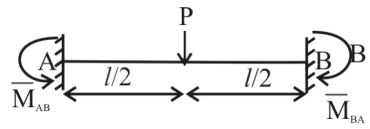
$\therefore$ distance between points contraflexure= 0.577 l

504. Fixed end moments for a beam fixed at both ends and centrally loaded with a concentrated load P are,

(a) $\frac{-Pl}{8}, \frac{Pl}{8}$ (b) $\frac{-Pl^2}{30}, \frac{Pl^2}{30}$
(c) $\frac{-Pl}{4}, \frac{Pl}{4}$ (d) $0, \frac{Pl}{4}$

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Ans. (a) :



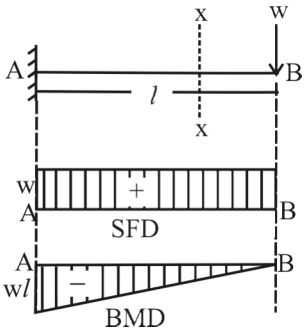
$\overline{M}_{AB} = \frac{-Pl}{8}, \overline{M}_{BA} = \frac{Pl}{8}$

505. If a cantilever beam is subjected to a point load at its free end, then the shear force under the point load is

(a) zero (b) less than the load
(c) equal to the load (d) more than the load

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Ans. (c) :



$S_x = +W$

$M_x = -W_x$

$M_{max} = -wl$

If cantilever beam is subjected to a point load at its free end, then the shear force under the point loads is equal to the load.

506. A beam with uniformly distributed load W, over the entire span of beam l, beam is supported at one end only. What will be bending moment at support?

(a) $\frac{Wl^2}{8}$ (b) $\frac{Wl}{8}$
(c) $\frac{Wl^2}{2}$ (d) $\frac{Wl}{8}$

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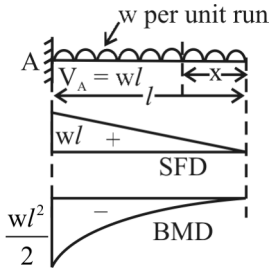
Ans. (c) :

$S_x = + w_x$

$M_x = -\frac{wx^2}{2}$

$M_{max} = -\frac{wl^2}{2}$

$S_{max} = +wl$



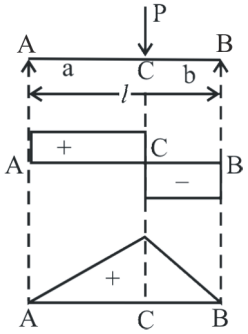
507. A simply supported beam AB, having eccentric point load P, placed at C, so that AC = a, CB = b, length of beam is *l*, bending moment at point C is.

(a) $\frac{Pab}{2l}$ (b) $\frac{2Pab}{l}$

(c) $\frac{Pab^2}{l}$ (d) $\frac{Pab}{l}$

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Ans. (d) : Simply supported beam carrying a point load placed eccentrically on the span–



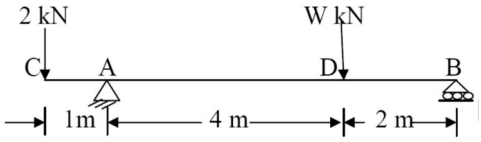
$S_x = +\frac{Pb}{l}$ (from A to C)

$S_x = -\frac{Pa}{l}$ (from C to B)

$M_x = +\frac{Pb}{l} \cdot x$ (from A to C) at a, distance 'x' from A

$$M_{max} = M_C = \frac{Pab}{l}$$

508. Beam AB is supported as shown at A and B and has an overhang AC; and it carries loads as shown in the figure. Which of the following will give the value of W so that $R_A = R_B$?

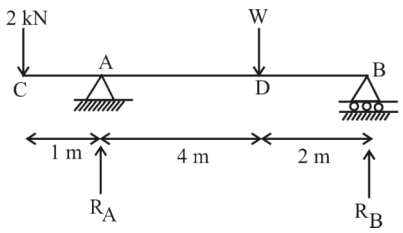


(a) 4 kN (b) 6 kN

(c) 8 kN (d) 10 kN

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Ans. (c) :



$\sum F_y = 0$

$R_A + R_B = 2 + W$ .....(i)

$(M_z)_A = 0$

$R_B \times 6 + 2 = 4\ W$ .....(ii)

then $R_A = R_B$ given

$2R_B = 2 + W$ .....(iii)

$3 \times (2R_B) + 2 = 4W$

$3 \times (2 + W) + 2 = 4W$

$6 + 3W + 2 = 4W$

$8 = 4W - 3W$

$W = 8$ kN

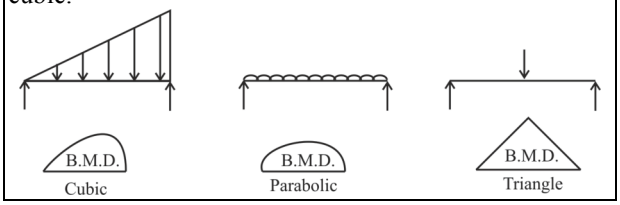
509. The variation of the bending moment in the portion of a beam carrying linearly varying load is :

(a) linear (b) parabolic

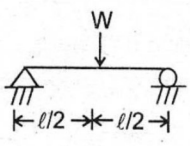
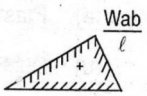
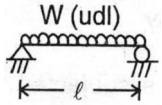
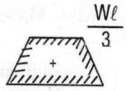
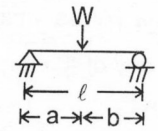
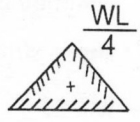
(c) cubic (d) constant

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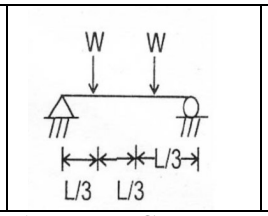
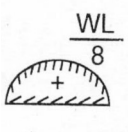
Ans. (c) : The variation of the bending moment in the portion of a beam carrying linearly varying load is cubic.



510. Match the List-I with List-II and select the correct answer by using codes given below the lists :

	List-I		List-II
	(Simply supported beams)		**Bending moment diagrams**
A		**1**	
B		**2**	
C		**3**	

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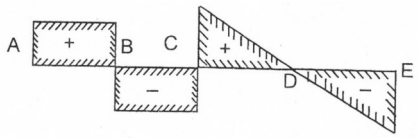
D		4	

	A	B	C	D
(a)	4	1	2	3
(b)	3	4	1	2
(c)	1	2	3	4
(d)	2	3	4	1

OPSC Civil Services Pre 2011

Ans. (b) :

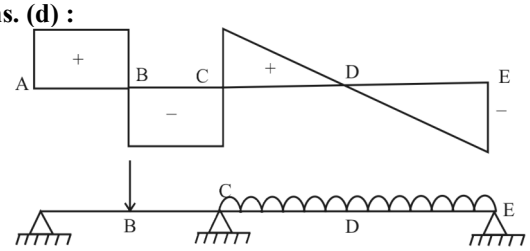
511. Consider the following statement with reference to a continuous beam supported at A, C and E for which shear force diagram is shown in fig. below
There is :
1. A concentrated load acting at point B
2. A concentrated load acting at point A
3. A uniformly distributed load acting in the portion AC
4. A uniformly distributed load acting in the portion CE



(a) 1 and 2 are correct (b) 2 and 3 are correct
(c) 3 and 4 are correct (d) 1 and 4 are correct

OPSC Civil Services Pre 2011

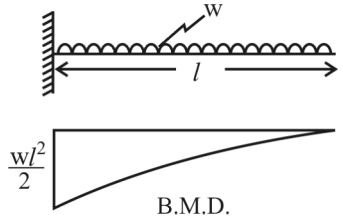
Ans. (d) :



512. What is the maximum bending moment in a cantilever beam of length *l* and subjected to UDL of w / unit length?
(a) w/*l* (b) w*l*/2
(c) wl^2/2 (d) wl^2/6

Rajasthan Nagar Nigam AE Shift-2, 2016

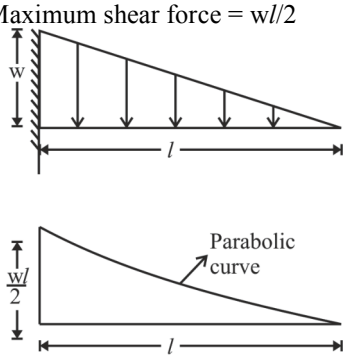
Ans. (c) : In a cantilever beam the maximum bending moment occurs at the fixed end. Moment at the free end is zero and maximum at the fixed end.



513. What is the maximum shear force in cantiliver beam subjected to uniformly varying load of w/unit length?
(a) wl^2/2 (b) w*l*
(c) wl^2/6 (d) w*l*/2

Rajasthan AE Nagar Nigam Shift-3, 2016

Ans. (d) : Maximum shear force = w*l*/2



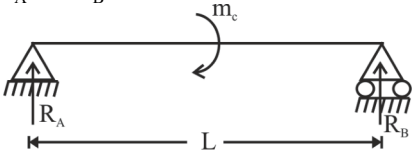
514. A simply supported beam is subjected to external moment of M_C at center of beam. What is the shear force at the point of application of moment?
(a) 2 M_C / L (b) M_C / L
(c) M_C / 2 (d) L / M_C

Rajasthan AE Nagar Nigam Shift-3, 2016

Ans. (b) : $\Sigma F_y = 0$

$R_A + R_B = 0$

$R_A = -R_B$



$\Sigma M_A = 0$

$R_B \times L - M_c = 0$

$R_B \times L = M_c$

$$R_B = \frac{M_c}{L}$$

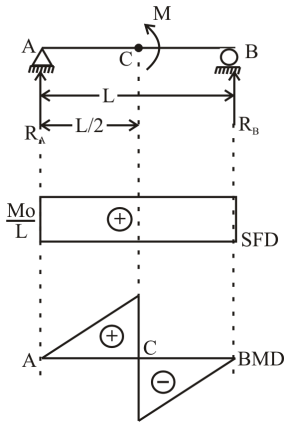
So, $R_A = -R_B = -M_c/L$

So, shear force at point of application of moment = M_c/L.

515. A simply supported beam, of effective span L, is subjected to a concentrated moment M at its center. Maximum bending moment in the beam is
(a) M (b) M/4
(c) (M × L)/4 (d) M/2

Sikkim PSC AE 2017

Ans. (d) :



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Due to symmetry,

$$R_A = R_B = \frac{M_o}{L}$$

$M_C - M_A$ = Area under SFD between A and C

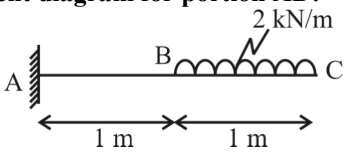
$$\Rightarrow \quad M_c = \frac{M_o}{L} \times \frac{L}{2}$$

$$\Rightarrow \quad \boxed{M_c = \frac{M_o}{2}}$$

From BMD, it is clear that, max. B.M. occurs at C,

$$\boxed{M_{max} = \frac{M_o}{2}}$$

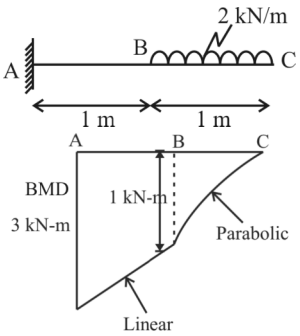
516. A cantilever beam is loaded as shown in the figure below. What is the shape of bending moment diagram for portion AB?



(a) Parabolic
(b) Linearly varying with maximum bending moment at B
(c) Constant bending moment from A to B
(d) Lineraly varying with maximum bending moment at A

TANGEDCO AE 2018
ESE 2005

Ans. (d) :



B.M. at point C, $M_C = 0$

$M_B = 2 \times \frac{1}{2} = 1$ kN-m

$M_A = 2 \times \left(1 + \frac{1}{2}\right) = 2 \times \frac{3}{2} = 3$ kN-m

517. For a overhanging beam the expression for bending moment is given as $2.5x - \frac{3x^2}{2}$ kNm. Then the location of maximum bending moment is

(a) $\frac{5}{3}$ (b) $\frac{2.5}{3}$

(c) $\frac{3}{2.5}$ (d) $\frac{3}{5}$

TRB Poly. Tech. Lect. 2012

Ans. (b) : Given

$$M = 2.5x - \frac{3}{2}x^2$$

For Max.B.m

$$\frac{dM}{dx} = 0$$

$$0 = 2.5 - \frac{3}{2} \times 2x$$

$$\Rightarrow 3x = 2.5$$

$$\boxed{x = \frac{2.5}{3}}$$

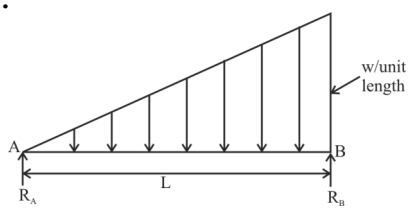
518. Simply supported beam of length L is loaded with a triangularly varying load intensity, starting from zero at the left end to W at the right end. Find the maximum shear force

(a) $\frac{WL}{3}$ (b) $\frac{WL}{6}$

(c) $\frac{2WL}{3}$ (d) WL

TSGENCO AE 2015

Ans. (a) :



$$\sum M_A = 0 \Rightarrow R_B \times L = \frac{WL}{2} \times \frac{2L}{3}$$

$$R_B = \frac{WL}{3} kN$$

$$R_A = \frac{WL}{2} - \frac{WL}{3} = \frac{WL}{6} kN$$

Maximum S.F. is at support $B = \frac{WL}{3}$ kN

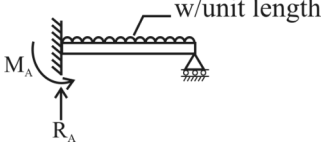
519. A propped cantilever beam of span *l* is loaded with uniformly distributed load of intensity w/unit length all through the span, bending moment at the fixed end is

(a) $\frac{wl^2}{8}$ (b) $\frac{wl^2}{2}$

(c) $\frac{wl^2}{12}$ (d) $\frac{wl^2}{24}$

UKPSC AE (Urban Engineering)-2011

Ans. (a) :



$$\frac{WL^4}{8EI} = \frac{R_B L^3}{3EI}$$

$$R_B = \frac{3WL}{8}$$

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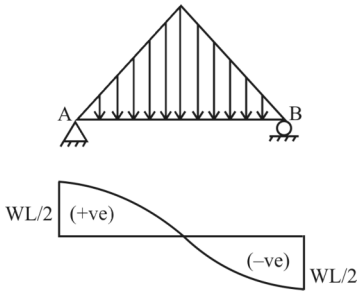
$$M_A = R_B \times (L) - W(L)(L/2)$$
$$= \frac{3WL^2}{8} - \frac{WL^2}{2}$$
$$M_A = \frac{1}{8}WL^2$$

520. A simply supported beam of span L which carries over its full span a load varying uniformly from zero at either end to W kg/m at mid-span, will have zero shear force at

(a) Ends
(b) Mid-span
(c) Both ends as well as mid-span
(d) Nowhere

Punjab PSC SDE 2016

Ans. (b) :



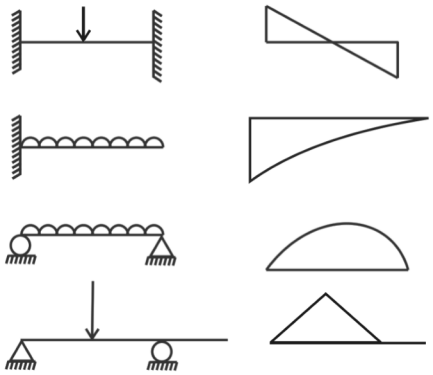
Shear force diagram

521. Which of the following beams will have point of contraflexure? (self-weight is negligible)

(a) A simply supported beam with UDL over entire span
(b) An overhanging beam with loading only over the supported span and not on the overhangs
(c) A fixed beam subjected to concentrated load
(d) A cantilever subjected to uniformly varying load

Punjab PSC SDE 2016
ESE 2002

Ans. (c) : Point of contraflexure or inflexion is that point where bending moment changes its sign.



522. Which of the following statements is correct?

(a) SF is the first derivative of BM
(b) SF is the first derivative of intensity of load
(c) Load intensity on a beam is the first derivative of BM
(d) BM is the first derivative of SF

Punjab PSC SDE 2016
ESE 1995

Ans. (a) :

$$V = \frac{dM}{dx} \qquad P = \frac{dV}{dx}$$

Where, V = shear force
P = load
M = bending moment

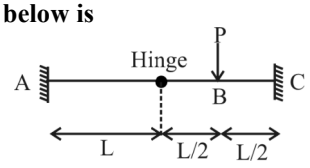
523. Area of S.F. diagram between two points of a span represents :

(a) Change in S.F. between the points
(b) Change in B.M. between the two points
(c) Change in load between the two points
(d) Maximum S.F. between the points

TRB Poly. Lect. Tech. 2017

Ans. (b) : Area of S.F. diagram between two points of a span represents change in B.M. between the two points.

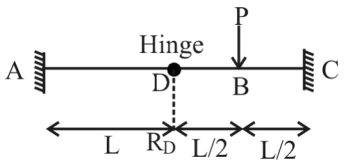
524. The bending moment at point A of a beam as shown below is



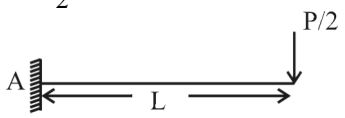
(a) zero
(b) $\frac{3PL}{2}$
(c) $\frac{PL}{2}$
(d) 2PL

UJVNL AE 2016

Ans. (c) :



$\Sigma M_C = 0$
$\Rightarrow R_D \times L = P \times L/2$
$\Rightarrow R_D = \frac{P}{2}$



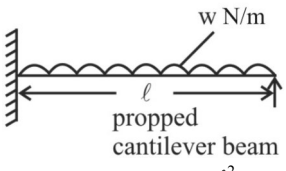
Bending moment at point A $= \frac{P}{2} \times L = \frac{PL}{2}$

525. A UDL of intensity w/length is applied on a propped cantilever of span length *l*. the bending moment at the fixed end will be

(a) $wl^2/24$
(b) $wl^2/16$
(c) $wl^2/12$
(d) $wl^2/8$

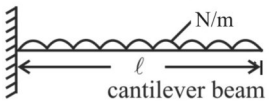
UPRVUNL AE 2016

Ans.: (d)

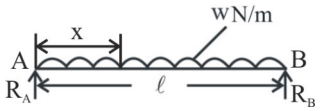


Bending moment at fixed end $= \frac{w\ell^2}{8}$

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Bending moment at fixed end $= w \times \ell \times \frac{\ell}{2} \Rightarrow \frac{w\ell^2}{2}$



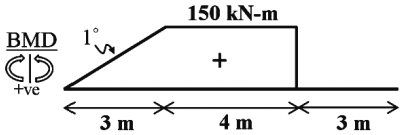
$$M_x = R_A.x - \frac{wx^2}{2} = \frac{wl}{2}x - \frac{wx^2}{2}$$

bending moment at middle–

$$M_C = \frac{wl}{2}\left(\frac{l}{2}\right) - \frac{w}{2}\left(\frac{l}{2}\right)$$

$$= \frac{wl^2}{4} - \frac{wl^2}{8} = \frac{wl^2}{8}$$

526. Beam for the below shown Bending Moment Diagram



(a) must be a Simply supported Beam
(b) must be a Cantilever beam
(c) can either be a Simply supported beam or a Cantilever beam
(d) can neither be a Simply Supported Beam nor a Cantilever Beam

GPSC AE June 2019

Ans. (c) : Can either be a simply supported beam or a cantilever beam.

527. Which of the following statements are correct?

(i) Algebraic sum of Area of Shear Force Diagram between two sections is equal to the difference between Bending Moment values at those sections.

(ii) Algebraic sum of Area of Bending Moment Diagram between two sections is equal to the difference between Shear Force values at those sections.

(iii) In Simply Supported Beams, algebraic sum of Area of Shear Force Diagram on left side of any section is same as that on the right side of that section.

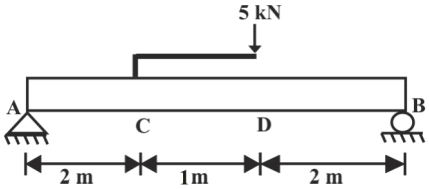
(iv) In Simply Supported Beams, algebraic sum of Area of Bending Moment Diagram on left side of any section is same as that on the right side of that section.

(a) (i) and (iii) (b) (i) and (iv)
(c) (ii) and (iii) (d) (ii) and (iv)

GPSC AE June 2019

Ans. (a) :
(i) Algebraic sum of Area of shear force diagram between two section is equal to the difference between Bending moment values at those sections.
(iii) In SSB, algebraic sum of Area of SFD on left side of any section is same as that on right side of that section.

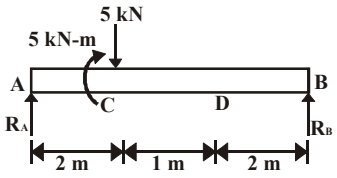
528. For the beam shown below, the vertical reactions at A and B are respectively



(a) 2 kN, 3 kN (b) 3 kN, 2 kN
(c) -1 kN, 1 kN (d) 1 kN, 1 kN

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) :



$\sum F_y = 0$

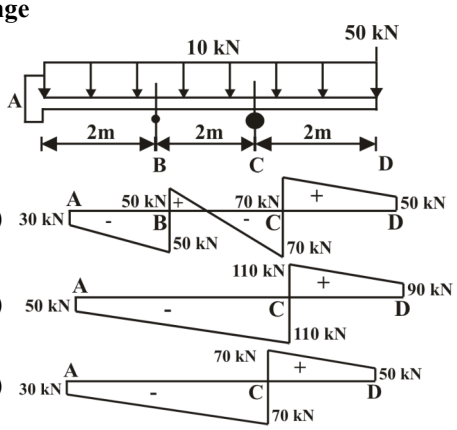
$R_A + R_B = 5$ kN

$\sum M_A = 0$

$R_B (5) = 5 \times 2 + 5$

$\therefore$ $R_B = 3$ kN and $R_A = 2$ kN

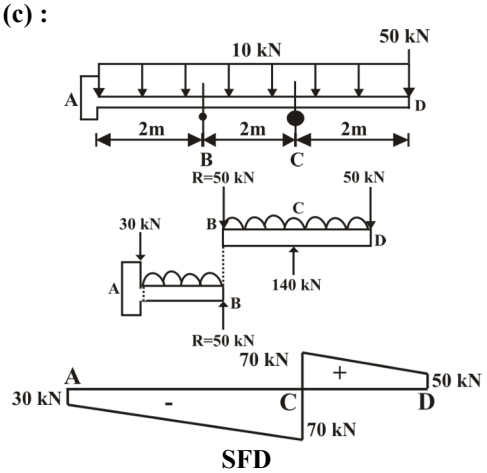
529. Out of the options given below, which one is the correct shear force diagram? B is an internal hinge



(d) None of the above

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (c) :



SFD

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530. A beam of rectangular section 200mm × 300mm carries certain loads such that bending moment at a section A is M and at another section B it is (M + ΔM). The distance between section A and B is 1 m and there are no external loads acting between A and B. If ΔM is 20 kNm, maximum shear stress in the beam section is

(a) 0.5 MPa (b) 1.0 MPa
(c) 1.5 MPa (d) 2.0 MPa

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) : $\frac{dM}{dx} = F$

$\frac{20}{1} = F$

F = 20 kN

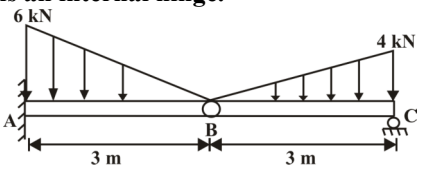
For rectangular cross-section

$\tau_{max} = \frac{3}{2}\left[\tau_{avg}\right]$

$\tau_{max} = \frac{3}{2}\left[\frac{F}{bd}\right]$

$= \frac{3}{2}\left[\frac{20\times10^3}{200\times300}\right] = 0.5 \text{ MPa}$

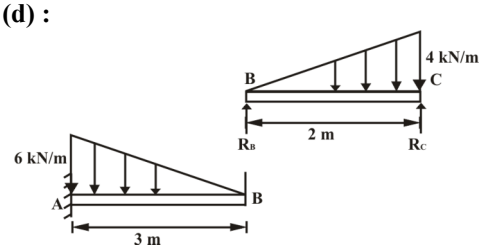
531. Calculate the shear force and bending moment at point B for the beam AB subjected to linearly varying load as shown in the figure. The value of the linearly varying load at the point is 6 kN/m and 4 kN/m, respectively. Point B is an internal hinge.



(a) 2.67 kN and 0 kN-m
(b) 4 kN and 0 kN-m
(c) 4 kN and 1.33 kN-m
(d) 1.33 kN and 0 kN-m

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (d) :



Considering part AB–

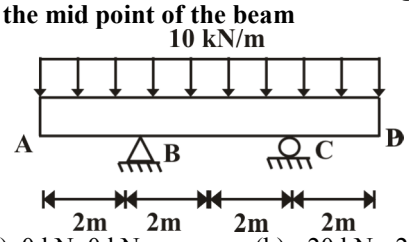
$S.F_B = \frac{wl}{6} = \frac{6\times3}{6} = 3 \text{ kN}$

Consider BC part

$SF_B = R_B = \frac{(4)(2)}{6} = 1.33 \ kN$

$BM_B = 0$ (at hinge moment is zero)

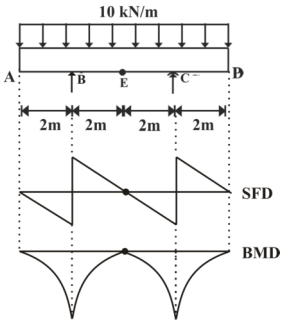
532. Calculate the shear force and bending moment at the mid point of the beam



(a) 0 kN, 0 kN-m (b) -20 kN, -20 kN-m
(c) 20 kN, -20 kN-m (d) 20 kN, 0 kN-m

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (a) :



Due to symmetry, R_B (or) R_C

$= \frac{\text{Total load}}{2} = \frac{10\times8}{2} = 40 \text{ kN}$

At mid span

$SF_E = 10 \times 4 - 40 = 0$

$BM_E = 40\ (2) - 10 \times 4 \times 2 = 0$

SF and BM both will be zero at mid point.

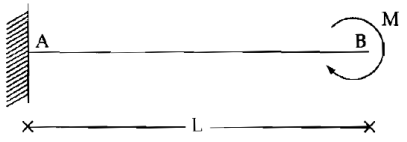
533. The behaviour of a beam column cross section is expressed by which of the following relationship?

(a) Moment - Curvature
(b) Moment - Axial compression
(c) Axial compression - Curvature
(d) Moment - Curvature - Axial compression

Maharashtra PSC (Mains) 2019 Paper–I

Ans. (d) : The behaviour of a beam column cross section is expressed by moment - curvature - axial compression.

534. A cantilever beam AB of span 'L' is subjected to a moment 'M' at the free end as shown in figure. What is the slope and deflection at free end B?
Consider same c/s and material. (i.e., EI is same)



(a) $\frac{ML}{EI}, \frac{ML^2}{2EI}$ (b) $\frac{M}{lEI}, \frac{ML^2}{EI}$
(c) $\frac{2ML}{EI}, \frac{2ML^2}{EI}$ (d) $\frac{ML}{EI}, \frac{2ML^2}{EI}$

Maharashtra PSC (Mains) 2019 Paper–I

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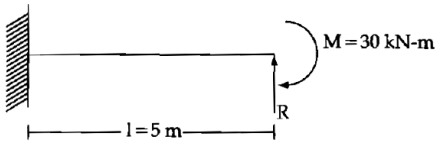
Ans. (a) : The deflection of free end of a cantilever subjected to moment M at free end.

$$\delta = \frac{ML^2}{2EI}$$

The slope at free end of a cantilever subjected to moment M at free end

$$\theta = \frac{ML}{EI}$$

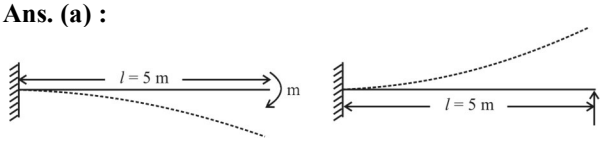
535. In the propped cantilever as shown in figure, the value of propped reaction 'R' will be:



(a) 9 kN (b) 6 kN
(c) 3 kN (d) 2 kN

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Ans. (a) :



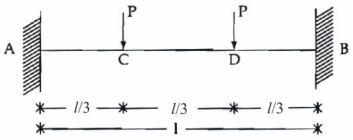
$$\Delta_1 = \frac{ML^2}{2EI} \qquad \Delta_2 = \frac{RL^3}{3EI}$$

$$\Delta_1 = \Delta_2$$

$$\frac{30 \times 5^2}{2\ EI} = \frac{R \times 5^3}{3\ EI}$$

$$R = \frac{30 \times 3}{2 \times 5} = 9 \text{ kN}$$

536. A fixed beam AB of length 'l' having constant flexural rigidity EI carries two loads P at its third points C and D as shown in figure.



Numerically, maximum bending moment will occur:

(a) at C and at D and will be equal to $\frac{2}{9}Pl$

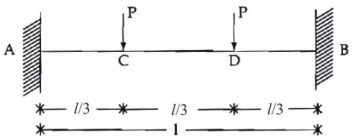
(b) between C and D and will be equal to $\frac{Pl}{9}$

(c) at A and at B and will be equal to $\frac{2}{9}Pl$

(d) between A and C and also between B and D and will be equal to $\frac{Pl}{9}$

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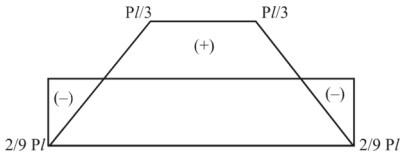
Ans. (c) :



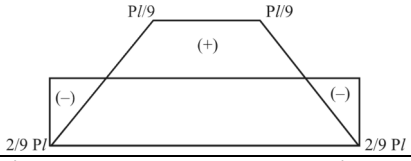
$$\text{FEM} = \frac{P(l/3)(2l/3)^2}{l^2} + \frac{P(2l/3)(l/3)^2}{l^2}$$

$$\text{FEM} = \frac{4}{27}Pl + \frac{2}{27}Pl = \frac{2pl}{9}$$

BMD



Net BMD

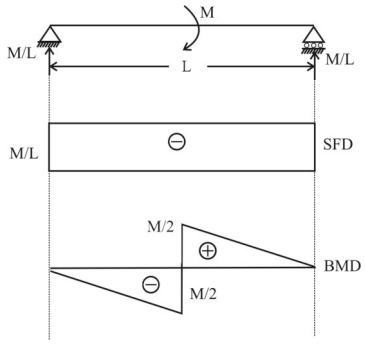


537. A simply supported beam carries couple at a point on its span, the shear force:

(a) Varies by cubic law
(b) Varies by parabolic law
(c) Varies linearly
(d) Is uniform throughout

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Ans. (d) :



Shear force diagram is uniform throughout.

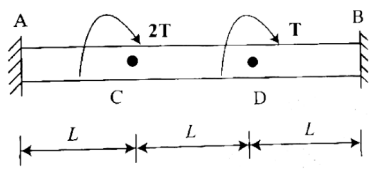
538. A fixed beam of uniform section is carrying a point load at its mid-span. If the moment of inertia of the middle half-length is reduced to half its previous value, then the fixed end moments will

(a) Increase (b) Decrease
(c) Remain changed (d) Change their direction

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Ans. (a) A fixed beam of uniform section is carrying a point load at its mid-span. If the moment of inertia of the middle half-length is reduced to half its previous value, then the fixed end moments will increase.

539. A circular steel bar of diameter d and length 3L fixed at both ends is loaded at intermediate section by torques T and 2T as shown in the following figure. The support reactions can be given as

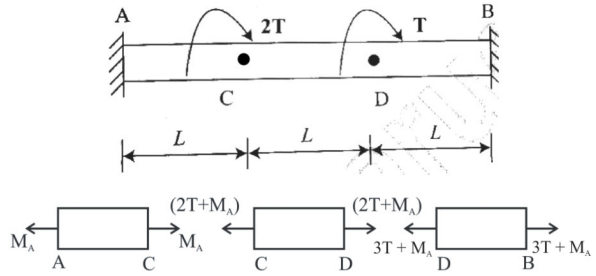


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(a) $M_A = T$ and $M_B = 2T$
(b) $M_A = 2T$ and $M_B = T$
(c) $M_A = 4T/3$ and $M_B = 5T/3$
(d) $M_A = 5T/3$ and $M_B = 4T/3$

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Ans. (d) :



$$\theta_{AB} = 0 = \frac{-M_A \times L}{GJ} = \frac{-(2T + M_A) \times L}{GJ} = \frac{-(3T + M_A)L}{GJ} = 0$$

$$M_A + 2T + M_A + 3T + M_A = 0$$

$3M_A + 5T = 0$

$$M_A = \frac{-5T}{3}$$

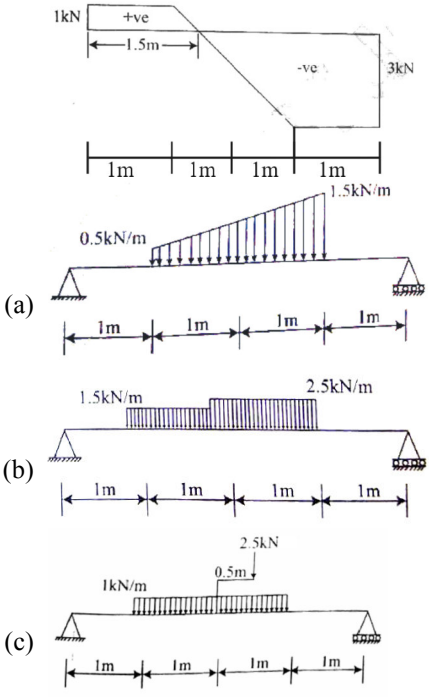
$$\boxed{M_A = \frac{5T}{3}} \text{ Clockwise}$$

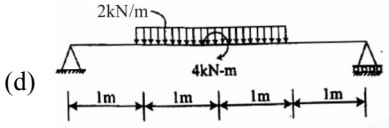
$M_B = 3T + M_A$

$$= 3T - \frac{5T}{3}$$

$$\boxed{M_B = \frac{4T}{3}} \text{ Anticlockwise}$$

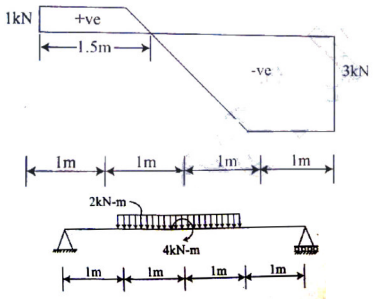
540. If the following figure represents the Shear Force Diagram of a beam, then the loading conditions of the beam would be?





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Ans. (d) :



$$-w = \frac{dv}{dx}$$

$$-w = \frac{1}{0.5}$$

$$\boxed{w = -2 \text{ kN/m}}$$

w = 2kN/m downward. For moment equilibrium, M = 4kN-m.

541. If the magnitude of shear force is constant, then the magnitude of the slope of bending moment curve is

(a) zero (b) increasing
(c) decreasing (d) constant

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Ans. (d) : If S.F. (V) = constant
Slope of bending moment curve,

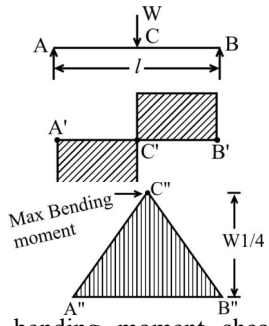
$$\frac{dM}{dx} = V = \text{constant}$$

542. For a maximum bending moment, shear force at that section should be

(a) zero (b) maximum
(c) minimum (d) None of the above

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Ans. (a)



For maximum bending moment shear force at the section should be zero.

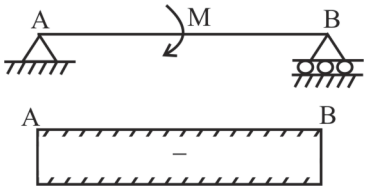
543. For uniform shear force throughout the span of a simply supported beam, it should carry

(a) a concentrated load at the mid-span
(b) a couple anywhere in the sections
(c) ud*l* over its entire span
(d) two concentrated loads equally spaced

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Ans. (b)



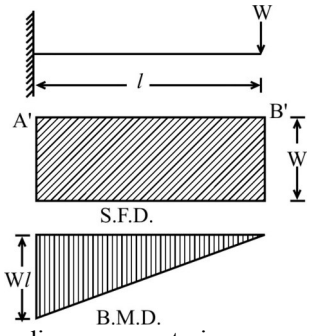
For uniform shear force throughout the span of a simply supported beam, it should carry a couple anywhere in the section.

544. Maximum bending moment in a cantilever carrying a concentrated load at the free end occurs

(a) at the fixed end (b) at the free end
(c) at the mid span (d) None of these

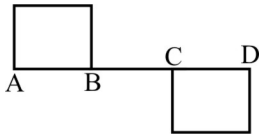
APPSC AEE (Civil/Mechanical) 2012

Ans. (a)



Maximum bending moment in a cantilever beam carrying a concentrated load at the free end occurs at the fixed end.

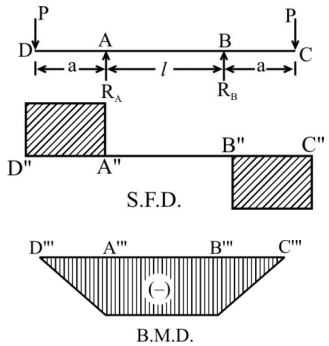
545. The given figure shows the shear force diagram for the beam ABCD. Bending moment in the portion BC of the beam



(a) is zero
(b) varies linearly from B to C
(c) parabolic variation between B and C
(d) is a non–zero constant

APPSC AEE (Civil/Mechanical) 2012

Ans. (d)



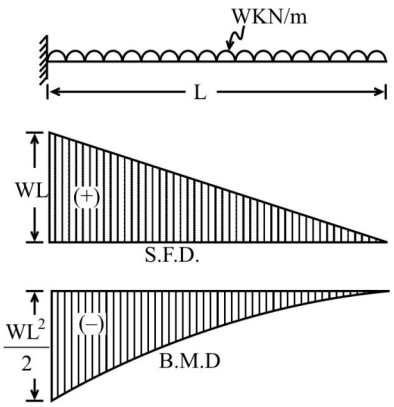
Bending moment in the portion BC of the beam is a non-zero constant.

546. A cantilever is subjected to udl throughout the length. If the maximum shear force is 200kN and maximum bending moment is 400kN, the span "L" of the beam in metres is

(a) 3 (b) 2
(c) 4 (d) 8

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Ans. (c)



Maximum shear force = 200kN

WL = 200kN ……..(i)

Maximum Bending moment

$\frac{WL^2}{2}$ =400kN …..(ii)

for equation (i) and (ii)

$$\frac{\frac{WL^2}{2}}{WL} = \frac{400}{200}$$

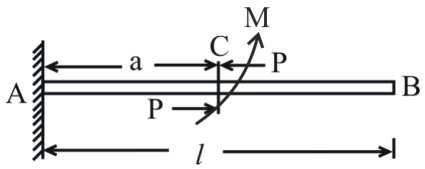
$$\frac{L}{2} = 2 \Rightarrow \boxed{L = 4\ m}$$

547. A cantilever beam AB of length *l* is subjected to an anticlockwise couple of 'M' at a section C, distance 'a' from support. Then the maximum shear force is equal to

(a) M (b) $\frac{M}{2}$
(c) Zero (d) Ma

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Ans. (c)



A cantilever beam AB of length *l* is subjected to an anticlockwise couple of M at a section C, distance 'a' from support, then the maximum shear force is equal to zero.

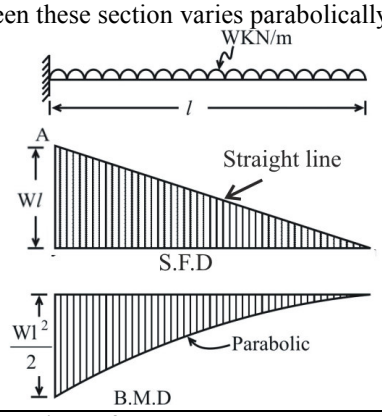
548. If SFD between two sections varies linearly, BM between these sections varies

(a) linearly (b) parabolically
(c) constant (d) None of these

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Ans. (b) If SFD between two section varies linearly, BM between these section varies parabolically



549. At section of a beam sudden change in BM indicates the action of

(a) point load (b) couple
(c) point load or couple (d) ud*l*

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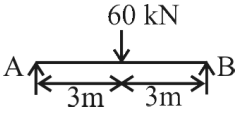
Ans. (b) At section of a beam sudden in BM indicates the action of couple.

550. A freely supported beam of span 6m is subjected to a point of 60 kN at mid span. The maximum BM is equal to

(a) 300 kN-m (b) 180 kN-m
(c) 90 kN-m (d) 270 kN-m

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Ans. (c)



$R_A + R_B = 60$ kN

$\Sigma M_B = 0$

$R_A \times 6 = 60 \times 3$

$R_A = 30$ kN

$R_B = 30$ kN

Max. bending moment $= R_A \times 3$

$= 90$ kN

551. If a freely supported beam is subjected to udl throughout the span, the shape of the BMD is

(a) rectangle (b) straight line
(c) equilateral triangle (d) parabola

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Ans. (d) A freely supported beam is subjected to udl throughout the span, acquire the shape of parabola for BMD.

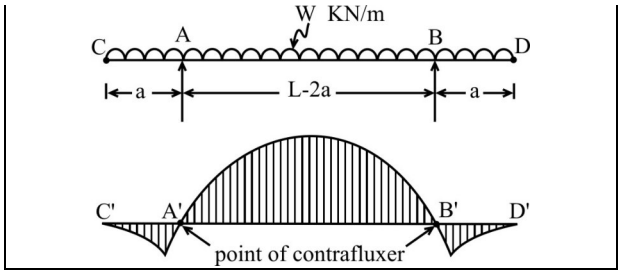
552. In a double overhanging beam carrying ud*l* throughout its length, the number of points of contraflexure are

(a) 1 (b) 2
(c) zero (d) 3

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Ans. (b) In a double overhanging beam carrying UDL throughout its length, the number of points of contraflexure are two.

Points of contraflexure:– Where the bending moment will change sign from negative to positive or vice versa. Such a point, where the bending moment change sign, is known as a point of contraflexure.



553. A beam ABCD is simply supported at B and C. The simply supported span BC=*l*. The overhangs are each 'a'. If the beam carries a ud*l* of w/metre over the entire length, the maximum BM at the centre is

(a) $\frac{w\ell^2}{8} - \frac{wa^2}{4}$ (b) $\frac{wa^2}{4}$

(c) $\frac{wa^2}{2}$ (d) $\frac{w\ell^2}{8} - \frac{wa^2}{2}$

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Ans. (c) $\frac{wa^2}{2}$

554. In a propped cantilever beam, the number of points of contra-flexure is

(a) 1 (b) 2
(c) 3 (d) 4

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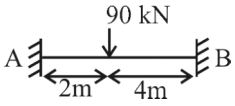
Ans. (a) In a propped cantilever beam, the number of point of contra-flexure is one.

555. A fixed beam 'AB' 6 m long carries a vertical load 90kN at 2 m from 'A'. The fixed end moments at 'A' and 'B' are

(a) 40 kN-m, 80 kN-m
(b) 40 kN-m, 120kN-m
(c) 80kN-m, 40 kN-m
(d) 120 kN-m, 80kN-m

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Ans. (c)



$$M_{AB} = \frac{Pab^2}{L^2} = \frac{90 \times 2 \times 16}{36} = 80\,\text{kN} - \text{m}$$

$$M_{BA} = \frac{Pa^2b}{L^2} = \frac{90 \times 4 \times 4}{36} = 40\,\text{kN} - \text{m}$$

556. Bending moment of simply supported beam carrying a load of w per meter and span L is

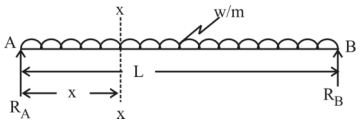
(a) $\frac{wL}{2}$ (b) $\frac{wL^2}{2}$

(c) $\frac{wL^2}{8}$ (d) $\frac{wL^2}{12}$

GPSC Assistant Prof. 2016

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Ans. (c) : Due to symmetry, $R_A = R_B = \frac{wL}{2}$



$$M_x = R_A \times x - w \times x \times \frac{x}{2}$$

$$M_x = \frac{wLx}{2} - \frac{wx^2}{2}$$

$$M_x = \frac{w}{2}\left[Lx - x^2\right]$$

at x = 0,

$$M_o = \frac{w}{2}\left[L(0) - 0^2\right] = 0$$

at x = L,

$$M_B = \frac{w}{2}\left[L \times L - L^2\right] = 0$$

at x = L/2

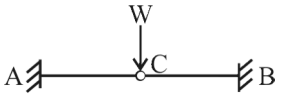
$$M_{L/2} = \frac{w}{2}\left[L \times \left(\frac{L}{2}\right) - \left(\frac{L}{2}\right)^2\right] = \frac{w}{2}\left[\frac{L^2}{2} - \frac{L^2}{4}\right]$$

$$= \frac{w}{2}\left[\frac{2L^2 - L^2}{4}\right]$$

$$M_{L/2} = \frac{w}{2}\left(\frac{L^2}{4}\right)$$

$$M_{L/2} = \frac{wL^2}{8}$$

557. The fixed beam AB has hinge C at mid span concentrated load W applied at C what is the fixed end moment at A:

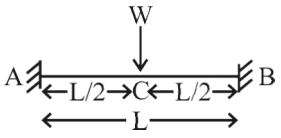


(a) $\frac{WL}{8}$ (b) $\frac{WL}{2}$

(c) $\frac{WL^2}{8}$ (d) $\frac{WL}{4}$

PTCUL AE 2016

Ans. (a) :



Fixed end moment at 'A'

$$\overline{M}_A = \frac{-wab^2}{L^2}$$

$$a = \frac{L}{2}, \; b = \frac{L}{2}$$

$$= \frac{-w\left(\frac{L}{2}\right)\cdot\left(\frac{L}{2}\right)^2}{L^2} = -\frac{WL}{8}$$

558. The shear force in a beam subjected to pure bending is

(a) Positive (b) Negative

(c) Zero (d) None of the above

UJVNL AE 2016

Ans. (c) : In case of pure bending shear force is zero.

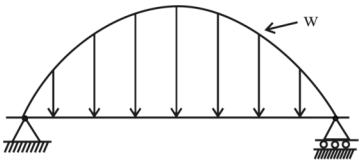
559. A simply supported beam has parabolic loading with maximum intensity 'w' at its centre, the maximum S.F. in the beam is equal to

(a) wL/2 (b) wL/4

(c) wL/3 (d) w

UJVNL AE 2016

Ans. (c) :



Area of parabola $= \frac{2}{3} \times w \times L = \frac{2}{3}wL$



Maximum shear force $= \frac{wL}{3}$

560. Bending moment at any section in a conjugate beam gives in the actual beam

(a) Slope (b) Deflection

(c) Curvature (d) Bending moment

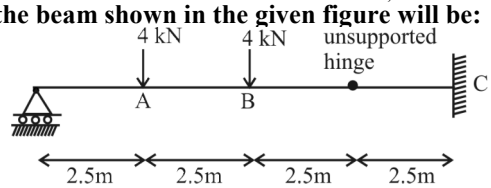
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Ans. (b)

Moment on conjugate beam = deflection on real beam

Shear on conjugate beam = slope on real beam

561. The bending moments at point A, B and C of the beam shown in the given figure will be:



(a) 10 kNm, 10 kNm and 10 kNm

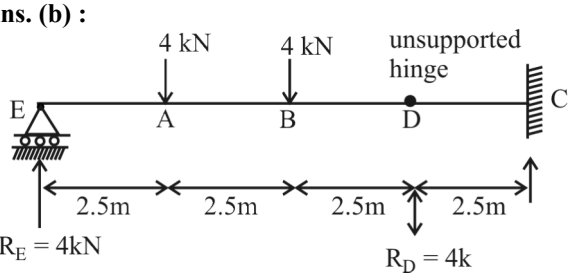
(b) 10 kNm, 10 kNm and –10 kNm

(c) 20 kNm, 10 kNm and –10 kNm

(d) 10 kNm, –10 kNm and 20 kNm

UJVNL AE 2016
ESE 1999

Ans. (b) :



B.M. at C = 4 × 2.5m = 10 kNm (anticlockwise)

B.M. at A = 4 × 2.5m = 10 kNm (anticlockwise)

B.M. at B = 4 × 2.5m = 10 kNm (anticlockwise)

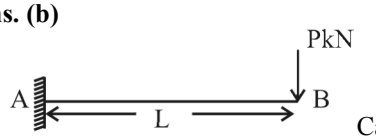
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562. The moment at the free end of a cantilever beam of length 'L' subjected to point load 'P' kN at the end is

(a) PL (b) 0
(c) (PL)/2 (d) P

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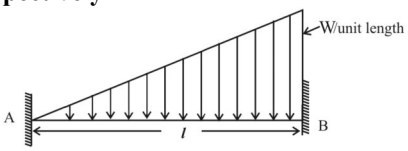
Ans. (b)



Cantilever beam

Moment at B = 0

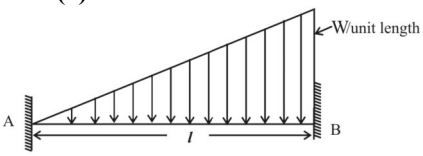
563. Fixed end moments at A and B for the fixed beam shown in the figure, subjected to the indicated uniformly varying load, are respectively



(a) $Wl^2/30$ and $Wl^2/20$ (b) $Wl^2/20$ and $Wl^2/30$
(c) $Wl^2$12 and $Wl^2/8$ (d) $Wl^2/8$ and $Wl^2/12$

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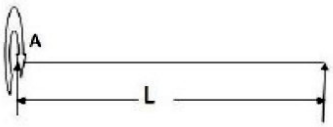
Ans. (a)



Fixed end moment at A $= +\dfrac{Wl^2}{30}$

Fixed end moment at B $= -\dfrac{Wl^2}{20}$

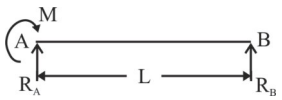
564. The bending moment diagram of the beam shown in figure is –



(a) a trapezium (b) a triangle
(c) a parabola (d) a rectangle

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Ans. (b) :



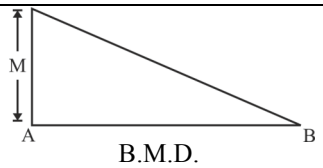
$\sum M_A = 0$

$-R_B \times L + M = 0$

$$\boxed{R_B = \frac{M}{L}}$$

$R_A - R_B$

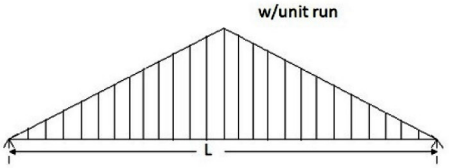
$$\boxed{R_A = \frac{-M}{L}}$$



B.M.D.

B.M. just before A = 0
B.M. just after A = M
B.M. at B = 0

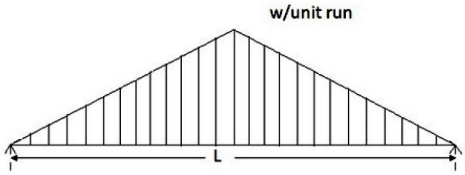
565. What is the maximum bending moment at mid span of the beam given below –



(a) $wL^2/6\sqrt{2}$ (b) $wL^2/6$
(c) $wL^2/12$ (d) $wL^2/2$

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UPPCL AE 2014

Ans. (c) :



Due to symmetry,

$$R_A = R_B = \frac{\frac{1}{2} \times w \times L}{2}$$

$$R_A = R_B = \frac{wL}{4}$$

Maximum B.M. at midspan

$$= R_A \times \frac{L}{2} - \left(\frac{1}{2} \times w \times \frac{L}{2}\right) \times \frac{1}{3} \times \frac{L}{2}$$

$$= \frac{wL}{4} \times \frac{L}{2} - \frac{wL^2}{24}$$

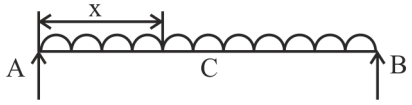
$$= \frac{3wL^2 - wL^2}{24} = \frac{wL^2}{12}$$

566. A simply supported beam with two equal spans carries uniformly distributed load 'W' on entire length of beam. If each span has length 'L' the bending moment at the central support is-

(a) $WL^2/12$ (b) $WL^2/4$
(c) $WL^2/2$ (d) $WL^2/8$

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Ans. (d) :



$$M_x = R_A.x - \frac{wx^2}{2}$$

$$= \frac{wl}{2}x - \frac{wx^2}{2}$$

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bending moment at middle–

$$M_C = \frac{wl}{2}\left(\frac{l}{2}\right) - \frac{w}{2}\left(\frac{l}{2}\right)$$

$$= \frac{wl^2}{4} - \frac{wl^2}{8} = \frac{wl^2}{8}$$

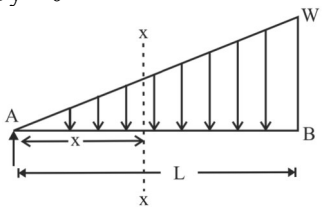
567. In a simply supported beam span 'L' with a triangular load varying from zero at one end to the maximum value 'W' at other end, maximum bending moment is –

(a) $WL^2/9\sqrt{3}$ (b) WL/3

(c) $WL/9\sqrt{3}$ (d) WL/4

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Ans. (a) : $\Sigma Fy = 0$



$$R_A + R_B = \frac{WL^2}{2}$$

$$\Sigma M_A = 0$$

$$\frac{WL}{2} \times \frac{2L}{3} - R_B \times L$$

$$\boxed{R_B = \frac{WL}{3}}$$

$$\boxed{R_A = \frac{WL}{6}}$$

$$B.M_{xx} = \frac{WLx}{6} - \frac{Wx^3}{6L}$$

$$\frac{\partial M_{xx}}{\partial_x} = 0 \Rightarrow x = \frac{L}{\sqrt{3}}$$

$$M_{max} = \frac{WL}{6} \times \frac{L}{\sqrt{3}} - \frac{WL^2}{18\sqrt{3}}$$

$$M_{Max} = \frac{WL^2}{9\sqrt{3}}$$

568. If a constant section beam is subjected to a uniform bending moment throughout, it bends as-

(a) zig zag (b) a catenary

(c) a circular arc (d) a parabolic arc

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Ans. (c) : In uniform bending, the beam bends into circular arch.

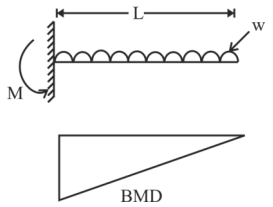
569. For a cantilever beam of span L carrying uniformly distributed load, w on its entire span, the maximum bending moment is .....

(a) $\frac{WL^2}{2}$ (b) $\frac{WL}{2}$

(c) $\frac{WL^2}{4}$ (d) $\frac{WL^2}{8}$

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Ans. (a) :



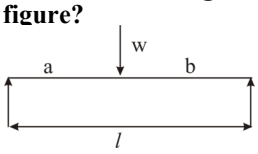
Reaction at support = wL

Taking moment at support

$$M_{max} = wL\frac{L}{2} - M = 0$$

$$\boxed{M = \frac{wL^2}{2}}$$

570. What is maximum bending moment for beam shown in figure?

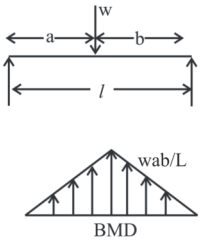


(a) $\frac{Wba^2}{l^2}$ (b) $\frac{Wba}{l}$

(c) $\frac{Wab^2}{l^2}$ (d) $\frac{Wb^2a^2}{l^2}$

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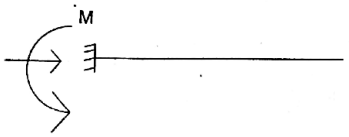
Ans. (b) :



In this case

$$\boxed{M_{max} = \frac{Wab}{L}}$$

571. Choose the correct option classifying the following structure.



(a) Indeterminate to 1st degree

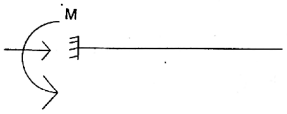
(b) Indeterminate to 2nd degree

(c) Determinate

(d) Unstable

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Ans. (c) :



Static indeterminacy = R – E

R = 3, E = 3(equilibrium eq^n)

(Support r × n) = 3 – 3 = 0

Statically determinate structure.

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572. A simply supported beam of length L is subject to two point loads, each of P unit, at A and B equidistant from supports. Select the correct statement

(a) There will be no shear force between A and B and no bending moment between A and B.

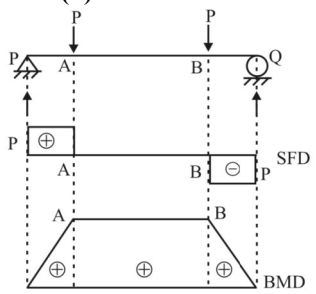
(b) There will be no shear force between A and B, but there will be uniform bending moment between A and B.

(c) There will be uniform shear force between A and B, but bending moment will gradually increase to maximum up to mid point of A and B.

(d) There will be uniform shear force between A and B, with maximum bending moment at A and B with gradual decrease to zero at mid point of A and B.

MPPSC AE 2014

Ans. (b)



From SFD & BMD, it is clear that, there is no S.F between A & B, But there is uniform bending moment between A & B

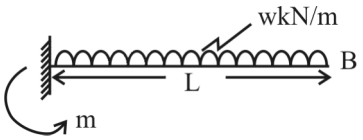
573. The absolute maximum bending moment in a prismatic cantilever beam of length *L* under a uniformly distributed load of *w* per unit length is _____:

(a) $\frac{wL^2}{2}$ (b) wL^2

(c) $\frac{wL^2}{4}$ (d) $\frac{wL^2}{8}$

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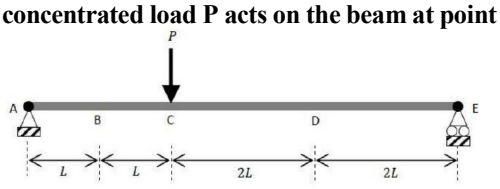
Ans. (a) :



Maximum BM occurs at point A

$M = WL \times \frac{L}{2} = \frac{WL^2}{2}$

574. For the beam shown below, support A is hinge and E is roller. The beam has a constant EI. A concentrated load P acts on the beam at point C.



Now consider the following two statements:

Statement I: Bending moment reaches maximum at point C

Statement II: Slop reaches maximum at point A

Which of the following is CORRECT.

(a) Both Statement I and Statement II are true

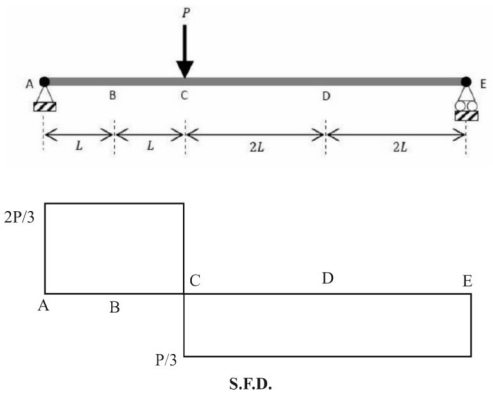
(b) Statement I is true, But Statement II is false

(c) Statement I is false, but Statement II is true

(d) Both Statement I and Statement II are false

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Ans. (a) :



$R_A + R_E = P$

$\Sigma M_E = 0;$

$R_A \times 6L = P \times 4L$

$$R_A = \frac{2P}{3}$$

$R_E = P - R_A = P - \frac{2P}{3}$

$$R_E = \frac{P}{3}$$

As the S.F at C is zero, so the maximum BM will occur at point C. (Statement I is true)

Under the point load the deflection will be maximum. So the slope at point of maximum deflection will always zero.

575. In a beam, the bending moment is maximum at the point where the shear force _____.

(a) changes sign

(b) is minimum

(c) half the total load on the beam

(d) is maximum

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Ans. (a) : In a beam, the bending moment is maximum at the point where the shear force changes sign.

576. The bending moment at the point where shear force is zero, is :

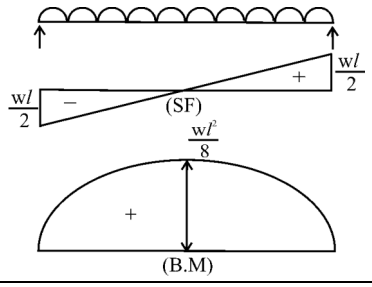
(a) 0 (b) Maximum

(c) 5 (d) 1

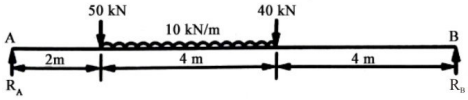
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Ans. (b) The point at which bending moment is maximum.



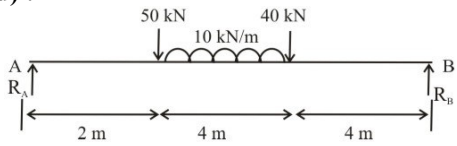
577. Calculate the reaction at support A for a simply supported beam shown in figure.



(a) 70 kN (b) 50 kN
(c) 90 kN (d) 80 kN

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Ans. (d) :



$R_A + R_B = 50 + 40 + 10 \times 4$

$R_A + R_B = 130$ kN

Moment taking above R_B

$R_B \times 0 + 40 \times 4 + 40 \times 6 + 50 \times 8 - R_A \times 10 = 0$

$160 + 240 + 400 - R_A \times 10 = 0$

$R_A \times 10 = 800$

$R_A = \frac{800}{10}$

$\boxed{R_A = 80}$ kN

$R_A + R_B = 130$

$80 + R_B = 130$

$R_B = 130 - 80$

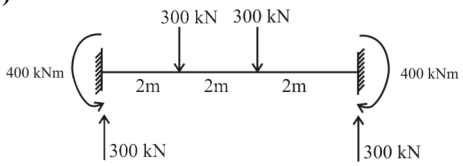
$\boxed{R_B = 50}$ kN

578. For a 6 m long fixed beam carrying two loads of 300 kN, each at 2 m from each support, the point of contraflexure will be situated from distance a from left support, where 'a' is

(a) 1.33 m (b) 3 m
(c) 2 m (d) 1.5 m

LMRC AM 2015

Ans (a)

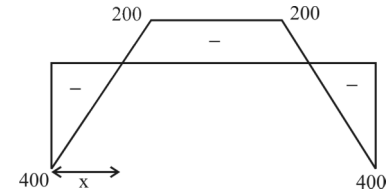


$$MF_{AB} = \frac{300 \times 2 \times 4^2}{6^2} + \frac{300 \times 4 \times 2^2}{6^2} = 400 \text{ kNm}$$

BMD $= 300x - 400$ $0 < x < 2$

$= 300x - 300(x-2) - 400$ $2 < x < 4$

$= 300x - 400$ $0 < x < 0$



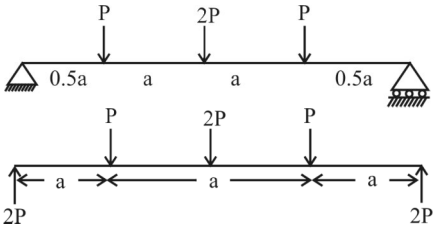
$$x = \frac{400}{300} = 1.33$$

579. A simply supported beam of length '3a' supports three loads separated by distance 'a'. The middle load is 2P, while the loads at distance 'a' from each support are P each. What will be the maximum shear force in the beam?

(a) 2P (b) 4P
(c) 6P (d) Zero

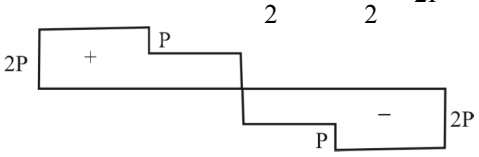
LMRC AM 2015

Ans (a)



Due to symmetry vertical reaction at support

$$= \frac{\text{Total load}}{2} = \frac{4P}{2} = 2P$$



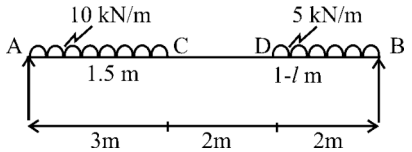
$(S.F.)_{max} = 2P$ (at support)

580. A simply supported beam of lengths 7 m is carrying uniformly distributed loaded as shown in the figure. The support reaction at B is :

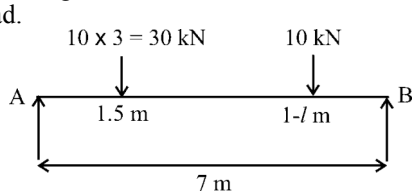
(a) 35 kN (b) 10 kN
(c) 20 kN (d) 15 kN

DMRC AM 2020

Ans. (d)



For calculating the reaction we can convert UDL into point load.



Now, $\Sigma M_A = 0$

$30 \times 1.5 + 10 \times 6 = R_B \times 7$

$\Rightarrow 105 = 7 R_B$

$R_B = 15$ kN

Hence reaction at B = 15 kN

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581. What is the bending moment at the mid span of a simply supported beam carrying a uniformly varying beam?

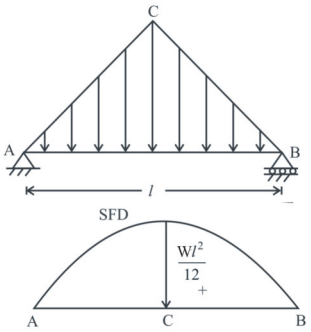
(a) $\frac{Wl^3}{8}$ (b) $\frac{Wl^3}{16}$

(c) $\frac{Wl^4}{48}$ (d) $\frac{Wl^2}{12}$

(e) $\frac{Wl^2}{24}$

CGPSC AE 2017

Ans. (d)



Reaction at support $= \frac{1}{2}\left(\frac{Wl}{2}\right) = \frac{Wl}{4}$

$$M_{max} = \frac{Wl}{4}\left(\frac{l}{2}\right) - \frac{Wl}{4}\left(\frac{l}{2}\right)\times\frac{1}{3}$$

$$= \frac{Wl^2}{8} - \frac{Wl^2}{24} = Wl^2\left(\frac{3-1}{24}\right)$$

$$\boxed{M_{max} = \frac{Wl^2}{12}}$$

582. A simply supported beam carrying a uniformly distributed load of 8 kN/m per unit over the whole span of 3 m. What will be the maximum bending moment of the beam?

(a) 8.34 kN-m (b) 9 kN-m

(c) 12.20 kN-m (d) 15.17 kN-m

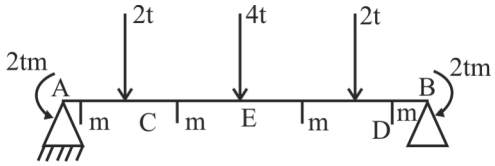
(e) 3 kN-m

CGPSC AE 2017

Ans. (b) Maximum bending moment $= \frac{wl^2}{8}$

$$= \frac{8\times 3^2}{8} = 9 \text{ kN} - \text{m}$$

583. A simply supported beam is loaded as shown in the given figure. The bending moment at E would be



(a) 6 tm (Sagging) (b) 4 tm (Hogging)

(c) 6 tm (Hogging) (d) 4 tm (Sagging)

Haryana PSC AE 2010

ESE 1995

Ans. (d) :

Reaction at A,

$$R_A = \frac{2\times 1 + 4\times 2 + 2\times 3}{4} = 4\text{t}$$

$R_B = 4$t

Bending moment at E,

$M_E = -2 + 4 \times 2 - 2 \times 1$

$= 4$ t-m (sagging)

584. In a simply supported beam of span (L+2a) with equal overhang (a) carries a uniformly distributed load over the whole length. Bending moment changes sign if –

(a) L = 2a (b) L < 2a

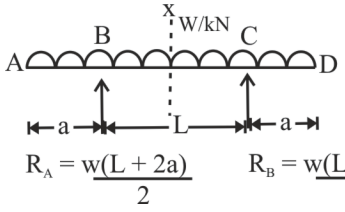
(c) L = 3a (d) L > 2a

UPPCL AE 2014

Haryana PSC AE 2010

Ans. (d) :

$$R_A = R_B = \frac{w(L+2a)}{2}$$



$R_A = \frac{w(L+2a)}{2}$ $R_B = \frac{w(L+2a)}{2}$

In span BC,

$$M_x = \frac{wL^2}{8} - \frac{wa^2}{2}$$

(i) For L = 2a, $M_x = 0$

$\because$ for L > 2a bending moment changes sign.

585. A beam is supported such that there are only three unknown reactions at the supports. These unknowns can be determined by using the following fundamental equation in statics:

(a) $\Sigma H = 0$

(b) $\Sigma V = 0$

(c) $\Sigma H = 0; \Sigma V = 0$

(d) $\Sigma H = 0; \Sigma V = 0; \Sigma M = 0$

(e) $\Sigma H = 0; \Sigma M = 0$

Haryana PSC AE 2018 Paper–I

Ans. (d) : Algebraic sum of all horizontal force ΣH = 0

Algebraic sum of all vertical force ΣV = 0

Algebraic sum of all moment force ΣM = 0

586. A rectangular log of wood is floating in water with a load of 100 N as its centre. The maximum shear force in the wooden log is:

(a) 50 N at each end

(b) 100 N at the centre

(c) 50 N at the centre

(d) 72.5 N at each end

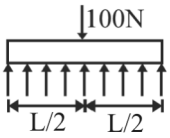
(e) None of these options

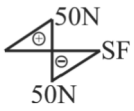
Haryana PSC AE 2018 Paper–I

Ans. (c) : Suppose length of block = L

Water will act as UDL in as shown in fig.

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At equilibrium WL = 100 N

$$W = \frac{100N}{L}$$

Consider the section xx at a distance x from one end.
Shear force in the wooden of S_{xx} = wx
The maximum shear force in the wooden log is at the center x = L/2

Maximum shear force center $= \frac{100}{L} \times \frac{L}{2} = 50N$

587. For a simply-supported beam of length L, the bending moment is described as $M = a(x - x^3/L^2), 0 \le x \le L$; where a is in kN. The shear force will be zero at:

(a) The supports (b) x = L/2
(c) $x = L/\sqrt{3}$ (d) x = L/3
(e) x = L/4

Haryana PSC AE 2018 Paper–I

Ans. (c) :



$$M = a\left(x - \frac{x^3}{L^2}\right) \qquad a \Rightarrow In\ kN$$

$$S.F. = \frac{dm}{x} = a\left(x - \frac{x^3}{L^2}\right) \quad 0 \le x \le L$$

$$= a\left(1 - \frac{3x^2}{L^2}\right)dx$$

Now $S.F. = a\left(1 - \frac{3x^2}{L^2}\right)dx$

Put S.F. = 0

$$a\left(1 - \frac{3x^2}{L^2}\right)dx = 0$$

$$1 - \frac{3x^2}{L^2} = 0$$

$$\frac{3x^2}{L^2} = 1$$

$$3x^2 = L^2$$

$$x^2 = \frac{L^2}{3}$$

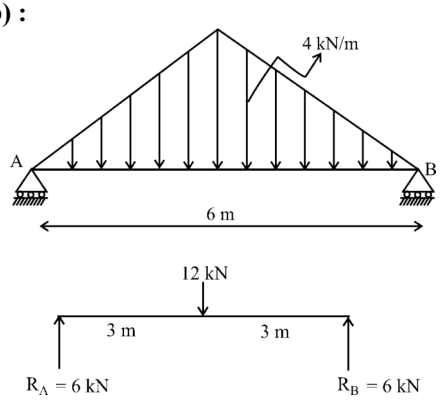
$$x = \frac{L}{\sqrt{3}}$$

588. A simply supported beam of span 6 m with a gradually varying load (GVL) of zero at each end and of 4kN/unit length at mid span. The maximum B.M. in kN/m :

(a) 18 (b) 12
(c) 6 (d) 24

UP Jal Nigam AE 2016

Ans. (b) :



Total load = $2\left[\frac{1}{2} \times 3 \times 4\right] = 12\,kN$

due to symmetry, $R_A = R_B = 6$ kN
BM. will be maximum at centre

B.M. = RA × 3 – $\frac{1}{2}(4 \times 3)\left(\frac{1}{3} \times 3\right)$

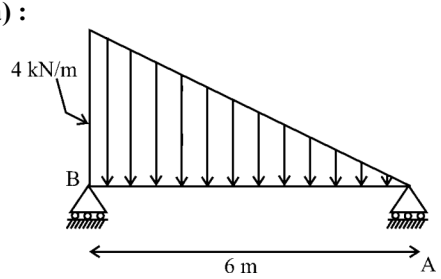
B.M. = 6 × 3 – 6
B.M. = 12 kN-m

589. A simply supported beam 6m carrying a Gradually varying Load (GVL) zero at one end A and 4kN/m at other end B then reaction at A and B in kN.

(a) 4, 8 (b) 8, 4
(c) 12, 12 (d) 3, 9

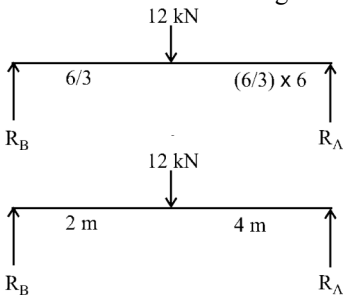
UP Jal Nigam AE 2016

Ans. (a) :



Total load = $\frac{1}{2} \times 6 \times 4 = 12$ kN

The total load act at the G.G. of Triangular load



$R_B = \frac{12 \times 4}{6}$ $R_A = \frac{12 \times 2}{6}$

R_B = 8 kN R_A = 4 kN

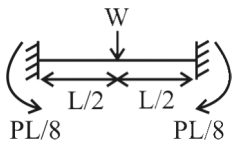
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590. If a point load acting at the mid-span of a fixed beam of uniform section produces fixed end moment of 84 kNm, then the same load when spread uniformly over the entire span will produce fixed end moment equal to

(a) 24 kNm (b) 38 kNm
(c) 56 kNm (d) 72 kNm

UPRVUNL AE 2014

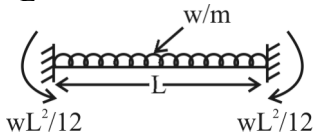
Ans. (c) :



Given, Fixed end moment = 84 kNm

So, $\frac{PL}{8} = 84\,kNm$.....(i)

$P = \frac{84\times 8}{L}$



as per question–

$w \times L = P$

$w \times L = \frac{84\times 8}{L}$

$wL^2 = 84 \times 8$

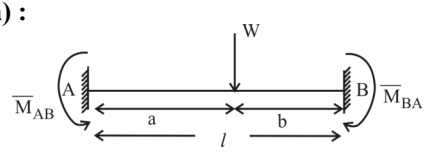
So, FEM due to UDL $\frac{wL^2}{12} = \frac{84\times 8}{12} = 56\,kNm$

591. Due to some point load on a fixed beam, the maximum free bending moment is M. What would be the sum of fixed end moments?

(a) M (b) M/2
(c) 2M (d) 4M

UPRVUNL AE 2014

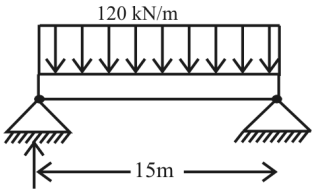
Ans. (a) :



$(\bar{M}_{AB})$ Fixed end moment at end A = $\frac{wab^2}{l^2}$

$(\bar{M}_{BA})$ Fixed end moment at end B = $\frac{wba^2}{l^2}$

592. The dimension of a steel beam are given follows:
Width = 120 mm
Height = 750 mm
This beam is loaded as per the figure given below. Determine bending moment for the beam.



(a) 3375 kN-m (b) 4740 kN-m
(c) 6750 kN-m (d) 8750 kN-m

UPRVUNL AE 2014

Ans. (a) : In the case of udl act at simply supported beam then,

The value of maximum bending moment = $\frac{wl^2}{8}$

Max. bending moment = $\frac{120\times 15^2}{8} = 3375$ kN-m

593. Which of the following is the point of zero bending moment where continuous curve of bending moment changes sign?
(i) Point of Contraflexure
(ii) Point of inflection
(iii) Point of virtual hinge
Choose the correct answer from the option given below.

(a) Only (ii) (b) (i) and (ii)
(c) Only (iii) (d) i, ii and iii

UPRVUNL AE 2014
ESE 2016

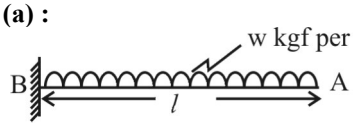
Ans. (d) : The point of zero bending moment where continuous curve of bending moment changes sign is point of contraflexure or point of simplification or point of virtual hinge.

594. Which of the following gives the correct expression for the shear force at the fixed end of a cantilever of length (*l*) carrying a uniformly distributed load of W kgf per unit length for the whole length (*l*)?

(a) (-)W*l* (b) Wl^2/2
(c) W*l*/2 (d) zero

UPRVUNL AE 2014

Ans. (a) :



$(S.F)_{x\text{-}x}$ right	$= +wx$	linear equation
$(S\text{-}F)_A^-$ $x = 0$	$= 0$	
$(S\text{-}F)_B^-$ $x = l$	$= -wL$	

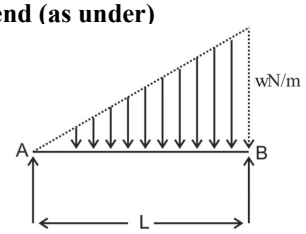
595. What is the value of shearing force on a section having maximum bending moment?

(a) Minimum (b) Maximum
(c) Zero (d) It changes sign

UPRVUNL AE 2014

Ans. (c) : At section having maximum bending moment when the value of shear force is zero.

596. A simply supported beam carries a varying load from zero at one end to w N/m at the other end (as under)



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The length of the beam is L. The shear force will be zero at a distance 'x' from A. Find 'x':

(a) $\frac{L}{2}$ (b) $\frac{L}{4}$

(c) $\frac{L}{\sqrt{3}}$ (d) $\frac{L}{3}$

RRB SSE Bilaspur : 21.12.2014

Ans : (c) Total load $= \frac{1}{2} \times L \times w = \frac{wL}{2}$

$R_A + R_B = \frac{wL}{2}$

$\Sigma M_B = 0$

$R_A \times L = \frac{wL}{2} \times \frac{L}{3}$

$R_A = \frac{wL}{6}$

$R_B = \frac{wL}{2} - \frac{wL}{6} = \frac{wL}{3}$

Let S.F at section C will be–

$S.F_C = \frac{wL}{6} - \frac{wx^2}{2L}$

$\frac{wL}{6} - \frac{wx^2}{2L} = 0$

$x = \frac{L}{\sqrt{3}}$

597. A mild steel flat of width 100 mm and thickness 12 mm is bent into an arc of a circle of radius 10 m by applying a pure moment M. If Young's modulus E = 200 GPa, then the magnitude of M is

(a) 72 Nm (b) 144 Nm

(c) 216 Nm (d) 288 Nm

APPSC AEE (CIVIL/MECHANICAL) 2019

Ans. (d) : Radius of curvature (R) = 10,000 mm

Modulus of elasticity (E) = 200×10^3 MPa

Thickness (t) = 12 mm

From bending equation

$\frac{M}{I} = \frac{E}{R}$

$\frac{M}{\left[\frac{100 \times 12^3}{12}\right]} = \frac{200 \times 10^3}{10,000}$

M = 288000 N-mm = 288 N-m

598. In a simply supported beam, the shear force is uniform throughout the length of span when it is subjected to :

(a) UDL over the span

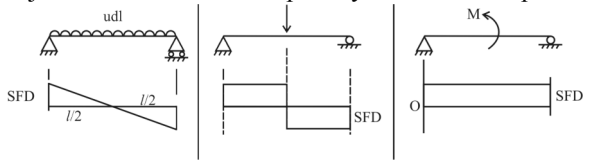
(b) a point load at mid-span

(c) a moment couple anywhere within the span

(d) two point loads kept such that these loads divide the beam in three equal parts

HPPSC Poly. Lect. 2016

Ans : (c) In a simply supported beam, the shear force is uniform throughout the length of span when it is subjected to A moment couple anywhere within span.



599. The free end of a real beam both SF and BM are zero will result in _____ end of the corresponding conjugate

(a) simple support (b) hinged

(c) fixed (d) free

IRCON AE 2017

Ans. (c) :

Real Beam	Conjugate Beam
Fixed	Free
Free	Fixed
Pinned/Hinged	Pinned/Hinged
Roller	Roller
Intermediate roller	Pinned

600. Relation between shear force(s) and loading rate(w) is:

(a) $\frac{dx}{ds} = w$ (b) $\frac{dw}{dx} = s$

(c) $\frac{dx}{dw} = s$ (d) $\frac{ds}{dx} = w$

IRCON AE 2017

Ans. (d) : Shear force (s) loading rate (w) bending moment (m).

$$\boxed{\frac{ds}{dx} = w}$$

$$\boxed{\frac{dm}{dx} = s}$$

slope of shear force diagram for a section represents loading rate.

601. A beam curved in plan is designed for

(a) bending moment and shear force

(b) shear force and torsion

(c) bending moment and torsion

(d) bending moment, shear force and torsion

WBPSC AE 2016

Ans. (d) : A beam curved in plan is designed for bending moment, shear force and torsion.

602. Rate of change of bending moment is equal to

(a) Shear force (b) Deflection

(c) Slope (d) Rate of loading

WBPSC AE 2012

Ans. (a) :

$\frac{dM}{dx} = F$

$\frac{dF}{dx} = W$

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603. The loading on the conjugate beam will be

(a) Loading on the real beam divided by S.F. diagram
(b) B.M. diagram divided by S.F. diagram
(c) B.M. diagram divided by EI
(d) None of these

WBPSC AE 2003

Ans. (c) : Conjugate beam is defined as the imaginary beam with the same dimensions (length) as that of the original beam but load at any point on the conjugate beam is equal to the bending moment at that point divided by EI.

604. A beam AB of length 10 m having both ends fixed is acted upon by a concentrated load of 10 tonnes at C at a distance of 4 m from A on the beam. The B.M. at C is

(a) Sagging and of magnitude 11.52 t.m.
(b) Hogging and of magnitude 11.52 t.m.
(c) Sagging and of magnitude 11.4 t.m.
(d) Hogging and of magnitude 9.6 t.m.

WBPSC AE 2001

Ans. (d) : $M_A = \frac{10 \times 4 \times 6^2}{10^2} = 14.4\,t-m$ (Hogging)

$M_B = \frac{10 \times 6 \times 4^2}{10^2} = 9.6$ (Hogging)

$M_C = R_A \times 4 - 14.4 \qquad (R_A = 6.1)$

$= 6 \times 4 - 14.4$

$= 24 - 14.4$

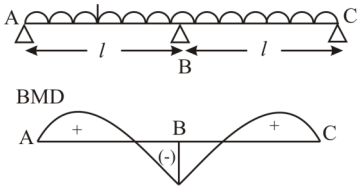
$= 9.6$ t-m (Sagging)

605. What kind of bending moments, (if any) are developed at the intermediate support of a continuous beam loaded by vertical loads

(a) Hogging
(b) Sagging
(c) No moment
(d) Mostly sagging, but sometimes zero

RPSC POLY. TECH. LECT. 2014

Ans : (a)



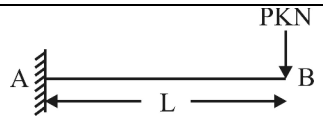
At the support B, negative i.e, hogging bending moment is developed.

606. The moment at the free end of a cantilever beam of length 'L' subjected to point load 'P' kN at the end is

(a) PL (b) 0
(c) (PL)/2 (d) P

APTRANSCO 2017

Ans : (b) :



Moment at free end = P × 0 = 0

Moment at point A, (M) = P × L

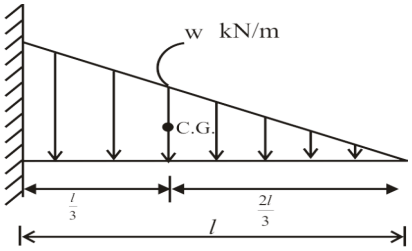
M = PL (clockwise)

607. A cantilever beam is subjected to uniformly varying load with intensity zero at free end and w kN/unit length at fixed end. The maximum bending moment is given by :

(a) $M = w\,l^2/8$ (b) $M = w\,l^2/6$
(c) $M = w\,l^2/4$ (d) $M = w\,l^2/2$

UPPCL AE 2019

Ans. (b) :



Maximum bending moment at fixed end

$= \frac{1}{2} \times w \times l \times \frac{l}{3}$

$\boxed{B.M = \frac{1}{6} w l^2}$ $\boxed{S.F. = \frac{wl}{2}}$

608. A cantilever beam of uniform cross-section is propped at the free end. The length of the beam is L. A total load W is acting on the beam in uniformly distributed manner over the entire span The support at free end is not allowed to settle. In that case the reaction at support will be

(a) $\frac{1}{2}W$ (b) $\frac{3}{4}W$
(c) $\frac{3}{16}W$ (d) $\frac{3}{8}W$

WBPSC AE (MCD) 1998

Ans. (d) : Total load due to UDL = W

$\delta_{UDL} = \delta_{R_B}$

$\frac{wL^4}{8EI} = \frac{R_B L^3}{3EI}$

$\boxed{R_B = \frac{3}{8}W}$ (as w.L = W)

609. Position of maximum bending moment in an arbitrarily loaded beam is were

(a) load is maximum (b) shear changes sign
(c) shear is maximum (d) slope is zero

WBPSC AE (Common) 1998

Ans. (b) : Position of maximum bending moment in an arbitrarily loaded beam is were SF changes sign.

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3. Bending and Shear Stress

610. Bending compressive and tensile stresses respectively are calculated based on :
(a) Net area and gross area
(b) Gross area and net area
(c) Net area in both cases
(d) Gross area in both cases

Nagaland PSC CTSE 2017 Paper–I

Ans. (b) : Bending compressive and tensile stresses respectively are calculated based on gross area & net area.

611. At the neutral axis of beam :
(a) The layers are subjected to maximum bending stress
(b) The layers are subjected to minimum bending stress
(c) The layers are subjected to compression
(d) The layers do not undergo any strain

Nagaland PSC CTSE 2017 Paper–I

Ans. (d) : At the neutral Axis of Beam, Bending stress is zero, so the Layer do not undergo any strain.

612. A beam of uniform strength has :
(a) Same cross-section throughout the beam
(b) Same bending stress at every section
(c) Same bending moment at every section
(d) Same shear stress at every section

Nagaland PSC CTSE 2017 Paper–I
(UPRVUNL AE 2015
RPSC A.En. 2013)

OR

A beam of uniform strength contains same -
(a) Bending Moment (b) Bending stress
(c) Deflection (d) Stiffness

RPSC AE 2018

Ans. (b) A beam of uniform strength contains same bending stress at every point and have different cross section.

613. When a cantilever beam is loaded at its free end the maximum compressive stress shall develop at :
(a) Bottom fibre (b) Top fibre
(c) Neutral axis (d) Centre of gravity

Nagaland PSC CTSE 2017 Paper–I

Ans. (a) : When a cantilever Beam is loaded at its free end, the maximum compressive stresses shall develop at Bottom fiber.

614. The bending equation is :
(a) $\frac{T}{J}=\frac{\tau}{r}=\frac{C\theta}{\ell}$ (b) $\frac{T}{J}=\frac{\tau}{r}=\frac{C\theta}{\ell}$
(c) $\frac{M}{I}=\frac{\sigma}{y}=\frac{E}{R}$ (d) $\frac{\ell}{r}=\frac{\tau}{J}=\frac{C\theta}{\ell}$

TNPSC AE 2019
Nagaland PSC CTSE 2017 Paper–I
Karnataka PSC AE 2015 Paper - II
(RPSC A.En. 2013)

Ans. (c) : Bending Equation (Flexure formula)

$$\boxed{\frac{M}{I}=\frac{\sigma}{y}=\frac{E}{R}}$$

615. In a simple bending of beam, the stress in the beam varies :
(a) Linearly (b) Parabolically
(c) Hyperbolically (d) Elliptically

Nagaland PSC CTSE 2017 Paper–I

Ans. (a) : In a simple Bending of Beam, the stresses in the Beam varies Linearly.

616. Rectangular beam is loaded transversely the maximum tensile stress is developed on the :
(a) Top layer (b) Bottom layer
(c) Neutral axis (d) Every cross-section

Nagaland PSC CTSE 2017 Paper–I

Ans. (b) : When a Rectangular Beam is loaded transversely, the maximum tensile stresses is developed on the bottom layer.

617. When a rectangular beam is loaded transversely the maximum compressive stress in developed on the :
(a) Top layer (b) Neutral axis
(c) Bottom layer (d) Every cross-section

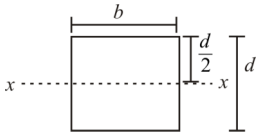
Nagaland PSC CTSE 2017 Paper–I

Ans. (a) : When a Rectangular Beam is loaded transversely, the maximum compressive stress is developed on the Top Layer.

618. The moment of inertia of a rectangle of width b and depth d, about an axis XX passing through its centre of gravity and parallel to the ends, is given as
(a) $I_{xx} = db^3/24$ (b) $I_{xx} = bd^3/12$
(c) $I_{xx} = db^3/64$ (d) $I_{xx} = db^3/32$

RPSC AE 2013

Ans : (b)

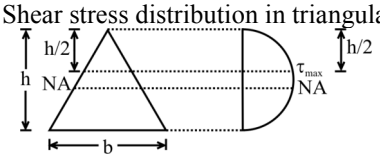


$$I_{xx}=\frac{bd^3}{12}$$

619. Shear stress in a beam having triangular cross-section is maximum at
(a) extreme top fibre (b) mid depth
(c) neutral axis (d) extreme bottom fibre

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (b) Shear stress distribution in triangular section:



$$\tau=\frac{12S}{bh^3}\left(hy-y^2\right)$$

for τ_{max} $\frac{d\tau}{dy}=0$

$\therefore \left(h-2y\right)=0$ $\quad y=\frac{h}{2}$

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620. The ratio of maximum and average shear stresses on a rectangular section is

ISRO Scientist/Engineer 2018
RPSC VPITI 2012

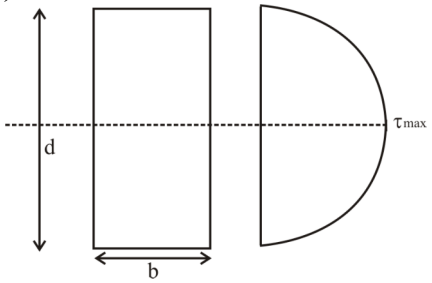
OR

The ratio of the maximum shear stress to average shear stress in a beam with rectangular cross-section is:

(a) 0.5 (b) 1.0
(c) 1.5 (d) 2.0

ISRO Scientist/Engineer 2013
UPPSC State Eng. A.E. 2007 (I)
WBPSC AE (Common) 1998

Ans : (c)



$$\tau = \frac{6S}{bd^3}\left(\frac{d^2}{4} - y^2\right)$$

Where S = shear force

for τ_{max}, $\frac{d_z}{d_y} = 0 \Rightarrow y = 0$

$$\tau_{max} = \frac{6S}{bd^3} \times \frac{d^2}{4} = \frac{3}{2}\frac{S}{bd}$$

we know

$$\frac{S}{bd} = \tau_{avg}$$

$$\frac{\tau_{max}}{\tau_{avg}} = 1.5$$

621. The maximum bending stress induced in a steel wire of modulus of elasticity 100 kN/mm^2 and diameter 2 mm when wound on a drum of diameter 2 m is approximately equal to

(a) 50 N/mm^2 (b) 100 N/mm^2
(c) 200 N/mm^2 (d) 400 N/mm^2

RPSC POLY. TECH. LECT. 2014

Ans : (b) Given, E=100 KN/mm^2;

$$\text{Radius of drum} = \frac{D}{2} = \frac{2000}{2} = 1000mm$$

$$\text{Distance of outer fiber in wire } y = \frac{d}{2} = \frac{2}{2} = 1\ mm$$

$$\text{Bending equation} = \frac{f}{y} = \frac{E}{R}$$

$$f = \frac{E.y}{R} = \frac{100 \times 1000 \times 1}{1000}$$

$$\boxed{f = 100\,N/mm^2}$$

622. The maximum magnitude of shear stress due to shear force F on a rectangular section of area A at the neutral axis is .....

(a) $\frac{F}{A}$ (b) $\frac{F}{2A}$
(c) $\frac{3F}{2A}$ (d) $\frac{2F}{3A}$

(UKPSC AE (Paper–I) 2007)

Ans : (c) $\tau_{max} = 1.5\tau_{avg}$ for rectangular section

$$\tau_{max} = 1.5 \times \frac{F}{A} = \frac{3F}{2A}$$

623. Which of the following represents a beam having maximum bending stress at all cross sections ?

(a) Beams of adjusting length
(b) Beams of varying strength
(c) Beams of uniform strength
(d) Beams of maximum shear strength

Rajasthan AE (Nagar Nigam) 2016

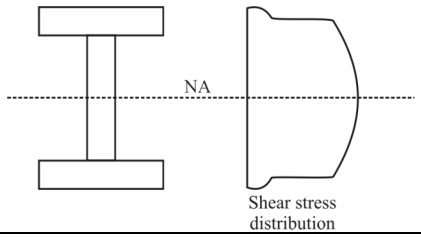
Ans. (c) : Hence, in beam of uniform cross section there is a considerable waste of materials. When a beam is suitably designed such that extreme fibre are loaded to the maximum permissible stress P_{max} by varying the cross section it will be known as a beam of uniform strength.

624. For a given shear force across a symmetrical 'I' section, the intensity of shear stress is maximum at the -

(a) junction of the flange and the web, but on web
(b) junction of the flange and the web, but on the flange
(c) centroid of the section
(d) extreme fibres

RPSC AE 2018

Ans. (c) Shear force across a symmetrical 'I' section, the intensity of shear stress is maximum at the centroid of the section.



625. Maximum shear stress in a triangular lamina is:

(a) 1.33 times the average shear stress at y = h/3
(b) 1.5 times the average shear stress at y = h/3
(c) 1.33 times the average shear stress at y = h/2
(d) 1.5 time the average shear stress at y = h/2

GPSC AE Class (1&2) Paper - 2 2017

Ans. (d) Maximum shear stress of a beam triangular section occurs at middle of height.
Maximum shear stress = 1.5 shear stress at neutral axis.

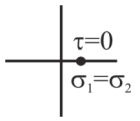
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626. The radius of Mohr's circle is zero when the state of stress is such that

(a) Shear stress is zero
(b) There is pure shear
(c) There is no shear stress but identical direct stresses in two mutually perpendicular directions.
(d) There is no shear stress but equal direct stresses, opposite in nature, in two mutually perpendicular directions.

GPSC AE (CLASS 1 & 2) 2019

Ans : (c) : When $\boxed{\text{Radius} = \tau_{max} = \frac{\sigma_1 - \sigma_2}{2}}$



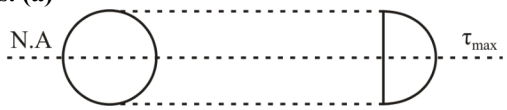
$o = \frac{\sigma_1 - \sigma_2}{2}$ $\boxed{\sigma_1 = \sigma_2}$

627. The maximum shear stress across a circular section is

(a) $\left(\frac{4}{3}\right)$ Average shear stress
(b) $\left(\frac{3}{2}\right)$ Average shear stress
(c) $\left(\frac{5}{4}\right)$ Average shear stress
(d) $\left(\frac{9}{5}\right)$ Average shear stress

ESE 2019
ISRO Scientist/Engineer 2015

Ans. (a)



In a circular section maximum shear stress.

$\left[\tau_{max} = \frac{4}{3}\tau_{avg}\right]$

- Maximum shear stress in rectangular section

$\left[\tau_{max} = \frac{3}{2}\tau_{avg}\right]$

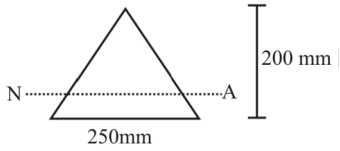
628. A beam of triangular cross-section is subjected to a shear force of 50 kN. The base width of the section is 250 mm and the height is 200 mm. The beam is placed with its base horizontal. The shear stress at neutral axis will be nearly.

(a) 2.2 N/mm^2 (b) 2.7 N/mm^2
(c) 3.2 N/mm^2 (d) 3.7 N/mm^2

ESE 2019

Ans. (b)

Shear force 'V' = 50 kN
B = 250mm
H = 200mm



For triangular section

$\tau_{NA} = \frac{4}{3}\left[\tau_{avg}\right]$

$\tau_{NA} = \frac{4}{3}\left[\frac{V}{Area}\right]$

$\tau_{NA} = \frac{4}{3}\left[\frac{\left(50\times10^3\right)}{\left(\frac{1}{2}\times250\times200\right)}\right]$

$\tau_{NA} = \frac{4}{3}\left[\frac{50\times10^3}{25000}\right]$

$\tau_{NA} = \frac{4}{3}\times2$

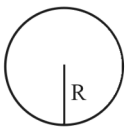
$\boxed{\tau_{NA} = 2.7\text{MPa or N/mm}^2}$

629. A circular ring of radius 42cm is cut and bent into the form of a rectangle whose sides are in the ratio of 6:5. The small side of the rectangle is :

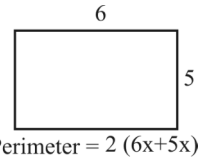
(a) 80 cm (b) 30 cm
(c) 120 cm (d) 60cm

ISRO Scientist/Engineer 2019

Ans : (d) : Circumference of circle = 2πR
Perimeter of rectangle = 2 (*l*+b)



Perimeter = 2πR



Perimeter = 2 (6x+5x)

$2\pi R = 2\times\left(6x + 5x\right)$

$2\times\frac{22}{7}\times42 = 2\times11x$

x = 12 cm

The small side of rectangle is 5x = 5 × 12 = 60 cm

630. The moment of inertia of a rectangular section 3 cm wide and 4 cm deep about an axis passing through its centre of gravity and parallel to width is

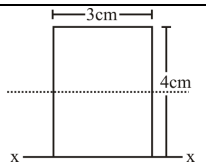
(a) 9 cm^4 (b) 12 cm^4
(c) 16 cm^4 (d) 20 cm^4

ISRO Scientist/Engineer 2017

Ans : (c)

Width of section (b) = 3cm
Depth of section (d) = 4cm

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So, the moment of inertia of the section about an axis passing through its centre of gravity and parallel to width is $\left[I_{xx} = \frac{bd^3}{12} \right]$

$$I_{xx} = \frac{3 \times (4)^3}{12}$$

$$I_{xx} = \frac{3 \times 16 \times 4}{12}$$

$$\boxed{I_{xx} = 16 cm^4}$$

631. Modulus of rigidity is the ratio of

(a) Linear stress to lateral strain
(b) Lateral strain to linear strain
(c) Linear stress to linear strain
(d) Shear stress to shear strain

ISRO Scientist/Engineer 2017

Ans : (d) :

■ **Modulus of Rigidity**

It is the ratio of shear stress to the displacement per unit sample length (Shear Strain).

$$\left[\text{Modulus of Rigidity} = \frac{\text{Shear Stress}}{\text{Shear Strain}} \right]$$

$$G = \frac{\tau}{\phi}$$

$$\text{Young's Modulus} = \frac{\text{Longitudinal Stress}}{\text{Longitudinal Strain}}$$

$$Y = \frac{F/A}{\Delta L/L}$$

$$\left[Y = \frac{FL}{A \Delta L} \right]$$

632. Bending equation is truly applicable for

(a) Hinge
(b) span with constant moment
(c) shear span
(d) ductile materials

Vizag steel MT 2017

Ans. (d) : The assumption consider to derive the bending equation is-

The Young's modulus in tension and compression is equal.

Note-Most of ductile metal have equal Young's modulus in both tension and compression.

633. The ratio of moment of inertia to extreme fibre distance is

(a) Poisson ratio (b) Shear stress
(c) Section modulus (d) None of the above

GPSC Poly. Tech. Lect. 2016

Ans. (c) :

$$\text{Section modulus (z)} = \frac{\text{moment of inertia}}{\text{extreme fibre distance from the neutral axis}}$$

$$\boxed{z = \frac{I}{y}}$$

634. Maximum shear stress in rectangular section is

(a) $\frac{2V}{bd}$ (b) $\frac{3V}{bd}$
(c) $\frac{3V}{2bd}$ (d) $\frac{2V}{3bd}$

GPSC Poly. Tech. Lect. 2016

Ans. (c) : For rectangular section beam, the shear stress distribution is parabolic and maximum shear stress is at neutral axis of the section.

$\tau_{max} = \frac{3}{2}\frac{V}{bd} = 1.5$ times the average shear stress.

635. If a circular shaft is subjected to a torque 'T' and bending moment 'M', the ratio of maximum bending stress to maximum shear stress is

(a) $\frac{M}{T}$ (b) $\frac{M}{2T}$
(c) $\frac{2M}{T}$ (d) $\frac{2M}{2T}$

GPSC Poly. Tech. Lect. 2016
Punjab PSC Sub divisional Engineer 2014
JPSC AE 2009

Ans. (c) : Max. bending stress = $\frac{32M}{\pi d^3}$

Max. shear stress = $\frac{16T}{\pi d^3}$

So, when you divide = $\frac{\frac{32M}{\pi d^3}}{\frac{16T}{\pi d^3}} = \frac{2M}{T}$

636. Bending compressive and tensile stresses respectively are calculated based on

(a) Bending moment
(b) Shear force
(c) Shear force and bending moment
(d) Torsion

UPSC JWM 2017

Ans. (a) : Bending compressive and tensile stress are calculated bond on bending moment.

$$\frac{M}{I} = \frac{\sigma}{y}$$

637. The neutral axis of a beam is subjected to ..... stress.

(a) zero
(b) maximum tensile
(c) minimum tensile

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(d) maximum compressive

GPSC R&B AE 2016

Ans. (a) : The neutral axis is the basis through a beam where the stress is zero, that is there is neither tension nor compression.
On applying load on beam, upper portion of NA goes to compression and bottom on is in tension.

638. The section modulus (z) of rectangular beam having width 'b' and depth 'd' in uniform and isotropic section is:

(a) $\frac{bd^3}{6}$ (b) $\frac{bd^2}{6}$

(c) $\frac{bd^3}{12}$ (d) $\frac{b^2d}{6}$

GPSC R&B AE 2016

Ans. (b & d) : Section modulus (z) = $\frac{I}{R}$

moment of inertia of rectangular section

$$I = \frac{bd^3}{12} \qquad R = \frac{b}{2} \text{ or } \frac{d}{2}$$

$$Z = \frac{bd^3}{12\left(d/2\right)} \text{ or } \frac{db^3}{12\left(b/2\right)}$$

$$\boxed{Z = \frac{bd^2}{6} \text{ or } \frac{db^2}{6}}$$

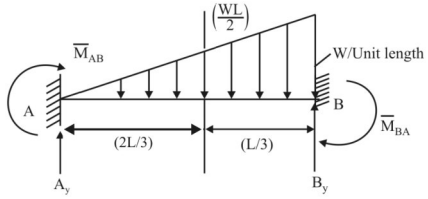
639. A fixed beam AB is subjected to a triangular load varying from zero at end A to W per unit length at end B, the ratio of fix end moments at A to B will be

(a) 1/2 (b) 2/3

(c) 4/9 (d) 1/4

MPPSC AE 2014

Ans. (b)



$$\boxed{A_y + B_y = \frac{1}{2} W \times L} \quad ---(1)$$

$\Sigma M_B = 0$

$$A_y \times L = \frac{WL}{2} \times \frac{L}{3} \Rightarrow A_y = \frac{WL}{6}$$

$$A_y + B_y = \frac{W \times L}{2}$$

$$B_y = \frac{WL}{2} - \frac{WL}{6}$$

$$B_y = \frac{WL}{3}$$

We know that,

$$\bar{M}_{AB} = \frac{-WL^2}{30}; \quad \bar{M}_{BA} = \frac{WL^2}{20}$$

$$\therefore \boxed{\frac{\bar{M}_{AB}}{\bar{M}_{BA}} = \frac{-WL^2/30}{WL^2/20} = \frac{2}{3}}$$

640. In which of the following cases, there exists the situation of Pure Bending in some part of the beam or along the entire beam?

(i) A simply supported beam subjected to two equally spaced downward concentrated loads

(ii) A simply supported beam subjected to two equally spaced opposite moments

(iii) A cantilever beam subjected to clockwise moment at free end

(iv) An overhanging beam with both side equal overhangs subjected to different concentrated loads at both the free ends

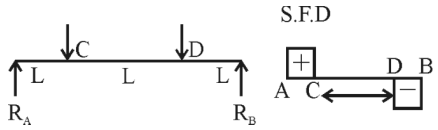
(a) (i) and (iv) (b) (i), (ii) and (iv)

(c) (i), (ii) and (iii) (d) (i) and (iii)

Gujarat PSC AE (N.W.R.) 2020

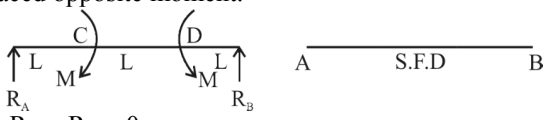
Ans. (c) :

- A simply supported beam subjected to two equally spaced downward concentrated loads.



Part CD of beam AB is in pure bending condition.

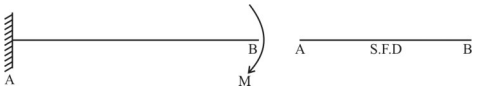
- A simply supported beam subjected to two equally spaced opposite moment.



As $R_A = R_B = 0$

Beam AB is in pure bending condition.

- A cantilever beam subjected to clockwise moment at free end.



Beam AB is in pure bending condition.

641. Section modulus of Hollow Circular Section having external Dia. (D) and internal Dia. (d) is

(a) $\frac{\pi(D-d)^4}{32}$ (b) $\frac{\pi\left(D^4-d^4\right)}{36D}$

(c) $\frac{\pi\left(D^4-d^4\right)}{32D}$ (d) $\frac{\pi(D-d)^4}{36}$

Gujarat PSC AE 2017

Ans : (c) Section modulus $Z = \frac{I}{Y}$

Moment of inertia of hollow circular section

$$= \frac{\pi\left(D^4-d^4\right)}{64}$$

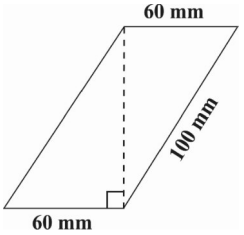
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$$Y = \frac{D}{2}$$

$$Z = \frac{\frac{\pi\left(D^4 - d^4\right)}{64}}{\frac{D}{2}}$$

$$Z = \frac{\pi\left(D^4 - d^4\right)}{32D}$$

642. Value of Elastic Section Modulus for the given section about horizontal centroidal axis is–



(a) 5000000 (b) 2560000
(c) 100000 (d) 64000

Gujarat PSC AE (N.W.R.) 2020

Ans. (d) :

$$\sqrt{100^2 - 60^2} = 80$$

$$I_C = \frac{1}{12}a^3 b(\sin\theta)^3$$

$$= \frac{1}{12} \times (100)^3 \times 60(0.8)^3$$

$$= 2560000$$

$$z = \frac{I_C}{y} = \frac{2560000}{40} = 64000$$

643. The moment of inertia of a circular section, about its axis, is given be:

(a) $\frac{\pi(D)^4}{64}$ (b) $\frac{\pi(D)^4}{32}$

(c) $\frac{\pi(D)^4}{16}$ (d) $\frac{\pi(D)^4}{8}$

KPSC AE 2016

Ans. (a) : Moment of inertia of different section are as follows –

i) Rectangular Section	$A = bh$ $I_x = \frac{1}{12}bh^3, I_y = \frac{1}{12}hb^3,$
ii) Triangular Section	$A = \frac{1}{2}bh, I_x = \frac{1}{36}bh^3$
iii) Circular Section	$A = \pi r^2$ $I_x = I_y = \frac{\pi}{64}D^4$

644. Radius of gyration of two rectangular columns is in a ratio of 3 :1. What shall be the ratio of their depths ?

(a) 3 : 1 (b) $1:\sqrt{3}$
(c) 1 : 3 (d) $\sqrt{3}:1$

Rajasthan JEn (Degree) 2016

Ans. (a) : Given, $\frac{r_1}{r_2} = \frac{3}{1}$

$$\frac{\sqrt{\frac{I_1}{A_1}}}{\sqrt{\frac{I_2}{A_2}}} = \frac{3}{1}$$

$$\frac{\sqrt{\frac{b_1 d_1^3}{12 b_1 d_1}}}{\sqrt{\frac{b_2 d_2^3}{12 b_2 d_2}}} = \frac{3}{1}$$

$$\frac{\sqrt{d_1^2}}{\sqrt{d_2^2}} = \frac{3}{1}$$

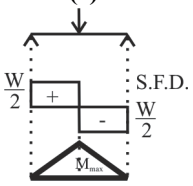
$$\boxed{\frac{d_1}{d_2} = \frac{3}{1}}$$

645. Transverse shear applied to a beam normally results in

(a) Normal stress only
(b) Shear stress only
(c) Normal and shear stress
(d) Axial stress only

GPSC AE (CLASS 1 & 2) 2019

Ans : (c) : Transverse shear applied to a



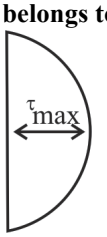
$$\frac{wl}{4} = M_{max}$$

$$\boxed{\tau = \frac{V}{bd}}$$

$$\boxed{\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}}$$

$$\boxed{\sigma = M\frac{y}{I}}$$

646. The following shear stress distribution profile belongs to which type of section?



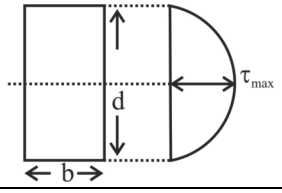
(a) I-section and rectangular beam
(b) T-section only

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(c) Rectangular and circular section
(d) Rectangular only

GPSC AE (Class 1 & 2) 2019

Ans : (c) :



647. The radius of gyration of section of area A and moment of inertia from the axis is :

(a) $\frac{A}{I}$ (b) $\frac{I}{A}$

(c) $\sqrt{\frac{I}{A}}$ (d) $\sqrt{\frac{A}{I}}$

ISRO Scientist/Engineer 2013
UKPSC AE (Paper–I) 2007

Ans : (c)

r is radius of gyration

$A \rightarrow$ area of cross section

$I \rightarrow$ moment of inertia

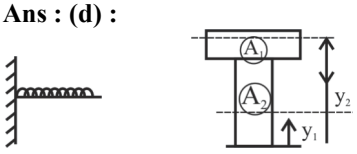
$\therefore \quad \boxed{r = \sqrt{\frac{I}{A}}}$

648. A cantilever beam of T cross-section carries uniformly distributed load. Where does the maximum magnitude of bending stress occur?

(a) At the top of cross-section
(b) At the junction of flange and web
(c) At the mid-depth point
(d) At the bottom of the section

GPSC AE (CLASS 1 & 2) 2019

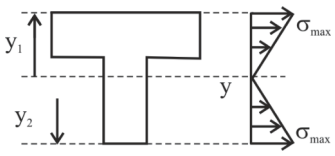
Ans : (d) :



$\boxed{\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}}$ $\boxed{\sigma \times y}$

As area above has flange section thus centre of it will be $(\bar{y})$ will be more nearer to that flange more area therefore ($y_2 >> y_1$)

$\boxed{\sigma_{max} \times \text{at } y_2 >>> \sigma_{max} \times \text{at } y_1}$



649. For a triangular cross section having horizontal base, Ratio of shear stress at mid height of the section to shear stress at neutral axis of the section is

(a) 3/4 (b) 8/9
(c) 9/8 (d) 4/3

Gujarat PSC AE (N.W.R.) 2020

Ans. (c) : Shear stress at mid height $= \frac{3}{2} \times \frac{S}{\frac{1}{2}bh}$

Shear stress at neutral axis $= \frac{8}{3} \times \frac{S}{bh}$

$$\frac{(\text{Shear stress})_{\text{mid height}}}{(\text{shear stress})_{\text{neutral axis}}} = \frac{\frac{3}{2} \times \frac{S}{\frac{1}{2}bh}}{\frac{8}{3} \times \frac{S}{bh}} = \frac{9}{8}$$

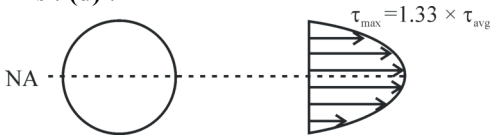
650. The ratio of average shear stress to maximum shear stress for a circular section is

(a) 2 (b) $\frac{2}{3}$

(c) $\frac{1}{2}$ (d) $\frac{3}{4}$

ASSAM PSC CCE (PRE) 2015

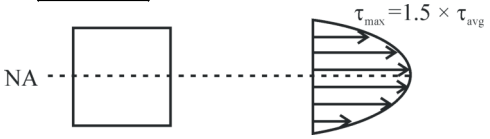
Ans : (d) :



For circular section

$$\tau_{max} = \frac{4}{3}\tau_{avg}$$

$$\boxed{\frac{\tau_{avg}}{\tau_{max}} = \frac{3}{4}}$$



For square section

$$\tau_{max} = \frac{3}{2}\tau_{avg}$$

$$\boxed{\frac{\tau_{avg}}{\tau_{max}} = \frac{2}{3}}$$

651. What will be the depth of a beam that is 20 cm wide and is subjected to shearing force of 10,000 kg. It may be noted that in this case the maximum shearing stress is 30 kg/cm^2.

(a) 15cm (b) 20 cm
(c) 25 cm (d) 30 cm

GESCOM (KARNATAKA) AE 2011 2011

Ans : (c) : Maximum shear stress of rectangle $= \frac{Fd^2}{8I}$

$$30 = \frac{10,000 \times d^2}{8 \times \frac{bd^3}{12}}$$

$$30 = \frac{10,000 \times 12}{8 \times 20 \times d}$$

$d = 25\text{cm}$

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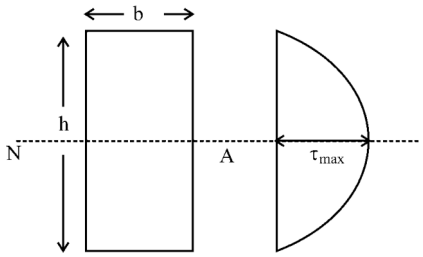
652. If V denotes the shear force acting at a particular section along the span of a beam having a rectangular cross-section of width b and depth h, then the shear stress at the neutral axis is :

(a) 0 (b) $\frac{3V}{2bh}$

(c) $\frac{3V}{4bh}$ (d) $\frac{V}{bh}$

KPSC AE 2020

Ans. (b) :



Rectangular cross-section **Shear stress Distribution**

Maximum shear stress occurs at the neutral axis.

$\tau_{max} = \frac{3}{2}\frac{v}{bh}$

$\tau_{max} = 1.5\ \tau_{avg}$

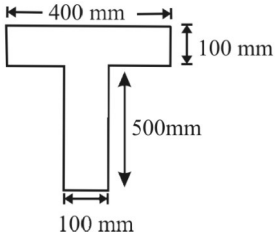
where,

v = shear force

b = width of section

h = depth of section

653. In the 'T' section as shown in figure, distance of neutral axis from top is–

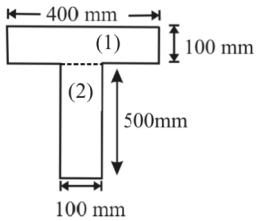


(a) 100 mm (b) 200 mm

(c) 300 mm (d) 216 mm

UKPSC AE 2013 Paper–I

Ans : (d)



Neutral axis from top flange

$$= \frac{400\times100\times50 + 500\times100\times350}{400\times100 + 500\times100}$$

$\bar{y} = 216\ mm$

654. Section modulus for hollow circular section of outer diameter 'D' and inner diameter 'd' is –

(a) $\frac{\pi}{64D}\left(D^4 - d^4\right)$ (b) $\frac{\pi\left(D^4 - d^4\right)}{32D}$

(c) $\frac{\pi\left(D^3 - d^3\right)}{64D}$ (d) $\frac{\pi\left(D^3 - d^3\right)}{32D}$

GPSC R&B AE 2016

UTTRAKHAND AE 2013

Ans : (b) Section modulus $= \frac{I}{y}$

$$I_{hollow} = \frac{\pi}{64}\left[D^4 - d^4\right]$$

$$y = \frac{D}{2}$$

$$\therefore \quad Z = \frac{I}{y} = \frac{\pi\left[D^4 - d^4\right]}{32D}$$

655. The shear stress on a beam section is maximum–

(a) On the extreme bottom surface fibers

(b) On the extreme top surface fibers

(c) At the free ends

(d) At the neural axis of the section

UKPSC AE 2013 Paper–I

Or

The shear stress on a beam section is maximum:

(a) At the centroid of the section

(b) On the extreme free surface fibers

(c) At the free edges

(d) At the neutral axis but not at the centre of the gravity

RPSC GWD A. En. 2014

RPSC ACF LECTURE 2011

Ans : (d) Shear stress a point is $q = \frac{FA\bar{y}}{Ib}$

Where $\bar{y}$ is distance from neutral axis.

q is maximum for neutral axis and zero for upper and lower edges.

656. A beam is said to have been subjected to a pure bending when :

(a) shear force is constant throughout the span

(b) load is applied as UDL throughout the span

(c) load is applied at the mid span only

(d) none of the above

(UKPSC AE (Paper–I) 2007)

Ans : (d) When shear force is zero and bending moment is constant, then the beam is said to have subjected to pure bending.

657. A beam of rectangular cross-section is 100 mm wide and 200 mm deep. If the section is subjected to a shear force of 20 kN, then the maximum shear stress in the section is

(a) 1 N/mm^2 (b) 1.125 N/mm^2

(c) 1.33 N/mm^2 (d) 1.5 N/mm^2

RPSC POLY. TECH. LECT. 2014

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Ans : (d) $\tau = 1.5\,\tau_{avg} = 1.5 \times \frac{F}{A}$,

Given, F = 20 kN; Area = 200 × 100

$\tau_{max} = 1.5 \times \frac{20 \times 1000}{200 \times 100} = 1.5\ N/mm^2$

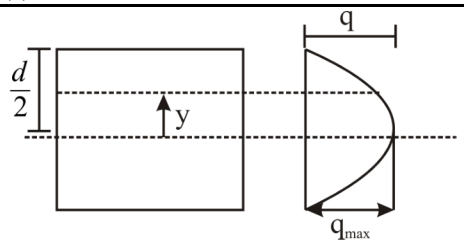
658. Maximum shear stress for rectangular section is (total transverse shear at the section = V; entire cross–sectional area = A)

(a) $\frac{3V}{A}$ (b) $\frac{2V}{3A}$

(c) $\frac{3V}{2A}$ (d) $\frac{V}{2A}$

MPPSC AE 2016

Ans : (c)



$q_{max} = 1.5 \times q_{av}$

Distribution of shear stress is parabolic in nature.

659. An important assumption in the derivation of the equation M/I=f/Y= E/R (notations have their usual meaning) for a beam cross section under flexure is :

(a) Neutral axis passes through its centroid
(b) Neutral axis lies above the centroid
(c) Neutral axis lies below the centroid
(d) None of these

(BPSC AE Paper V 2006)

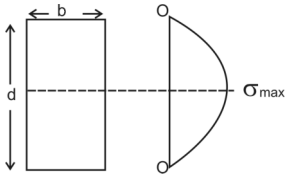
Ans : (a) In derivation of Bending equation $\frac{M}{I} = \frac{f}{y} = \frac{E}{R}$, is an important assumption that its neutral axis passes through its centroid.

660. Shear stress at a section in a beam is zero at

(a) extreme top and bottom fiber
(b) neutral axis
(c) mid depth
(d) none of these

[UPPSC State Eng. A.E. 2004 (I)]

Ans : (a)



The shear stress at a distance y from N.A. is given by

$$\tau = \frac{S(A\bar{Y})}{Ib}$$

S = shear force

$A\bar{Y}$ = (NA) moment of area above the level of y above NA

$\bar{Y}$ = Distance of centroid of area from NA

I =MOI about NA

since shear force is an internal resistance which is zero at outer surfaces. Hence shear stresses at top and bottom will be zero.

661. The variation of shear stress at a section of a beam is

(a) constant (b) linear
(c) parabolic (d) cubic

(UPRVUNL AE 2015)
(UKPSC AE (Paper–I) 2007)
(UPPSC State Eng. A.E. 2004(I))
(UPPSC State Eng. A.E. 2003)

Ans : (c)

(i) Rectangular section:

$$\tau = \frac{6S}{bd^3}\left(\frac{d^2}{4} - y^2\right)$$ ..... Parabolic

(ii) Circular section:

$$\tau = \frac{4S}{3\pi R^4}\left(R^2 - y^2\right)$$ ..... Parabolic

(ii) Triangular section:

$$\tau = \frac{12S}{bh^3}\left(hy - y^2\right)$$ ..... Parabolic

where S= shear force.

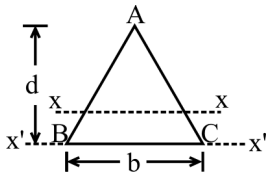
662. Moment of inertia of a triangular cross-section of height d and base width b about its centroid axis parallel to base is:

(a) $\frac{bd^3}{36}$ (b) $\frac{bd^3}{24}$

(c) $\frac{bd^3}{12}$ (d) $\frac{bd^3}{6}$

[UPPSC State Eng. A.E. 2007 (II)]

Ans : (a)



$I'_{x'x'} = \frac{bd^3}{12}$, $I_{xx} = \frac{bd^3}{36}$

663. A solid shaft of diameter 'D' carries a twisting moment that develops maximum shear stress. If the shaft is replaced by a hollow one of outside diameter 'D' and inside diameter $\frac{'D'}{2}$, then the maximum shear stress will be-

(a) 1.067 τ (b) 1.143 τ
(c) 1.33 τ (d) 2 τ

Maharashtra PSC (Mains) 2017 Paper–I

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Ans. (a) : $\tau = \frac{16T}{\pi D^3}$, $\tau_1 = \frac{T \times \frac{D}{2}}{\frac{\pi\left[D^4 - \left(\frac{D}{2}\right)^4\right]}{32}} = \frac{16 \times 16 \times T}{15 \times \pi D^3}$

$T = \frac{\tau \times \pi d^3}{16}$, $\tau_1 = \frac{16 \times 16 \times \pi D^3}{15 \times \pi D^3 \times 16} \times \tau = 1.067\,\tau$

664. A fixed beam of uniform section is carrying a point load at its mid span. If the moment of inertia of the middle half length is reduced to half of its previous value, then the fixed end moments will-

(a) Decrease (b) Increase
(c) Change their direction (d) Remain constant

OPSC AEE 2015 PAPER–I

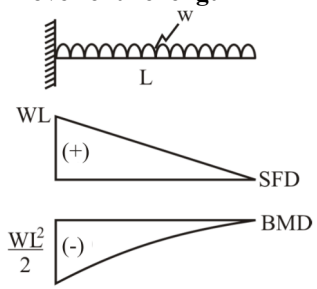
Ans. (b) : A fixed beam of uniform section is carrying a point load at its mid span. If the moment of inertia of the middle half length is reduced to half of its previous value, then the fixed end moments will increase.

665. The shear force and bending moment is zero at the free end of a cantilever beam, if it carries-

(a) Point load at the free end
(b) Point load at the middle of its length
(c) Uniformly distributed load over the whole length
(d) None of these

OPSC AEE 2015 PAPER–I

Ans. (c) : UDL over entire length

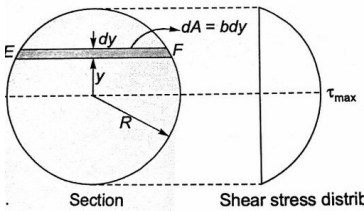


666. Maximum shear stress in a beam of circular section is _____ times the average stress-

(a) 1.25 (b) 1.33
(c) 1.5 (d) 1.67

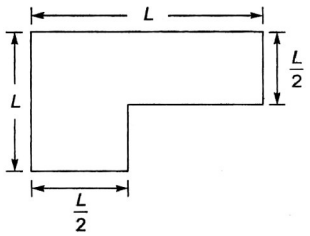
BPSC AE Pre 2018
BPSC AE Paper V 2001

Ans. (b) :



Maximum shear stress occurs at neutral axis & y = 0.
Maximum Shear stress = 4/3 [F/A].
Where, F/A is average shear stress.
The maximum shear stress distribution is 33% more than average shear stress

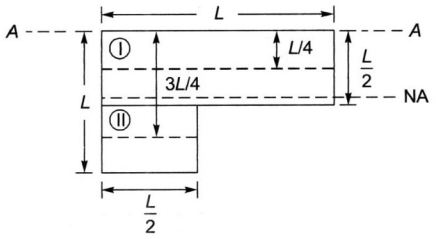
667. For the section shown in the figure, the neutral axis lines at :



(a) $\frac{L}{3}$ from the top (b) $\frac{L}{12}$ from the top
(c) $\frac{5L}{12}$ from the top (d) $\frac{7L}{12}$ from the top

[HPPSC 2016-II]

Ans. (c) :



At neutral axis $A\bar{x} = A_1\bar{x}_1 + A_2\bar{x}_2$
Let (I)–(I) be reference section:-

$$\left(L \times \frac{L}{2} + \frac{L}{2} \times \frac{L}{2}\right)\bar{x} = L \times \frac{L}{2} \times \frac{L}{4} + \frac{L}{2} \times \frac{L}{2} \times \left(\frac{L}{2} + \frac{L}{4}\right)$$

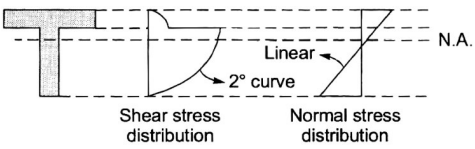
$$\Rightarrow \bar{x} = \frac{5L}{12}$$

668. For a beam having cross-section as T, which is a correct statement ?

(a) Shear stress variation is parabolic below Neutral axis and normal stress is linear below Neutral axis
(b) Shear stress variation is linear and normal stress is parabolic below Neutral axis
(c) Both shear and normal stresses are linear along the cross-section
(d) Both shear and normal stresses are parabolic along the cross-section

[HPPSC AE 2016]

Ans. (a) : Distribution of shear and normal stress in a T-beam is shown be low:-



669. A right-angled triangle with base 50 mm and height 75 mm is placed with its base horizontal, subjected to shear force of F. If moment of inertia is 'I' about the axis passing through centroid parallel to the base, then the shear stress at the axis passing through the centroid is:

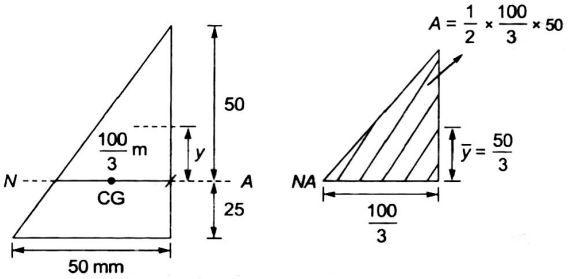
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(a) $416.67\frac{F}{I}$ (b) $449.67\frac{F}{I}$

(c) $316.67\frac{F}{I}$ (d) $492.2\frac{F}{I}$

Karnataka PSC AE 2017 Paper - II

Ans. (a) :



$$\tau_{NA} = \frac{VA\bar{y}}{Ib}$$

$$= \frac{Fx\left(\frac{1}{2}\times\frac{100}{3}\times 50\right)\times\left(\frac{50}{3}\right)}{I\times\frac{100}{3}}$$

$$= 416.67\frac{F}{I}$$

670. Consider following for bending stress induced in a beam :

(i) Directly proportional to modulus of elasticity

(ii) Inversely proportional to curvature and

(iii) Inversely proportional to radius of curvature.

Which one of the following is correct answer ?

(a) (i) only (b) (i) and (ii) only

(c) (i) and (iii) only (d) All

Rajasthan JEN (Degree) Shift - II 2016

Ans. (c) Bending equation-

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

where,

σ = bending stress in (N/m^2)

M = Moment about neutral axis (N-m)

y = Perpendicular distance to neutral axis in (m)

I = second moment of neutral axis in (m^4)

E = Modulus of Elasticity

R = radius of curvature

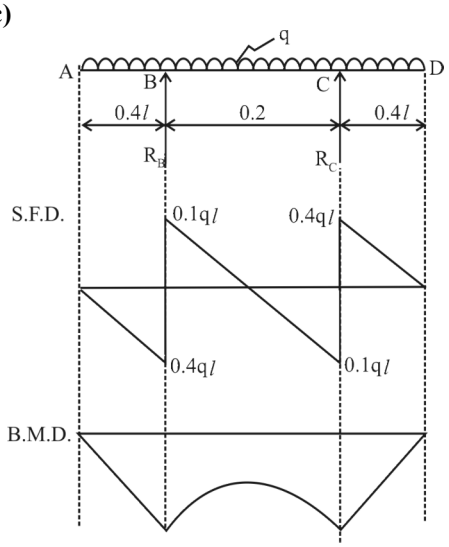
671. A simply supported prismatic beam of span L, and section modulus Z, has an overhang of 0.4 L is loaded with uniformly distributed load q, throughout the span. The maximum bending stress induced is :

(a) $\pm 0.02\frac{qL^2}{Z}$ (b) $\pm 0.04\frac{qL^2}{Z}$

(c) $\pm 0.08\frac{qL^2}{Z}$ (d) $\pm 0.085\frac{qL^2}{Z}$

Rajasthan JEN (Degree) Shift - II 2016

Ans. (c)



$$R_B + R_C = 0.5ql$$

B.M. at middle = $0.5ql\times\frac{l}{4} + 0.5ql\times 0.1l$

$$\left(-\frac{1}{3}+\frac{1}{20}\right)ql^2$$

B.M. at B or C = $0.4l\times q\times\frac{1}{2}\times 0.4l$

$$= \frac{2}{25}ql^2$$

Maximum bending moment at B or C

$$\frac{M}{I} = \frac{\sigma}{y}$$

$$\sigma = \frac{M}{Z}$$

$$= \frac{2}{25}\times\frac{ql^2}{Z}$$

$$= 0.08\frac{ql^2}{Z}$$

672. The ratio of average shear stress to maximum shear stress in a prismatic beam of rectangular cross section is :

(a) 3/4 (b) 4/3

(c) 3/2 (d) 2/3

Rajasthan JEN (Degree) Shift - II 2016

Ans. (d) $\tau_{max} = \frac{3}{2}\ \tau_{average}$

$$\frac{\tau_{average}}{\tau_{max}} = \frac{2}{3}$$

673. The intensity of u.d.l. which, when it acts over the entire span of 1 m of a cantilever beam of rectangular cross-section of width of 100 mm and depth 200m. would produce a maximum shear stress of 1.5 N/mm^2, is -

(a) 30 kN/m (b) 26.6 kN/m

(c) 20 kN/m (d) 36.6 kN/m

RPSC AE 2018, ESE 2015

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Ans. (c) : Length of cantilever beam = 1 m = 1000 mm
section of cantilever beam = b × d
= 100 mm × 200 mm
Maximum shear force in cantilever beam = wL
= 1000 w N

$$\tau_{max} = \frac{3}{2}\tau_{av}$$

$$\tau_{av} = \frac{2}{3}\tau_{max}$$

$$= \frac{2}{3}\times 1.5$$

$$\tau_{av} = 1.0 \text{ N/mm}^2$$

$$\frac{V}{b\times d} = \tau_{av}$$

$1000 w = 1 \times 100 \times 200$
$w = 20$ N/mm
$w = 20$ kN/m

674. Two beams of same material have equal cross-sectional area. If one beam has square cross-section and the other has circular cross section-
(a) Both the beam will be equally strong
(b) Circular section will be stronger
(c) Square section will be stronger
(d) Strength depends on loading condition

RPSC AE 2018

Ans. (c) :
Area of square section = Area of circular section

$$b\times b = \frac{\pi d^2}{4}$$

$$\left(\frac{b}{d}\right) = \frac{\sqrt{\pi}}{2} \quad .....(1)$$

$$\frac{(z)\text{square}}{(z)\text{circular}} = \frac{\left(\frac{b^4}{\frac{12}{b/2}}\right)}{\left(\frac{\pi d^4}{64}\right)\times\left(\frac{2}{d}\right)}$$

$$= \frac{b^3}{6\times\pi}\times\frac{32}{d^3}$$

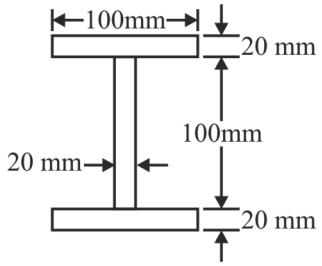
Put the value in eq^n (1)

$$= \frac{32}{6\pi}\times\left(\frac{\sqrt{\pi}}{2}\right)^3$$

$$\frac{(z)\text{square}}{(z)\text{circular}} = 1.18 \qquad [(z)\text{square} > (z)\text{circular}]$$

Square section will be more stronger.

675. The beam shown in the following figure is made of three plants nailed together. The nailing is done at a spacing of 25 mm. If the vertical shear in the beam is 640 N and the moment of inertia (I) is 16×10^{-6} m^4, the shear flow (in N/m) on the lower surface of the upper plank will be :



(a) 3600 (b) 5200
(c) 4500 (d) 4800
(e) 4550

CGPSC AE 2014 Shift–I

Ans. (d) : Shear flow (N/m) $q_w = \frac{V(D^2-d^2)}{8I}\times B$

$$= \frac{640(140^2-100^2)\times 100}{8\times 10^{-6}}$$

$$\boxed{q_w = 4800}$$

676. Which one of the following statements specifies shear flow?
(a) Flow of shear force along the beam
(b) It is the product of the shear stress at any level and the corresponding width b of the section
(c) Unbalanced force on any side of given section divided by area of section
(d) The deformation at any level due to sudden variation in shear stress

ESE 2019

Ans. (b) : shear flow = product of shear stress multiplied with width

$$\text{Shear flow} = \tau b = \frac{VA\bar{y}}{I}$$

Shear flow is a measure of force per unit length along a longitudinal axis of a beam.

677. The ratio of maximum shear stress to the average shear stress in case of a rectangular beam is equal to

APPSC AEE (Civil/Mechanical) 2012

OR

In the case of a rectangular beam subjected to a transverse shearing force, the ratio of maximum shear stress to average shear stress is
(a) 0.75 (b) 1.00
(c) 1.25 (d) 1.50

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